

## MetaTrader 5 trading platform

### MetaTrader 5 Trading Platform

The guide for installing, configuring and working with MetaTrader 5.

MetaTrader 5 is an advanced complete-cycle multi-asset trading platform for organizing brokerage services.

You traders can perform operations on different markets and exchanges using a single account. The flexible trading systems enables the highly efficient management of all asset types, including currencies, futures and stocks. In addition to essential trading features, you can offer professional technical and fundamental analysis tools to your traders, as well as the ability to use algorithmic trading robots.

The distributed platform architecture can handle high loads allowing companies to provide services to hundreds of thousands of online users. The system has built-in failover mechanisms and protection against possible external attacks.

With the advanced back-office system, you can organize the workflow to match your business expectations.

### Installation

Prepare hardware in accordance with the [system requirements](#). We recommend using at least two dedicated servers hosted by different providers to implement the main and the backup infrastructures. [Prepare the system](#) and proceed to MetaTrader 5 [installation](#). After that, [activate the license](#) and configure your platform.

Use the [quick deployment](#) feature to install additional platform components. If you need to move the platform or reconfigure the network infrastructure, use [console commands](#).

### Platform Setup

MetaTrader 5 trading platform administration tasks cover the management of trading servers and of all intermediate components (access points, backup servers, history data servers, etc.). The relevant setup data is provided in the appropriate sections.

Before you proceed to platform configuration, please read about the [basic trading concepts](#) and underlying principles explaining the [arrangement of price charts and Depth of Market features](#).



[General information about the Administrator](#)



[Built-in client management system](#) which aggregates



[ECN](#) to implement your own trading network with

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|                                                                                     |                                                                                                                                                                                     |                                                                                     |                                                                                                                                                            |                                                                                       |                                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     | terminal,<br>instructions for<br>working with<br>controls,<br><u>symbols</u> and<br>groups                                                                                          |                                                                                     | trader data:<br>trading<br>accounts,<br>personal<br>details,<br>documents,<br>history and<br>other<br>information                                          |                                                                                       | order matching<br>and liquidity<br>aggregated from<br>different sources                                                                     |
|    | The <u>home page</u><br>of each server<br>contains data on<br>its current<br>status: license<br>validity period,<br>number of<br>clients and<br>groups for<br>account<br>allocation |    | <u>Managers</u><br>users with<br>special<br>permissions to<br>manage the<br>system: edit<br>accounts,<br>configure<br>quotes,<br>generate<br>reports, etc. |    | <u>Funds and ETFs</u><br>create an<br>investment fund<br>with automated<br>performance<br>calculations,<br>commissions and<br>payments      |
|  | <u>Network</u><br>configure and<br>manage all<br>platform<br>components,<br>including<br>access, history<br>and backup<br>servers                                                   |  | <u>Orders</u> access<br>and manage<br>the history of<br>all orders on<br>the server as<br>well as all<br>active orders                                     |  | <u>Symbols</u><br>settings of the<br>financial<br>instruments<br>available to your<br>clients: currency<br>pairs, futures,<br>stocks, bonds |
|  | <u>Integration</u> with<br>instant<br>messaging<br>systems, mail<br>services, KYC<br>service<br>providers and<br>business<br>analytics<br>systems                                   |  | <u>Deals</u> work<br>with the history<br>of deals<br>performed by<br>your traders<br>and manage<br>balance<br>operations                                   |  | <u>Spreads</u><br>configure margin<br>discounts for<br>oppositely<br>directed positions<br>on correlated<br>financial<br>instruments        |
|  | <u>Security</u> set<br>up built-in<br>firewall,<br>external<br>Anti-DDoS<br>services and                                                                                            |  | <u>Positions</u><br>manage<br>traders'<br>positions and<br>access<br>historical                                                                            |  | <u>Charts</u> manage<br>data underlying<br>financial symbol<br>charts. Traders<br>analyze the charts<br>to make trading                     |

## MetaTrader 5 trading platform

security certificates; access general information about the platform security system

operations, from the initiating order up to the latest volume modification

decisions

Automations of routine operations based on predefined scenarios. Configure conditions and relevant actions to be triggered upon these conditions



Routing of incoming client trading requests depending on various conditions. Create and manage the rules



Ticks manage quotes delivered into the platform. The availability of the high-quality tick history enables the testing of trading strategies in close-to-real conditions



Time synchronize time, configure daylight saving time settings, set up server operation schedules



Gateways integrate the MetaTrader 5 platform with external trading systems. Access almost any global exchange via gateways



Synchronization with other MetaTrader 4 and MetaTrader 5 servers enables quick and efficient response to missing data issues



Holidays configure holidays for financial instrument groups and individual trading symbols



Data feeds to deliver into the platform quotes and news items from various data providers



Subscriptions start offering high-quality market data, personal manager services, and other paid service to your traders directly via client trading terminals.



Groups to categorize trading accounts for convenient and



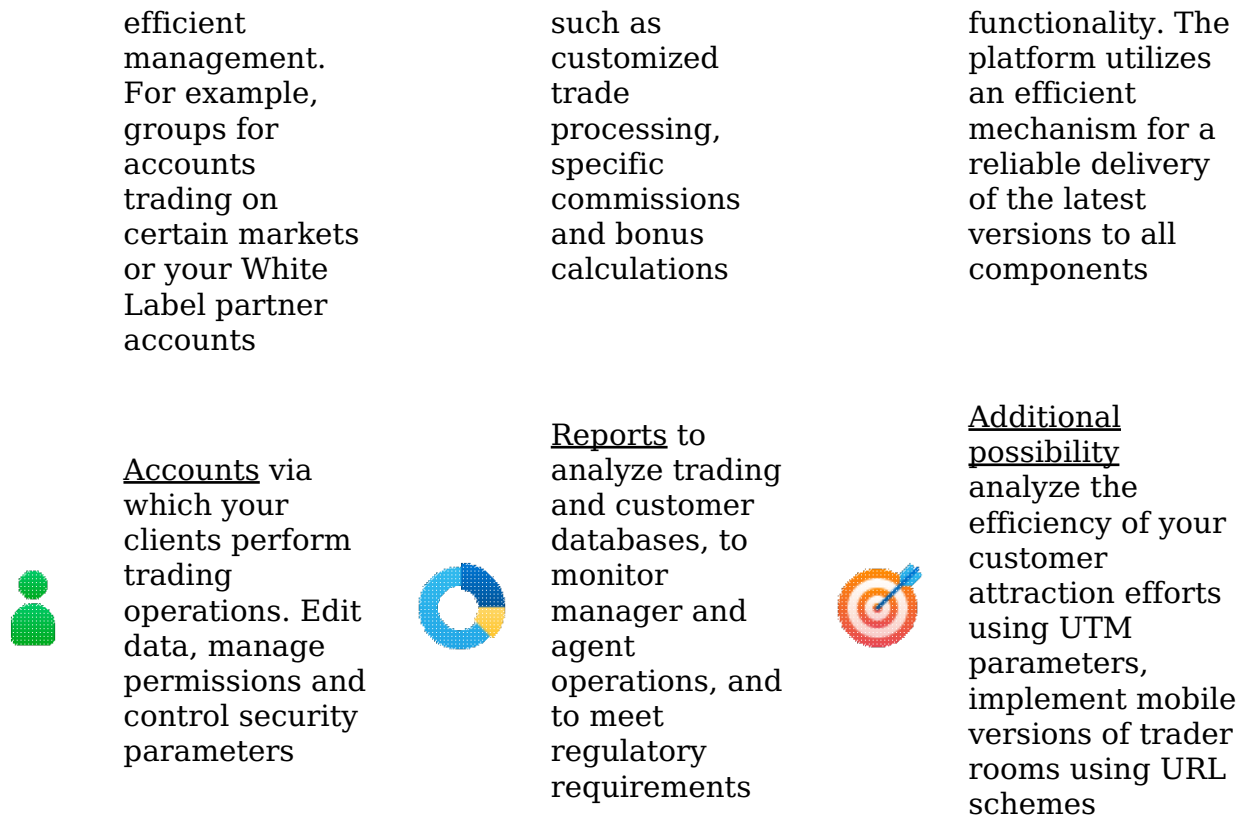
Plugins to extend the trading platform functionality,



Updates increase security and enhance MetaTrader 5 platform



## MetaTrader 5 trading platform



## Platform Components

The MetaTrader 5 trading platform is a distributed system. The main functions are implemented on different components to enable efficient load balancing, allowing companies to offer services to hundreds of thousands of traders.

The components interact via the main trading server which stores all platform settings.



### Trading Server

Two types of trading servers exist in the platform: one main server and additional ones. The main trading server serves trading operations and manages the entire system configuration.



### Access Server

Access servers act as proxy servers and firewalls for the MetaTrader 5 platform. They process incoming client connections, pack and send authorization requests to the trade server, and protect it from attacks and overload.



### History Server

The history server processes price and news data. It receives and filters quotes from gateways and data feeds, as well as stores and provides trading history in the form of one-minute bars and ticks to other platform components.

## MetaTrader 5 trading platform

### components



#### Backup Server

It creates backup copies of data that can be used in case of failure of a trading or history server. One or more backup servers are associated with each trading server. They synchronize their states with the main server in real-time mode, and thus you can instantly switch to them in case of failure. Also, backup servers create periodic copies of databases which can be used to restore the system state as of the required day



#### Data Feeds

Deliver quotes and news into the platform. The trading platform includes more than a dozen out-of-the-box data feeds which enable connections to popular data providers



#### Gateways

The gateways integrate the MetaTrader 5 platform with external trading systems and liquidity providers. They can forward trading operations to external systems, as well as broadcast quotes and news from them



#### WebTerminal

The web terminal allows trading in financial markets from any web browser. The rich set of functions that are available in the desktop platform can be accessed directly on the web, without downloading or installing any application. To connect the web terminal, request it from the App Store on the technical support site

## MetaTrader 5 Trading Platform / Platform Installation

### Platform Installation

The choice and preparation of hardware, as well as the correct installation of the MetaTrader 5 trading platform are very important steps, on which the reliability of the platform operation and the quality of your customer service will depend. This section of the user manual provides information on trading platform installation and preparation:

- **System Requirements** recommendations on what hardware to use for the platform servers.
- **System Preparations** recommendations on how to prepare the operating system before platform installation.
- **Installation** initial installation and recommendations on further platform extension.
- **Activation** before you start using the platform, it needs to be activated in order to remove all functional limitations.
- **White Label** how to order a White Label, and what data is needed for that.
- **Fast Deployment** instructions on fast installation of additional platform components.
- **Console Setup** instructions on platform configuration using the command line.
- **Platform Migration** how to move the platform to new hardware.
- **Platform Uninstall** if necessary, the trading platform can be uninstalled from the system.

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### MetaTrader 5 Trading Platform / Platform Installation / System Requirements

#### System Requirements

The MetaTrader 5 platform components are recommended to be installed on dedicated servers rented from hosting companies.

#### Hardware

|                  | <b>Minimum requirements</b>                                   | <b>Recommended requirements</b>                                                                                                      |
|------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| CPU              | Intel i7 12xx or AMD Ryzen 9 5900 series, quad-core or higher | Intel Xeon Silver 43XX or AMD EPYC 40XX                                                                                              |
| RAM              | No less than 16GB                                             | No less than 32GB                                                                                                                    |
| Disk             | RAID-1 array with 2 x 1 TB disks                              | RAID-1 array with two 1 TB SSD/NVMe disks for the trade server and the history server, RAID-1 array with 2 x 1 TB for backup servers |
| Network          | Both download and upload speed no less than 100 Mbps          | Both download and upload speed no less than 1 Gbps                                                                                   |
| Operating System | Windows Server 2022 Standard x64 or newer                     | Windows Server 2025 Standard / Server Core edition                                                                                   |

- The processor must support AVX2 instructions to enable platform installation.
- Choose a server configuration depending on the number of clients, financial instruments and the density of quotes stream. When databases grow, causing an increased load on the trading platform, even the initially recommended configuration may be fail to cope with such load.
- Due to the fact that the backup server is basically a duplicate of the main trading server, it is desirable that the configuration of the backup server is similar or even identical to the configuration of the main server. It's not recommended to place the backup server at the same hosting company as the main server, as it may happen that the entire network of the provider will be unavailable. Locating the servers at different hosting companies increases data security and the resiliency of the system.
- History server processes and stores a huge volume of information. Due to this fact, important criteria for it are disk volume and disk read/write speed.

#### Additional Requirements

- No third-party software for system time synchronization is allowed.
- Use virtualization if you have control over the hypervisor and can ensure enough resources for the virtual machine running MetaTrader 5. At least 8

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logical cores must be available to the virtual machine (virtual processors are specified here, unlike physical ones used for dedicated servers). The machine should support the Over-Provisioning technology and have the disk speed of no less than 30 MB/s.

- Configure BIOS on the server to run low-latency applications: turn all the energy saving features off (including CPU C-state), configure memory settings to the 'High Performance' state.

### Recommended basic cluster configuration

We recommend renting at least two servers. It is advisable to have servers that are physically located in different data centers. One server is to be used as the main one. The second server is used as the backup one. The main server data is to be replicated to it in real time. Thus, even if one of the hosting providers has troubles, you will be able to restore the platform on the backup server. This will allow you to minimize risks, while the probability of failure in both data centers is very small.

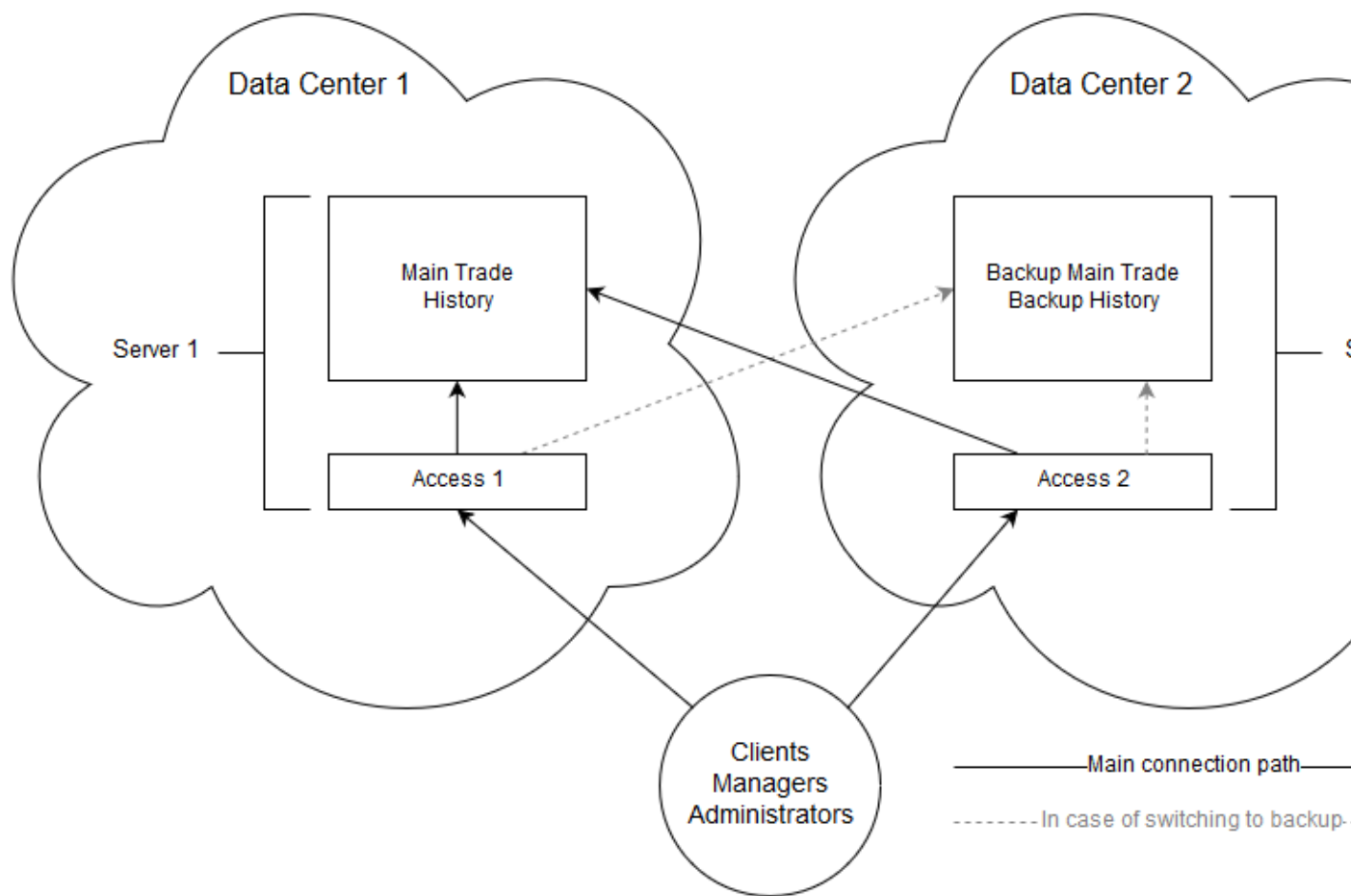
The following platform components should be installed on the main and backup servers:

| <b>Main server</b> | <b>Standby server</b>                   |
|--------------------|-----------------------------------------|
| Main trade server  | Backup server for the main trade server |
| History server     | Backup server for the history server    |
| Access server      | Access server                           |

Components can be installed in different physical servers within one data center, in order to provide a greater platform performance.



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In MetaTrader 5 Administrator, suggested configuration looks as follows:

The screenshot displays the MetaTrader 5 Administrator window. The left sidebar shows a hierarchical tree structure under 'Administrator' > 'Servers'. The main panel shows a table of server configurations with columns: Type, Address, CPU, Thre..., Me..., Soc..., and Conn....

| Type              | Address           | CPU | Thre... | Me...  | Soc... | Conn... |
|-------------------|-------------------|-----|---------|--------|--------|---------|
| Main Trade Server | 14.140.80.42:1951 | 3%  | 125     | 34 157 | 91     | 7       |
| Trade Server      | 14.140.80.42:1952 | 2%  | 127     | 34 157 | 91     | 0       |
| Access Server     | 14.140.80.42:2000 | 6%  | 87      | 34 157 | 91     | 2       |
| Access Server     | 14.140.80.42:1950 | 1%  | 97      | 34 157 | 91     | 2       |
| History Server    | 14.140.80.42:1953 | 3%  | 89      | 34 157 | 91     | 8       |
| Backup Server     | 14.140.80.42:1954 | 4%  | 29      | 34 157 | 91     |         |
| Backup Server     | 14.140.80.42:1955 | 3%  | 37      | 34 157 | 91     |         |

The bottom status bar shows 'For Help, press F1' and a network status indicator '1055 / 0 Kb'.

It is strongly recommended to install the history server close to trade servers in order to avoid delays when delivering quotes and sending trading operations to external systems through gateways.

MetaTrader 5 Trading Platform / Platform Installation / System Preparation

System Preparation

An important issue that directly affects the security and stability of the trading platform is the configuration and optimization of your computer with the operating system. This section highlights the main stages of the process:

- Configuring the user interface
- Disabling server roles
- Removing unneeded programs
- Configuring system updates
- Disabling time synchronization
- Configuring network connections
- Configuring Windows firewall
- Disabling sounds
- Configuring general performance parameters
- Configuring files deletion without moving them to the Recycle Bin.
- Configuring remote access
- Configuring antivirus software

- After completing configurations, restart the server.
- Please note that these are only recommendations. Use them at your discretion depending on your specific circumstances and needs.

Graphical interface

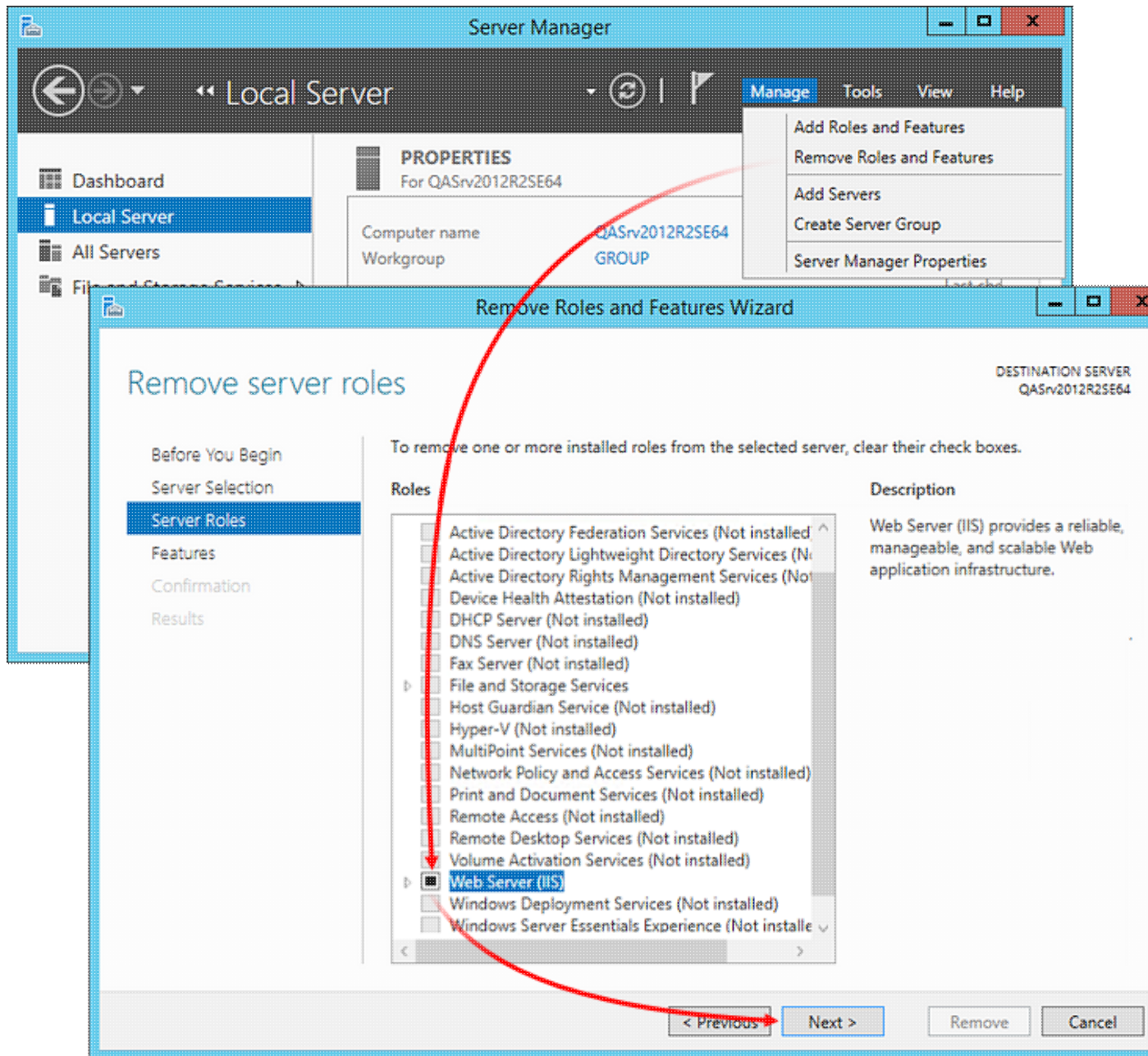
First, it is recommended to configure a graphical user interface of the operating system. You should disable anti-aliasing, shadows, menus, transition effects, etc. These settings allow for fast working with the server in remote access.

Additionally, set blank page as a home page in the Internet Explorer options.

Disabling server roles

Disable unnecessary server roles using the server configuration wizard. Launch it via the Administration section and pass through all the steps clicking Next till you see the list of server roles. If there is any active role, select it and click "Next." Skip the next step for configuring the system components by clicking Next. Check the list of removed roles on the confirmation step and click Remove.

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## Removing unneeded programs

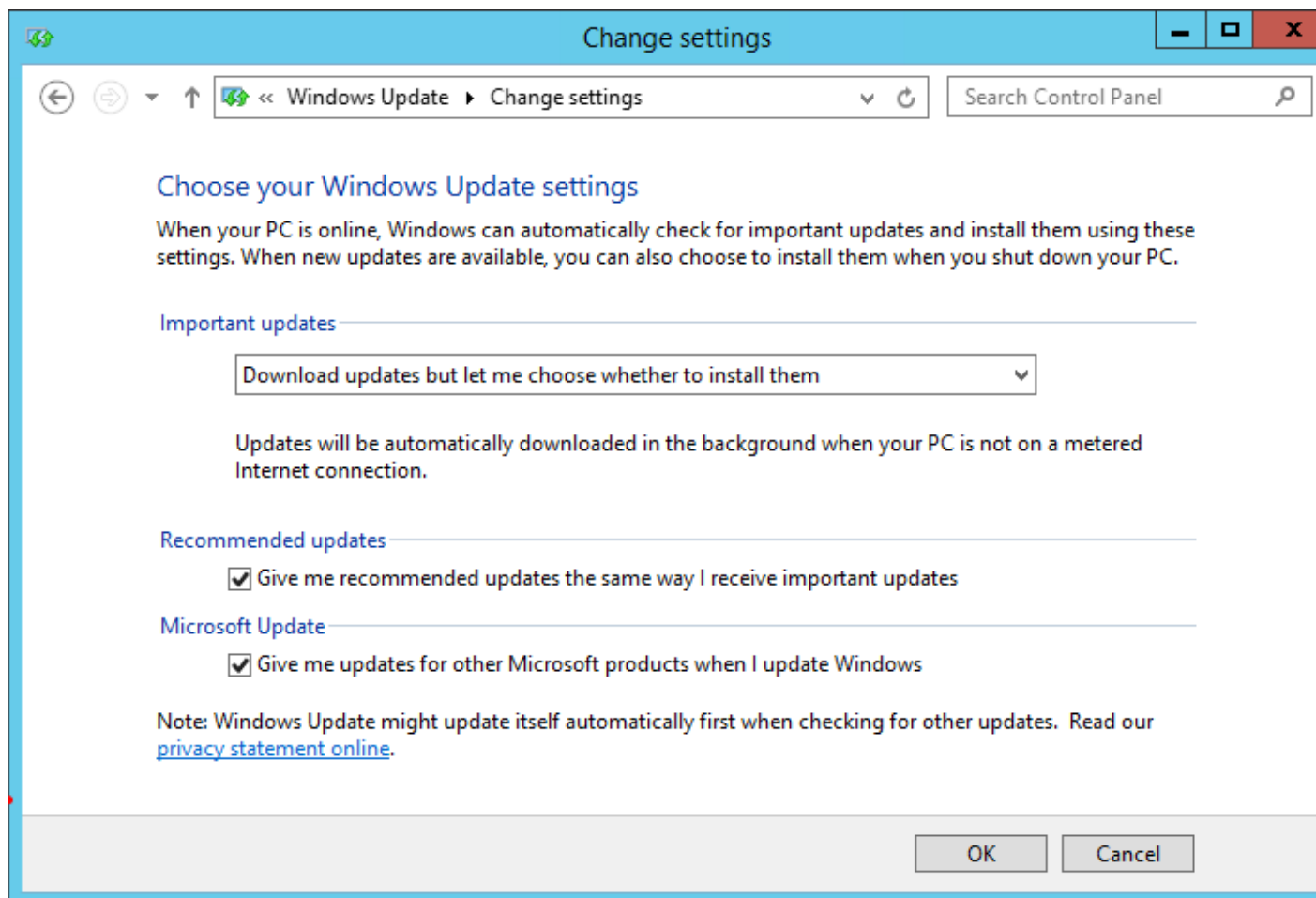
Uninstall any unnecessary programs that can slow down the trading platform or disrupt its security:

- Web server (Internet Information Server, Apache, etc.);
- Mail server, DNS server, SNMP, etc.;
- Databases (Oracle, MSSQL, etc.);
- Various development environments (IDE), compilers, etc.;
- .NET and Java environments;
- Various control agents from the server manufacturer (usually, this is an entire set of default programs) designed to remotely monitor the server.

Launch the wizard for installing and deleting programs. Move to "Uninstall or change a program" and remove unnecessary programs. Remove programs as you see fit depending on your needs and resource efficiency.

## Configuring system updates

Enable auto update with manual installation confirmation in the Control Panel. Fully automated mode is not recommended since Windows does not allow selecting the update installation day. It is recommended to install updates on weekends when the server load is minimal.



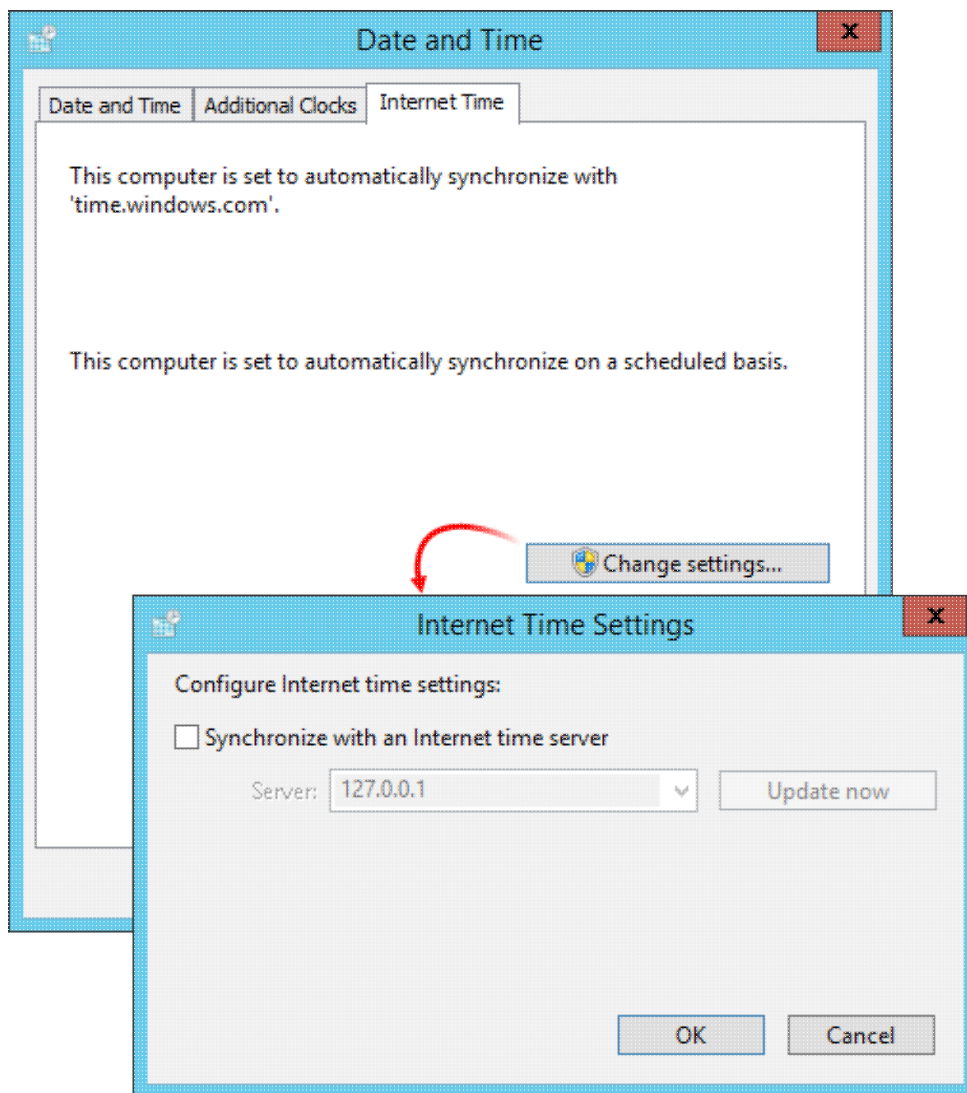
Before installing the platform, you should install all necessary updates for your operating system.

### Disabling time synchronization

System time synchronization should be performed only by the MetaTrader 5 main trade server on the server where the trading platform is installed. Windows Time should be disabled.

The MetaTrader 5 built-in time synchronization service verifies and corrects time (if necessary) once per hour.

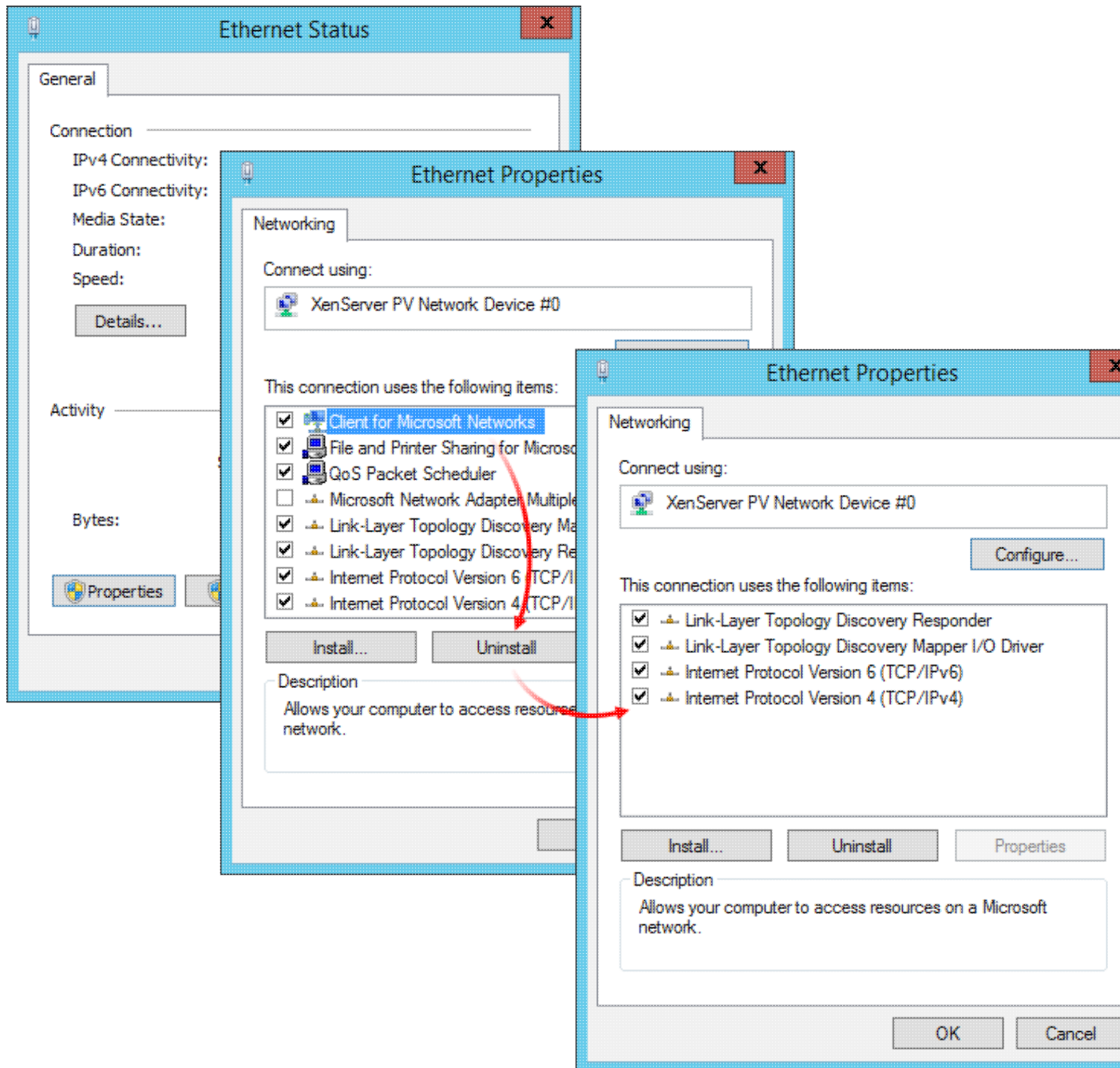
In order to disable time synchronization in the operating system, open the "Date and Time" section of the Control Panel. Go to the "Internet Time" tab, enter 127.0.0.1 to the Server field and unflag the automated synchronization:



If the address of the time synchronization server is not specified in the time settings of the platform, the Windows Time service is automatically disabled in the operating system when MetaTrader 5 servers are launched.

### Configuring network connections

Right-click on the active network connection and select Properties:





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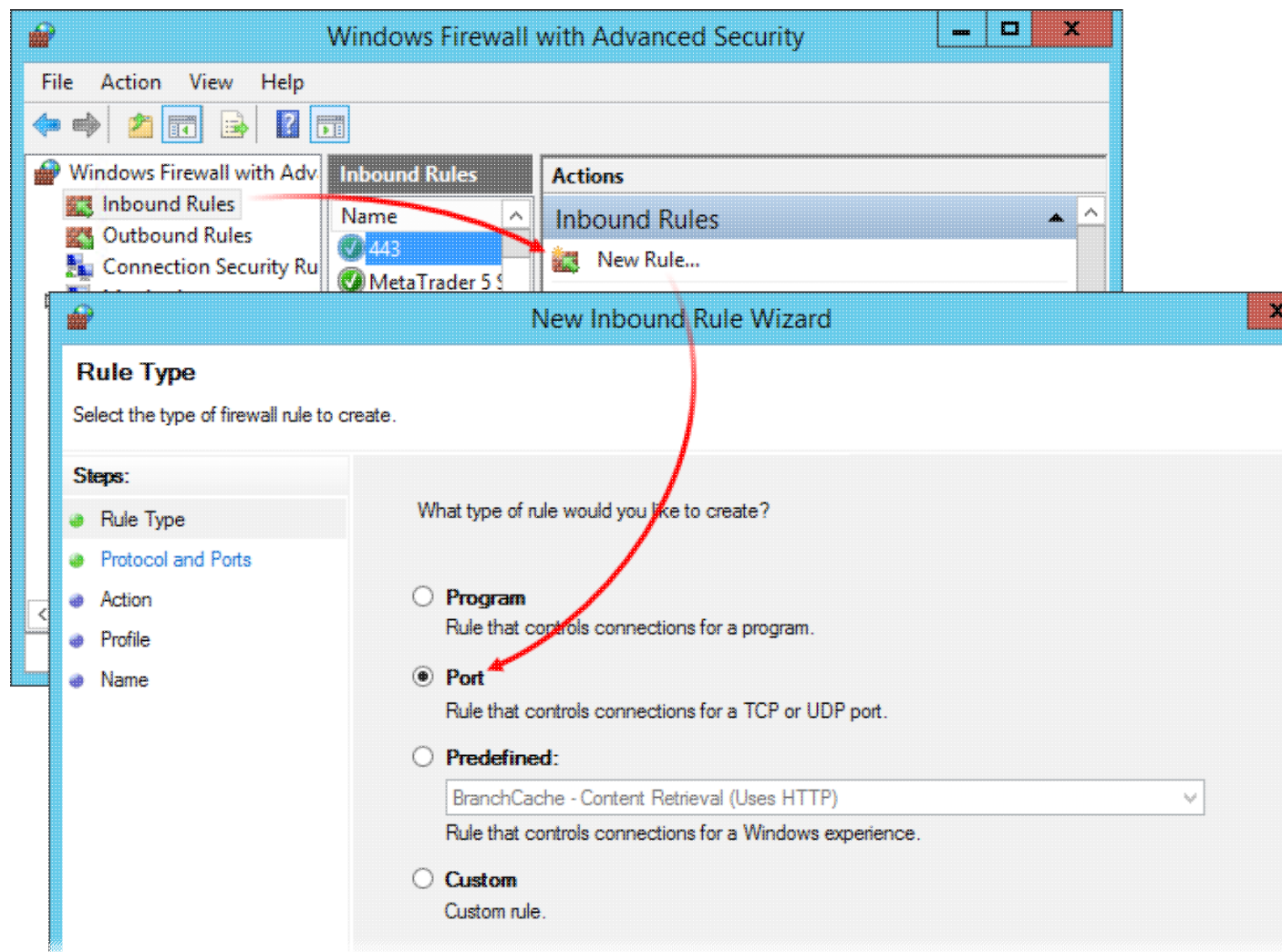
On the Networking tab, remove all components except for TCP/IP v4, TCP/IP v6 and Link-Layer Topology Discovery\* protocols. They are helpful when configuring the routing.

### Configuring Windows firewall

The platform installer (including the one used during the quick deployment) automatically adds permissions for the necessary ports to the Windows firewall.

If an additional port should be opened for your platform configuration, go to the Control Panel → Windows Firewall → Additional settings. Create a new inbound rule and select:

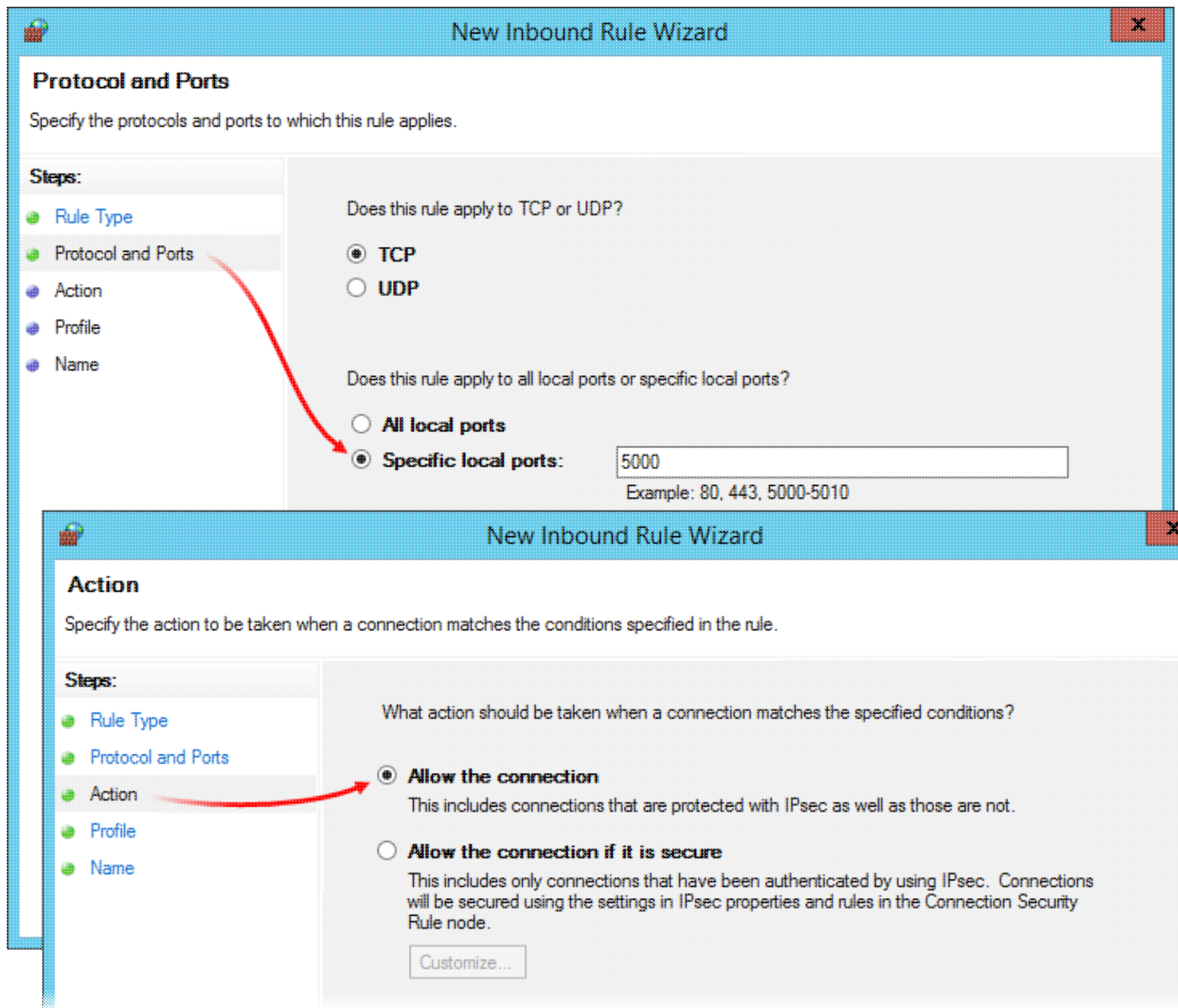
- **Program** in case you want to allow connections for a specific application, for example a gateway.
- **Port** if you want to open certain ports regardless of an application.





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Next, set the port number and select "Allow the connection":



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Depending on the location of your PC, specify the profile the rule is applied for: domain, private network or public network. At the last stage, enter a name for the rule.

For each component of the platform, the list of ports that should be allowed is different:

### Access Server

| <b>Ports for outgoing connections</b>                | <b>Reason</b>                                                                                                                                                                 |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ports, on which <u>trade servers</u> work.           | Access servers rout clients' connections to trade servers. In case with the main trade server, additionally configuration files are updated and <u>time is synchronized</u> . |
| Port, a <u>history server</u> works on.              | Receiving news, quotes and <u>updates</u> .                                                                                                                                   |
| Port 443.                                            | Connection to MetaQuotes' Updates server ( <a href="https://updates.metaquotes.net">https://updates.metaquotes.net</a> ).                                                     |
| <b>Ports for incoming connections</b>                | <b>Reason</b>                                                                                                                                                                 |
| Port the access server works on ( <u>Bindings</u> ). | These ports will be listened to for receiving clients' connections.                                                                                                           |

### Backup Server

| <b>Ports for outgoing connections</b>               | <b>Reason</b>                                                                                                             |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Port of the <u>main trade server</u> .              | Through the main server, configuration files are updated, as well time is synchronized.                                   |
| Port of the server, whose <u>backups are made</u> . | Backup copying of the server data.                                                                                        |
| Port, on which a <u>history server</u> works.       | Receiving updates.                                                                                                        |
| Ports, on which <u>gateways</u> are running.        | Backing up gateway data.                                                                                                  |
| Port 443.                                           | Connection to MetaQuotes' Updates server ( <a href="https://updates.metaquotes.net">https://updates.metaquotes.net</a> ). |
| <u>Access server</u> ports                          | Monitoring the availability of the backup server if <u>automatic switch</u> to the backups server is enabled.             |

### Trading Server

|  | <b>Reason</b> |
|--|---------------|
|--|---------------|

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| <b>Ports for outgoing connections</b>        |                                                                                                                                                     |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Port 25.                                     | Sending <u>reports</u> to the mail server.                                                                                                          |
| Port 37.                                     | <u>Time synchronization</u> by the TIME protocol (for the main server).                                                                             |
| Port 123.                                    | Time synchronization by the NTP protocol (for the main server).                                                                                     |
| Port of the main trade server.               | Receipt of configuration files and time synchronization (for servers other than the main).                                                          |
| Port, on which a history server works.       | Receiving quotes and updates.                                                                                                                       |
| Ports, on which <u>gateways</u> are running. | Work with gateways: receiving quotes, trading.                                                                                                      |
| Port 443.                                    | Connection to MetaQuotes' Updates server ( <a href="https://updates.metaquotes.net">https://updates.metaquotes.net</a> ).                           |
| <b>Ports for incoming connections</b>        | <b>Reason</b>                                                                                                                                       |
| Port where the trade server works.           | This port is required for the main trade server to connect all other components. For non-main servers it is required for connecting access servers. |

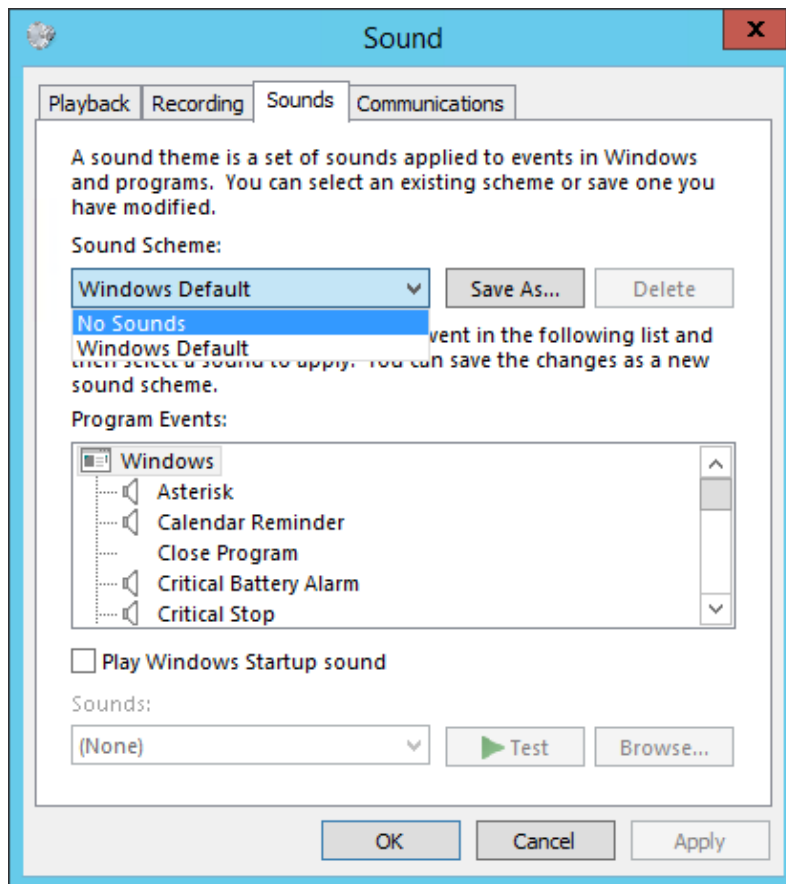
### History Server

| <b>Ports for outgoing connections</b>   | <b>Reason</b>                                                                                                                                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port 443.                               | Connection to MetaQuotes' Updates server ( <a href="https://updates.metaquotes.net">https://updates.metaquotes.net</a> ).                                                                                      |
| Port of the main trade server.          | Receipt of configuration files and time synchronization.                                                                                                                                                       |
| Ports, on which data feeds work.        | Ports for <u>data feeds</u> that establish connection to remote servers. For example, <u>MetaTrader4Feeder</u> (port 443 is used on default) and <u>TCNewsFeeder</u> (FTP ports 20 or 21 are used on default). |
| Ports, on which remote data feeds work. | Port numbers depend on the <u>data feeds</u> that are connected.                                                                                                                                               |
| <b>Ports for incoming</b>               | <b>Reason</b>                                                                                                                                                                                                  |

| Ports for outgoing connections                  | Reason                                                                                                                                                                    |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| connections                                     |                                                                                                                                                                           |
| Port, on which the <u>history server</u> works. | This port is required for connecting other components of the platform.                                                                                                    |
| Ports, on which data feeds work.                | Ports, on which <u>data feeds</u> work, that receive external connections from remote servers. For example, <u>DJPrimeTassNewsFeeder</u> (port 20000 is used on default). |

### Disabling sounds

Disable sounds for the server the trading server is installed at. Open the Sound section in the Control Panel:

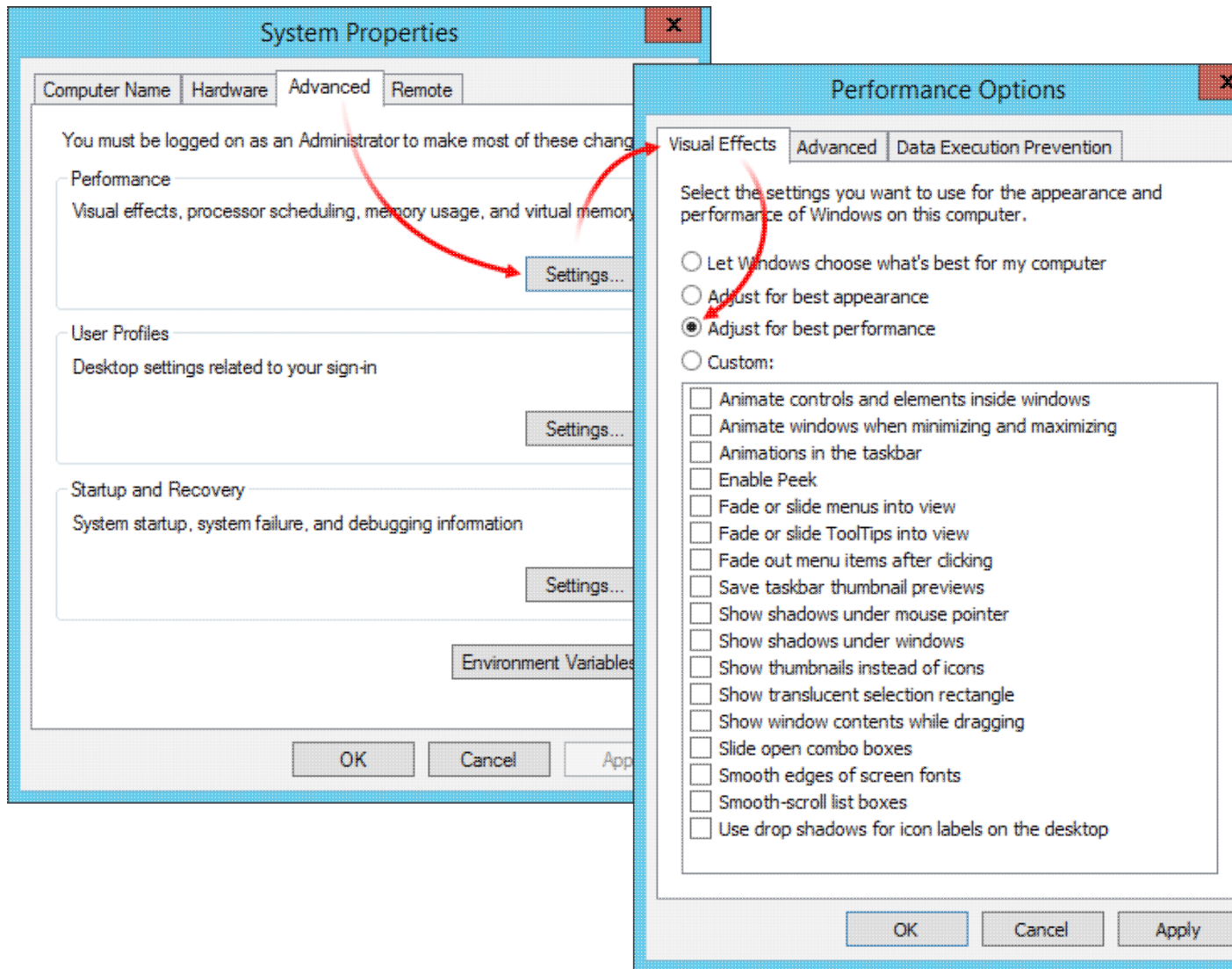


## MetaTrader 5 trading platform

Select "No Sounds" in the list of sound schemes and click OK.

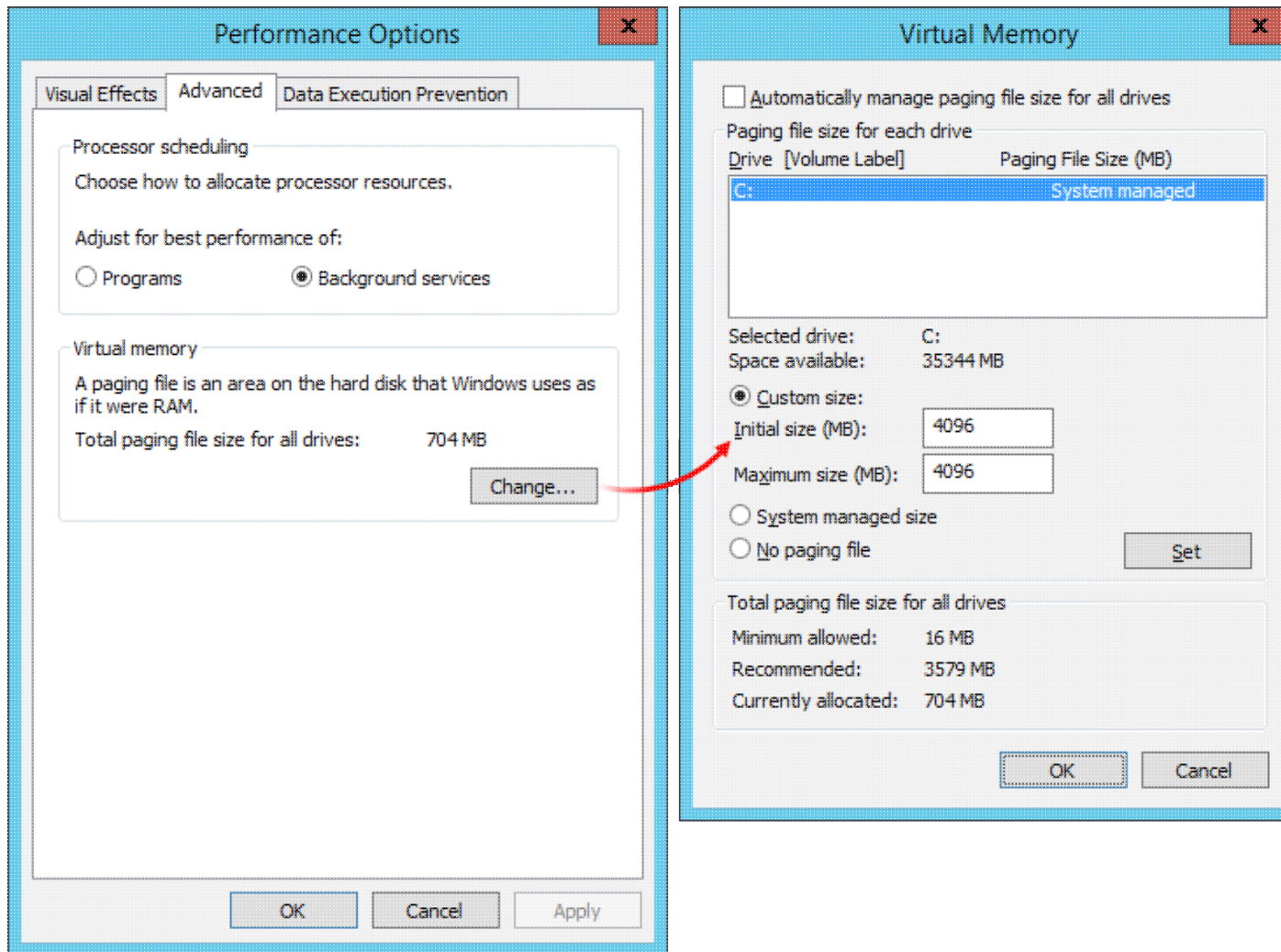
### Configuration general performance parameters

Open the Control Panel → System → System Properties → Advanced → Performance settings. Select "Adjust for best performance" on the "Visual Effects" tab:



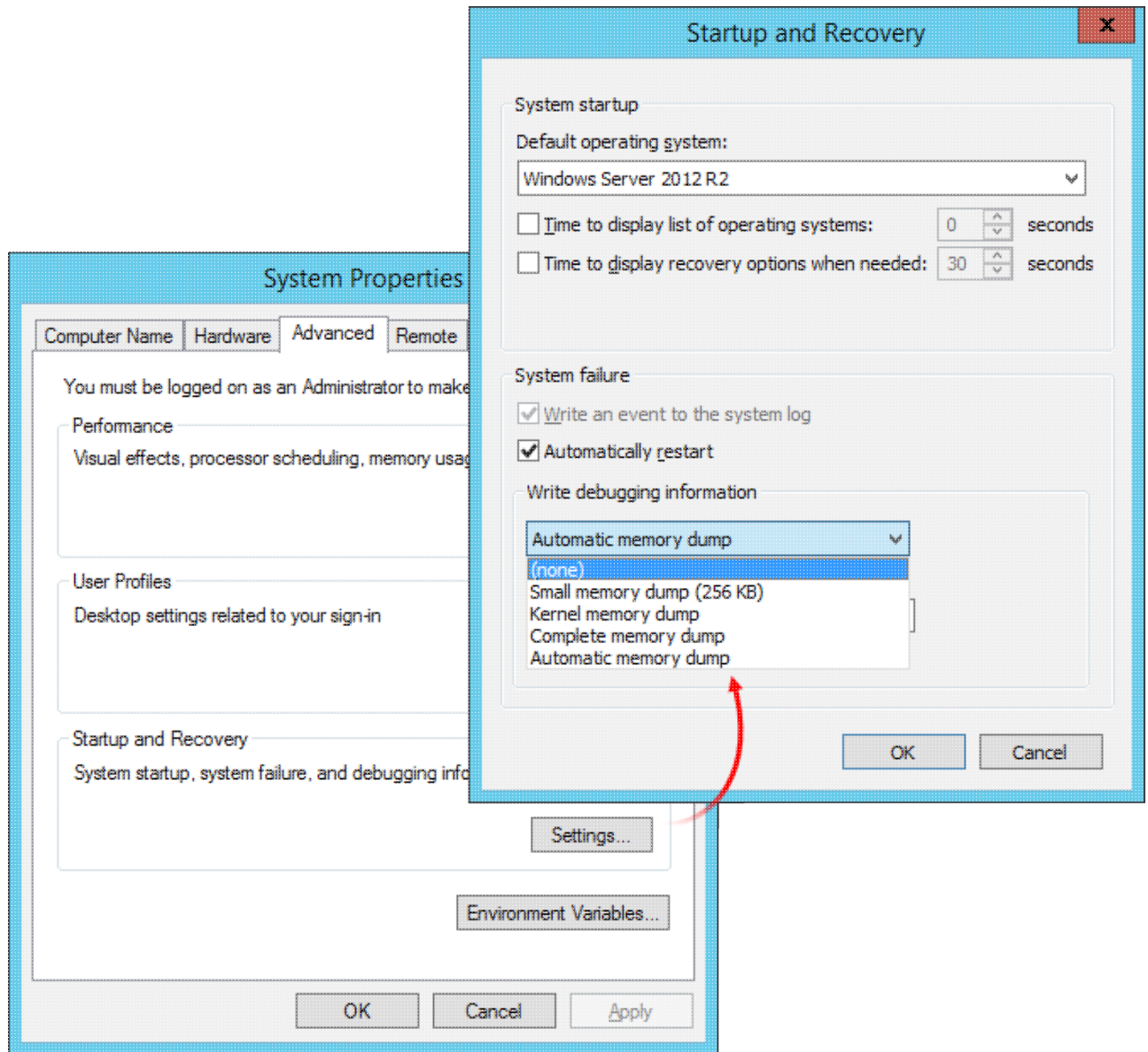
## MetaTrader 5 trading platform

On the Advanced tab, click Change and enter the value of the initial swap file size equal to its maximum value.



## MetaTrader 5 trading platform

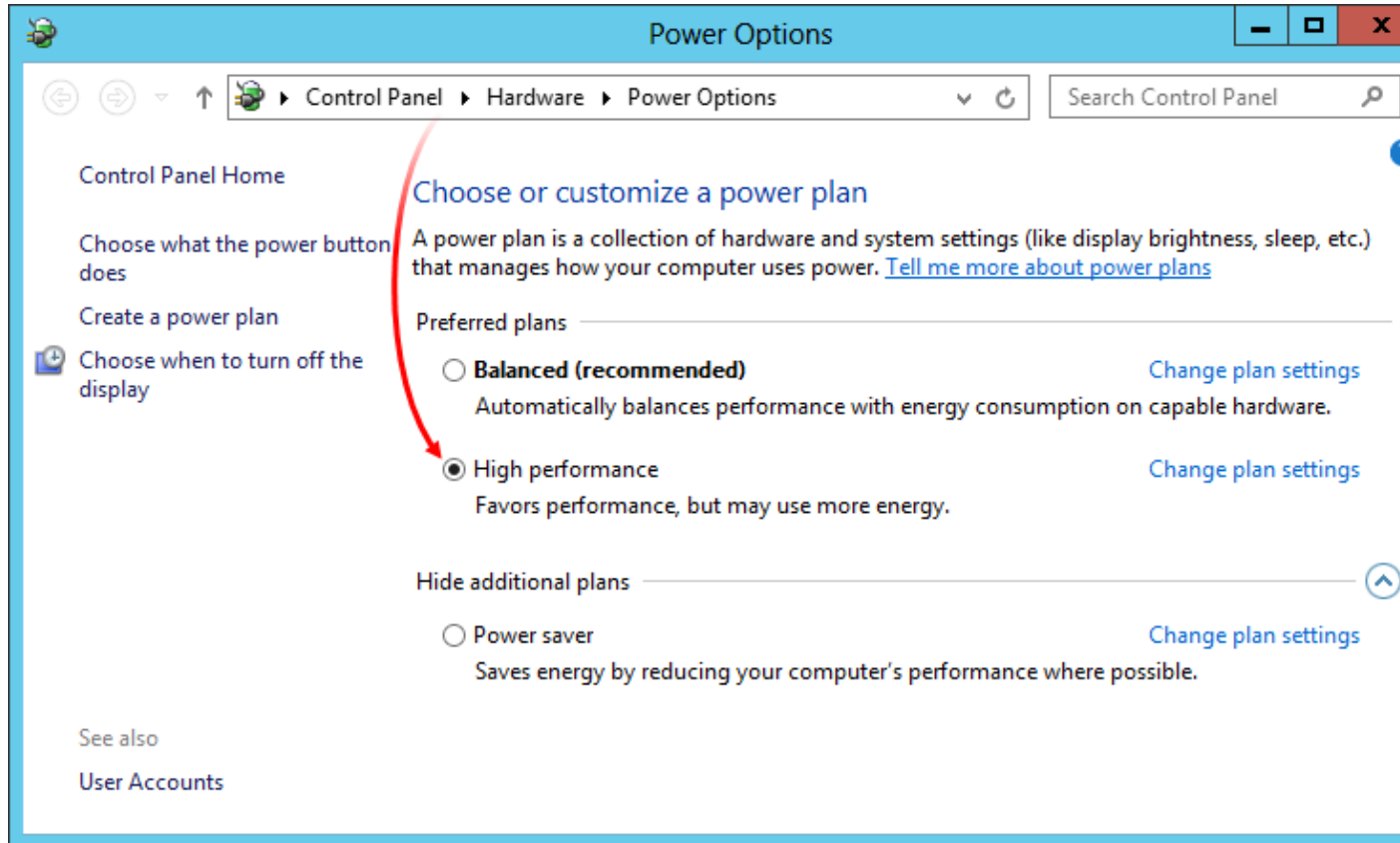
Return to the System Properties Advanced and open the startup and recovery settings:



## MetaTrader 5 trading platform

Disable the "Time to display list of operating systems" option. Set "Write debugging information" to None.

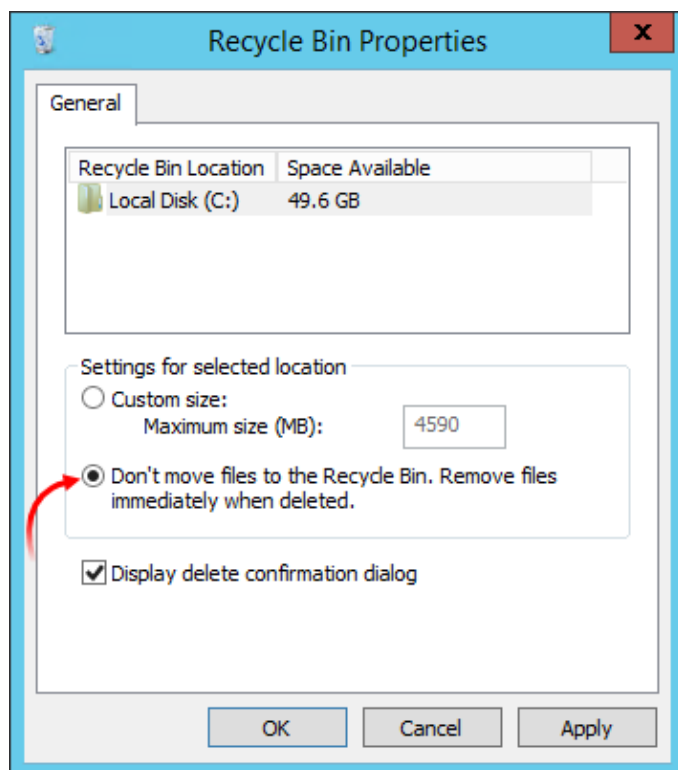
From the control panel, navigate to Hardware > Power Options. Select maximum performance mode:





## Configuring files deletion without moving them to the Recycle Bin

In order to ensure the files are deleted immediately, click Properties in the bin's context menu. Select "Don't move files to the Recycle Bin. Remove files immediately when deleted":



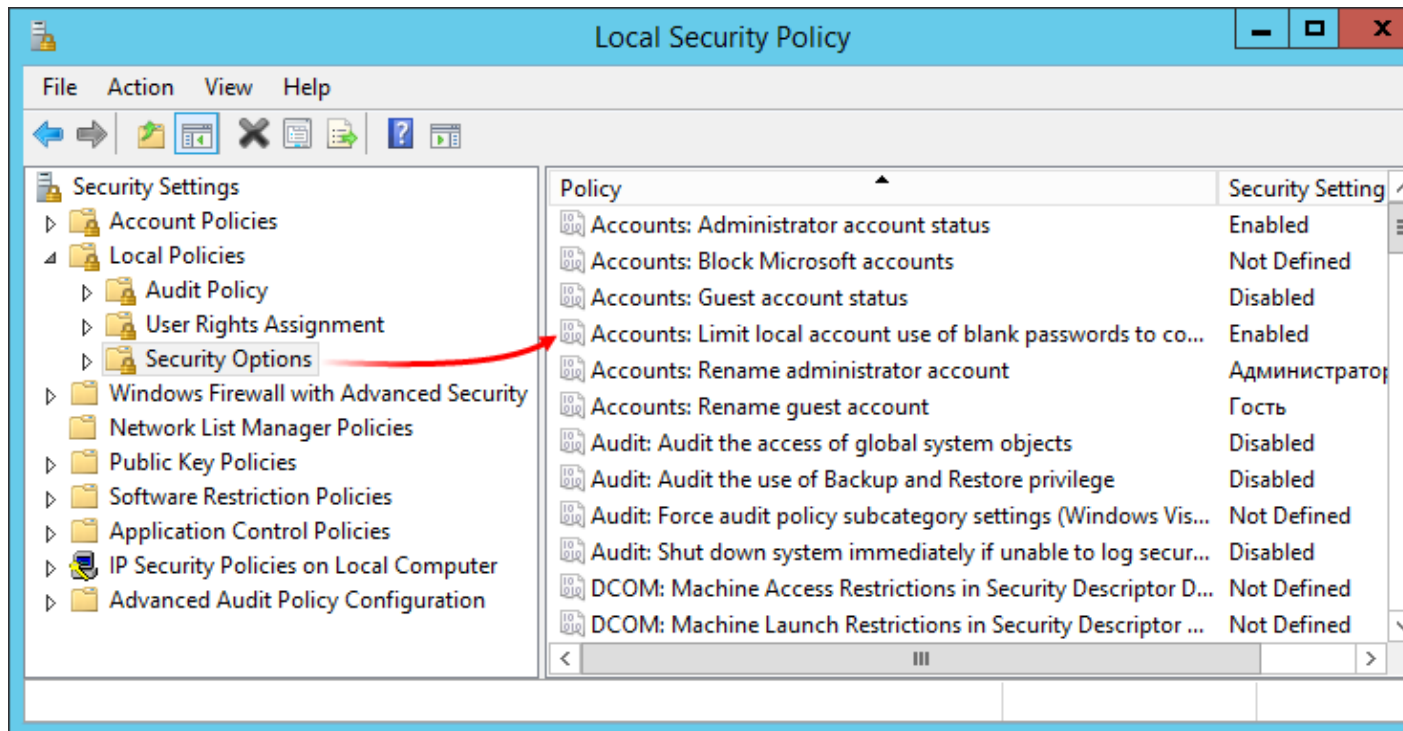
## Server Configurator

The "Server Configurator" program is specially designed to facilitate the server preliminary configuration process by automating the manual work of terminating various services, system configuration via the registry and deleting temporary and unnecessary information from the disks. You can download this utility from the following link <https://support.metaquotes.net/spfiles/srvcfg.exe>. The program does not require installation, it can be simply launched through the EXE file. Go through all the steps enabling all the flags. A more detailed description of the Server Configurator can be found in the article "[Using the Server Configurator](#)".

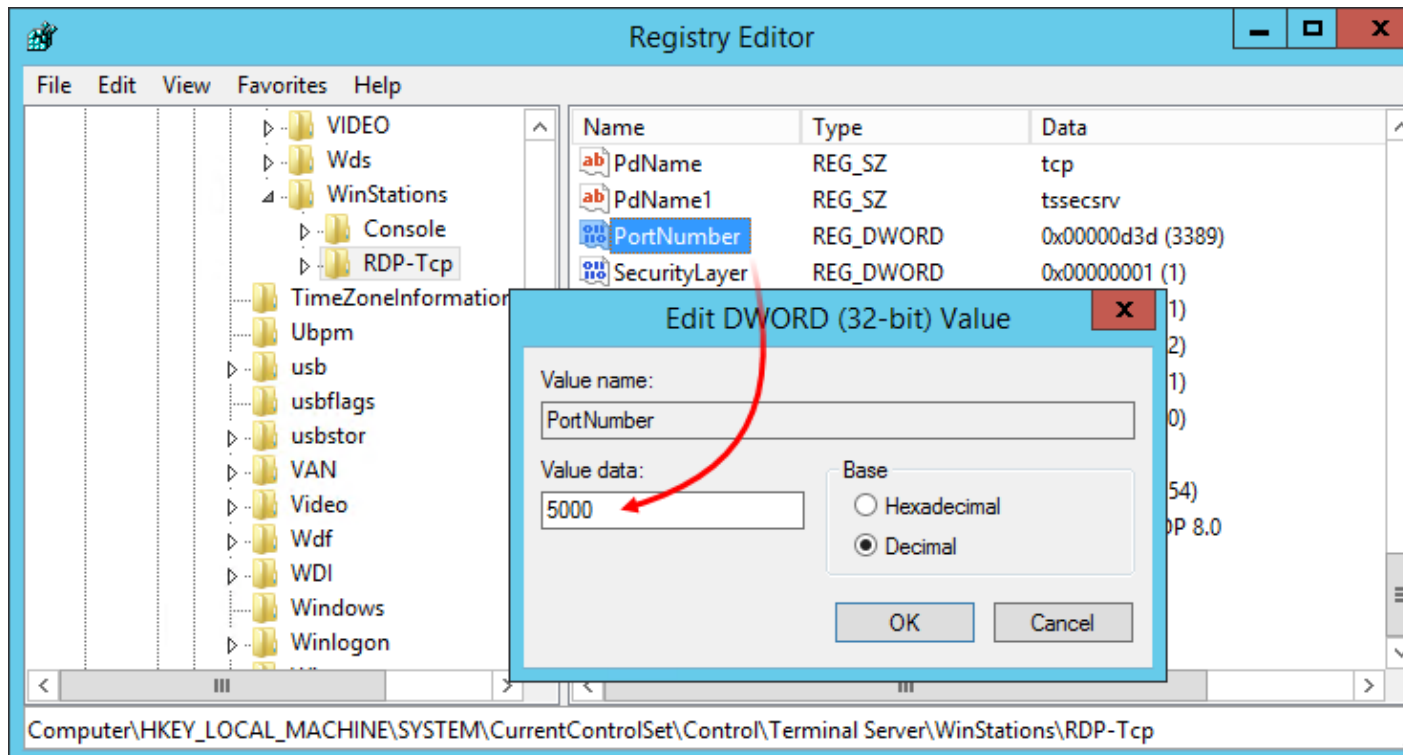
## Configuring remote access

Keep the data for the remote access to the server in a safe place and do not disclose them to anyone without particular reason.

**Blocking RDP connections for accounts having no password.** Accounts with no entry password should not have permission for remote access to the server. Open the Control Panel Administrative Tools Local Security Policy Security Options. Enable the parameter "Accounts: Limit local account use of blank passwords to console logon only".



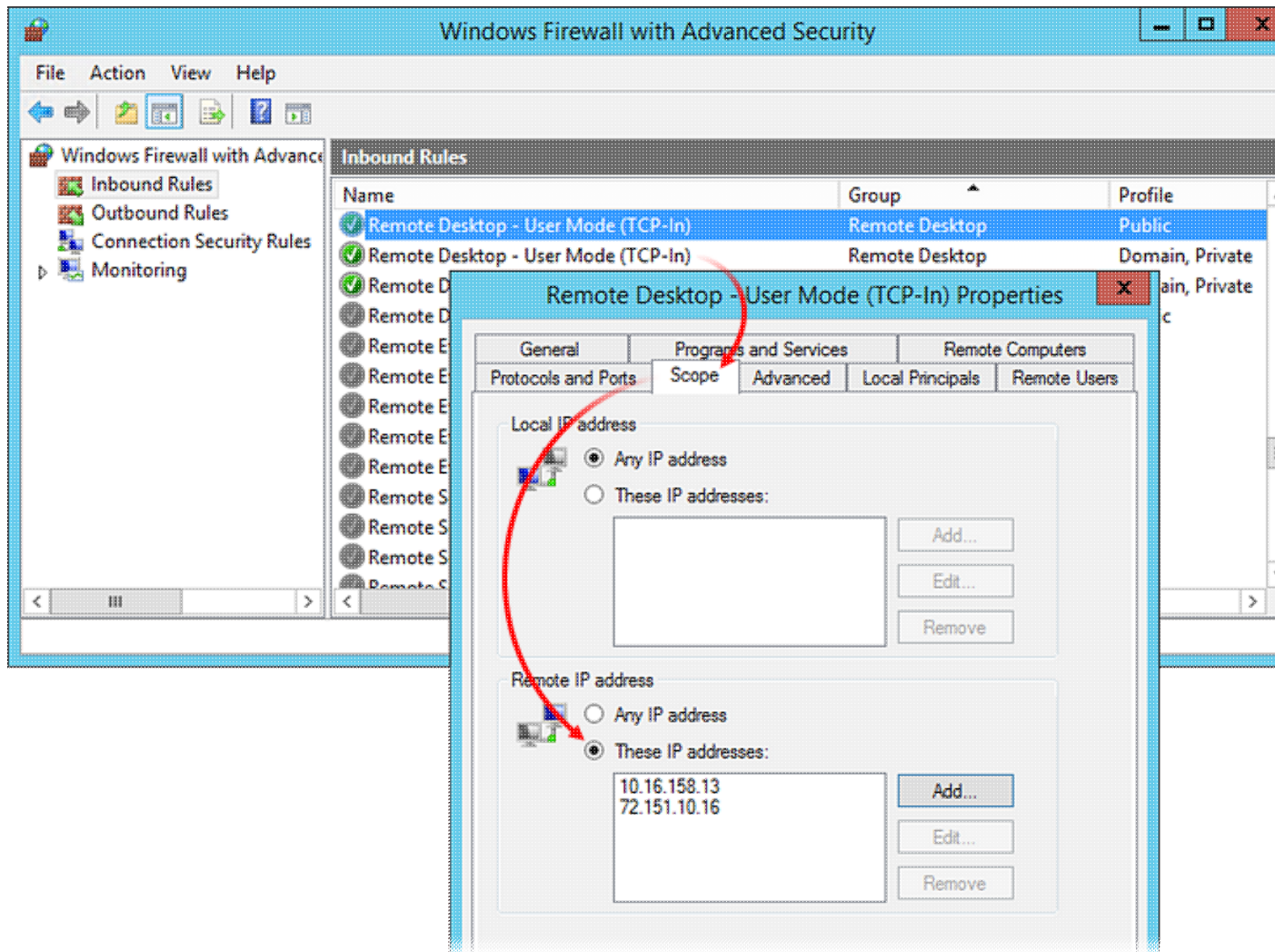
**Changing a standard port for remote connection.** In order to protect against the attacks monitoring "well-known" ports, change the port for the Remote Desktop Protocol. Open the registry editor (regedit), go to HKEY\_LOCAL\_MACHINE\SYSTEM\CurrentControlSet\Control\Terminal Server\WinStations\RDP-Tcp and find the Port Number parameter. Double-click on it, select the decimal system in the new window and set the necessary port as the value.



- The specified port should be permitted in the Windows Firewall.
- In order to remotely connect to the server, specify the new port after an IP address separated by a colon. For example, 10.59.162.3:5000.

**Limitation of IP address list for remote connection.** As an additional security measure, you can restrict the list of IP addresses that can be used to remotely connect to the server. For example, you can allow connection only from your office's IP addresses.

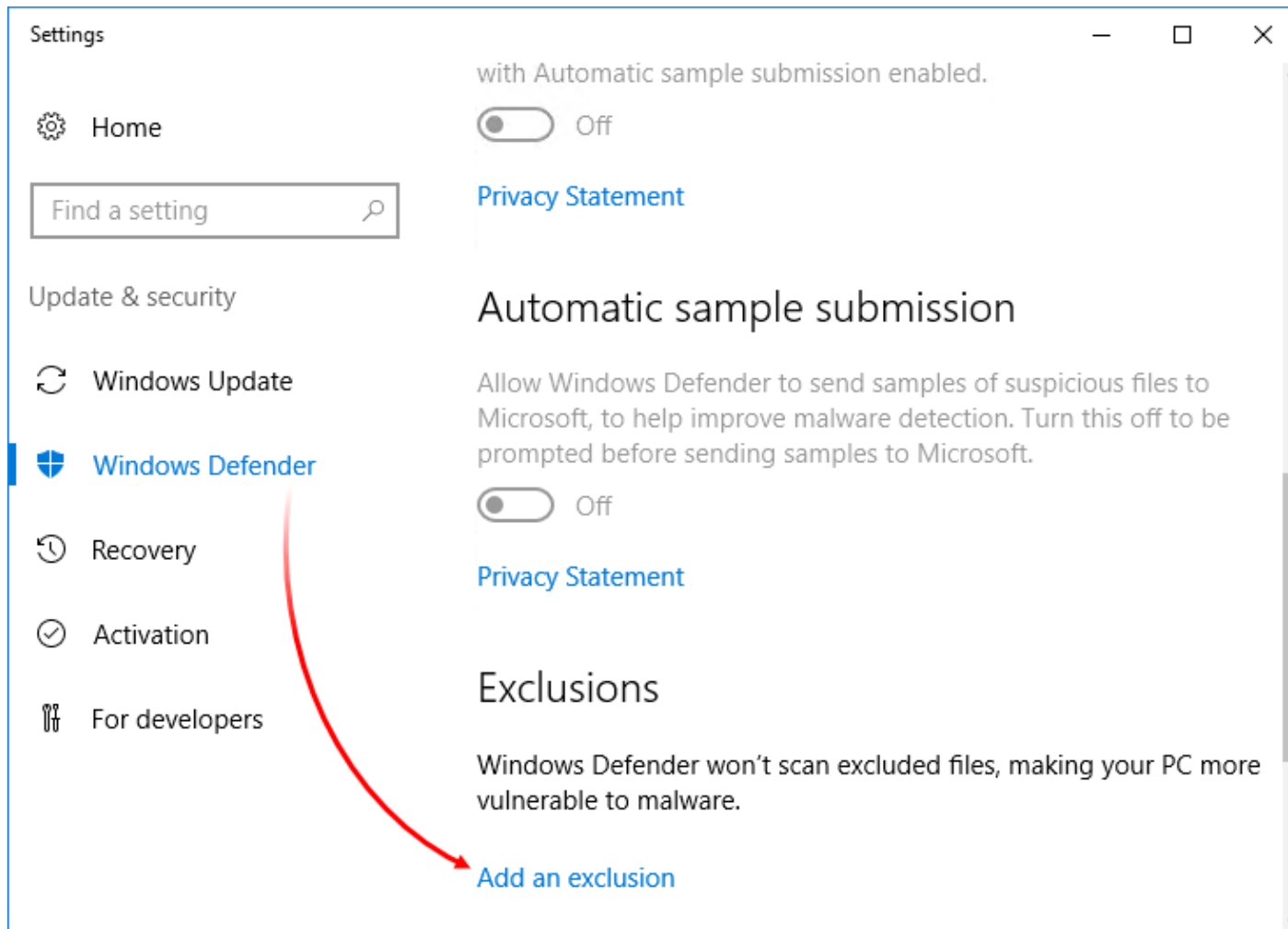
Open Windows Firewall and find "Remote Desktop - User Mode (TCP-In)" in the list of inbound rules. There may be several such rules depending on the number of network profiles. Open the one you need and go to Scope tab. In the "Remote IP address" section, select "These IP addresses" and add the necessary ones to the list:



## Configuring antivirus software

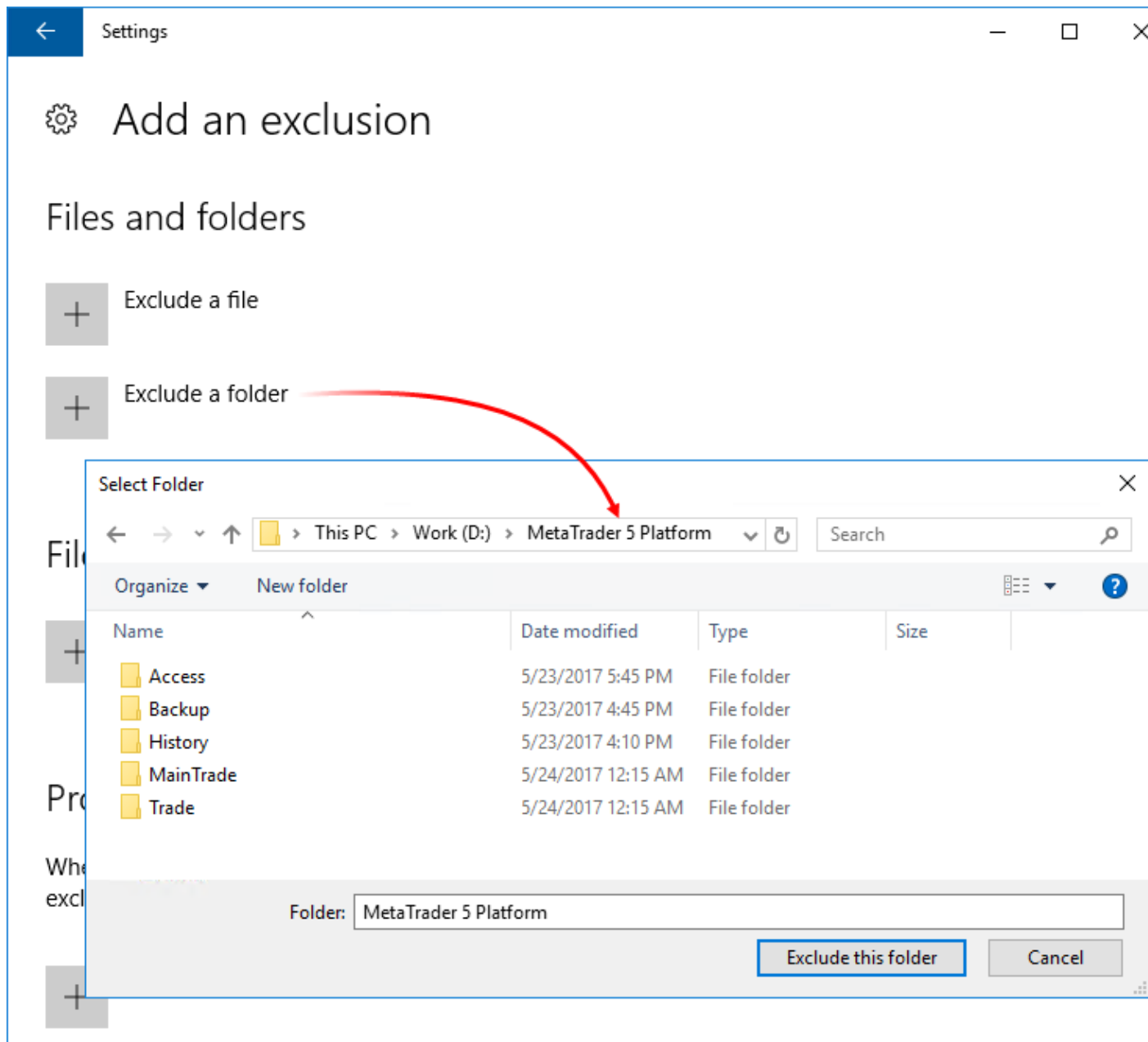
If antivirus software is used on your server, add to its exclusions the system processes of the MetaTrader 5 platform, as well as its installation directory. Constant monitoring by antivirus software reduces platform performance.

Here is an example of adding exclusions in Windows Defender, which is a built-in antivirus software used on Windows Server 2016. Open Windows Settings → Updates and Security → Windows Defender. Go to "Exclusions" and click "Add an exclusion".

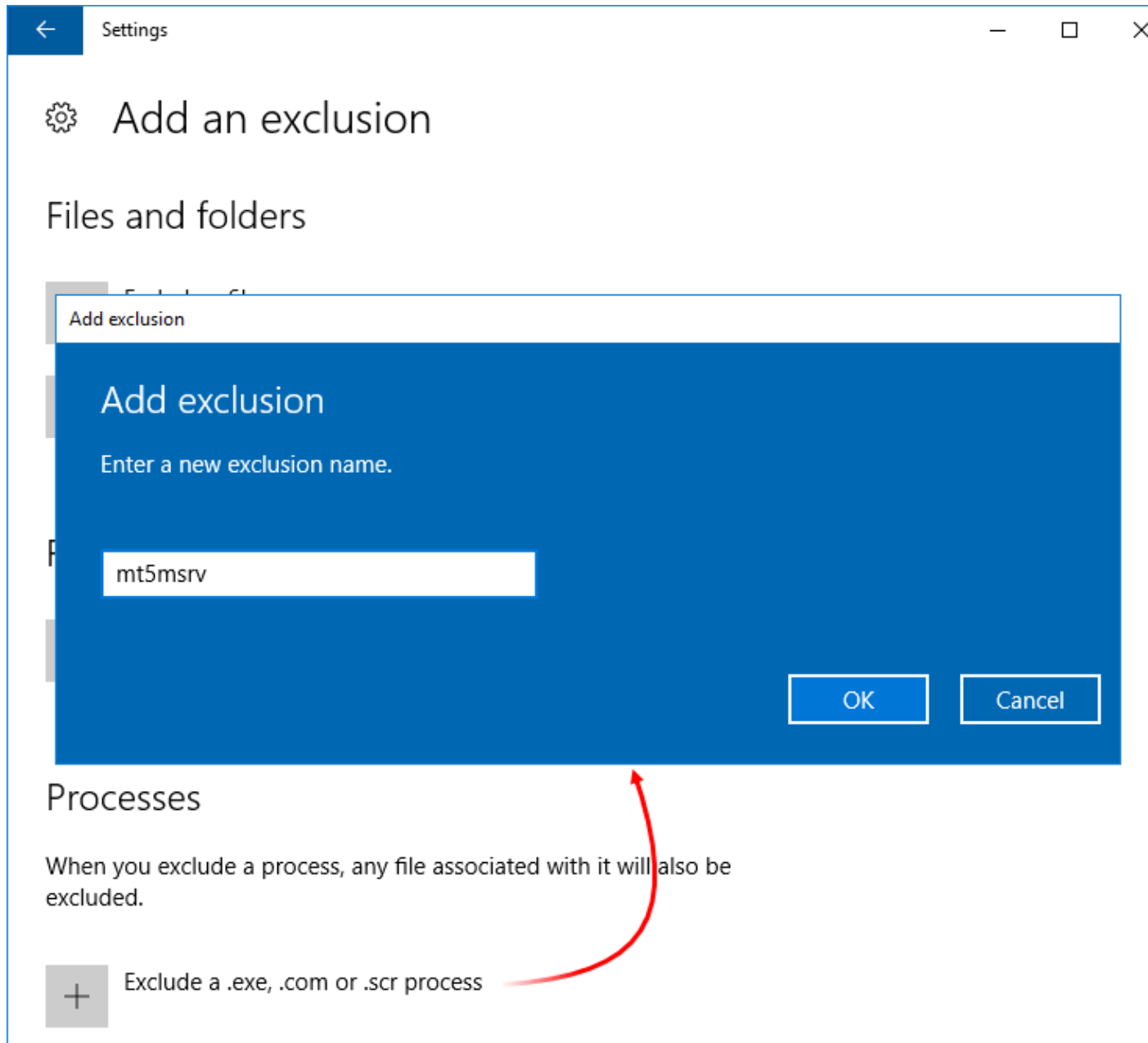


## MetaTrader 5 trading platform

Select "Exclude folder" and choose the platform installation directory.



Click "Exclude process .exe, .com or .src". Specify the name of the platform component process in the window that opens. You can specify the name of the server executable file (for example mt5trade64.exe) or the name of the process (for example mt5msrv).



## MetaTrader 5 trading platform

Similar exclusions should be added for all installed components of the cluster. The default names of the services are:

- mt5msrv the main trade server
- mt5tsrv an additional trade server
- mt5asrv an access server
- mt5bsrv a backup server
- mt5hsrv a history server

If several components of the same type are installed, a digit is added to the service name. For example, mt5tsrv5. The list of installed processes can be viewed in the task manager.

## MetaTrader 5 Trading Platform / Platform Installation / Installation

### Installation of the MetaTrader 5 Platform

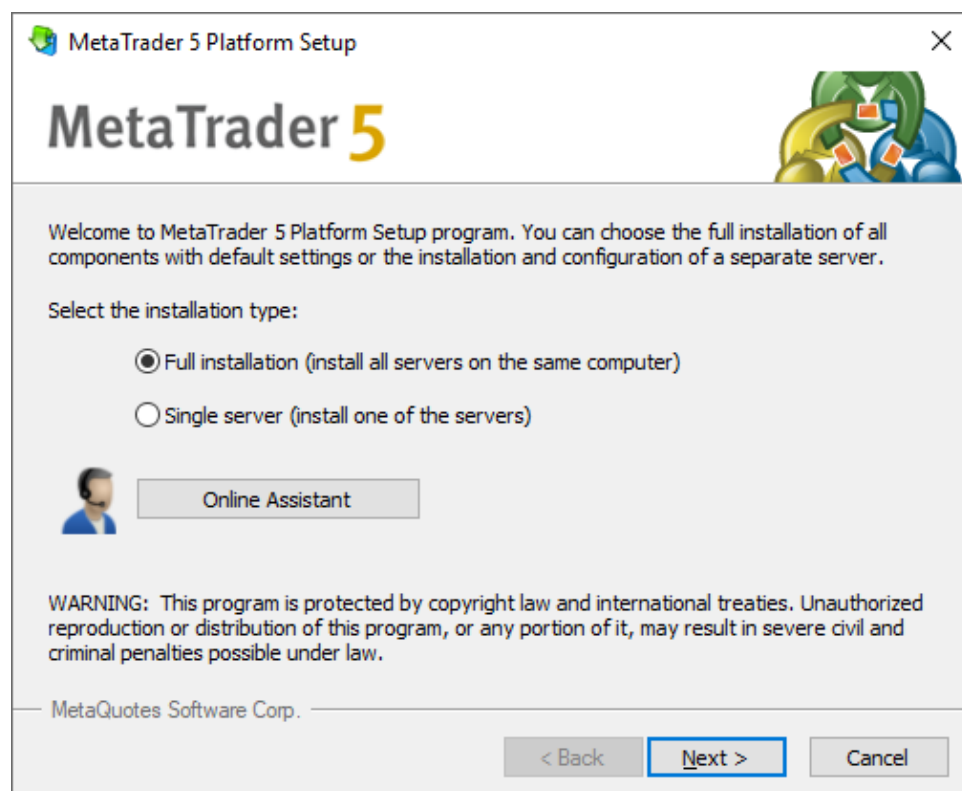
Before you start platform installation, you need to rent servers from hosting providers keeping in mind system requirements. Required preparatory works should be performed on these servers. After that you can proceed to platform installation:

- Complete platform installation on the main server
- First connection to the platform and necessary actions
- Installation of additional backup servers
- Additional access servers
- Additional trade servers
- Installation of separate platform components using the installer
- Installation of a platform for testing new features and developing own solutions

### Complete platform installation on the main server

The platform installer allows you to deploy a required set of servers on one local computer, including the main trade server, its backup server, the history server and the access server. This option should be used during the first platform installation on the main server. After that, you will be able to connect to the platform using the MetaTrader 5 Administrator and proceed to the installation and configuration of additional components.

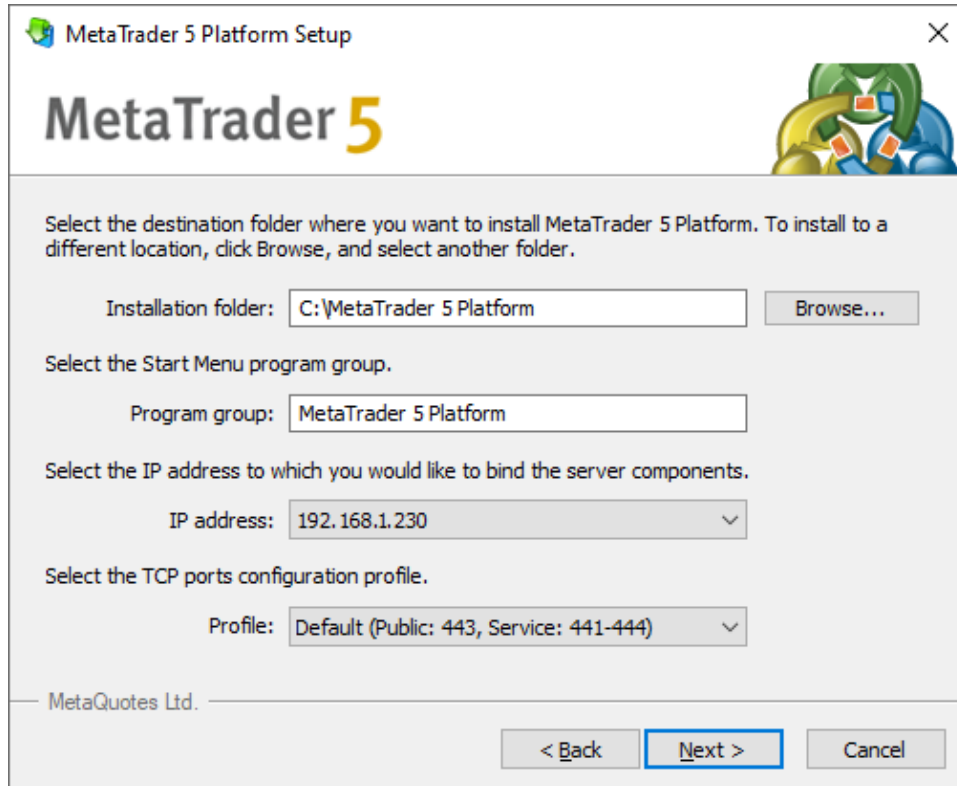
Run the **mt5srvsetup.exe** platform installer and select "Full installation":





## MetaTrader 5 trading platform

Read the license agreement and, if you agree, check "Yes, I agree with all terms of this license agreement". Next, specify the path to the platform license \*.lic. Information from the license is displayed at this stage, including the product (license expiration date), the company for which the license has been issued, and the domain name.



At the next stage, specify the installation parameters:

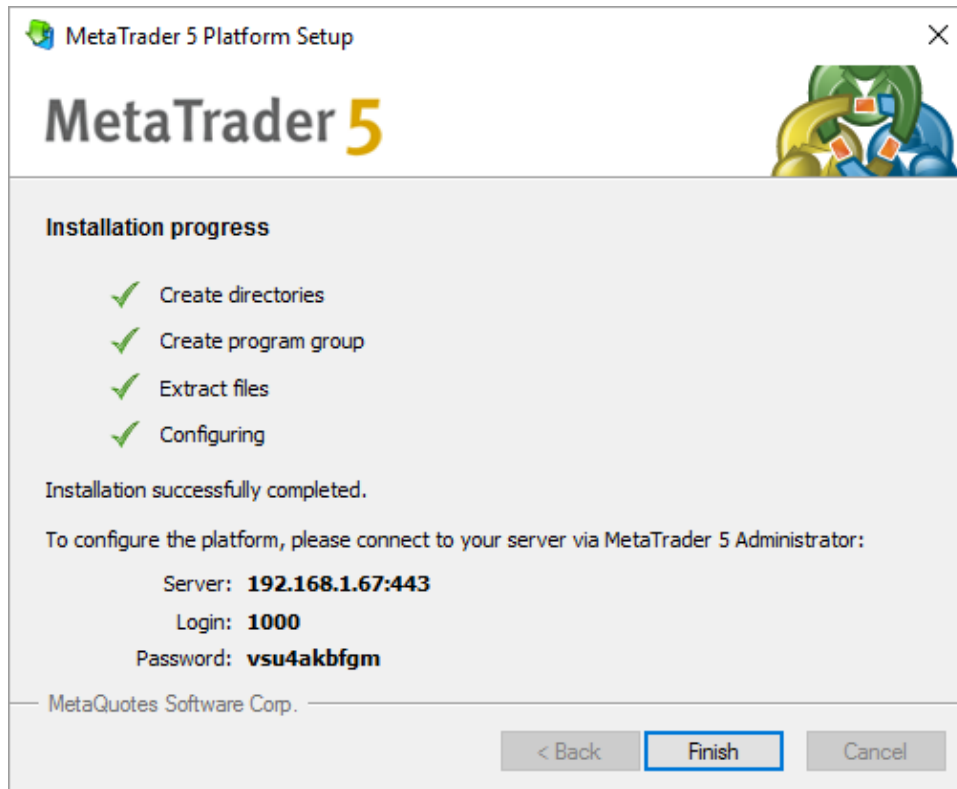
- **Installation folder** click "Browse" and select the folder to which you want to install the trading platform.
- **Program group** specify the group name for the trading platform; this name will be displayed in the Start menu.
- **IP address** choose one of the IP addresses available on the server. Connection to the platform will be established using this address.
- **Profile** choose one of the port profiles for connection. The 441-444 range is used by default, and the port 443 will be used for public access (access server). Other ports are allocated for the trade server, the history server, and the backup server. There are also some additional profiles. For example, the range of 1950-1953, in which port 1950 is used for the access server.

- If all ports specified in the profiles are occupied by other applications, you should install the platform components one by one. In this case, you can specify any other port manually.
- Full Installation is not possible if at least one of the ports from the selected

## MetaTrader 5 trading platform

range (profile) is busy. In this case, it does not matter what network interface it is used for. The installer checks all available interfaces to prevent any network conflicts. If you receive a message about a busy port during Full Installation, install the platform components one by one.

A click on the Next button will launch the platform installation process:



## MetaTrader 5 trading platform

At the end of the installation procedure, a dialog box will appear showing details for the first connection to the platform using MetaTrader 5 Administrator: a login and password for the automatically created administrator entry, and the address for connection. Be sure to save this data in a safe place, otherwise you will not be able to connect to the platform without this information.

To complete platform installation, press "Finish". After that, a program group specified during installation, with the commands to launch and stop platform components will appear in the Start menu.

### First connection to the platform and necessary actions

One administrator account with login "1000" and a random password is automatically created during platform installation. These data, as well as the address of the server to connect to, are displayed on the last step of the platform installation. Be sure to save this data in a safe place, because without it you will not be able to connect to the platform using the MetaTrader 5 Administrator.

The password to the automatically created administrator account is also saved in the log file of the main trade server (the /Logs folder).

Add the newly installed platform to the MetaTrader 5 Administrator, and then connect to it using the administrator account. For security reasons, you will immediately need to change your password.

At the next step, you need to activate licenses, in order to remove all functional limitations of the platform.

The password generated for the first administrator account is also used by all components of the platform for internal authentication. If necessary, change it on the Common tab of each server.

### Installation of additional backup servers

Once the main platform part is installed, it is recommended to configure appropriate backup servers. The ability to ensure uninterrupted business operation is one of the main requirements set by financial regulators.

The MetaTrader 5 platform provides a separate component responsible for data security, which is the backup server. It runs in parallel as an exact copy of the primary trade or history server. The state of the backup server is constantly synchronized with the primary server, allowing any time to switch to the backup server. However, the backup creation process is resource efficient:

- Critical and frequently changing data are synchronized in real time. These include client and trade databases, platform settings, configurations and databases of gateway trade executions.
- The remaining data are backed up every hour, including mail and news databases, files, executable files and gateways and data feeds, plug-ins, etc.

- Always install backup servers it is a mandatory component of the platform. The trade and history servers do not create backups of data bases.
- Several backup servers can be created for each trade and history server, which reduces the risk of system failure to a minimum.

During the complete platform installation, a backup server is only created for the main trade server. It is installed on the same computer/disk, where the main components are installed. It only provides the minimum level of protection, for example against the accidental deletion of the main server databases. However, in case of hardware failure, the main and the backup server will be unavailable.

In order to provide a reliable platform operation, it is recommended to manually add backup servers for the history server, as well as additional trade servers (if any). Such backup servers must be installed separately from the main servers, and always on separate computers.

The easiest way to install the server is to configure it first in the MetaTrader 5 Administrator terminal, and then to use the deployment function.

Before deploying a server, make sure to configure the server network parameters: the password that will be used by other cluster components for connection to the backup server, as well as the IP addresses, through which the new server will send and receive data. All these parameters will be registered in the server installation package. Thus, all remaining cluster components will be able to connect to the backup server immediately after installation, while you will be able to manage the server via the Administrator terminal.

Make sure that network parameters are correctly configured and all the required ports are open. It is important, because during deployment, an instant connection to all the required servers of the cluster will be required.

If a component of the MetaTrader 5 cluster is deployed on another subnet (different from the one where the main trade server and the history server are installed), the main and the history servers must be accessible via the internet. Otherwise, the newly installed component will not be able to connect to them.

All other parameters can be configured after installation.

## MetaTrader 5 trading platform

**Server** [?] [X]

Common Network Backup SQL Export SQL Options Folders Service

Please specify common server settings: type, name, internal identifier, and password.

Type: Backup Server

Name: MetaTrader 5 Main Backup Server

ID: 4 Password: .....

**Server** [?] [X]

Common Network Backup SQL Export SQL Options Folders Service

Please specify network settings of the server: outgoing address, public addresses, and bindings.

The following IPv4 and IPv6 addresses are used for outgoing connections from this server.

Address: 18.140.18.42:2000 [2a00:1111:1:20::2a]:2000

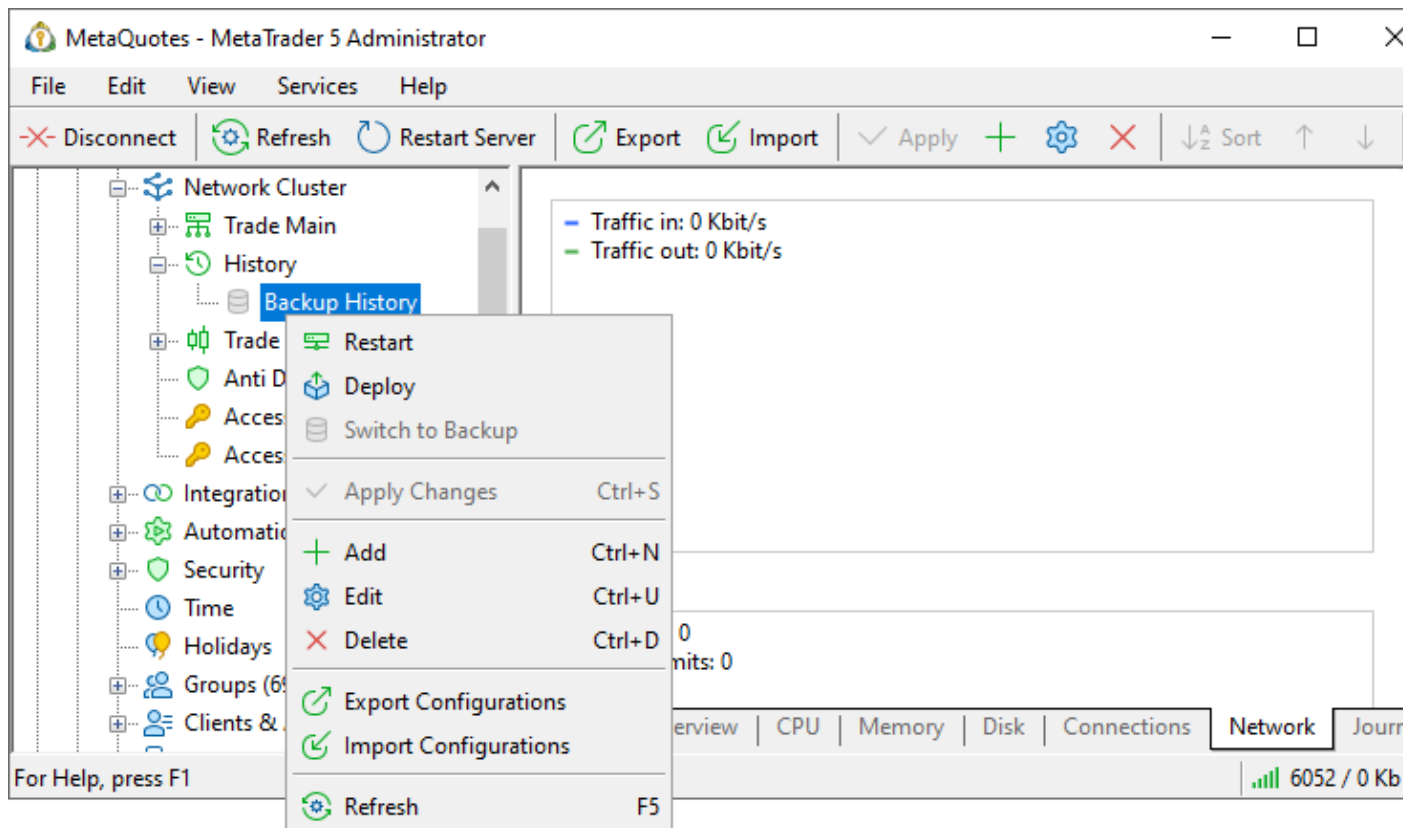
Listen on these IP addresses ☒ Automatically use as public addresses

| Address                   | Port |
|---------------------------|------|
| 18.140.18.42:2000         | 2000 |
| [2a00:1111:1:20::2a]:2000 | 2000 |

OK Cancel Help

## MetaTrader 5 trading platform

After creating a configuration, run the server deployment command:



After that you will receive an executable file. Copy it to the directory to which you want to install the new server and then run from the command line with the /install key. For example:

```
D:\MetaTrader 5 Platform\Backup History\Deploy_5_Backup_Server.exe /install
```

The file will install the backup server with the parameters that were pre-configured in the MetaTrader 5 Administrator. Then you should configure the rest backup parameters:

Server

Common Network Backup SQL Export SQL Options Folders Service

Backup server is used for backup of trade and history servers. Please specify the name of server to be reserved and backup settings.

Master server: Trade Main, 1

☒ Use this backup server for failover

☒ Synchronize journals with master server

☒ Enable backups

Backups path: F:\MT5Backups\Main

Full backup time: 00:17 Additional backups: Every hour

Keep backups: 1 day

OK Cancel Help

When configuring parameters of backup servers, you should find balance between the frequency of backups and the available disk space on the computer where the backup server is installed:

- **Backup time** time of creating file copies. Apart from synchronizing with the main server in real time, the backup server creates database file copies on the disk on a daily basis. Such copies allow restoring the main server status on a specific day.
- **Additional backups** frequency of creating additional file copies. By default, file copies are created every 24 hours, but you can set this to happen more often once per hour or once per 4 hours. Keep in mind that this will require more server resources and more disk space. This option is only available for the backup of trading servers; backup copies of history servers are always created once every 24 hours.
- **Keep backups** file copies storage period. All backup copies older than a period specified in this field are automatically deleted.
- **Enable tick backups** this parameter should be enabled with caution, because tick data take up too much space. If you back up tick data, set short copy storing period so that they do not consume all the available disk space.

- Repeat the procedure and install backup servers for other platform components, including the history server and all other trade servers. Install backup servers separately from the main ones, and always on separate computers.
- Install access servers close to the backup ones. These access servers will be used for client connections in case the platform switches to the backup

servers.

### Additional access servers

It is recommended to install additional access servers in order to improve system performance and reliability. Access servers process client connections, cache data, and protect the platform against DDOS attacks. They also monitor the availability of the trade and history servers to enable the operation of the automatic failover system.

Connection to the entire trading platform is only available through access servers. If you decide to use one server, and it fails, neither administrators nor traders will be available to connect to the platform. In addition, the entire connection processing load will be handled by one server, which may affect negatively the system performance.

It is recommended to install several access servers on separate computers. The platform analyzes the current load on each server, and effectively distributes client connections between them.

Access servers can be installed using the earlier described deployment procedure. After the installation, configure all other parameters in the Network cluster section.

### Additional trade servers

If the company becomes larger and the number of active clients increases, you can install additional trade servers in order to remove some load from the main server. It is recommended to install an additional trade server in the same data center, where the main and history servers are installed. Trade servers should not be far from the history server, in order to avoid delays when delivering quotes and sending trading operations to external systems through gateways.

After the installation of an additional trade server, it is recommended to install its backup server in a different data center, where the backup servers of the main and history servers are running.

Additional trade servers can be installed using the earlier described deployment procedure. After the installation, configure all other parameters in the Network cluster section.

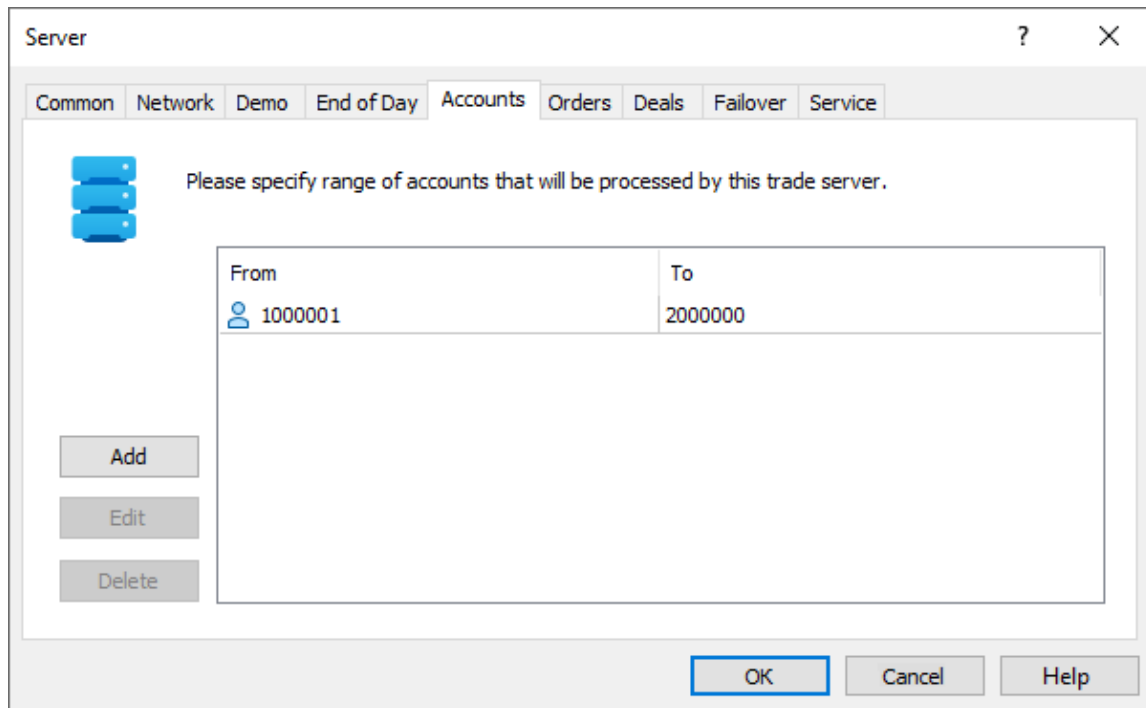
### Automatic creation of the first administrator account

The platform architecture allows an administrator/manager to only work with accounts and databases on the trade server, where the administrator or manager account (its group) is located. For example, when connected to the main server, the administrator/manager cannot create accounts on additional trade servers. In order to connect to the additional trade server and manage accounts on it, the administrator or manager needs to have an account on this server.

During the installation of an additional trade server, one administrator account is automatically created on it. The first free login available in the range of accounts set for the server will be given to the administrator account:



## MetaTrader 5 trading platform



In addition to the administrator account, groups linked to this account are also created automatically on the server:

- demo\demoforex-ID
- managers\dealers-ID
- managers\administrators-ID
- real\real-ID

Here ID is the internal identifier of the added trade server.

- The administrator account and groups can only be created after you specify the range of accounts. It can be specified when adding a server or when editing it. In other words, once a trade server with the specified account range is added, groups and a manager account are automatically created for this server. If you configure the range before running the deployment, the administrator account and the group will be created during server installation.
- A password for the manager account is generated randomly. The login and password of the created account are written to the server's log file (the /Logs folder). The following entry is added: default manager with login '1000' and password 'lfd5fircvs' added
- When you first connect using the automatically created administrator account, you will need to complete the forced password change procedure.

### Launching the server

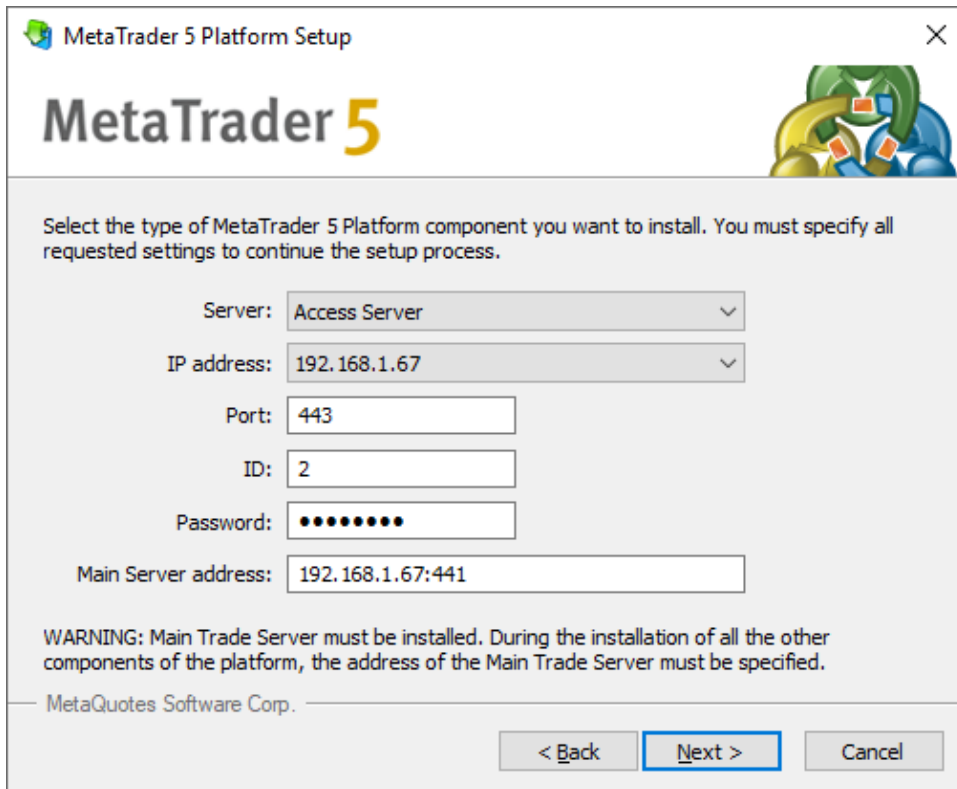
For the newly added server to start functioning, you need to restart the entire platform (the main trade server). After the system reboot, the following operations are performed automatically:

## MetaTrader 5 trading platform

- An account in the manager\administrators-ID group is created for the automatically generated administrator account. If the group does not exist for some reasons, an account will be created in another manager group available on the server.
- If additional accounts were created in the Manager section before restarting the system, and the logins of these accounts fall into the account range of the added server, accounts in appropriate groups will also be created for them.

### Installation of separate platform components using the installer

The simplest way to install individual platform components is to use the deployment procedure. Another available option is to use the platform installer. Select "Single server" and the desired server type.



The image shows the 'MetaTrader 5 Platform Setup' window. It features the MetaTrader 5 logo at the top left and a decorative graphic of three interlocking rings at the top right. The main text instructs the user to select the type of component to install. Below this, there are several input fields: 'Server' (a dropdown menu set to 'Access Server'), 'IP address' (a dropdown menu set to '192.168.1.67'), 'Port' (a text box with '443'), 'ID' (a text box with '2'), 'Password' (a masked text box with dots), and 'Main Server address' (a text box with '192.168.1.67:441'). A warning message is displayed below the fields, stating that the Main Trade Server must be installed and its address specified. At the bottom, there are three buttons: '< Back', 'Next >' (which is highlighted with a blue border), and 'Cancel'. The MetaQuotes Software Corp. logo is visible in the bottom left corner.

MetaTrader 5 Platform Setup

**MetaTrader 5**

Select the type of MetaTrader 5 Platform component you want to install. You must specify all requested settings to continue the setup process.

Server: Access Server

IP address: 192.168.1.67

Port: 443

ID: 2

Password: ••••••••

Main Server address: 192.168.1.67:441

WARNING: Main Trade Server must be installed. During the installation of all the other components of the platform, the address of the Main Trade Server must be specified.

MetaQuotes Software Corp.

< Back Next > Cancel

## MetaTrader 5 trading platform

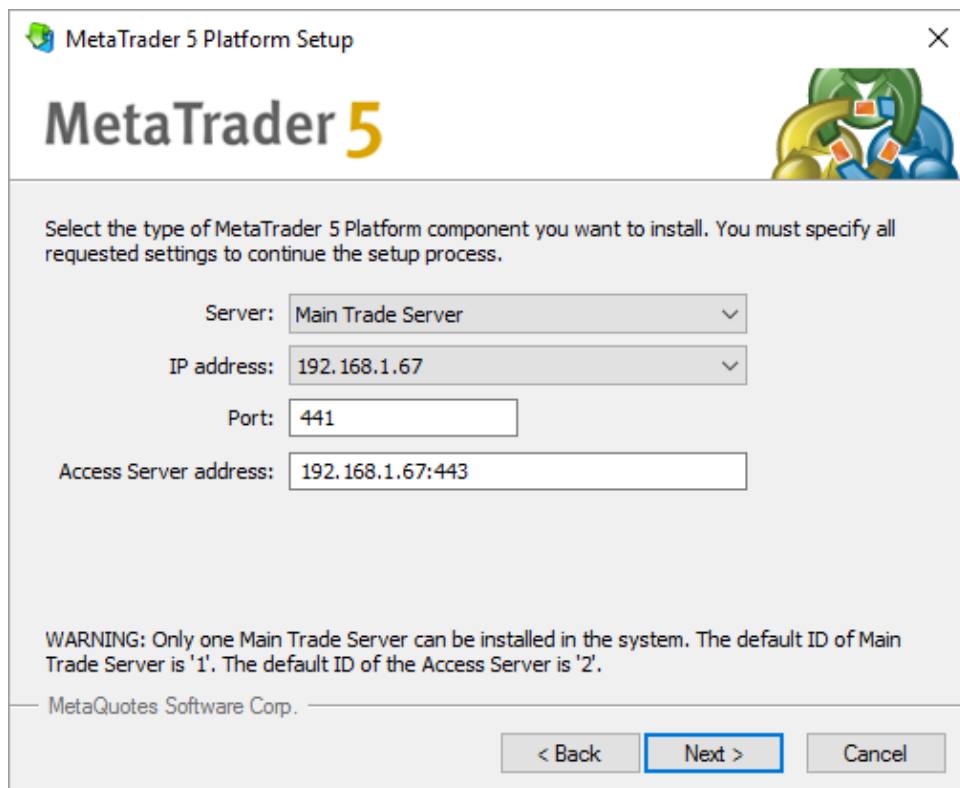
Then you need to specify server parameters, all of them are required:

- **IP address** choose one of the available IP addresses to install the server.
- **Port** specify the port for connection to the server. It must not be occupied by other applications. Port 443 is used by default.
- **ID** will be used for the internal identification of the server.
- **Password** an internal password used for connection between components and the access server.
- [when installing the main server] **Access Server address** the IP address and port number of an existing or future access server, separated by a colon.
- [when installing other servers] **Main Server address** IP address and port of the main trade server separated by a colon.

- Only one main trade server can be installed in the system. The main server is responsible for connections between all other components.
- Install the main trade server before installing platform components.
- After installing an additional server, add its configuration in the Network cluster section and then configure the required parameters.

A separate installation of the main trade server and its access server

Install the main trade server before installing platform components.



MetaTrader 5 Platform Setup

**MetaTrader 5**

Select the type of MetaTrader 5 Platform component you want to install. You must specify all requested settings to continue the setup process.

Server: Main Trade Server

IP address: 192.168.1.67

Port: 441

Access Server address: 192.168.1.67:443

WARNING: Only one Main Trade Server can be installed in the system. The default ID of Main Trade Server is '1'. The default ID of the Access Server is '2'.

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< Back Next > Cancel

## MetaTrader 5 trading platform

A minimum working configuration of the trading platform includes the main trading server and an access server. Without an access server, connection to the trade server from the terminals (client, manager or administrator) will not be possible. That is why, when installing the main trade server for the first time, the parameters of the future access server are added to its configuration files to enable the first connection through the administrator terminal:

| Parameter       | The main trade server | The access server        |
|-----------------|-----------------------|--------------------------|
| <b>ID</b>       | 1                     | 2                        |
| <b>Password</b> | A random password     | The same random password |

In the installation dialog, the future IP address and port of the access server is specified. These data are written to the main server configuration files to enable recognition of the installed access server.

The identifier "2" and a random password for the future access server is also written to the main server configuration. These data will be used for the authentication of the access server on the main server. The same password is used for the server and for the first administrator account with the login 1000. This password will be displayed on the last installation step. It will also be available in the main trade server log in [installation directory\logs\]. The internal identifier "1" is assigned to the main trade server.

You will need to specify these parameters when installing an access server. These details can be changed after connecting to the platform using the administrator terminal.

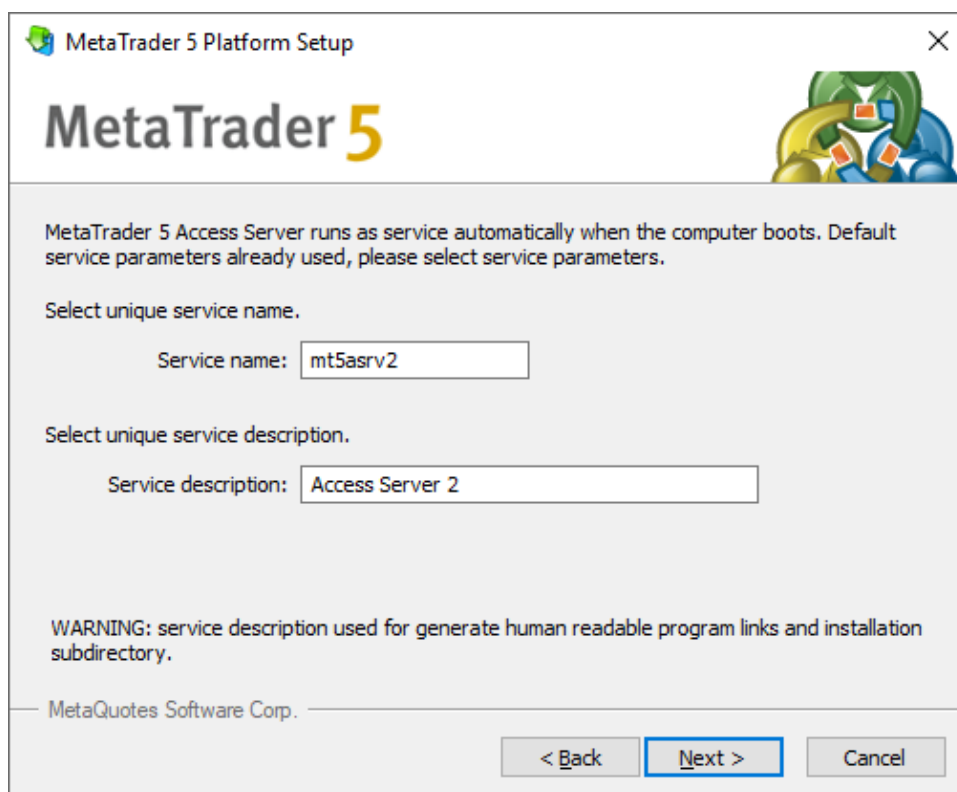
### Unique service names for platform components

Trade platform settings run as services in the operating system. These services must have unique names. The following names are assigned to the services by default:

- mt5msrv the main trade server
- mt5tsrv an additional trade server
- mt5asrv an access server
- mt5bsrv a backup server
- mt5hsrv a history server

If a server of a certain type is already installed in the operating system, you will need to specify the unique service name and description when installing additional servers of the same type:

## MetaTrader 5 trading platform



When deploying an additional server, a number corresponding to the server ID in the platform is automatically added to the server name. For example: mt5tsrv5.

Installation of a platform for testing new features and developing own solutions

In addition to the main operating platform, you can install a test (developer) platform for free. Use it to test new platform settings and develop applications using MetaTrader 5 API, without affecting workflow processes and real client operations. It is also recommended to use this platform for installing updates released in beta mode. This will allow you to explore new features in advance and test the operation of third-party applications before updating your main platform.

Please contact Service Desk to receive the test platform license.

A test license is a complete analog of the main license (including the limitation on the number of accounts), but it has several specific features:

- The server name in the license contains "Test". This name is displayed in all terminals.
- Activations of servers installed under the test license cannot be set as main.

The following limitations apply to servers installed under the test license:

- The server is not included in the list of available servers in the desktop terminal. To connect to the server, specify the access server address manually.
- Connection to the server from mobile terminals and the web terminal is not possible.

## MetaTrader 5 trading platform

- The server does not allow sending push notifications through the Manager terminal and API.
- Virtual hosting cannot be allocated for an account opened on the server with the test license.
- An account opened on the server with the test license cannot be registered as a trading signal.

For application development using MetaTrader 5 API, the platform provides debug versions of the trading and history servers (regardless of the license type).

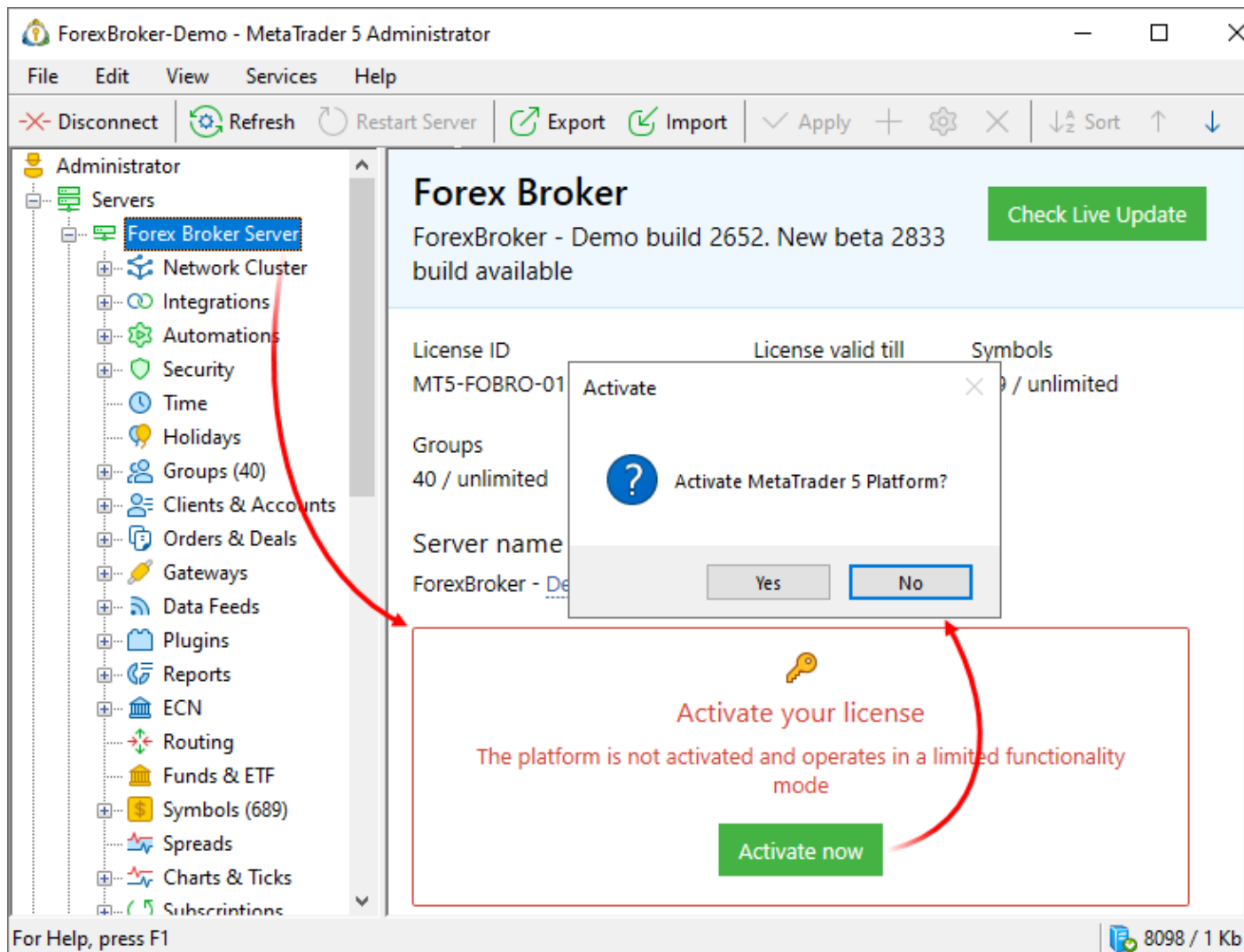
MetaTrader 5 Trading Platform / Platform Installation / Activation

### Activation

A license given for installation with the platform distribution is non-activated and has several limitations:

- Only one trade server (main) can be installed in the system;
- Not more than 100 users can be created;
- Not more than 5 groups can be created, including four groups created on default.

After installing the platform and connecting to the server via the administrator terminal, go to the start page and click on "Activate now":



## MetaTrader 5 trading platform

For launching the activation process, you can also use the "🔑 Activate" command of the [Services](#) menu. The following actions are performed during activation:

- A request is sent to the update server of the developer company;
- In case of successful license checking, a new activated license is generated; it is bound to the configuration of server where the platform is installed;
- Activated [license](#) is sent back to the trade server.

The number of license activations is limited. Normally, up to five activations are allowed for one license.

### Activation Management

To view your activations, visit the "[App Store\Licenses](#)" section of the technical support website:

News Articles Documentation Workshops Certification App Store Billing Analytics <sup>NEW</sup> Support

## Licenses of Forex Broker



### MetaTrader 5

Forex Broker

forex.broker.net

created: 2009.11.03

license expiration: 2024.01.20

| <input type="checkbox"/> | Server name      | IP            | Build | Created    | Active    |
|--------------------------|------------------|---------------|-------|------------|-----------|
| <input type="checkbox"/> | ForexBroker-Temp | 10.01.01.126  | 2515  | 2020.02.01 | 2020.07.1 |
| <input type="checkbox"/> | ForexBroker-Demo | 75.123.197.45 | 2537  | 2019.06.01 | 2020.07.1 |

A list of activations of each platform license contains the following data:

- **Server name** the name displayed in client terminals (in the program name, in the Navigator window, etc.). The first part of the name is taken from the appropriate [White Label](#), the second one is used from platform settings (specified on the [start page](#)).
- **IP** the IP address from which the platform was activated.
- **Build** current platform build. If the build is shown in red, this means there is a [new version](#) available for the platform. We recommend installing the latest version to ensure the operation stability and to access all the new features of the platform.
- **Created** platform activation date.
- **Last active** the date of the last platform data update (including the platform build). Platforms send related service information to the server every 24 hours, during [optimization time](#). The last activity time is updated each time the data is sent. If no data is received for one month, the date is shown in red, which is a warning about server inactivity. If a platform does not update data for three months, it is automatically removed from the list of available servers in terminals, as well as ["Signals"](#) and ["Virtual Hosting"](#) services.



## MetaTrader 5 trading platform

Two types of platform activations are available: main (shown in bold) and non-main.

- Client terminals can only connect to a platform with the main activation.
- To enable connections of client terminals, the platform sends information about its access points (installed access servers) to the update server every hour. Information is sent to the server regardless of the activation type, but only access points of the main activation are transmitted to terminals.
- Servers with non-main activation are not shown in the broker selection dialogs when opening accounts through terminals.

To set a platform activation as main, please contact Service Desk.

- If you move the platform to another server and activate it, the new activation will appear in the list as non-main. As soon as you complete the platform preparation, be sure to contact Service Desk to change the activation to main. Otherwise, traders may have problems connecting to the new server, since access points will not be known to client terminals.
- Desktop terminals receive up-to-date information about access points during each account connection. It is recommended to keep at least one old access point operating during 1-2 weeks after migrating the platform to other equipment/hosting provider, so that terminals can receive such information.
- Activation is bound to server hardware. In case you change computer configuration, a new activation from the same IP address can appear in the list. If the list of access points has not changed, this will not affect operation with client terminals. However, we recommend that you contact the support team to switch the new activation to the main one.

Outdated activation can be removed by clicking . If you do not have enough permissions to delete activations or do not see the server management section, please contact Service Desk.

MetaTrader 5 Trading Platform / Platform Installation / White Label

## White Label

White Label refers to the branding of the desktop terminal of the trading platform for a specific brokerage company. The White Label includes the company's name, contact details, logos, and other MetaTrader 5 terminal settings.

The client-side part of the platform also includes MetaTrader 5 mobile terminals for iPhone and for Android, as well as the a web terminal that provides account access via a web browser. If these components were not purchased together with the platform, you can ordered then separately at any time. Please note that no appearance modifications (branding/white labeling) are applied to these components.

White Label applies exclusively to the MetaTrader 5 desktop client terminal. The Manager and Administrator terminals remain standard and are not subject to customization.

## White Label for the Client Terminal

After the platform purchase, a support ticker will be automatically created in the Service Desk section of the technical support site. This ticket will include an online form used for creating the White Label. The ticket will allow you to track the delivery status and provide additional information if necessary.

The online form is located in the body of the ticket, directly beneath the subject line.

## MetaTrader 5 trading platform

### About window

About

Click to  
upload logo

24-bit color BMP  
image file. Size should  
not exceed 131 px  
width and 151 px  
height

Forex Broker MetaTrader 5  
Version: 5.00 Build 2037  
30 Apr 2020

Forex Broker

Legal address of the company

Phone

[www.forexbroker.dev](http://www.forexbroker.dev)

[support@forexbroker.dev](mailto:support@forexbroker.dev)

T

Copyright 2001-2020, MetaQuotes Software Corp.

Close

### Icons

Upload  
icon

Bro

[How](#)

White Label

Custom files

Servers

Documents

Server name:

ForexBroker

Short name:

Forex Broker MT5

**Account types:** Keep in mind, for demo accounts onboarding you will need to set up allocation groups on MetaTrader 5

Copy details to clipboard

- The fields marked in red are mandatory and must be completed. These include: company address, website, name, short name, server name, and logos.
- All terminal information (branding, images, server name) must match the company's name or its legally registered brand. Promotional slogans are not allowed in images unless the company holds a valid legal registration or license for their use.
- If you do not yet have ready-made logos, you may use the default MetaQuotes logos. Logos and other details can be updated later. To do this, simply open the White Label editing form from the [Download](#) page. The corresponding link appears when you hover your cursor over the terminal name.
- The "Custom Files," "Servers," and "Documents" sections are available only when editing the White Label via the [technical support website](#). These sections cannot be completed during the initial platform order.

#### The About window

In this tab, specify your company name, contact details, logo, and program name. This information will be visible to traders in the "About" window of your terminal. You can edit the relevant fields directly within the "About" window. When you hover the cursor over each field, a tooltip will appear explaining what information should be entered.

| Element      | Description                                                                                                                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company name | The name of the company for which the White Label is being created. By default, this field is populated with the company name provided during contract signing. Scanned copies of company documents can be uploaded in the Documents tab.                                                                                 |
| Address      | The registered office address of the company, as specified in the KYC form. The address can be specified in two lines. Use the special character \n to insert a line break.                                                                                                                                               |
| Phone        | <p>Phone number. You may provide additional contact information beyond just the company phone number. The data can be split into two lines using the special character \n. For example:</p> <p>Tel : +1 xxx xxxxxx\nFax: +1 xxx xxxxxx</p> <p>Phone numbers should be specified in the standard international format.</p> |
| Site         | <p>The URL of your company's website, for example:</p> <p>www.metaquotes.net</p>                                                                                                                                                                                                                                          |

| Element      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Do not include the "https" protocol prefix; it will be added automatically. Only one website address can be entered; additional links will not be functional.                                                                                                                                                                                                                                                                                                                                |
| E-mail       | The e-mail address of your company's technical support service.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Logo         | <p>A BMP image with 24-bit color depth. The image dimensions must not exceed 131 pixels in width and 151 pixels in height.</p> <p>To upload a logo, click the Select button and choose an image file in the dialog window. After uploading, you can center the image by clicking the corresponding button.</p> <p>To use a standard MetaQuotes logo, click the Default button (<a href="#">archive with MetaQuotes logos</a>).</p>                                                           |
| Program name | <p>The name is generated based on the company name or registered trademark (brand). It is used as:</p> <ul style="list-style-type: none"> <li>• ♦ the title of the client terminal window</li> <li>♦ the name of the terminal installation directory and Start menu group</li> <li>♦ the label under the desktop shortcut</li> </ul> <p>If you want to use the word "MetaTrader" in the program name, it must be written exactly as: MetaTrader (as one word, with capital "M" and "T").</p> |

The About window always displays the MetaQuotes Ltd copyright notice, a link to the company's website, and a link to the End-User License Agreement (EULA).

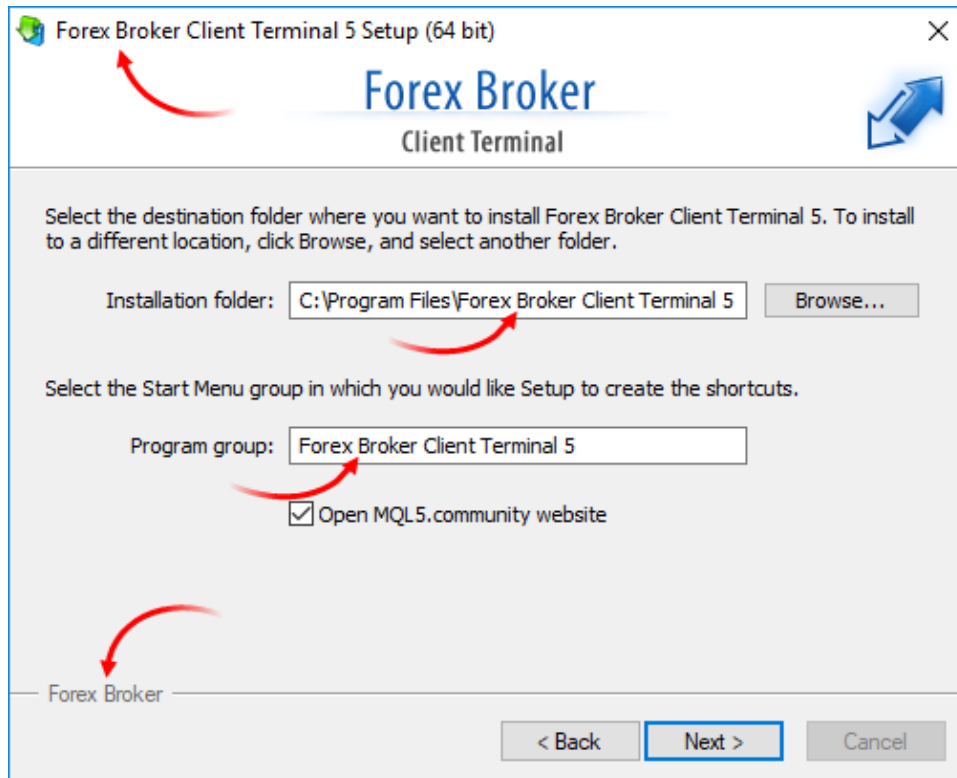
### The Installer window

In this section, you can specify your company logo to be displayed in the terminal installation window, as well as the icon under which your terminal will operate.

| Element | Description                                                                                                                                                                                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Banner  | <p>A BMP image with 8-bit color depth (256 colors), with exact dimensions of 500 pixels wide by 60 pixels high.</p> <p>You can create a banner by combining the default MetaQuotes logo with your company or program name.</p>                                                         |
| Icon    | <p>An ICO file containing a set of frames sized 16x16, 32x32, 48x48, and 64x64 pixels, each with 8-bit and 32-bit color depth.</p> <p>This icon will appear in the main window of the client terminal, on the desktop, and in the Start menu. The icon file (terminal.ico) is also</p> |

| Element | Description                                                                                                                                                                                                                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>included in the directory of the installed terminal.</p> <p>Icons can be created using various free programs, such as <a href="#">IcoFX</a> and others. A detailed guide on creating an icon using "Icon Studio" is available in the answer to the question: "<a href="#">How to create an icon for the client terminal?</a>"</p> |

In addition to these elements, the company name is also used in the installer. It is displayed in the window title, installation paths, and footers:



## Custom files

In this tab, you can upload files that will be automatically included in the client terminal build and placed into their respective folders.

Custom files should be taken from the corresponding folders of the client terminal. To prepare them, open the terminal, configure it as desired, close it, and then copy the necessary files. These can then be uploaded using the custom file upload interface. Do not include any files whose purpose you do not clearly understand. If you wish to keep all default settings and the appearance of the client terminal unchanged, you may skip this tab entirely.

| Folder                       | File types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| config                       | <p>Here you can upload files that store terminal settings. Upon first launch after installation, the client terminal will apply these configurations. For example, terminal.ini stores data related to the position and size of the client terminal windows (note: this does not include chart windows, which are saved in the profile - see below). Keep in mind that the coordinates are stored in absolute values, which may result in different window positions on different screens. We do not recommend including this file in the terminal build. The file named trade.ini contains trading settings of the terminal (found under Tools - Options in the terminal menu). The common.ini file contains general settings (also under Tools - Options).</p> <p>It is strictly prohibited to upload a common.ini file that contains the value AllowDllImport=1. For security reasons, we do not allow DLL imports to be enabled by default for end users.</p> |
| bases\default\symbols        | <p>You can upload price history files here, along with a symbol set file that defines which instruments will appear in the Market Watch window on the terminal's first launch. For details please see "<a href="#">How to change the default list of symbols in the client terminal label?</a>"</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MQL5                         | <p>This folder and its subfolders may contain custom indicators, scripts, or Expert Advisors (EAs) that will be available in the client terminal.</p> <p>It is strictly forbidden to include any .DLL files, or .EX5 files that reference external DLLs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MQL5\profiles\charts\default | <p>Here, you can add a default terminal profile that will be applied on the first startup. For details please see "<a href="#">How to change the default set of charts in the client terminal label?</a>"</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## MetaTrader 5 trading platform

| Folder                   | File types                                                                                                                                                                                                                                      |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MQL5\profiles\symbolsets | This directory stores symbol set files (*.set) that can be selected from the context menu in the Market Watch window. You may create your own sets based on the types of instruments offered.                                                   |
| MQL5\profiles\templates  | You can upload your own chart templates (*.tpl) and terminal report templates (*.htm). You may also modify built-in report templates such as ReportTrade.htm, ReportHistory.htm, and ReportTester.htm, and upload the customized versions here. |

### Servers

In this tab, select the servers that should appear in the list when opening a new account in the terminal. Example:

**Open an Account**

**List of brokers**  
Select a broker you are to trade through

add new broker like 'BrokerName' or address 'company.com' Find your broker

[Forex Broker](#) [ForexBroker](#)

**Open an Account**

**Opening a demo account**  
A demo account allows trading with virtual money and learn without risk

First name:

Second name:

Email:

Phone:

☒ Use hedge in trading

Server:

Account type:

Deposit:  EUR, virtual money to be deposited to your account

Leverage:  virtual credit funds to be provided by the broker for trading



## Documents

Upload documents confirming the registration details of the company for which the White Label is being issued. A list of required documents can be found in the article "[Guide to easily meet KYC requirements](#)". Only color scans are accepted.

If you are editing an existing White Label and your company details have not changed, re-uploading the documents is not required.

## Submitting the final version

Carefully review all entered information before submitting the final version. After submission, changes can only be made once the client terminal build has been completed. To temporarily save your progress and return later, click "Save as a draft".

Once all data is confirmed, click "Send final version" in the bottom-right corner of the form.

## What happens next

After your order is confirmed (the automatically created ticket will change status to "Completed"), the client terminal White Label will be built using the information provided in the form. The download link for the [web installer](#) will appear under [App Store / White Labels / MetaTrader 5](#).

To modify the White Label later, click the edit button in the block containing the desktop terminal link:

[Documentation](#) [Workshops](#) [Certification](#) [App Store](#) [Billing](#) [Analytics](#) [Support](#)

## MetaTrader 5 White Labels of Broker

☐ Show links on [download.metatrader.com](https://download.metatrader.com)

Broker 

Desktop Terminal

<https://download.mql5.com/cdn/web/broker/mt5/mt5setup.exe>

We strongly recommend that brokers use distribution links provided in the [App Store / White Labels / MetaTrader 5](#) section of this website when sharing installers with traders. If your corporate policy prohibits linking to external websites, you may host the files on your own server, just be sure to periodically check for and apply updates. Always keep your applications up to date!

### Platform configuration for White Label

Ensure that after the White Label is registered, your trading server receives the updated license file. By default, the trading server checks for license updates at the end of the trading day. To apply the update immediately, run a [manual platform update](#). Note that this will restart your trading server.

If the server does not restart, it indicates an update error. Verify that the server machine can access: <https://updates.mt5quotes.net>. For further details, check the [History server log](#) filtered by keyword "Update" or in the "logs\mt5srvupdater.log" file located in the [History server directory](#).

If the license file is not updated, clients will see outdated company information in their terminals when connected to your server. Also, outdated data will be displayed in client statements.

### White Label for the Mobile Terminal

Mobile terminals allow your clients to manage their trading accounts without a desktop PC. Using smartphones, they can analyze market conditions and execute trades at any time.

Ordering a mobile terminal gives you access to several versions for different device types:

- MetaTrader 5 for iPhone for iPhone, iPod Touch and iPad powered by iOS 15.0 and higher;
- MetaTrader 5 for Android for smartphones and tablet PCs powered by Android OS 4.0 or higher.

Clients can connect to any trading server owned by a company that has purchased a mobile White Label.

To order the mobile terminal, select MetaTrader 5 Mobile in the [Buy section](#) of the technical support website. In the popup window, specify your company name. It must exactly match the name used in your desktop terminal's White Label.

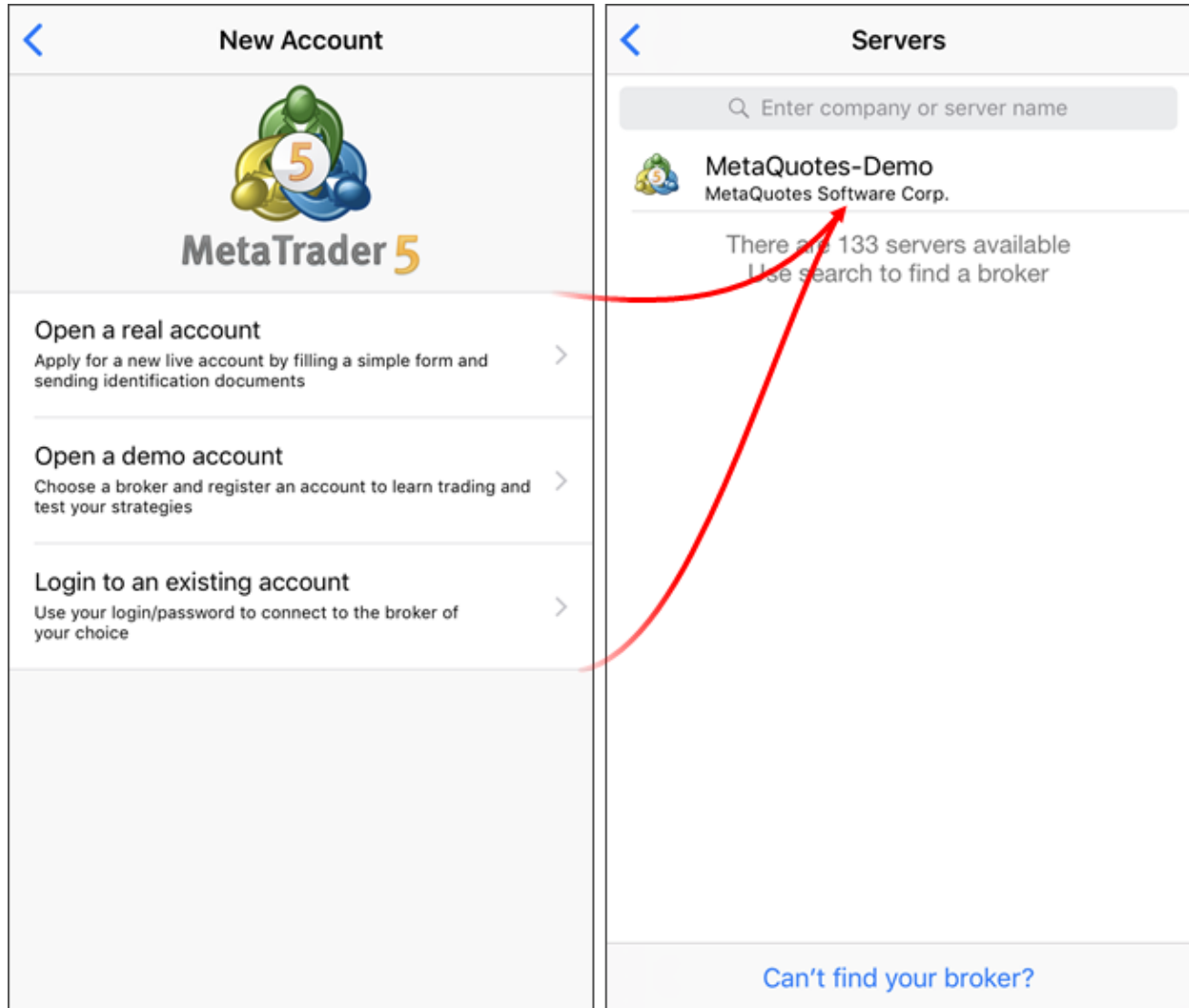
Once the order is confirmed, a support ticket will be automatically created in the [Service Desk](#) with your order number in the title. This ticket will help you track the status of your order and provide any additional information if required. Once the order status changes to "Completed", the mobile White Label will be assembled for you.

The demo and live account opening settings configured in the [Account Allocation](#) section will be applied to the mobile terminal just like the desktop terminal. The server icon shown in the list will match the one used in the desktop terminal.

## MetaTrader 5 trading platform

MetaTrader 5 for iPhone can be downloaded via [iTunes](#) or from the App Store on iPod Touch/iPhone/iPad. MetaTrader 5 for Android can be downloaded from the [Google Play](#) website or Google Play mobile application.

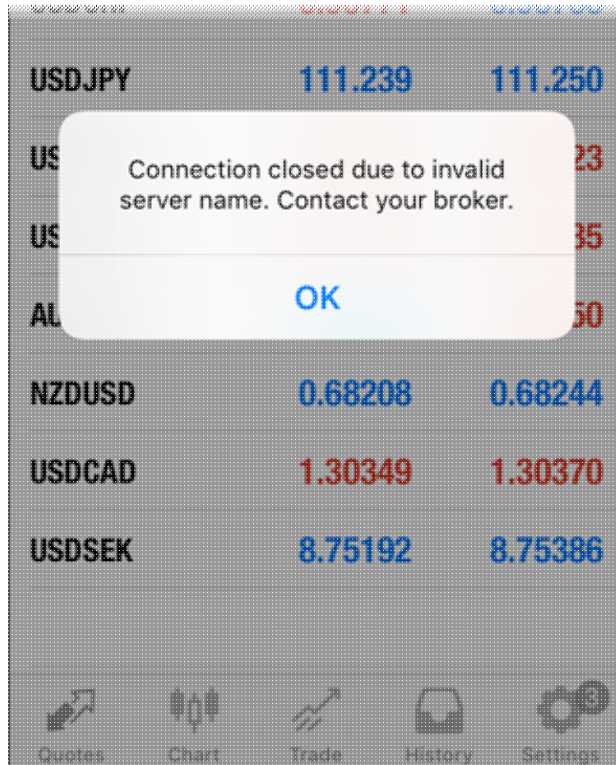
Your server will be accessible for connecting both existing and new accounts.



## MetaTrader 5 trading platform

By default, only the MetaQuotes-Demo server appears in the list. To find another server, users can simply type the first few letters of its name or the associated company. Server search results are filtered based on the account type - demo servers for demo accounts, real servers for real accounts.

If MetaTrader 5 mobile users encounter the error "invalid server name", it likely means that the Company field in the group settings is incorrect or does not match the company name associated with the mobile White Label.



## MetaTrader 5 trading platform

You can always find the exact company name in the original White Label request ticket, on the [White Labels](#) page, or in the [edit form](#).

MetaTrader 5 Trading Platform / Platform Installation / Fast Deployment

## Fast Deployment

The fast deployment function simplifies installation of the platform's additional components. During a conventional server installation, you should specify its settings twice: during the installation itself and when creating its configuration via the Administrator terminal. Instead, you can first create a correct server configuration and then receive a setup file with all the previously specified settings.

Before the deployment, configure the server network parameters: ID and password to be used by other cluster components for connection to the new server, as well as IP addresses to be used by the new server for sending and receiving data. All these parameters, as well as data for connecting to the platform's main server, are specified in the installation package. Thus, all remaining cluster components are able to connect to the new server immediately after the installation allowing you to manage the server via the Administrator terminal.

The function of deployment is available only for access, backup and additional trade servers.

## Creating and Configuring the Server

Create a server configuration in the Network section by clicking + Add. Specify an ID and a password to be used by other cluster components for connection to the new server on the Common tab. Specify IP addresses to be used by the new server to send and receive data on the Network tab.

Make sure that your network parameters are properly configured and all required ports are open, because instant connection to all required servers within the cluster will be needed during deployment.

If any of the MetaTrader 5 cluster components needs to be deployed on a different subnet (different from the net where the main trade server and the history server are installed), the main and history servers must be accessible via the Internet. Otherwise, the installed component will not be able to connect to these servers.

## MetaTrader 5 trading platform

**Server**

Common Network Backup SQL Export SQL Options Folders Service

Please specify common server settings: type, name, internal identifier, and password.

Type: Backup Server

Name: MetaTrader 5 Main Backup Server

ID: 4 Password: .....

**Server**

Common Network Backup SQL Export SQL Options Folders Service

Please specify network settings of the server: outgoing address, public addresses, and bindings.

The following IPv4 and IPv6 addresses are used for outgoing connections from this server.

Address: 18.140.18.42:2000 [2a00:1111:1:20::2a]:2000

Listen on these IP addresses ☒ Automatically use as public addresses

| Address                   | Port |
|---------------------------|------|
| 18.140.18.42:2000         | 2000 |
| [2a00:1111:1:20::2a]:2000 | 2000 |

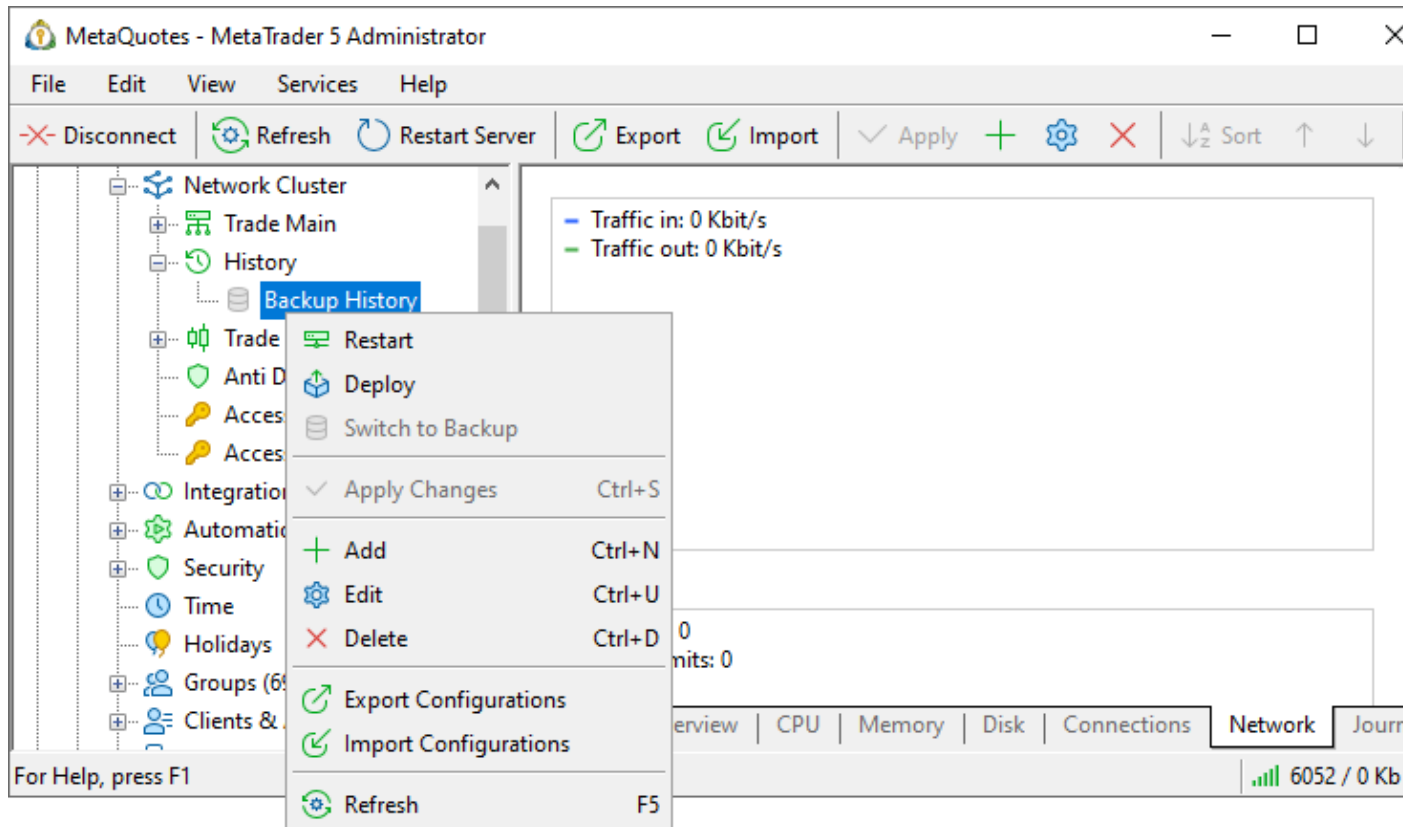
OK Cancel Help

## MetaTrader 5 trading platform

The created server will be marked by a gray icon (for example, ) , which means that the server service is not running in the operating system. In our case, such a service is simply not installed.

### Deployment

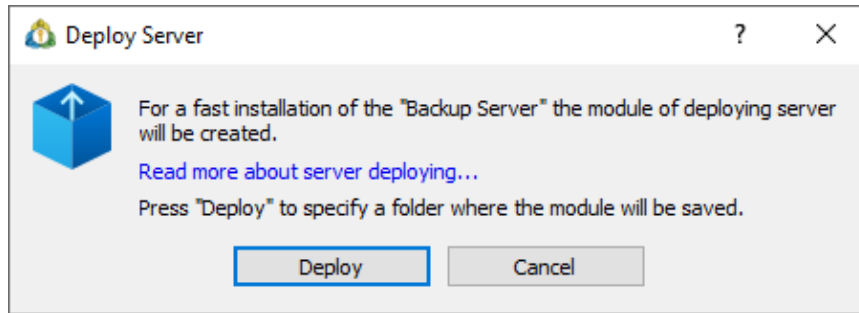
Click " Deploy..." in the new server context menu:





## MetaTrader 5 trading platform

This will open the deployment dialog. Click Deploy and select a folder where the setup file is to be saved.



A file of the following form will be saved to the specified folder: Deploy\_ID\_ServerName.exe. Here ID is the internal ID of the server, ServerName name of the server. Copy the file to the required computer, to the directory from which the installed server will operate. After that run it from the command line with the /install key. For example:

```
D:\MetaTrader 5 Platform\Backup History\Deploy_5_Backup_Server.exe /install
```

The file will install the server with the parameters pre-configured in the MetaTrader 5 Administrator. The results of deployment are written in a text file Deploy\_ID\_ServerName.txt, located in the same directory.

```
D:\MetaTrader 5 Platform\Backup History>Deploy_7_Backup_Server.exe /install
0      2017.05.29 15:02:43.380
0      2017.05.29 15:02:43.380      MetaTrader 5 Server Deployer (build 1602, 25 May 2017)
0      2017.05.29 15:02:43.381      Copyright 2001-2017, MetaQuotes Software Corp.
0      2017.05.29 15:02:43.381
0      2017.05.29 15:02:44.008 Deployer      Type      = Backup Server
0      2017.05.29 15:02:44.008 Deployer      ID      = 7
0      2017.05.29 15:02:44.011 Deployer      Name      = Backup Server
0      2017.05.29 15:02:44.011 Deployer      Own Address      = 192.168.9.67:447
0      2017.05.29 15:02:44.012 Deployer      Main Address      = 192.168.9.67:441
0      2017.05.29 15:02:44.013 Deployer      History Address      = 192.168.9.67:442
0      2017.05.29 15:02:44.013 Deployer      Service      = mt5bsrv7
0      2017.05.29 15:02:44.014 Deployer      Service Display      = mt5bsrv7 7
0      2017.05.29 15:02:44.015 Downloader      connect 192.168.9.67:442 success
0      2017.05.29 15:02:44.025 License information updated
0      2017.05.29 15:02:44.316 Downloader      'mt5bs64' build 1590 loaded
0      2017.05.29 15:02:44.379 Downloader      'mt5hsu64' build 1590 loaded
0      2017.05.29 15:02:44.381 Downloader      downloaded 20444 kb in 0 sec (55825.36 kbps)
2      2017.05.29 15:02:44.922 Deployer      'mt5bs64' unpacked
2      2017.05.29 15:02:44.947 Deployer      'mt5hsu64' unpacked
0      2017.05.29 15:02:45.028 Deployer      server 'Backup Server' enabled in firewall
0      2017.05.29 15:02:45.033 Deployer      server 'Backup Server' enabled in TCP/IP filtering
0      2017.05.29 15:02:46.015 Deployer      server 'Backup Server' installed as 'mt5bsrv7' service
0      2017.05.29 15:02:46.890 Deployer      server 'Backup Server' configured
0      2017.05.29 15:02:51.219 Deployer      server 'Backup Server' started
0      2017.05.29 15:02:51.220 Deployer      Service 'mt5bsrv7' check initialized
0      2017.05.29 15:02:51.222 Deployer      Service 'mt5bsrv7' checked
D:\MetaTrader 5 Platform\Backup History>
```

## MetaTrader 5 trading platform

During installation, the availability of a network connection to the main and historical servers of the platform is checked. When installing a backup server, the system additionally checks the connection to the primary server. If the connection cannot be established, the installation is aborted and the relevant message is added to the log. To ignore errors and to install the component anyway, restart the installer with the additional `/nochecks` switch.

After installation you can configure other parameters of the server.

### Additional Parameters of Installation

The deployer file can be run with additional parameters specified in the command line:

- **/nogui** in an operating system with UAC (User Account Control) enabled, running the deployer file may require higher privilege (administrator rights) and evoke a window with the corresponding request. When running the deployer file with this parameter, an attempt to acquire higher privilege is not made.
- **/main:address:port** using this parameter, you can redefine the address of the main trade server in the configuration of the server installed.
- **/history:address:port** using this parameter, you can redefine the address of the history server in the configuration of the server installed.
- **/name:name** using this parameter, you can define a short name of the service of the server installed.
- **/display:display** using this parameter, you can define a full name of the service of the server installed.
- **/desc:description** using this parameter, you can define a description of the service of the server installed.

An example of running the file with additional parameters: `Deploy_5_AccessSever.exe /install /main:192.168.1.135:433 /name:mt5accesssrv`.

- The server will be automatically configured according to the settings specified for it in the administrator terminal.
- The server is installed in the folder, from which the deployer was started.
- After deploying a server you should restart the main trade server.
- If a component is installed in another subnetwork (different from the one, where the main trade and the history servers are installed), the main trade server and the history server must be accessible via the Internet. Otherwise, the newly installed component will not be able to connect to them.
- In case of deploying a trade server, a manager account is created on it. It is required for the first connection to the server. The account login and password are saved in the log file of the server (`/Logs` folder). The log record looks as following: default manager with login '1000' and password 'lfd5fircvs' added

### Removing the Server

## MetaTrader 5 trading platform

To remove the service of the installed server from the operating system, you can use the corresponding console command of the server (/uninstall). After that, the folder where the server was installed, can be removed physically from the disk.

## MetaTrader 5 Trading Platform / Platform Installation / Console Setup

### Console Setup

Every executable file (\*.exe) of the trading platform servers possesses a certain set of console commands. Using them you can perform common actions with appropriate services in the system, as well set up connection of components when connection via the administrator terminal is impossible.

To start executing console commands, open the "Run" window of the operating system and execute the "cmd" command. In the command line go to the directory where the executable file of one of the servers is located, enter its name and specify one of the console commands. For example:

```
C:\MetaTrader 5 Platform\Access>mt5access.exe /stop
```

### Common Commands

The following set of console commands perform common actions and is available for executable files of all servers:

- **/start** start the service that is connected with this executable file of the server;
- **/stop** stop the service that is connected with this executable file of the server;
- **/restart** restart the service that is connected with this executable file of the server;
- **/console** start the service in the console mode;
- **/install [/name:service\_name /display:displayed\_name /description:service\_description]** install a service associating it with this executable file. For additional parameters you can specify the service name for the operating system, extended (displayed) service name for users, as well its description. If you do not specify any of parameters, the service name will be set by default;
- **/uninstall [/name:service\_name]** uninstall a service associated with this executable file, For an additional parameter, you can specify the service name to be uninstalled.
- **/info** show information about the server: internal identifier (ID), own address, address of the main server and binding addresses (for access servers).

Additional parameters are specified without square brackets. For example: /install /name:mt5access /display:Access server

### Setup of Servers

If incorrect settings were specified during the installation of any of components, and connection to the platform via the administrator terminal is impossible, you can re-configure the components using the console command **/config**. It can be executed on any server except for the main trade server. This command has the following parameters:

## MetaTrader 5 trading platform

- **/main:address:port** (for trade servers) or **/main\_trade:address:port** (for other servers) IP address and port of the main trade server, separated by a colon.
- **/own:address:port** IP address and port of the server you are configuring, separated by a colon.
- **/history:address:port** the IP address and port number of the history server. The parameter is only used when configuring access servers. It enables immediate specification of the history server address, without waiting for it from the main server. This improves the speed and reliability server deployment.
- **/id:identifier** the internal identifier of the server you are configuring.
- **/password:password** the internal password of the server you are configuring.

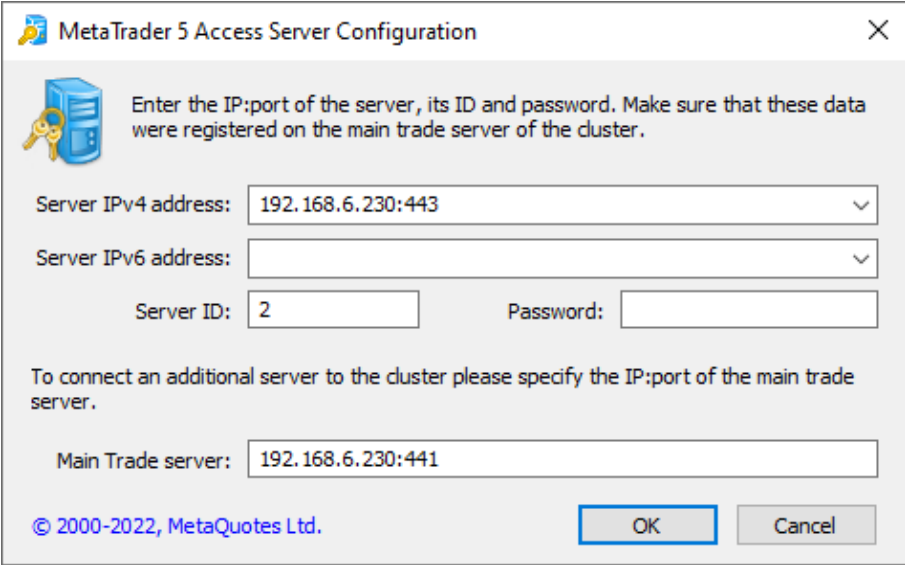
You can specify multiple addresses separated by commas in the 'main', 'own' and 'history' parameters. For example: /main:192.168.0.1:440,[2a00:1987:1:30::2a]:440.

These parameters are analogous to those specified in the "Network" section in the administrator terminal. After the command execution, the earlier specified parameters for the server will be overwritten.

Example: /config /main:192.168.0.1:440 /own:192.168.0.1:443 /id:3  
/password:3jdaLjsQ

### Setup of Servers in the Graphical Mode

In order to start setting up a server in the graphical mode, you should run its executable file with the **/gui** key from the command line. For example, mt5access64.exe /gui.



The screenshot shows the 'MetaTrader 5 Access Server Configuration' dialog box. It contains the following fields and text:

- Header: MetaTrader 5 Access Server Configuration
- Icon: A blue server icon with a key.
- Instruction: Enter the IP:port of the server, its ID and password. Make sure that these data were registered on the main trade server of the cluster.
- Server IPv4 address: 192.168.6.230:443 (dropdown menu)
- Server IPv6 address: (empty dropdown menu)
- Server ID: 2 (text input)
- Password: (empty text input)
- Additional instruction: To connect an additional server to the cluster please specify the IP:port of the main trade server.
- Main Trade server: 192.168.6.230:441 (text input)
- Footer: © 2000-2022, MetaQuotes Ltd.
- Buttons: OK and Cancel

Parameters that are set here are the same as in the console mode:

- **Server IPv4\IPv6 address** colon separated IP address and port of the configured server;
- **Server ID** the internal identifier of the server you are configuring;
- **Password** the internal password of the server you are configuring;
- **Main Trade server** IP address and port of the main trade server, separated by a colon.

- When launching the backup server setup, the Restore command becomes available in the configuration window. It can be used to restore a server that was backed up by the selected server. Find more details about the server recovery [here](#).
- Use only the /gui\_main key to configure the main trade server. When using the /gui key, the main server becomes a secondary one.

### Linking to NUMA Nodes

You can link specific servers to the required [NUMA nodes](#). How to link:

- Stop the server.
- Call the command 'mt5trade64.exe /modify /numa\_node:X' indicating the node number instead of X. Repeat for other servers.
- Start the server.

### Setup of the Main Trade Server

To configure the main trade server, a separate console command **/config\_main** with the below parameters is used:

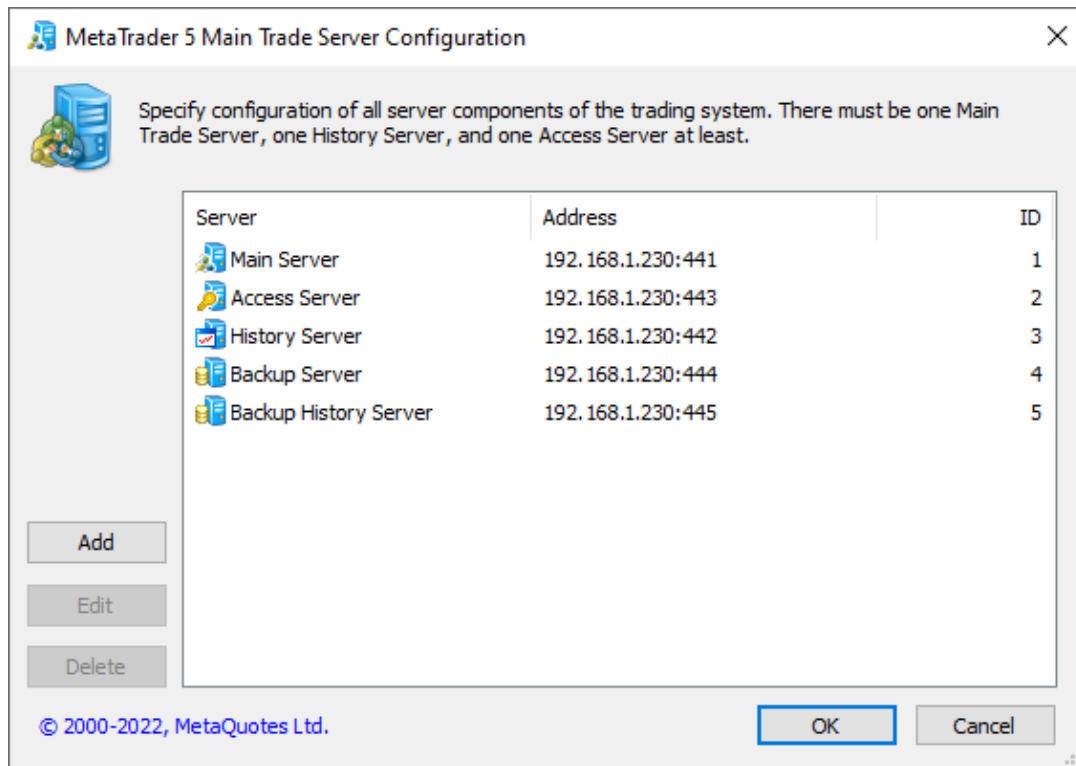
- **/main:address:port** IP address and port of the [main trade server](#), separated by a colon;
- **/access:address:port** IP address and port of an [access server](#), separated by a colon;
- **/history:address:port** IP address and port of a [history server](#), separated by a colon;
- **/backup:address:port** IP address and port of a [backup server](#), separated by a colon;
- **/password:password** the internal password that will be set to all the above mentioned servers.

In addition, [internal IDs](#) will be automatically assigned to all servers: "1" to the trade server, "2" to the access server, "3" to the history server, and "4" to the backup server.

### Setup of the Main Trade Server in the Graphical Mode

To start setting up a main trade server in the graphical mode, you should start its executable file with the **/gui\_main** key from the command line. For example, mt5trade64.exe /gui\_main.

## MetaTrader 5 trading platform



The appeared window contains the list of servers which settings are already present at the main trade server. Three commands are provided for managing the servers:

- **Add** add a server configuration;
- **Edit** change configuration of a selected server;
- **Delete** delete configuration of a selected server. When deleting a configuration, the physical deletion of the server form the computer is not performed.

The windows of settings contain the main parameters of servers, that are set through the administrator terminal in the "Network" section:

## MetaTrader 5 trading platform

Server

Common Network Permissions Servers

Please specify common server settings: type, name, internal identifier, and password.

Type: Access Server

Name: Access Server

ID: 2 Password:

OK Cancel

Additional column Used is displayed for the ranges of accounts, orders and deals. It displays the current use of the range.

Server

Common Network Accounts Range Orders Range Deals Range

Please specify range of accounts that will be processed by this trade server.

| From | To      | Used        |
|------|---------|-------------|
| 1000 | 1000000 | 1000 - 1092 |

Add Edit Delete

OK Cancel



You can stop using the current range by specifying the last value from Used field in To field. After that, create a new range.

Parameters set when configuring the main trade server are checked for correctness:

- Adding a duplicate main or history server there can be only one server of each type in a cluster.
- Specifying incorrect ranges of logins, orders and deals.
- Creating more trade servers than allowed by the license. Such servers will not work.
- Deleting main server (deleting directly or changing its type).
- Deleting a working trade server.
- Changing networks settings so that the main server will become inaccessible to the administrator. For example, deleting the last access server.

In all the cases mentioned above the administrator will get the corresponding warning when trying to save the changes.

### Rebinding a Cluster to Another IP Address

When changing an IP address on the server the MetaTrader 5 cluster is installed at, rebinding to a new address is done as follows:

- Execute `/gui_main` command for the main trading server and replace the addresses of all cluster components with new ones.
- Execute `/gui` command for the history server and replace the addresses of the history and main servers with new ones.
- Execute `/gui` for the access server and replace the addresses of the access and the main servers with new ones.

## MetaTrader 5 Trading Platform / Platform Installation / Platform Moving

### Platform migration to new hardware

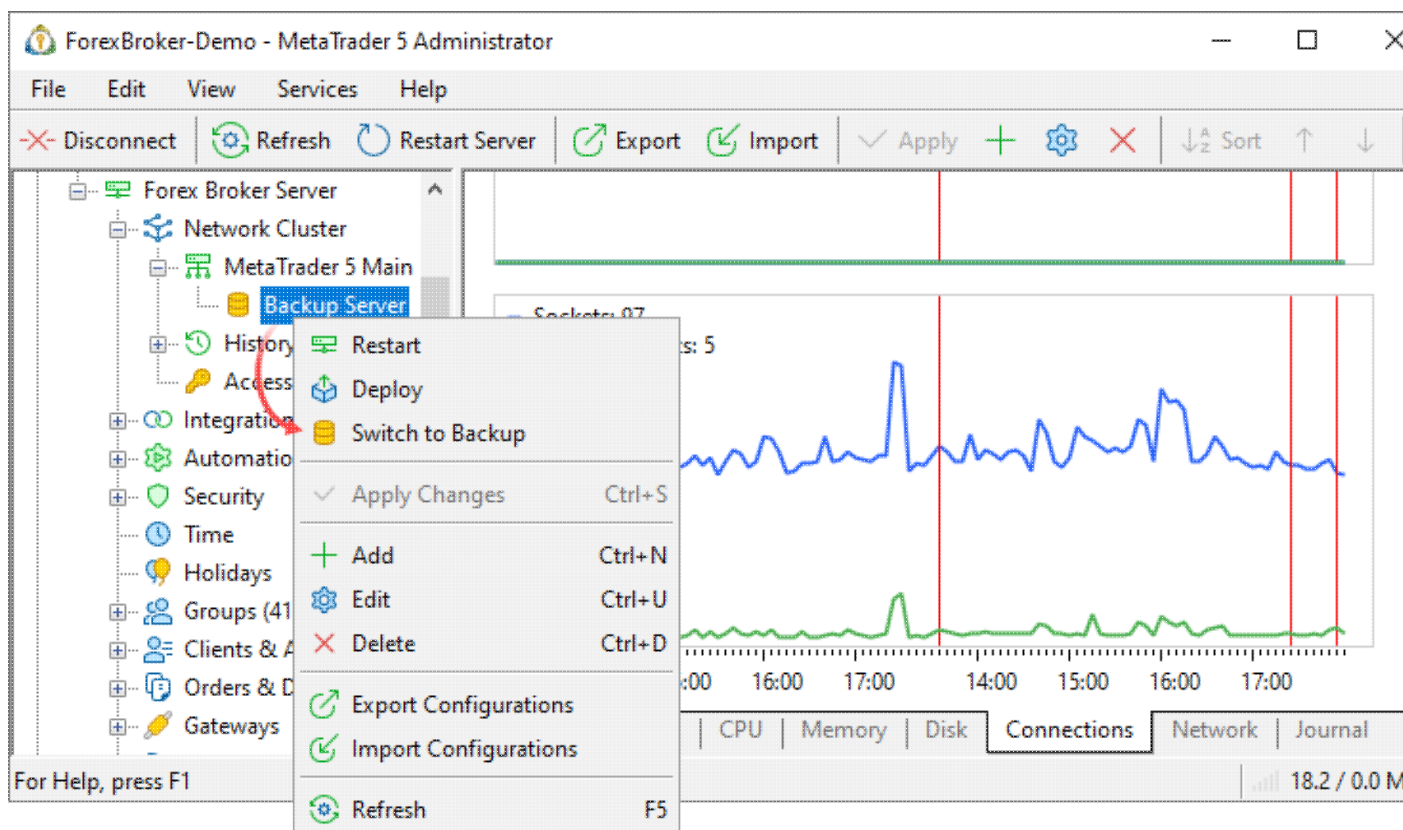
If you move the platform to other hardware/hosting provider, in addition to proper platform installation and configuration, it is important to perform a number of actions to ensure uninterrupted service for traders.

### Migration via a backup server

The back-up server replicates the main server data in real time and features its full copy. At any time you can switch to using it manually this is a quick and automatic procedure. In addition to emergency cases, the procedure can be used for migrating servers to new hardware.

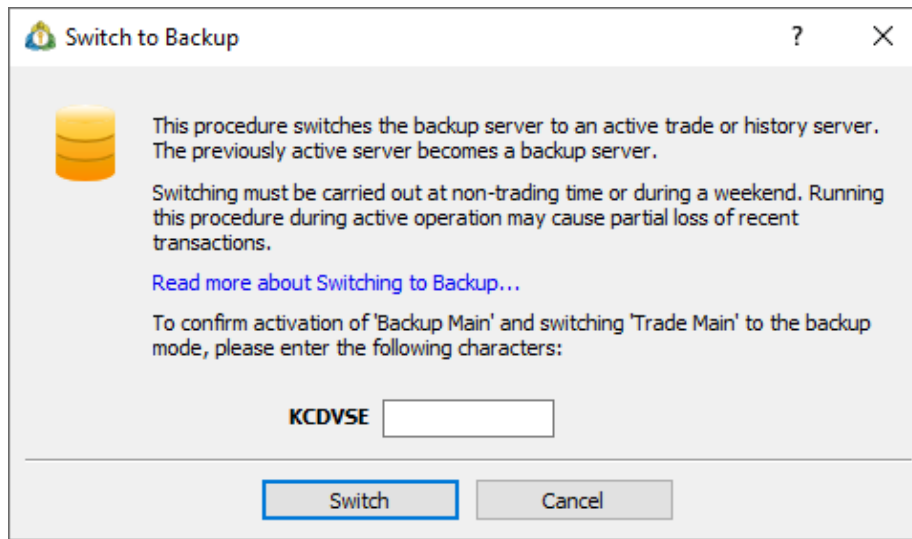
Install the backup server on the computer, to which you plan to migrate the server. After installation, run it for a few days on the new hardware to make sure it operates well.

Restart the backup server right before you switch to it. This will help avoid loss of data backed up once an hour. As long as the server copies data, its icon will display 🟢 (the icon will look like 🟡). Wait until the process is completed and run the "🟡Switch to backup server" command in the context menu:



## MetaTrader 5 trading platform

To avoid accidental switching, the platform requires an additional confirmation. In the dialog that appears, enter the required characters and click "Switch".



After the procedure is complete, you will see that the main server and the backup server have swapped their places in the Network section:

Run the same procedure for all other history and trade servers. Then install new backup servers and access servers. This can be conveniently done using the fast deployment procedure.

- Desktop terminals receive up-to-date information about access points during each account connection. It is recommended to keep at least one old access point operating during 1-2 weeks after migrating the platform to other equipment/hosting provider, so that terminals can receive such information.
- Server migration must only be performed in non-trading hours. When switching to the backup server, it does not copy data which the main server continues to receive.
- In order to prevent important data from being lost, trading operations and changes in the client base are not allowed on the main server right after the start of switching to the backup server. The ban is valid for one minute. If the platform fails to switch to a backup server within this period, the ban is removed.

### Platform activation after migration

When migration is complete, go to "Services" menu and click "Start Live Update". Go to the App Store\Licenses section and make sure the activation is available in the list. Two types of platform activations are available: main and non-main.

- Client terminals can only connect to a platform with the main activation.
- To enable connections of client terminals to the platform, the platform sends information about its access points (installed access servers) to the server every hour. Information is sent to the server regardless of the activation type, but only access points of the main activation are transmitted to terminals.

## MetaTrader 5 trading platform

- Servers with non-main activation are not shown in the broker selection dialogs when opening accounts through terminals.

Thus, to ensure full-featured platform operation, you need to set the new activation as main. To do this, please contact [Service Desk](#) after completing platform installation.

### Update of Installers

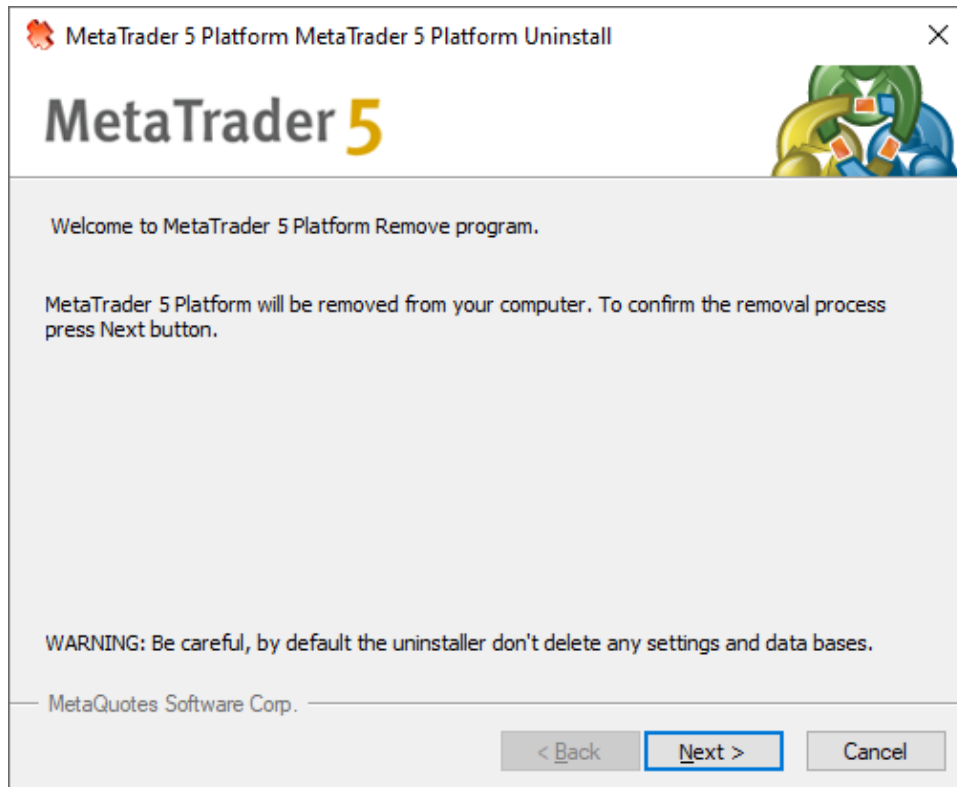
When switching an activation to the main type, support specialists will recompile the client terminal installer for you, i.e. they will include new access points. If you distribute the installer file by yourself, be sure to download its new version from the [Download](#) section and update the file on your resources.

To allow your traders to download the latest installer version any time, we recommend providing them direct terminal download links from the "Download" section. Thus, you will not need to update the files on your resources manually.

MetaTrader 5 Trading Platform / Platform Installation / Platform Uninstall

Platform Uninstall

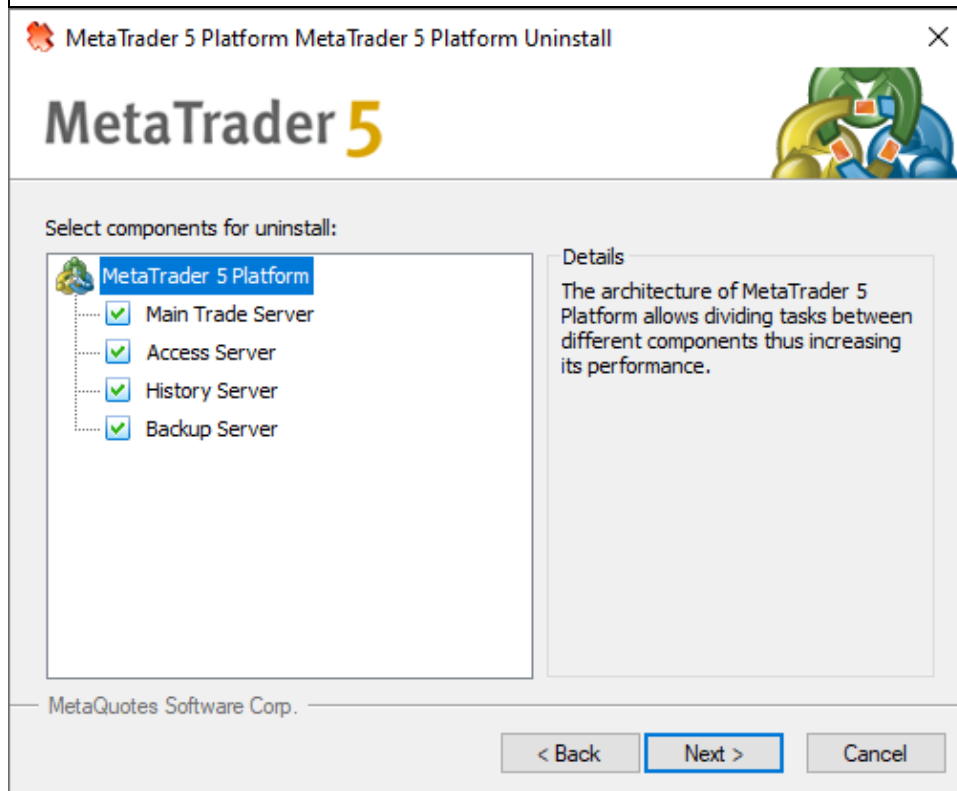
To remove the entire platform or one of its components, start file **uninstall.exe** located in the program directory or execute the "**Uninstall**" command in the corresponding group of programs in the "Start" menu. The following window will appear after that:



## MetaTrader 5 trading platform

In the greeting window, read the deinstallation warning. If you are sure that want to continue the deinstallation process, press "Next".

When a platform is removed, configurations and databases are not removed on default. In order to perform the full removal, enable the corresponding option on the next deinstallation stage.



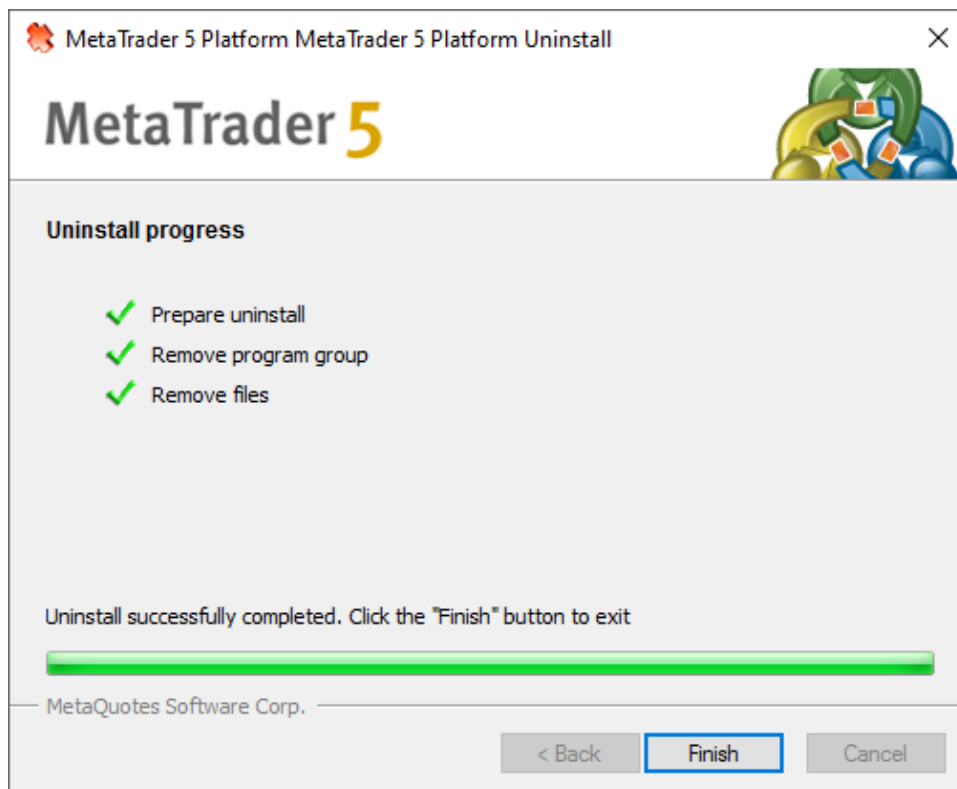
## MetaTrader 5 trading platform

All currently installed platform components are shown in the left part of the window. Tick off ☒ servers that you want to remove. Information about the component selected in the tree-like list is shown in the right part of the window. The following details are shown:

- **Server address** IP address of a selected server;
- **Server port** port of a selected server;
- **Server ID** ID of a selected server used for its internal identification;
- **Service name** name of the service, under which the server works in the operating system.

- To delete the server configurations and data bases, select "Delete server data bases". Be maximally attentive with this option, because in this case data recovery will be impossible. However, if data bases were not deleted, the platform can be recovered then with all the data and configurations.
- The option of data base deletion is enabled separately for each separate server.

After you press "Next" all the selected platform components will be deleted.



## MetaTrader 5 trading platform

To complete the deinstallation process press "Done".



## MetaTrader 5 Trading Platform / Platform Components

### Platform Components

The trading platform consists of the following components:

- **Main Trade Server**

Besides the tasks of maintaining trading, this server is designed for managing the entire platform. It means that the administrator of the main trade server can control the entire system.

- **Trade Servers**

Besides the main trade server, the unlimited number of trade servers can be installed. They are intended for storing and managing all client and trade databases; trading activities are performed through them.

- **Access Server**

Access servers are proxy servers and firewalls of the system at the same time. They check clients' connections, collect authorization information and route clients' connections. Besides that access servers cache the largest part of all data transmitted to a client (including history data and terminal updates).

- **History Server**

History server receives, processes and stores price data, news and updates, and transmits them to other components of the system.

- **Backup Server**

Backup servers create backups of both history and trade servers.

- **Data Feeds**

Data feeds are special components of the platform that are implemented as separate executable files that enable the receipt of news and quotes from different providers.

- **Gateways**

Gateways are intended for integration of the MetaTrader 5 platform with ECNs and exchanges. Gateways allow bringing out trade operation to external systems as well as translating quotes and news from them.

- **Administrator Terminal**

This terminal enables the remote control of the entire MetaTrader 5 platform. It allows changing any platform settings, managing client and trade databases and perform other operation.

- **Manager Terminal**

This terminal is used for working with the broker's clients: managing databases, servicing trading operations, reporting and managing risks.

- **Client Terminal for Windows**

The trader's main workstation enabling traders to analyze quotes and to execute trading operations. It features algo trading and strategy testing facilities.

- **Mobile Terminal for iOS**

It enables traders to manage their trading accounts, to view symbol charts and to perform trading operations using iPhone or iPad.

- **Mobile Terminal for Android**

The terminal enables traders to manage their trading accounts, to view symbol charts and to perform trading operations using Android powered smartphones

## MetaTrader 5 trading platform

and tablets.

- **WebTerminal**

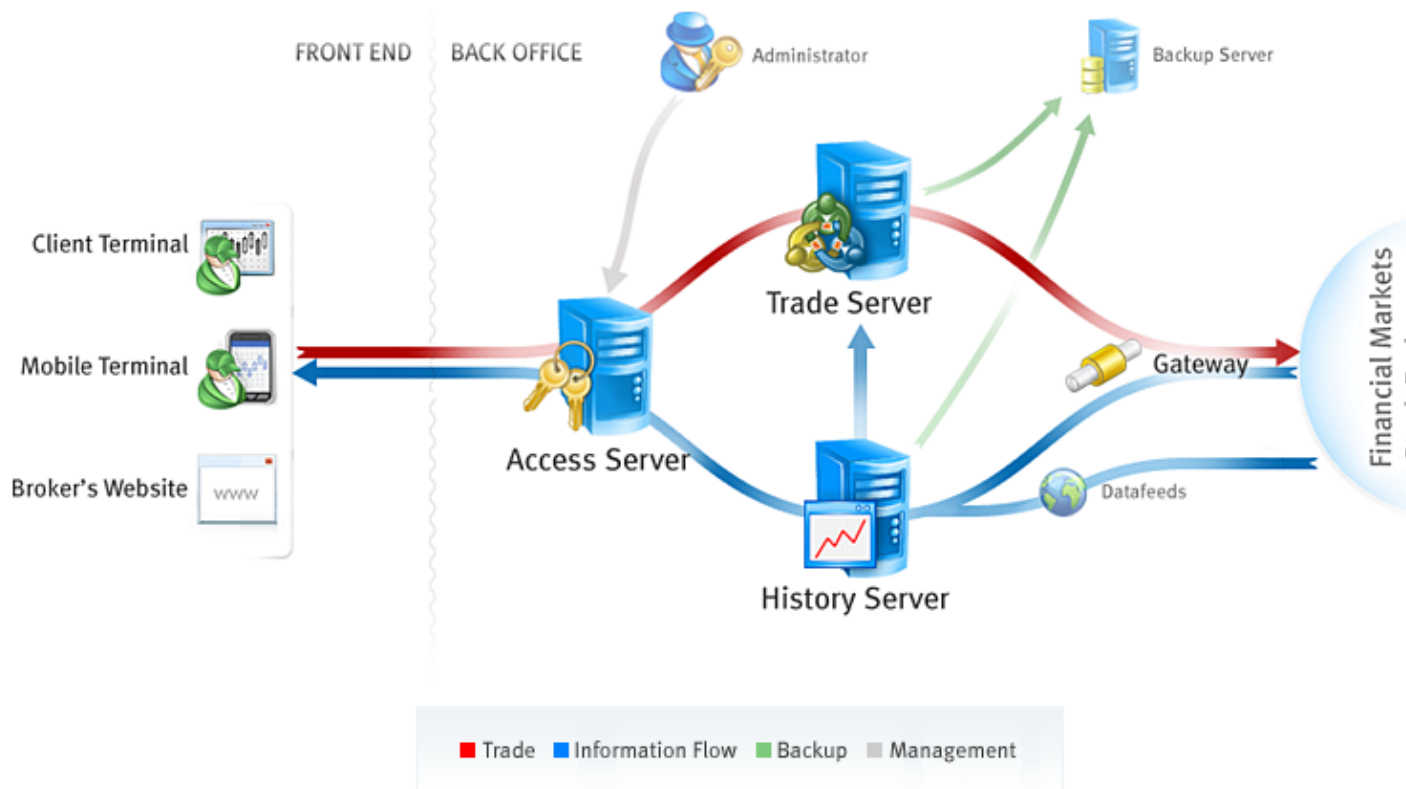
MetaTrader 5 WebTerminal allows trading in the financial markets via a web browser. It works in all operating systems and browsers requiring no additional software.

- **MetaTrader 5 API**

It is a toolkit which allow further expansion of the platform capabilities, integration with other trading systems and back-office components, as well as platform customization for specific business needs.

### Interaction of Components

The MetaTrader 5 trading platform is a distributed system. It is divided into separate components that are connected to each other via the main trade server. Configurations of all the platform components are stored on the main server. Other components refer to the main trade server, read the configurations and operate then in accordance with them.



## MetaTrader 5 trading platform

The internal identification of servers is performed using IDs and passwords set during their installation. You can view or change identifiers and passwords in the "Network" section. The main trade server identifies a component that has referred to it by its ID and passes to it configurations set for this component.

The minimal working configuration of the platform is the installed main trade server and an access server. The access server allows the connection to the main server using terminals (client, manager or administrator terminal).

MetaTrader 5 Trading Platform / Platform Components / Trade Server

## Trading Server

The system has two types of trading servers: one main server and additional ones. The main trading server serves trading operations and manages the entire system configuration.

The trading server performs the following functions:

- Storing and management of clients' records.
- Authentication and authorization of client connections.
- Storing and management of trade records.
- Check, management and execution of trade requests.
- Management of the internal mailing system.

- One main trading server and the unlimited number of additional ones can be configured in the system.
- For server configuration details, please see the "Network cluster" section.

MetaTrader 5 Trading Platform  
/ Platform Components / Trade  
Server / Structure of  
Directories and Files

### Structure of Directories and Files

The trade server is installed to the "*Trade*" folder. It contains the following executable files:

- **mt5srvupdater64.exe** the executable file of the live update system of the trade server;
- **mt5Trade64.exe** the executable file of the trade server;
- **mt5sendmail64.exe** the executable file of an utility for sending reports via email.

The main server directory contains five folders: *archive*, *bases*, *config*, *logs*, *templates*.

The directory ***archive*** contains the database of accounts moved to the archive:

| Files                         | Description                                         |
|-------------------------------|-----------------------------------------------------|
| <b>users_archive_YYYY.dat</b> | Databases of archive users by years.                |
| <b>users_archive_YYYY.idx</b> | Index files of databases of archive users by years. |

The directory ***bases*** contains all databases of the trade server:

| Files and folders | Description                                                                       | Files                      | Description                                          |
|-------------------|-----------------------------------------------------------------------------------|----------------------------|------------------------------------------------------|
| <b>deals\</b>     | Database of <u>deals</u> .                                                        | <b>deals_YYYY.mm.dat</b>   | Databases of deals by months.                        |
|                   |                                                                                   | <b>deals_YYYY.mm.idx</b>   | Index files of databases of deals by months.         |
| <b>history\</b>   | Database of <u>order history</u> (filled, rejected, canceled and expired orders). | <b>history_YYYY.mm.dat</b> | Databases of order history by months.                |
|                   |                                                                                   | <b>history_YYYY.mm.idx</b> | Index files of databases of order history by months. |
| <b>mail\</b>      | Email database.                                                                   | <b>mail_YYYY.mm.dat</b>    | Email databases by month.                            |
|                   |                                                                                   | <b>mail_YYYY.mm.idx</b>    | Index files of email databases by months.            |
| <b>daily\</b>     |                                                                                   | <b>daily_YYYY.mm.dat</b>   |                                                      |

## MetaTrader 5 trading platform

| Files and folders       | Description                                                                                                                                                                                                            | Files                    | Description                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------|
|                         | The database of <u>daily reports</u> .                                                                                                                                                                                 |                          | Databases of daily reports by months.                |
|                         |                                                                                                                                                                                                                        | <b>daily_yyyy.mm.idx</b> | Index files of the daily report databases by months. |
| <b>crm\</b>             | <u>Client</u> database. The client database itself, as well as the databases of <u>documents</u> , <u>comments</u> and files are stored in subdirectories.                                                             |                          |                                                      |
| <b>performance\</b>     | Monthly trade server performance databases and data indexes, which are displayed on the <u>Monitoring</u> tab.                                                                                                         |                          |                                                      |
| <b>confirmation.dat</b> | Database of <u>verified phone numbers and emails</u> .                                                                                                                                                                 |                          |                                                      |
| <b>liveupdate.dat</b>   | Database of <u>live updates</u> .                                                                                                                                                                                      |                          |                                                      |
| <b>orders.dat</b>       | Database of active <u>orders</u> (placed orders).                                                                                                                                                                      |                          |                                                      |
| <b>orders.idx</b>       | The index file of the database of active orders.                                                                                                                                                                       |                          |                                                      |
| <b>positions.dat</b>    | Database of <u>positions</u> .                                                                                                                                                                                         |                          |                                                      |
| <b>positions.idx</b>    | The index file of the database of positions.                                                                                                                                                                           |                          |                                                      |
| <b>users.dat</b>        | Database of <u>accounts</u> .                                                                                                                                                                                          |                          |                                                      |
| <b>daily.dat</b>        | Database of <u>daily reports</u> .                                                                                                                                                                                     |                          |                                                      |
| <b>daily.idx</b>        | Index file of the daily report database.                                                                                                                                                                               |                          |                                                      |
| <b>certificates.dat</b> | Database of client certificates generated by the server for the <u>extended authorization</u> . The server stores only the public part of the certificate, while the private part is stored only at the client's side. |                          |                                                      |
| <b>certificates.idx</b> | Index file of the certificate database.                                                                                                                                                                                |                          |                                                      |
| <b>executions.dat</b>   | Database of trade executions. More detailed information is available in the <u>MetaTrader 5 Gateway API user guide</u> .                                                                                               |                          |                                                      |
| <b>kyc.dat</b>          | Database for storing the result of client verifications through external <u>KYC service providers</u> .                                                                                                                |                          |                                                      |
| <b>geo.dat</b>          | Database for the operation of <u>geo services</u> .                                                                                                                                                                    |                          |                                                      |

The **config** directory contains all the server configurations in the form of encrypted \*.ini files:

| Files             | Description                                 |
|-------------------|---------------------------------------------|
| <b>access.ini</b> | Settings of <u>access by IP addresses</u> . |

| Files                    | Description                                          |
|--------------------------|------------------------------------------------------|
| <b>common.ini</b>        | Common server settings.                              |
| <b>feeders.ini</b>       | Settings of <u>data feeds</u> .                      |
| <b>groups.ini</b>        | Settings of <u>groups</u> .                          |
| <b>history_sync.ini</b>  | Settings of <u>synchronization of history data</u> . |
| <b>holidays.ini</b>      | Settings of <u>holidays</u> .                        |
| <b>managers.ini</b>      | Settings of <u>managers</u> .                        |
| <b>performance.ini</b>   | Settings of the <u>server performance</u> display.   |
| <b>requests.ini</b>      | Settings of request <u>routing rules</u> .           |
| <b>server.ini</b>        | Individual server settings.                          |
| <b>servers.ini</b>       | Settings of the internal <u>network of servers</u> . |
| <b>symbol_groups.ini</b> | Individual settings of <u>symbols for groups</u> .   |
| <b>symbols.ini</b>       | Settings of <u>symbols</u> .                         |
| <b>time.ini</b>          | <u>Time</u> settings.                                |
| <b>license.lic</b>       | The license file.                                    |

The **logs** directory contains the files of the server operation journal, as well as crash logs:

| Folders and files        | Description                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Crash\crash.log.*</b> | The /crash directory contains server crash files. These files are automatically sent to the software developing company for detecting reasons of the crash and eliminating them.                    |
| <b>yyyymmdd.log</b>      | <u>Journal</u> files that contain all the information about events that occur on the trade server. Server logs are stored in separate files for each working day. Here yyyy year, mm month, dd day. |
| <b>mt5srvupdater.log</b> | Journal files of the platform <u>updates</u> .                                                                                                                                                      |

The **plugins** directory contains DLL files of plugins for the trade server written using the MetaTrader 5 Server API.

The **templates** directory contains two folders: "Certificate", "Confirmation", "Greeting" and "Statement". They contain templates of emails for certificate confirmation, templates of welcoming letters and templates for daily and monthly reports.

| Subfolders | Description | Files | Description |
|------------|-------------|-------|-------------|
|------------|-------------|-------|-------------|

| Subfolders           | Description                                                                                         | Files                     | Description                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Default\</b>      | Folder of default email templates.                                                                  | <b>Default.html</b>       | Templates of default emails, that are used if templates corresponding to the user's language were not found.             |
| <b>Users folder\</b> | Folder of templates of emails that can be sent to separate groups only according to their settings. | <b>language_name.html</b> | templates of emails by languages that are sent in accordance with the language specified by the user. E.g. English.html. |

The **reports** directory contains the DLL files of reports created using MetaTrader 5 Report API.

The **confirms** directory is used to prepare HTML files of daily reports before sending them to clients.

The **settings** directory stores dashboard and filter settings used in manager terminals.



MetaTrader 5 Trading Platform / Platform Components / Trade Server /  
Mail Templates

## Mail Templates

Several types of emails that are automatically sent to clients are implemented in the the MetaTrader 5 platform:

- Welcoming email at account opening;
- Email requesting the certificate confirmation;
- Daily report on trade activity;
- Monthly report on trade activity.

- All templates must be of the Unicode format (UTF-16/UCS-2 Little Endian).
- In order to start using a new template, the main trade server must be restarted.

## Welcoming Emails

Welcome messages are sent to clients through the internal mail system when an account is opened. It does not matter how the account is opened: by the trader via the client terminal or by the broker via the Manager/Administrator terminal. The welcome email contains the account number and password, as well as general information about the MetaTrader 5 platform.

Welcome email templates are located in the following directory:

- **Trader server directory\templates\greeting\default\\*.htm** default templates used for groups;
- **Trader server directory\templates\greeting\custom folder\\*.htm** templates of emails that can be sent only to separate groups with account of their settings;
- **Trader server directory\templates\greeting\preliminary\\*.htm** templates of emails that are sent when a preliminary account is opened from the client terminal.

An unlimited number of folders with special templates can be create. The name of the folder with templates that will be used for a group are specified in the "Company" tab in its settings.

## Certificate Confirmation Email

These letters are sent if the extended authorization and certificate confirmation mode is enabled for this group. Such an email contains information on how to confirm a certificate. Templates of such emails are stored in the following directory:

- **Trader server directory\templates\certificate\default\\*.htm** default templates used for groups;

- **Trader server directory\templates\certificate\custom folder\\*.htm**  
templates of emails that are sent only to separate groups with account of their settings.

### Phone and Email Verification

These emails\messages are sent to verify phone numbers and emails, specified during registration of demo and preliminary accounts from client terminals.

- **Trader server directory\templates\verify\_email\default\\*.htm** email templates as HTM files.
- **Trader server directory\templates\verify\_phone\default\\*.htm** SMS templates as text files. Do not use too long messages. If the allowed length is exceeded (depending on the provider), a message may be cropped or split into multiple SMS.

The <!--CONFIRMATION\_CODE--> macro is used for confirmation emails. This code adds the generated confirmation code to the text.

Templates from the "default" folder are used on default for all groups. If you need to use custom templates for a specific group, create another folder in the same directory and place the template files to it. Then specify the folder name in the group settings.

### Daily Report on Trade Activity

Daily reports on trade activity of clients are sent if the "Send daily statements by email" option is enabled in the group settings. Templates of such emails are stored in the following:

- **Trader server directory\templates\confirmation\default\\*.htm** default templates used for groups;
- **Trader server directory\templates\confirmation\custom folder\\*.htm** templates of reports that are sent only to separate groups with account of their settings.

### Monthly Report on Trade Activity

Monthly reports on trade activity of clients are sent if the "Send daily statements by email" option is enabled in the group settings. Templates of such emails are stored in the following:

- **Trader server directory\templates\statement\default\\*.htm** default templates used for groups;
- **Trader server directory\templates\statement\custom folder\\*.htm** templates of reports that are sent only to separate groups with account of their settings.

### Email templates in different languages

All template files should be named according to the language they are written in. A file name is used by the platform to determine which users the template is to be applied

to. The user's language is defined in the [account settings](#).

Languages are specified in the format standard for the family of Windows operating systems, though without the specification of regional dialect specifics. E.g. English.htm, Russian.htm and so on. The only exceptions are Chinese language templates. In case of simplified Chinese, a template should be named chinese.htm, while in case of traditional Chinese, it is named chinese\_traditional.htm.

If no template was found for the language specified in the account, the default.htm template is used. If a separate folder is specified for a group, the default template is first searched for in this folder, and if it is not found an email according to the default.htm template located in the "Default" folder is sent.

### The Format of Files

A template file can contain a plain text, HTML tags, as well as CSS design elements. To insert images, set them on your web server and specify the appropriate links in ``. The web server must operate using the HTTPS protocol. Images with URLs starting with http:// will not be displayed.

Besides, the templates can be equipped with special macros for inserting various data depending on an account an email is sent to.

- Common macros
- `<!--ACCOUNT-->` account number.
- `<!--LOGIN-->` account number.
- `<!--GROUP-->` user group.
- `<!--PASSWORD-->` account master password.
- `<!--INVESTOR-->` account investor password.
- `<!--NAME-->` user first name (obsolete macro).
- `<!--USERNAME-->` user first name (obsolete macro).
- `<!--FIRST_NAME-->` user first name.
- `<!--LAST_NAME-->` user last name.
- `<!--MIDDLE_NAME-->` user middle name.
- `<!--COUNTRY-->` country of residence.
- `<!--CITY-->` city of residence.
- `<!--STATE-->` state (region) of residence.
- `<!--ZIPCODE-->` zip code.
- `<!--ADDRESS-->` residential address.
- `<!--PHONE-->` phone number.
- `<!--EMAIL-->` email address.
- `<!--COMMENT-->` account comment.
- `<!--ID-->` ID.
- `<!--STATUS-->` residency status.
- `<!--PHONEPASS-->` phone password.
- `<!--AGENT-->` agent account associated with the user.
- `<!--CURRENCY-->` account deposit currency.
- `<!--COMPANY-->` [company name](#) from the account group settings.
- `<!--LEVERAGE-->` recipient's current leverage.

Daily and monthly report macros

- **<!--DATE-->** date the report is generated for, YYYY.MM.DD.
- **<!--TIME-->** time the report is generated for, HH::MM:SS.
- **<!--FULLTIME-->** date and time the report is generated for, YYYY.MM.DD HH::MM:SS.
- **<!--BALANCE-->** balance at the time of the report generation.
- **<!--CREDIT-->** credit funds at the time of the report generation.
- **<!--INTERESTRATE-->** client's annual interest rate.
- **<!--COMMISSION\_DAILY-->** amount of standard commissions charged from a client for the day the report is generated for.
- **<!--COMMISSION\_MONTHLY-->** amount of standard commissions charged from a client for the month the monthly report is generated for (or for the current month in case of a daily report).
- **<!--AGENT\_DAILY-->** agent commissions charged from a client for the day the report is generated for.
- **<!--AGENT\_MONTHLY-->** agent commissions charged from client's operations for the month the monthly report is generated for (or for the current month in case of a daily report).
- **<!--PREV\_BALANCE\_DAILY-->** client balance at the end of the previous trading day.
- **<!--PREV\_BALANCE\_MONTHLY-->** client balance at the end of the previous month.
- **<!--PREV\_EQUITY\_DAILY-->** client equity at the end of the previous trading day.
- **<!--PREV\_EQUITY\_MONTHLY-->** client equity at the end of the previous month.
- **<!--PREV\_DATE-->** the date of the last but one closure of the trading day.
- **<!--PREV\_TIME-->** the time of the last but one closure of the trading day.
- **<!--PREV\_FULLTIME-->** the date and time of the last but one closure of the trading day.
- **<!--MARGIN-->** money required to cover open positions as of the end of the day/month.
- **<!--MARGIN\_FREE-->** amount of free margin volume as of the end of the day/month.
- **<!--MARGIN\_LEVEL-->** margin level as of the end of the day/month.
- **<!--FLOATING\_PROFIT-->** floating profit/loss on all open positions at the time of the report generation.
- **<!--FLOATING\_STORAGE-->** size of swaps charged for client's positions for a day, but not yet reflected in the balance.
- **<!--FLOATING\_COMMISSION-->** floating client commission blocked on the account but not yet reflected in the balance at the time of the report generation.
- **<!--FLOATING\_EQUITY-->** client equity volume at the time of the report generation.
- **<!--FLOATING\_PL-->** total client's floating profit/loss at the time of the report generation. Calculated as **<!--FLOATING\_PROFIT-->** + **<!--FLOATING\_STORAGE-->** + **<!--FLOATING\_COMMISSION-->**.
- **<!--FLOATING\_LIABILITIES-->** amount of client liabilities at the time of report generation. It is only used for the Exchange risk management model.
- **<!--FLOATING\_ASSETS-->** amount of client assets at the time of report generation. It is only used for the Exchange risk management model.
- **<!--CLOSED\_PROFIT-->** total closed profit/loss at all deals per day/month.

- **<!--CLOSED\_STORAGE-->** size of swaps charged for a client's positions for a day/month.
- **<!--CLOSED\_DEPOSIT-->** total funds deposited/withdrawn from the account per day/month.
- **<!--CLOSED\_CREDIT-->** credit funds deposited/withdrawn from the account per day/month.
- **<!--CLOSED\_CHARGE-->** other depositions/withdrawals from the client's balance per day/month.
- **<!--CLOSED\_CORRECTION-->** corrective balance operations performed on the client's account per day/month.
- **<!--CLOSED\_BONUS-->** bonus funds deposited/withdrawn from the account per day/month.
- **<!--CLOSED\_COMMISSION\_INSTANT-->** standard commissions withdrawn from a client's account per day/month instantly (during a trade).
- **<!--CLOSED\_COMMISSION\_ROUND-->** commission by orders and positions accumulated during a day/month. Depending on the settings (specified for the group in the administrator terminal), preliminary commission calculation is performed during a day/month and the appropriate funds are blocked in the account and displayed here. Final commission calculation is performed at the end of a day/month and the appropriate sum is withdrawn from the account by the balance operation.
- **<!--CLOSED\_FEE-->** total fees charged from the client for the day/month.
- **<!--CLOSED\_AGENT-->** agent commissions charged from the client for the day the report is generated for.
- **<!--CLOSED\_INTEREST-->** annual interest rate accruals per day/month the report is generated for.
- **<!--CLOSED\_DIVIDEND-->** dividends received by the client per day/month the report is generated for.
- **<!--CLOSED\_TAX-->** taxes charged per day/month the report is generated for.
- **<!--CLOSED\_PL-->** total profit/loss at a client's account per day/month the report is generated for. Calculated as **<!--CLOSED\_PROFIT-->** + **<!--CLOSED\_STORAGE-->** + **<!--CLOSED\_COMMISSION\_INSTANT-->**.
- **<!--CLOSED\_ADDITIONAL-->** financial result of other transactions conducted on the account for the day/month the report is generated for. These are additional charges, corrections, bonuses, agent commissions, annual interests, dividends and taxes
- **<!--CLOSED\_TOTAL-->** total financial result of the client's account for the day/month the report is generated for. Calculated as the sum of all the above values with the **<!--CLOSED\_\*-->** prefix apart from **<!--CLOSED\_PL-->** and **<!--CLOSED\_ADDITIONAL-->**.
- **<!--CLOSED\_SO\_COMPENSATION-->** the sum of balance operations connected with the negative balance compensation after Stop Out.
- **<!--CLOSED\_COST-->** the total amount of costs for all deals for the day/month for which the report is generated. The value calculation does not depend on group settings. If a macro is included in a report template, its value will be calculated and substituted.
- **<!--PREV\_EQUITY\_DIFF\_PERC\_DAILY-->** a change in equity as compared to the previous day value. Indicated as a percentage.
- **<!--PREV\_EQUITY\_DIFF\_PERC\_MONTHLY-->** a change in equity as compared to the previous month value. Indicated as a percentage.

## Macros of Trading Operations

Daily reports include blocks of trading operations performed by a client during a day/month. Each of these blocks begins with a macro corresponding to the operation type:

- **<!--MQTABLE=Closed Orders-->** closed orders.
- **<!--MQTABLE=Closed Deals-->** executed deals.
- **<!--MQTABLE=Positions-->** current open positions.
- **<!--MQTABLE=Orders-->** current open orders.

Each of these blocks ends with the **<!--MQTABLE-->** macro. Information about trading operations is displayed using macros inside the blocks. Depending on the block in which a macro is contained, it substitutes information about the appropriate operation type, i.e. order, deal or positions.

- **<!--Ticket-->** the ticket of a trading operation.
- **<!--Type-->** the type of a trading operation (buy, sell or a pending order).
- **<!--Size-->** the volume of a trading operation. The initial and filled volume is additionally displayed for orders.
- **<!--Item-->** the name of a trading instrument.
- **<!--ISIN-->** International Securities Identification Number (ISIN).
- **<!--Price-->** the price at which a trading operation was executed.
- **<!--SL-->** the Stop Loss level.
- **<!--TP-->** the Take Profit level.
- **<!--Open Time-->** operation time.
- **<!--Close Time-->** order or position closing time.
- **<!--State-->** order status (filled, rejected, canceled).
- **<!--Comment-->** a comment on the operation.
- **<!--Entry-->** trade direction (in, out, in/out).
- **<!--Commission-->** commission charged for the operation.
- **<!--Fee-->** the amount of fees for the operation.
- **<!--Swap-->** operation swap.
- **<!--Cost-->** the amount of costs incurred when performing deals relative to the current mid-point spread cost. The value calculation does not depend on group settings. If a macro is included in a report template, its value will be calculated and substituted.
- **<!--Profit-->** profit received from the operation.
- **<!--Market-->** the market price of a trading instrument at the time of report generation.
- **<!--Color-->** a macro for alternating the background color of even and odd rows (a row with a white background, the next one has a gray background, etc).

MetaTrader 5 Trading Platform / Platform Components / Trade Server / Daily Reports

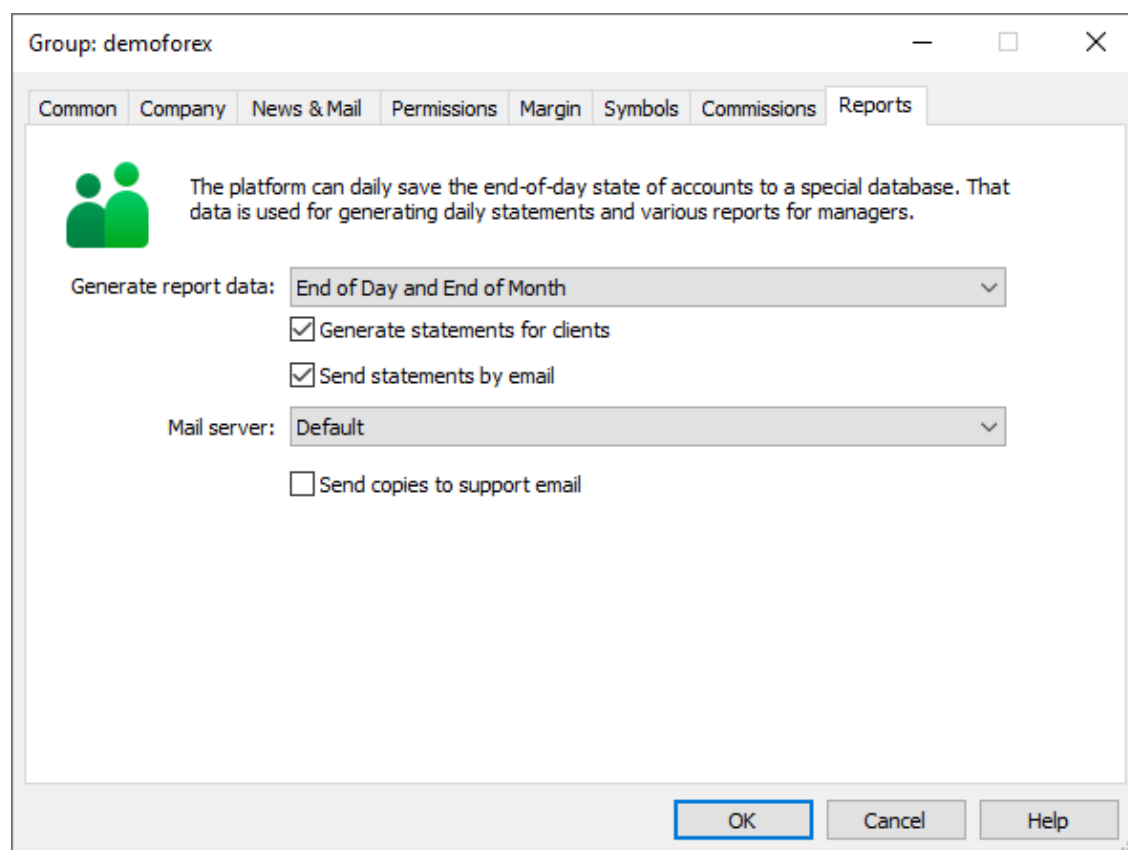
## Daily Reports

At the end of each working day and month, the trade server prepares reports on client deals and places them to [trade server directory]/confirms/YYYYMMDD", where YYYYMMDD is the current date). Each report is an HTML file having the name "login\_mail.htm" (for example, "123\_mail.htm"). After creating the reports, the trade server sends them using SendMail utility.

The database of all generated reports is saved in the file [trade server directory]/bases/daily.dat. The data from the database can be requested via the Manager terminal, for example, by using a report from Daily Report standard delivery.

## Enabling/Disabling Daily Reports

Generation of reports can be enabled/disabled for each client group separately.



## MetaTrader 5 trading platform

Here you can also configure sending reports to clients and (if necessary) the technical support via email.

### Customizing Report Appearance

Email templates with daily and monthly trading activity reports are located in the /templates/confirmation and /templates/statement folders of the trade server.

### Configuring Daily Report Generation Time

Daily and monthly report generation time is defined by the trade server settings:

Server: Trade Main - ID 1

Common Network Demo **End of Day** Accounts Orders Deals Failover Service

Please specify trade server settings: end of day time, schedule, statements generation modes.

End of day time: 23:59

End of day schedule: Monday, Tuesday, Wednesday, Thursday, Friday

Rollover schedule: Monday, Tuesday, Wednesday, Thursday, Friday

Daily statements: End of day

Last end of day: 2022.09.22 23:59:59 Previous end of day: 2022.09.21 23:59:59

Monthly statements: Last day of month

Last end of month: 2022.08.31 23:59:59 Previous end of month: 2022.07.31 23:59:59

OK Cancel Help



The reports are affected by the following parameters:

- **End of day time** report generation start time.
- **End of day schedule** report generation days. The option does not affect monthly reports.
- **Daily statements** daily report generation time: at the end of a trading day (before charging swaps, annual interest, and commissions) or at the start of a trading day (after charging swaps, annual interest, and commissions).
- **Monthly statements** monthly report generation time: at the end of the last day of month (before charging swaps, annual interest, and commissions) or at the start of a first day of the next month (after charging swaps, annual interest, and commissions).

## Report Data

| Name              | Description                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Datetime          | Daily report generation date and time.                                                                |
| Login             | Initial key. The login of a client for whom the daily report is generated.                            |
| DatetimePrev      | Daily report previous generation date and time.                                                       |
| Name              | The name of a client in a daily report.                                                               |
| Group             | Client group in a daily report.                                                                       |
| Currency          | Client's deposit currency in a daily report.                                                          |
| Company           | The company serving the client in a daily report.                                                     |
| EMail             | An email of a client in a daily report.                                                               |
| Balance           | The size of a client's balance in a daily report.                                                     |
| Credit            | The amount of a client's credit funds in a daily report.                                              |
| InterestRate      | The annual interest rate of a client in a daily report.                                               |
| CommissionDaily   | The amount of commissions charged from a client for a day in the report.                              |
| CommissionMonthly | The amount of commissions charged from a client for the current month in a report.                    |
| AgentDaily        | The size of agent commissions charged for a client's trade operations for a day, from a daily report. |
| AgentMonthly      | The amount of agent commissions charged for a client's trade operations for the current month.        |
| BalancePrevDay    | Client's balance as of the end of the previous day.                                                   |
| BalancePrevMonth  | Client's balance as of the end of the previous trading month.                                         |

# MetaTrader 5 trading platform

| Name            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EquityPrevDay   | A client's equity as of the end of the previous day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| EquityPrevMonth | The value of a client's equity as of the end of the previous trading month.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Margin          | <p>Size of a client's margin in a daily report. The report does not contain data on the margin charged for each position or order because in some cases, it is not possible to calculate the margin value for the following reasons:</p> <ul style="list-style-type: none"> <li>• On hedging accounts, multiple oppositely directed positions can exist for the same instrument. If their volumes do not match, the <u>hedged margin</u> is calculated based on the aggregate positions.</li> <li>• If on a netting account the margin is charged for pending orders and there are two oppositely directed orders for the same instrument, it is unknown in which of them the margin should be indicated. It is not known in advance which of the orders will be used to open a position and which one to close it.</li> <li>• When using <u>spreads</u>, the margin depends on the aggregate orders and positions.</li> </ul> |
| MarginFree      | A client's free margin in a daily report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MarginLevel     | The margin level of a client in the daily report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| MarginLeverage  | The margin leverage of a client in the daily report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Profit          | The size of the current profit for all open positions of a client in a daily report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ProfitStorage   | The current size of swaps charged for a client's open positions for a day, but not yet reflected in the balance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ProfitEquity    | The amount of the current floating equity of a client in a daily report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DailyProfit     | The amount of a client's daily profit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DailyBalance    | The amount accrued to a client's balance during the reported day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DailyCredit     | The amount of credit given to a client during the reported day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DailyCharge     | The amount of other charges to the client's balance during the reported day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DailyCorrection | The amount of corrective balance operations for a reported day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DailyBonus      | The amount of bonuses added to the client's balance for the reported day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| <b>Name</b>      | <b>Description</b>                                                                          |
|------------------|---------------------------------------------------------------------------------------------|
| DailyStorage     | The amount of swaps calculated for the client for the reported day.                         |
| DailyCommInstant | The amount of instant commissions charged from the client for a reported day.               |
| DailyCommRound   | The amount of turnover commissions charged from the client for a reported day.              |
| DailyCommFee     | The fee amount charged for the client's deals for the reported day.                         |
| DailyAgent       | The size of agent commissions charged for a client's trade operations for the reported day. |
| DailyInterest    | The amount accrued to a client as part of the annual interest rate for the reported day.    |

MetaTrader 5 Trading Platform / Platform Components / Trade Server /  
SendMail Utility

## SendMail Utility

MetaTrader 5 SendMail is a command-line utility and a component of MetaTrader 5 platform. It is used by a trade server for sending reports and emails. This application can be used for sending reports manually.

## Report Generation

The trade server prepares reports on client deals at the end of a working day placing them to the separate directory ("*MetaTraderServer/confirm/YYMMDD*", where *YYMMDD* is current date). Each report is an HTML file having the name "*login\_mail.htm*" (for example, "*123\_mail.htm*"). After generating all reports, the trade server saves the configuration file ("*mail.cfg*") containing descriptions of account groups with e-mail settings in the same directory and then launches MetaTrader 5 SendMail utility.

## Operation Principles

After launching MetaTrader 5, SendMail reads the configuration file and starts sending emails to account groups. Description of each group contains data that is sufficient for authorization on the specified SMTP server. In case of an authorization error, the program simply moves to the next group sending the appropriate error message to the journal. If authorization is successful, reports are consistently sent to these group's accounts.

Two types of errors may occur when sending reports. First, connection to the SMTP server may be lost. In this case, an attempt to handle the next group is made. Second, the server may report that it is not able to send a report to the specified client address (for example, if the latter is incorrect). In this case, the report is marked as unsent and handling is moved to the next one.

After all MetaTrader 5 groups are handled, SendMail records the results in the same configuration file. Then all the reports and the configuration file are compressed into a ZIP file. ZIP file's name is based on the name of a low-level folder in the path to the reports (for example, if the path is "*C:\mt5\trade\_server\confirm\20130101*", then ZIP file's name is "*20130101.zip*"). After the successful archiving, all source files are deleted.

Attention: We strongly recommend to use your own mail server (at least having the *mail.your\_company.domain* name) instead of a provider's or even a public one. That ensures sender's (your company's) authenticity and allows adjusting the mail server's parameters for efficient mailing. Currently, mail servers and providers introduce strict rules concerning mail sending and receiving. That often leads to some brokers having issues when performing mass mailing.

To simplify the task, you may configure your own mail server, so that it is able to accept any mail from the trade server's IP address without limitations. If you do not have access to your mail server's settings, ask your provider to configure it appropriately.

## Sending Reports Manually

Reports may be sent manually in two modes: group and individual ones. To choose the mode, one of the keys should additionally be specified in the command line when launching MetaTrader 5 SendMail:

- `mt5sendmail64.exe /mail:[path]` individual mailing.
- `mt5sendmail64.exe /group:[path]` group mailing.
- `/archive` specify this key after `/mail:[path]` or `/group:[path]` to force the SendMail archive the reports after sending.

## Individual Mailing

MetaTrader 5 SendMail with "mail" key should be launched to send a report to a single user:

```
mt5sendmail64.exe /mail:"path" /archive
```

Path is the path to the directory containing *folder.cfg* mailing configuration file (the file should have exactly the name mentioned above). Do not add a backslash at the end of the path:

```
mt5sendmail64.exe /mail: "C:\MetaTrader 5 Platform\MainTrade\confirms\2021.05.11.daily"
```

Here is an example of a configuration file structure:

```
from=abc@company.net
name=ABC Company
subject=Trade Report
to=johnsmith@mail.net
to_name=John Smith
charset=utf-8
body=D:\Reports\John_Smith\mail.htm
attachments=D:\Reports\John_Smith\balance.jpg
smtp_srv=abc@company.net
smtp_login=mailer
smtp_pass=mailerpassword
```

The following parameters are specified in the configuration file:

- **from** e-mail address, from which the report is sent.
- **name** sender's name.
- **subject** email subject.
- **to** recipient's email address.
- **to\_name** recipient's name.
- **charset** email's character set.
- **body** path to the HTM file containing the email's contents.

## MetaTrader 5 trading platform

- **attachments** path to the file that is to be attached to the email. If you want to attach several files, specify paths to them divided by tab. The line length, including the parameter name, should not exceed 256 characters.
- **smtp\_srv** SMTP server address used for sending messages.
- **smtp\_login** login for authorization on the mail server. In most cases, it is a mailbox, for example, "*your\_name@mail.ru*".
- **smtp\_pass** password for authorization on the mail server (mailbox password).

- Up to 8 attachments can be added to an email.
- Attachment size may not exceed 4 MB.
- The password for authorization on the SMTP server is stored in the clear. This is done in order to modify the configuration file easily. The administrator can change the configuration (for example, a password) and launch SendMail manually. For security purposes, it is recommended to limit access to report configuration files.

### Group Mailing

MetaTrader 5 SendMail with "group" key should be launched to send a report to a group of users:

```
mt5sendmail64.exe /group:"path" /archive
```

Path is a path to *mail.cfg* file (the file should have exactly the name mentioned above) containing description of mailing settings.

Report files being sent should be in the same directory with *mail.cfg* configuration file. The report for each user should be in the form of an HTM file named "*login\_mail.htm*" (for example, "*123\_mail.htm*").

Here is an example of a configuration file structure:

```
<group>
name=demoforex
company=MetaQuotes Software Corp.
email=reporter@metaquotes.ru
subject=Trade Report
smtp_srv=mail.metaquotes.ru
smtp_login=reporter
smtp_pass=securepass

  101  1  John Smith  johnsmith@mail.ru
  102  1  Ivan Ivanov  ivan@mail.ru
  103  2  Larisa Ivanovna  larisa@mail.ru
</group>

<group>
name=forever
company=MetaQuotes Software Corp.
```

```
email=mailer@metaquotes.ru
smtp_srv=mail.metaquotes.ru
smtp_login=mailer
smtp_pass=mailerpassword

151 0 Alisa alisa@pole.ru
152 0 Brom brom@fix.com
153 0 SirX sirx@fix.com
</group>
```

Each target user group is described by a couple of <group> tags. Description of each group contains several mandatory fields:

- **name** group name.
- **company** company name that will be specified as a sender's name.
- **email** e-mail address, on behalf of which the report is sent.
- **subject** email subject.
- **smtp\_srv** SMTP server address used for sending messages.
- **smtp\_login** login for authorization on the mail server. In most cases, it is a mailbox, for example, "your\_name@mail.ru".
- **smtp\_pass** password for authorization on the mail server (mailbox password).

The password for authorization on the SMTP server is stored in the clear. This is done in order to modify the configuration file easily. The administrator can change the configuration (for example, a password) and launch SendMail manually. For security purposes, it is recommended to limit access to report configuration files.

The group's description is followed by the list of users that should receive the reports. The description and the list should be divided by an empty line used as a separator.

Each entry in the user list consists of the four fields: login, mailing status, recipient name and email. Tab character is used as a field separator. The first field may contain spaces before its contents. Let's examine a sample entry in more details:

```
102 1 Ivan Ivanov ivan@mail.ru
```

Status ID may have the following values:

- **0** report has not been sent or is on hold.
- **1** report has been sent successfully.
- **2** report delivery error, invalid recipient e-mail address.

After the reports have been sent, mailing status is updated in the same configuration file.

## MetaTrader 5 Trading Platform / Platform Components / Trade Server / Return Errors

### Return Errors

When an attempt is made to perform invalid operations, a trade server returns errors to terminals. They are displayed on the "Journal" tab of the "Toolbox" window, as well as in journals of servers.

### Common Errors

| Error                    | Description                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Common error             | Error not included to any of categories listed below.                                                                                      |
| Invalid parameters       | Incorrect parameters are specified at the attempt to change any of configurations.                                                         |
| Disk error               | This error occurs when it is impossible to write information on a disk.                                                                    |
| Memory error             | This error indicates that there is not enough memory.                                                                                      |
| Network error            | This error is shown when a failure occurs in the network interaction of the platform components.                                           |
| Not enough permissions   | This error occurs when a manager/administrator attempts to perform an action, for which he or she doesn't have enough <u>permissions</u> . |
| Operation timeout        | The error means that the operation fulfillment waiting period is over.                                                                     |
| No connection            | Connection to the server couldn't be established.                                                                                          |
| Service is not available | This error can appear when information is requested from one of the platform components that is currently unavailable.                     |
| Too frequent requests    | The error occurs if requests to the server are made too often.                                                                             |
| Not found                | Requested information was not found.                                                                                                       |

### Authorization Errors

| Error                 | Description                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Invalid terminal type | This error can occur at the attempt to connect using a <u>manager</u> group account via the client terminal or vice versa - connect from manager/administrator terminal using an account from a common <u>group</u> . |
| Invalid account       | This error appears if invalid account number or incorrect password are specified during <u>authorization</u> .                                                                                                        |



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| Error                                   | Description                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unknown account                         | This error appears if a user tries to authorize using an account that is created on a trade server, connection to which is <u>prohibited for the access server</u> . Due to it, the access server cannot identify the account and denies its connection.                                                                                                  |
| Account disabled                        | The error appears at the attempt to connect using a <u>disabled</u> account.                                                                                                                                                                                                                                                                              |
| Advanced authorization                  | This message indicates that the procedure of <u>extended authorization</u> must be performed for connecting.                                                                                                                                                                                                                                              |
| Certificate required                    | This message is shown at the attempt to to connect using the account that requires a SSL certificate to be generated in order to connect.                                                                                                                                                                                                                 |
| Invalid certificate                     | This message appears at the attempt to connect using a certificate that is invalid for this account. For example, if it was <u>reset</u> on the server.                                                                                                                                                                                                   |
| Certificate is not confirmed            | The message is shown at the attempt to connect using an <u>unconfirmed</u> certificate.                                                                                                                                                                                                                                                                   |
| Attempt to connect to non-access server | This error means that an attempt is made to connect to one of the platform components directly, bypassing the <u>access server</u> .                                                                                                                                                                                                                      |
| Invalid or fake server                  | The error indicates that the server, to which there was an attempt to connect, is invalid.                                                                                                                                                                                                                                                                |
| Only updates available                  | This error is shown when an attempt is made to connect to the <u>history server</u> at the time when only connecting to it for obtaining updates is allowed. This situation can appear during the platform update, when the server has been updated and is operating, but its configurations are not yet synchronized with the <u>main trade server</u> . |
| Old version                             | It indicates that the version of the component is old.                                                                                                                                                                                                                                                                                                    |
| Account doesn't have manager config     | The error occurs at the attempt to connect using the account for which no <u>manager</u> configurations have been created.                                                                                                                                                                                                                                |
| IP address unallowed for manager        | The error occurs at the attempt of a manager connection from an <u>unallowed IP address</u> .                                                                                                                                                                                                                                                             |
| Group is not initialized                | The error occurs at the attempt to create an account in the <u>group</u> that hasn't been initialized yet. In order to initialize a group after it has been created, the trade server needs to be <u>restarted</u> .                                                                                                                                      |
|                                         |                                                                                                                                                                                                                                                                                                                                                           |

| Error                           | Description                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certificate generation disabled | The error occurs if the terminal requests generation of a new SSL certificate, but the possibility of their automatic generation is disabled on the server. |

## Configuration Management Errors

| Error                              | Description                                                                                                                                                                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Last admin config deleting         | The error appears at the attempt to delete the last <u>manager configuration</u> .                                                                                                                                                                  |
| Last admin group cannot be deleted | The error appears at the attempt to delete the last administrator <u>group</u> .                                                                                                                                                                    |
| Accounts or trades in group        | The error appears at the attempt to delete a <u>group</u> that contains <u>accounts</u> or <u>trade operations</u> .                                                                                                                                |
| Invalid accounts or trades ranges  | The error appears at the attempt to set the range of <u>accounts</u> , <u>orders</u> or <u>deals</u> for a trade server that coincide with the same ranges on other servers.                                                                        |
| Account is not from manager group  | The error occurs at the attempt to create a <u>manager configuration</u> based on the account that does not belong to the manager group.                                                                                                            |
| Built-in protected config          | The error occurs at the attempt to delete a built-in configuration of collection of the server <u>performance parameters</u> .                                                                                                                      |
| Configuration duplicate            | This error is shown when an attempt is made to add the second <u>main trade server</u> or a <u>history server</u> in the "Network" section.                                                                                                         |
| Configuration limit reached        | This error appears at the attempt to create a new configuration, when the limit to their creation is reached. This can happen if the license was <u>not activated</u> . In this case the number of <u>groups</u> , for example, is limited to five. |
| Invalid network configuration      | This error appears at the attempt to save such <u>network settings</u> that the administrator will not be able to connect to the main trade server.                                                                                                 |

## Account Management Errors

| Error                            | Description                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Last admin account deleting      | This error appears at the attempt to delete the last <u>manager</u> account.                                         |
| Logins range exhausted           | The error appears at the attempt to create an account when the <u>range of allowed accounts</u> was exhausted.       |
| Login reserved at another server | This error appears at the attempt to <u>create an account</u> via a manager or administrator terminal with the login |

## MetaTrader 5 trading platform

| Error                                               | Description                                                                                                                                                     |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | that already exists on another server.                                                                                                                          |
| Account already exists                              | The error appears at the attempt to <u>create an account</u> with already existing login.                                                                       |
| Attempt of self-deletion                            | The error appears if an administrator is trying to delete his or her own account via the administrator terminal.                                                |
| Invalid account password                            | The error appears if incorrect password was specified during <u>password verification</u> .                                                                     |
| Users limit reached                                 | The error appears at the attempt to create a new <u>account</u> when their limit is over. The situation is possible when the license was <u>not activated</u> . |
| Account has open trades                             | The error appears at the attempt to delete an account with <u>open positions</u> .                                                                              |
| Attempt to move account to different server         | The error appears at the attempt to move an <u>account</u> to the group that belongs to <u>another trade server</u> .                                           |
| Attempt to move account to different currency group | The error appears at the attempt to move an account to the group with a different <u>deposit currency</u> .                                                     |

### State of Trade Requests

| Error                   | Description                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Request on the way      | The request has been sent to a server but it hasn't accepted it yet.                                                                     |
| Request accepted        | The request has been accepted by the server and is waiting to be processed.                                                              |
| Request processed       | The server has started to process the request.                                                                                           |
| Requote                 | Requoting as a reply to a client's request to execute a trade operation.                                                                 |
| Prices                  | Sending prices.                                                                                                                          |
| Request rejected        | The request is rejected by the server or dealer.                                                                                         |
| Request canceled        | the request was canceled by the server, dealer or client.                                                                                |
| Order placed            | The order has been placed and is waiting till the specified processing conditions appear.                                                |
| Request executed        | The requested operation has been executed.                                                                                               |
| Request executed partly | The requested operation has been executed partially. E.g. a <u>deal</u> on the part of volume specified in the <u>order</u> is executed. |
| Request error           | Error of request execution.                                                                                                              |

| Error                          | Description                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Request timeout                | Time of request execution waiting is over.                                                                                                                               |
| Invalid request                | The request didn't pass the common validation procedure.                                                                                                                 |
| Invalid volume                 | Incorrect volume is specified in the request.                                                                                                                            |
| Invalid price                  | Incorrect price is specified in the request.                                                                                                                             |
| Invalid stops                  | Incorrect Take Profit and Stop Loss levels are specified in the request.                                                                                                 |
| Trade disabled                 | Trading is <u>disabled</u> for this account.                                                                                                                             |
| Market closed                  | The error appears at the attempt to execute trade operations beyond the allowed period (both <u>common trading hours</u> and <u>trade sessions</u> of separate symbols). |
| No money                       | Attempt to execute a trade operations, when there is not enough money for its execution.                                                                                 |
| Price changed                  | The error appears when the <u>maximal deviation</u> of the current price at the request price is exceeded.                                                               |
| No prices                      | The error appears when there is no thread of quotes.                                                                                                                     |
| Invalid expiration             | The error occurs at the attempt to place an order with an incorrect <u>expiration</u> date.                                                                              |
| Order has been changed already | The error appears at the attempt to modify an order, which has been simultaneously modified.                                                                             |
| Too many trade requests        | The error appears if trade requests are sent too often to the server.                                                                                                    |
| AutoTrading disabled by server | This error indicates that trading of <u>Expert Advisors</u> on this account is prohibited.                                                                               |
| AutoTrading disabled by client | This error indicates that trading of Expert Advisors is disabled in the client terminal.                                                                                 |

MetaTrader 5 Trading Platform / Platform Components / Access Server

## Access Server

Access servers are proxy servers and the platform firewalls at the same time. They perform the following functions:

- Processing of incoming client connections.
- Packing authorization requests and sending them to the trade server.
- Checking activity of client connections protecting the trade server from attacks and overload (antiflood control).
- Saving history data, depth of market and news, and translate them to clients, thus reducing the load to the history server.
- Cashing and providing Live Update to terminals.
- Monitoring the operation of the history and trade servers.

The unlimited number of access servers can exist for each trade server. Terminals are switched between them automatically, depending on the priority settings.

- For server configuration details, see the "Network cluster" section.
- Use the Access Server hosting from MetaQuotes to quickly and safely deploy an access server in the desired region.

## MetaTrader 5 Trading Platform / Platform Components / Access Server / Structure of Directories and Files

### Structure of Directories and Files

The access server is installed to folder "access\_server". This folder contains the following executable files:

- **mt5srvupdater64.exe** the executable file of the live update system of the access server;
- **mt5access64.exe** the executable file of the access server.

The main directory of the access server contains five folders: *bases*, *config*, *history*, *liveupdate*, *logs*.

The **bases** directory contains the news databases, as well as data on the server performance.

| Files                  | Description                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>performance\</b>    | Monthly access server performance databases and data indexes, which are displayed on the <u>Monitoring</u> tab.     |
| <b>news.dat</b>        | Database of news sent to clients.                                                                                   |
| <b>news.idx</b>        | The index file of the news database.                                                                                |
| <b>performance.dat</b> | Data about the access server performance that are displayed on the " <u>Monitor</u> " tab are written to this file. |

The **config** directory contains configuration files:

| Files              | Description                                          |
|--------------------|------------------------------------------------------|
| <b>server.ini</b>  | Individual settings of the access server.            |
| <b>servers.ini</b> | Settings of the internal <u>network of servers</u> . |
| <b>symbols.ini</b> | Configurations of <u>symbols</u> .                   |
| <b>time.ini</b>    | <u>Time</u> settings.                                |

The **history** directory contains the base of history data by symbols, which was received from the history server:

| Files and folders          | Files           | Description                                                                                                                                                                                                                                                                  |
|----------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[2 chars]\[symbol]\</b> | <b>yyyy.hsc</b> | Minute data for the symbol, divided by years. '2 chars' are the first two characters in the instrument name; 'symbol' is the name of the instrument. Arranging symbol data in different directories reduces the load on the file system and provides faster data operations. |

| Files and folders   | Files    | Description                                                                                                                                                                                                                                                                |
|---------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [2 chars]\[symbol]\ | yyyy.tkc | Tick data for the symbol, divided by years. '2 chars' are the first two characters in the instrument name; 'symbol' is the name of the instrument. Arranging symbol data in different directories reduces the load on the file system and provides faster data operations. |
| tickers.dat         |          | Data by tickers.                                                                                                                                                                                                                                                           |

The **liveupdate** directory contains the latest updates of the client, manager and administrator terminals:

| Files           | Description                                                                               |
|-----------------|-------------------------------------------------------------------------------------------|
| mt5adm.build    | Live update of the administrator terminal. The build number is specified after the point. |
| mt5clw.build    | Live update of the client terminal.                                                       |
| mt5ckwide.build | Live Update of MetaEditor.                                                                |
| mt5clwmql.build | Live Update of the MQL5 compiler.                                                         |
| mt5man.build    | Live update of the manager terminal.                                                      |

The **logs** directory keeps files of the access server operation journal, as well crash logs:

| Files and folders | Description                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crash\crash.log.* | The /crash directory contains server crash files. These files are automatically sent to the software developing company for detecting reasons of the crash and eliminating them.                     |
| yyyymmdd.log      | <u>Journal</u> files that contain all the information about events that occur on the access server. Server logs are stored in separate files for each working day. Here yyyy year, mm month, dd day. |
| mt5srvupdater.log | Journal files of the platform <u>updates</u> .                                                                                                                                                       |

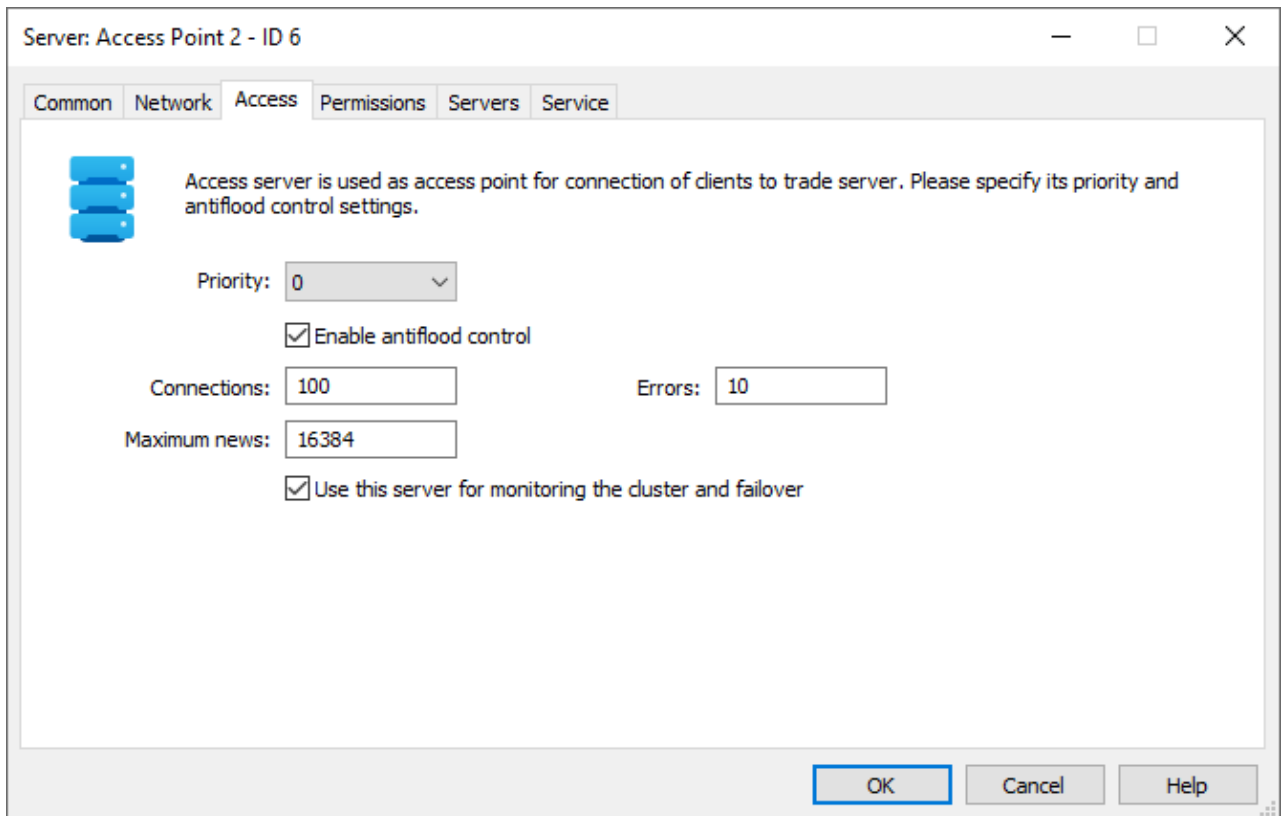
MetaTrader 5 Trading Platform / Platform Components / Access Server /  
Antiflood Control

## Antiflood Control

The antiflood control system works on access servers. It allows protecting the trading platform from external harmful attacks. This protection system collects the database of users who send incorrect requests to the server (e.g. attempt to authorize with an incorrect login or password), as well as users who send too often requests.

Antiflood control works with all types of connections to the server including manager ones (via the manager terminal and Manager API).

The number of connections and incorrect request per time unit can be set up on the "Access" tab of the access server:



The screenshot shows a window titled "Server: Access Point 2 - ID 6" with a standard Windows interface (minimize, maximize, close buttons). The "Access" tab is selected, showing a configuration panel for an access server. The panel includes a description of the access server's role and a set of configuration options. The "Priority" is set to 0. The "Enable antiflood control" checkbox is checked. The "Connections" field is set to 100, and the "Errors" field is set to 10. The "Maximum news" field is set to 16384. The "Use this server for monitoring the cluster and failover" checkbox is also checked. At the bottom right, there are "OK", "Cancel", and "Help" buttons.

Server: Access Point 2 - ID 6

Common Network **Access** Permissions Servers Service

Access server is used as access point for connection of clients to trade server. Please specify its priority and antiflood control settings.

Priority: 0

☒ Enable antiflood control

Connections: 100 Errors: 10

Maximum news: 16384

☒ Use this server for monitoring the cluster and failover

OK Cancel Help



## System of Operation

The antiflood control system monitors user activity. Users are identified by their IP addresses as well as by the ID linked to their computers and operating systems. The following two activity types are controlled:

- **The total number of connections from one user**

The system tracks if a user creates too many connections. When a user connects to the server, their connection counter is incremented by one. If the next connection occurs in less than 3 seconds, the counter is incremented. If the specified time interval is exceeded, the counter is reset. When the counter reaches the number specified in the "Connections" field, the user is blocked for 5 minutes. The blocking period increases if the connections limit is reached again. The maximum blocking period is one hour.

- **Number of invalid packets from one user**

The system blocks brute-force attacks by analyzing authentication errors, which imply multiple login attempts with incorrect data. Also, the system detects garbage flood packets, which can be sent to the server by third-party utilities in an effort to reduce the server performance (i.e. a DoS attack). When a user sends an invalid packet to the server, the user's error counter is incremented by one. If the next invalid packet is sent in less than 5 minutes, the counter is incremented. If the specified time interval is exceeded, the counter is reset. When the counter reaches the number specified in the "Errors" field, the user is blocked for 5 minutes. The blocking period increases if the connections limit is reached again. The maximum blocking period is one hour.

The following records appear in the access server operation Journal when a user is blocked:

- *IP* is blocked after *N* connections [intruder] a user is blocked by IP if the number of connections is exceeded;
- *CID* is blocked after *N* errors [intruder] a user is blocked by CID (computer ID) if the number of errors is exceeded.

- It is strongly recommended to keep the antiflood control system enabled.
- IP addresses added to the "permit always" list in the corresponding section, are not checked by the antiflood control system.

### Retrieve IP address from X-Forwarded-For

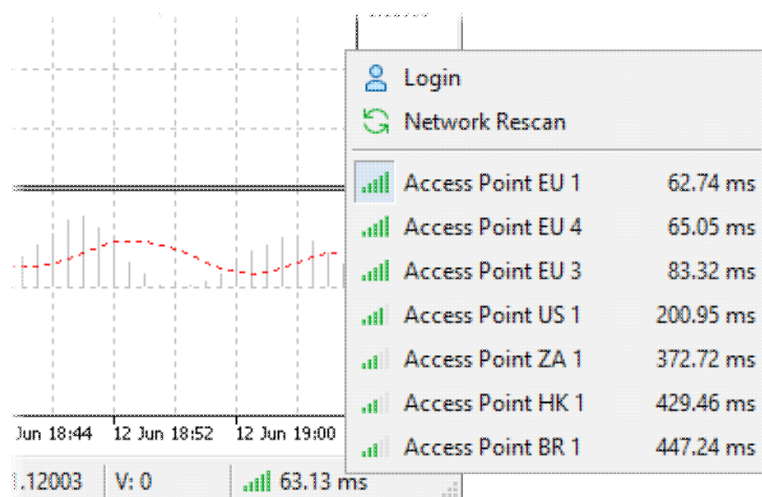
Clients can connect to the platform through public services from various proxy servers. This may happen, for example, if you use an Anti DDoS system, such as Cloudflare. In this case, the proxy server address will be displayed for the client's IP address in account details and logs. The same address will be used for anti-flood control system checks. To avoid this and provide real data, you can add proxy server addresses to the firewall whitelist (the "always allow" rule). In this case, the access server will try to retrieve the user's real IP address from the X-Forwarded-For header provided by the proxy server upon connection.

Please make sure the IP addresses you are adding to the whitelist belong to trusted services, which may not transmit false addresses in requests.

## MetaTrader 5 Trading Platform / Platform Components / Access Server / Priority

### Priority

The preference of an access server for client terminals to connect to a trade server is defined by its priority and connection quality. The lower the value of priority is, the more preferable the access server is. A base priority (from 0 to 15) can be specified in its settings. It defines the server preference if all other conditions are equal. The final analysis of an access server is conducted upon the ping and the current priority, which depends on the basic priority and the number of current connections. Also, the quality of connection to the server is shown in the client, manager and administrator terminals based on the same data:



The current priority is calculated according to the following formula: *Current Priority* = (*Base Priority* + *Connections* / 1000),

where:

- **Current Priority** is the priority at the server current moment;
- **Base Priority** is the base priority set in its parameters;
- **Connections** the number of current connections.

Every 1000 client connections increase the current priority of a server by one. The value of the current priority of access servers is available on the "Network" tab.

MetaTrader 5 Trading Platform / Platform Components / History Server

## History Server

The history server processes price and news data. This server performs the following functions:

- Receiving and filtering price and news data from gateways and datafeeds.
- Packing price and news data.
- Storing and providing price history in the form of 1-minute bars and ticks to other components of the platform.
- Storing and providing the news thread.
- Receiving, checking and distributing Live Updates among the MetaTrader 5 platform components.

|                                                                               |
|-------------------------------------------------------------------------------|
| For server configuration details, see the " <u>Network cluster</u> " section. |
|-------------------------------------------------------------------------------|

## MetaTrader 5 Trading Platform / Platform Components / History Server / Structure of Directories and Files

### Structure of Directories and Files

The history server is installed to folder "*history\_server*". It contains the following executable files:

- **mt5srvupdater64.exe** the executable file of the live update system of the history server. This component has a number of console commands;
- **mt5history64.exe** the executable file of the history server.

The main directory of the history server contains the following folders: bases, config, datafeed, gateway, history, liveupdate, logs, plugins.

The **bases** directory contains different data bases:

| Files and folders | Description         | Files and folders                  | Description                                                                           |
|-------------------|---------------------|------------------------------------|---------------------------------------------------------------------------------------|
| ecn\              | ECN data directory. | executions\                        | Bases and indexes of ECN trade executions by trade servers.                           |
|                   |                     | history\                           | Bases and indexes related to the history of client order execution in ECN, by months. |
|                   |                     | symbols\{Symbol}\matching.dat      | Symbol databases related to client orders placed in ECN for matching.                 |
|                   |                     | symbols\{Symbol}\filling_items.dat | Symbol databases of matching operations processed in ECN.                             |

|                   |                                                                                                                  | Files and folders                    | Description                                                     |
|-------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------|
|                   |                                                                                                                  |                                      |                                                                 |
|                   |                                                                                                                  | symbols\{Symbol}\books\yyyymmdd.book | ECN Market Depth journals by days.                              |
|                   |                                                                                                                  | filling_orders.dat                   | Databases of internal ECN order which are executed on gateways. |
| Files and folders | Description                                                                                                      |                                      |                                                                 |
| performance\      | Monthly history server performance databases and data indexes, which are displayed on the <u>Monitoring</u> tab. |                                      |                                                                 |
| news.dat          | News data base.                                                                                                  |                                      |                                                                 |
| news.idx          | The index file of the news database.                                                                             |                                      |                                                                 |

The **config** directory contains different configurations as \*.ini files:

| Files                    | Description                                          |
|--------------------------|------------------------------------------------------|
| <b>common.ini</b>        | Common History Server settings.                      |
| <b>ecn_symbols.ini</b>   | <u>ECN</u> symbol settings.                          |
| <b>mt5srvupdater.ini</b> | <u>Update</u> settings.                              |
| <b>history_sync.ini</b>  | Settings of <u>history data synchronization</u> .    |
| <b>server.ini</b>        | Individual settings of the access server.            |
| <b>servers.ini</b>       | Settings of the internal <u>network of servers</u> . |
| <b>symbol_groups.ini</b> | Individual settings of <u>symbols for groups</u> .   |
| <b>symbols.ini</b>       | <u>Symbol</u> settings.                              |
| <b>time.ini</b>          | <u>Time</u> settings.                                |

The **datafeed** directory contains files for working with data feeds:

| Files                                    | Description                                                                                                                                                                                                                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[datafeed_name]\logs\yyyymmdd.log</b> | Journal files in which records regarding data feed operation are stored. For each data feed which has been added through the <u>relevant section</u> of the Administrator terminal, a separate journal file is created. The file name is set in accordance with the data feed name. |

| Files                                             | Description                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[datafeed_name]\*.dat</b>                      | Data files with data feed settings.                                                                                                                                                                                                                                                                        |
| <b>*.exe</b>                                      | Data feed executables. It is not allowed to have several datafeed executable files with the same name in the history server directory. If you place several identical files in different subdirectories, this may lead to conflicts in the operation and display of modules in the Administrator terminal. |
| <b>MT5APIGateway.dll,<br/>MT5APIGateway64.dll</b> | Libraries for data feed operation.                                                                                                                                                                                                                                                                         |

The **gateway** directory contains files for working with gateways:

| Files and folders                                                    | Description                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[gateway_name]\[gateway configuration name]\logs\yyyymmdd.log</b> | Journal files in which gateway operation logs are stored. For each data feed which has been added through the <u>relevant section</u> of the Administrator terminal, a separate journal file is created. The file name is set in accordance with the data feed name.                                     |
| <b>[gateway_name]\[gateway configuration name]\*.dat</b>             | Data files with gateway settings.                                                                                                                                                                                                                                                                        |
| <b>*.exe</b>                                                         | Gateway executables. It is not allowed to have several datafeed executable files with the same name in the history server directory. If you place several identical files in different subdirectories, this may lead to conflicts in the operation and display of modules in the Administrator terminal. |
| <b>MT5APIGateway.dll,<br/>MT5APIGateway64.dll</b>                    | Libraries for gateway operation.                                                                                                                                                                                                                                                                         |

The **history** folder contains history data divided by symbols:

| Folders                    | Files           | Description                                                                                                                                                                                                                                                                |
|----------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[2 chars]\[symbol]\</b> | <b>yyyy.hsc</b> | History data on a symbol, divided by years. '2 chars' are the first two characters in the instrument name; 'symbol' is the name of the instrument. Arranging symbol data in different directories reduces the load on the file system and provides faster data operations. |
| <b>[2 chars]\[symbol]\</b> | <b>yyyy.tkc</b> | Tick data on a symbol, divided by years. '2 chars' are the first two characters in the instrument name;                                                                                                                                                                    |

## MetaTrader 5 trading platform

| Folders | Files | Description                                                                                                                                                     |
|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |       | 'symbol' is the name of the instrument. Arranging symbol data in different directories reduces the load on the file system and provides faster data operations. |

The **liveupdate** folder contains the latest updates of all the platform components:

| Files                  | Description                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| <b>mt5adm.build</b>    | Live update of the administrator terminal. The build number is specified after the point. |
| <b>mt5as.build</b>     | Live update of the access server.                                                         |
| <b>mt5bs.build</b>     | Live update of the backup server.                                                         |
| <b>mt5clw.build</b>    | Live update of the client server.                                                         |
| <b>mt5clwide.build</b> | Live Update of MetaEditor.                                                                |
| <b>mt5clwmql.build</b> | Live Update of the MQL5 compiler.                                                         |
| <b>mt5hs.build</b>     | Live update of the history server.                                                        |
| <b>mt5hsu.build</b>    | Live update of the update system of the history server.                                   |
| <b>mt5man.build</b>    | Live update of the manager server.                                                        |
| <b>mt5ts.build</b>     | Live update of the trade server.                                                          |

The **logs** folder contains log files of the history server operation, as well as crash logs:

| Files and folders        | Description                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Crash\crash.log.*</b> | The /crash directory contains server crash files. These files are automatically sent to the software developing company for detecting reasons of the crash and eliminating them.                      |
| <b>yyyymmdd.log</b>      | <u>Journal</u> files that contain all the information about events that occur on the history server. Server logs are stored in separate files for each working day. Here yyyy year, mm month, dd day. |
| <b>mt5srvupdater.log</b> | Journal files of the platform <u>updates</u> .                                                                                                                                                        |



MetaTrader 5 Trading Platform / Platform Components / History Server /  
Interaction with Quote Providers

## Interaction with Quote Providers

In MetaTrader 5, gateways and data feeds can be used as providers of quotes. Their interaction with the history server is identical. This interaction can be analyzed in terms of the physical connection and at the level of quotes streaming.

### Physical Connection

The physical connection must be established for each source of quotes enabled in the appropriate settings of the platform. The physical connection details can be configured on the "Timeouts" tab of gateways and data feeds. Let's consider the following configuration example:

- Interval between reconnections = 5 seconds.
- Number of reconnection attempts = 10.
- Interval between series of reconnections = 60 seconds.

If a gateway/data feed loses connection with an external server, a reconnection attempt is made in 5 seconds. If it fails, another one is made in 5 seconds. The total number of attempts is 10. If unable to reconnect, a series of attempts is repeated after a pause of 60 seconds.

### Stream of Quotes

A stream of prices for several symbols goes through each physical connection to a quote provider.

At each point of time, the history server accepts the stream of prices for a certain symbol only from one quote provider, while other price streams of the same symbol are ignored. The source selected for the stream of prices for a symbol is considered a current (active) source for this symbol.

During operation the active source for an instrument may change. It is changed in accordance with priority settings. The priority of data feeds and gateways is determined by their position in the list.

The priority of gateways, if they are used as a source of quotes, is always higher than that of data feeds.

The stream of prices switches to the source with a higher priority as soon as the first quotes for a symbol is received from that source.

It switches to the source with a lower priority by a timeout. In case no quotes are received from the active quote source during a certain time period (it is specified in the "Datafeeds timeout" parameter in history server settings), then the history server switches to a source with the lower priority that provides quotes for the same symbol.

The time to wait for a quote for a symbol is defined in the "Datafeeds timeout" parameter in [history server settings](#).

After a new quotes source is selected, it is considered active. All cases of server switching to streams from other sources are reflected in the [journal](#) in the form of the following entry:

2011.03.10 10:11:05 Ticks datafeed 4: CHFJPY activation

Here the entry means that for CHFJPY, a stream of quotes from the fourth data feed (a position in the list of data feeds at the moment the entry is made) is selected.

MetaTrader 5 Trading Platform / Platform Components / History Server /  
Quotes Filtration

Quotes Filtration

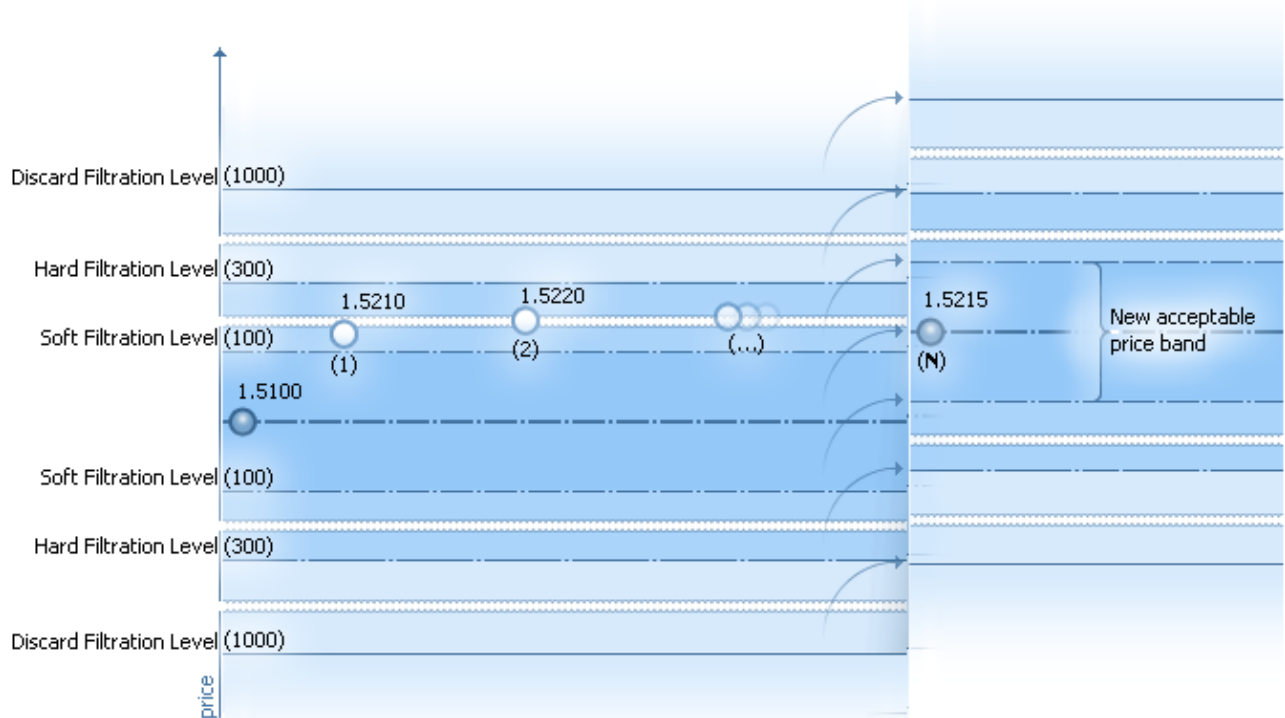
The filtration system is intended for controlling the correctness of quotes on financial symbols from data feeds. Filters can be set upon the "Quotes" tab of each symbol.

- Filters cannot be applied to instruments with the enabled Depth of Market **and** with the one of the following properties: Exchange calculation type (begins with "Exchange") **or** Last price based charting mode. Also, filtering is not applied to splice symbols.
- To completely disable filtering, set 0 for all levels: Soft, Hard and Discard.
- The platform automatically filters out negative and zero Bid and Ask prices of OTC instruments, regardless of whether their Market Depth is enabled or not. OTC (over-the-counter or off-exchange) symbols include financial instruments whose calculation type does not start with "Exchange".
- Analysis and filtration are performed separately for Bid and Ask.
- Filters are not applied to the first tick after a break in the quotes stream: after platform restart, after a break in quoting sessions, after off-hours and after holidays. Filtration cannot be applied because it is not known in advance what was the price preceding the break. This rule does not apply to other price checks, including allowable spread.

Three levels of quote filtration are available:

- **Soft filtration** the soft filtration level is the first border of the channel of allowed symbol prices. If a new price (Bid or Ask) differs from the previous one by more than the specified value (in points), it is deleted from the thread translated to clients. However, if such a price difference appears again the number of times specified in the "Filter" field, the new price level is accepted, and the filtration level is shifted by the specified value. Such quotes will be translated again.
- **Hard filtration** the hard filtration level is the second border of the price channel. If a received quote exceeds the level both of the soft and of the hard filtration, it is cut out from the thread translated to clients. For a new level of accepted price to be set, the quotes must be repeat the number of times specified in both levels;
- **Discard filtration level** if the difference between prices of the previous and new quote exceed the specified value, such new prices are definitely removed from the thread.

## MetaTrader 5 trading platform



- channel of allowed prices
- soft filtration zone
- hard filtration zone
- discard filtration zone
- new quote
- N** number of quotes (set in the "Filter" filed) required to move to a new level
- previous quote

It is considered that there is a certain acceptable channel of price data fluctuation. This channel is limited by the set level of soft filtration. If the Bid or Ask price of the newly received quote differs from appropriate prices of the previous quote by the value of the specified soft filtration level, it is discarded. If several following quotes (the number is specified in the "Filter" parameter) also exceed the soft filtration level, the new price channel is set.

### Example:

The last EURUSD quote was 1.50213/1.50231 (Bid/Ask). The soft filtration level is equal to 150, "Filter" parameter is equal to 3. The following quotes are received: 1.50373/1.50391, 1.50370/1.50388, 1.50372/1.50390, 1.50374/1.50392. In this case the first three quotes will be filtered away, because they exceed the previous one by more than 150 points. The last quote will be let in, and the new filtration level will be set  $1.50374/1.50392 \pm 150$ .

The hard filtration level is an additional way to protect from incorrect quotes. If a new quote differs from the previous one by the value that is higher than the specified hard filtration level, it will be filtered away. The additional hard filtration protection is a more complicated mechanism of setting the new price channel. To confirm the new

level, first the soft filter (specified number of quotes that exceed this value) must be passed, and then the hard one.

The discard filtration implies the unconditional filtering away of quotes that differ by this value. Such prices are deliberately considered incorrect.

### Filtration of Similar Quotes

The trading platform filters similar quotes received from data sources. If the platform receives the same quote as the previous one within a minute, it skips the quote. If the time interval is greater than one minute, the platform accepts the quote. The same quote is also accepted if the minute has changed. It allows plotting the charts correctly on a low liquidity market.

Example:

- 13:01:15 a quote is received
- 13:01:32 the same quote is received, it is not accepted
- 13:01:50 the same quote is received, it is not accepted
- 13:02:01 the same quote is received, it is accepted (the interval between quotes is less than 60 seconds, but the minute has changed)

### Spread Control

The Minimum Spread and Maximum Spread parameters in the Quotes tab are provided for additional protection. If the difference between the Bid and Ask prices in the incoming quote does not fall within the specified values, such a quote is removed from the stream.

The check only applies to over-the-counter (OTC) financial instruments with the floating spread. OTC instruments have one of the following calculation types: Forex, Futures, CFD, CFD Leverage or CFD Index.

MetaTrader 5 Trading Platform / Platform Components / History Server /  
Console Commands

Console Commands

The history server component mt5srvupdater64.exe has several console commands that allow updating and activating the platform if servers are unavailable for connection via the administrator terminal.

In order to execute these commands, start file mt5srvupdater64.exe from the root directory of the history server, specifying corresponding keys:

- **/update** download updated files of all the system components, if there are such files, from the developer's update server;
- **/upgrade** install updates of all the platform components. This command can be executed only if the updates have been downloaded;
- **/activate** activate the platform.

For example, after you execute command "mt5srvupdater64.exe /update /upgrade", the platform updates will be downloaded and installed.

The protocol of the update process is kept in mt5srvupdater.txt, located in folder /history/logs/.

MetaTrader 5 Trading Platform / Platform Components / Backup Server

## Backup Server

Servers of this type are used for creating backup copies of data that will be used in case of the trade or history server failure. They perform the following functions:

- They provide the real-time backuping of the trade server and the history server. Each server is associated with one or more separate backup server instances which can replace it at any time.
- They create backups of all databases every day, besides they perform regular backups of client and trade bases.
- When backuping a history server they create real time backups of data required for correct restoring of gateways: custom settings that can be stored (at developer's discretion) in the settings.dat file in a gateway work folder, and trade executions database.
- They provide automatic failover in case the primary server becomes unavailable.
- Using backup servers, you can easily migrate servers to new hardware.
- Backup servers enable you to quickly restore server operation in the manual mode even if the main trade server is unavailable, and connection to the platform via MetaTrader 5 Administrator is not possible.
- Replicate information to database managed by MySQL, MariaDB, PostgreSQL, Firebird, MSSQL or Oracle.

|                                                                                      |
|--------------------------------------------------------------------------------------|
| For server configuration details, please see the " <u>Network cluster</u> " section. |
|--------------------------------------------------------------------------------------|

MetaTrader 5 Trading Platform / Platform Components / Backup Server / Backup Features

## Backup Features

All critical trade and history server data is backed up in real time. Some non-critical trade server data, as well as history server data are backed up every hour. Critical and non-critical data are displayed in the table below:

| Data                                   | Trade server                                                                                                                                                                                                              | History server                                                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Critical (real time backup)            | User base<br>Order base<br>Deal base<br>Configuration databases                                                                                                                                                           | Trade execution base (executions) and custom settings (settings.dat) of running gateways<br><br>Tick flow                                                                                                |
| Non-critical (backup every hour)       | mt5sendmail64.exe and mt5trade64.exe files<br><br>Plugins (DLL, INI, DAT files including subdirectories)<br><br>Reports (DLL, INI files without subdirectories)<br><br>Templates (HTML, HTM files without subdirectories) | mt5history(64).exe file<br><br>Gateways and data feeds (EXE, DLL, INI, DAT, JAR, CMD, BAT, PS1, VBS and PY files including subdirectories)<br><br>Plugins (DLL, INI, DAT files including subdirectories) |
|                                        | User directories and files specified in the <u>Folders</u> section in backup server settings                                                                                                                              |                                                                                                                                                                                                          |
| Non-critical (backup every five hours) | Archive databases<br><br>Mail database<br><br>Daily report database                                                                                                                                                       | Minute bar history                                                                                                                                                                                       |
| Non-critical (backup every 24 hours)   |                                                                                                                                                                                                                           | Tick history                                                                                                                                                                                             |

The installation directory of the backup server contains the same folders as the server, whose backups it creates, except for the *logs* folder that contains the journal entries and crash logs of the backup server itself. In addition, the following executable files are available in the directory:



- **mt5srvupdater64.exe** the executable file of the live update system of the backup server;
- **mt5backup64.exe** the executable file of the backup server.

### Periodic file backup

Apart from synchronizing with the main server in real time, the backup server creates database file copies on the disk for certain points in time. The path for saving files and the creation time are set in the backup server settings.

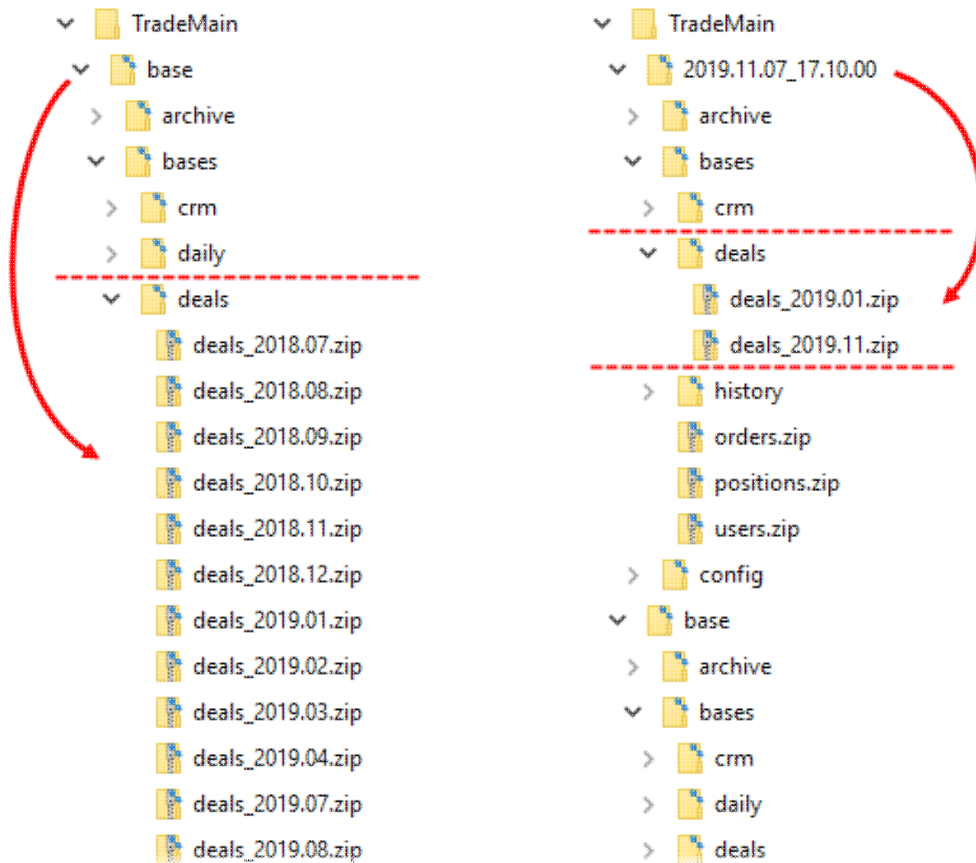
To save time and disk space, backups are performed incrementally:

- The first copy is made for all data of the backed up server. That copy is placed to ["Backups path"]\base\. The folder and file structure there fully matches the one of the backed up server.
- Each subsequent day (hour of 4 hours, depending on the "Additional backups"), an incremental copy of the data is created. Such a copy contains only the changes as compared to the first (base) copy. Such copies are placed to ["Backups path"]\[YYYY.MM.DD\_HH.MM.SS]\.

Thus, the server does not need to copy all the data during each backup.

Let us consider an example with a trade database. Backup was enabled on 2019.11.06. The server created a database copy on this day. It is shown on the left in the figure below. The entire history of deals is contained under the 'base' directory. It is divided by months and is presented as ZIP files. During the next day, traders performed deals through the platform, as well as the administrator edited a trade which was executed in January 2019. The next day, the backup server created an incremental copy in the catalog 2019.11.07\_17.10.00 (on the right). You can see that only two ZIP archives are included in the incremental copy directory, i.e. those which have changed relative to the base copy:

## MetaTrader 5 trading platform

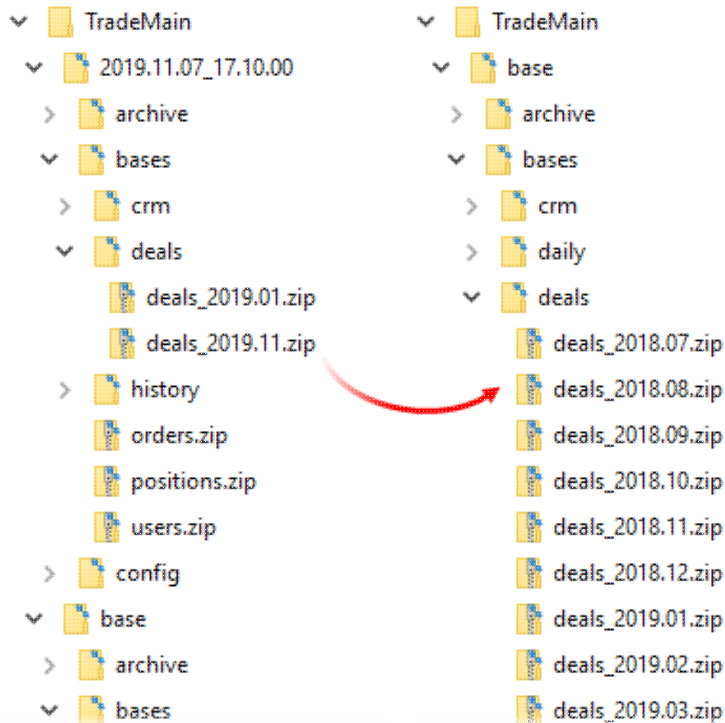


The server is able to automatically delete old copies to ensure that the backups do not take up all the free disk space over time. The storage depth is set by the "Keep backups" parameter in the backup server settings. When deleting an old backup copy, it is preliminarily applied to the main one. In fact, its files are combined with the files from the "base" directory. After that, the "base" directory will have the same status as the databases on the server at the time of creation of the deleted incremental copy. Subsequent incremental copies will already be created relative to the new base copy. This approach also reduces the amount of data copied during backup.

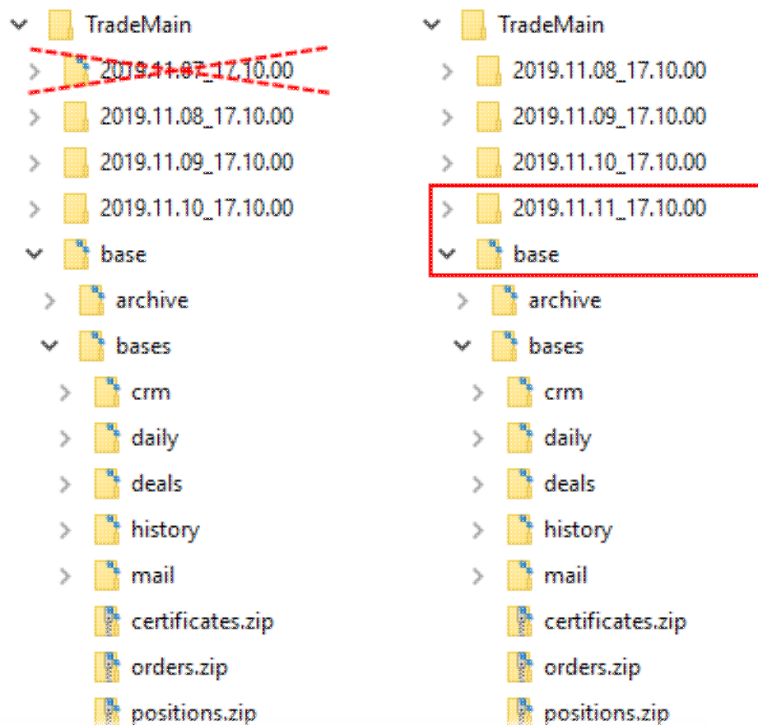
Let us consider this procedure using the previous example. Suppose, the "Keep backups" parameter is set to "3 days".

- The backup was enabled on 2019.11.06 and the server created a backup copy ("base" folder). the following three days, the server was creating incremental copies in separated directories.
- On 2019.11.11, when it is time to create the next copy, the server copies data from the incremental archive of 2019.11.07 into the base copy.

## MetaTrader 5 trading platform



Then, the incremental archive of 2019.11.07 is deleted, while the incremental archive dated 2019.11.11 is created based on the updated base copy:



In addition to backup copies, the server saves the "mt5backupinfo.txt" file under the directory [backup server installation directory]\base\. The file contains backup service information. Do not delete or modify this file. The file is not required for database recovery, so there is no need to copy it.

The "Enable backups" parameter allows you to completely disable creating backup copies on the disk. The backup server status synchronization with the main one is not disabled. The backup server still remains its full copy. Disabling creation of copies on the disk may be needed if the backup server is used only to export data to SQL.

### Working with file copies

If you want to save the database status as of a certain point in time (for example, in order to move it to the long-term archive on another server), simply copy the basic and incremental copies for the desired date. Together, they represent the complete database status of the backed up server as of the certain point in time.

To request data from a file copy, go to the necessary section of MetaTrader 5 Administrator (Orders, Deals, Positions, Accounts) and select a copy for the desired date in the query line. The terminal collects and presents data from the basic and selected incremental copies. Such requests actually restore the state of the database as at the selected point in time.

### Database recovery for a specific date

To restore server databases as on a specific date, stop all the servers within the cluster. Next, unzip and copy the necessary databases from the \base\ directory of the backup server to the corresponding directories of the primary server.

## MetaTrader 5 trading platform

MetaTrader5Backup > TradeMain > base > bases > deals >

| Name              | Date modified        |
|-------------------|----------------------|
| deals_2018.07.zip | 2019-11-07 4:25 PM   |
| deals_2018.08.zip | 2019-11-07 4:25 PM   |
| deals_2018.09.zip | 2019-11-07 4:25 PM   |
| deals_2018.10.zip | 2019-11-07 4:25 PM   |
| deals_2018.11.zip | 2019-11-07 4:25 PM   |
| deals_2018.12.zip | 2019-11-07 4:25 PM   |
| deals_2019.01.zip | 2019-11-07 4:25 PM   |
| deals_2019.02.zip | 2019-11-07 4:25 PM   |
| deals_2019.03.zip | 2019-11-07 4:25 PM   |
| deals_2019.04.zip | 2019-11-07 4:25 PM   |
| deals_2019.07.zip | 2019-11-07 4:25 PM   |
| deals_2019.08.zip | 2019-11-07 4:25 PM   |
| deals_2018.07.dat | 2018-11-01 9:50 AM   |
| deals_2018.08.dat | 2018-11-01 9:50 AM   |
| deals_2018.09.dat | 2018-11-01 9:50 AM   |
| deals_2018.10.dat | 2018-11-01 9:50 AM   |
| deals_2018.11.dat | 2018-12-03 9:06 AM   |
| deals_2018.12.dat | 2018-12-14 9:11 AM   |
| deals_2019.01.dat | 2019-01-24 6:05 PM   |
| deals_2019.02.dat | 2019-02-18 6:02 PM   |
| deals_2019.03.dat | 2019-03-21 6:05 PM   |
| deals_2019.04.dat | 2019-04-17 7:05 PM   |
| deals_2019.07.dat | 2019-07-31 11:47 ... |
| deals_2019.08.dat | 2019-08-21 10:33 ... |

MetaTrader 5 Platform > MainTrade > bases > deals

| Name              | Date modified        |
|-------------------|----------------------|
| deals_2018.07.dat | 2018-11-01 9:50 AM   |
| deals_2018.08.dat | 2018-11-01 9:50 AM   |
| deals_2018.09.dat | 2018-11-01 9:50 AM   |
| deals_2018.10.dat | 2018-11-01 9:50 AM   |
| deals_2018.11.dat | 2018-12-03 9:06 AM   |
| deals_2018.12.dat | 2018-12-14 9:11 AM   |
| deals_2019.01.dat | 2019-01-24 6:05 PM   |
| deals_2019.02.dat | 2019-02-18 6:02 PM   |
| deals_2019.03.dat | 2019-03-21 6:05 PM   |
| deals_2019.04.dat | 2019-04-17 7:05 PM   |
| deals_2019.07.dat | 2019-07-31 11:47 ... |
| deals_2019.08.dat | 2019-08-21 10:33 ... |

Then unzip and copy the required data from the incremental copy directory for the desired date (its format is [YYYY.MM.DD\_HH.MM.SS]) to the relevant directories of the main server.

| MetaTrader5Backup > TradeMain > 2019.11.07_17.10.00 > bases > deals > |                    | MetaTrader 5 Platform > MainTrade > bases > deals > |                    |
|-----------------------------------------------------------------------|--------------------|-----------------------------------------------------|--------------------|
| Name                                                                  | Date modified      | Name                                                | Date modified      |
| deals_2019.01.zip                                                     | 2019-11-07 5:10 PM | deals_2018.07.dat                                   | 2018-11-07 5:10 PM |
| deals_2019.11.zip                                                     | 2019-11-07 5:10 PM | deals_2018.08.dat                                   | 2018-11-08 5:10 PM |
| deals_2019.01.dat                                                     | 2019-11-07 5:10 PM | deals_2018.09.dat                                   | 2018-11-09 5:10 PM |
| deals_2019.11.dat                                                     | 2019-11-07 5:10 PM | deals_2018.10.dat                                   | 2018-11-10 5:10 PM |
|                                                                       |                    | deals_2018.11.dat                                   | 2018-11-11 5:10 PM |
|                                                                       |                    | deals_2018.12.dat                                   | 2018-11-12 5:10 PM |
|                                                                       |                    | deals_2019.01.dat                                   | 2019-11-01 5:10 PM |
|                                                                       |                    | deals_2019.02.dat                                   | 2019-11-02 5:10 PM |
|                                                                       |                    | deals_2019.03.dat                                   | 2019-11-03 5:10 PM |
|                                                                       |                    | deals_2019.04.dat                                   | 2019-11-04 5:10 PM |
|                                                                       |                    | deals_2019.07.dat                                   | 2019-11-07 5:10 PM |
|                                                                       |                    | deals_2019.08.dat                                   | 2019-11-08 5:10 PM |
|                                                                       |                    | deals_2019.11.dat                                   | 2019-11-11 5:10 PM |

It is not recommended to restore trading databases separately. Orders, deals and positions are interrelated and should match each other.

## Moving the current day logs

The backup server is a copy of the backed up one since it stores the same data and features the same file and directory structure. In particular, it stores its logs in directories and files with the same names (Logs\\*.log)

The special mechanism prevents mixing of the both servers' current day logs when switching to the backup server.

When switching the servers, the current day log file of the previous backed up server is renamed by adding the .failover extension to its name. The backup server to replace it starts keeping the current day log from scratch. The \*.failover file is generated only at the previous backed up server. The server, to which the switching is performed, simply creates its log file from scratch.

If a reverse switching is performed the same day, the .failover file is renamed back to .log and the backup server goes on keeping the log in it.

Let's consider an example of switching the main trade server to the backup one and vice versa.

## Contents of \MainTrade\logs\

|                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Current day log: 20170330.log<br><br>first entry: 10:56:16.265 Startup service start initialized<br>last entry: 10:57:55.421 Exit shutdown finished                                                                                                                                                                                                                                                  | Current da<br><br>first entry<br>last entry:                                                    |
| Switching the servers, the backup server is active here now                                                                                                                                                                                                                                                                                                                                          | Switching                                                                                       |
| The previous trade server log is renamed to 20170330.log.failover:<br><br>first entry: 10:56:16.265 Startup service start initialized<br>last entry: 10:57:55.421 Exit shutdown finished<br><br>The new file of the backup server log: 20170330.log<br><br>first entry: 10:57:52.507 Startup service start initialized<br>last entry: 10:57:55.401 192.168.0.131 Config: network config synchronized | The new fi<br><br>first entry<br>last entry:                                                    |
| Reverse switching, the trade server is active here again                                                                                                                                                                                                                                                                                                                                             | Reverse sv                                                                                      |
| 20170330.log.failover is renamed back to 20170330.log and the new trade server entries are added to it:<br><br>first entry: 10:56:16.265 Startup service start initialized<br>10:57:55.421 Exit shutdown finished<br>11:02:25.454 Startup service start initialized<br>last entry: 11:04:01.157 192.168.0.131 Config: network config synchronized                                                    | The previo<br><br>first entry<br>last entry:<br><br>The backu<br><br>first entry<br>last entry: |

### Emergency Situations During the Backup Process

Creation of backup copies by the server can be tracked by its journal. This section contains examples of journal entries describing emergency situations during backup.

**2012.06.13 17:19:57 TradeOrders: invalid index header timestamp**

Such a journal entry means that orders base index (order.idx) does not correspond to the orders base data (order.dat). A possible reason for such entry may be a backup server abnormal shutdown (for example, power cut or server manual shutdown).

This situation is not dangerous. The backup server will reconstruct the index and continue its operation. The data in the database remains fully intact. However, the reason of the abnormal server shutdown should be found out.

**2012.06.13 17:19:57 TradeDeals: deals\_2012.06.dat: base has invalid hash**

This entry indicates that the deals base on a backup server differs from the one on a trading server after synchronization. In this case, the backup server will remove the indicated base and synchronize it again.

## MetaTrader 5 trading platform

Such backup server journal entries are permissible for deals base (deals\_\*) after weekends (when databases optimization compaction are performed on the main server). The reason is that a backup server cannot always synchronize a month deals base in saving mode (delete the entries that were deleted during a trading server optimization) after optimizing the base on a trading server. In that case, the backup server performs the full database synchronization.

For the rest of the databases such a situation is not acceptable in any time.



MetaTrader 5 Trading Platform / Platform Components / Backup Server /  
Switching to Backup Server

Switching to Backup Server

The MetaTrader 5 platform can automatically monitor the availability of trade and history servers. The monitoring is performed by the backup server itself as well as by access servers. Each server monitors the availability of the main server and polls other monitoring servers to check whether the main server is available to them.

If the main server is unavailable for some time, the platform will automatically switch to the backup server. The downtime is minimal, while switching usually takes less than a minute.

In the platform, you can also switch to a backup server manually. In case of a failure of the history server or a non-main trade server, you can quickly switch to the backup server in the automated mode. The same procedure provides for an easy migration of servers to new hardware. You will only need to properly setup the backup server via MetaTrader 5 Administrator, use the fast deployment procedure and switch to the newly installed backup server afterwards.

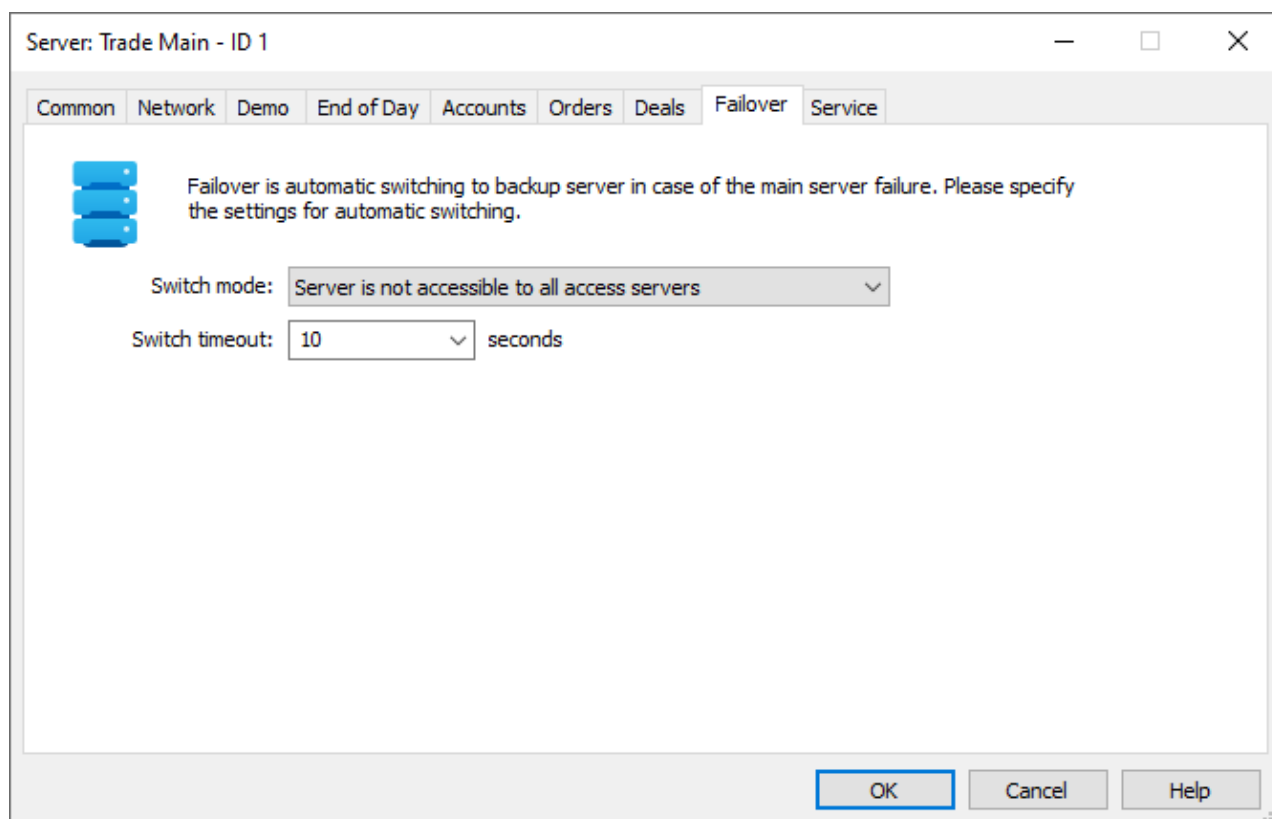
All critical trade and history server data is backed up in real time. Some non-critical trade server data, as well as history server data are backed up every hour. When switching to the backup server, critical data obtained during the procedure may be lost (the procedure itself usually takes less than a minute). Therefore, we strongly recommend that you switch to the backup server only outside of working hours.

Switching to the Backup Server Automatically

Automatic switching to a backup server allows minimizing the platform unavailability time in case of emergency situations. The platform automatically monitors the performance of its components and switches to backup servers if necessary.

The necessity to switch to the backup server is defined by the monitoring ("witness") servers. The backup server itself and access servers (with monitoring mode enabled) act as the monitoring ones. The backup server monitors the availability of the master server in real time mode and checks if it is available for the access servers as well.

Automatic switching can be enabled in the master server's settings (either a trade or a history one):



There are two scenarios for determining the unavailability of the main server:

- Server is not accessible to most access servers – the number of the monitoring servers unable to access the master server should exceed the ones able to access it at least by one for the switch to occur. If you have configured five monitoring servers, the main server should be unavailable to at least three of them, or to four of six monitoring servers. If you are using two monitoring servers (the minimum allowed number), the main server should be unavailable for both of them.
- Server is not accessible to all access servers – the master server should be unavailable for all monitoring servers for the switch to occur. If you have configured five monitoring servers, the main server should be unavailable to all of them.

If several backup servers are used for one main server, then each of the backup servers will start switching to the master server mode in case of the main server failure. The last switched server will be used as the main server, while all the rest of them will switch back to backup mode.

If a certain backup server should not be used for automatic switching, disable in its settings the following option: "Use this backup server for failover". This may be needed if the backup server is used only for exporting data to SQL database.

- The number of monitoring servers should not be less than 2.

- It is recommended to install access servers on different computers (different data centers), separate from the main and history servers. This will provide the most relevant server monitoring data.

In "Switch timeout" parameter, you can specify the time (in seconds) during which the server should be unavailable for monitoring servers to start switching to the back-up server. Also, after this time period, other platform components start their attempts to connect to the access points of the current backup server (trying to connect to it as to the master one).

When measuring the time of the master server's unavailability, its cause is considered. In case of a manual restart of the server, the unavailability time is increased according to the time required for restart.

- Unexpected failure switching is performed after the specified timeout.
- The server or the system service or the operating system is restarted by the administrator switch time is increased by the time spent for restart. The minimum time is 30 seconds and the maximum time is 300 seconds.
- Server restart during update (LiveUpdate) switch time is increased by the time spent for restart. The minimum time is 180 seconds and the maximum time is 300 seconds.

To make an access server a monitoring one, enable "Use this server for monitoring the cluster and failover" option in the server's settings:

Server: Access Point 2 - ID 6

Common Network **Access** Permissions Servers Service

Access server is used as access point for connection of clients to trade server. Please specify its priority and antiflood control settings.

Priority: 0

☒ Enable antiflood control

Connections: 100 Errors: 10

Maximum news: 16384

☒ Use this server for monitoring the cluster and failover

OK Cancel Help

- If the access server is unavailable for the backup one, that does not mean that the master server is also unavailable. In this case, the number of monitoring servers is reduced.
- In case of the master and other servers' simultaneous failure, the master server is restored first.

### Features of Connecting to Monitoring Servers

The backup server uses the following algorithm for connecting to the monitoring servers:

- As soon as the main server becomes unavailable for the backup server, the backup server checks all monitoring servers one by one and tries to connect to them.
- First, the backup server tries to connect to the monitoring server via the local address if available. Connection to a local address is performed if listen addresses of the backup and access servers are located in 10.\*, 172.16.\* 172.31.\* or 192.168.\* subnet. The first three octets in their addresses should coincide. In that case, both servers are deemed to be located in a single subnet, and the backup server tries to connect directly to the listen address of the access server. Example: the backup server has 192.168.0.100:1951 listen address, while the access one - 192.168.0.105:1950.
- If connection via the local address has failed, the backup server uses public points of the access server.

In order for the switch to occur as fast as possible, all access servers should be available. There should be no disabled servers among the monitoring ones. The backup server spends 5 seconds trying to connect to a non-existent address.

### Using Several Backup Servers

If several backup servers are used for a single main one, then each of the backup servers starts switching to the master server mode in case of the main server's failure. The last switched server is used as the master one, while all the rest of them switch back to backup server mode.

### Logging Monitoring Results

You can request the backup server's log using "Failover" keyword to control the process of monitoring the master server. Sample entries:

|                     |          |                                                                    |
|---------------------|----------|--------------------------------------------------------------------|
| 2013.09.09 09:07:33 | Failover | master server '1' - 'Trade Main' is available                      |
| 2013.09.09 09:07:33 | Failover | master server '1' - 'Trade Main' is available <b>for</b> witness   |
| 2013.09.09 09:07:34 | Failover | master server '1' - 'Trade Main' is available <b>for</b> witness   |
| 2013.09.09 09:07:54 | Failover | witness access server '7' is not available                         |
| 2013.09.09 09:07:54 | Failover | master server '1' - 'Trade Main' is available <b>for</b> witness   |
| 2013.09.09 09:07:54 | Failover | master server '1' - 'Trade Main' is available <b>for</b> 4 witness |

These entries mean as follows:

## MetaTrader 5 trading platform

- Master server with identifier 1 is available.
- Master server with identifier 1 is available for monitoring server 2 named Access Point 1.
- Master server with identifier 1 is available for monitoring server 6 named Access Point 2.
- Monitoring server with identifier 7 is unavailable for the backup server.
- Master server with identifier 1 is available for monitoring server 11 named Access Point 3.
- Master server is available for 4 witnesses and not available to 0 witnesses. The time, during which the server has been unavailable, is shown in brackets.

### Working after Switching to the Backup Server

After the backup server has switched to the trading one, the client terminals scan public access points of the access server in order to connect to it. The time of going through the access points depends on the following factors:

- Actual accessibility of the public point for a client. If an address is not available for the client (for example, a local IP address is specified in the settings as a public access point), the terminal spends 10 seconds trying to connect to it. An attempt to connect to the next access point is made only in 10 seconds.
- The number of the access server addresses unavailable for clients. For example, if two local addresses 192.168.0.100 and 192.168.0.101, as well as an external one - access.server.com (available external address of the server) are specified among the public access points, the client terminals will first spend 20 seconds trying to connect to local addresses before successfully connecting to the external one.
- Availability of the trade and history servers for the access one. The access server goes through the access points of the trade and history servers according to their network settings. Correctness of the network settings defines how quickly the access server becomes ready for work.

It is recommended that the access servers having local IP addresses in the list of public points are made available only for the administrators and managers working in the same local network. To do this, uncheck all options except "Allow administrator connection" and "Allow manager connection" ones in the Permissions tab of the access server.

### Manual Switching to a Backup Server and Migration of Servers

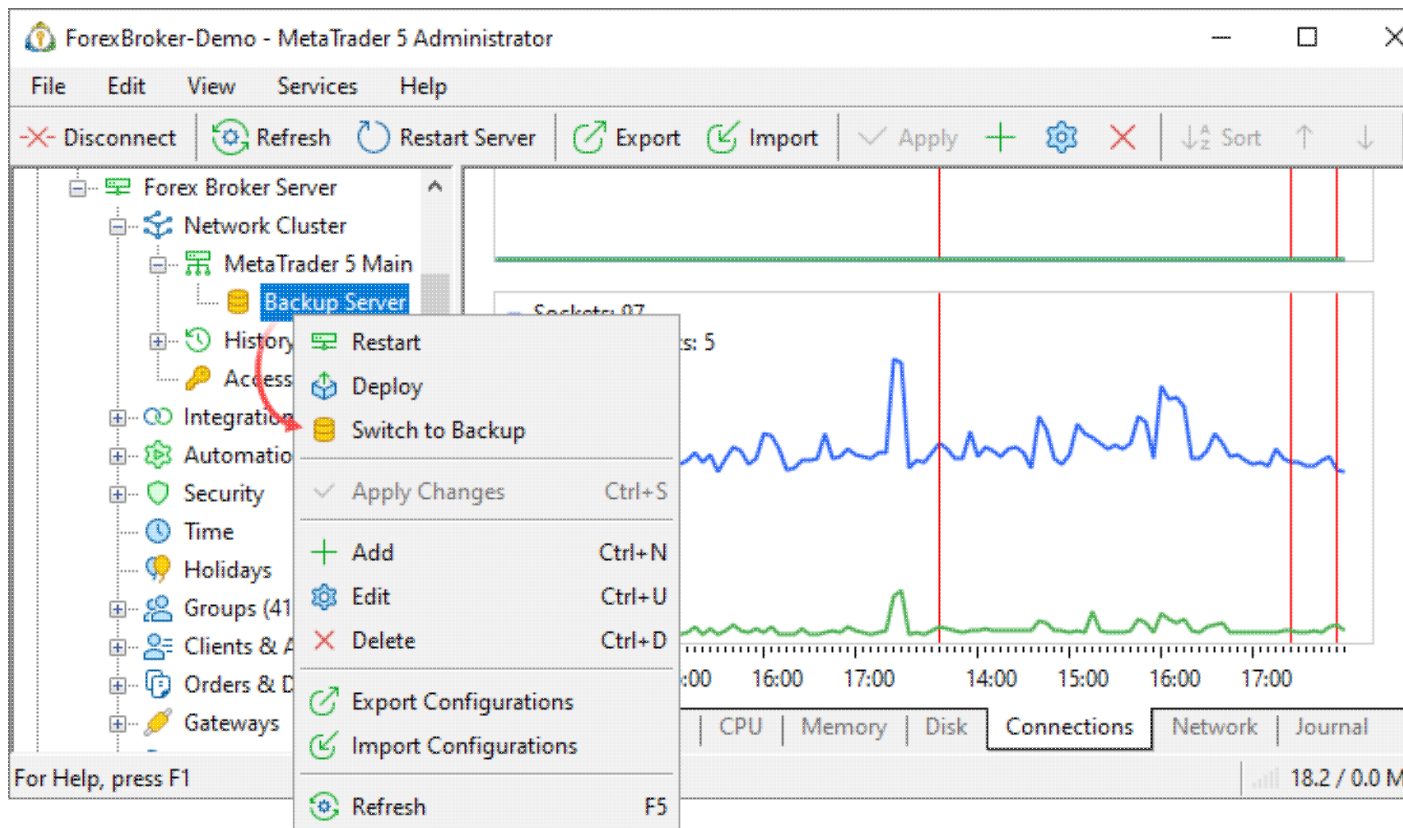
In the platform, it is possible switch to a backup server from the main one manually. It is a quick and automatic procedure. In addition to emergency cases, the procedure can be used for migrating servers to new hardware.

Install the backup server on the computer, to which you plan to migrate the server. After installing and launching the backup server, it is recommended to let it operate for a few days on the new machine to make sure it operates well.

## MetaTrader 5 trading platform

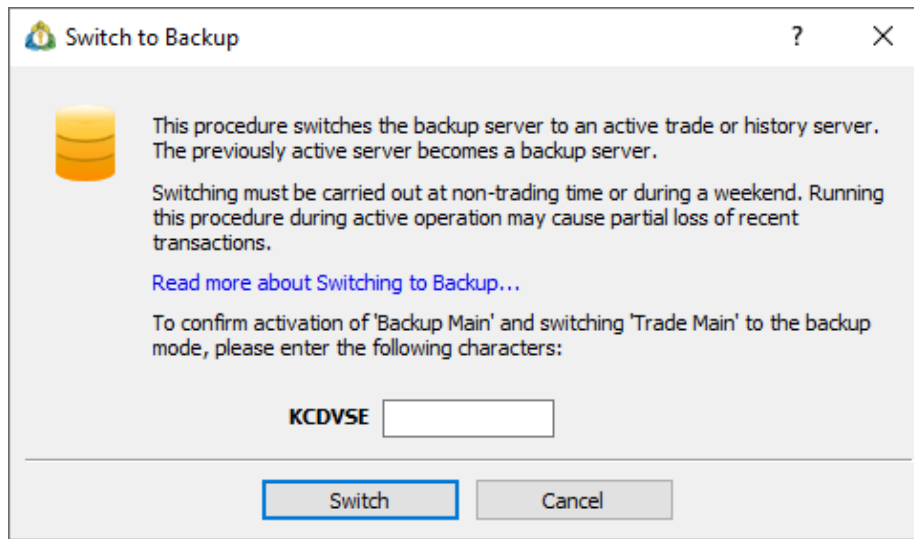
- If you want to avoid the loss of data that is backed up every hour, the backup server should be restarted before switching to it. The server creates the latest data backup. While the server is busy backing up data, the following sign is displayed on its icon 🔄 (the icon itself has the following look 🔄). Start switching to the backup server only after the backup process is complete.
- Server migration must only be performed in non-trading hours. During the switching procedure, the main server continues to receive data, which will not be copied to the backup server.
- In order to prevent important information from being lost, trading and changes in the client base are not allowed on the main server right after the start of switching to the backup server. The ban is valid for one minute. If the platform fails to switch to a backup server within this period, the ban is removed.

Execute "🔄 Switch to backup server" command:



## MetaTrader 5 trading platform

To avoid accidental switching, the platform requires an additional confirmation. In the dialog that appears, enter the required characters and click "Switch".



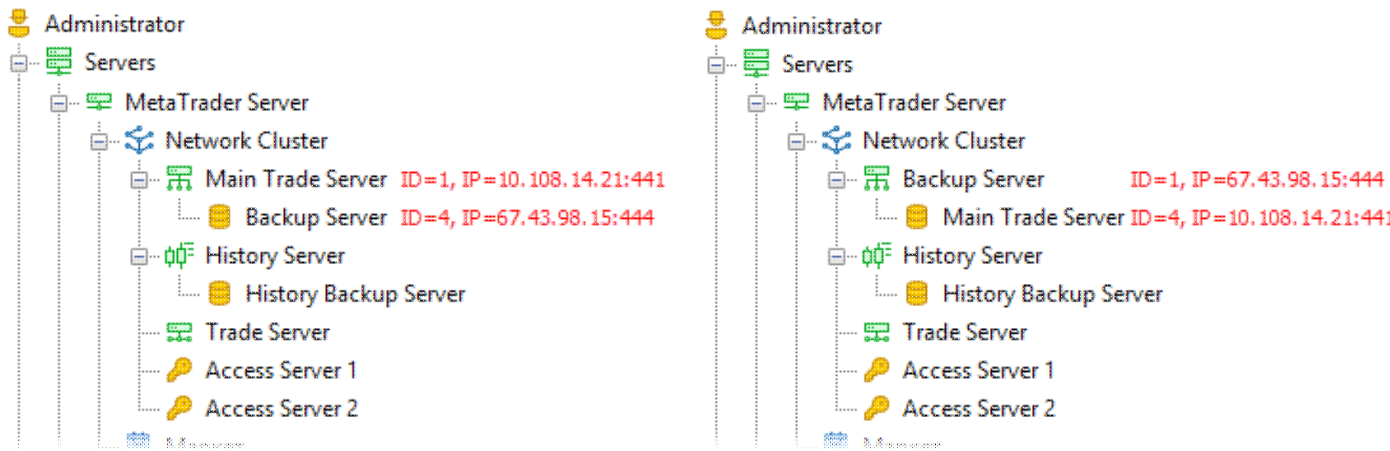
After the procedure is complete, you will see that the trade/history server has changed places with the backup server in Network section.

### The Features of the Switching Procedure

On the backup server's side, the switching process is performed as follows:

- The backup server stops Windows service.
- It also updates the network settings of the platform. The servers exchange roles, while the main server becomes the backup one, and the backup server starts operating as the main trade server:
  - ◆ The network settings (IP addresses, outgoing address, public points) and the name of the former backup server are set for the main server in the platform configuration.
  - ◆ The IP address and the name of the former main server are set for the backup server in the platform configuration.
  - ◆ On the servers, only their internal IDs change. The ID of the former main server is set for the new backup server, and the ID of the former backup server is set for the new main one.

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- The master server's Windows service is installed
- Notification is sent to the master server, waiting for confirmation.
- The master server's Windows service is launched.
- The former backup server's Windows service is deleted.

If the backup server has been active during the switch, it is switched to the backup mode:

- The master server's Windows service is stopped.
- Network settings are updated similar to how it is done for the backup server.
- The backup server's Windows service is installed and launched.
- The former master server's Windows service is deleted.

- If the former master server was inactive during the switch to the backup one (for example, the server computer was shut down) and the copy restored from backup is already working by the moment the server is restored, the former master server switches to the backup mode automatically (unless switching to a backup server is prohibited in its settings).
- Physically (on the hard drive), the new trade/history server operates from the former backup server's directory, while the new backup server operates from the trade/history server's one.



MetaTrader 5 Trading Platform / Platform Components / Backup Server / Restoring Server

## Restoring Server

If the main trade server fails, you will not be able to switch to the backup server manually using the MetaTrader 5 Administrator, because you will not be able to connect to the server. Managing other cluster components will also be impossible. The system of automatic switching to the backup server allows avoiding such a situation. However, if the system was not enabled, you will need to recover the server manually:

1. Go to the computer where the main trade server is installed, and stop the system service if it runs. The default name of the main trade server service is mt5tmsrv. It can be stopped in the Control Panel Administrative Tools Services, as well as using the command line:

```
net stop mt5tmsrv  
or  
D:\MetaTrader 5 Platform\Main Trade\mt5trade64.exe /stop.
```

If the computer is down (is unavailable), skip this step. However, once the computer is back up, please make sure that the old trade server service has not been restarted.

2. Go to the computer where the backup server is installed, and stop the system service. The default name of the backup server service is mt5bsrv. It can also be stopped using the Control Panel or the command line:

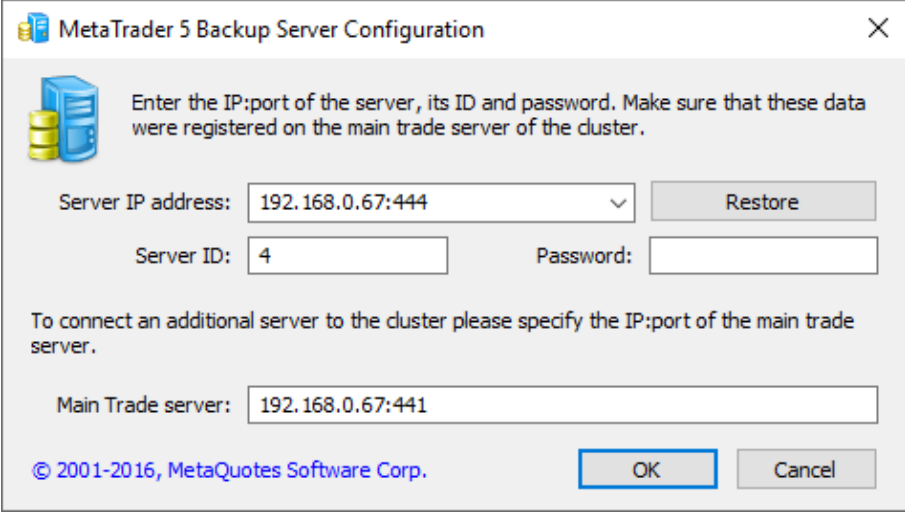
```
net stop mt5bsrv  
or  
D:\MetaTrader 5 Platform\Backup Main Trade\mt5backup64.exe /stop.
```

3. Run the backup server file from the command line with the /gui parameter. For example:

```
D:\MetaTrader 5 Platform\Backup Main Trade\mt5backup64.exe /gui.
```

Click Restore in the window that appears.

## MetaTrader 5 trading platform



MetaTrader 5 Backup Server Configuration

Enter the IP:port of the server, its ID and password. Make sure that these data were registered on the main trade server of the cluster.

Server IP address: 192.168.0.67:444 Restore

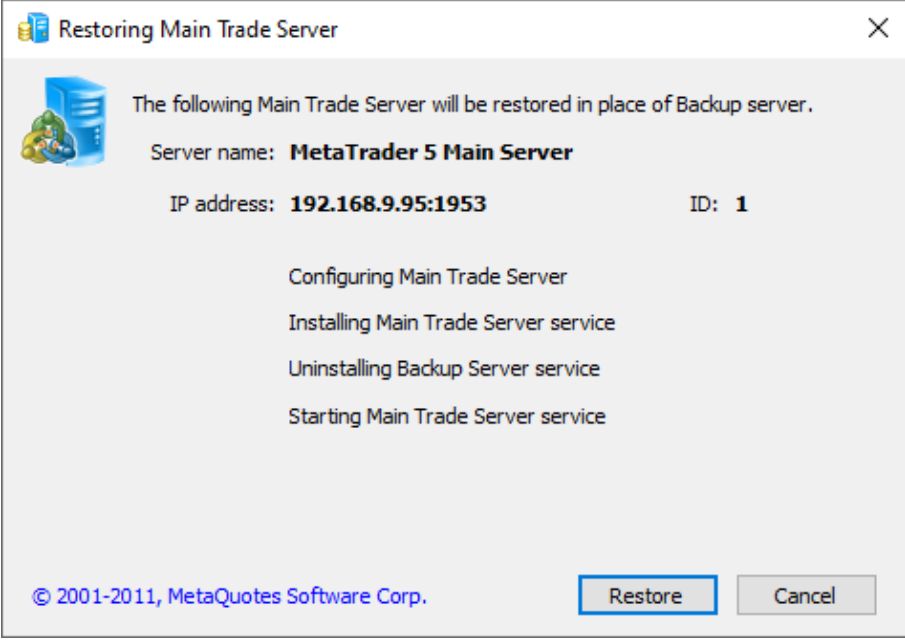
Server ID: 4 Password:

To connect an additional server to the cluster please specify the IP:port of the main trade server.

Main Trade server: 192.168.0.67:441

© 2001-2016, MetaQuotes Software Corp. OK Cancel

After that, the recovery process is started. Other components of the cluster will automatically switch to the new main server.



Restoring Main Trade Server

The following Main Trade Server will be restored in place of Backup server.

Server name: **MetaTrader 5 Main Server**

IP address: **192.168.9.95:1953** ID: **1**

Configuring Main Trade Server  
Installing Main Trade Server service  
Uninstalling Backup Server service  
Starting Main Trade Server service

© 2001-2011, MetaQuotes Software Corp. Restore Cancel

The process of restoring is run automatically in several steps:

- **Configuring Main Trader Server**

At this stage, the IP address and port of the main server are replaced with those of the backup server, which are displayed in the upper part of the window. The password and identifier are not changed.

- **Installing Main Trade Server service**

- **Uninstalling Backup Server service**

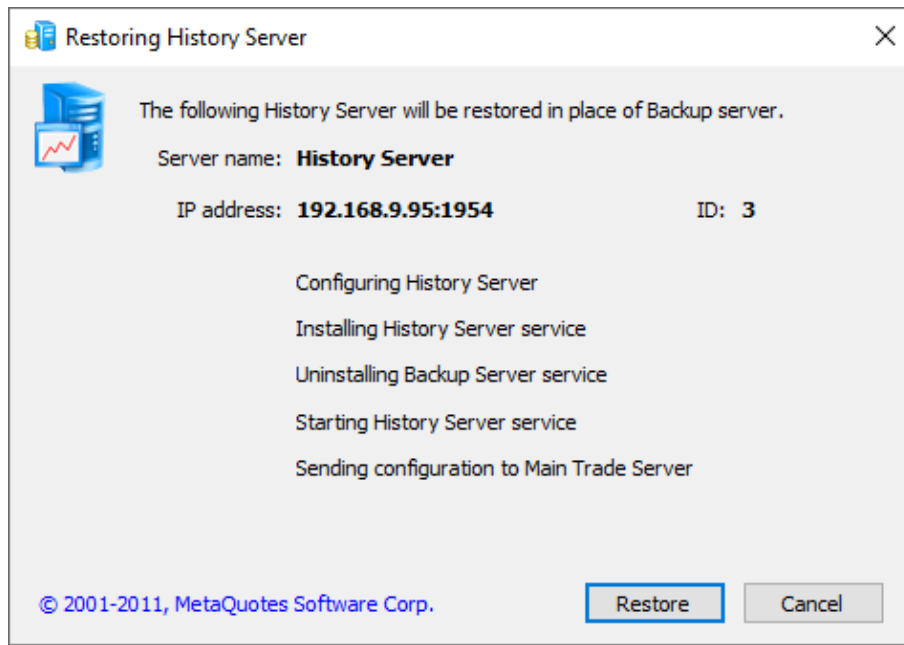
- **Starting Main Trade Server service**

- Before you launch the backup server, make sure the main server is stopped. Otherwise, your clients may start working with different servers. For example, your main server and access server are located at the same provider. The provider has issues with the internet connection and the servers became unavailable. The backup server is deployed in this case. After a while, connection to the main server is restored causing two servers to work simultaneously. In this case, contact your provider and request the immediate disabling of the main server.
- If one of the stages cannot be completed, all the changes made during restoring will be rolled back.
- As soon as restoring is finished, it is recommended to setup a new backup server.

After the recovery of the main trade server, you will be able to connect to the cluster via MetaTrader 5 Administrator. Other components can be switched to backup servers in a regular way through the interface. For example, if the history server was installed on the same machine, you can switch to the backup server by running the appropriate command in its context menu.

The described procedure can also be used to restore other servers, including additional trade servers and history server. When restoring servers, new configuration data will be sent to the main trade server. A restored server will connect to the cluster without additional settings, and you will be able to control it via MetaTrader 5 Administrator.

## MetaTrader 5 trading platform



MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export

## SQL Export

The MetaTrader 5 trading platform provides standard options for the real-time data export to MySQL, Microsoft SQL Server, FireBird, Oracle, MariaDB and PostgreSQL databases. This option enables the quick and easy deployment of data export to an external DBMS for using the platform data in any popular programming language and third-party applications. Thus, it is possible to create an intermediate layer between a trade server and broker's program services that regularly access trading data. This reduces the load on the trade server when it receives the current trading data.

The export function is enabled by simple specification of settings for connection to DBMS via MetaTrader 5 Administrator. After that, the backup server will immediately perform an initial synchronization of data from the DBMS. Further, new data will be exported to the DBMS in real time. The following data is exported:

- Information about clients (general information and trading status)
- Current active orders and positions
- The history of orders and deals
- Current prices
- Virtually all settings of the trading platform (except for the working time, synchronization and spreads)

The description of installation and setup of popular databases is provided in appropriate subsections:

- MySQL Server 5.7 (supported versions: 5.1, 5.5, 5.6, 5.7, 8.0, 8.1)
- MariaDB 10.2 (all version supported)
- Microsoft SQL Express 2012 (supported versions: 2005, 2008, 2008 R2, 2012, 2014, 2016, 2017, 2019, 2022)
- Oracle Database Express Edition 11g (supported versions: 11g/11g Express Edition, 12c, 18c, 19c, 21c, 23ai)
- PostgreSQL (supported versions: 8.4 17)

## Operation Principle

Export mechanism has been implemented on the side of backup servers replicating data from the trade servers.

Below are the general steps of synchronization in a backup server:

- The backup server connects to the appropriate trade server and is synchronized with it.
- After synchronization with the trade server is successfully complete, the backup server synchronizes an external DBMS.
- The backup server applies changes to its databases and the external DBMS via transactions from the trade server.

The backup server performs initial synchronization based on time stamps (Timestamp field in the tables) during each connection to DBMS:

- All logs with different time stamps in the external DBMS are replaced with backup server logs.
- All entries that are absent at the backup server are removed from the external DBMS.

A time stamp is used for checking the identity of the backup and the external DBMS logs. If a time stamp on the backup is similar to the one at the external DBMS, a backup server considers that all other log fields are similar and the log update is not required.

After databases are synchronized, the backup server applies change transactions on the external DBMS. Besides, prices and profit values for active orders (mt5\_orders), positions (mt5\_positions) and related trading accounts (mt5\_accounts) are also periodically updated.

A detailed description of exported tables can be found in the following subsections:

- **mt5\_symbols** symbols' configurations.
- **mt5\_symbols\_sessions** symbols' trade and quotation sessions.
- **mt5\_groups** groups' configurations.
- **mt5\_groups\_symbols** individual symbol settings for groups.
- **mt5\_commissions** commission settings for groups.
- **mt5\_commissions\_tiers** commission level settings.
- **mt5\_managers** manager accounts.
- **mt5\_clients** client database.
- **mt5\_documents** database of client documents.
- **mt5\_users** account database.
- **mt5\_orders** open order database.
- **mt5\_positions** position database.
- **mt5\_orders\_history** closed order database.
- **mt5\_deals** deal database.
- **mt5\_accounts** trade account state database.
- **mt5\_prices** database of prices.
- **mt5\_daily** database of daily reports.
  
- **mt5\_daily\_orders** data on the status of open orders at the end of the trading day.
- **mt5\_daily\_positions** data on the status of position at the end of the trading day.
  
- **mt5\_holidays** holiday configurations.
- **mt5\_network** general settings of servers.
- **mt5\_network\_access\_servers** access servers settings.
- **mt5\_network\_history\_servers** history server settings.
- **mt5\_network\_trade\_servers** trade servers settings.
- **mt5\_network\_backup\_servers** backup servers settings.
- **mt5\_network\_backup\_folders** backed up custom folders.
- **mt5\_firewall** firewall settings.

## MetaTrader 5 trading platform

- **mt5\_routing** settings of routing rules.
- **mt5\_routing\_dealers** settings of dealers/gateways in routing rules.
- **mt5\_routing\_conds** additional conditions in routing rules
- **mt5\_feeders** settings of data feeds.
- **mt5\_feeder\_translates** conversion settings on data feeds.
- **mt5\_feeder\_params** additional settings of data feeds.
  
- **mt5\_feeder\_symbols** symbol settings for data feeds.
  
- **mt5\_reports** report settings.
- **mt5\_report\_params** additional settings of reports.
- **mt5\_plugins** plugin settings
- **mt5\_plugin\_params** additional settings of plugins.
- **mt5\_time** platform trading time settings.
- **mt5\_time\_weekdays** working time schedule of the platform, by days.
- **mt5\_gateways** gateway settings.
- **mt5\_gateways\_params** additional gateway settings.
- **mt5\_gateways\_translates** translation settings in gateways.
  
- **mt5\_gateways\_symbols** symbol settings for gateways.

Your own data in MetaTrader 5 tables

In the platform, you can create your own tables and databases, add your own fields in mt5\_\* tables, which are used for data export, as well as create stored procedures and triggers.

- The backup server does not recreate data, but only adds or updates existing records. A table entry is only deleted if the appropriate entry is deleted on the platform side. For example, if a user is added to the mt5\_users tables, the appropriate user record will exist in the database until the user is deleted from MetaTrader 5.
- The backup server only works with its own tables and fields.
- The backup server does not create default indexes. Each index is an extra load on the database, which can slow down exports and information updates.
- When you create your own fields in mt5\_\* tables, do not forget to set default values for them (or allow NULL). Otherwise, the backup server will not be able to add new entries to the tables.

Additional information

Find additional recommendations on handling SQL databases in the articles:

- [Export to SQL database takes too much time](#)
- [How to present account permissions in MySQL](#)

MetaTrader 5 Trading Platform  
/ Platform Components /  
Backup Server / SQL Export /  
Installation and Setup of  
MySQL

Installation and Setup of MySQL Server

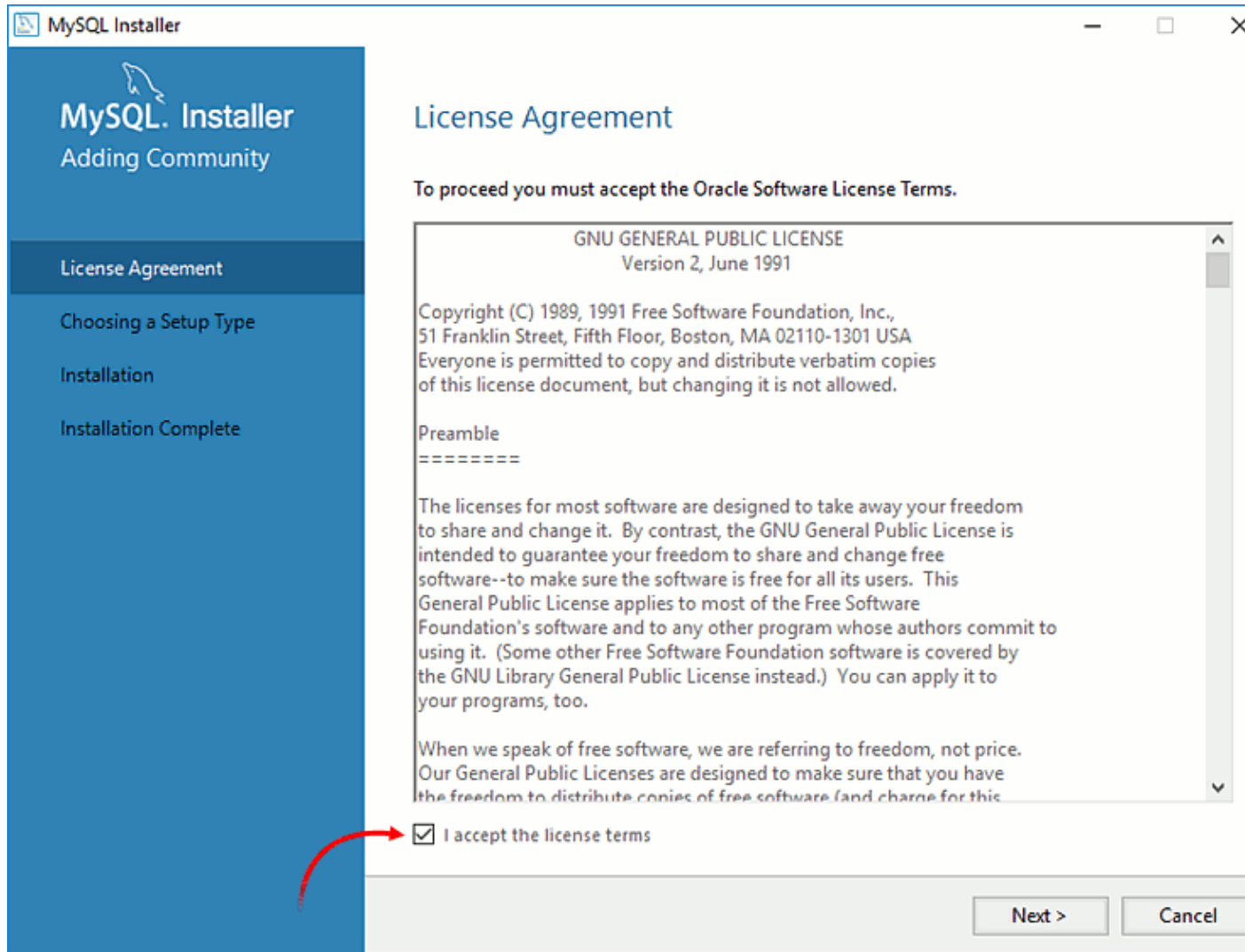
Download one of the distribution kits from MySQL website  
(<http://dev.mysql.com/downloads/mysql>) to start installation.

Select 32 or 64-bit considering potential volumes of databases. 64-bit server version is more preferable, as it makes possible to have data caches of more than 2 Gb and has better scalability due to increasing RAM on the server.

Downloaded distribution kit should be installed in accordance with the instructions:

- at the first step, agree to the terms of use and click "Next"
- at the second step, select the installation type ("Server only" is recommended)
- at the third step, click "Execute" and wait for the installation to complete
- at the fourth and fifth steps, click "Next" to go to the MySQL configuration

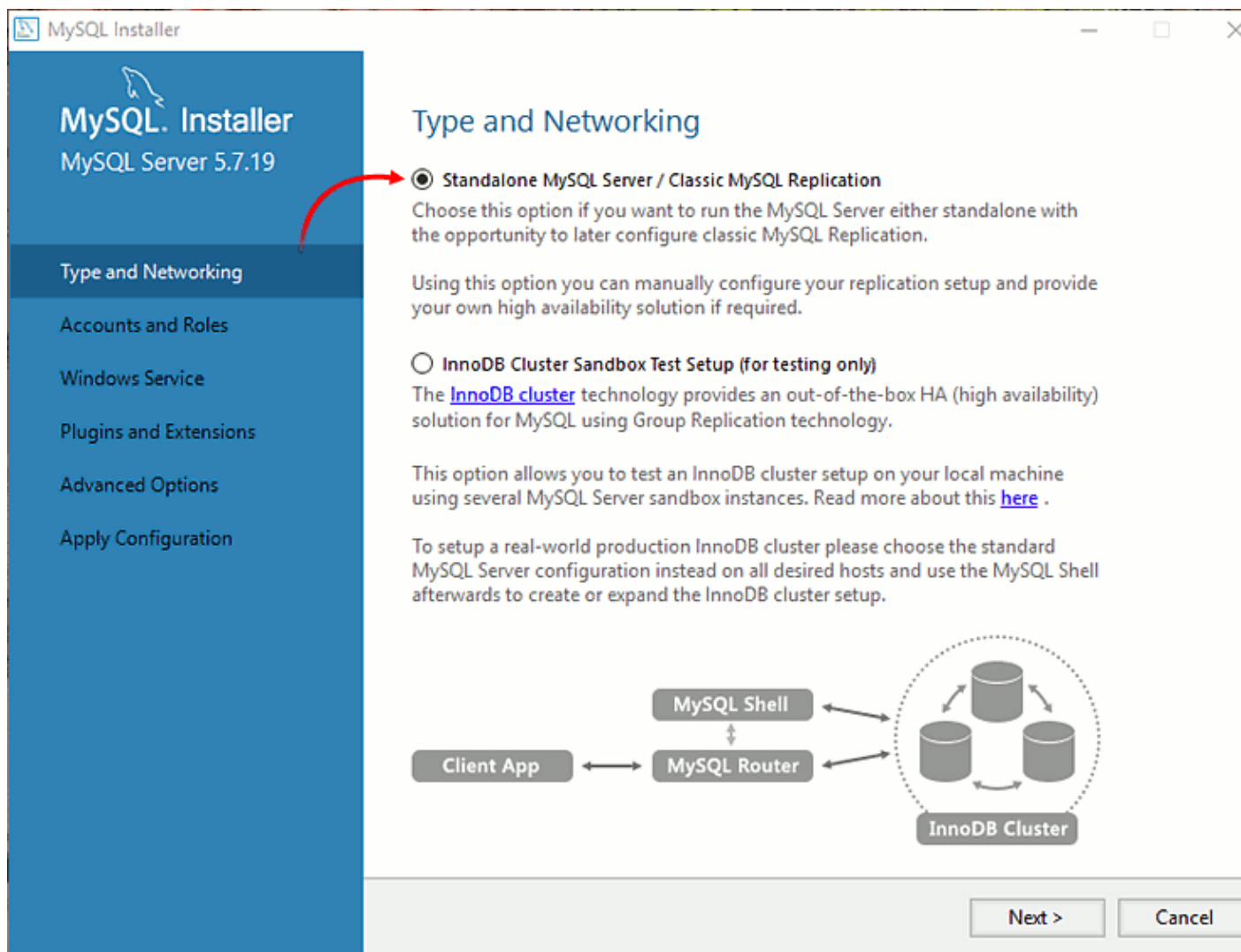




During the first configuration step, select the installation type "Standalone MySQL Server /Classic MySQL Replication". Next, select the type of the computer on which MySQL is installed. Based on this information, the configuration Wizard sets efficient parameters for memory, hard drive and CPU consumption. Developer Machine should not be used in real work, therefore, select one of the following types:

- Server Machine is used in case other services (for example, web server) are present on the computer. In this case, MySQL will consume a "moderate" amount of memory.
- Dedicated MySQL Server Machine should be selected if the computer will be used exclusively for MySQL server. In this case, MySQL will consume "maximum available" amount of memory for faster performance.

In addition to the computer type, ports and communication channels can also be configured from here, although you can use default values for these parameters.



At the next step, specify the password for the root login (administrator account). Here you can also create additional accounts and configure permissions for them.

MySQL Installer  
MySQL Server 5.7.19

Type and Networking  
**Accounts and Roles**  
Windows Service  
Plugins and Extensions  
Apply Configuration

### Accounts and Roles

**Root Account Password**  
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password Strength: **Strong**

Please specify the username, password, and database role.

 Username:

Host:

Role:

Authentication: ☒ MySQL

Password:

Confirm Password:

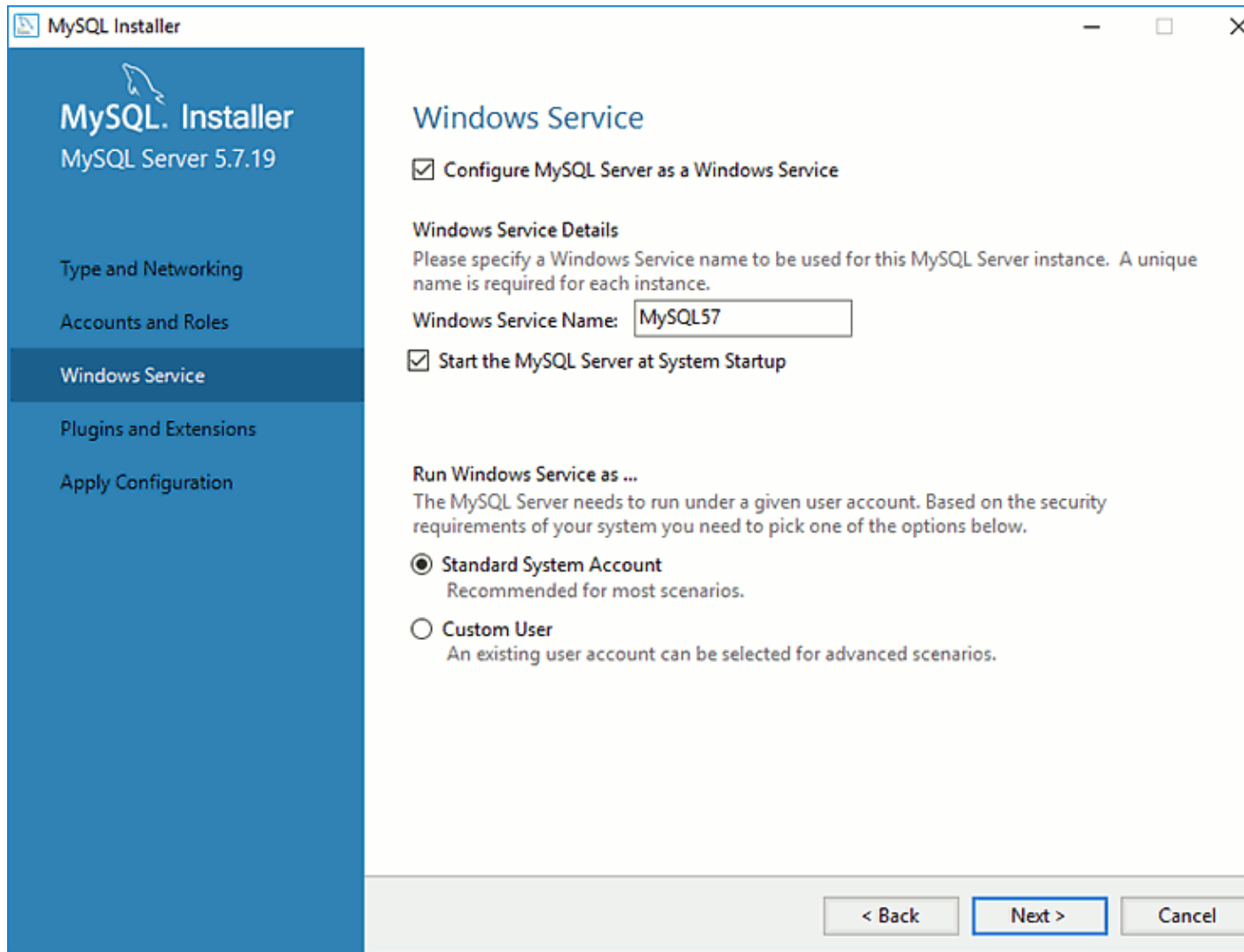
Password Strength: **Strong**

**Add User**  
Edit User  
Delete

< Back   Next >   Cancel

## MetaTrader 5 trading platform

Further you will be asked to configure the integration of the MySQL server with the operating system, as well as the server extensions. You can use default settings for the above parameters. At the last step, click "Execute" and wait for the configuration to complete.



To increase MySQL server performance, edit the my.ini file located in the MySQL installation directory, i.e. C:\Program Files\MySQL\MySQL Server 5.7\my.ini for the 64-bit version and C:\Program Files (x86)\MySQL\MySQL Server 5.7\my.ini for the 32-bit version. The file can also be located in C:\Program Data\MySQL\MySQL Server 5.7.

```
# InnoDB, unlike MyISAM, uses a buffer pool to cache both indexes and  
# row data. The bigger you set this the less disk I/O is needed to  
# access data in tables. On a dedicated database server you may set this  
# parameter up to 80% of the machine physical memory size. Do not set it  
# too large, though, because competition of the physical memory may  
# cause paging in the operating system. Note that on 32bit systems you  
# might be limited to 2-3.5G of user level memory per process, so do not  
# set it too high.  
innodb_buffer_pool_size=2046M
```

This parameter should be selected considering the server's bit count and available physical memory:

- for a 32-bit server this value should not exceed 80 % of physical memory and it also should not exceed 2046;
- for a 64-bit server this value should not exceed 80 % of physical memory.

When exporting data, the platform uses zero date if a date is not specified. Thus in my.ini you need to turn off the SQL modes that prohibit zero dates: NO\_ZERO\_IN\_DATE and NO\_ZERO\_DATE. Find the "sql\_mode" parameter and delete that modes from it:

```
sql_mode = "STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_D"
```

The last step is creation of the schema, to which MySQL data is exported, and performing configuration on the side of MetaTrader 5.

To do this, launch MySQL 5.7 Command Line Client in Start - All Programs - MySQL - MySQL Server 5.7 and enter root password. The following script should be executed after that:

```
create schema schema_name;
```

For example:

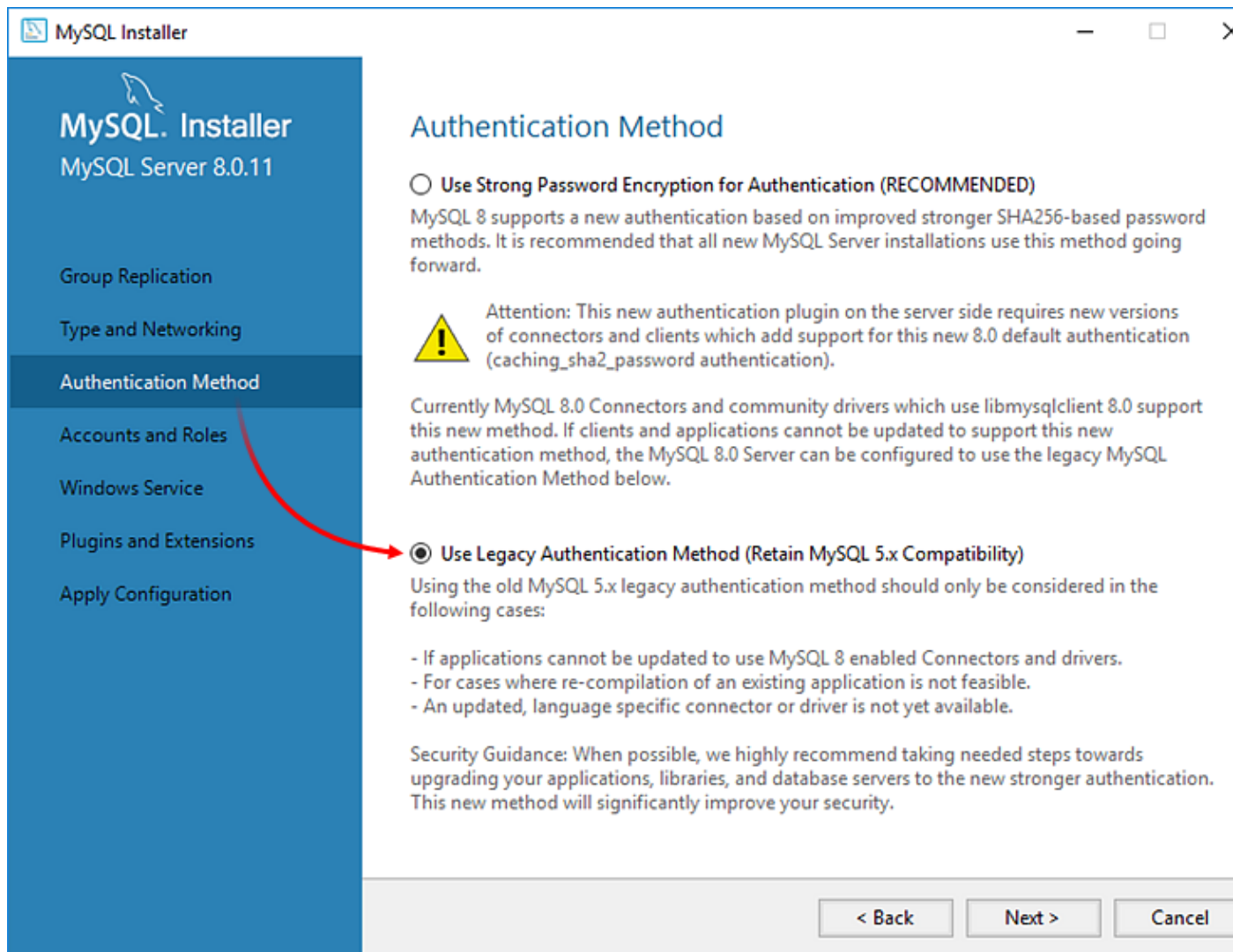
```
create schema metatrader5;
```

Installation of MySQL version 8.0 and above

MySQL versions 8.0 and higher support new authentication type, which includes an improved SHA256 encryption of passwords, additional caching on the server side and a number of other new functions. The previous authentication version (Legacy) is also supported, but the new one is used by default.

The new authentication version is not supported in MetaTrader 5. To enable export of platform data to MySQL version 8.0 or higher, choose the previous authentication

method "Use Legacy Authentication Method" during installation:



## MetaTrader 5 trading platform

If MySQL is already installed with the new authentication method, you may switch to the Legacy method as described below. Open my.ini in C:\ProgramData\MySQL\MySQL Server 8.0\ and find the following line:

```
default_authentication_plugin=caching_sha2_password
```

Change the values of the parameter:

```
default_authentication_plugin=mysql_native_password
```

Next, change the authentication type for the user, under which you are connecting to the database. Open 'Command Line Client' and run the following command:

```
ALTER USER 'username'@'localhost' IDENTIFIED WITH mysql_native_password BY 'password'
```

Here 'username' is the name of the database user, and password is the new password for the user.

After that you will be able to connect to the database using the Legacy authentication method and to export data from MetaTrader 5 into it.

Configuration on MetaTrader 5 side

In case backup and MySQL servers are installed on the same computer, connection settings will be as follows:

The screenshot shows the 'Server' dialog box with the 'SQL Export' tab selected. The dialog contains the following fields and options:

- Type:** A dropdown menu showing 'MySQL 5.x & MariaDB 5.x - 10.x'.
- Server:** A text field containing '127.0.0.1'.
- Login:** A text field containing 'root'.
- Password:** A text field with masked characters (dots).
- Data Folder:** A text field containing 'metatrader5'.

At the bottom of the dialog, there are three buttons: 'OK' (highlighted with a blue border), 'Cancel', and 'Help'.

In case backup and MySQL servers are located on different computers:

Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: MySQL 5.x & MariaDB 5.x - 10.x

Server: 192.168.9.150

Login: root Password: .....

Data Folder: metatrader5

OK Cancel Help

It is recommended to install MySQL and backup servers on the same computer for faster export. It should also be kept in mind that the backup server itself also consumes resources. Also, remember that it is possible to deploy several backups for a trade server. These backups will work close to necessary DBMS.

Additional export parameters can be selected at SQL Options tab.

After all settings are specified, the backup server will immediately start export into the external DBMS. Request the journal of the appropriate backup server using SQL keyword to track the export process:

```
19:00:16 SQL Export MySQL: successfully connect to '127.0.0.1:metatrader5' for 'root'
19:00:16 SQL Export MySQL: server version 5.7.19
19:00:16 SQL Export MySQL: table 'mt5_symbols' creating started
19:00:16 SQL Export MySQL: table 'mt5_symbols' created
19:00:16 SQL Export MySQL: primary key for 'mt5_symbols' adding started
19:00:16 SQL Export MySQL: primary key for 'mt5_symbols' added
19:00:16 SQL Export MySQL: table 'mt5_symbols_sessions' creating started
19:00:16 SQL Export MySQL: table 'mt5_symbols_sessions' created
19:00:16 SQL Export MySQL: primary key for 'mt5_symbols_sessions' adding started
19:00:16 SQL Export MySQL: primary key for 'mt5_symbols_sessions' added
19:00:16 SQL Export Symbols: 0 symbols and 0 sessions loaded in 312 msecs
19:00:16 SQL Export Symbols: synchronization started
...
```



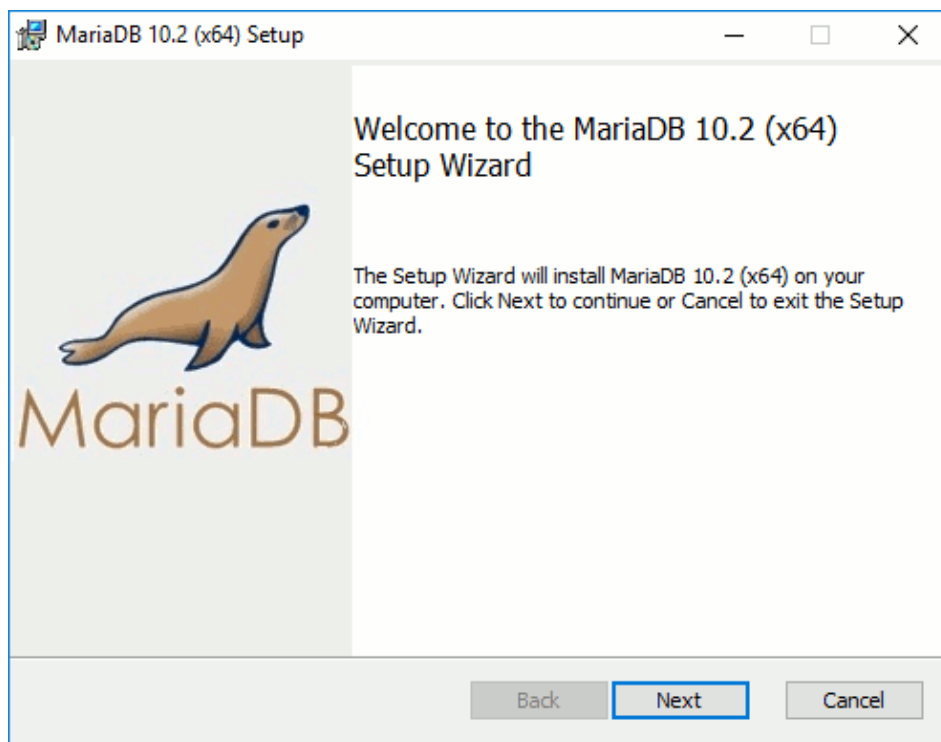
MetaTrader 5 Trading Platform  
/ Platform Components /  
Backup Server / SQL Export /  
Installation and Setup of  
MariaDB

Installation and Setup of MariaDB

To install MariaDB, download the required version from the official product website <https://downloads.mariadb.org/>. At the time, the platform supports all available MariaDB versions.

Choose between 32-bit and 64-bit versions based on the evaluated potential volume of data bases. The 64-bit server version is more preferable, as it makes possible to have data caches of more than 2 Gb and has better scalability due to increasing RAM on the server.

Run the installation file and follow the on-screen instructions. The below example shows the installation of MariaDB 10.2 (x64).

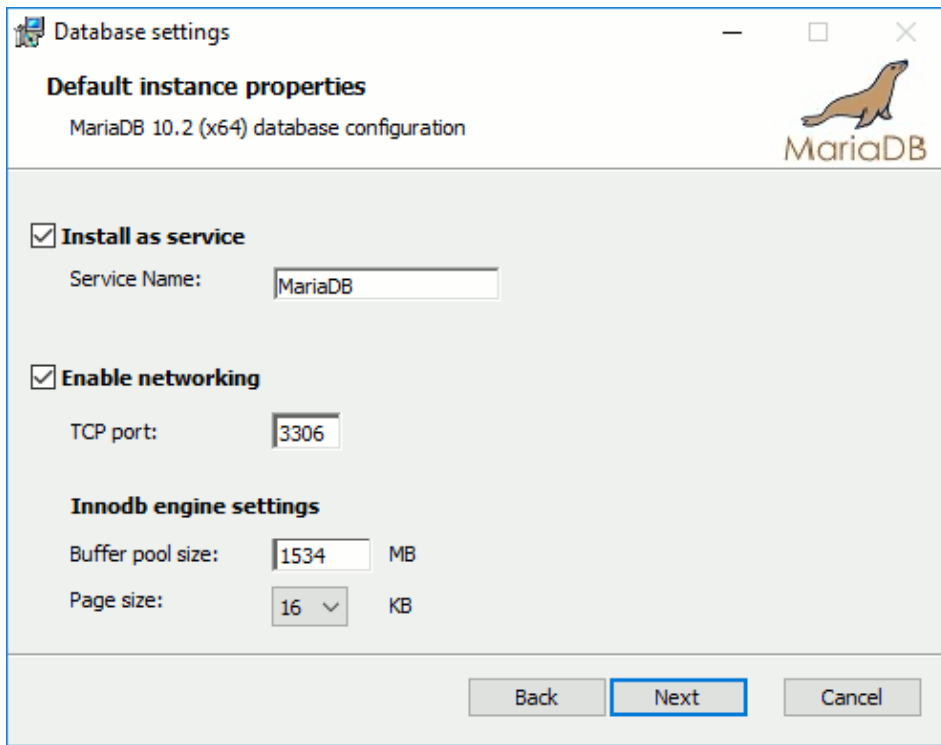


## MetaTrader 5 trading platform

After selecting components for the installation (you can install with default settings), specify the password for the "root" administrator account. For security reasons, it is recommended that you do not allow the use of the "root" account during remote connections (leave the "Enable root access from remote machines" option disabled). Instead, after installing the MariaDB server and creating a scheme (database) for working with MetaTrader 5, create a separate account with access only to the desired schema, and allow remote connections to it.

It is also recommended to enable the "Use UTF8 as default server's character set" option.

Then, specify network settings (default settings can be used) and the parameter for sending anonymous reports on the DBMS operation (can be disabled). After that, start the installation process and wait for it to finish.



**Database settings**

**Default instance properties**  
MariaDB 10.2 (x64) database configuration

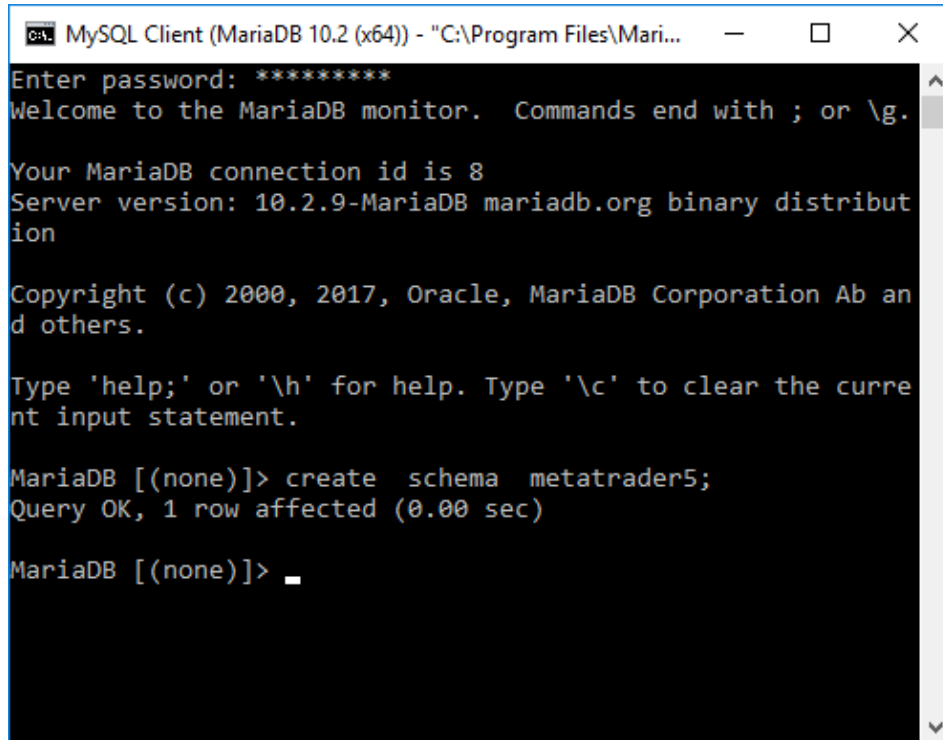
☒ **Install as service**  
Service Name:

☒ **Enable networking**  
TCP port:

**InnoDB engine settings**  
Buffer pool size:  MB  
Page size:  KB

## MetaTrader 5 trading platform

Now create a schema (a database) to which data from MetaTrader 5 will be exported. To do this, launch the MySQL Client (MariaDB) program from the Start menu. In the window that appears, enter the password for the "root" account and press Enter. Then enter a command to create a data base, e.g. "create schema metatrader5;". Here "metatrader5" is the name of the schema. You can use any other name.

A screenshot of a Windows command prompt window titled "MySQL Client (MariaDB 10.2 (x64)) - 'C:\Program Files\Mari...". The window has a black background with white text. It shows the login process for MariaDB, including a password prompt, a welcome message, connection ID, server version, and copyright information. The user enters the command "create schema metatrader5;" and receives the response "Query OK, 1 row affected (0.00 sec)". The prompt then returns to "MariaDB [(none)]>".

```
MySQL Client (MariaDB 10.2 (x64)) - "C:\Program Files\Mari...
Enter password: *****
Welcome to the MariaDB monitor.  Commands end with ; or \g.

Your MariaDB connection id is 8
Server version: 10.2.9-MariaDB mariadb.org binary distribut
ion

Copyright (c) 2000, 2017, Oracle, MariaDB Corporation Ab an
d others.

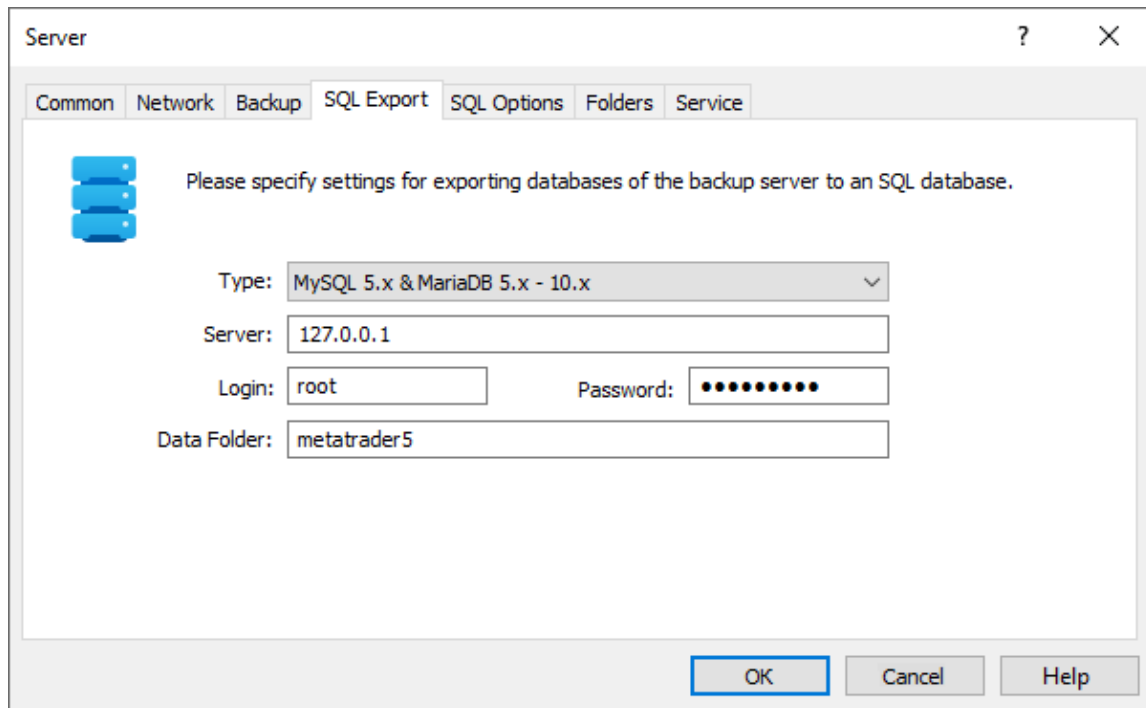
Type 'help;' or '\h' for help. Type '\c' to clear the curre
nt input statement.

MariaDB [(none)]> create schema metatrader5;
Query OK, 1 row affected (0.00 sec)

MariaDB [(none)]> _
```

Next, configure data export on the MetaTrader 5 side. If the backup server and the MariaDB server are installed on the same computer, the following settings will be used:

## MetaTrader 5 trading platform



Server

Common Network Backup SQL Export SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: MySQL 5.x & MariaDB 5.x - 10.x

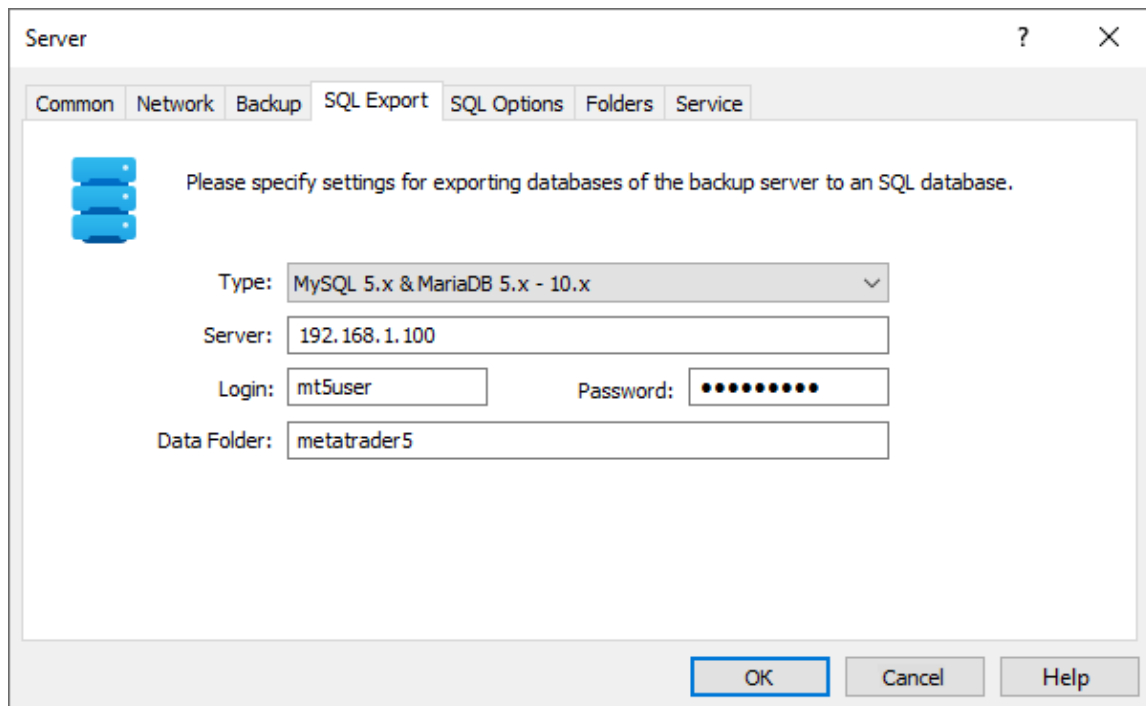
Server: 127.0.0.1

Login: root Password: ●●●●●●●●

Data Folder: metatrader5

OK Cancel Help

If the database is installed separately from the backup server, specify its address in the "Server" field. In the Login and Password fields, specify the details of the account that has access to the required schema and is allowed for use during remote connection.



Server

Common Network Backup SQL Export SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: MySQL 5.x & MariaDB 5.x - 10.x

Server: 192.168.1.100

Login: mt5user Password: ●●●●●●●●

Data Folder: metatrader5

OK Cancel Help

It is recommended to install MariaDB and the backup server on the same computer for faster export. However, it should also be kept in mind that the backup server consumes resources. Also, remember that it is possible to deploy several backups for a trade server. These backup servers will run close to necessary DBMS.

Additional export parameters can be selected at [SQL Options](#) tab.

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export / Installation and Setup of MS SQL

Installation and Setup of Microsoft SQL Server

Download one of the distribution kits from Microsoft website (<http://www.microsoft.com/en-us/download/details.aspx?id=29062>) to install free version of Microsoft SQL Express 2012.

You need to download ENU\x86\SQLEXPRESS\_x86\_ENU.exe or ENU\x64\SQLEXPRESS\_x64\_ENU.exe.

- MetaTrader 5 supports data export both to Microsoft SQL Server 2005/2008/2012/2014/2016/2017/2019/2022 and to their Express versions.
- Microsoft SQL Server Express 2005 has a limit of 4 GB on user data. Microsoft SQL Server Express 2008 R2 / 2012 has a limit of 10 GB. Therefore, it is recommended to use 2008 R2 or 2012 as Express version. If the limit is exceeded, new data will not be added to the database.
- Operating system Windows 8 or higher is required for the Microsoft SQL 2016 installation.
- Select 32 or 64-bit considering potential volumes of databases. 64-bit server version is more preferable, as it makes possible to have data caches of bigger volume and has better scalability due to increasing RAM on the server.
- By default, the backup server connects to the database using the Windows system driver MS OLE DB Provider for SQL Server (SQLOLEDB). This driver is pre-installed in all operating systems, however it is outdated and does not support the TLS 1.1 and TLS 1.2 protocols. If your company's policy does not allow using TLS protocols below 1.1. and they are disabled in the system, the SQLOLEDB driver will not be able to provide backup server connection to the database. In this case you should install an additional driver MSOLEDBSQL, which supports protocols TLS 1.1 and 1.2. After that the backup server will automatically start using the new driver for connecting to the database.

Downloaded distribution kit should be installed in accordance with the instructions. First, select "New SQL Server stand-alone installation or add features to an existing installations" to install a new copy of Microsoft SQL Server. Check the "I accept the license terms" option at "License Terms" step. Leave default parameters at the following steps up to "Server Configuration". Set Automatic mode for "SQL Server Browser" at "Server Configuration" step:

## MetaTrader 5 trading platform

The screenshot shows the 'Service Accounts' tab of the SQL Server Enterprise Setup wizard. It includes a recommendation from Microsoft and a table for configuring services.

Microsoft recommends that you use a separate account for each SQL Server service.

| Service                    | Account Name             | Password | Startup Type |
|----------------------------|--------------------------|----------|--------------|
| SQL Server Database Engine | NT Service\MSSQL\$SQL... |          | Automatic    |
| SQL Server Browser         | NT AUTHORITY\LOCAL ...   |          | Automatic    |

Select "Mixed Mode (SQL Server authentication and Windows authentication)" and specify the administrator password at "Database Engine Configuration" step:

The screenshot shows the 'Server Configuration' tab of the SQL Server Enterprise Setup wizard. It guides the user through setting the authentication mode and administrator password.

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

Enter password: [password field]

Confirm password: [password field]

Specify SQL Server administrators

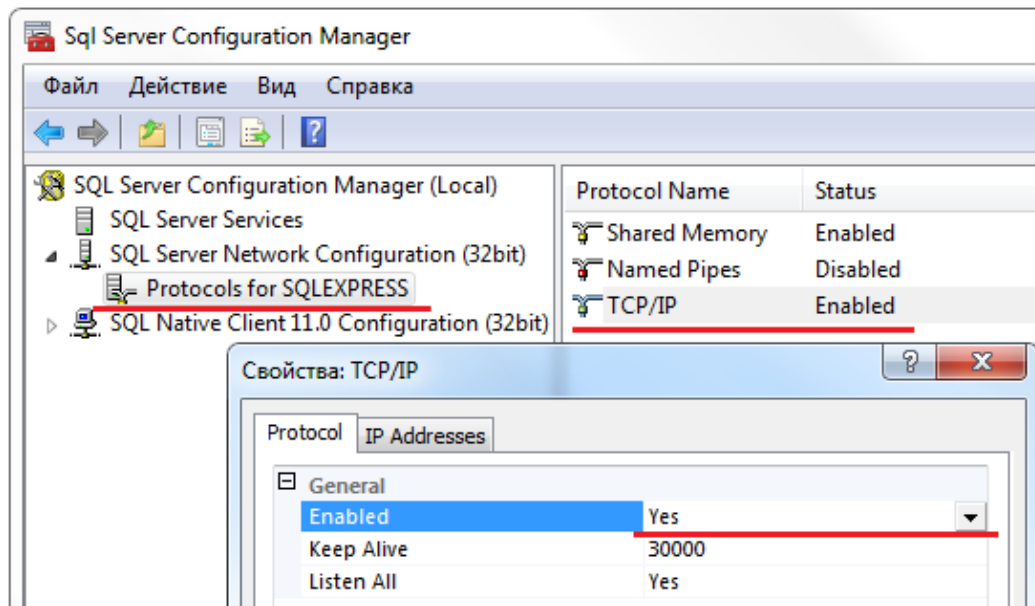
Anton (Anton)

SQL Server administrators have unrestricted access to the Database Engine.

Buttons: Add Current User, Add..., Remove

Leave default parameters at the following steps.

TCP/IP should also be enabled for access to SQL Server from the network. To do this, launch "SQL Server Configuration Manager" from the start menu:



Restart SQL Server (SQLEXPRESS) service in SQL Server Services section.

Launch SQLCMD.exe application ("C:\Program Files (x86)\Microsoft SQL Server\110\Tools\Binn\OSQL.EXE" or "C:\Program Files (x86)\Microsoft SQL Server\110\Tools\Binn\OSQL.EXE") with the following parameters after installation to create a database:

```
SQLCMD.EXE -S localhost\SQLEXPRESS -U sa -P password
```

The password is the same as at "Database Engine Configuration" step. Execute the following command in SQLCMD.exe console:

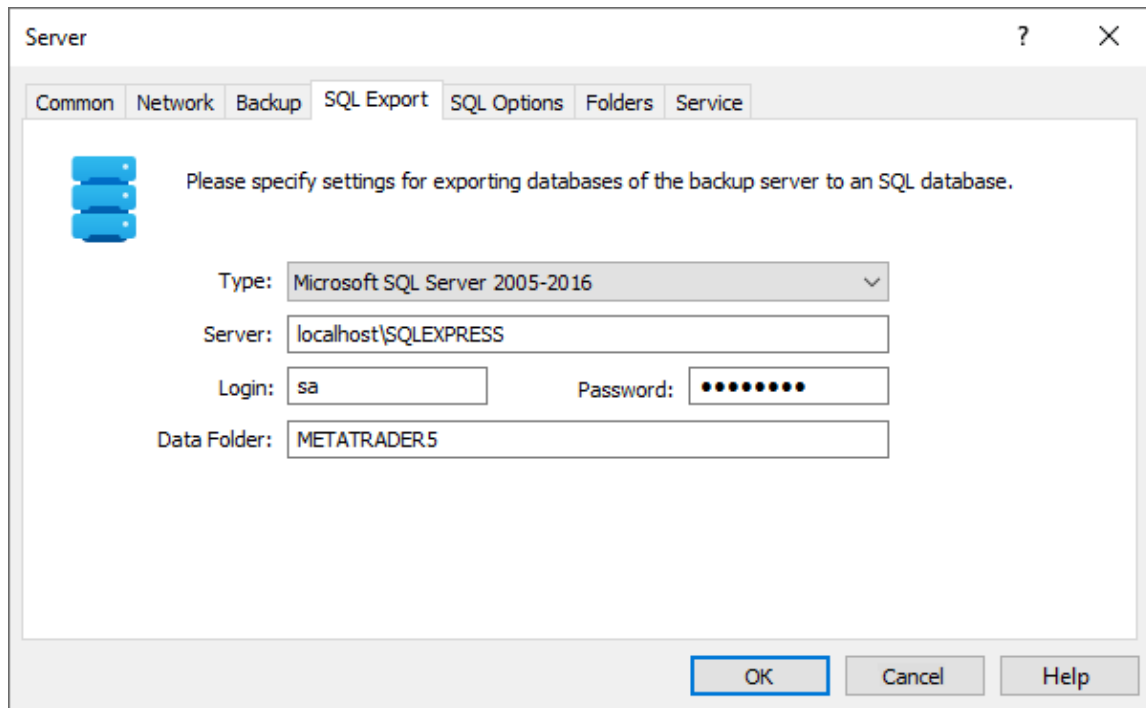
```
CREATE DATABASE METATRADER5  
GO
```

The database will be generated.

In case backup and MS SQL servers are installed on the same computer, connection settings will be as follows:



## MetaTrader 5 trading platform



Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: Microsoft SQL Server 2005-2016

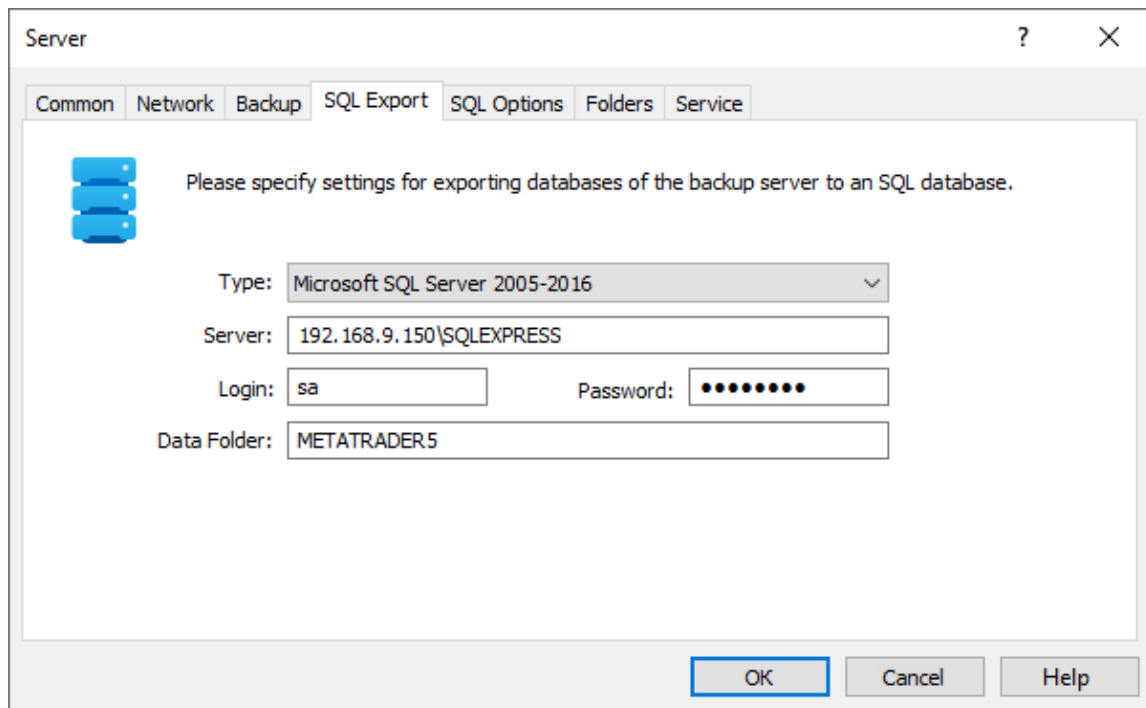
Server: localhost\SQLEXPRESS

Login: sa Password: .....

Data Folder: METATRADER5

OK Cancel Help

In case backup and MS SQL servers are located on different computers:



Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: Microsoft SQL Server 2005-2016

Server: 192.168.9.150\SQLEXPRESS

Login: sa Password: .....

Data Folder: METATRADER5

OK Cancel Help

It is recommended to install MS SQL and backup servers on the same computer for faster export. It should also be kept in mind that the backup server itself also consumes resources. Also, remember that it is possible to deploy several backups for a trade server. These backups will work close to necessary DBMS.

Additional export parameters can be selected at [SQL Options](#) tab.

The backup server will immediately start export into the external DBMS after all settings are specified. Request the log of the appropriate backup server using SQL keyword to track the export process:

```
13:24:40 SQL Export MSSQL: successfully connected to database 'mt5' at 'localhost\SQL
13:24:40 SQL Export MSSQL: server version 11.0.2100.60 RTM (Express Edition)
13:24:40 SQL Export MSSQL: table 'mt5_symbols' creating started
13:24:40 SQL Export MSSQL: table 'mt5_symbols' created
13:24:40 SQL Export MSSQL: primary key for 'mt5_symbols' adding started
13:24:40 SQL Export MSSQL: primary key for 'mt5_symbols' added
13:24:40 SQL Export MSSQL: table 'mt5_symbols_sessions' creating started
13:24:40 SQL Export MSSQL: table 'mt5_symbols_sessions' created
13:24:40 SQL Export MSSQL: primary key for 'mt5_symbols_sessions' adding started
13:24:40 SQL Export MSSQL: primary key for 'mt5_symbols_sessions' added
13:24:40 SQL Export Symbols: 0 symbols and 0 sessions loaded in 344 msec
13:24:40 SQL Export Symbols: synchronization started
...
```

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export / Installation and Setup of Oracle

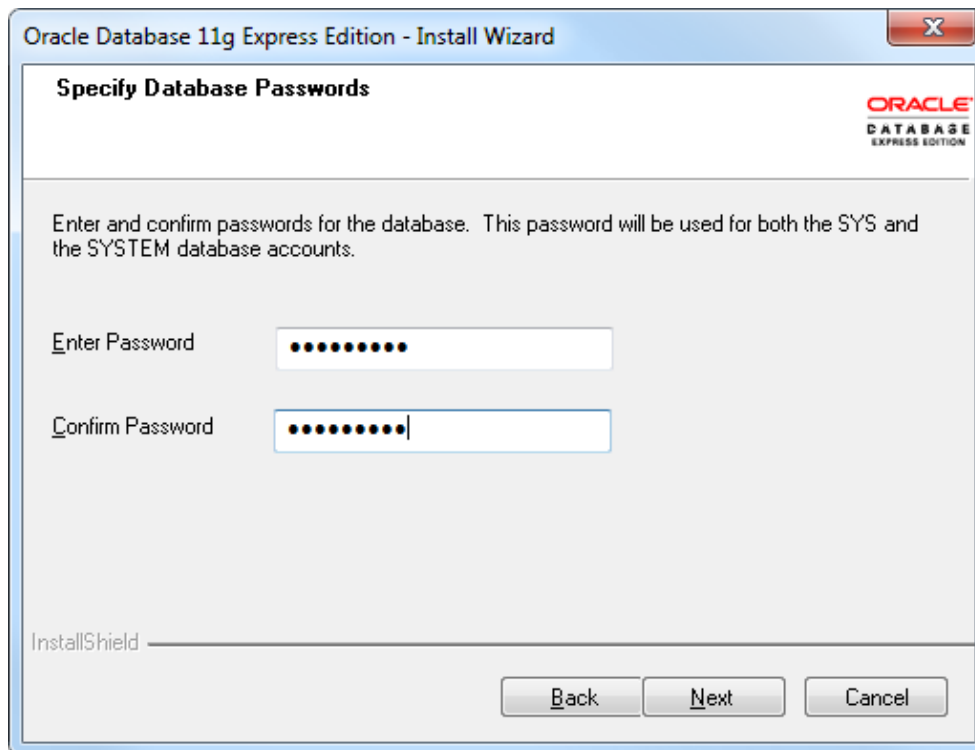
Installation and Setup of Oracle Database

Download the distribution kit from the Oracle's website (<http://www.oracle.com/technetwork/products/express-edition/downloads/index.html>) to install free version of Oracle Database Express Edition. Oracle allows the free use of only the 32-bit version of Express. Therefore, agree to the license terms and download Oracle Database Express Edition 11g Release 2 for Windows x32.

- "Update for Visual C++ 2013 and Visual C++ Redistributable Package" must be installed on the computer where the backup server is installer. Otherwise expert to Oracle databases will not be possible.
- Oracle Database Express Edition is limited to 11 GB of user data. If the limit is exceeded, new data will not be added to the database.

Distribution kit consists of a ZIP archive that should be unpacked into a temporary folder. Then launch DISK1\setup.exe and follow the instructions:

- accept the license terms at License Agreement stage;
- specify administrator password at Specify Database Passwords stage:



Then click Next at the following stages and Install at the last one.

Create the client that will be used for connection. To do that, launch sqlplus.exe:

```
sqlplus.exe system/password
```

Use the same password that was specified at Specify Database Passwords stage during installation. For example:

```
sqlplus.exe system/MyAdminPassword
```

Then execute the following command to create the user:

```
create user user_name identified by password default tablespace users temporary tablespace
```

And allow connection for it:

```
grant connect,resource to user_name;
```

For example:

```
create user mt5user identified by mt5password default tablespace users temporary tablespace  
grant connect,resource to mt5user;
```

In case backup and Oracle servers are installed on the same computer, connection settings will be as follows:

The screenshot shows a window titled "Server" with a tabbed interface. The "SQL Export" tab is selected. The window contains a message: "Please specify settings for exporting databases of the backup server to an SQL database." Below this message are several input fields: "Type" (a dropdown menu showing "Oracle 11g - 12c"), "Server" (a text box with "localhost"), "Login" (a text box with "mt5user"), "Password" (a text box with masked characters "••••••••"), and "Data Folder" (a text box with "XE"). At the bottom right, there are three buttons: "OK", "Cancel", and "Help". The "OK" button is highlighted with a blue border.

In case backup and Oracle servers are located on different computers:

Server

Common Network Backup **SQL Export** SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: Oracle 11g - 12c

Server: 192.168.9.150

Login: mt5user Password: .....

Data Folder: XE

OK Cancel Help

- Oracle Database Express Edition has only one database named XE. Therefore, that is the database that should be specified in Data Folder parameter.
- It is recommended to install Oracle and backup servers on the same computer for faster export. It should also be kept in mind that the backup server itself also consumes resources. Also, remember that it is possible to deploy several backups for a trade server. These backups will work close to necessary DBMS.

Additional export parameters can be selected at SQL Options tab.

The backup server will immediately start export into the external DBMS after all settings are specified. Request the log of the appropriate backup server using SQL keyword to track the export process:

```

2012.10.18 16:33:36 SQL Export Oracle: successfully connected to database 'XE' at 'local
2012.10.18 16:33:36 SQL Export Oracle: server version Oracle Database 11g Express Ed
2012.10.18 16:33:36 SQL Export Oracle: table 'mt5_symbols' creating started
2012.10.18 16:33:36 SQL Export Oracle: table 'mt5_symbols' created
2012.10.18 16:33:36 SQL Export Oracle: primary key for 'mt5_symbols' adding started
2012.10.18 16:33:36 SQL Export Oracle: primary key for 'mt5_symbols' added
2012.10.18 16:33:36 SQL Export Oracle: table 'mt5_symbols_sessions' creating started
2012.10.18 16:33:36 SQL Export Oracle: table 'mt5_symbols_sessions' created
2012.10.18 16:33:36 SQL Export Oracle: primary key for 'mt5_symbols_sessions' adding
2012.10.18 16:33:36 SQL Export Oracle: primary key for 'mt5_symbols_sessions' added
2012.10.18 16:33:36 SQL Export Symbols: 0 symbols and 0 sessions loaded in 203 msec
2012.10.18 16:33:36 SQL Export Symbols: synchronization started
    
```

...

---

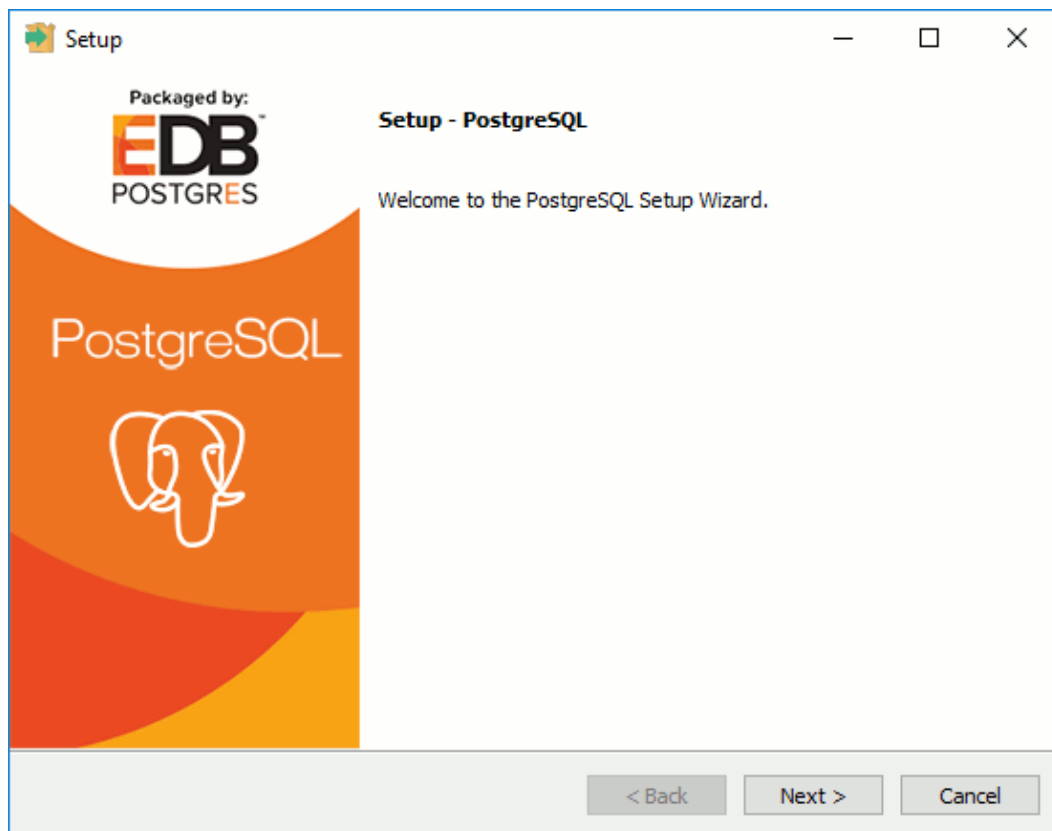
MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export / Installation and Setup of PostgreSQL

## Installation and Setup of PostgreSQL

Download the required version of PostgreSQL installer from the [official website](#).

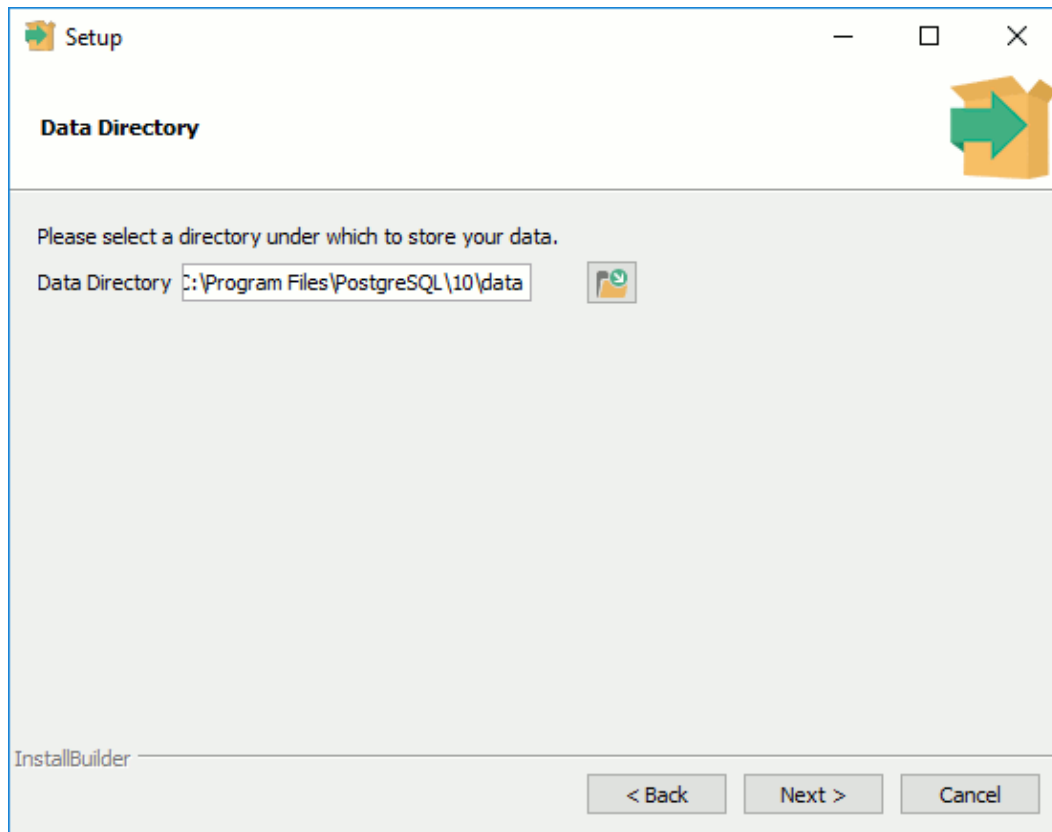
"Update for Visual C++ 2013 and Visual C++ Redistributable Package" must be installed on the computer where the backup server is installed. Otherwise export to PostgreSQL databases will not be possible.

Run the installation file and follow the on-screen instructions. Choose an installation directory and a set of components:



## MetaTrader 5 trading platform

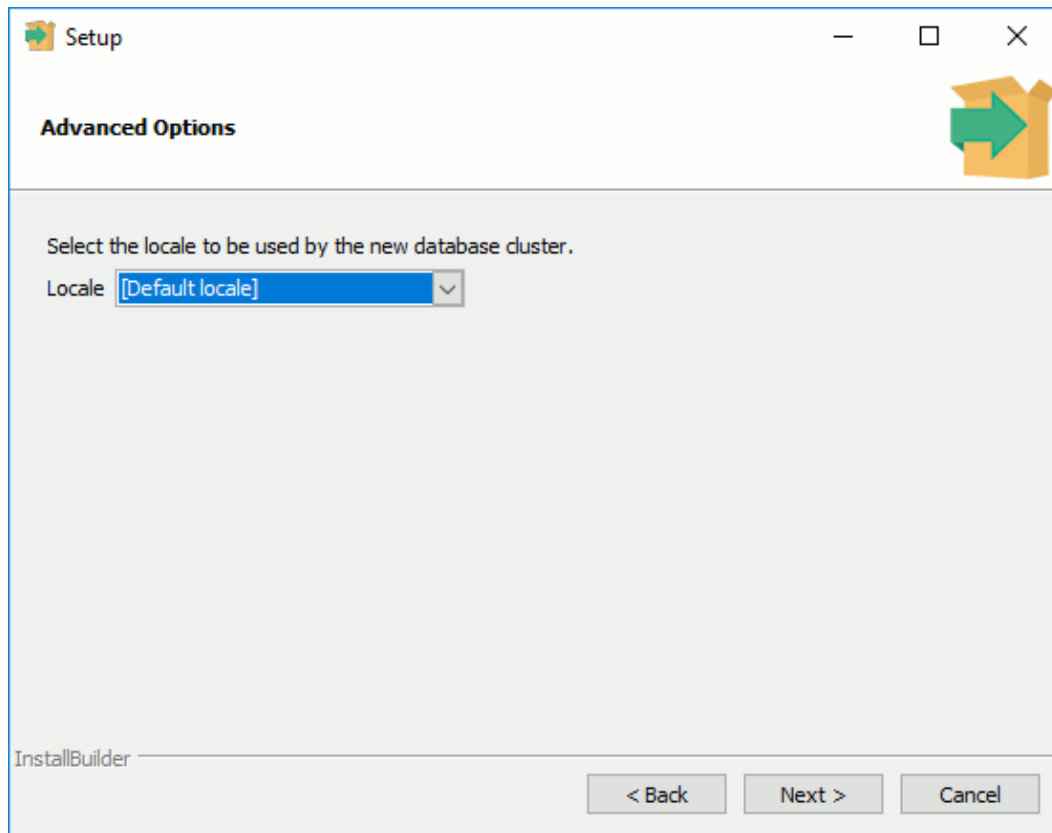
Next, select the directory, in which the databases will be stored, set a password for the administrator account "postgres". Then specify the port at which the PostgreSQL server will listen to incoming connections.



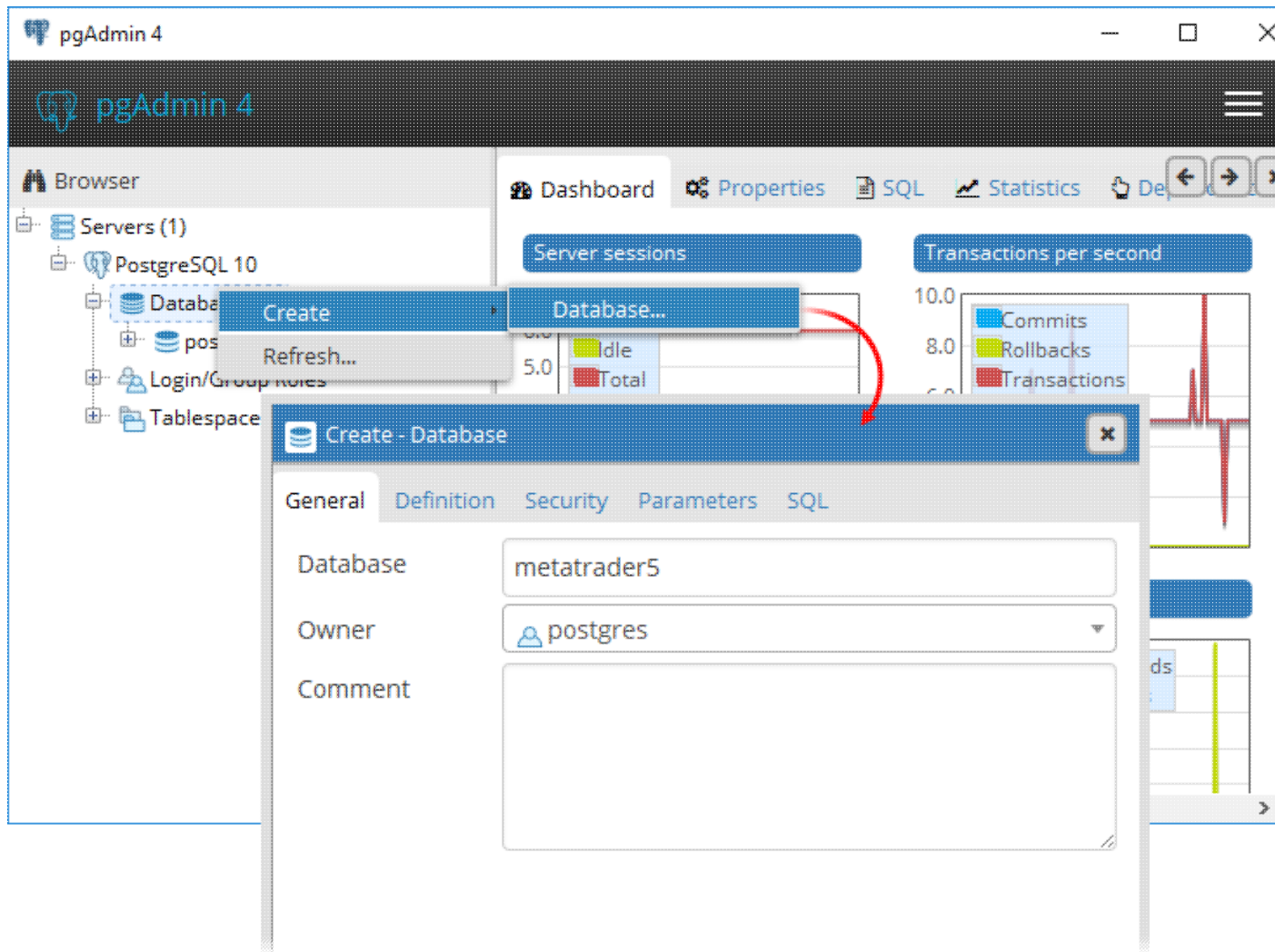
Next, set the locale to be used by the database cluster. The locale is inherited from the operating system by default. Locale affects the operation of some SQL requests. In the next steps, click "Next", and wait for the installation to complete.



## MetaTrader 5 trading platform

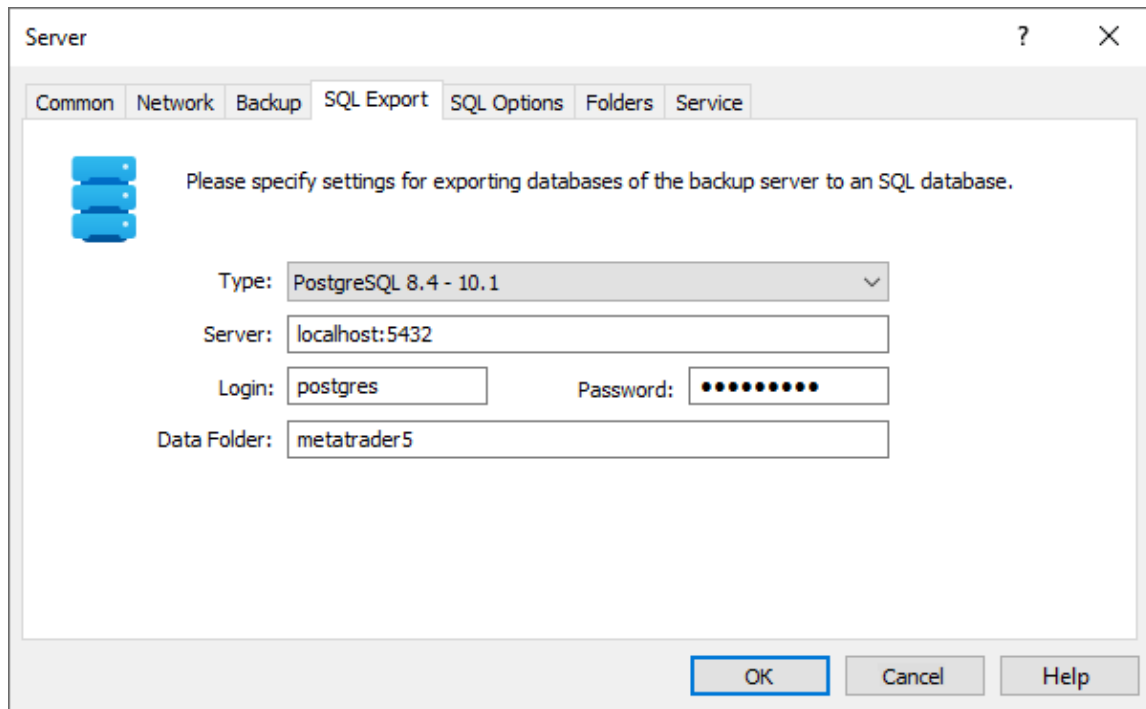


After installation, run the pgAdmin utility from the Start menu. Connect to the server using the postgres login and the password specified during installation. Next, create a database to export information from MetaTrader 5.



## MetaTrader 5 trading platform

Then configure data export on the platform site. In case the backup server and PostgreSQL servers are installed on the same computer, connection parameters should be configured as follows:



The 'Server' dialog box is shown with the 'SQL Export' tab selected. The 'Type' dropdown is set to 'PostgreSQL 8.4 - 10.1'. The 'Server' text field contains 'localhost:5432'. The 'Login' text field contains 'postgres' and the 'Password' text field is masked with dots. The 'Data Folder' text field contains 'metatrader5'. The 'OK' button is highlighted with a blue border.

Server

Common Network Backup SQL Export SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: PostgreSQL 8.4 - 10.1

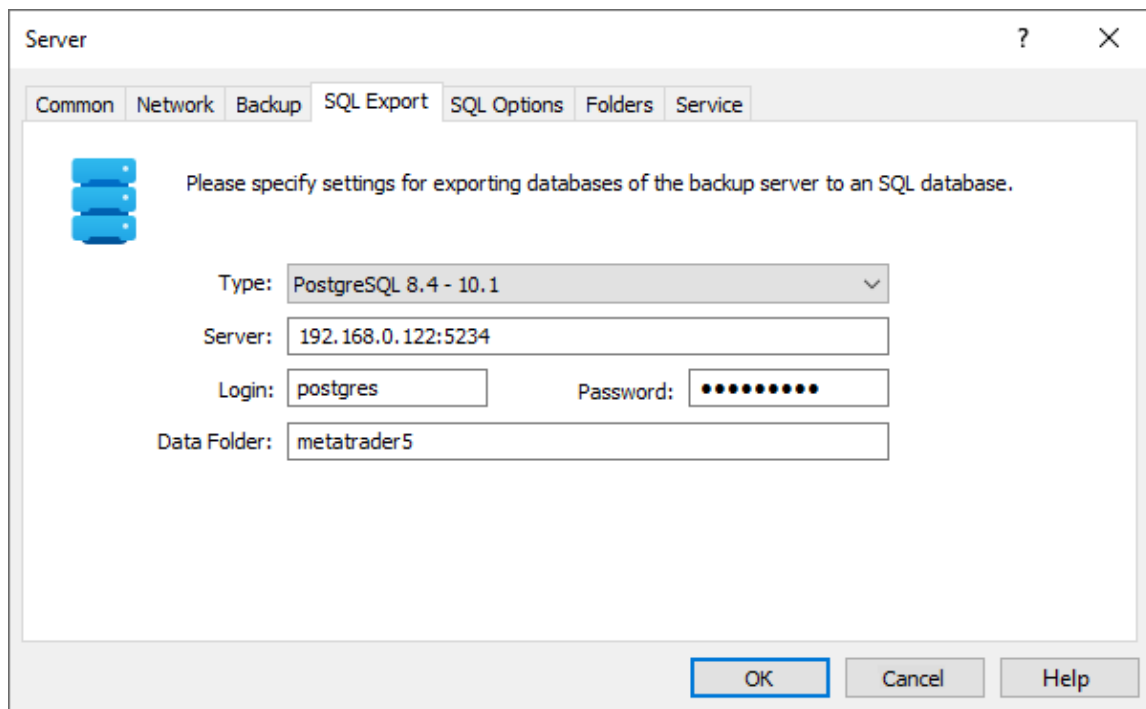
Server: localhost:5432

Login: postgres Password: .....

Data Folder: metatrader5

OK Cancel Help

In case the backup server and PostgreSQL servers are installed on different computers, connection parameters should be configured as follows:



The 'Server' dialog box is shown with the 'SQL Export' tab selected. The 'Type' dropdown is set to 'PostgreSQL 8.4 - 10.1'. The 'Server' text field contains '192.168.0.122:5234'. The 'Login' text field contains 'postgres' and the 'Password' text field is masked with dots. The 'Data Folder' text field contains 'metatrader5'. The 'OK' button is highlighted with a blue border.

Server

Common Network Backup SQL Export SQL Options Folders Service

Please specify settings for exporting databases of the backup server to an SQL database.

Type: PostgreSQL 8.4 - 10.1

Server: 192.168.0.122:5234

Login: postgres Password: .....

Data Folder: metatrader5

OK Cancel Help

- It is recommended to install PostgreSQL and the backup server on the same computer for faster export. However, it should also be kept in mind that the backup server consumes resources. Also, remember that it is possible to deploy several backups for a trade server. These backup servers will run close to necessary DBMS.
- All authentication methods are supported for connecting to PostgreSQL, including scram-sha-256.

Additional export parameters can be selected at [SQL Options](#) tab.

The backup server will immediately start export into the external DBMS after all settings are specified. Request the log of the appropriate backup server using SQL keyword to track the export process:

```
2017.12.21 15:17:17.154      SQL Export      PostgreSQL: server version 10.1
2017.12.21 15:17:17.156      SQL Export
PostgreSQL: successfully connected to database 'metatrader5' at 'localhost' as user 'postgres'
2017.12.21 15:17:17.170      SQL Export
PostgreSQL: table 'mt5_symbols_sessions' creating started
2017.12.21 15:17:17.180      SQL Export
PostgreSQL: table 'mt5_symbols_sessions' created
...
```

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_symbols

mt5\_symbols

'[Symbols](#)' configurations are exported to the table. The table contains the following fields:

| Name          | Type    | Description                                                                                                                                                                    |
|---------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symbol_ID     | Integer | Primary key. Unique symbol ID for more efficient request of the symbol data from the database. Assigned automatically during the export.                                       |
| Timestamp     | Integer | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed. |
| Symbol        | String  | Symbol name.                                                                                                                                                                   |
| Path          | String  | Path to a symbol.                                                                                                                                                              |
| ISIN          | String  | International Securities Identification Number (ISIN) of a symbol.                                                                                                             |
| Description   | String  | Symbol description.                                                                                                                                                            |
| International | String  | The international symbol name.                                                                                                                                                 |
| Category      | String  | The name of the category or sector to which the symbol belongs.                                                                                                                |
| Exchange      | String  | The name of the exchange in which the security is traded.                                                                                                                      |
| CFI           | String  | Instrument classification in accordance with the ISO 10962 standard.                                                                                                           |
| Sector        | Integer | The economic sector the instrument belongs to. Passed as a value of the <a href="#">EnSectors</a> enumeration.                                                                 |
| Industry      | Integer | The industry branch the instrument belongs to. Passed as a value of the <a href="#">EnIndustries</a> enumeration.                                                              |
| Country       | String  | The country of the company whose shares are traded on the stock                                                                                                                |

# MetaTrader 5 trading platform

| Name                 | Type     | Description                                                                                                                        |
|----------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
|                      |          | exchange.                                                                                                                          |
| Basis                | String   | The underlying asset of a derivative financial instrument.                                                                         |
| Source               | String   | The name of the source symbol whose quotes are used for the current financial instrument.                                          |
| Page                 | String   | The address of the web page of a symbol.                                                                                           |
| CurrencyBase         | String   | The base currency of a symbol.                                                                                                     |
| CurrencyBaseDigits   | Integer  | The accuracy of conversion into the base currency.                                                                                 |
| CurrencyProfit       | String   | The profit currency for a symbol.                                                                                                  |
| CurrencyProfitDigits | Integer  | The accuracy of conversion into the profit currency.                                                                               |
| CurrencyMargin       | String   | The symbol margin currency.                                                                                                        |
| CurrencyMarginDigits | Integer  | The accuracy of conversion into the margin currency.                                                                               |
| Color                | COLORREF | The color of the symbol in the "Market Watch" window of the terminals.                                                             |
| ColorBackground      | COLORREF | The color of the symbol background in the "Market Watch" window of the terminals.                                                  |
| Digits               | Integer  | The number of decimal places in the price of the symbol.                                                                           |
| Point                | Float    | Point size. Calculated as $1/10^{\text{Digits}}$ .                                                                                 |
| Multiply             | Float    | The value to multiply the price to, to get the number of points. Calculated as $10^{\text{Digits}}$ .                              |
| TickFlags            | Integer  | Options for working with tick data. Passed as a value of the <u>EnTicksFlags</u> enumeration (sum of values of appropriate flags). |
| TickBookDepth        | Integer  | The range of the Depth of Market.                                                                                                  |
| FilterSoft           | Integer  | The soft level of price filtering.                                                                                                 |
| FilterSoftTicks      | Integer  |                                                                                                                                    |

# MetaTrader 5 trading platform

| Name               | Type    | Description                                                                                                                                                                |
|--------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |         | The value of the ticks counter for the soft filtering.                                                                                                                     |
| FilterHard         | Integer | The hard level of price filtering.                                                                                                                                         |
| FilterHardTicks    | Integer | The value of the ticks counter for the hard filtering.                                                                                                                     |
| FilterDiscard      | Integer | The discard level of price filtering.                                                                                                                                      |
| FilterSpreadMax    | Integer | The maximum allowed spread value.                                                                                                                                          |
| FilterSpreadMin    | Integer | The minimum allowed spread.                                                                                                                                                |
| SubscriptionsDelay | Integer | The delivery delay for the quotes provided by subscription. Indicated in minutes.                                                                                          |
| TradeMode          | Integer | The symbol trading mode. Passed in a value of the <u>EnTradeMode</u> enumeration.                                                                                          |
| CalcMode           | Integer | The mode of margin and profit calculation. Passed in a value of the <u>EnCalcMode</u> enumeration.                                                                         |
| ExecMode           | Integer | Execution mode of a symbol. Passed in a value of the <u>EnExecutionMode</u> enumeration.                                                                                   |
| GTCMode            | Integer | Types of orders that can be set for the symbol. Passed as a value of the <u>EnGTCMode</u> enumeration (sum of values of appropriate flags).                                |
| FillFlags          | Integer | Types of filling allowed for the symbol. Passed as a value of the <u>EnFillingFlags</u> enumeration (sum of values of appropriate flags).                                  |
| ExpirFlags         | Integer | Available types of order expiration for a symbol. Passed as a value of the <u>EnExpirationFlags</u> enumeration (sum of values of appropriate flags).                      |
| Spread             | Integer | Symbol spread size.                                                                                                                                                        |
| SpreadBalance      | Integer | Symbol spread balance. Spread balance is set a shift from the equal distribution of the spread value between Bid and Ask prices. For example, if the spread is equal to 10 |

| Name              | Type    | Description                                                                                                                                                                                                                                                           |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |         | and it is distributed as -5 Bid/+5 Ask, then the spread balance value is 0. The -6 Bid/+4 Ask ratio corresponds to value -1, ratio -4 Bid/+6 Ask corresponds to value 1.                                                                                              |
| SpreadDiff        | Integer | Symbol spread difference. This parameter returns the base value of the spread, which is actually equal to 0. To work with spread difference of a particular group, the <u>corresponding parameter of the group</u> should be used.                                    |
| SpreadDiffBalance | Integer | Spread balance difference. This parameter returns the base value of the balance of spread difference, which is actually equal to 0. To work with the balance of spread difference of a certain group, the <u>corresponding parameter of the group</u> should be used. |
| TickValue         | Float   | The price of one tick of a symbol.                                                                                                                                                                                                                                    |
| TickSize          | Float   | The size of one tick of a symbol.                                                                                                                                                                                                                                     |
| ContractSize      | Float   | The contract size for the symbol.                                                                                                                                                                                                                                     |
| StopsLevel        | Integer | The price band, within which placing stop orders is not allowed.                                                                                                                                                                                                      |
| FreezeLevel       | Integer | The price band, within which it is not allowed to modify orders and positions.                                                                                                                                                                                        |
| QuotesTimeout     | Integer | The time to wait for quotes in seconds, after which trading is automatically disabled for the symbol.                                                                                                                                                                 |
| VolumeMin         | Integer | The minimum volume of trade operations for a symbol. One unit corresponds to 1/10000 lot.                                                                                                                                                                             |
| VolumeMinExt      | Integer | The minimum volume of trade operations for the symbol for the group with extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                                  |
| VolumeMax         | Integer |                                                                                                                                                                                                                                                                       |



| Name                   | Type    | Description                                                                                                                                             |
|------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |         | The maximum volume of trade operations for a symbol. One unit corresponds to 1/10000 lot.                                                               |
| VolumeMaxExt           | Integer | The maximum volume of trade operations for the symbol for the group with extended accuracy. One unit corresponds to 1/100000000 lot.                    |
| VolumeStep             | Integer | The volume change step for trade operations for a symbol. One unit corresponds to 1/10000 lot.                                                          |
| VolumeStepExt          | Integer | The volume change step allowed for trade operations for the symbol, with extended accuracy. One unit corresponds to 1/100000000 lot.                    |
| VolumeLimit            | Integer | The maximum allowed aggregate volume of positions and orders for a symbol in one direction. One unit corresponds to 1/10000 lot.                        |
| VolumeLimitExt         | Integer | The maximum aggregate volume (with extended accuracy) of positions and orders for the symbol in one direction. One unit corresponds to 1/100000000 lot. |
| MarginFlags            | Integer | Additional margin checking modes. Passed in a value of the <a href="#"><u>EnMarginFlags</u></a> enumeration.                                            |
| MarginInitial          | Float   | The size of the initial margin.                                                                                                                         |
| MarginMaintenance      | Float   | The size of the maintenance margin.                                                                                                                     |
| MarginInitialBuy       | Float   | The initial margin rate for market Buy orders.                                                                                                          |
| MarginInitialSell      | Float   | The initial margin rate for market Sell orders.                                                                                                         |
| MarginInitialBuyLimit  | Float   | The initial margin rate for Buy Limit orders.                                                                                                           |
| MarginInitialSellLimit | Float   | The initial margin rate for Sell Limit orders.                                                                                                          |
| MarginInitialBuyStop   | Float   | The initial margin rate for Buy Stop orders.                                                                                                            |

# MetaTrader 5 trading platform

| Name                           | Type    | Description                                                                                                       |
|--------------------------------|---------|-------------------------------------------------------------------------------------------------------------------|
| MarginInitialSellStop          | Float   | The initial margin rate for Sell Stop orders.                                                                     |
| MarginInitialBuyStopLimit      | Float   | The initial margin rate for Buy Stop Limit orders.                                                                |
| MarginInitialSellStopLimit     | Float   | The initial margin rate for Sell Stop Limit orders.                                                               |
| MarginMaintenanceBuy           | Float   | The maintenance margin rate for market Buy orders.                                                                |
| MarginMaintenanceSell          | Float   | The maintenance margin rate for market Sell orders.                                                               |
| MarginMaintenanceBuyLimit      | Float   | The maintenance margin rate for Buy Limit orders.                                                                 |
| MarginMaintenanceSellLimit     | Float   | The maintenance margin rate for Sell Limit orders.                                                                |
| MarginMaintenanceBuyStop       | Float   | The maintenance margin rate for Buy Stop orders.                                                                  |
| MarginMaintenanceSellStop      | Float   | The maintenance margin rate for Sell Stop orders.                                                                 |
| MarginMaintenanceBuyStopLimit  | Float   | The maintenance margin rate for Buy Stop Limit orders.                                                            |
| MarginMaintenanceSellStopLimit | Float   | The maintenance margin rate for Sell Stop Limit orders.                                                           |
| MarginHedged                   | Float   | The hedged margin value.                                                                                          |
| SwapMode                       | Integer | The swap calculation mode for a symbol. Passed in a value of the <u>EnSwapMode</u> enumeration.                   |
| SwapLong                       | Float   | The swap size for long positions.                                                                                 |
| SwapShort                      | Float   | The swap size for short positions.                                                                                |
| SwapYearDay                    | Integer | The number of days in a year used in calculating swap percent. Passed by the <u>EnSwapDays</u> enumeration value. |
| SwapFlags                      | Integer | Additional swap settings. Passed by the <u>EnSwapFlags</u> enumeration value.                                     |
| SwapRateSunday                 | Float   | Swap multiplier for Sundays.                                                                                      |
| SwapRateMonday                 | Float   | Swap multiplier for Mondays.                                                                                      |

# MetaTrader 5 trading platform

| Name              | Type    | Description                                                                                                                                                                 |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SwapRateTuesday   | Float   | Swap multiplier for Tuesdays.                                                                                                                                               |
| SwapRateWednesday | Float   | Swap multiplier for Wednesdays.                                                                                                                                             |
| SwapRateThursday  | Float   | Swap multiplier for Thursdays.                                                                                                                                              |
| SwapRateFriday    | Float   | Swap multiplier for Fridays.                                                                                                                                                |
| SwapRateSaturday  | Float   | Swap multiplier for Saturdays.                                                                                                                                              |
| TimeStart         | Integer | The start date of trading for a symbol. It is considered that there is no time limitation for trading by a symbol if both TimeStart and TimeExpiration are equal to 0.      |
| TimeExpiration    | Integer | The date of trading expiration for a symbol. It is considered that there is no time limitation for trading by a symbol if both TimeStart and TimeExpiration are equal to 0. |
| REFlags           | Integer | The request execution flags. Passed as a value of the <u>EnRequestsFlags</u> enumeration (sum of values of appropriate flags).                                              |
| RETimeout         | Integer | Time in seconds during which the price issued by a dealer in the request execution mode is valid.                                                                           |
| IECheckMode       | Integer | Check mode for instant execution. Passed in a value of the <u>EnInstantMode</u> enumeration.                                                                                |
| IETimeout         | Integer | The maximum allowed difference between the time of arrival of the price, at which the client places an order, and the time of the last price.                               |
| IESlipProfit      | Integer | The maximum allowed slippage in the profitable direction during instant execution.                                                                                          |
| IESlipLosing      | Integer | The maximum allowed slippage in the loss direction during instant execution.                                                                                                |
| IEVolumeMax       | Integer | The maximum volume of a trade operation that can be executed in the instant execution mode. One unit corresponds to 1/10000 lot.                                            |

| Name                | Type    | Description                                                                                                                                                                                                            |
|---------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEVolumeMaxExt      | Integer | The maximum volume (with extended accuracy) of a trade operation that can be executed in the instant execution mode. One unit corresponds to 1/100000000 lot.                                                          |
| PriceSettle         | Float   | The clearing price of the previous session.                                                                                                                                                                            |
| PriceLimitMax       | Float   | The maximum allowed price of the symbol.                                                                                                                                                                               |
| PriceLimitMin       | Float   | The minimum allowed price of the symbol.                                                                                                                                                                               |
| TradeFlags          | Integer | The trade flags of the symbol. Passed in a value of the <u>EnTradeFlags</u> enumeration.                                                                                                                               |
| OrderFlags          | Integer | The flags of order types that are allowed for the symbol. Passed in a value of the <u>EnOrderFlags</u> enumeration (sum of values of appropriate flags).                                                               |
| MarginRateLiquidity | Float   | The liquidity rate of the symbol. It determines the amount of the current value of an asset for the specified financial instrument, which will be taken into account as collateral (accounted for in client's equity). |
| MarginRateCurrency  | Float   | The margin currency rate (rate change radius of the currency, a futures contract is denominated in, relative to the Russian ruble).                                                                                    |
| FaceValue           | Float   | The face value of a bond.                                                                                                                                                                                              |
| AccruedInterest     | Float   | The accrued interest of a bond.                                                                                                                                                                                        |
| SpliceType          | Integer | The futures contract splicing type.                                                                                                                                                                                    |
| SpliceTimeType      | Integer | The date of splicing of the futures contracts.                                                                                                                                                                         |
| SpliceTimeDays      | Integer | The offset of splicing of the futures contracts.                                                                                                                                                                       |
| OptionMode          | Integer | <u>Option</u> type and style:                                                                                                                                                                                          |

| Name           | Type    | Description                                                                                                                                                                                                      |
|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |         | <ul style="list-style-type: none"> <li>• 0 means a European call option</li> <li>• 1 means a European put option</li> <li>• 2 means an American call option</li> <li>• 3 means an American put option</li> </ul> |
| PriceStrike    | Float   | The price, at which an option gives the right to buy or sell an asset (the strike price).                                                                                                                        |
| FilterGap      | Integer | The difference between the previous and the next quote, starting from which a <u>gap</u> is considered to be formed.                                                                                             |
| FilterGapTicks | Integer | The number of ticks for disabling the <u>gap mode</u> . If no new gap occurs within the specified number of quotes, the mode is disabled.                                                                        |
| TickChartMode  | Integer | The mode of creation of the symbol chart: 0 using the Bid price, 1 using the Last price.                                                                                                                         |

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## Enumerations

To pass information about symbols the following enumerations are used:

- [EnFillingFlags](#)
- [EnExpirationFlags](#)
- [EnTradeMode](#)
- [EnExecutionMode](#)
- [EnCalcMode](#)
- [EnGTCMode](#)
- [EnTickFlags](#)
- [EnMarginFlags](#)
- [EnSwapMode](#)
- [EnSwapDays](#)
- [EnSwapFlags](#)
- [EnInstantMode](#)
- [EnRequestFlags](#)
- [EnTradeFlags](#)
- [EnOrderFlags](#)
- [EnMarginTypes](#)
- [EnSpliceType](#)
- [EnSpliceTimeType](#)
- [EnSectors](#)
- [EnIndustries](#)

### EnFillingFlags

The order filling methods allowed for a symbol are enumerated in EnFillingFlags.

| ID              | Value | Description                                                                                                                                                           |
|-----------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FILL_FLAGS_NONE | 0     | All filling methods are disabled.                                                                                                                                     |
| FILL_FLAGS_FOK  | 1     | The "Fill or Kill" mode. The order must be filled completely or canceled. This type of filling is automatically set for the instant and request execution.            |
| FILL_FLAGS_IOC  | 2     | Immediate or Cancel. An order can be filled partially and the residual volume is canceled. This type of filling is only available for the stock and market execution. |

### EnExpirationFlags

Types of orders allowed for the symbol are enumerated in EnExpirationFlags.

| ID              | Value | Description                        |
|-----------------|-------|------------------------------------|
| TIME_FLAGS_NONE | 0     | All expiration types are disabled. |

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| ID                       | Value | Description                                                                                                                                                                                                                             |
|--------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TIME_FLAGS_GTC           | 1     | Orders are good till canceled.                                                                                                                                                                                                          |
| TIME_FLAGS_DAY           | 2     | Orders are effective only during the current trading day.                                                                                                                                                                               |
| TIME_FLAGS_SPECIFIED     | 4     | Orders are effective till the date specified by the trader.                                                                                                                                                                             |
| TIME_FLAGS_SPECIFIED_DAY | 8     | Orders that expire at the specified day. An order expires at 00:00 of the specified day or the nearOrders with expiration at a specified day. An order expires at 00:00 of a specified day or at a nearest trade time.est trading time. |

EnTradeMode

Symbol trading modes are enumerated in EnTradeMode.

| ID              | Value | Description                       |
|-----------------|-------|-----------------------------------|
| TRADE_DISABLED  | 0     | Trade is disabled.                |
| TRADE_LONGONLY  | 1     | Only long positions are allowed.  |
| TRADE_SHORTONLY | 2     | Only short positions are allowed. |
| TRADE_CLOSEONLY | 3     | Only closure is allowed.          |
| TRADE_FULL      | 4     | Full trading access.              |

EnExecutionMode

Execution types are enumerated in EnExecutionMode.

| ID                 | Value | Description              |
|--------------------|-------|--------------------------|
| EXECUTION_REQUEST  | 0     | Request execution mode.  |
| EXECUTION_INSTANT  | 1     | Instant execution mode.  |
| EXECUTION_MARKET   | 2     | Market execution mode.   |
| EXECUTION_EXCHANGE | 3     | Exchange execution mode. |

EnCalcMode

Types of profit and margin calculation for a symbol are enumerated in EnCalcMode.

| ID                 | Value | Description                   |
|--------------------|-------|-------------------------------|
| TRADE_MODE_FOREX   | 0     | The Forex calculation mode.   |
| TRADE_MODE_FUTURES | 1     | The Futures calculation mode. |

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| ID                             | Value | Description                                                                                                                                                                                               |
|--------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRADE_MODE_CFD                 | 2     | The CFD calculation mode.                                                                                                                                                                                 |
| TRADE_MODE_CFDINDEX            | 3     | The CFD Index calculation mode.                                                                                                                                                                           |
| TRADE_MODE_CFDLEVERAGE         | 4     | The CFD Leverage calculation mode.                                                                                                                                                                        |
| TRADE_MODE_FOREX_NO_LEVERAGE   | 5     | The Forex No Leverage calculation mode.                                                                                                                                                                   |
| TRADE_MODE_EXCH_STOCKS         | 32    | The Exchange Stocks calculation mode.                                                                                                                                                                     |
| TRADE_MODE_EXCH_FUTURES        | 33    | The Exchange Futures calculation mode.                                                                                                                                                                    |
| TRADE_MODE_EXCH_FORTS          | 34    | The Exchange FORTS calculation mode for Derivatives Market of the Moscow Exchange.                                                                                                                        |
| TRADE_MODE_EXCH_OPTIONS        | 35    | The Exchange Options calculation mode.                                                                                                                                                                    |
| TRADE_MODE_EXCH_OPTIONS_MARGIN | 36    | The Exchange Margin Options calculation mode.                                                                                                                                                             |
| TRADE_MODE_EXCH_BONDS          | 37    | The Exchange Bonds calculation mode.                                                                                                                                                                      |
| TRADE_MODE_SERV_COLLATERAL     | 64    | Non-tradable instruments of this type are used as client's assets to provide the required margin for open positions of other instruments. For these instruments the margin and profit are not calculated. |

EnGTCMode

Types of order expiration are enumerated in EnGTCMode.

| ID                    | Value | Description                                                                                                                                                                                    |
|-----------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDERS_GTC            | 0     | "Good till canceled" mode. As a trade day changes, pending orders are preserved.                                                                                                               |
| ORDERS_DAILY          | 1     | "Good till today including SL/TP" mode. Orders are effective only within one trading day. As soon as it is over, all Stop Loss and Take Profit levels, as well all pending orders are deleted. |
| ORDERS_DAILY_NO_STOPS | 2     |                                                                                                                                                                                                |



| ID | Value | Description                                                                                                                                              |
|----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |       | "Good till today excluding SL/TP" mode. As a trading day changes, only pending orders are deleted, while Stop Loss and Take Profit levels are preserved. |

EnTickFlags

Options of working with the symbol's tick data are enumerated in EnTickFlags.

| ID              | Value | Description                                                                                                                                                                                                                                                                     |
|-----------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TICK_REALTIME   | 1     | Allow real-time quotes from data feeds.                                                                                                                                                                                                                                         |
| TICK_COLLECTRAW | 2     | Enable keeping of raw prices.                                                                                                                                                                                                                                                   |
| TICK_FEED_STATS | 4     | Receive market statistics (Ask High, Bid Low, etc.) directly from data feeds without calculation on the history server. If this flag is not set, the statistical information is calculated by the history server.<br><br>This flag can only be set together with TICK_REALTIME. |
| TICK_NONE       | 0     | Beginning of enumeration. Corresponds to the absence of rights.                                                                                                                                                                                                                 |

EnMarginFlags

The additional margin checks are enumerated in EnMarginFlags.

| ID                         | Value | Description                                                                                                                                                                                                                               |
|----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARGIN_FLAGS_NONE          | 0     | The standard mode of margin checking. The margin is checked when any order is placed and when a pending orders triggers.                                                                                                                  |
| MARGIN_FLAGS_CHECK_PROCESS | 1     | If this flags is enabled, another check of margin is added to those described above: the margin is checked before executing an order after it is confirmed (checked) by the server (at automated execution), by a dealer or by a gateway. |
| MARGIN_FLAGS_CHECK_SLTP    | 2     | This flag enables an additional check of margin before a position is closed by stop loss or take profit. If the position close results in reducing the margin to a level insufficient to maintain open positions and orders, the stop     |

| ID                           | Value | Description                                                                                                                                                               |
|------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |       | loss/take profit will not trigger, the position will stay open. This check must be enabled in case the trade operations are transmitted to an external system (exchange). |
| MARGIN_FLAGS_HEDGE_LARGE_LEG | 4     | If the flag is enabled, the margin for hedged positions is calculated using larger leg (the total volume of users positions and orders opened in the same direction).     |

### EnSwapMode

Types of swap calculation are enumerated in EnSwapMode. The swap size is specified using the SwapLong and SwapShort parameters.

| ID                         | Value | Description                                                                                                                                                                                                                     |
|----------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWAP_DISABLED              | 0     | Swap charging is disabled.                                                                                                                                                                                                      |
| SWAP_BY_POINTS             | 1     | Swap charging in points of a symbol price.                                                                                                                                                                                      |
| SWAP_BY_SYMBOL_CURRENCY    | 2     | Swap charging in the base currency of a symbol.                                                                                                                                                                                 |
| SWAP_BY_MARGIN_CURRENCY    | 3     | Swap charging in the margin currency of a symbol.                                                                                                                                                                               |
| SWAP_BY_GROUP_CURRENCY     | 4     | Swap charging in the group (deposit) currency.                                                                                                                                                                                  |
| SWAP_BY_INTEREST_CURRENT   | 5     | Swap charging as a per cent from the price of a symbol at calculation of swap.                                                                                                                                                  |
| SWAP_BY_INTEREST_OPEN      | 6     | Swap charging as a per cent from the open price of a position by a symbol.                                                                                                                                                      |
| SWAP_REOPEN_BY_CLOSE_PRICE | 7     | Swap charging by reopening position. At the end of a trading day position is closed. Next day it is reopened by the close price +/- number of points specified using the <u>SwapLong</u> and <u>SwapShort</u> parameters.       |
| SWAP_REOPEN_BY_BID         | 8     | Swap charging by reopening position. At the end of a trading day position is closed. Next day it is reopened by the current Bid price +/- number of points specified using the <u>SwapLong</u> and <u>SwapShort</u> parameters. |

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| ID                      | Value | Description                                       |
|-------------------------|-------|---------------------------------------------------|
| SWAP_BY_PROFIT_CURRENCY | 9     | Swap charging in the profit currency of a symbol. |

### EnSwapDays

Triple swap charging days are enumerated in EnSwapDays.

| ID                 | Value | Description                |
|--------------------|-------|----------------------------|
| SWAP_DAY_SUNDAY    | 0     | Sunday.                    |
| SWAP_DAY_MONDAY    | 1     | Monday.                    |
| SWAP_DAY_TUESDAY   | 2     | Tuesday.                   |
| SWAP_DAY_WEDNESDAY | 3     | Wednesday.                 |
| SWAP_DAY_THURSDAY  | 4     | Thursday.                  |
| SWAP_DAY_FRIDAY    | 5     | Friday.                    |
| SWAP_DAY_SATURDAY  | 6     | Saturday.                  |
| SWAP_DAY_DISABLED  | 7     | Triple swaps are disabled. |

### EnSwapFlags

Additional swap settings are enumerated in EnSwapFlags.

| ID                           | Value | Description                                                                                                                                                                                                                |
|------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWAP_FLAGS_NONE              | 0     | Additional settings are not used.                                                                                                                                                                                          |
| SWAP_FLAGS_CONSIDER_HOLIDAYS | 1     | Account for holidays when calculating swaps. If the flag is enabled, the platform checks all <u>holiday</u> configurations. The day before the holiday, the swap is doubled. No swap is charged on the day of the holiday. |

### EnInstantMode

Types of check for the instant execution mode are enumerate in EnInstantMode.

| ID                   | Value | Description                           |
|----------------------|-------|---------------------------------------|
| INSTANT_CHECK_NORMAL | 0     | The normal mode of instant execution. |

### EnRequestFlags

Options of the Request execution mode are enumerated in EnRequestFlags.

| ID                  | Value | Description                   |
|---------------------|-------|-------------------------------|
| REQUEST_FLAGS_NONE  | 0     | No flags.                     |
| REQUEST_FLAGS_ORDER | 1     | Additional confirmation mode. |

EnTradeFlags

Trade flags of symbols are enumerated in EnTradeFlags.

| ID                           | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRADE_FLAGS_NONE             | 0     | Beginning of enumeration.<br>Corresponds to the absence of flags.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TRADE_FLAGS_PROFIT_BY_MARKET | 1     | <p>This flag is used for the Forex type symbols only (<u>TRADE_MODE_FOREX</u>).</p> <p>By default for the conversion of profit/loss from the profit currency to the deposit currency, the price at which a deal (exit from position) is performed is used, the profitability/unprofitability of the deal is not taken into consideration.</p> <p>If this flag is enabled, the conversion is performed using the current Bid/Ask price depending of the profitability/unprofitability of a deal. The Bid price is taken for calculations for profitable deals, because as a result of a profitable deal, a trader obtains a certain amount of the profit currency and needs to sell it for the deposit currency. The Ask price is taken for losing deals, because as a result of a losing deal, a trader needs to buy a certain amount of currency for the deposit currency.</p> |
| TRADE_FLAGS_ALLOW_SIGNALS    | 2     | If this flag is not enabled, clients will not be able to copy trade operations by this symbol using the <u>Signals</u> service. Trading signals can also be disabled on a client group level ( <u>EnTradeFlags</u> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

EnOrderFlags

Flags of orders that are allowed for the symbol are enumerated in EnOrderFlags.

| Identifier | Value | Description |
|------------|-------|-------------|
|------------|-------|-------------|

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|                        |    |                                                                                                                                                                                                                                                               |
|------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_FLAGS_NONE       | 0  | Beginning of enumeration. Corresponds to the absence of flags.                                                                                                                                                                                                |
| ORDER_FLAGS_MARKET     | 1  | Market orders Buy and Sell are allowed.                                                                                                                                                                                                                       |
| ORDER_FLAGS_LIMIT      | 2  | Limit orders Buy Limit and Sell Limit are allowed.                                                                                                                                                                                                            |
| ORDER_FLAGS_STOP       | 4  | Stop orders Buy Stop and Sell Stop are allowed.                                                                                                                                                                                                               |
| ORDER_FLAGS_STOP_LIMIT | 8  | Stop Limit orders Buy Stop Limit and Sell Stop Limit are allowed.                                                                                                                                                                                             |
| ORDER_FLAGS_SL         | 16 | Stop Loss orders are allowed.                                                                                                                                                                                                                                 |
| ORDER_FLAGS_TP         | 32 | Take Profit orders are allowed.                                                                                                                                                                                                                               |
| ORDER_FLAGS_CLOSEBY    | 64 | Orders to close a position by an opposite one ( <u>OP_CLOSE_BY</u> ) are allowed. Close By permission of a symbol does not mean that this type of orders will be available on netting accounts. Close By orders can only be used on <u>hedging accounts</u> . |

### EnMarginTypes

Types of orders used for specified margin rates are enumerated in EnMarginTypes.

| Identifier             | Value | Description                    |
|------------------------|-------|--------------------------------|
| MARGIN_BUY             | 0     | Market buy order.              |
| MARGIN_SELL            | 1     | Market sell order.             |
| MARGIN_BUY_LIMIT       | 2     | Buy Limit pending order.       |
| MARGIN_SELL_LIMIT      | 3     | Sell Limit pending order.      |
| MARGIN_BUY_STOP        | 4     | Buy Stop pending order.        |
| MARGIN_SELL_STOP       | 5     | Sell Stop pending order.       |
| MARGIN_BUY_STOP_LIMIT  | 6     | Buy Stop Limit pending order.  |
| MARGIN_SELL_STOP_LIMIT | 7     | Sell Stop Limit pending order. |

### EnSpliceType

Futures splicing types are available in EnSpliceType.

| Identifier        | Value | Description  |
|-------------------|-------|--------------|
| SPLICE_NONE       | 0     | No splicing. |
| SPLICE_UNADJUSTED | 1     |              |

| Identifier      | Value | Description                                                                                                                                                                                                                                                                     |
|-----------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |       | Splicing futures quotes "as is" the price level of the previous contract is not adjusted to the current (front) contract prices.                                                                                                                                                |
| SPLICE_ADJUSTED | 2     | In this mode, a difference between the last quote of the previous contract and the first quote of the front contract is calculated. All quotes of the previous contract are then adjusted by this value. This makes a spliced chart look smooth without gaps between contracts. |

EnSpliceTimeType

Futures splicing dates are available in EnSpliceTimeType.

| Identifier             | Value | Description                                                          |
|------------------------|-------|----------------------------------------------------------------------|
| SPLICE_TIME_EXPIRATION | 0     | Splicing at the very moment of instrument expiration TimeExpiration. |

EnSectors

EnSectors lists economic sectors a trading instrument may belong to.

| ID                            | Value | Description            |
|-------------------------------|-------|------------------------|
| SECTOR_UNDEFINED              | 0     | Undefined              |
| SECTOR_BASIC_MATERIALS        | 1     | Basic materials        |
| SECTOR_COMMUNICATION_SERVICES | 2     | Communication services |
| SECTOR_CONSUMER_CYCLICAL      | 3     | Consumer cyclical      |
| SECTOR_CONSUMER_DEFENSIVE     | 4     | Consumer defensive     |
| SECTOR_ENERGY                 | 5     | Energy                 |
| SECTOR_FINANCIAL              | 6     | Finance                |
| SECTOR_HEALTHCARE             | 7     | Healthcare             |
| SECTOR_INDUSTRIALS            | 8     | Industrials            |
| SECTOR_REAL_ESTATE            | 9     | Real estate            |
| SECTOR_TECHNOLOGY             | 10    | Technology             |
| SECTOR_UTILITIES              | 11    | Utilities              |
| SECTOR_CURRENCY               | 12    | Currency               |
| SECTOR_CURRENCY_CRYPTOC       | 13    | Crypto currency        |
| SECTOR_INDEXES                | 14    | Indices                |

| ID                 | Value | Description |
|--------------------|-------|-------------|
| SECTOR_COMMODITIES | 15    | Commodities |

## EnIndustries

EnIndustries lists industry branches a trading instrument may belong to.

| ID                             | Value | Description                                                                                     |
|--------------------------------|-------|-------------------------------------------------------------------------------------------------|
| INDUSTRY_UNDEFINED             | 0     | Undefined                                                                                       |
| Basic materials                |       |                                                                                                 |
| INDUSTRY_AGRICULTURAL_INPUTS   | 1     | Agricultural inputs                                                                             |
| INDUSTRY_ALUMINIUM             | 2     | Aluminium                                                                                       |
| INDUSTRY_BUILDING_MATERIALS    | 3     | Building materials                                                                              |
| INDUSTRY_CHEMICALS             | 4     | Chemicals                                                                                       |
| INDUSTRY_COKING_COAL           | 5     | Coking coal                                                                                     |
| INDUSTRY_COPPER                | 6     | Copper                                                                                          |
| INDUSTRY_GOLD                  | 7     | Gold                                                                                            |
| INDUSTRY_LUMBER_WOOD           | 8     | Lumber and wood production                                                                      |
| INDUSTRY_INDUSTRIAL_METALS     | 9     | Other industrial metals and mining                                                              |
| INDUSTRY_PRECIOUS_METALS       | 10    | Other precious metals and mining                                                                |
| INDUSTRY_PAPER                 | 11    | Paper and paper products                                                                        |
| INDUSTRY_SILVER                | 12    | Silver                                                                                          |
| INDUSTRY_SPECIALTY_CHEMICALS   | 13    | Specialty chemicals                                                                             |
| INDUSTRY_STEEL                 | 14    | Steel                                                                                           |
| INDUSTRY_BASIC_MATERIALS_FIRST | 1     | Beginning of the basic materials types enumeration. Corresponds to INDUSTRY_AGRICULTURAL_INPUTS |
| INDUSTRY_BASIC_MATERIALS_LAST  | 14    | End of the basic materials types enumeration. Corresponds to INDUSTRY_STEEL                     |
| INDUSTRY_BASIC_MATERIALS_END   | 50    | Basic materials types enumeration                                                               |
| Communication services         |       |                                                                                                 |
| INDUSTRY_ADVERTISING           | 51    | Advertising agencies                                                                            |
| INDUSTRY_BROADCASTING          | 52    | Broadcasting                                                                                    |
| INDUSTRY_GAMING_MULTIMEDIA     | 53    | Electronic gaming and multimedia                                                                |

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| ID                             | Value | Description                                                                               |
|--------------------------------|-------|-------------------------------------------------------------------------------------------|
| INDUSTRY_ENTERTAINMENT         | 54    | Entertainment                                                                             |
| INDUSTRY_INTERNET_CONTENT      | 55    | Internet content and information                                                          |
| INDUSTRY_PUBLISHING            | 56    | Publishing                                                                                |
| INDUSTRY_TELECOM               | 57    | Telecom services                                                                          |
| INDUSTRY_COMMUNICATION_FIRST   | 51    | Beginning of the communication services enumeration. Corresponds to INDUSTRY_ADVERTISING. |
| INDUSTRY_COMMUNICATION_LAST    | 57    | End of the communication services enumeration. Corresponds to INDUSTRY_TELECOM.           |
| INDUSTRY_COMMUNICATION_END     | 100   | Communication services types enumeration limit.                                           |
| Consumer cyclical              |       |                                                                                           |
| INDUSTRY_APPAREL_MANUFACTURING | 101   | Apparel manufacturing                                                                     |
| INDUSTRY_APPAREL_RETAIL        | 102   | Apparel retail                                                                            |
| INDUSTRY_AUTO_MANUFACTURERS    | 103   | Auto manufacturers                                                                        |
| INDUSTRY_AUTO_PARTS            | 104   | Auto parts                                                                                |
| INDUSTRY_AUTO_DEALERSHIP       | 105   | Auto and truck dealerships                                                                |
| INDUSTRY_DEPARTMENT_STORES     | 106   | Department stores                                                                         |
| INDUSTRY_FOOTWEAR_ACCESSORIES  | 107   | Footwear and accessories                                                                  |
| INDUSTRY_FURNISHINGS           | 108   | Furnishing, fixtures and appliances                                                       |
| INDUSTRY_GAMBLING              | 109   | Gambling                                                                                  |
| INDUSTRY_HOME_IMPROV_RETAIL    | 110   | Home improvement retail                                                                   |
| INDUSTRY_INTERNET_RETAIL       | 111   | Internet retail                                                                           |
| INDUSTRY_LEISURE               | 112   | Leisure                                                                                   |
| INDUSTRY_LODGING               | 113   | Lodging                                                                                   |
| INDUSTRY_LUXURY_GOODS          | 114   | Luxury goods                                                                              |
| INDUSTRY_PACKAGING_CONTAINERS  | 115   | Packaging and containers                                                                  |
| INDUSTRY_PERSONAL_SERVICES     | 116   | Personal services                                                                         |
| INDUSTRY_RECREATIONAL_VEHICLES | 117   | Recreational vehicles                                                                     |
| INDUSTRY_RESIDENT_CONSTRUCTION | 118   | Residential construction                                                                  |



| ID                             | Value | Description                                                                                                                               |
|--------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| INDUSTRY_RESORTS_CASINOS       | 119   | Resorts and casinos                                                                                                                       |
| INDUSTRY_RESTAURANTS           | 120   | Restaurants                                                                                                                               |
| INDUSTRY_SPECIALTY_RETAIL      | 121   | Specialty retail                                                                                                                          |
| INDUSTRY_TEXTILE_MANUFACTURING | 122   | Textile manufacturing                                                                                                                     |
| INDUSTRY_TRAVEL_SERVICES       | 123   | Travel services                                                                                                                           |
| INDUSTRY_CONSUMER_CYCL_FIRST   | 101   | Beginning of enumeration of industries related to production of goods and cyclical demand. Corresponds to INDUSTRY_APPAREL_MANUFACTURING. |
| INDUSTRY_CONSUMER_CYCL_LAST    | 123   | End of enumeration of industry branches related to production of goods and cyclical demand. Corresponds to INDUSTRY_TRAVEL_SERVICES.      |
| INDUSTRY_CONSUMER_CYCL_END     | 150   | Limit of the enumeration of industries related to production of goods and cyclical demand.                                                |
| Consumer defensive             |       |                                                                                                                                           |
| INDUSTRY_BEVERAGES_BREWERS     | 151   | Beverages - Brewers                                                                                                                       |
| INDUSTRY_BEVERAGES_NON_ALCO    | 152   | Beverages - Non-alcoholic                                                                                                                 |
| INDUSTRY_BEVERAGES_WINERIES    | 153   | Beverages - Wineries and distilleries                                                                                                     |
| INDUSTRY_CONFECTIONERS         | 154   | Confectioners                                                                                                                             |
| INDUSTRY_DISCOUNT_STORES       | 155   | Discount stores                                                                                                                           |
| INDUSTRY_EDUCATION_TRAINING    | 156   | Education and training services                                                                                                           |
| INDUSTRY_FARM_PRODUCTS         | 157   | Farm products                                                                                                                             |
| INDUSTRY_FOOD_DISTRIBUTION     | 158   | Food distribution                                                                                                                         |
| INDUSTRY_GROCERY_STORES        | 159   | Grocery stores                                                                                                                            |
| INDUSTRY_HOUSEHOLD_PRODUCTS    | 160   | Household and personal products                                                                                                           |
| INDUSTRY_PACKAGED_FOODS        | 161   | Packaged foods                                                                                                                            |
| INDUSTRY_TOBACCO               | 162   | Tobacco                                                                                                                                   |
| INDUSTRY_CONSUMER_DEF_FIRST    | 151   | Beginning of enumeration of industries related to production of consumer goods and services. Corresponds to INDUSTRY_BEVERAGES_BREWERS.   |

| ID                              | Value | Description                                                                                                                    |
|---------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------|
| INDUSTRY_CONSUMER_DEF_LAST      | 162   | End of enumeration of industry branches related to production of consumer goods and services. Corresponds to INDUSTRY_TOBACCO. |
| INDUSTRY_CONSUMER_DEF_END       | 200   | Limit of the enumeration of industry branches related to production of consumer goods and services.                            |
| Energy                          |       |                                                                                                                                |
| INDUSTRY_OIL_GAS_DRILLING       | 201   | Oil and gas drilling                                                                                                           |
| INDUSTRY_OIL_GAS_EP             | 202   | Oil and gas extraction and processing                                                                                          |
| INDUSTRY_OIL_GAS_EQUIPMENT      | 203   | Oil and gas equipment and services                                                                                             |
| INDUSTRY_OIL_GAS_INTEGRATED     | 204   | Oil and gas integrated                                                                                                         |
| INDUSTRY_OIL_GAS_MIDSTREAM      | 205   | Oil and gas midstream                                                                                                          |
| INDUSTRY_OIL_GAS_REFINING       | 206   | Oil and gas refining and marketing                                                                                             |
| INDUSTRY_THERMAL_COAL           | 207   | Thermal coal                                                                                                                   |
| INDUSTRY_URANIUM                | 208   | Uranium                                                                                                                        |
| INDUSTRY_ENERGY_FIRST           | 201   | Beginning of enumeration of energy industry types. Corresponds to INDUSTRY_OIL_GAS_DRILLING.                                   |
| INDUSTRY_ENERGY_LAST            | 208   | End of enumeration of energy industry types. Corresponds to INDUSTRY_URANIUM.                                                  |
| INDUSTRY_ENERGY_END             | 250   | Limit of the energy industry types enumeration.                                                                                |
| Finance                         |       |                                                                                                                                |
| INDUSTRY_EXCHANGE_TRADED_FUND   | 251   | Exchange traded fund                                                                                                           |
| INDUSTRY_ASSETS_MANAGEMENT      | 252   | Assets management                                                                                                              |
| INDUSTRY_BANKS_DIVERSIFIED      | 253   | Banks - Diversified                                                                                                            |
| INDUSTRY_BANKS_REGIONAL         | 254   | Banks - Regional                                                                                                               |
| INDUSTRY_CAPITAL_MARKETS        | 255   | Capital markets                                                                                                                |
| INDUSTRY_CLOSE_END_FUND_DEBT    | 256   | Closed-End fund - Debt                                                                                                         |
| INDUSTRY_CLOSE_END_FUND_EQUITY  | 257   | Closed-end fund - Equity                                                                                                       |
| INDUSTRY_CLOSE_END_FUND_FOREIGN | 258   | Closed-end fund - Foreign                                                                                                      |
| INDUSTRY_CREDIT_SERVICES        | 259   | Credit services                                                                                                                |

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| ID                                | Value | Description                                                                                        |
|-----------------------------------|-------|----------------------------------------------------------------------------------------------------|
| INDUSTRY_FINANCIAL_CONGLOMERATE   | 260   | Financial conglomerates                                                                            |
| INDUSTRY_FINANCIAL_DATA_EXCHANGE  | 261   | Financial data and stock exchange                                                                  |
| INDUSTRY_INSURANCE_BROKERS        | 262   | Insurance brokers                                                                                  |
| INDUSTRY_INSURANCE_DIVERSIFIED    | 263   | Insurance - Diversified                                                                            |
| INDUSTRY_INSURANCE_LIFE           | 264   | Insurance - Life                                                                                   |
| INDUSTRY_INSURANCE_PROPERTY       | 265   | Insurance - Property and casualty                                                                  |
| INDUSTRY_INSURANCE_REINSURANCE    | 266   | Insurance - Reinsurance                                                                            |
| INDUSTRY_INSURANCE_SPECIALTY      | 267   | Insurance - Specialty                                                                              |
| INDUSTRY_MORTGAGE_FINANCE         | 268   | Mortgage finance                                                                                   |
| INDUSTRY_SHELL_COMPANIES          | 269   | Shell companies                                                                                    |
| INDUSTRY_FINANCIAL_FIRST          | 251   | Beginning of enumeration of the financial services types. Corresponds to INDUSTRY_EXCHANGE_TRADED. |
| INDUSTRY_FINANCIAL_LAST           | 269   | End of enumeration of the financial services types. Corresponds to INDUSTRY_SHELL_COMPANIES.       |
| INDUSTRY_FINANCIAL_END            | 300   | Limit of the financial services type enumeration.                                                  |
| Healthcare                        |       |                                                                                                    |
| INDUSTRY_BIOTECHNOLOGY            | 301   | Biotechnology                                                                                      |
| INDUSTRY_DIAGNOSTICS_RESEARCH     | 302   | Diagnostics and research                                                                           |
| INDUSTRY_DRUGS_MANUFACTURERS      | 303   | Drugs manufacturers - general                                                                      |
| INDUSTRY_DRUGS_MANUFACTURERS_SPEC | 304   | Drugs manufacturers - Specialty a                                                                  |
| INDUSTRY_HEALTHCARE_PLANS         | 305   | Healthcare plans                                                                                   |
| INDUSTRY_HEALTH_INFORMATION       | 306   | Health information services                                                                        |
| INDUSTRY_MEDICAL_FACILITIES       | 307   | Medical care facilities                                                                            |
| INDUSTRY_MEDICAL_DEVICES          | 308   | Medical devices                                                                                    |
| INDUSTRY_MEDICAL_DISTRIBUTION     | 309   | Medical distribution                                                                               |
| INDUSTRY_MEDICAL_INSTRUMENTS      | 310   | Medical instruments and supplies                                                                   |
| INDUSTRY_PHARM_RETAILERS          | 311   | Pharmaceutical retailers                                                                           |
| INDUSTRY_HEALTHCARE_FIRST         | 301   | Beginning of enumeration of health                                                                 |

| ID                                   | Value | Description                                                                      |
|--------------------------------------|-------|----------------------------------------------------------------------------------|
|                                      |       | services types. Corresponds to INDUSTRY_BIOTECHNOLOGY.                           |
| INDUSTRY_HEALTHCARE_LAST             | 311   | End of enumeration of healthcare types. Corresponds to INDUSTRY_PHARM_RETAILERS. |
| INDUSTRY_HEALTHCARE_END              | 350   | Limit of the healthcare services type enumeration.                               |
| Industrials                          |       |                                                                                  |
| INDUSTRY_AEROSPACE_DEFENSE           | 351   | Aerospace and defense                                                            |
| INDUSTRY_AIRLINES                    | 352   | Airlines                                                                         |
| INDUSTRY_AIRPORTS_SERVICES           | 353   | Airports and air services                                                        |
| INDUSTRY_BUILDING_PRODUCTS           | 354   | Building products and equipment                                                  |
| INDUSTRY_BUSINESS_EQUIPMENT          | 355   | Business equipment and supplies                                                  |
| INDUSTRY_CONGLOMERATES               | 356   | Conglomerates                                                                    |
| INDUSTRY_CONSULTING_SERVICES         | 357   | Consulting services                                                              |
| INDUSTRY_ELECTRICAL_EQUIPMENT        | 358   | Electrical equipment and parts                                                   |
| INDUSTRY_ENGINEERING_CONSTRUCTION    | 359   | Engineering and construction                                                     |
| INDUSTRY_FARM_HEAVY_MACHINERY        | 360   | Farm and heavy construction machinery                                            |
| INDUSTRY_INDUSTRIAL_DISTRIBUTION     | 361   | Industrial distribution                                                          |
| INDUSTRY_INFRASTRUCTURE_OPERATIONS   | 362   | Infrastructure operations                                                        |
| INDUSTRY_FREIGHT_LOGISTICS           | 363   | Integrated freight and logistics                                                 |
| INDUSTRY_MARINE_SHIPPING             | 364   | Marine shipping                                                                  |
| INDUSTRY_METAL_FABRICATION           | 365   | Metal fabrication                                                                |
| INDUSTRY_POLLUTION_CONTROL           | 366   | Pollution and treatment controls                                                 |
| INDUSTRY_RAILROADS                   | 367   | Railroads                                                                        |
| INDUSTRY_RENTAL_LEASING              | 368   | Rental and leasing services                                                      |
| INDUSTRY_SECURITY_PROTECTION         | 369   | Security and protection services                                                 |
| INDUSTRY_SPECIALTY_BUSINESS_SERVICES | 370   | Specialty business services                                                      |
| INDUSTRY_SPECIALTY_MACHINERY         | 371   | Specialty industrial machinery                                                   |
| INDUSTRY_STUFFING_EMPLOYMENT         | 372   | Stuffing and employment services                                                 |
| INDUSTRY_TOOLS_ACCESSORIES           | 373   | Tools and accessories                                                            |

| ID                               | Value | Description                                                                                                  |
|----------------------------------|-------|--------------------------------------------------------------------------------------------------------------|
| INDUSTRY_TRUCKING                | 374   | Trucking                                                                                                     |
| INDUSTRY_WASTE_MANAGEMENT        | 375   | Waste management                                                                                             |
| INDUSTRY_INDUSTRIALS_FIRST       | 351   | Beginning of enumeration of industries types. Corresponds to INDUSTRY_AEROSPACE_DEFENSE.                     |
| INDUSTRY_INDUSTRIALS_LAST        | 375   | End of enumeration of industry types. Corresponds to INDUSTRY_WASTE_MANAGEMENT.                              |
| INDUSTRY_INDUSTRIALS_END         | 400   | Limit of the industry types enumeration.                                                                     |
| Real estate                      |       |                                                                                                              |
| INDUSTRY_REAL_ESTATE_DEVELOPMENT | 401   | Real estate - Development                                                                                    |
| INDUSTRY_REAL_ESTATE_DIVERSIFIED | 402   | Real estate - Diversified                                                                                    |
| INDUSTRY_REAL_ESTATE_SERVICES    | 403   | Real estate services                                                                                         |
| INDUSTRY_REIT_DIVERSIFIED        | 404   | REIT - Diversified                                                                                           |
| INDUSTRY_REIT_HEALTHCARE         | 405   | REIT - Healthcare facilities                                                                                 |
| INDUSTRY_REIT_HOTEL_MOTEL        | 406   | REIT - Hotel and motel                                                                                       |
| INDUSTRY_REIT_INDUSTRIAL         | 407   | REIT - Industrial                                                                                            |
| INDUSTRY_REIT_MORTGAGE           | 408   | REIT - Mortgage                                                                                              |
| INDUSTRY_REIT_OFFICE             | 409   | REIT - Office                                                                                                |
| INDUSTRY_REIT_RESIDENTIAL        | 410   | REIT - Residential                                                                                           |
| INDUSTRY_REIT_RETAIL             | 411   | REIT - Retail                                                                                                |
| INDUSTRY_REIT_SPECIALITY         | 412   | REIT - Specialty                                                                                             |
| INDUSTRY_REAL_ESTATE_FIRST       | 401   | Beginning of enumeration of the real estate services types. Corresponds to INDUSTRY_REAL_ESTATE_DEVELOPMENT. |
| INDUSTRY_REAL_ESTATE_LAST        | 412   | End of enumeration of the real estate services types. Corresponds to INDUSTRY_REIT_SPECIALITY.               |
| INDUSTRY_REAL_ESTATE_END         | 450   | Limit of the real estate services types enumeration.                                                         |
| Technology                       |       |                                                                                                              |
| INDUSTRY_COMMUNICATION_EQUIPMENT | 451   | Communication equipment                                                                                      |
| INDUSTRY_COMPUTER_HARDWARE       | 452   | Computer hardware                                                                                            |

| ID                                    | Value | Description                                                                                     |
|---------------------------------------|-------|-------------------------------------------------------------------------------------------------|
| INDUSTRY_CONSUMER_ELECTRONICS         | 453   | Consumer electronics                                                                            |
| INDUSTRY_ELECTRONIC_COMPONENTS        | 454   | Electronic components                                                                           |
| INDUSTRY_ELECTRONIC_DISTRIBUTION      | 455   | Electronics and computer distribu                                                               |
| INDUSTRY_IT_SERVICES                  | 456   | Information technology services                                                                 |
| INDUSTRY_SCIENTIFIC_INSTRUMENTS       | 457   | Scientific and technical instrumen                                                              |
| INDUSTRY_SEMICONDUCTOR_EQUIPMENT      | 458   | Semiconductor equipment and ma                                                                  |
| INDUSTRY_SEMICONDUCTORS               | 459   | Semiconductors                                                                                  |
| INDUSTRY_SOFTWARE_APPLICATION         | 460   | Software - Application                                                                          |
| INDUSTRY_SOFTWARE_INFRASTRUCTURE      | 461   | Software - Infrastructure                                                                       |
| INDUSTRY_SOLAR                        | 462   | Solar                                                                                           |
| INDUSTRY_TECHNOLOGY_FIRST             | 451   | Beginning of enumeration of high-<br>industry types. Corresponds to<br>INDUSTRY_COMMUNICATION_E |
| INDUSTRY_TECHNOLOGY_LAST              | 462   | End of enumeration of high-tech in<br>types. Corresponds to INDUSTRY_                           |
| INDUSTRY_TECHNOLOGY_END               | 500   | Limit of the high-tech industry typ<br>enumeration.                                             |
| Utilities                             |       |                                                                                                 |
| INDUSTRY_UTILITIES_DIVERSIFIED        | 501   | Utilities - Diversified                                                                         |
| INDUSTRY_UTILITIES_POWERPRODUCERS     | 502   | Utilities - Independent power prod                                                              |
| INDUSTRY_UTILITIES_RENEWABLE          | 503   | Utilities - Renewable                                                                           |
| INDUSTRY_UTILITIES_REGULATED_ELECTRIC | 504   | Utilities - Regulated electric                                                                  |
| INDUSTRY_UTILITIES_REGULATED_GAS      | 505   | Utilities - Regulated gas                                                                       |
| INDUSTRY_UTILITIES_REGULATED_WATER    | 506   | Utilities - Regulated water                                                                     |
| INDUSTRY_UTILITIES_FIRST              | 501   | Beginning of enumeration of utiliti<br>types. Corresponds to<br>INDUSTRY_UTILITIES_DIVERSIFI    |
| INDUSTRY_UTILITIES_LAST               | 506   | End of enumeration of utilities ser<br>Corresponds to<br>INDUSTRY_UTILITIES_REGULATI            |
| INDUSTRY_UTILITIES_END                | 550   | Limit of the utilities services types<br>enumeration.                                           |
| Commodities                           |       |                                                                                                 |

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| <b>ID</b>                         | <b>Value</b> | <b>Description</b>                                                                                 |
|-----------------------------------|--------------|----------------------------------------------------------------------------------------------------|
| INDUSTRY_COMMODITIES_AGRICULTURAL | 551          | Agriculture                                                                                        |
| INDUSTRY_COMMODITIES_ENERGY       | 552          | Energy                                                                                             |
| INDUSTRY_COMMODITIES_METALS       | 553          | Metals                                                                                             |
| INDUSTRY_COMMODITIES_PRECIOUS     | 554          | Precious metals                                                                                    |
| INDUSTRY_COMMODITIES_FIRST        | 551          | Beginning of enumeration of commodity types<br>Corresponds to<br>INDUSTRY_COMMODITIES_AGRICULTURAL |
| INDUSTRY_COMMODITIES_LAST         | 554          | End of enumeration of commodity types<br>Corresponds to<br>INDUSTRY_COMMODITIES_PRECIOUS           |
| INDUSTRY_COMMODITIES_END          | 600          | Limit of the commodity type enumeration                                                            |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export / mt5\_symbols\_sessions

mt5\_symbols\_sessions

Data on symbols' trade and quoting sessions is exported to the table. The table contains the following fields:

| Name       | Type    | Description                                                                                                                  |
|------------|---------|------------------------------------------------------------------------------------------------------------------------------|
| Session_ID | Integer | Primary key. Unique session ID. Assigned automatically during the export.                                                    |
| Symbol_ID  | Integer | Unique <u>ID of the symbol</u> , to which the session is applied.                                                            |
| Type       | Integer | Session type: 0 - quoting, 1 - trade.                                                                                        |
| Day        | Integer | Day of the week from 0 to 6. 0 - Sunday, 6 - Saturday.                                                                       |
| Open       | Integer | The opening time of a trading or quoting session of a symbol in minutes elapsed since 00:00. For example, 100 denotes 01:40. |
| Close      | Integer | The closing time of a trading or quoting session of a symbol in minutes elapsed since 00:00.                                 |



MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_groups

mt5\_groups

Groups' configurations are exported to the table. The table contains the following fields:

| Name                | Type    | Description                                                                                                                                                                    |
|---------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group_ID            | Integer | Primary key. Unique group ID for more efficient request of the group data from the database. Assigned automatically during the export.                                         |
| Timestamp           | Integer | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed. |
| Group               | String  | The name of a group, including a path to it in accordance with the hierarchy.                                                                                                  |
| Server              | Integer | The ID of the trade server, to which the group is linked.                                                                                                                      |
| PermissionFlags     | Integer | Flags of group permissions. Passed as a value of the <u>EnPermissionsFlags</u> enumeration (sum of values of appropriate flags).                                               |
| AuthMode            | Integer | Authorization mode for accounts in the group. Passed in a value of the <u>EnAuthMode</u> enumeration.                                                                          |
| AuthPasswordMin     | Integer | The minimum password length for accounts in the group.                                                                                                                         |
| Company             | String  | Name of the company that services the group.                                                                                                                                   |
| CompanyPage         | String  | The website address of the company that services the group.                                                                                                                    |
| CompanyEmail        | String  | The email address of the company that services the group.                                                                                                                      |
| CompanySupportPage  | String  | The technical support website address of the company that services the group.                                                                                                  |
| CompanySupportEmail | String  | The technical support email address of the company that services the group.                                                                                                    |
| CompanyCatalog      | String  | The name of the subdirectory that stores the templates of reports, emails, etc. for the company that services this group.                                                      |

# MetaTrader 5 trading platform

| Name               | Type                     | Description                                                                                                                                                               |
|--------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Currency           | String                   | The group deposit currency.                                                                                                                                               |
| CurrencyDigits     | Integer                  | The number of digits after the decimal point in the group deposit currency.                                                                                               |
| ReportsMode        | Integer                  | Report generation modes. Passed in a value of the <u>EnReportsMode</u> enumeration.                                                                                       |
| ReportsFlags       | Integer                  | Report sending options. Passed as a value of the <u>EnReportFlags</u> enumeration (sum of values of appropriate flags).                                                   |
| ReportsEmail       | String                   | <u>The mail server</u> used for sending reports to clients from the group.                                                                                                |
| ReportsSMTP        | String                   | Address of SMTP server for sending reports. The field is obsolete and is not updated.                                                                                     |
| ReportsSMTPLogin   | String                   | A login for the authorization on the SMTP server that is used for sending reports. The field is obsolete and is not updated.                                              |
| NewsMode           | Integer                  | The mode of news sending to the clients from the group. Passed in a value of the <u>EnNewsMode</u> enumeration.                                                           |
| NewsCategory       | String                   | The categories of news received by the group. Use the backslash character "\" to specify subcategories.                                                                   |
| NewsLangs          | Array of integer numbers | The array of languages, in which the group receives news. The language is specified in the LANGID format used in the <u>MS Windows</u> (value from Prim.lang.identifier). |
| MailMode           | Integer                  | The mode of operation of the internal mail system for the group. Passed in a value of the <u>EnMailMode</u> enumeration.                                                  |
| TradeFlags         | Integer                  | Trade options of the group. Passed as a value of the <u>EnTradeFlags</u> enumeration (sum of values of appropriate flags).                                                |
| TradeInterestrates | Float                    | The annual interest rate on deposits of the group accounts.                                                                                                               |
| TradeVirtualCredit | Float                    | The amount of additional funds that a brokerage company can provide to a client for opening a position with a volume larger than allowed by the client's current funds.   |
| MarginFreeMode     | Integer                  | The mode of using of floating profit/loss in the free margin. Passed in a value of the                                                                                    |

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| Name                 | Type    | Description                                                                                                                                                |
|----------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |         | <u>EnFreeMarginMode</u> enumeration.                                                                                                                       |
| MarginSOMode         | Integer | The mode of checking the levels of Stop Out and Margin Call. Passed in a value of the <u>EnStopOutMode</u> enumeration.                                    |
| MarginCall           | Float   | The level of Margin Call. Units are determined by the MarginSOMode parameter.                                                                              |
| MarginStopOut        | Float   | The level of Stop Out. Units are determined by the MarginSOMode parameter.                                                                                 |
| MarginFreeProfitMode | Integer | The mode of using the profit/loss fixed during a trade day in the free margin.                                                                             |
| MarginMode           | Integer | The risk management model of the group.                                                                                                                    |
| MarginFlags          | Integer | Margin calculation flags.                                                                                                                                  |
| DemoLeverage         | Integer | The default credit leverage for demo accounts opened in the group.                                                                                         |
| DemoDeposit          | Float   | The default amount of deposit for demo accounts opened in the group.                                                                                       |
| LimitHistory         | Integer | The maximum number of days, for which the group can request data on conducted trade operation. Passed in a value of the <u>EnHistoryLimit</u> enumeration. |
| LimitOrders          | Integer | The maximum number of orders that can be simultaneously placed by an account from this group.                                                              |
| LimitSymbols         | Integer | The maximum number of symbols, for which an account can simultaneously receive quotes.                                                                     |
| LimitPositions       | Integer | The maximum number of open positions which the client can have on the account at the same time.                                                            |
| LimitPositionsVolume | Float   | Currently the field is not used.                                                                                                                           |
| TradeTransferMode    | Integer | The mode of transferring funds between accounts.                                                                                                           |

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## Enumerations

To pass information about groups the following enumerations are used:

- [EnPermissionsFlags](#)
- [EnAuthMode](#)
- [EnReportsMode](#)
- [EnReportsFlags](#)
- [EnNewsMode](#)
- [EnMailMode](#)
- [EnHistoryLimit](#)
- [EnFreeMarginMode](#)
- [EnStopOutMode](#)
- [EnTradeFlags](#)
- [EnMarginFreeProfitFlags](#)

### EnPermissionsFlags

Flags of permissions for groups are listed in EnPermissionsFlags.

| ID                           | Value      | Description                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PERMISSION_NONE              | 0x00000000 | No permissions. Default value.                                                                                                                                                                                                                                                                                                              |
| PERMISSION_CERT_CONFIRM      | 0x00000001 | Enable confirmation of certificates.                                                                                                                                                                                                                                                                                                        |
| PERMISSION_ENABLE_CONNECTION | 0x00000002 | Allow client connections.                                                                                                                                                                                                                                                                                                                   |
| PERMISSION_RESET_PASSWORD    | 0x00000004 | Force users to change their master password at first login. A user will not be able to take any actions before changing the password.                                                                                                                                                                                                       |
| PERMISSION_FORCED_OTP_USAGE  | 0x00000008 | In some countries, regulators require use of additional account security measures, such as the use of OTP. When this flag is enabled, all clients in this group will need to use one-time passwords to connect. Otherwise, clients can bind their accounts to the generator or use the default authentication method. Before you enable it, |

| ID                            | Value      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               |            | <p>please inform your clients about the new OTP option. Be extremely careful when enabling this option for the only one manager group on the trading server.</p> <p>For generating one-time passwords, the mobile terminal MetaTrader 5 for iPhone is used for generating one-time passwords.</p>                                                                                                                                                                                                                                                                   |
| PERMISSION_RISK_WARNING       | 0x00000010 | <p>If this flag is enabled, when a client connects in the trading terminal, a warning about the risks associated with operations on the financial markets appears. Trade operations on a client's account are not allowed until the client confirms that he or she has read the warning and is aware of the risks. To confirm, the client should check "I am aware of the risks and I wish to trade high risk investment products". This warning is displayed once per session of the terminal. The next time it will appear after the restart of the terminal.</p> |
| PERMISSION_REGULATION_PROTECT | 0x00000020 | <p>Enforce country-specific regulatory restrictions for retail clients.</p> <p>Every country has regulators state bodies that supervise the activities of financial institutions (including brokers). Each regulator has its own set of requirements applied to trading currencies, securities and other instruments.</p>                                                                                                                                                                                                                                           |

| ID                       | Value      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |            | <p>In most cases, clients of brokerage firms are individuals from different countries. The platform provides the means of meeting the requirements of certain regulators without interfering with the work of traders not covered with these requirements. Restrictions applied to trader accounts depend on the client's country and other conditions set by that country's regulator.</p> <p>For example, the National Securities Market Commission (Comisión Nacional del Mercado de Valores) of Spain compels brokers to warn clients using the leverage of 1:10 or higher about potential risks in a special way. Therefore, if a client's country is Spain and they use a leverage of 1:10 or higher, an additional warning is displayed in the trading dialog</p> <p>Currently, only one regional limitation is used in the platform. The list is to be expanded later.</p> |
| PERMISSION_NOTIFY_DEALS  | 0x00000040 | Allow accounts to subscribe to <u>server push notifications</u> about deals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| PERMISSION_NOTIFY_ORDERS | 0x00000080 | Allow accounts to subscribe to server push notifications about orders.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| ID                         | Value      | Description                                                                        |
|----------------------------|------------|------------------------------------------------------------------------------------|
| PERMISSION_NOTIFY_BALANCES | 0x00000100 | Allow accounts to subscribe to server push notifications about balance operations. |

### EnAuthMode

Types of authorization of clients in the group are listed in EnAuthMode.

| ID            | Value | Description                                      |
|---------------|-------|--------------------------------------------------|
| AUTH_STANDARD | 0     | Standard authorization.                          |
| AUTH_RSA1024  | 1     | Extended authorization with 1024-bit encryption. |
| AUTH_RSA2048  | 2     | Extended authorization with 2048-bit encryption. |

### EnReportsMode

Report generation modes are listed in EnReportsMode.

| ID                 | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REPORTS_DISABLED   | 0     | Reports are disabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| REPORTS_FULL       | 1     | <p>The platform can save the end-of-day and/or end-of-month states of accounts to a special database. The information includes balance, equity, margin, and other details. The database is located on the trade server, in the bases\daily\daily_*.dat file. Data from the file is used in end-of-day and end-of-month trading reports sent to clients, as well as in some manager reports. If the mode is enabled, the platform will save information on accounts from the selected group to the database.</p> <p>This flag enables the generation of both end-of-day and end-of-month data.</p> |
| REPORTS_DAY_ONLY   | 2     | Enable data generation for reports only at the end of the day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| REPORTS_MONTH_ONLY | 3     | Enable data generation for reports only at the end of the month.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### EnReportsFlags

Report generation options are listed in IMTConGroup::EnReportsFlags.

| ID                | Value | Description                    |
|-------------------|-------|--------------------------------|
| REPORTSFLAGS_NONE | 0     | No additional options enabled. |

## MetaTrader 5 trading platform

| ID                      | Value | Description                                                                                                                                                                                                                                                |
|-------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REPORTSFLAGS_EMAIL      | 1     | Enables sending of generated HTML report files to clients by email. Addresses from <u>trading accounts</u> are used for sending.                                                                                                                           |
| REPORTSFLAGS_SUPPORT    | 2     | Enables sending of report copies to a <u>technical support email address</u> .                                                                                                                                                                             |
| REPORTSFLAGS_STATEMENTS | 4     | Enables account state report generation. The HTML report files are created using templates from the \templates\confirmation\ and \templates\statement\ folders on the trade server. To generate reports, the <u>REPORTS_STANDARD</u> mode must be enabled. |

### EnNewsMode

Modes of news sending are listed in EnNewsMode.

| ID                 | Value | Description               |
|--------------------|-------|---------------------------|
| NEWS_MODE_DISABLED | 0     | News sending is disabled. |
| NEWS_MODE_HEADERS  | 1     | Only news headers.        |
| NEWS_MODE_FULL     | 2     | Full package.             |

### EnMailMode

Modes of using the internal mail system are listed in EnMailMode.

| ID                 | Value | Description                       |
|--------------------|-------|-----------------------------------|
| MAIL_MODE_DISABLED | 0     | Disable the internal mail system. |
| MAIL_MODE_FULL     | 1     | Enable the internal mail system.  |

### EnHistoryLimit

The intervals of trading history available to clients in the group are listed in EnHistoryLimit.

| ID                     | Value | Description         |
|------------------------|-------|---------------------|
| TRADE_HISTORY_ALL      | 0     | The entire history. |
| TRADE_HISTORY_MONTHS_1 | 1     | One month.          |
| TRADE_HISTORY_MONTHS_3 | 2     | Three months.       |
| TRADE_HISTORY_MONTHS_6 | 3     | Six months.         |



## MetaTrader 5 trading platform

| ID                   | Value | Description  |
|----------------------|-------|--------------|
| TRADE_HISTORY_YEAR_1 | 4     | One year.    |
| TRADE_HISTORY_YEAR_2 | 5     | Two years.   |
| TRADE_HISTORY_YEAR_3 | 6     | Three years. |

EnFreeMarginMode

Modes of using the floating profit/loss in the free margin are listed in EnFreeMarginMode.

| ID                     | Value | Description                        |
|------------------------|-------|------------------------------------|
| FREE_MARGIN_NOT_USE_PL | 0     | Do not use unrealized profit/loss. |
| FREE_MARGIN_USE_PL     | 1     | Use unrealized profit/loss.        |
| FREE_MARGIN_PROFIT     | 2     | Use unrealized profit.             |
| FREE_MARGIN_LOSS       | 3     | Use unrealized loss.               |

EnStopOutMode

Modes for checking Margin Call and Stop Out are listed in EnStopOutMode.

| ID              | Value | Description                                                 |
|-----------------|-------|-------------------------------------------------------------|
| STOPOUT_PERCENT | 0     | The levels of Margin Call and Stop Out in percentage terms. |
| STOPOUT_MONEY   | 1     | The levels of Margin Call and Stop Out in money terms.      |

EnTradeFlags

Group trade options are enumerated in EnTradeFlags.

| ID                     | Value      | Description                                                       |
|------------------------|------------|-------------------------------------------------------------------|
| TRADEFLAGS_NONE        | 0x00000000 | Options are disabled.                                             |
| TRADEFLAGS_SWAPS       | 0x00000001 | Allow charging of swaps.                                          |
| TRADEFLAGS_TRAILING    | 0x00000002 | Enable trailing stop.                                             |
| TRADEFLAGS_EXPERTS     | 0x00000004 | Enable trading using Expert Advisors.                             |
| TRADEFLAGS_EXPIRATION  | 0x00000008 | Enable order expiration.                                          |
| TRADEFLAGS_SIGNALS_ALL | 0x00000010 | Allow using the <u>"Signals"</u> service in the client terminals. |
| TRADEFLAGS_SIGNALS_OWN | 0x00000020 |                                                                   |

| ID                         | Value      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |            | Allow using the <u>signals</u> from over-the-counter servers only. If this flag is set, only signals created in this groups will be able to be used. Signals created on the basis of accounts opened by your brokerage company. Signals created on the basis of other accounts will not be displayed in client terminals.                                                                                                                                                                                                                                                                                                                                                  |
| TRADEFLAGS_SO_COMPENSATION | 0x00000040 | Automatically execute on a client account the special "stop-out compensation" operation, which increases the balance and sets it to zero, if the balance has become negative after a position was closed by Stop Out. For more details please read the <u>corresponding section</u> .                                                                                                                                                                                                                                                                                                                                                                                      |
| TRADEFLAGS_SO_FULLY_HEDGED | 0x00000080 | If the flag is enabled, then Stop Out will be performed on accounts with open positions, the zero margin (positions are covered) and negative equity. If the option is disabled, orders and positions will not be forcibly closed in above cases. The flag can be only used on <u>hedging accounts</u> .                                                                                                                                                                                                                                                                                                                                                                   |
| TRADEFLAGS_FIFO_CLOSE      | 0x00000100 | <p>The position closing mode by FIFO rule. If the flag is enabled, then positions for each instrument can only be closed only in the order in which they were opened: the oldest one should be closed first, then the next one, etc. The option is only for hedging accounts, in which traders can have multiple positions for the same financial instrument. There are three main methods to close a position; the flag behavior will be different for each of the methods:</p> <ul style="list-style-type: none"> <li>• Closing from the client terminal: the trader closes position manually, using trading robot, based on the Signals service subscription</li> </ul> |

| ID                        | Value      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |            | <p>etc. In case of an attempt to close a position, which does not meet the FIFO rule, the trader will receive an appropriate error.</p> <ul style="list-style-type: none"> <li>• Closing upon Stop Loss or Take Profit activation: the orders are processed on the server side, so the position closure is not requested from the trader (terminal) side, but initiated by the server. If a Stop Loss or Take Profit triggers a position, and this position does not comply with the FIFO rule (there is an older position from the same symbol), the position will not be closed. An appropriate message will be printed to the log: "position close prohibited by FIFO rule. position #100448219 buy EURUSD 1.11544 sl: 1.11549 tp: 1.11549 take profit activation skipped".</li> <li>• Closing upon Stop Out triggering: such operation is also processed on the server side. In a normal mode, it is disabled, in case of <u>Stop Out</u> positions are closed starting with the one having the least loss. If this option is enabled, the open time will be additionally checked for all positions. The server determines losing positions for each symbol, finds the oldest position for each symbol, and then closes the one which has the greatest loss among the found positions.</li> </ul> |
| TRADEFLAGS_HEDGE_PROHIBIT | 0x00000200 | Prohibit opening of opposite positions and placing of opposite orders. If this option is enabled, accounts are not allowed to have oppositely directed positions and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| ID                                | Value      | Description                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   |            | orders for the same financial instrument. For example, if the account has a Buy position, the user cannot open a Sell position or place a pending sell order for the same symbol. If such an attempt is made, the user will receive an error. The option is only valid for groups with the <u>hedging</u> position accounting mode. |
| TRADEFLAGS_DEAL_COST              | 0x00000400 | Calculate deal execution <u>costs</u> and display them in client terminals. NFA regulated brokers should enable this option.                                                                                                                                                                                                        |
| TRADEFLAGS_SO_COMPENSATION_CREDIT | 0x00000800 | Works as an addition to the TRADEFLAGS_SO_COMPENSATION flag. If enabled, the credit fund of the account will be set to zero and the negative balance compensation operation. <u>Credit funds</u> are withdrawn in a separate balance operation with the "so credit compensation" type.                                              |

### EnMarginFreeProfitFlags

Modes of using of the profit/loss fixed during a trade day in the free margin are enumerated in EnMarginFreeProfitFlags.

| ID                      | Value | Description                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FREE_MARGIN_PROFIT_PL   | 0     | Include both profit and loss fixed during a day in the free margin.                                                                                                                                                                                                                                                                                                                                    |
| FREE_MARGIN_PROFIT_LOSS | 1     | Include only loss fixed during a day in the free margin. The profits of the client during a trading day are accumulated in the <u>BlockedProfit</u> field of the trade account and are not included in the free margin. At the end of the trading day, the accumulated profits are credited to the balance and the value of IMTAccount::BlockedProfit is reset. fixed during a day in the free margin. |

### EnMarginMode

The models of risk management are enumerated in EnMarginMode. The model defines the type of pre-trade control and the position accounting system used.

| <b>ID</b>                     | <b>Value</b> | <b>Description</b>                                                                                                                                                                                   |
|-------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARGIN_MODE_RETAIL            | 0            | Used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Netting position accounting system is used.                                               |
| MARGIN_MODE_EXCHANGE_DISCOUNT | 1            | Used for the exchange market. Margin calculation is based on the discounts specified in symbol settings. Discounts are set by the broker, however they cannot be lower than the exchange set values. |
| MARGIN_MODE_RETAIL_HEDGED     | 2            | Used for the OTC market. Margin calculation is based on the type of instrument, as well as group settings. Hedging position accounting system is used.                                               |

EnMarginFlags

EnMarginFlags contains the margin calculation flags.

| <b>ID</b>              | <b>Value</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MARGIN_FLAGS_NONE      | 0            | No flags.                                                                                                                                                                                                                                                                                                                                                                                            |
| MARGIN_FLAGS_CLEAR_ACC | 1            | The flag is available only in the <u>FREE_MARGIN_PROFIT_LOSS</u> mode. If enabled, the profit accumulated by a client will be released (and thus included in the free margin) at the end of trade day. If this flag is disabled, then it will be possible to release the accumulated profit only using an external application (for example, a gateway). The server will not perform this operation. |

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mt5\_groups\_symbols

Individual symbol settings for groups are exported to the table. The table contains the following fields:

| Name              | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symbol_ID         | Integer | Primary key. Unique symbol ID for the group.                                                                                                                                                                                                                                                                                                                                                                                          |
| Group_ID          | Integer | <u>ID of the group</u> , to which symbol settings are applied.                                                                                                                                                                                                                                                                                                                                                                        |
| Timestamp         | Integer | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                                                                                                                                                        |
| Path              | String  | The path to a symbol or group of symbols that are subject to the special group settings.                                                                                                                                                                                                                                                                                                                                              |
| TradeMode         | Integer | The symbol trading mode for the group. Passed in a value of the <u>EnTradeMode</u> enumeration.                                                                                                                                                                                                                                                                                                                                       |
| ExecMode          | Integer | The symbol execution mode for the group. Passed in a value of the <u>EnExecutionMode</u> enumeration.                                                                                                                                                                                                                                                                                                                                 |
| FillFlags         | Integer | Types of filling allowed for the symbol in this group. Passed as a value of the <u>EnFillingFlags</u> enumeration (sum of values of appropriate flags).                                                                                                                                                                                                                                                                               |
| ExpirFlags        | Integer | Types of order expiration allowed for the symbol in this group. Passed as a value of the <u>EnExpirationFlags</u> enumeration (sum of values of appropriate flags).                                                                                                                                                                                                                                                                   |
| SpreadDiff        | Integer | Difference between the symbol spread for the group and the default spread.                                                                                                                                                                                                                                                                                                                                                            |
| SpreadDiffBalance | Integer | The balance of spread difference set for the group. The balance of spread difference is set a shift from the equal distribution of the spread difference value between Bid and Ask prices. For example, if the spread difference is equal to 4 and it is distributed as -2 Bid/+2 Ask, then the balance of spread difference value is 0. The -3 Bid/+1 Ask ratio corresponds to value -1, ratio -1 Bid/+3 Ask corresponds to value 1. |
| StopsLevel        | Integer | The price band, within which the group is not allowed to place stop orders for a symbol.                                                                                                                                                                                                                                                                                                                                              |

## MetaTrader 5 trading platform

| Name              | Type    | Description                                                                                                                                         |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| FreezeLevel       | Integer | The price band, within which it is not allowed to modify orders and positions for the group.                                                        |
| VolumeMin         | Integer | The minimum volume of trade operations for a symbol for the group. One unit corresponds to 1/10000 lot.                                             |
| VolumeMinExt      | Integer | The minimum volume (with extended accuracy) of trade operations for a symbol for the group. One unit corresponds to 1/100000000 lot.                |
| VolumeMax         | Integer | The maximum volume of trade operations for a symbol for the group. One unit corresponds to 1/10000 lot.                                             |
| VolumeMaxExt      | Integer | The maximum volume (with extended accuracy) of trade operations for a symbol for the group. One unit corresponds to 1/100000000 lot.                |
| VolumeStep        | Integer | The step of change of trade operations volume for a symbol for the group. One unit corresponds to 1/10000 lot.                                      |
| VolumeStepExt     | Integer | The step of change of trade operations volume (with extended accuracy) for a symbol for the group. One unit corresponds to 1/100000000 lot.         |
| VolumeLimit       | Integer | The maximum allowed aggregate volume of positions and orders for a symbol in one direction for this group. One unit corresponds to 1/10000 lot.     |
| VolumeLimitExt    | Integer | The maximum aggregate volume (with extended accuracy) of positions and orders for a symbol for this group. One unit corresponds to 1/100000000 lot. |
| MarginFlags       | Integer | The additional modes of symbol margin checking for the group. Passed in a value of the <a href="#">EnMarginFlags</a> enumeration.                   |
| MarginInitial     | Float   | The size of initial symbol margin for the group.                                                                                                    |
| MarginMaintenance | Float   | The size of symbol maintenance margin for the group.                                                                                                |
| MarginLong        | Float   | The group margin ratio for long positions and orders for a symbol.                                                                                  |
| MarginShort       | Float   | The group margin ratio for short positions and orders for a symbol.                                                                                 |
| MarginLimit       | Float   | The group margin ratio of limit orders for a symbol.                                                                                                |
| MarginStop        | Float   | The group margin ratio of stop orders for a symbol.                                                                                                 |

MetaTrader 5 trading platform

| Name              | Type    | Description                                                                                                                                                                        |
|-------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MarginStopLimit   | Float   | The group margin ratio for stop-limit orders for a symbol.                                                                                                                         |
| MarginHedged      | Float   | The hedged margin value.                                                                                                                                                           |
| SwapMode          | Integer | The swap calculation mode for a certain symbol for the group. Passed in a value of the <u>EnSwapMode</u> enumeration.                                                              |
| SwapLong          | Float   | The long position swap for a symbol for the group.                                                                                                                                 |
| SwapShort         | Integer | The short position swap for a symbol for the group.                                                                                                                                |
| SwapYearDays      | Integer | The number of days in a year used in calculating swap percent for a given group. Passed by the <u>EnSwapDays</u> enumeration value.                                                |
| SwapFlags         | Integer | Additional swap settings by symbol for the given group. Passed by the <u>EnSwapFlags</u> enumeration value.                                                                        |
| SwapRateSunday    | Float   | Sunday swap multiplier in symbol settings for the given group.                                                                                                                     |
| SwapRateMonday    | Float   | Monday swap multiplier in symbol settings for the given group.                                                                                                                     |
| SwapRateTuesday   | Float   | Tuesday swap multiplier in symbol settings for the given group.                                                                                                                    |
| SwapRateWednesday | Float   | Wednesday swap multiplier in symbol settings for the given group.                                                                                                                  |
| SwapRateThursday  | Float   | Thursday swap multiplier in symbol settings for the given group.                                                                                                                   |
| SwapRateFriday    | Float   | Friday swap multiplier in symbol settings for the given group.                                                                                                                     |
| SwapRateSaturday  | Float   | Saturday swap multiplier in symbol settings for the given group.                                                                                                                   |
| RETimeout         | Integer | Time in seconds during which the price issued by a dealer in the request execution mode is valid.                                                                                  |
| IECheckMode       | Integer | The mode of checking during instant execution set for a group. Passed in a value of the <u>EnInstantMode</u> enumeration.                                                          |
| IETimeout         | Integer | The maximum allowed difference between the time of arrival of the price, at which the client places an order, and the time of the last price. The timeout is specified in seconds. |



# MetaTrader 5 trading platform

| Name                                                                                                                            | Type    | Description                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IESlipProfit                                                                                                                    | Integer | The maximum allowed slippage in the profitable direction during instant execution.                                                                                                                                                   |
| IESlipLosing                                                                                                                    | Integer | The maximum allowed slippage in the loss direction during instant execution.                                                                                                                                                         |
| IEVolumeMax                                                                                                                     | Integer | The maximum volume of a trade operation that can be executed in the instant execution mode. One unit corresponds to 1/10000 lot.                                                                                                     |
| IEVolumeMaxExt                                                                                                                  | Integer | The maximum volume (with extended accuracy) of a trade operation that can be executed in the instant execution mode. One unit corresponds to 1/100000000 lot.                                                                        |
| OrderFlags                                                                                                                      | Integer | The flags of order types that are allowed for the symbol. Passed in a value of the <u>EnOrderFlags</u> enumeration (sum of values of appropriate flags).                                                                             |
| MarginRateLiquidity                                                                                                             | Float   | The liquidity rate of the symbol for the group. It determines the amount of the current value of an asset for the specified financial instrument, which will be taken into account as collateral (accounted for in client's equity). |
| REFlags                                                                                                                         | Integer | The flags of request execution for the group.                                                                                                                                                                                        |
| NULL value in the fields beginning from TradeMode means that the appropriate setting is inherited from the <u>base symbol</u> . |         |                                                                                                                                                                                                                                      |

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mt5\_commissions

Commission settings for groups are exported to this table. The table contains the following fields:

| Name             | Type    | Purpose                                                                                                                                                                                                              |
|------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commission_ID    | Integer | Initial key. A unique <u>commission setting ID</u> .                                                                                                                                                                 |
| Group_ID         | Integer | <u>ID of the group</u> , a commission setting belongs to.                                                                                                                                                            |
| Name             | String  | Commission setting name (up to 64 characters).                                                                                                                                                                       |
| Description      | String  | Commission setting description (up to 64 characters).                                                                                                                                                                |
| Path             | String  | Path to a symbol or a group of symbols covered by a commission setting.                                                                                                                                              |
| Mode             | Integer | Commission type: 0 standard, 1 agent.                                                                                                                                                                                |
| ModeRange        | Integer | Commission <u>levels type</u> : <ul style="list-style-type: none"> <li>• 0 volume</li> <li>• 1 turnover in money</li> <li>• 2 turnover in volume</li> </ul>                                                          |
| ModeCharge       | Integer | Commission <u>charge mode</u> : <ul style="list-style-type: none"> <li>• 0 daily</li> <li>• 1 monthly</li> <li>• 2 instant</li> </ul>                                                                                |
| TurnoverCurrency | String  | Currency, in which the <u>money turnover</u> is calculated.                                                                                                                                                          |
| ModeEntry        | Integer | Commission calculation mode depending on the trade direction: <ul style="list-style-type: none"> <li>• 0 all trades regardless of direction.</li> <li>• 1 only entry deals.</li> <li>• 2 only exit deals.</li> </ul> |
| ModeAction       | Integer | Commission calculation mode depending on the trade type: <ul style="list-style-type: none"> <li>• 0 all trades regardless of type.</li> <li>• 1 only Buy deals.</li> <li>• 2 only Sell deals.</li> </ul>             |
| ModeProfit       | Integer | Commission calculation modes depending on the deal profit:                                                                                                                                                           |

# MetaTrader 5 trading platform

| Name       | Type    | Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |         | <ul style="list-style-type: none"> <li>• 0 all deals.</li> <li>• 1 only profitable deals.</li> <li>• 2 only losing deals.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ModeReason | Integer | <p>Commission calculation modes depending on the reason for the deal.</p> <ul style="list-style-type: none"> <li>• 0x00000000 no commission will be charged for any trades.</li> <li>• 0x00000001 the deal was performed by the client manually via the client terminal.</li> <li>• 0x00000002 the deal was performed by the client using an Expert Advisor.</li> <li>• 0x00000004 the deal was performed by a dealer via the Manager terminal.</li> <li>• 0x00000008 the deal was performed from an external trading system.</li> <li>• 0x00000010 the deal was performed via the MetaTrader 5 mobile terminal for Android or iPhone.</li> <li>• 0x00000020 the deal was performed via the web terminal.</li> <li>• 0x00000040 the deal was performed as a result of copying of a trading signal, in accordance with the subscription, in the client terminal.</li> </ul> |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SQL Export / mt5\_commissions\_tiers

mt5\_commissions\_tiers

Commission levels are exported to this table. The table contains the following fields:

| Name          | Type    | Purpose                                                                                                                                                                                                                                                              |
|---------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier_ID       | Integer | Initial key. Unique <u>commission level ID</u> .                                                                                                                                                                                                                     |
| Commission_ID | Integer | <u>ID of the commission setting</u> the level belongs to.                                                                                                                                                                                                            |
| Mode          | Integer | Commission calculation unit: <ul style="list-style-type: none"> <li>• 0 account currency</li> <li>• 1 base currency</li> <li>• 2 profit currency</li> <li>• 3 margin currency</li> <li>• 4 points</li> <li>• 5 percentage</li> <li>• 6 specified currency</li> </ul> |
| Type          | Integer | Commission charge type: <ul style="list-style-type: none"> <li>• 0 per trade</li> <li>• 1 per volume</li> </ul>                                                                                                                                                      |
| Value         | Float   | Commission sum. Commission units depend on the commission calculation method (Mode).                                                                                                                                                                                 |
| RangeFrom     | Float   | The minimum deal volume (turnover), from which the commission will be charged.                                                                                                                                                                                       |
| RangeTo       | Float   | The maximum trade volume (turnover), from which the commission will be charged.                                                                                                                                                                                      |
| Minimal       | Float   | The minimum amount of commission. The value is specified in the group deposit currency.                                                                                                                                                                              |
| Currency      | String  | Commission calculation currency (if "Specified currency" is selected in the Mode field).                                                                                                                                                                             |

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mt5\_managers

Data about manager accounts is exported to this table. The table contains the following fields:

| Name                | Type    | Description                                                                                                                                                                                                                                                           |
|---------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login               | Integer | Initial key. The user login, based on which the manager account is created.                                                                                                                                                                                           |
| Timestamp           | Integer | A unique values within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of record has changed, it means that the record has changed.                                                                                                   |
| Name                | String  | The name of the manager.                                                                                                                                                                                                                                              |
| Mailbox             | String  | The name of the manager's mailbox in the internal mailing system.                                                                                                                                                                                                     |
| Server              | Integer | The ID of the trade server to which the manager belongs.                                                                                                                                                                                                              |
| RequestLimitLogs    | Integer | <p>The time period of system logs that are available to a manager:</p> <ul style="list-style-type: none"> <li>• 0 unlimited</li> <li>• 1 1 month</li> <li>• 2 3 months</li> <li>• 3 6 months</li> <li>• 4 1 year</li> <li>• 5 2 years</li> <li>• 6 3 years</li> </ul> |
| RequestLimitReports | Integer | <p>The time period of reports that are available to a manager:</p> <ul style="list-style-type: none"> <li>• 0 unlimited</li> <li>• 1 1 month</li> <li>• 2 3 months</li> <li>• 3 6 months</li> <li>• 4 1 year</li> <li>• 5 2 years</li> <li>• 6 3 years</li> </ul>     |
| Groups              | String  | The list of groups processed by the manager. The groups are separated by commas, for example: "demo\forex-netting,demo\forex-netting".                                                                                                                                |
| Access              | String  |                                                                                                                                                                                                                                                                       |

# MetaTrader 5 trading platform

| Name                                                                                                                                                  | Type    | Description                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                       |         | The list of IP addresses, from which a manager is allowed to connect to the platform. Example: 192.168.0.1-192.168.0.10,192.168.0.12-192.168.0.2 |
| Further fields in the table describe <u>manager permissions</u> . A value of 1 means that the right is granted to the manager, 0 means no permission. |         |                                                                                                                                                  |
| Right_Admin                                                                                                                                           | Integer | Connection using the administrator terminal.                                                                                                     |
| Right_Manager                                                                                                                                         | Integer | Connection using the manager terminal.                                                                                                           |
| Right_Cfg_Servers                                                                                                                                     | Integer | Network configuration                                                                                                                            |
| Right_Cfg_Access                                                                                                                                      | Integer | Configuration of the list of IP access.                                                                                                          |
| Right_Cfg_Time                                                                                                                                        | Integer | Configuration of the server working time.                                                                                                        |
| Right_Cfg_Holidays                                                                                                                                    | Integer | Configuration of holidays.                                                                                                                       |
| Right_Cfg_Groups                                                                                                                                      | Integer | Configuration of groups.                                                                                                                         |
| Right_Cfg_Managers                                                                                                                                    | Integer | Configuration of manager rights.                                                                                                                 |
| Right_Cfg_Requests                                                                                                                                    | Integer | Configuration of the routing table.                                                                                                              |
| Right_Cfg_Gateways                                                                                                                                    | Integer | Configuration of gateways.                                                                                                                       |
| Right_Cfg_Plugins                                                                                                                                     | Integer | Configuration of plugins.                                                                                                                        |
| Right_Cfg_Datafeeds                                                                                                                                   | Integer | Configuration of data feeds.                                                                                                                     |
| Right_Cfg_Reports                                                                                                                                     | Integer | Configuration of reports.                                                                                                                        |
| Right_Cfg_Symbols                                                                                                                                     | Integer | Configuring the symbols.                                                                                                                         |
| Right_Cfg_Hst_Sync                                                                                                                                    | Integer | Configuration of synchronization.                                                                                                                |
| Right_Cfg_ECN                                                                                                                                         | Integer | ECN configuration.                                                                                                                               |
| Right_Cfg_VPS                                                                                                                                         | Integer | Configuring Sponsored VPS for traders.                                                                                                           |
| Right_Cfg_Web_Services                                                                                                                                | Integer | Configuring integration with web services: SSL certificates and addresses for callback requests.                                                 |
| Right_Cfg_Funds                                                                                                                                       | Integer | Configuring investment funds in the Administrator terminal                                                                                       |
| Right_Cfg_Messengers                                                                                                                                  | Integer | Configuring integration with SMS providers and messengers in the Administrator terminal                                                          |
| Right_Cfg_KYC                                                                                                                                         | Integer | Configuring integration with KYC services in the Administrator terminal                                                                          |
| Right_Cfg_Automations                                                                                                                                 | Integer | Configuring automatic actions for specified scenarios in the Administrator terminal.                                                             |
| Right_Cfg_Allocations                                                                                                                                 | Integer |                                                                                                                                                  |

# MetaTrader 5 trading platform

| Name                  | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |         | Accessing the Allocations section of the Administrator terminal. The section allows configuring groups, in which traders are able to open demo and preliminary real accounts directly from client terminals.                                                                                                                                                                                                                                       |
| Right_Cfg_Corporates  | Integer | Access to <u>corporate link</u> settings.                                                                                                                                                                                                                                                                                                                                                                                                          |
| Right_Cfg_Payments    | Integer | Configuring integration with payment systems.                                                                                                                                                                                                                                                                                                                                                                                                      |
| Right_Cfg_Mails       | Integer | Configuring integration with email services in the Administrator terminal.                                                                                                                                                                                                                                                                                                                                                                         |
| Right_Srv_Journals    | Integer | Access to server journals.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Right_Srv_Reports     | Integer | Receiving automatic server reports.                                                                                                                                                                                                                                                                                                                                                                                                                |
| Right_Charts          | Integer | Editing history data on the server.                                                                                                                                                                                                                                                                                                                                                                                                                |
| Right_Email           | Integer | Sending internal emails.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Right_News            | Integer | Permission to send newsletters. An administrator or manager can only send newsletters if his or her account belongs to a group created on the main trading server.                                                                                                                                                                                                                                                                                 |
| Right_Export          | Integer | Permission to export data.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Right_Techsupport     | Integer | Access to the technical support tab in the administrator and manager terminals. The permission is obsolete and is no longer used.                                                                                                                                                                                                                                                                                                                  |
| Right_Market          | Integer | Permission to access the Market of applications in the MetaTrader 5 Administrator. The permission is obsolete and is no longer used.                                                                                                                                                                                                                                                                                                               |
| Right_Accountant      | Integer | Permission to work with funds on accounts.                                                                                                                                                                                                                                                                                                                                                                                                         |
| Right_Acc_Read        | Integer | Access to accounts.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Right_Acc_Technical   | Integer | When combined with the " <u>Show to regular managers</u> " permission, provides greater convenience when working with various testing and technical accounts. Disable the "Enable visibility for regular managers" permission for all technical accounts, and then disable access to technical accounts for the managers who do not configure the platform. Otherwise, such technical accounts can be confusing for managers working with clients. |
| Right_Acc_Tech_Modify | Integer | Allows enabling and disabling the " <u>Show to regular managers</u> " and " <u>Include in server reports</u> " options for a trading account. Without this permission, the manager can only see the states of the relevant options, without the ability to change them                                                                                                                                                                             |

# MetaTrader 5 trading platform

| Name                    | Type    | Description                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |         | (read-only).                                                                                                                                                                                                                                                                                                                                                                |
| Right_Acc_Details       | Integer | Access to personal details of accounts.                                                                                                                                                                                                                                                                                                                                     |
| Right_Acc_Manager       | Integer | Account editing.                                                                                                                                                                                                                                                                                                                                                            |
| Right_Acc_Delete        | Integer | Deleting client accounts via the administrator and manager terminals, and via the Manager API. The Right_Acc_Manager permission is required in order to enable this permission.                                                                                                                                                                                             |
| Right_Acc_Online        | Integer | Getting the current client connections.                                                                                                                                                                                                                                                                                                                                     |
| Right_Confirm_Actions   | Integer | By default the manager terminal displays a confirmation dialog when performing balance operations on client accounts and closing multiple orders. A manager needs to enter a randomly generated sequence of characters in order to confirm the appropriate action. If this permission is disabled the above actions will be performed immediately without any confirmation. |
| Right_Notifications     | Integer | Permission to send push notifications to clients' mobile devices from the manager terminal. Messages are sent based on MetaQuotes ID, which is a unique user identifier. To obtain the ID, a user needs to install MetaTrader 5 Mobile for <a href="#">iPhone</a> and <a href="#">Android</a> . For more information please read the MetaTrader 5 Manager user guide.       |
| Right_Trades_Read       | Integer | Viewing trading orders, deals and positions. This right affects the possibility to enable Right_Trades_Manager and Right_Trades_Dealer permissions.                                                                                                                                                                                                                         |
| Right_Trades_Manager    | Integer | Changing any fields of orders, deals and positions in the administrator terminal, and changing position open prices in the manager terminal.                                                                                                                                                                                                                                |
| Right_Trades_Delete     | Integer | Deleting any orders, deals and positions via the administrator and manager terminals, and via the Manager API. The Right_Trades_Manager permission is required in order to enable this permission.                                                                                                                                                                          |
| Right_Trades_Dealer     | Integer | The possibility to perform trading and dealing operations in the manager terminal.                                                                                                                                                                                                                                                                                          |
| Right_Trades_Supervisor | Integer | This right allows the manager to view the entire queue of requests received from groups of clients available to the manager, as well as to track processing of requests by other dealers in the manager terminal. Manager works in the "Supervisor                                                                                                                          |



| Name                    | Type    | Description                                                                                                                                                                                                                                                                  |
|-------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |         | mode without connecting as a dealer in the manager terminal. After connecting as a dealer, the manager will only see the requests that are forwarded to him/her for processing in accordance with the routing rules.                                                         |
| Right_Quotes_Raw        | Integer | If this permission is enabled, the "Show raw quotes" command will appear in the context menu of the Market Watch window in the manager terminal. With this permission, the manager can view quotes without considering spread difference settings set for the manager group. |
| Right_Quotes            | Integer | Permission to throw in quotes                                                                                                                                                                                                                                                |
| Right_Symbol_Details    | Integer | Permission to change spread and execution mode.                                                                                                                                                                                                                              |
| Right_Risk_Manager      | Integer | Permission to receive information about client's aggregate positions and company's coverage positions.                                                                                                                                                                       |
| Right_Group_Margin      | Integer | Permission to configure margin for groups in MetaTrader 5 Manager.                                                                                                                                                                                                           |
| Right_Group_Commission  | Integer | Permission to configure commissions for groups in MetaTrader 5 Manager.                                                                                                                                                                                                      |
| Right_Reports           | Integer | Permission to request and receive various reports of client operations.                                                                                                                                                                                                      |
| Right_Finteza_Access    | Integer | Access to the <a href="#">Finteza Analytics</a> section.                                                                                                                                                                                                                     |
| Right_Finteza_Websites  | Integer | View Finteza data relating to websites in the Analytics section of the Manager terminal.                                                                                                                                                                                     |
| Right_Finteza_Campaigns | Integer | View Finteza data relating to marketing campaigns in the Analytics section of the Manager terminal.                                                                                                                                                                          |
| Right_Finteza_Reports   | Integer | The permission is currently not used.                                                                                                                                                                                                                                        |
| Right_Clients_Access    | Integer | Access to the Clients section in the Administrator and Manager terminals.                                                                                                                                                                                                    |
| Right_Clients_Create    | Integer | Permission to create new client records manually.                                                                                                                                                                                                                            |
| Right_Clients_Edit      | Integer | Permission to edit client data, except for documents.                                                                                                                                                                                                                        |
| Right_Clients_Delete    | Integer | Permission to delete client records.                                                                                                                                                                                                                                         |
| Right_Clients_KYC       | Integer | Permission to launch automated validation of client data via <a href="#">integrated KYC services</a> . The permission affects the launching of verification from Manager and Administrator terminals, as well as from API.                                                   |

MetaTrader 5 trading platform

| <b>Name</b>                  | <b>Type</b> | <b>Description</b>                                                                                                                            |
|------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Right_Documents_Access       | Integer     | Permission to view client documents.                                                                                                          |
| Right_Documents_Create       | Integer     | Permission to add general information about documents in client records.                                                                      |
| Right_Documents_Edit         | Integer     | Permission to edit general information about documents in client records.                                                                     |
| Right_Documents_Delete       | Integer     | Permission to delete general information about documents from client records.                                                                 |
| Right_Documents_Files_Add    | Integer     | Permission to add document files in client records.                                                                                           |
| Right_Documents_Files_Delete | Integer     | Permission to delete document files from client records.                                                                                      |
| Right_Comments_Access        | Integer     | Permission to read comments to clients and their documents.                                                                                   |
| Right_Comments_Create        | Integer     | Permission to write comments to clients and their documents.                                                                                  |
| Right_Comments_Delete        | Integer     | Permission to delete comments to clients and their documents.                                                                                 |
| Right_Admin_Computer         | Integer     | Access to the server machine administration menu the Network section of the Administrator terminal                                            |
| Right_Subscriptions_View     | Integer     | Permission to view existing settings in the Subscriptions section in the Administrator terminal as well as access to subscription statistics. |
| Right_Subscriptions_Edit     | Integer     | Permission to create, edit and remove settings in the Subscriptions section of the Administrator terminal                                     |
| Right_Payments_Access        | Integer     | Permission to view current and processed <u>payment</u> and payment accounts.                                                                 |
| Right_Payments_Process       | Integer     | <u>Permission to confirm and reject payments</u> process manually via the Manager terminal.                                                   |
| Right_Payments_Edit          | Integer     | Permission to edit current and processed payments and payment accounts.                                                                       |
| Right_Payments_Delete        | Integer     | Permission to delete current and processed payment and payment accounts.                                                                      |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_clients

mt5\_clients

Data about clients is exported to this table. The table contains the following fields:

| Name                     | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ClientID                 | Integer | Initial key. Unique entry ID.                                                                                                                                                                                                                                                                                                                                                                      |
| Timestamp                | Integer | A unique values within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has changed.                                                                                                                                                                                                                              |
| ClientType               | Integer | Client type: <ul style="list-style-type: none"> <li>• 0 Not specified</li> <li>• 1 Individual</li> <li>• 2 Corporate</li> <li>• 3 Fund</li> </ul>                                                                                                                                                                                                                                                  |
| ClientStatus             | Integer | Client status: <ul style="list-style-type: none"> <li>• 0 Not registered</li> <li>• 1 Registered</li> <li>• 2 not interested</li> <li>• 3 Not completed</li> <li>• 4 Completed</li> <li>• 5 Information</li> <li>• 6 Rejected</li> <li>• 7 Approved</li> <li>• 8 Financed</li> <li>• 9 Active</li> <li>• 10 Inactive</li> <li>• 11 Suspended</li> <li>• 12 Closed</li> <li>• 13 Deleted</li> </ul> |
| AssignedManager          | Integer | The login of the assigned manager.                                                                                                                                                                                                                                                                                                                                                                 |
| DateCreated              | Integer | Client creation date.                                                                                                                                                                                                                                                                                                                                                                              |
| DateModified             | Integer | Client's last modification date.                                                                                                                                                                                                                                                                                                                                                                   |
| Comment                  | String  | A comment to the client.                                                                                                                                                                                                                                                                                                                                                                           |
| ComplianceApprovedBy     | Integer | The login of the manager by whom the client was approved.                                                                                                                                                                                                                                                                                                                                          |
| ComplianceClientCategory | String  |                                                                                                                                                                                                                                                                                                                                                                                                    |

# MetaTrader 5 trading platform

| Name                      | Type     | Description                                                                                                                                          |
|---------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |          | Client compliance category. Currently not used.                                                                                                      |
| ComplianceDateApproval    | DateTime | The date when the client was approved, in the format of YYYY-MM-DD HH:MM:SS                                                                          |
| ComplianceDateTermination | DateTime | The date when the provision of services to the client was discontinued, in the format of YYYY-MM-DD HH:MM:SS                                         |
| LeadCampaign              | String   | The website from which the client came (lead source).                                                                                                |
| LeadSource                | String   | The name of the marketing campaign, as a result of which the client came (lead campaign).                                                            |
| Introducer                | String   | The login of the user by whom the client was introduced.                                                                                             |
| PersonTitle               | String   | Client's title.                                                                                                                                      |
| PersonName                | String   | First name and last name                                                                                                                             |
| PersonMiddleName          | String   | Middle name.                                                                                                                                         |
| PersonBirthDate           | DateTime | Date of birth, in the format of YYYY-MM-DD HH:MM:SS                                                                                                  |
| PersonCitizenship         | String   | Citizenship.                                                                                                                                         |
| PersonGender              | Integer  | Gender: <ul style="list-style-type: none"> <li>• 0 Not specified</li> <li>• 1 male</li> <li>• 2 Female</li> </ul>                                    |
| PersonTaxID               | String   | Client's Tax ID.                                                                                                                                     |
| PersonDocumentType        | String   | Document type of document: passport, driver's license, etc.                                                                                          |
| PersonDocumentNumber      | String   | Document number.                                                                                                                                     |
| PersonDocumentDate        | DateTime | Document issue date, in the format of YYYY-MM-DD HH:MM:SS                                                                                            |
| PersonDocumentExtra       | String   | Additional document information.                                                                                                                     |
| PersonEmployment          | Integer  | Employment status: <ul style="list-style-type: none"> <li>• 0 Unemployed</li> <li>• 1 Employed</li> <li>• 2 Entrepreneur or self-employed</li> </ul> |

MetaTrader 5 trading platform

| Name                | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |         | <ul style="list-style-type: none"> <li>• 3 Retired</li> <li>• 4 Student</li> <li>• 5 Other</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| PersonIndustry      | Integer | <p>Employment area:</p> <ul style="list-style-type: none"> <li>• 0 Not specified</li> <li>• 1 Agriculture, Food and Natural Resources</li> <li>• 2 Architecture and Construction</li> <li>• 3 Business Administration and Management</li> <li>• 4 Art, Audio/Video Technology and Communication</li> <li>• 5 Education and Training</li> <li>• 6 State and Administrative Management</li> <li>• 7 Health</li> <li>• 8 Tourism and Hospitality</li> <li>• 9 Information Technology</li> <li>• 10 Legal and Public Safety, Correction and Protection Services</li> <li>• 11 Manufacturing</li> <li>• 12 Marketing and Sales</li> <li>• 13 Science and Technology</li> <li>• 14 Engineering and Mathematics</li> <li>• 15 Transportation, Distribution and Logistics</li> <li>• 16 other</li> </ul> |
| PersonEducation     | Integer | <p>Education:</p> <ul style="list-style-type: none"> <li>• 0 Not specified</li> <li>• 1 Secondary</li> <li>• 2 Bachelor's degree or equivalent</li> <li>• 3 Master's degree or equivalent</li> <li>• 4 PhD or equivalent</li> <li>• 5 other</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| PersonWealthSource  | Integer | <p>Source of income:</p> <ul style="list-style-type: none"> <li>• 0 Employment/business activity</li> <li>• 1 Savings or investments</li> <li>• 2 Gift or inheritance</li> <li>• 3 other</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| PersonAnnualIncome  | Float   | Annual income.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| PersonNetWorth      | Float   | Net assets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PersonAnnualDeposit | Float   | Annual deposit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

# MetaTrader 5 trading platform

| Name                    | Type     | Description                                                                                                                                                                              |
|-------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CompanyName             | String   | Company name.                                                                                                                                                                            |
| CompanyRegNumber        | String   | Company registration number.                                                                                                                                                             |
| CompanyRegDate          | String   | company registration date.                                                                                                                                                               |
| CompanyRegAuthority     | String   | Company registration authority.                                                                                                                                                          |
| CompanyVat              | String   | VAT number.                                                                                                                                                                              |
| CompanyLei              | String   | LEI number for EMIR reports.                                                                                                                                                             |
| CompanyLicenseNumber    | String   | Company license number.                                                                                                                                                                  |
| CompanyLicenseAuthority | String   | Licensing authority.                                                                                                                                                                     |
| CompanyCountry          | String   | Country of incorporation.                                                                                                                                                                |
| CompanyAddress          | String   | Company's legal address.                                                                                                                                                                 |
| CompanyWebsite          | String   | Company's webiste.                                                                                                                                                                       |
| ContactPreferred        | Integer  | Preferred method of communication: <ul style="list-style-type: none"> <li>• 0 Not specified</li> <li>• 1 Email</li> <li>• 2 Telephone</li> <li>• 3 SMS</li> <li>• 4 Messenger</li> </ul> |
| ContactLanguage         | String   | Client's language.                                                                                                                                                                       |
| ContactEmail            | String   | Client's email.                                                                                                                                                                          |
| ContactPhone            | String   | Phone number.                                                                                                                                                                            |
| ContactMessengers       | String   | Messengers.                                                                                                                                                                              |
| ContactSocialNetworks   | String   | Accounts in social networks.                                                                                                                                                             |
| ContactLastDate         | DateTime | Last contact date, in the format of YYYY-MM-DD HH:MM:SS                                                                                                                                  |
| AddressCountry          | String   | Client's country.                                                                                                                                                                        |
| AddressPostcode         | String   | Client's postal code.                                                                                                                                                                    |
| AddressStreet           | String   | Client's address.                                                                                                                                                                        |
| AddressState            | String   | State/region of residence.                                                                                                                                                               |
| AddressCity             | String   | City.                                                                                                                                                                                    |
| ExperienceFX            | Integer  | Forex trading experience, number of years.                                                                                                                                               |
| ExperienceCFD           | Integer  | CFD trading experience, number of years.                                                                                                                                                 |

# MetaTrader 5 trading platform

| Name              | Type    | Description                                                                                                                                                                                                                                                     |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ExperienceFutures | Integer | Futures trading experience, number of years.                                                                                                                                                                                                                    |
| ExperienceStocks  | Integer | Stock trading experience, number of years.                                                                                                                                                                                                                      |
| ClientOrigin      | Integer | How the client record was created: <ul style="list-style-type: none"> <li>• 0 manually</li> <li>• 1 based on a demo account</li> <li>• 2 based on a contest account</li> <li>• 3 based on a preliminary account</li> <li>• 4 based on a real account</li> </ul> |
| ClientOriginLogin | Integer | The number of the account, based on which the client record was created.                                                                                                                                                                                        |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_documents

mt5\_documents

Data about client documents is exported to this table. The table contains the following fields:

| Name            | Type     | Description                                                                                                                                                                                                                                                                                                                       |
|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DocumentID      | Integer  | Initial key. Unique document ID.                                                                                                                                                                                                                                                                                                  |
| Timestamp       | Integer  | A unique values within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has changed.                                                                                                                                                             |
| RelatedClient   | Integer  | The ID of the client to whom the document belongs. Corresponds to Client ID from the <u>mt5_clients</u> table.                                                                                                                                                                                                                    |
| ApprovedDate    | DateTime | Document approval date, in the format of YYYY-MM-DD HH:MM:SS                                                                                                                                                                                                                                                                      |
| ApprovedBy      | Integer  | The login of the manager by whom the document was approved.                                                                                                                                                                                                                                                                       |
| DateIssue       | DateTime | Document issue date, in the format of YYYY-MM-DD HH:MM:SS                                                                                                                                                                                                                                                                         |
| DateExpiration  | DateTime | Document expiration date, in the format of YYYY-MM-DD HH:MM:SS                                                                                                                                                                                                                                                                    |
| DocumentType    | Integer  | Document type: <ul style="list-style-type: none"> <li>• 0 Other</li> <li>• 1 Proof of identity</li> <li>• 2 Proof of address</li> <li>• 3 Registration address</li> <li>• 4 CEO's ID document</li> <li>• 5 Certificate of Registration</li> <li>• 6 Certificate of Directors</li> <li>• 7 Certificate of good standing</li> </ul> |
| DocumentName    | String   | Document name.                                                                                                                                                                                                                                                                                                                    |
| DocumentComment | String   | Comment to the document.                                                                                                                                                                                                                                                                                                          |
| DocumentStatus  | Integer  | Document status: <ul style="list-style-type: none"> <li>• 0 New</li> <li>• 1 Approved</li> <li>• 2 Rejected</li> <li>• 3 Archived</li> </ul>                                                                                                                                                                                      |



MetaTrader 5 trading platform

| Name                                                                               | Type | Description |
|------------------------------------------------------------------------------------|------|-------------|
|                                                                                    |      | • 4 Deleted |
| Only document data is exported to SQL. Document files themselves are not exported. |      |             |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_users

mt5\_users

Data on account database is exported to the table. The table contains the following fields:

| Name             | Type     | Description                                                                                                                                                                                                                                                                                                                  |
|------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login            | Integer  | Primary key. The login of a user.                                                                                                                                                                                                                                                                                            |
| Timestamp        | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                                               |
| TimestampTrade   | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                                               |
| Group            | String   | User group.                                                                                                                                                                                                                                                                                                                  |
| CertSerialNumber | Integer  | The number of a last used certificate for user authorization.                                                                                                                                                                                                                                                                |
| Rights           | Integer  | Flags of the users permissions. Passed using a value of the <u>EnUserRights</u> enumeration (sum of values of appropriate flags).                                                                                                                                                                                            |
| Registration     | DateTime | Time of a client record generation in the YYYY-MM-DD HH:MM:SS format.                                                                                                                                                                                                                                                        |
| LastAccess       | DateTime | The date of the last connection using an account in the YYYY-MM-DD HH:MM:SS format. This field is not updated in real time, in order to save traffic and reduce the load on the platform. The value is only updated when the user connects to the platform, if more than 24 hours have passed since the previous connection. |
| LastPassChange   | DateTime | The date of the last password change.                                                                                                                                                                                                                                                                                        |
| LastIP           | String   | The IP address from which the user last connected to the server.                                                                                                                                                                                                                                                             |
| Name             | String   | The name of the user. Obsolete field.                                                                                                                                                                                                                                                                                        |
| FirstName        | String   | The first name of the client.                                                                                                                                                                                                                                                                                                |
| LastName         | String   | The last name of the client.                                                                                                                                                                                                                                                                                                 |
| MiddleName       | String   | The middle name of the client.                                                                                                                                                                                                                                                                                               |
| Company          | String   | The name of user's company.                                                                                                                                                                                                                                                                                                  |
| Account          | String   | The number of a user's account in an external bank.                                                                                                                                                                                                                                                                          |

MetaTrader 5 trading platform

| Name              | Type     | Description                                                                                                               |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------|
| Country           | String   | The user's country of residence.                                                                                          |
| Language          | Integer  | User's language in the format LANGID used in <u>MS Windows</u> (value from Prim.lang.identifier).                         |
| ClientID          | Integer  | The identifier of the <u>client</u> , to whom the trading account corresponds.                                            |
| City              | String   | The user's city of residence.                                                                                             |
| State             | String   | The user's state (region) of residence.                                                                                   |
| ZIPCode           | String   | The user's zip code.                                                                                                      |
| Address           | String   | The address of the user.                                                                                                  |
| Phone             | String   | The user's phone number.                                                                                                  |
| EMail             | String   | The email address of the user.                                                                                            |
| ID                | String   | The number of a user's identity document.                                                                                 |
| Status            | String   | Client's status.                                                                                                          |
| Comment           | String   | A comment to the user.                                                                                                    |
| Color             | COLORREF | The color of the user. This is the color of the user's request shown when handling the requests via the manager terminal. |
| PhonePassword     | String   | The user's phone password.                                                                                                |
| Leverage          | Integer  | The size of a user's leverage.                                                                                            |
| Agent             | Integer  | Agent account number of the user.                                                                                         |
| Balance           | Float    | The current balance of a user.                                                                                            |
| Credit            | Float    | The current amount of funds credited to the user.                                                                         |
| InterestRate      | Float    | The amount accrued for the current month calculated based on the annual interest rate.                                    |
| CommissionDaily   | Float    | The amount of commissions charged from the user for a current month.                                                      |
| CommissionMonthly | Float    | The total amount of commissions charged from the user for the current month.                                              |
| BalancePrevDay    | Float    | The value of the user's balance as of the end of the previous trading day.                                                |
| BalancePrevMonth  | Float    | The value of a user's balance as of the end of the previous trading month.                                                |
| EquityPrevDay     | Float    | The user's equity as of the end of the previous day.                                                                      |
| EquityPrevMonth   | Float    | The value of the user's equity as of the end of the previous trading month.                                               |

# MetaTrader 5 trading platform

| Name           | Type    | Description                                                                                                                                                                                                                                                                                                                    |
|----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TradeAccounts  | String  | Account numbers in external trading systems and gateway identifiers used for working with that systems. The string format is:<br>gateway_ID=account_number gateway_ID=account_num                                                                                                                                              |
| MQID           | String  | MetaQuotes ID of the user.                                                                                                                                                                                                                                                                                                     |
| LeadCampaign   | String  | Name of the <u>marketing campaign</u> a client was attracted b                                                                                                                                                                                                                                                                 |
| LeadSource     | String  | <u>Lead source</u> (address of the website a client has come fro                                                                                                                                                                                                                                                               |
| ApiData        | String  | User data which can be added via MetaTrader 5 API. Sample user data entry:<br>[{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.000000<br>It specifies the user data index, the ID of the application t<br>added it, as well as the data of three types: Int, UInt and<br>double. The string may contain up to 16 such entries. |
| LimitOrders    | Integer | The maximum number of active (placed) pending orders allowed on the account.                                                                                                                                                                                                                                                   |
| LimitPositions | Integer | Maximum value of open positions allowed on the account                                                                                                                                                                                                                                                                         |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / [mt5\\_users](#) / Enumerations

## Enumerations

To pass information about users the following enumerations are used:

- [EnUsersRights](#)
- [EnUsersPasswords](#)
- [EnUsersConnectionTypes](#)
- [EnSoActivation](#)

## EnUsersRights

Permissions that can be given to a user are enumerated in EnUsersRights.

| ID                        | Value              | Description                                                                                                                  |
|---------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| USER_RIGHT_NONE           | 0x0000000000000000 | No permissions.                                                                                                              |
| USER_RIGHT_ENABLED        | 0x0000000000000001 | The user is allowed to connect.                                                                                              |
| USER_RIGHT_PASSWORD       | 0x0000000000000002 | The user is allowed to change the password.                                                                                  |
| USER_RIGHT_TRADE_DISABLED | 0x0000000000000004 | Trading is disabled for the user.                                                                                            |
| USER_RIGHT_INVESTOR       | 0x0000000000000008 | Service value for internal use.                                                                                              |
| USER_RIGHT_CONFIRMED      | 0x0000000000000010 | User's certificate is confirmed.                                                                                             |
| USER_RIGHT_TRAILING       | 0x0000000000000020 | The user is allowed to use trailing stop.                                                                                    |
| USER_RIGHT_EXPERT         | 0x0000000000000040 | The user is allowed to use Expert Advisors.                                                                                  |
| USER_RIGHT_OBSOLETE       | 0x0000000000000080 | The flag is obsolete and is not used.                                                                                        |
| USER_RIGHT_REPORTS        | 0x0000000000000100 | The user is allowed to receive daily reports. If the permission is not enabled, daily reports are neither generated nor sent |

MetaTrader 5 trading platform

| ID                           | Value               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |                     | for the account.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| USER_RIGHT_READONLY          | 0x00000000000000200 | Service value for internal use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| USER_RIGHT_RESET_PASS        | 0x00000000000000400 | The user must change password during the next connection.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| USER_RIGHT_OTP_ENABLED       | 0x00000000000000800 | The user can use OTP authentication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| USER_RIGHT_SPONSORED_HOSTING | 0x00000000000002000 | Brokers can pay the <u>virtual hosting</u> fee for their customers. The service is extremely important for traders, and the opportunity to receive a VPS for free can give them a good reason to choose your company over competitors. The availability of a broker-sponsored VPS is controlled at the individual account level. Only if this flag is enabled, the appropriate payment plan will be shown to the trader in the client terminal. For more details, please read the <u>appropriate section</u> . |
| USER_RIGHT_API_ENABLED       | 0x00000000000004000 | The user is allowed to connect via the Web API.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| USER_RIGHT_TECHNICAL         | 0x00000000000010000 | Permission for convenient work with technical accounts. Disable it for testing accounts                                                                                                                                                                                                                                                                                                                                                                                                                        |

| ID                         | Value               | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                     | to hide them from all managers who do not have special <u>access to technical accounts</u> . Such technical accounts can be confusing for the managers working with clients, in which case hiding them can be useful.<br><br>This permission affects the account visibility in the general account list in the Administrator and Manager terminals, as well as in the list of online accounts in the Manager terminal. |
| USER_RIGHT_EXCLUDE_REPORTS | 0x00000000000020000 | Allows excluding an account from <u>server reports</u> . Like the previous permission, it is used for convenient management of technical accounts.                                                                                                                                                                                                                                                                     |

#### EnUsersPasswords

Types of passwords are enumerated in EnUsersPasswords.

| ID                 | Value | Description            |
|--------------------|-------|------------------------|
| USER_PASS_MAIN     | 0     | The master password.   |
| USER_PASS_INVESTOR | 1     | The investor password. |
| USER_PASS_API      | 2     | API password.          |

#### EnUsersConnectionTypes

Types of client connections are enumerated in EnUsersConnectionTypes:

| ID | Value | Description |
|----|-------|-------------|
|----|-------|-------------|

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| ID                          | Value | Description                                                                |
|-----------------------------|-------|----------------------------------------------------------------------------|
| USER_TYPE_CLIENT            | 0     | Connection from a client terminal.                                         |
| USER_TYPE_CLIENT_WINMOBILE  | 1     | Connection from a mobile terminal for Windows Mobile (under development).  |
| USER_TYPE_CLIENT_WINPHONE   | 2     | Connection from a mobile terminal for Windows Phone 7 (under development). |
| USER_TYPE_CLIENT_API_WEB    | 3     | Connection via the client Web API.                                         |
| USER_TYPE_CLIENT_IPHONE     | 4     | Connection from a mobile terminal for iPhone.                              |
| USER_TYPE_CLIENT_ANDROID    | 5     | Connection from a mobile terminal for Android (under development).         |
| USER_TYPE_CLIENT_BLACKBERRY | 6     | Connection from a mobile terminal for BlackBerry (under development).      |
| USER_TYPE_ADMIN             | 32    | Connection from an administrator terminal.                                 |
| USER_TYPE_MANAGER           | 33    | Connection from a manager terminal.                                        |
| USER_TYPE_MANAGER_API       | 34    | Connection via the manager interface of the Manager API.                   |
| USER_TYPE_ADMIN_API         | 36    | Connection via the administrator interface of the Manager API.             |
| USER_TYPE_MANAGER_API_WEB   | 37    | Connection via the manager Web API.                                        |

### EnSoActivation

The account status as per the minimum amount of funds on the account required to maintain trading positions are enumerated in EnSoActivation.

| ID                     | Value | Description  |
|------------------------|-------|--------------|
| ACTIVATION_NONE        | 0     | None.        |
| ACTIVATION_MARGIN_CALL | 1     | Margin call. |
| ACTIVATION_STOP_OUT    | 2     | Stop out.    |



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mt5\_orders

Data on open orders is exported to the table. The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                    |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Order          | Integer  | Primary key. Order ticket.                                                                                                                                                     |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed. |
| ExternalID     | String   | The order ID in external trading systems.                                                                                                                                      |
| Login          | Integer  | The login of the client, to whom the order belongs.                                                                                                                            |
| Dealer         | Integer  | The login of a dealer, who has processed an order.                                                                                                                             |
| Symbol         | String   | The symbol of an order.                                                                                                                                                        |
| Digits         | Integer  | The number of decimal places in the price of an order.                                                                                                                         |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client who has placed the order.                                                                                      |
| ContractSize   | Float    | The contract size of the symbol, for which an order was placed.                                                                                                                |
| State          | Integer  | The current state of an order. Passed as a value of the <u>EnOrderMode</u> enumeration.                                                                                        |
| Reason         | Integer  | The reason for placing the order. Passed as a value of the <u>EnOrderReason</u> enumeration.                                                                                   |
| TimeSetup      | DateTime | Order placement time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeSetupMsc   | DateTime | Order placement time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |
| TimeExpiration | DateTime | Order expiration time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                       |
| TimeDone       | DateTime | Order execution time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeDoneMsc    | DateTime | Order execution time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |
| ModifyFlags    | Integer  | The order modification flags. Passed as a value of the <u>EnTradeModifyFlags</u> enumeration (sum of values of appropriate flags).                                             |
| Type           | Integer  | Order type. Passed as a value of the <u>EnOrderType</u> enumeration.                                                                                                           |

| Name             | Type     | Description                                                                                                                                                                                                 |
|------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TypeFill         | Integer  | Order filling type. Passed as a value of the <u>EnOrderFilling</u> enumeration.                                                                                                                             |
| TypeTime         | Integer  | Order expiration type. Passed in a value of the <u>EnOrderTime</u> enumeration.                                                                                                                             |
| PriceOrder       | Float    | Order price.                                                                                                                                                                                                |
| PriceTrigger     | Float    | The order triggering price.                                                                                                                                                                                 |
| PriceCurrent     | Float    | The current price of the symbol, for which an order has been placed.                                                                                                                                        |
| PriceSL          | Float    | The Stop Loss level of an order.                                                                                                                                                                            |
| PriceTP          | Float    | The Take Profit level of an order.                                                                                                                                                                          |
| VolumeInitial    | Integer  | The initial order volume. One unit corresponds to 1/10000 lot.                                                                                                                                              |
| VolumeInitialExt | Integer  | The initial order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                |
| VolumeCurrent    | Integer  | The current unfilled volume of an order. One unit corresponds to 1/10000 lot.                                                                                                                               |
| VolumeCurrentExt | Integer  | The current unfilled order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                       |
| ExpertID         | Integer  | The ID of the Expert Advisor that has placed the order.                                                                                                                                                     |
| PositionID       | Integer  | The position ID (ticket) set in the order.                                                                                                                                                                  |
| PositionByID     | Integer  | The opposite position ID (ticket) set in the order. The property is set for Close By operations ( <u>OP_CLOSE_BY</u> ). The ticket of the position that is closed by the opposite one is set in PositionID. |
| Comment          | String   | A comment to an order.                                                                                                                                                                                      |
| ActivationMode   | Integer  | Order activation type. Passed in a value of the <u>EnOrderActivation</u> enumeration.                                                                                                                       |
| ActivationTime   | DateTime | Order activation time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                                                    |
| ActivationPrice  | Float    | The price, at which the order was activated.                                                                                                                                                                |
| ActivationFlags  | Integer  | Order activation flags. Passed as a value of the <u>EnTradeActivationFlags</u> enumeration (sum of values of appropriate flags).                                                                            |
| RateMargin       | Float    | The rate of conversion of the margin currency of the symbol to the deposit currency of the user, which is used when calculating the margin requirements for the order.                                      |
| ApiData          | String   | User data which can be added via MetaTrader 5 API. Sample                                                                                                                                                   |

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| Name | Type | Description                                                                                                                                                                                                                                                                      |
|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |      | <p>user data entry:<br/> [{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.00000000}]<br/> It specifies the user data index, the ID of the application that added it, as well as the data of three types: Int, UInt and double. The string may contain up to 16 such entries.</p> |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / [mt5\\_orders](#) / Enumerations

## Enumerations

To pass information about orders the following enumerations are used:

- [EnOrderType](#)
- [EnOrderFilling](#)
- [EnOrderTime](#)
- [EnOrderState](#)
- [EnOrderActivation](#)
- [EnOrderReason](#)
- [EnTradeActivationFlags](#)
  
- [EnTradeModifyFlags](#)

### EnOrderType

Types of trade orders are listed in EnOrderType.

| ID number          | Value | Description                                                                                                                                                                                                   |
|--------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OP_BUY             | 0     | A Buy order.                                                                                                                                                                                                  |
| OP_SELL            | 1     | A Sell order.                                                                                                                                                                                                 |
| OP_BUY_LIMIT       | 2     | A Buy Limit order.                                                                                                                                                                                            |
| OP_SELL_LIMIT      | 3     | A Sell Limit order.                                                                                                                                                                                           |
| OP_BUY_STOP        | 4     | A Buy Stop order.                                                                                                                                                                                             |
| OP_SELL_STOP       | 5     | A Sell Stop order.                                                                                                                                                                                            |
| OP_BUY_STOP_LIMIT  | 6     | A Buy Stop Limit .                                                                                                                                                                                            |
| OP_SELL_STOP_LIMIT | 7     | A Sell Stop Limit order.                                                                                                                                                                                      |
| OP_CLOSE_BY        | 8     | A close by order closing two oppositely directed positions at a single symbol. This type of operations is used only for the hedging position accounting system ( <a href="#">MARGIN_MODE_RETAIL_HEDGED</a> ). |

### EnOrderFilling

Types of order filling are listed in EnOrderFilling.

| ID number      | Value | Description                                                                                                                                     |
|----------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_FILL_FOK | 0     | Fill or Kill. The order must be filled completely or canceled. This type of filling is automatically set for the instant and request execution. |

| ID number         | Value | Description                                                                                                                                                           |
|-------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_FILL_IOC    | 1     | Immediate or Cancel. An order can be filled partially and the residual volume is canceled. This type of filling is only available for the stock and market execution. |
| ORDER_FILL_RETURN | 2     | Return the remainder to the queue. This mode is intended only for pending orders.                                                                                     |

EnOrderTime

Types of order expiration are listed in EnOrderTime.

| ID number                | Value | Description                                                                                |
|--------------------------|-------|--------------------------------------------------------------------------------------------|
| ORDER_TIME_GTC           | 0     | Good till Canceled.                                                                        |
| ORDER_TIME_DAY           | 1     | Intraday.                                                                                  |
| ORDER_TIME_SPECIFIED     | 2     | Specified time.                                                                            |
| ORDER_TIME_SPECIFIED_DAY | 3     | Specified day. An order expires at 00:00 of the specified day or the nearest trading time. |

EnOrderState

Possible states of orders are listed in EnOrderState.

| ID number                  | Value | Description                                                                                                                                           |
|----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_STATE_STARTED        | 0     | Started.                                                                                                                                              |
| ORDER_STATE_PLACED         | 1     | Placed.                                                                                                                                               |
| ORDER_STATE_CANCELED       | 2     | Canceled.                                                                                                                                             |
| ORDER_STATE_PARTIAL        | 3     | Partially filled.                                                                                                                                     |
| ORDER_STATE_FILLED         | 4     | Filled.                                                                                                                                               |
| ORDER_STATE_REJECTED       | 5     | Rejected.                                                                                                                                             |
| ORDER_STATE_EXPIRED        | 6     | Expired.                                                                                                                                              |
| ORDER_STATE_REQUEST_ADD    | 7     | The order passed (by the gateway) to be placed. This state is used for notifying that a request for placing the order is being already processed.     |
| ORDER_STATE_REQUEST_MODIFY | 8     | The order passed (by the gateway) to be modified. This state is used for notifying that a request for modifying the order is being already processed. |

| ID number                  | Value | Description                                                                                                                                         |
|----------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_STATE_REQUEST_CANCEL | 9     | The order passed (by the gateway) to be deleted. This state is used for notifying that a request for deleting the order is being already processed. |

EnOrderActivation

Types of order activation are listed in EnOrderActivation.

| ID number             | Value | Description                                   |
|-----------------------|-------|-----------------------------------------------|
| ACTIVATION_NONE       | 0     | Not activated.                                |
| ACTIVATION_PENDING    | 1     | Activation of a pending order.                |
| ACTIVATION_STOPLIMIT  | 2     | Activation of a Stop Limit order.             |
| ACTIVATION_EXPIRATION | 3     | Cancellation of an order upon expiration.     |
| ACTIVATION_STOPOUT    | 4     | Order is being removed because of a stop out. |

EnOrderReason

Types of reasons for order placing are listed in EnOrderReason.

| ID number                    | Value | Description                                                    |
|------------------------------|-------|----------------------------------------------------------------|
| ORDER_REASON_CLIENT          | 0     | Order placed by a client manually through the client terminal. |
| ORDER_REASON_EXPERT          | 1     | Order placed by a client with using an Expert Advisor.         |
| ORDER_REASON_DEALER          | 2     | Order placed by a dealer through the manager terminal.         |
| ORDER_REASON_SL              | 3     | Order placed as a result of Stop Loss activation.              |
| ORDER_REASON_TP              | 4     | Order placed as a result of Take Profit activation.            |
| ORDER_REASON_SO              | 5     | Order placed when the client reached the Stop-Out level.       |
| ORDER_REASON_ROLLOVER        | 6     | Order placed when reopening a position for charging swaps.     |
| ORDER_REASON_EXTERNAL_CLIENT | 7     | Order placed by a client from an external trading system.      |
| ORDER_REASON_VMARGIN         | 8     | Order placed for accruing variation margin.                    |

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| ID number                     | Value | Description                                                                                                                                                                          |
|-------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORDER_REASON_GATEWAY          | 9     | Order placed by a MetaTrader 5 gateway that had connected to the trading platform.                                                                                                   |
| ORDER_REASON_SIGNAL           | 10    | Order placed as a result of copying <u>a trade signal</u> according to a subscription in the client terminal.                                                                        |
| ORDER_REASON_SETTLEMENT       | 11    | Order placed as a result of performing operations connected with the settlement of a futures contract/option. Not used at the moment.                                                |
| ORDER_REASON_TRANSFER         | 12    | Order placed due to transferring a position at the settlement price to a new symbol with the same underlying asset. Not used at the moment.                                          |
| ORDER_REASON_SYNC             | 13    | Order placed as a result of synchronization of an account's trade state with an external system.                                                                                     |
| ORDER_REASON_EXTERNAL_SERVICE | 14    | Order placed from an external trading system for technical reasons (for example, to correct the trade state of a client).                                                            |
| ORDER_REASON_MIGRATION        | 15    | Order created as a result of importing trade operations from a MetaTrader 4 server.                                                                                                  |
| ORDER_REASON_MOBILE           | 16    | Order created via the MetaTrader 5 mobile terminal for Android or iPhone.                                                                                                            |
| ORDER_REASON_WEB              | 17    | Order created via the web terminal.                                                                                                                                                  |
| ORDER_REASON_SPLIT            | 18    | Order created as a result of a symbol split.                                                                                                                                         |
| ORDER_REASON_CORPORATE_ACTION | 19    | Order created as a result of a corporate action, such as consolidating or renaming securities, transferring a client to a different account, etc. API applications set this flag for |

| ID number | Value | Description                                                                                                     |
|-----------|-------|-----------------------------------------------------------------------------------------------------------------|
|           |       | service operations so that the platform does not account for such corporate actions in commission calculations. |

### EnTradeActivationFlags

Types of activation flags that can be assigned to orders upon forming a trade execution are listed in EnTradeActivationFlags:

| ID number                 | Value | Description                                       |
|---------------------------|-------|---------------------------------------------------|
| ACTIV_FLAGS_NO_LIMIT      | 0x01  | Do not handle reaching of the Limit level.        |
| ACTIV_FLAGS_NO_STOP       | 0x02  | Do not handle the reaching of the stop level.     |
| ACTIV_FLAGS_NO_SLIMIT     | 0x04  | Do not handle reaching of the Stop-Limit level.   |
| ACTIV_FLAGS_NO_SL         | 0x08  | Do not handle activation upon Stop Loss.          |
| ACTIV_FLAGS_NO_TP         | 0x10  | Do not handle activation upon Take Profit.        |
| ACTIV_FLAGS_NO_SO         | 0x20  | Do not handle activation upon Stop-Out.           |
| ACTIV_FLAGS_NO_EXPIRATION | 0x40  | Do not handle order cancellation upon expiration. |
| ACTIV_FLAGS_NONE          | 0x00  | No flags.                                         |

Flags of orders are inherited by positions created as a result of their execution.

### EnTradeModifyFlags

EnTradeModifyFlags lists the flags assigned to orders when they are changed by an administrator, manager or API:

| ID number                | Value      | Description                                            |
|--------------------------|------------|--------------------------------------------------------|
| MODIFY_FLAGS_ADMIN       | 0x00000001 | Order changed by an administrator.                     |
| MODIFY_FLAGS_MANAGER     | 0x00000002 | Open price has been modified by a manager.             |
| MODIFY_FLAGS_POSITION    | 0x00000004 | Flag not used for orders.                              |
| MODIFY_FLAGS_RESTORE     | 0x00000008 | Order restored.                                        |
| MODIFY_FLAGS_API_ADMIN   | 0x00000010 | Order changed via Manager API administrator interface. |
| MODIFY_FLAGS_API_MANAGER | 0x00000020 | Order changed via Manager API manager interface.       |



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| <b>ID number</b>         | <b>Value</b> | <b>Description</b>             |
|--------------------------|--------------|--------------------------------|
| MODIFY_FLAGS_API_SERVER  | 0x00000040   | Order changed via Server API.  |
| MODIFY_FLAGS_API_GATEWAY | 0x00000080   | Order changed via Gateway API. |
| ACTIV_FLAGS_NONE         | 0x00000000   | No flags.                      |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_orders\_history

mt5\_orders\_history

Data on closed orders is exported to the table. If "[Export history orders and deals into separate tables by years](#)" option is enabled in settings, closed orders for each year are exported to a separate table. A year is specified in the table heading, for example, mt5\_orders\_2012. If the option is disabled, all closed orders are exported to a single mt5\_orders\_history table.

The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                    |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Order          | Integer  | Primary key. Order ticket.                                                                                                                                                     |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed. |
| ExternalID     | String   | The order ID in external trading systems.                                                                                                                                      |
| Login          | Integer  | The login of the client, to whom the order belongs.                                                                                                                            |
| Dealer         | Integer  | The login of a dealer, who has processed an order.                                                                                                                             |
| Symbol         | String   | The symbol of an order.                                                                                                                                                        |
| Digits         | Integer  | The number of decimal places in the price of an order.                                                                                                                         |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client who has placed the order.                                                                                      |
| ContractSize   | Float    | The contract size of the symbol, for which an order was placed.                                                                                                                |
| State          | Integer  | The current state of an order. Passed as a value of the <a href="#">EnOrderMode</a> enumeration.                                                                               |
| Reason         | Integer  | The reason for placing the order. Passed as a value of the <a href="#">EnOrderReason</a> enumeration.                                                                          |
| TimeSetup      | DateTime | Order placement time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeSetupMsc   | Integer  | Order placement time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |
| TimeExpiration | DateTime | Order expiration time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                       |
| TimeDone       | DateTime | Order execution time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeDoneMsc    | Integer  | Order execution time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |

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| Name             | Type              | Description                                                                                                                                                                                                 |
|------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type             | Integer           | Order type. Passed as a value of the <u>EnOrderType</u> enumeration.                                                                                                                                        |
| TypeFill         | Integer           | Order filling type. Passed as a value of the <u>EnOrderFilling</u> enumeration.                                                                                                                             |
| TypeTime         | Integer           | Order expiration type. Passed in a value of the <u>EnOrderTime</u> enumeration.                                                                                                                             |
| PriceOrder       | Float             | Order price.                                                                                                                                                                                                |
| PriceTrigger     | Float             | The order triggering price.                                                                                                                                                                                 |
| PriceCurrent     | Float             | The current price of the symbol, for which an order has been placed.                                                                                                                                        |
| PriceSL          | Float             | The Stop Loss level of an order.                                                                                                                                                                            |
| PriceTP          | Float             | The Take Profit level of an order.                                                                                                                                                                          |
| VolumeInitial    | Integer           | The initial order volume. One unit corresponds to 1/10000 lot.                                                                                                                                              |
| VolumeInitialExt | Integer           | The initial order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                |
| VolumeCurrent    | Integer           | The current unfilled volume of an order. One unit corresponds to 1/10000 lot.                                                                                                                               |
| VolumeCurrentExt | Integer           | The current unfilled order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                       |
| ExpertID         | Integer           | The ID of the Expert Advisor that has placed the order.                                                                                                                                                     |
| PositionID       | Integer           | The position ID (ticket) set in the order.                                                                                                                                                                  |
| PositionByID     | Integer           | The opposite position ID (ticket) set in the order. The property is set for Close By operations ( <u>OP_CLOSE_BY</u> ). The ticket of the position that is closed by the opposite one is set in PositionID. |
| Comment          | String            | A comment to an order.                                                                                                                                                                                      |
| ActivationMode   | Integer number    | Order activation type. Passed in a value of the <u>EnOrderActivation</u> enumeration.                                                                                                                       |
| ActivationTime   | DateTime          | Order activation time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                                                    |
| ActivationPrice  | Fractional number | The price, at which the order was activated.                                                                                                                                                                |
| ActivationFlags  | Integer number    | Order activation flags. Passed as a value of the <u>EnTradeActivationFlags</u> enumeration (sum of values of appropriate flags).                                                                            |
| RateMargin       | Fractional        | The rate of conversion of the margin currency of the symbol                                                                                                                                                 |

# MetaTrader 5 trading platform

| Name    | Type   | Description                                                                                                                                                                                                                                                                                                                     |
|---------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | number | the deposit currency of the user, which is used when calculating the margin requirements for the order.                                                                                                                                                                                                                         |
| ApiData | String | User data which can be added via MetaTrader 5 API. Sample user data entry:<br>[{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.00000000}]<br>It specifies the user data index, the ID of the application that added it, as well as the data of three types: Int, UInt and double. The string may contain up to 16 such entries. |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_positions

mt5\_positions

Data on positions is exported to the table. The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Position_ID    | Integer  | Primary key. Internal position identifier used by the backup server to export data. Use the Position field to identify positions in your reports and databases. The Position field contains the real identifier (ticket) assigned to the position in MetaTrader 5. The Position_ID field is only used for internal purposes and is not related with trading operation tickets. |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                                                                                                 |
| Login          | Integer  | The login of the client, to whom the trade position belongs.                                                                                                                                                                                                                                                                                                                   |
| Symbol         | String   | The symbol of a trade position.                                                                                                                                                                                                                                                                                                                                                |
| Action         | Integer  | Position type. Passed in a value of the <a href="#">EnPositionAction</a> enumeration.                                                                                                                                                                                                                                                                                          |
| Digits         | Integer  | The number of decimal places in the price of a position.                                                                                                                                                                                                                                                                                                                       |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client who has opened the position.                                                                                                                                                                                                                                                                                   |
| Reason         | Integer  | The reason for position opening. Passed in a value of the <a href="#">EnPositionReason</a> enumeration.                                                                                                                                                                                                                                                                        |
| ContractSize   | Float    | The contract size of the symbol, for which a position is opened.                                                                                                                                                                                                                                                                                                               |
| Position       | Integer  | The ticket (unique identifier) of a trade position in a MetaTrader 5 platform.                                                                                                                                                                                                                                                                                                 |
| ExternalID     | String   | The position ticket (unique number) in an external trading system.                                                                                                                                                                                                                                                                                                             |
| TimeCreate     | DateTime | Time of position creation, in seconds that have elapsed since 01.01.1970.                                                                                                                                                                                                                                                                                                      |
| TimeUpdate     | DateTime | Time of the last modification of a position, in seconds that have elapsed since 01.01.1970. The modification time of a position is the time of the last modification of its volume. Virtually, it is the time of the last deal performed by the financial instrument that corresponds to that position.                                                                        |
| TimeCreateMsc  | DateTime | Position generation time in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                                                                                                                                                                                                                                 |

| Name             | Type     | Description                                                                                                                                                                                                                                                                              |
|------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimeUpdateMsc    | DateTime | Time of a trading position last change in the YYYY-MM-DD HH:MM:SS.MS format. The modification time of a position is the time of the last modification of its volume. Virtually, it is the time of the last deal performed by the financial instrument that corresponds to that position. |
| PriceOpen        | Float    | The weighted average open price of a position. Calculated by the following formula: (price of deal 1 * volume of deal 1 + ... + price of deal N * volume of deal N) / (volume of deal 1 + ... + volume of deal N).                                                                       |
| PriceCurrent     | Float    | The current price of the symbol, for which a trade position has been opened.                                                                                                                                                                                                             |
| PriceSL          | Float    | The Stop Loss level of a trade position.                                                                                                                                                                                                                                                 |
| PriceTP          | Float    | The Take Profit level of a trade position.                                                                                                                                                                                                                                               |
| Volume           | Integer  | The volume of a trade position. One unit corresponds to 1/10000 lot.                                                                                                                                                                                                                     |
| VolumeExt        | Integer  | The trade position volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                                                                                            |
| Profit           | Float    | Returns of a trade position in deposit currency.                                                                                                                                                                                                                                         |
| Storage          | Float    | The swap size for a position in deposit currency.                                                                                                                                                                                                                                        |
| RateProfit       | Float    | The exchange rate of the profit currency of a position to the deposit currency of a client group.                                                                                                                                                                                        |
| RateMargin       | Float    | The exchange rate of the margin currency of a position to the client's deposit currency.                                                                                                                                                                                                 |
| ExpertID         | Integer  | The ID of the Expert Advisor that has opened the position.                                                                                                                                                                                                                               |
| ExpertPositionID | Integer  | Position ID.                                                                                                                                                                                                                                                                             |
| Comment          | String   | A comment to a position.                                                                                                                                                                                                                                                                 |
| Dealer           | Integer  | The login of a dealer, who has processed an order, which opened the position.                                                                                                                                                                                                            |
| ActivationMode   | Integer  | Position activation type. Passed in a value of the <u>EnActivation</u> enumeration.                                                                                                                                                                                                      |
| ActivationTime   | DateTime | Position activation time in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                                                                                                                                           |
| ActivationPrice  | Float    | Position activation price.                                                                                                                                                                                                                                                               |
| ActivationFlags  | Integer  | Position activation flags. Passed as a value of the <u>EnTradeActivationFlags</u> enumeration (sum of values of appropriate flags).                                                                                                                                                      |

# MetaTrader 5 trading platform

| Name    | Type   | Description                                                                                                                                                                                                                                                                                                                   |
|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ApiData | String | User data which can be added via MetaTrader 5 API. Sample user data entry:<br>[{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.00000000}]. It specifies the user data index, the ID of the application that added it, as well as the data of three types: Int, UInt and double. The string may contain up to 16 such entries. |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / [mt5\\_positions](#) / Enumerations

## Enumerations

To pass information about positions the following enumerations are used:

- [EnPositionAction](#)
- [EnActivation](#)
- [EnTradeActivationFlags](#)
- [EnPositionReason](#)

### EnPositionAction

Types of positions are listed in EnPositionAction.

| ID number     | Value | Description |
|---------------|-------|-------------|
| POSITION_BUY  | 0     | Buy.        |
| POSITION_SELL | 1     | Sell.       |

### EnActivation

Types of position activation are listed in the EnActivation enumeration.

| ID number          | Value | Description |
|--------------------|-------|-------------|
| ACTIVATION_NONE    | 0     | None.       |
| ACTIVATION_SL      | 1     | Stop Loss.  |
| ACTIVATION_TP      | 2     | Take Profit |
| ACTIVATION_STOPOUT | 3     | Stop Out.   |

### EnTradeActivationFlags

Flags of trade position activation are listed in the EnTradeActivationFlags enumeration:

| ID number             | Value | Description                                     |
|-----------------------|-------|-------------------------------------------------|
| ACTIV_FLAGS_NO_LIMIT  | 0x01  | Do not handle reaching of the Limit level.      |
| ACTIV_FLAGS_NO_STOP   | 0x02  | Do not handle the reaching of the stop level.   |
| ACTIV_FLAGS_NO_SLIMIT | 0x04  | Do not handle reaching of the Stop-Limit level. |
| ACTIV_FLAGS_NO_SL     | 0x08  | Do not handle activation upon Stop Loss.        |
| ACTIV_FLAGS_NO_TP     | 0x10  | Do not handle activation upon Take Profit.      |



| ID number                 | Value | Description                                       |
|---------------------------|-------|---------------------------------------------------|
| ACTIV_FLAGS_NO_SO         | 0x20  | Do not handle activation upon Stop-Out.           |
| ACTIV_FLAGS_NO_EXPIRATION | 0x40  | Do not handle order cancellation upon expiration. |
| ACTIV_FLAGS_NONE          | 0x00  | No flags.                                         |

Activation flags are inherited from the orders, as a result of which the position is created.

## EnPositionReason

Reasons for position opening are enumerated in EnPositionReason:

| ID number                       | Value | Description                                                                                                          |
|---------------------------------|-------|----------------------------------------------------------------------------------------------------------------------|
| POSITION_REASON_CLIENT          | 0     | Position opened manually by a client from the client terminal.                                                       |
| POSITION_REASON_EXPERT          | 1     | Position opened by a client using an Expert Advisor.                                                                 |
| POSITION_REASON_DEALER          | 2     | Position opened by a dealer through the manager terminal.                                                            |
| POSITION_REASON_SL              | 3     | Not used for positions.                                                                                              |
| POSITION_REASON_TP              | 4     | Not used for positions.                                                                                              |
| POSITION_REASON_SO              | 5     | Not used for positions.                                                                                              |
| POSITION_REASON_ROLLOVER        | 6     | Position reopened to charge swaps.                                                                                   |
| POSITION_REASON_EXTERNAL_CLIENT | 7     | Position opened from an external trading system.                                                                     |
| POSITION_REASON_VMARGIN         | 8     | Not used for positions.                                                                                              |
| POSITION_REASON_GATEWAY         | 9     | Position opened by a MetaTrader 5 gateway connected to the platform.                                                 |
| POSITION_REASON_SIGNAL          | 10    | Position opened as a result of copying a <u>trading signal</u> according to the subscription in the client terminal. |
| POSITION_REASON_SETTLEMENT      | 11    | Position opened as a result of operations associated with a futures/option delivery date. It is currently not used.  |
| POSITION_REASON_TRANSFER        | 12    |                                                                                                                      |

| ID number                        | Value | Description                                                                                                                                                                                                                                                                                             |
|----------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |       | Position opened as a result of transferring a position with a calculated price to a new symbol with the same underlying asset.                                                                                                                                                                          |
| POSITION_REASON_SYNC             | 13    | Position opened while synchronizing a trading account state with an external system.                                                                                                                                                                                                                    |
| POSITION_REASON_EXTERNAL_SERVICE | 14    | Position opened in the external trading system for service purposes (e.g. to correct a trading state).                                                                                                                                                                                                  |
| POSITION_REASON_MIGRATION        | 15    | Position opened as a result of import of clients' trading operations from the MetaTrader 4 server.                                                                                                                                                                                                      |
| POSITION_REASON_MOBILE           | 16    | Position opened via the MetaTrader 5 mobile terminal for Android or iPhone.                                                                                                                                                                                                                             |
| POSITION_REASON_WEB              | 17    | Position opened via the web terminal.                                                                                                                                                                                                                                                                   |
| POSITION_REASON_SPLIT            | 18    | Position opened as a result of a symbol split.                                                                                                                                                                                                                                                          |
| POSITION_REASON_CORPORATE_ACTION | 19    | Position created as a result of a corporate action, such as consolidating or renaming securities, transferring a client to a different account, etc. API applications set this flag for service operations so that the platform does not account for such corporate actions in commission calculations. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_deals

mt5\_deals

Data on deals is exported to the table. If "Export history orders and deals into separate tables by years" option is enabled in settings, deals for each year are exported to a separate table. A year is specified in the table heading, for example, mt5\_deals\_2012. If the option is disabled, all deals are exported to a single mt5\_deals table.

The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                                                                                                             |
|----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deal           | Integer  | Primary key. The ticket of a deal.                                                                                                                                                                                                                                      |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                          |
| Login          | Integer  | The login of the client, to whom the deal belongs.                                                                                                                                                                                                                      |
| Dealer         | Integer  | The login of a dealer, who has processed a deal.                                                                                                                                                                                                                        |
| Order          | Integer  | The ticket of the order, as a result of which a deal was executed.                                                                                                                                                                                                      |
| Action         | Integer  | Type of action performed with a deal. Passed in a value of the <u>EnDealAction</u> enumeration.                                                                                                                                                                         |
| Entry          | Integer  | Deal direction. Passed in a value of the <u>EnEntryFlags</u> enumeration.                                                                                                                                                                                               |
| Digits         | Integer  | The number of decimal places in the price of a deal.                                                                                                                                                                                                                    |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client has executed the deal.                                                                                                                                                                                  |
| ContractSize   | Float    | The contract size of the symbol, for which a deal was executed.                                                                                                                                                                                                         |
| Time           | DateTime | Trade execution time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                                                                                                                 |
| Symbol         | String   | The symbol, for which a deal is executed.                                                                                                                                                                                                                               |
| Price          | Float    | The price of the deal.                                                                                                                                                                                                                                                  |
| PriceSL        | Float    | The Stop Loss level of a deal. Stop Loss values for entry and reversal deals are set in accordance with the Stop Loss of orders, which initiated these deals. The Stop Loss values of appropriate positions as of the time of position closing are used for exit deals. |
| PriceTP        | Float    | Take Profit values for entry and reversal deals are set in accordance with the Take Profit of orders, which initiated these deals. The Take Profit values of appropriate positions as of the time of position closing are used for exit deals.                          |
| Volume         | Integer  | The deal volume. One unit corresponds to 1/10000 lot.                                                                                                                                                                                                                   |

## MetaTrader 5 trading platform

| Name            | Type    | Description                                                                                                                                                                                                                      |
|-----------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VolumeExt       | Integer | The deal volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                                              |
| VolumeClosed    | Integer | The position volume that was closed by the deal. One unit corresponds to 1/10000 lot.                                                                                                                                            |
| VolumeClosedExt | Integer | The extended accuracy volume of a position that was closed by the deal. One unit corresponds to 1/100000000 lot.                                                                                                                 |
| Profit          | Float   | Profit from a deal.                                                                                                                                                                                                              |
| Value           | Float   | The deal value in client deposit currency.                                                                                                                                                                                       |
| Storage         | Float   | The swap size for a deal.                                                                                                                                                                                                        |
| Commission      | Float   | The amount of <u>commission</u> charged for a deal.                                                                                                                                                                              |
| Fee             | Float   | <u>Fee</u> per deal.                                                                                                                                                                                                             |
| RateProfit      | Float   | The exchange rate of the profit currency of a deal to the deposit currency of a client group.                                                                                                                                    |
| RateMargin      | Float   | The exchange rate of the margin currency of a deal to the client deposit currency.                                                                                                                                               |
| ExpertID        | Integer | The ID of the Expert Advisor that has executed a deal.                                                                                                                                                                           |
| PositionID      | Integer | The position identifier (ticket) for a deal.                                                                                                                                                                                     |
| Comment         | String  | Comment to a deal.                                                                                                                                                                                                               |
| ProfitRaw       | Float   | The amount of return resulting from a deal. Return is specified in the profit currency of the symbol, for which the deal is executed.                                                                                            |
| PricePosition   | Float   | The price of the position closed with this deal.                                                                                                                                                                                 |
| TickValue       | Float   | The tick value price for a deal.                                                                                                                                                                                                 |
| TickSize        | Float   | The tick size for a deal.                                                                                                                                                                                                        |
| Flags           | Integer | The common flags of a deal. This parameter is reserved for future use.                                                                                                                                                           |
| Reason          | Integer | The reason for performing a deal. Passed in a value of the <u>EnDealReason</u> enumeration.                                                                                                                                      |
| Gateway         | String  | The ID of a gateway, using which a deal was performed.                                                                                                                                                                           |
| PriceGateway    | Float   | The price that was actually used for performing a deal through the gateway in an external trading system without taking into account its price transformation settings of the gateway.                                           |
| MarketBid       | Float   | The market Bid price as at the time of deal execution by the server. The field is only filled for the deals which were created after the platform was updated to build 2890 or higher. For earlier deals the value will be zero. |

# MetaTrader 5 trading platform

| Name       | Type     | Description                                                                                                                                                                                                                                                                                                         |
|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MarketAsk  | Float    | The market Ask price as at the time of deal execution by the s<br>The field is only filled for the deals which were created after t<br>platform was updated to build 2890 or higher. For earlier deal<br>value will be zero.                                                                                        |
| MarketLast | Float    | The market Last price as at the time of deal execution by the s<br>The field is only filled for the deals which were created after t<br>platform was updated to build 2890 or higher. For earlier deal<br>value will be zero.                                                                                       |
| TimeMsc    | DateTime | Trade execution time in the YYYY-MM-DD HH:MM:SS.MS form                                                                                                                                                                                                                                                             |
| ApiData    | String   | User data which can be added via MetaTrader 5 API. Sample v<br>entry:<br>[{"pos":0,"app_id":1,"valInt":500,"valUInt":500,"valDbl":0.000<br>It specifies the user data index, the ID of the application that a<br>as well as the data of three types: Int, UInt and double. The st<br>contain up to 16 such entries. |

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## Enumerations

To pass information about deals the following enumerations are used:

- [EnDealAction](#)
- [EnEntryFlags](#)
- [EnDealReason](#)

### EnDealAction

Types of actions performed by a deal are enumerated in EnDealAction.

| ID number               | Value | Description                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEAL_BUY                | 0     | A Buy deal.                                                                                                                                                                                                                                                                                                                                                                                        |
| DEAL_SELL               | 1     | A Sell deal.                                                                                                                                                                                                                                                                                                                                                                                       |
| DEAL_BALANCE            | 2     | Balance operation.                                                                                                                                                                                                                                                                                                                                                                                 |
| DEAL_CREDIT             | 3     | Credit operation.                                                                                                                                                                                                                                                                                                                                                                                  |
| DEAL_CHARGE             | 4     | Additional charges/withdrawals.                                                                                                                                                                                                                                                                                                                                                                    |
| DEAL_CORRECTION         | 5     | Correcting operations.                                                                                                                                                                                                                                                                                                                                                                             |
| DEAL_BONUS              | 6     | Bonuses.                                                                                                                                                                                                                                                                                                                                                                                           |
| DEAL_COMMISSION         | 7     | Commission.                                                                                                                                                                                                                                                                                                                                                                                        |
| DEAL_COMMISSION_DAILY   | 8     | Daily commission.                                                                                                                                                                                                                                                                                                                                                                                  |
| DEAL_COMMISSION_MONTHLY | 9     | Monthly commission.                                                                                                                                                                                                                                                                                                                                                                                |
| DEAL_AGENT_DAILY        | 10    | Daily agent commission.                                                                                                                                                                                                                                                                                                                                                                            |
| DEAL_AGENT_MONTHLY      | 11    | Daily agent commission.                                                                                                                                                                                                                                                                                                                                                                            |
| DEAL_INTERESTRATE       | 12    | Accrual of annual interest.                                                                                                                                                                                                                                                                                                                                                                        |
| DEAL_BUY_CANCELED       | 13    | A canceled Buy deal. Using the IMTExecution::TE_DEAL_CANCEL trade execution, the Gateway API can notify the platform about the cancellation of a previously executed deal in the external trading system. In this case the type of the earlier executed Buy trade is replaced with this one. The profit/loss of a trade is cleared. Then the client's position is recalculated and the appropriate |

| ID number             | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |       | profit/loss is added/subtracted as a separate balance operation. Deal cancellation does not change the client's order history. Deal cancellation does not entail changes in client's orders history. A deal of the DEAL_BUY_CANCELED type is not included into the calculation of the financial state of account and is not taken into account in recalculated positions.                                                                                                                                                                                                                                                                                                                                                                                                      |
| DEAL_SELL_CANCELED    | 14    | A canceled Sell deal. Using the IMTExecution::TE_DEAL_CANCEL trade execution, the Gateway API can notify the platform about the cancellation of a previously executed deal in the external trading system. In this case the type of the earlier executed Buy trade is replaced with this one. The profit/loss of a trade is cleared. Then the client's position is recalculated and the appropriate profit/loss is added/subtracted as a separate balance operation. Deal cancellation does not change the client's order history. Deal cancellation does not entail changes in client's orders history. A deal of the DEAL_SELL_CANCELED type is not included into the calculation of the financial state of account and is not taken into account in recalculated positions. |
| DEAL_DIVIDEND         | 15    | Dividend operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DEAL_DIVIDEND_FRANKED | 16    | Franked (non-taxable) dividend operations (tax is paid by a company, not a client).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DEAL_TAX              | 17    | Charging a tax.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DEAL_AGENT            | 18    | Charging an agent commission. Used in case of an instant commission charge to an agent (every time the agent's client performs a deal).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DEAL_SO_COMPENSATION  | 19    | An operation connected with the <u>compensation of a negative account</u> after the Stop Out event.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| ID number                   | Value | Description                                                                      |
|-----------------------------|-------|----------------------------------------------------------------------------------|
| DEAL_SO_COMPENSATION_CREDIT | 20    | <u>Withdrawing credit funds</u> after a negative balance compensation operation. |

EnEntryFlags

Types of actions performed by a deal with respect to positions are enumerated in EnEntryFlags.

| ID number    | Value | Description                                                                                                                                               |
|--------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENTRY_IN     | 0     | Entering the market or adding the volume.                                                                                                                 |
| ENTRY_OUT    | 1     | Exit from the market or partial closure.                                                                                                                  |
| ENTRY_INOUT  | 2     | Reversal.                                                                                                                                                 |
| ENTRY_OUT_BY | 3     | Close by a simultaneous closure of two opposite positions of the same financial instrument. This operation type is only used in the <u>hedging mode</u> . |

EnDealReason

Types of reasons for order placing are listed in EnDealReason.

| ID number                   | Value | Description                                                                                                                                                 |
|-----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEAL_REASON_CLIENT          | 0     | Deal performed by a client manually through the client terminal.                                                                                            |
| DEAL_REASON_EXPERT          | 1     | Deal performed by a client with using an Expert Advisor.                                                                                                    |
| DEAL_REASON_DEALER          | 2     | Deal performed by a dealer through the manager terminal.                                                                                                    |
| DEAL_REASON_SL              | 3     | Deal performed as a result of Stop Loss activation.                                                                                                         |
| DEAL_REASON_TP              | 4     | Deal performed as a result of Take Profit activation.                                                                                                       |
| DEAL_REASON_SO              | 5     | Deal performed when the client reached the Stop-Out level.                                                                                                  |
| DEAL_REASON_ROLLOVER        | 6     | Deal performed when reopening a position for charging swaps.                                                                                                |
| DEAL_REASON_EXTERNAL_CLIENT | 7     | Deal performed by a client from an external trading system. For this type of deals the commission is charged as distinct from DEAL_REASON_EXTERNAL_SERVICE. |



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| ID number                    | Value | Description                                                                                                                                                                                                                                                                                                |
|------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEAL_REASON_VMARGIN          | 8     | Deal performed for accruing variation margin.                                                                                                                                                                                                                                                              |
| DEAL_REASON_GATEWAY          | 9     | Deal performed by a MetaTrader 5 gateway that had connected to the trading platform.                                                                                                                                                                                                                       |
| DEAL_REASON_SIGNAL           | 10    | Deal performed as a result of copying a <u>trade signal</u> according to a subscription in the client terminal.                                                                                                                                                                                            |
| DEAL_REASON_SETTLEMENT       | 11    | Deal performed to compulsory close a position due to the settlement of a futures contract/option.                                                                                                                                                                                                          |
| DEAL_REASON_TRANSFER         | 12    | Deal performed due to transferring a position at the settlement price to a new symbol with the same underlying asset.                                                                                                                                                                                      |
| DEAL_REASON_SYNC             | 13    | Deal performed as a result of synchronization of an account's trade state with an external system.                                                                                                                                                                                                         |
| DEAL_REASON_EXTERNAL_SERVICE | 14    | Deal performed from an external trading system for technical reasons (for example, to correct the trade state of a client). For this type of deals the commission is not charged.                                                                                                                          |
| DEAL_REASON_MIGRATION        | 15    | Deal created as a result of importing trade operations from a MetaTrader 4 server.                                                                                                                                                                                                                         |
| DEAL_REASON_MOBILE           | 16    | The deal is conducted via the MetaTrader 5 mobile terminal for Android or iPhone.                                                                                                                                                                                                                          |
| DEAL_REASON_WEB              | 17    | The deal is conducted via the web terminal.                                                                                                                                                                                                                                                                |
| DEAL_REASON_SPLIT            | 18    | The deal is conducted as a result of a symbol split.                                                                                                                                                                                                                                                       |
| DEAL_REASON_CORPORATE_ACTION | 19    | The deal is created as a result of a corporate action, such as consolidating or renaming securities, transferring a client to a different account, etc. API applications set this flag for service operations so that the platform does not account for such corporate actions in commission calculations. |

MetaTrader 5 trading platform

| ID number | Value | Description |
|-----------|-------|-------------|
|           |       |             |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_accounts

mt5\_accounts

Data on the state of trading accounts is exported to the table. The table contains the following fields:

| Name              | Type    | Description                                                                                                                                                                              |
|-------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login             | Integer | Primary key. The login of the client, to whom the trading account belongs.                                                                                                               |
| CurrencyDigits    | Integer | The number of digits after the decimal point in the account deposit currency.                                                                                                            |
| Balance           | Float   | The balance of a trade account.                                                                                                                                                          |
| Credit            | Float   | The current amount of credit given to an account.                                                                                                                                        |
| Margin            | Float   | The current margin of the account.                                                                                                                                                       |
| MarginFree        | Float   | The free margin of an account.                                                                                                                                                           |
| MarginLevel       | Float   | The margin level as a percentage. It is calculated as a percentage of the current account equity (Equity) to the margin volume (Margin);                                                 |
| MarginLeverage    | Integer | Margin leverage.                                                                                                                                                                         |
| MarginInitial     | Float   | The current size of the initial margin of positions on a trading account.                                                                                                                |
| MarginMaintenance | Float   | The current size of the maintenance margin of positions on a trading account.                                                                                                            |
| Profit            | Float   | The size of the current profit for all open positions.                                                                                                                                   |
| Storage           | Float   | The current size of swaps charged for open positions on the account.                                                                                                                     |
| Floating          | Float   | The size of floating profit/loss of open positions on the account. The floating profit/loss is calculated as the sum of Profit, Storage and Commission of open positions on the account. |
| Equity            | Float   | The account equity calculated as a sum of Balance, Credit and Floating.                                                                                                                  |
| BlockedCommission | Float   | The amount of the standard commission locked on the account, which has been accumulated during the day/month.                                                                            |
| BlockedProfit     | Float   | The amount of intraday profit locked on the account.                                                                                                                                     |

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| Name        | Type  | Description                                                   |
|-------------|-------|---------------------------------------------------------------|
| Assets      | Float | The current total amount of assets on a trading account.      |
| Liabilities | Float | The current total amount of liabilities on a trading account. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_prices

mt5\_prices

Price data of financial instruments is exported to the table. The table contains the following fields:

| Name     | Type     | Description                                                                                                                       |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Price_ID | Integer  | Primary key. Unique symbol ID for more efficient request of the data from the database. Assigned automatically during the export. |
| Symbol   | String   | Symbol name.                                                                                                                      |
| Digits   | Integer  | The number of decimal places in the price.                                                                                        |
| Time     | DateTime | The time of quotes coming in the format YYYY-MM-DD HH:MM:SS.                                                                      |
| BidLast  | Float    | The Bid price.                                                                                                                    |
| BidLow   | Float    | The lowest bid price for the current day .                                                                                        |
| BidHigh  | Float    | The highest bid price for the current day.                                                                                        |
| BidDir   | Integer  | The direction of change of the bid price relative to its previous state (0 unchanged, 1 upwards, 2 downwards).                    |
| AskLast  | Float    | The Ask price.                                                                                                                    |
| AskLow   | Float    | The lowest ask price for the current day.                                                                                         |
| AskHigh  | Float    | The highest ask price for the current day.                                                                                        |
| AskDir   | Integer  | The direction of change of the ask price relative to its previous state (0 unchanged, 1 upwards, 2 downwards).                    |
| LastLast | Float    | The price of the last committed transaction.                                                                                      |
| LastLow  | Float    | The lowest price, at which a deal was executed during the current day.                                                            |
| LastHigh | Float    | The highest price, at which a deal was executed during the current day.                                                           |
| LastDir  | Integer  | The change direction of the price of the last deal relative to its previous state (0 unchanged, 1 upwards, 2 downwards).          |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_daily

mt5\_daily

Daily reports generated for clients are exported to the table. The table contains the following fields:

| Name              | Type     | Description                                                                                                                                                                                                                                                                                                                                               |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Datetime          | DateTime | The date and time of the daily report generation.                                                                                                                                                                                                                                                                                                         |
| Login             | Integer  | The login of the client for whom the daily report is generated.                                                                                                                                                                                                                                                                                           |
| Timestamp         | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                                                                            |
| DatetimePrev      | DateTime | Date and time of the previous daily report generation in the YYYY-MM-DD HH:MM:SS.                                                                                                                                                                                                                                                                         |
| Name              | String   | The name of a client in the daily report                                                                                                                                                                                                                                                                                                                  |
| Group             | String   | The group of a client in a daily report.                                                                                                                                                                                                                                                                                                                  |
| Currency          | String   | The client's deposit currency in a daily report.                                                                                                                                                                                                                                                                                                          |
| Company           | String   | The company serving the client in a daily report.                                                                                                                                                                                                                                                                                                         |
| EMail             | String   | An email of a client in a daily report.                                                                                                                                                                                                                                                                                                                   |
| Balance           | Float    | The size of a client's balance in a daily report.                                                                                                                                                                                                                                                                                                         |
| Credit            | Float    | The amount of a client's credit funds in a daily report.                                                                                                                                                                                                                                                                                                  |
| InterestRate      | Float    | The amount of accumulated annual interest. Annual interest is calculated every day in accordance with the <u>group settings</u> and is accumulated in a separate account field. At the end of each month, the accumulated amount is credited to the account balance using an <u>Interest rate</u> operation, and the InterestRate value is reset to zero. |
| CommissionDaily   | Float    | The amount of a client's commissions for a day in a report.                                                                                                                                                                                                                                                                                               |
| CommissionMonthly | Float    | The total amount of a client's commissions for the current month in a report.                                                                                                                                                                                                                                                                             |
| AgentDaily        | Float    | The amount of agent commission charged for a client's trade operations for a reported day.                                                                                                                                                                                                                                                                |

## MetaTrader 5 trading platform

| Name              | Type    | Description                                                                                                                     |
|-------------------|---------|---------------------------------------------------------------------------------------------------------------------------------|
| AgentMonthly      | Float   | The amount of agent commission charged for a client's trade operations for the current month.                                   |
| BalancePrevDay    | Float   | The value of a client's balance as of the end of the previous day.                                                              |
| BalancePrevMonth  | Float   | The value of a client's balance as of the end of the current month.                                                             |
| EquityPrevDay     | Float   | The value of a client's equity as of the end of the previous day.                                                               |
| EquityPrevMonth   | Float   | The value of a client's equity as of the end of the previous trading month.                                                     |
| Margin            | Float   | The size of a client's margin in a daily report.                                                                                |
| MarginFree        | Float   | A client's free margin in a daily report.                                                                                       |
| MarginLevel       | Float   | The margin level of a client in a daily report.                                                                                 |
| MarginLeverage    | Integer | The margin leverage of a client in a daily report.                                                                              |
| Profit            | Float   | The size of the current profit for all open positions of a client in a daily report.                                            |
| ProfitStorage     | Float   | The current size of swaps charged for a client's open positions for a day, but not yet reflected in the balance.                |
| ProfitEquity      | Float   | The amount of the current floating equity of a client in a daily report.                                                        |
| ProfitAssets      | Float   | The current amount of a client's assets in a daily report. It is only used for the <u>Exchange risk management model</u> .      |
| ProfitLiabilities | Float   | The current amount of a client's liabilities in a daily report. It is only used for the <u>Exchange risk management model</u> . |
| DailyProfit       | Float   | The amount of a client's daily profit.                                                                                          |
| DailyBalance      | Float   | The amount accrued to a client's balance during the reported day.                                                               |
| DailyCredit       | Float   | The amount of credit given to a client during the reported day.                                                                 |
| DailyCharge       | Float   | The amount of other charges to the client's balance during the reported day.                                                    |
| DailyCorrection   | Float   | The amount of corrective balance operations for a reported day.                                                                 |

MetaTrader 5 trading platform

| Name                | Type  | Description                                                                                |
|---------------------|-------|--------------------------------------------------------------------------------------------|
| DailyBonus          | Float | The amount of bonuses added to the client's balance for the reported day.                  |
| DailyStorage        | Float | The amount of swaps calculated for the client for a reported day.                          |
| DailyCommInstant    | Float | The amount of instant commissions charged from the client for a reported day.              |
| DailyCommRound      | Float | The amount of turnover commissions charged from the client for a reported day.             |
| DailyCommFee        | Float | The <u>fee</u> amount charged for the client's deals for the reported day.                 |
| DailyDividend       | Float | The amount of dividends accrued to the client for a reported day.                          |
| DailyTaxes          | Float | The amount of taxes withheld from the client for the reported day.                         |
| DailySOCompensation | Float | The amount of negative balance compensation accrued to the client for a reported day.      |
| DailyAgent          | Float | The amount of agent commission charged for a client's trade operations for a reported day. |
| DailyInterest       | Float | The amount accrued to a client as part of the annual interest rate for the reported day.   |



MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_daily\_orders

mt5\_daily\_orders

Data on the end-of-day status of open orders is exported to this table. The table is generated only if the backup server settings have the "Export additional columns for daily reports \ Orders from daily reports" option enabled. The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                    |
|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Datetime       | DateTime | The date for which the order status was saved.                                                                                                                                 |
| Order          | Integer  | Primary key. Order ticket.                                                                                                                                                     |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed. |
| ExternalID     | String   | The order ID in external trading systems.                                                                                                                                      |
| Login          | Integer  | The login of the client, to whom the order belongs.                                                                                                                            |
| Dealer         | Integer  | The login of a dealer, who has processed an order.                                                                                                                             |
| Symbol         | String   | The symbol of an order.                                                                                                                                                        |
| Digits         | Integer  | The number of decimal places in the price of an order.                                                                                                                         |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client who has placed the order.                                                                                      |
| ContractSize   | Float    | The contract size of the symbol, for which an order was placed.                                                                                                                |
| State          | Integer  | The current state of an order. Passed as a value of the <u>EnOrderMode</u> enumeration.                                                                                        |
| Reason         | Integer  | The reason for placing the order. Passed as a value of the <u>EnOrderReason</u> enumeration.                                                                                   |
| TimeSetup      | DateTime | Order placement time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeSetupMsc   | DateTime | Order placement time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |
| TimeExpiration | DateTime | Order expiration time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                       |
| TimeDone       | DateTime | Order execution time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                        |
| TimeDoneMsc    | DateTime | Order execution time in milliseconds in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                     |
| ModifyFlags    | Integer  | The order modification flags. Passed as a value of the <u>EnTradeModifyFlags</u> enumeration (sum of values of                                                                 |

| Name             | Type     | Description                                                                                                                                                                                                 |
|------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |          | appropriate flags).                                                                                                                                                                                         |
| Type             | Integer  | Order type. Passed as a value of the <u>EnOrderType</u> enumeration.                                                                                                                                        |
| TypeFill         | Integer  | Order filling type. Passed as a value of the <u>EnOrderFilling</u> enumeration.                                                                                                                             |
| TypeTime         | Integer  | Order expiration type. Passed in a value of the <u>EnOrderTime</u> enumeration.                                                                                                                             |
| PriceOrder       | Float    | Order price.                                                                                                                                                                                                |
| PriceTrigger     | Float    | The order triggering price.                                                                                                                                                                                 |
| PriceCurrent     | Float    | The current price of the symbol, for which an order has been placed.                                                                                                                                        |
| PriceSL          | Float    | The Stop Loss level of an order.                                                                                                                                                                            |
| PriceTP          | Float    | The Take Profit level of an order.                                                                                                                                                                          |
| VolumeInitial    | Integer  | The initial order volume. One unit corresponds to 1/10000 lot.                                                                                                                                              |
| VolumeInitialExt | Integer  | The initial order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                |
| VolumeCurrent    | Integer  | The current unfilled volume of an order. One unit corresponds to 1/10000 lot.                                                                                                                               |
| VolumeCurrentExt | Integer  | The current unfilled order volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                       |
| ExpertID         | Integer  | The ID of the Expert Advisor that has placed the order.                                                                                                                                                     |
| PositionID       | Integer  | The position ID (ticket) set in the order.                                                                                                                                                                  |
| PositionByID     | Integer  | The opposite position ID (ticket) set in the order. The property is set for Close By operations ( <u>OP_CLOSE_BY</u> ). The ticket of the position that is closed by the opposite one is set in PositionID. |
| Comment          | String   | A comment to an order.                                                                                                                                                                                      |
| ActivationMode   | Integer  | Order activation type. Passed in a value of the <u>EnOrderActivation</u> enumeration.                                                                                                                       |
| ActivationTime   | DateTime | Order activation time in the YYYY-MM-DD HH:MM:SS format.                                                                                                                                                    |
| ActivationPrice  | Float    | The price, at which the order was activated.                                                                                                                                                                |
| ActivationFlags  | Integer  | Order activation flags. Passed as a value of the <u>EnTradeActivationFlags</u> enumeration (sum of values of appropriate flags).                                                                            |
| RateMargin       | Float    |                                                                                                                                                                                                             |

# MetaTrader 5 trading platform

| Name    | Type   | Description                                                                                                                                                                                                                                                                                                                     |
|---------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |        | The rate of conversion of the margin currency of the symbol to the deposit currency of the user, which is used when calculating the margin requirements for the order.                                                                                                                                                          |
| ApiData | String | User data which can be added via MetaTrader 5 API. Sample user data entry:<br>[{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.00000000}]<br>It specifies the user data index, the ID of the application that added it, as well as the data of three types: Int, UInt and double. The string may contain up to 16 such entries. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_daily\_positions

mt5\_daily\_positions

Data on the end-of-day status of positions is exported to this table. The table is generated only if the backup server settings have the "Export additional columns for daily reports \ Orders from daily reports" option enabled. The table contains the following fields:

| Name           | Type     | Description                                                                                                                                                                                                                                                                                             |
|----------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Datetime       | DateTime | The date for which the position status was saved.                                                                                                                                                                                                                                                       |
| Timestamp      | Integer  | Unique record within the table. It is used for internal purposes of MetaTrader 5 servers. If the timestamp is changed for a record, it means that the record has been changed.                                                                                                                          |
| Login          | Integer  | The login of the client, to whom the trade position belongs.                                                                                                                                                                                                                                            |
| Symbol         | String   | The symbol of a trade position.                                                                                                                                                                                                                                                                         |
| Action         | Integer  | Position type. Passed in a value of the <u>EnPositionAction</u> enumeration.                                                                                                                                                                                                                            |
| Digits         | Integer  | The number of decimal places in the price of a position.                                                                                                                                                                                                                                                |
| DigitsCurrency | Integer  | The number of decimal places the deposit currency of the client who has opened the position.                                                                                                                                                                                                            |
| Reason         | Integer  | The reason for position opening. Passed in a value of the <u>EnPositionReason</u> enumeration.                                                                                                                                                                                                          |
| ContractSize   | Float    | The contract size of the symbol, for which a position is opened.                                                                                                                                                                                                                                        |
| Position       | Integer  | The ticket (unique identifier) of a trade position in a MetaTrader 5 platform.                                                                                                                                                                                                                          |
| ExternalID     | String   | The position ticket (unique number) in an external trading system.                                                                                                                                                                                                                                      |
| TimeCreate     | DateTime | Time of position creation, in seconds that have elapsed since 01.01.1970.                                                                                                                                                                                                                               |
| TimeUpdate     | DateTime | Time of the last modification of a position, in seconds that have elapsed since 01.01.1970. The modification time of a position is the time of the last modification of its volume. Virtually, it is the time of the last deal performed by the financial instrument that corresponds to that position. |
| TimeCreateMsc  | DateTime | Position generation time in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                                                                                                                                                          |
| TimeUpdateMsc  | DateTime | Time of a trading position last change in the YYYY-MM-DD HH:MM:SS.MS format. The modification time of a position is                                                                                                                                                                                     |

| Name             | Type     | Description                                                                                                                                                                                                        |
|------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |          | the time of the last modification of its volume. Virtually, it is the time of the last deal performed by the financial instrument that corresponds to that position.                                               |
| PriceOpen        | Float    | The weighted average open price of a position. Calculated by the following formula: (price of deal 1 * volume of deal 1 + ... + price of deal N * volume of deal N) / (volume of deal 1 + ... + volume of deal N). |
| PriceCurrent     | Float    | The current price of the symbol, for which a trade position has been opened.                                                                                                                                       |
| PriceSL          | Float    | The Stop Loss level of a trade position.                                                                                                                                                                           |
| PriceTP          | Float    | The Take Profit level of a trade position.                                                                                                                                                                         |
| Volume           | Integer  | The volume of a trade position. One unit corresponds to 1/10000 lot.                                                                                                                                               |
| VolumeExt        | Integer  | The trade position volume with an extended accuracy. One unit corresponds to 1/100000000 lot.                                                                                                                      |
| Profit           | Float    | Returns of a trade position in deposit currency.                                                                                                                                                                   |
| Storage          | Float    | The swap size for a position in deposit currency.                                                                                                                                                                  |
| RateProfit       | Float    | The exchange rate of the profit currency of a position to the deposit currency of a client group.                                                                                                                  |
| RateMargin       | Float    | The exchange rate of the margin currency of a position to the client's deposit currency.                                                                                                                           |
| ExpertID         | Integer  | The ID of the Expert Advisor that has opened the position.                                                                                                                                                         |
| ExpertPositionID | Integer  | Position ID.                                                                                                                                                                                                       |
| Comment          | String   | A comment to a position.                                                                                                                                                                                           |
| Dealer           | Integer  | The login of a dealer, who has processed an order, which opened the position.                                                                                                                                      |
| ActivationMode   | Integer  | Position activation type. Passed in a value of the <u>EnActivation</u> enumeration.                                                                                                                                |
| ActivationTime   | DateTime | Position activation time in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                                                                     |
| ActivationPrice  | Float    | Position activation price.                                                                                                                                                                                         |
| ActivationFlags  | Integer  | Position activation flags. Passed as a value of the <u>EnTradeActivationFlags</u> enumeration (sum of values of appropriate flags).                                                                                |
| ApiData          | String   | User data which can be added via MetaTrader 5 API. Sample user data entry:                                                                                                                                         |

# MetaTrader 5 trading platform

| Name | Type | Description                                                                                                                                                                                                                                                   |
|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |      | <p>[{pos:0,app_id:1,valInt:500,valUInt:500,valDbl:0.00000000}].</p> <p>It specifies the user data index, the ID of the application that added it, as well as the data of three types: Int, UInt and double. The string may contain up to 16 such entries.</p> |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_holidays

mt5\_holidays

Data about holidays is exported to this table. The table contains the following fields:

| Name        | Type    | Description                                                                                                                                                               |
|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Year        | Integer | The year of a holiday. 0 stands for yearly holidays.                                                                                                                      |
| Month       | Integer | Month of a holiday (1 January, 12 December).                                                                                                                              |
| Day         | Integer | Day of a holiday.                                                                                                                                                         |
| From        | Integer | Holiday start time. Specified in a number of minutes from 00:00. For example, 600 corresponds to 10:00.                                                                   |
| To          | Integer | Holiday end time. Specified in a number of minutes from 00:00. For example, 1200 corresponds to 20:00.                                                                    |
| Description | String  | Holiday description (no more than 128 symbols).                                                                                                                           |
| Timestamp   | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed. |
| Mode        | Integer | Holiday mode. 0 holiday disabled, 1 enabled.                                                                                                                              |
| Symbols     | String  | The list of financial instruments or their groups, for which a holiday is valid. Instruments and groups are separated by comma, for example: "EURUSD,CFD\*".              |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_network

mt5\_network

Data on the platform servers' general settings is exported to this table. The table contains the following fields:

| Name            | Type    | Description                                                                                                                                                                                                                        |
|-----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login           | Integer | Server ID.                                                                                                                                                                                                                         |
| Timestamp       | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.                                                          |
| Type            | Integer | Server type.                                                                                                                                                                                                                       |
| Name            | String  | Server name.                                                                                                                                                                                                                       |
| Address         | String  | Server address.                                                                                                                                                                                                                    |
| Port            | String  | Server port.                                                                                                                                                                                                                       |
| Adapter         | String  | Name of the currently used network controller.                                                                                                                                                                                     |
| ServiceTime     | Integer | Service time (time of optimization) when various operations aimed at increasing the performance and reliability of the platform are conducted. Specified in a number of minutes from 00:00. For example, 600 corresponds to 10:00. |
| FailoverMode    | Integer | Automatic <u>failover modes</u> : <ul style="list-style-type: none"> <li>• 0 failover disabled</li> <li>• 1 server is unavailable for most access servers</li> <li>• 2 server is unavailable for all access servers</li> </ul>     |
| FailoverTimeout | Integer | Time in seconds, during which the server should be unavailable for monitoring servers to start switching to the backup server.                                                                                                     |
| Adapters        | String  | List of all available network controllers on PC (comma-separated).                                                                                                                                                                 |
| Addresses       | String  | List of available addresses for outgoing connections from this server (comma-separated).                                                                                                                                           |
| Binds           | String  | List of listen addresses (comma-separated).                                                                                                                                                                                        |
| Points          | String  | List of public access points, via which                                                                                                                                                                                            |



# MetaTrader 5 trading platform

| Name                | Type     | Description                                                                                                                                                                                     |
|---------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |          | connections are to be accepted.                                                                                                                                                                 |
| Version             | Integer  | Server version.                                                                                                                                                                                 |
| Build               | Integer  | Server build.                                                                                                                                                                                   |
| BuildDate           | String   | Server build date.                                                                                                                                                                              |
| SysConnection       | Integer  | Status of a server connection to the main trade server.                                                                                                                                         |
| SysLastBoot         | DateTime | Time of the last server boot in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                             |
| SysOsName           | String   | Operating system of the computer running the server.                                                                                                                                            |
| SysCpuName          | String   | Processor type of the computer that is running the server.                                                                                                                                      |
| SysCpuNumber        | Integer  | Number of CPU cores.                                                                                                                                                                            |
| SysBits             | Integer  | Operating system bits: <ul style="list-style-type: none"> <li>• 32 32 bits</li> <li>• 64 64 bits</li> <li>• 0 other</li> </ul>                                                                  |
| SysMemoryTotal      | Integer  | The total amount of RAM in megabytes.                                                                                                                                                           |
| SysMemoryFree       | Integer  | The amount of free memory in megabytes.                                                                                                                                                         |
| SysMemoryCritical   | Integer  | The critical amount of free memory in megabytes.                                                                                                                                                |
| SysHddSize          | Integer  | Total volume of a disk in megabytes.                                                                                                                                                            |
| SysHddFree          | Integer  | Free memory on the disk in megabytes.                                                                                                                                                           |
| SysHddCritical      | Integer  | Critical amount of free memory on the disk in megabytes.                                                                                                                                        |
| SysHddFragmentation | Integer  | The current level of fragmentation of the server files in percentage.                                                                                                                           |
| SysHddFragCritical  | Integer  | The critical level of fragmentation of the server files in percentage.                                                                                                                          |
| SysDefragRecommend  | Integer  | The flag indicating that the operating system recommends defragmenting the disk: <ul style="list-style-type: none"> <li>• 0 no recommendation</li> <li>• 1 recommendation is present</li> </ul> |
| SysHddReadSpeed     | Integer  |                                                                                                                                                                                                 |

MetaTrader 5 trading platform

| Name                 | Type    | Description                                                                                       |
|----------------------|---------|---------------------------------------------------------------------------------------------------|
|                      |         | The current speed of data reading from the disk in megabytes per second.                          |
| SysHddReadCritical   | Integer | The critical speed of data reading from the disk in megabytes per second.                         |
| SysHddWriteSpeed     | Integer | The current speed of saving data to the disk in megabytes per second.                             |
| SysHddWriteCritical  | Integer | The critical speed of saving data to the disk in megabytes per second.                            |
| PerfConnectsMax      | Integer | The maximum number of simultaneous connections to a server that has been achieved during the day. |
| PerfConnectsCritical | Integer | Critical number of simultaneous connections to the server.                                        |
| PerfCpuMax           | Integer | Get the maximum level of CPU usage in percentage for the current day.                             |
| PerfCpuCritical      | Integer | Get the critical level of CPU usage in percentage.                                                |
| PerfMemoryMin        | Integer | The minimum size of free random access and virtual memory in megabytes per day.                   |
| PerfMemoryCritical   | Integer | The critical amount of free memory in megabytes.                                                  |
| PerfMemBlockMin      | Integer | The minimum value of the maximum memory block in megabytes per day.                               |
| PerfMemBlockCritical | Integer | The critical value of the maximum memory block in megabytes per day.                              |
| PerfNetworkMax       | Integer | The maximum level of network usage in kilobytes per second for the current day.                   |
| PerfNetworkCritical  | Integer | The critical level of network usage in kilobytes per second.                                      |
| PerfSocketsMax       | Integer | The maximum number of active sockets per day.                                                     |
| PerfSocketsCritical  | Integer | The critical number of active sockets.                                                            |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_network\_access\_servers

mt5\_network\_access\_servers

Data on the access servers settings is exported to this table. The table contains the following fields:

| Name                 | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login                | Integer | Server ID.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Priority             | Integer | Base priority of the access server from 0 to 15. The special priority 255 (idle) designed to create backup access servers is possible as well.                                                                                                                                                                                                                                                                                                                        |
| AntifloodEnable      | Integer | Antiflood control: 0 disabled, 1 enabled.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| AntifloodConnects    | Integer | The maximum number of connections from one IP address for a certain period of time, after which the address is temporarily blocked.                                                                                                                                                                                                                                                                                                                                   |
| AntifloodErrors      | Integer | The maximum number of incorrect connections, after which the IP address is temporarily blocked.                                                                                                                                                                                                                                                                                                                                                                       |
| NewsMaxCount         | Integer | The maximum number of news that can be stored on the access server.                                                                                                                                                                                                                                                                                                                                                                                                   |
| BalancingConnections | Integer | The current number of connections.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BalancingPriority    | Integer | The current priority of the access server.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| AccessMask           | Integer | <p>The allowed types of connection to the access server. Set as a sum of flag values:</p> <ul style="list-style-type: none"> <li>• 1 client connections.</li> <li>• 2 manager connections.</li> <li>• 4 administrator connections.</li> <li>• 8 connections via the Client API.</li> <li>• 16 connections via the Manager API.</li> <li>• 32 connections via the Web API.</li> </ul> <p>For example, the value of 63 means that all connection types are allowed.</p> |
| AccessFlags          | Integer | <p>Additional server accessing flags:</p> <ul style="list-style-type: none"> <li>• 1 the access server is <u>hidden from all terminals</u> but is available for connection.</li> </ul>                                                                                                                                                                                                                                                                                |
| Servers              | String  | IDs of trade servers (comma-separated), the connection to which is implemented through this                                                                                                                                                                                                                                                                                                                                                                           |

MetaTrader 5 trading platform

| Name | Type | Description    |
|------|------|----------------|
|      |      | access server. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_network\_history\_servers

mt5\_network\_history\_servers

Data on the history server settings is exported to this table. The table contains the following fields:

| Name             | Type    | Description                                                                 |
|------------------|---------|-----------------------------------------------------------------------------|
| Login            | Integer | Server ID.                                                                  |
| DatafeedsTimeout | Integer | Timeout of data feeds before switching to other ones. Specified in seconds. |
| NewsMax          | Integer | The maximum number of news that can be stored on the history server.        |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_network\_trade\_servers

mt5\_network\_trade\_servers

Data on the trade servers settings is exported to this table. The table contains the following fields:

| Name              | Type     | Description                                                                                                                                                                                                                                                                                                                                   |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login             | Integer  | Server ID.                                                                                                                                                                                                                                                                                                                                    |
| DemoMode          | Integer  | Mode of demo account allocation: <ul style="list-style-type: none"> <li>• 0 creation of demo accounts is disabled.</li> <li>• 1 prolong the period of demo accounts after connection.</li> <li>• 2 demo accounts with a fixed expiration date.</li> </ul>                                                                                     |
| DemoPeriod        | Integer  | The validity of demo accounts.                                                                                                                                                                                                                                                                                                                |
| OvernightMode     | Integer  | The overnight mode.                                                                                                                                                                                                                                                                                                                           |
| OvernightTime     | Integer  | The time of transition to the next day in minutes after 00:00.                                                                                                                                                                                                                                                                                |
| OvernightTimeLast | DateTime | The time of the last transition to the next day in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                        |
| OvernightTimePrev | DateTime | The time of the penultimate transition to the next day in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                 |
| OvernightDays     | Integer  | Schedule of operations related to the trading day closure. Set as a sum of flags: <ul style="list-style-type: none"> <li>• 0x00000001 Sunday</li> <li>• 0x00000002 Monday</li> <li>• 0x00000004 Tuesday</li> <li>• 0x00000008 Wednesday</li> <li>• 0x00000010 Thursday</li> <li>• 0x00000020 Friday</li> <li>• 0x00000040 Saturday</li> </ul> |
| OvermonthMode     | Integer  | The overmonth mode: <ul style="list-style-type: none"> <li>• 0 on the last day of the month.</li> <li>• 1 on the first day of the month.</li> </ul>                                                                                                                                                                                           |
| OvermonthTimeLast | DateTime | The time of the last transition to the next month in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                      |
| OvermonthTimePrev | DateTime |                                                                                                                                                                                                                                                                                                                                               |

# MetaTrader 5 trading platform

| Name               | Type    | Description                                                                                     |
|--------------------|---------|-------------------------------------------------------------------------------------------------|
|                    |         | The time of the penultimate transition to the next month in the YYYY-MM-DD HH:MM:SS.MSC format. |
| TotalUsers         | Integer | The total number of client accounts on the trade server.                                        |
| TotalUsersReal     | Integer | The total number of real clients on the trade server.                                           |
| TotalDeals         | Integer | The total number of deals executed on the trade server.                                         |
| TotalOrders        | Integer | The total number of active orders placed on the trade server.                                   |
| TotalOrdersHistory | Integer | The total number of orders in the history on the trade server.                                  |
| TotalPositions     | Integer | The total number of positions on the trade server.                                              |
| LoginsRange        | String  | Account ranges on the trade server. For example, 1000-1000000,2000001-2999001.                  |
| LoginsRangeUsed    | String  | Ranges of actually used logins from LoginsRange. For example, 1000-1004,0-0.                    |
| OrdersRange        | String  | Order ticket ranges on the trade server. For example, 1-1000000,2000001-2999001.                |
| OrdersRangeUsed    | String  | Ranges of actually used order tickets from OrdersRange. For example, 1-53,0-0.                  |
| DealsRange         | String  | Trade ticket ranges on the trade server. For example, 1-1000000,2000001-2999001.                |
| DealsRangeUsed     | String  | Ranges of actually used trade tickets and DealsRange. For example, 1-56,0-0.                    |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_network\_backup\_servers

mt5\_network\_backup\_servers

Data on the [backup servers settings](#) is exported to this table. The table contains the following fields:

| Name              | Type     | Description                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login             | Integer  | Server ID.                                                                                                                                                                                                                                                                                                                                                                                                |
| PairrServer       | Integer  | ID of the server to backup.                                                                                                                                                                                                                                                                                                                                                                               |
| BackupFlags       | Integer  | <u>Backup settings</u> . The settings are specified as a sum of flags: <ul style="list-style-type: none"> <li>• 1 enables backup.</li> <li>• 2 enables the backup of tick data.</li> <li>• 4 enables the possibility to use the server for the automatic Failover, with which the system switches to this backup server.</li> <li>• 8 enables synchronization of logs with the primary server.</li> </ul> |
| BackupPath        | String   | The path to save backups.                                                                                                                                                                                                                                                                                                                                                                                 |
| BackupPeriod      | Integer  | Backup frequency: <ul style="list-style-type: none"> <li>• 0 no periodic backups</li> <li>• 1 every 15 minutes</li> <li>• 2 every 30 minutes</li> <li>• 3 every hour</li> <li>• 4 every 4 hours</li> <li>• 5 every day</li> </ul>                                                                                                                                                                         |
| BackupTtl         | Integer  | Period to keep backups: <ul style="list-style-type: none"> <li>• 1 one day</li> <li>• 2 three days</li> <li>• 3 one week</li> <li>• 4 one month</li> <li>• 5 three months</li> <li>• 6 six months</li> </ul>                                                                                                                                                                                              |
| BackupTimeFull    | Integer  | Time of creating full backup copies in minutes since 00:00.                                                                                                                                                                                                                                                                                                                                               |
| BackupLastStartup | DateTime | The last backup copy creation time when launching the server in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                                                                       |
| BackupLastFull    | DateTime |                                                                                                                                                                                                                                                                                                                                                                                                           |



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| Name              | Type     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |          | The last full backup copy creation time in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| BackupLastArchive | DateTime | The last increment backup copy creation time in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| BackupLastSync    | DateTime | The time of the last successful synchronization with the backed up server in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                                                                                                                                                                                                                                                                                                        |
| SqlMode           | Integer  | <p>Mode of exporting data to an SQL database:</p> <ul style="list-style-type: none"> <li>• 0 export disabled</li> <li>• 1 export to Microsoft SQL Server</li> <li>• 2 export to FireBird</li> <li>• 3 export to MySQL</li> <li>• 4 export to Oracle</li> </ul>                                                                                                                                                                                                                                                          |
| SqlServer         | String   | The address of the server the database is installed on.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| SqlFolder         | String   | The name of the SQL database the data is exported to.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| SqlFlags          | Integer  | <p>Additional settings of data export to an SQL database. Defined by the sum of flags:</p> <p>1 export trade history to separate tables.</p> <p>2 do not export accounts and trade operations of demo groups</p> <p>For example, the value of 3 means that both settings are enabled.</p>                                                                                                                                                                                                                               |
| SqlPeriod         | Integer  | The frequency of price and profit data export.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SQLExportLastSync | Integer  | The time of the last full synchronization of databases and platform configurations with the SQL database. Specified in seconds since 01.01.1970. If data export is disabled or synchronization is in progress, the value is 0. The full synchronization is launched when a backup server is started or after connection to the SQL database or to the trading server is lost. After synchronization, the SQL database is updated in real time in accordance with the transactions of changes in the platform databases. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_network\_backup\_folders

mt5\_network\_backup\_folders

Data on the backed up custom directories is exported to this table. The table contains the following fields:

| Name      | Type    | Description                                                                            |
|-----------|---------|----------------------------------------------------------------------------------------|
| Folder_ID | Integer | Unique entry ID.                                                                       |
| Login     | Integer | Backup server ID.                                                                      |
| Folder    | String  | The path to the backed up folder relative to the backup server installation directory. |
| Masks     | String  | List of copied files (comma-separated). Files can be specified by masks.               |
| Filter    | String  | List of ignored files (comma-separated). Files can be specified by masks.              |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_firewall

mt5\_firewall

Data on the firewall settings is exported to this table. The table contains the following fields:

| Name      | Type    | Description                                                                                                                                                                   |
|-----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timestamp | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.     |
| Action    | Integer | The type of actions undertaken in accordance with the firewall rule: <ul style="list-style-type: none"> <li>• 0 block</li> <li>• 1 allow</li> <li>• 2 always allow</li> </ul> |
| From      | String  | Beginning of the range of the IP addresses the firewall rule is applied to.                                                                                                   |
| To        | String  | End of the range of the IP addresses the firewall rule is applied to.                                                                                                         |
| Comment   | String  | A comment to the firewall rule.                                                                                                                                               |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_routing

mt5\_routing

Data about trade request routing rules is exported to this table. The table contains the following fields:

| Name           | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name           | String  | The name of a routing rule.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Timestamp      | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mode           | Integer | The state of a routing rule: 0 disabled, 1 enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Request        | Integer | Types of requests for which the rule is applicable. The types are passed using the <u>EnRouteFlags</u> enumeration (as a sum of flags).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Type           | Integer | Types of orders for which the rule is applicable. The types are passed using the <u>EnTypeFlags</u> enumeration (as a sum of flags).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Flags          | Integer | Currently not used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ActionType     | Integer | <p>The value type for Action. Based on this field, it is possible to determine the field in which the Action value is contained: ActionValueInt, ActionValueUInt, ActionValueFloat or ActionValueString.</p> <p>Possible values:</p> <ul style="list-style-type: none"> <li>• 0 the current parameter does not have values (for example, <u>ACTION_CLEAR_TP</u>)</li> <li>• 1 the value is located in the ActionValueString field, and its type is string</li> <li>• 2 the value is located in the ActionValueInt field, and its type is int</li> <li>• 3 the value is located in the UInt field, and its type is uint</li> <li>• 4 the value is located in the ActionValueFloat field, and its type is float</li> </ul> |
| Action         | Integer | The type of action that is applied to a request in accordance with a rule. Passed as a value of the <u>EnRouteAction</u> enumeration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ActionValueInt | Integer | An int value for the action applied to the rule. For example, for the rule "pass to online dealers", the 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <b>Name</b>       | <b>Type</b> | <b>Description</b>                                                                        |
|-------------------|-------------|-------------------------------------------------------------------------------------------|
|                   |             | value means that the additional option "skip this rule if no dealers online" is disabled. |
| ActionValueUInt   | Integer     | An uint value for the action applied to the rule.                                         |
| ActionValueFloat  | Fraction    | A float value for the action applied to the rule.                                         |
| ActionValueString | String      | A string value for the action applied to the rule.                                        |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / [mt5\\_routing](#) / Enumerations

## Enumerations

The following enumerations are used for passing information about trade routing rules:

- [EnRouteFlags](#)
- [EnTypeFlags](#)
- [EnRouteAction](#)
- [EnRouteCondition](#)
- [EnConditionRule](#)

### EnRouteFlags

Conditions for applying a rule based on the request type are enumerated in EnRouteFlags.

| ID                | Value      | Description                                                                                                                    |
|-------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| REQUEST_NONE      | 0x00000000 | Conditions by the request type are not specified.                                                                              |
| REQUEST_PRICE     | 0x00000001 | Price request (for request execution).                                                                                         |
| REQUEST_REQUEST   | 0x00000002 | Confirmation of order execution as a dealer's type in the request execution mode (with the order confirmation option enabled). |
| REQUEST_INSTANT   | 0x00000004 | Placing an order in the instant execution mode.                                                                                |
| REQUEST_MARKET    | 0x00000008 | Placing an order in the market execution mode.                                                                                 |
| REQUEST_EXCHANGE  | 0x00000010 | Placing an order in the exchange execution mode.                                                                               |
| REQUEST_PENDING   | 0x00000020 | Placing a pending order.                                                                                                       |
| REQUEST_SLTP      | 0x00000040 | Modification of Stop Loss and Take Profit of a position.                                                                       |
| REQUEST_MODIFY    | 0x00000080 | Modification of a pending order.                                                                                               |
| REQUEST_REMOVE    | 0x00000100 | Deleting a pending order.                                                                                                      |
| REQUEST_ACTIVATE  | 0x00000200 | Activation (triggering) of a pending order.                                                                                    |
| REQUEST_STOPLIMIT | 0x00000400 | Activation of a Stop Limit order.                                                                                              |

| ID                          | Value      | Description                                                                                                                    |
|-----------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| REQUEST_SL                  | 0x00000800 | Triggering of a Stop Loss order.                                                                                               |
| REQUEST_TP                  | 0x00001000 | Triggering of a Take Profit order.                                                                                             |
| REQUEST_STOPOUT_ORDER       | 0x00002000 | A request to delete a pending order in case of reaching the stop-out level (if margin requirements are set for pending orders) |
| REQUEST_STOPOUT_POSITION    | 0x00004000 | A request to close a position when reaching stop-out.                                                                          |
| REQUEST_EXPIRATION          | 0x00008000 | Cancellation of an order upon expiration.                                                                                      |
| REQUEST_DEALER_POS_EXECUTE  | 0x00010000 | Position opening and closing by a dealer.                                                                                      |
| REQUEST_DEALER_ORD_PENDING  | 0x00020000 | Placing of a pending order by a dealer.                                                                                        |
| REQUEST_DEALER_POS_MODIFY   | 0x00040000 | Position modification by a dealer.                                                                                             |
| REQUEST_DEALER_ORD_MODIFY   | 0x00080000 | Order modification by a dealer.                                                                                                |
| REQUEST_DEALER_ORD_REMOVE   | 0x00100000 | Order deletion by a dealer.                                                                                                    |
| REQUEST_DEALER_ORD_ACTIVATE | 0x00200000 | Order activation by a dealer.                                                                                                  |
| REQUEST_DEALER_ORD_SLIMIT   | 0x00400000 | Activation of a Stop Limit order by a dealer. After this action is performed, the order turns into a limit order.              |
| REQUEST_DEALER_CLOSE_BY     | 0x00800000 | Close By. An operation of closing two oppositely directed positions at a single symbol performed by a dealer.                  |
| REQUEST_CLOSE_BY            | 0x01000000 | Close By. An operation of closing two oppositely directed positions at a single symbol performed by a client.                  |

### EnTypeFlags

Conditions for applying a rule based on the order type are enumerate in EnTypeFlags.

| ID | Value | Description |
|----|-------|-------------|
|----|-------|-------------|

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| <b>ID</b>            | <b>Value</b> | <b>Description</b>               |
|----------------------|--------------|----------------------------------|
| TYPE_NONE            | 0x0000       | No conditions by the order type. |
| TYPE_BUY             | 0x0001       | A Buy order.                     |
| TYPE_SELL            | 0x0002       | A Sell order.                    |
| TYPE_BUY_LIMIT       | 0x0004       | A limit Buy order.               |
| TYPE_SELL_LIMIT      | 0x0008       | A limit Sell order.              |
| TYPE_BUY_STOP        | 0x0010       | A stop Buy order.                |
| TYPE_SELL_STOP       | 0x0020       | A stop Sell order.               |
| TYPE_BUY_STOP_LIMIT  | 0x0040       | A limit Buy Stop order.          |
| TYPE_SELL_STOP_LIMIT | 0x0080       | A limit Sell Stop order.         |

EnRouteAction

Types of actions that are applied to requests are listed in EnRouteAction.

| <b>ID</b>            | <b>Value</b> | <b>Description</b>                                                                                                                                                                                                                                |
|----------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTION_DELAY_TIME    | 0            | Delay request execution by the specified number of milliseconds. After applying this action to a request, its execution continues in accordance with the created rules located below in the list. The delay is indicated in a separate parameter. |
| ACTION_DELAY_TICK    | 1            | Delay request execution by the specified number of ticks. After applying this action to a request, its execution continues in accordance with the created rules located below in the list. The delay is indicated in a separate parameter.        |
| ACTION_CLEAR_TP      | 2            | Clear the Take Profit level set in the order.                                                                                                                                                                                                     |
| ACTION_CLEAR_SL      | 3            | Clear the Stop Loss level set in the order.                                                                                                                                                                                                       |
| ACTION_CLEAR_SLTP    | 4            | Clear the Stop Loss and Take Profit levels set in the order.                                                                                                                                                                                      |
| ACTION_DEALER        | 1001         | Enqueue the request to be processed by the specified dealer. The flag of action omission in case there are no dealers online, is specified by an additional parameter.                                                                            |
| ACTION_DEALER_ONLINE | 1002         | Pass the request to dealers that are currently online. The flag of action omission in case there are no dealers online, is specified by an additional parameter.                                                                                  |



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| ID                    | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTION_REJECT         | 1003  | Reject a request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ACTION_REQUOTE        | 1004  | Send current market prices in response to the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ACTION_CONFIRM_CLIENT | 1005  | Confirm the execution of an order at a price requested in it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ACTION_CONFIRM_MARKET | 1006  | Confirm the execution of an order at the current market price.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ACTION_CANCEL_ORDER   | 1007  | <p>Cancel a pending order during its activation or modification. For example, if a pending order has triggered, but the client has already reached the maximum position volume and a new position cannot be opened, the routing rule will remove this order. Otherwise, the order would have continued to trigger on each new tick. When removing an order by this rule, "deleted [by dealer]" is added to the order comment. An entry about the routing rule that canceled the order is also added to the server journal. The server returns error code MT_RET_REQUEST_REJECT_CANCEL.</p> <p>The action can only be applied during pending order activation or modification (including modification buy a dealer). The rule does not affect other trade requests.</p> |

## EnRouteCondition

Additional conditions for activation of a routing rule are listed in IMTConCondition::EnRouteCondition.

| ID                 | Value | Description                                                                                                                  |
|--------------------|-------|------------------------------------------------------------------------------------------------------------------------------|
| CONDITION_DATETIME | 0     | Using this parameter you can compare date and time of a request with that specified in the "Value" field.                    |
| CONDITION_SYMBOL   | 1     | This parameter is used for specifying a symbol or a group of symbols requests for which will be subject to the routing rule. |
| CONDITION_VOLUME   | 2     | Deal volume requested in an order (in lots). This parameter is used for configuring a rule                                   |

| ID                         | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |       | depending on the request volume, e.g. automatic processing of requests less than 1 lot.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| CONDITION_MARKET_DEVIATION | 3     | This condition is applicable only with the instant execution mode. It takes into account the difference between the price of a client's request and the current market price. For Buy trades the deviation is calculated as ("Current Ask price" - "Client's request price"), for Sell trades it is equal to ("Client's request price" - "Current Bid price"). For example, if a client wants to buy at 1.2000, and the current Ask is 1.2008, then the deviation is equal to $1.2008 - 1.2000 = 8$ points. |
| CONDITION_TIME             | 4     | This parameter can be used for comparing the time or a request arrival (in minutes since 00:00) with the value specified in the "Value" field.                                                                                                                                                                                                                                                                                                                                                              |
| CONDITION_WEEKDAY          | 5     | This parameter allows to route requests depending on a day of the week.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| CONDITION_COMMENT          | 6     | This parameter allows to compare a request comment with a specified one. If "=" condition is specified, exact match of a comment is checked. If ">" or ">=" conditions are set, a specified substring is searched in a comment string. If "<" or "<=" conditions are set, comment substring is searched in a specified string.                                                                                                                                                                              |
| CONDITION_EXPERT           | 7     | This parameter allows to route requests placed by MQL5 programs.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| CONDITION_SIGNAL           | 8     | All operations copied by the client terminal in accordance with the subscription to a <u>trading</u>                                                                                                                                                                                                                                                                                                                                                                                                        |

| ID                             | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |       | <u>signal</u> are marked with a special flag. This parameter allows routing trade requests created by trade signals. If this condition is enabled, the rule will trigger for all signal operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| CONDITION_DEALER_LOGIN         | 9     | This parameter allows applying the routing rules depending on a dealer (or gateway) identifier specified in an order or position. The dealer identifier is specified in an order after it has been confirmed (processed) by the dealer/gateway. Due to it, this rule can be applied only when modifying/deleting an order, and not for newly created orders as they do not have a dealer identifier.. For positions, the dealer identifier is specified according to the dealer identifier of the order, whose execution resulted in the position opening. This parameter can be used when processing trade operations for a symbol through several gateways simultaneously. An order or position created through a specific gateway must be further processed through the same gateway. |
| CONDITION_SOURCE_LOGIN         | 10    | The parameter allows routing requests by a login of a dealer who set a request on a client's behalf.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| CONDITION_MARKET_DEVIATION_SPR | 11    | This condition works when executing orders in the Instant or Market modes, as well as when pending orders and Stop Loss/Take Profit orders are triggered. It takes into account the difference between the price of a client's request and the current market price. During a market execution, when a client does not set a price in the order,                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| ID            | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |       | <p>the difference between the market price during the request and the current market price is taken into account.</p> <p>The deviation is set in spreads. For floating-spread symbols, the current spread valid during the request check is used. For fixed-spread symbols, a spread value from the symbol settings is used.</p> <p>For Buy trades the deviation is calculated as ("Current Ask price" - "Client's request price"), for Sell trades it is equal to ("Client's request price" - "Current Bid price"). For example, if a client wants to buy at 1.2000, and the current Ask is 1.2008, then the deviation is equal to <math>1.2008 - 1.2000 = 8</math> points. The current spread is divided by this value and the result is compared with the value in the rule.</p> <p>When setting the condition, keep in mind that if the deviation is positive, opening at the request price is performed in the client's favor, if the deviation is negative, opening is performed against the client. Another example: if we set <math>&lt; -1</math>, the condition corresponds to the buy requests where a request price exceeds the current price by more than 1 spread.</p> |
| CONDITION_GAP | 12    | <p>This parameter allows processing trade requests in a special way under the market conditions that differ from normal ones. For example, after a gap, client requests can be rejected or requoted during a certain number of subsequent ticks. The gap mode is defined separately</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| ID                       | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |       | <p>for each symbol according to its settings.</p> <p>The parameter may take two values true or false (enabled/disabled). If the gap mode is active when checking a request according to the selected symbol routing rule, actions set in this rule are applied to it.</p> <p>The gap mode is checked by an instrument's Bid and Ask prices. If a gap is detected at least on one of the prices, the rule is triggered.</p> |
| CONDITION_LOGIN          | 1000  | The number of a client's account. This parameter allows creating individual rules for accounts.                                                                                                                                                                                                                                                                                                                            |
| CONDITION_GROUP          | 1001  | The group to which the client's account is included. This parameter is used for configuring rules for separate account groups.                                                                                                                                                                                                                                                                                             |
| CONDITION_COUNTRY        | 1002  | In this parameter a client's country can be specified. The specified rule will be applied to all clients living in this country.                                                                                                                                                                                                                                                                                           |
| CONDITION_CITY           | 1003  | Use this parameter to apply the rule to all clients living in a specified city.                                                                                                                                                                                                                                                                                                                                            |
| CONDITION_COLOR          | 1004  | Use this parameter to apply the rule for clients that are marked with the specified color.                                                                                                                                                                                                                                                                                                                                 |
| CONDITION_LEVERAGE       | 1005  | Use this parameter to apply the rule for clients with the specified leverage.                                                                                                                                                                                                                                                                                                                                              |
| CONDITION_COMMENT_CLIENT | 1006  | Use this parameter to apply the rule for clients with the specified comment.                                                                                                                                                                                                                                                                                                                                               |
| CONDITION_MARGIN         | 2000  | Use this parameter to set up rule application depending on the margin volume that is currently                                                                                                                                                                                                                                                                                                                             |

| ID                           | Value | Description                                                                                                                                         |
|------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |       | reserved (in the deposit currency).                                                                                                                 |
| CONDITION_MARGIN_LEVEL       | 2001  | This parameter allows using rules depending on the current margin level (in percents).                                                              |
| CONDITION_MARGIN_FREE        | 2002  | This parameter allows using rules depending on the current amount of free margin (in the deposit currency).                                         |
| CONDITION_EQUITY             | 2003  | This parameter allows using rules depending on the current equity on a client's account (in the deposit currency).                                  |
| CONDITION_BALANCE            | 2004  | This parameter allows using rules depending on the current balance of a client (in the deposit currency).                                           |
| CONDITION_PROFIT             | 2005  | This parameter allows using rules depending on the current floating profit of a client.                                                             |
| CONDITION_DAILY_DEALS        | 3000  | This parameter allows using rules depending on the number of deals of a client for the current and previous days (including weekends and holidays). |
| CONDITION_DAILY_DEALS_PERIOD | 3001  | The frequency of deals for a day. Calculated on the basis of the last 8 deals (the average time between deals).                                     |
| CONDITION_DAILY_PROFIT       | 3002  | The profit of the client, whose request is being handled, for the current and previous days (including weekends and holidays).                      |
| CONDITION_POSITION_VOLUME    | 4000  | The current volume of a position for the symbol, for which a request has arrived.                                                                   |
| CONDITION_POSITION_PROFIT    | 4001  | The current profit of a position for the symbol, for which a request has arrived.                                                                   |
| CONDITION_POSITION_AGE       | 4002  |                                                                                                                                                     |

| ID                              | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |       | Using this parameter you can specify time in seconds elapsed since position opening for the symbol a request for which is currently being handled. This parameter allows to track positions based on the time they have been held.                                                                                                                                                                                                   |
| CONDITION_POSITION_MODIFY_TIME  | 4003  | Using this parameter you can specify time in seconds elapsed since the last modification of a position for the symbol a request for which is currently being handled. Position modification means increase of its volume, partial closure, and modification of Stop Loss and Take Profit levels. This parameter allows to prevent evasion of the previous rule through manipulating one position, increasing or reducing its volume. |
| CONDITION_POSITION_AVERAGE_TIME | 4004  | This parameter allows to track positions based on the time the average age of the position for the symbol for which a request is being handled. The average position age is calculated as follows: $\text{Current time} - ((\text{Open time} + \text{Modification time})/2)$ .                                                                                                                                                       |
| CONDITION_POSITION_TOTAL        | 4005  | This parameter allows tracking the total number of a client's open positions on all symbols. For example, you can set the platform to reject trade requests to open new positions, if the client has reached the specified limit.                                                                                                                                                                                                    |
| CONDITION_POSITION_TOTAL_SYMBOL | 4006  | The parameter allows tracking the number of positions on the symbol that is specified in the current trade request. For example, if a client has placed an order on EURUSD, this condition will check the current number of the client's open positions on                                                                                                                                                                           |

| ID                            | Value | Description                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               |       | EURUSD.                                                                                                                                                                                                                                                                                                                                                                                         |
| CONDITION_ORDER_TOTAL         | 4007  | This parameter allows tracking the total number of a client's orders on all symbols. All orders are taken into account, including pending and history orders. Each opening and closing of a position (including partial closure), as well as placing of a pending order increases this counter.                                                                                                 |
| CONDITION_ORDER_TOTAL_SYMBOL  | 4008  | The parameter allows tracking the number of orders on the symbol that is specified in the current trade request. For example, if a client has placed an order on EURUSD, this condition will check the current number of the client's orders (both active and history) on EURUSD.                                                                                                               |
| CONDITION_POSITION_SL_TOUCHED | 4009  | The condition is triggered when the market price touches the stop loss of a position (while the stop loss may not be activated yet). Possible values true or false.                                                                                                                                                                                                                             |
| CONDITION_POSITION_TP_TOUCHED | 4010  | The condition is triggered when the market price touches the take profit of a position (while the take profit may not be activated yet). Possible values true or false.                                                                                                                                                                                                                         |
| CONDITION_ORDER_SL_TOUCHED    | 4011  | The condition is triggered when the market price touches the stop loss of a pending order. Possible values true or false. In combination with the REQUEST_ACTIVATE condition, it allows you to track the simultaneous breakthrough of a pending order (trigger) and its stop loss level. This may occur during a release of important news or after a weekend when a large price gap is formed. |
| CONDITION_ORDER_TP_TOUCHED    | 4012  | The condition is triggered when the market price touches the stop                                                                                                                                                                                                                                                                                                                               |



| <b>ID</b> | <b>Value</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                              |
|-----------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |              | loss of a pending order. Possible values true or false. In combination with the REQUEST_ACTIVATE condition, it allows you to track the simultaneous breakthrough of a pending order (trigger) and its take profit level. This may occur during a release of important news or after a weekend when a large price gap is formed. |

EnConditionRule

Types of parameter and value comparison are enumerated in IMTConCondition::EnConditionRule.

| <b>ID</b>        | <b>Value</b> | <b>Description</b>               |
|------------------|--------------|----------------------------------|
| RULE_EQ          | 0            | Condition of equality.           |
| RULE_NOT_EQ      | 1            | Condition of inequality.         |
| RULE_GREATER     | 2            | Condition of "greater than".     |
| RULE_NOT_LESS    | 3            | Condition of "not less than".    |
| RULE_LESS        | 4            | Condition of "less than".        |
| RULE_NOT_GREATER | 5            | Condition of "not greater than". |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_routing\_dealers

mt5\_routing\_dealers

Data about dealers/gateways to whom trade requests are forwarded for processing in accordance with a routing rule, is exported to this table.

| Name        | Type    | Description                                                            |
|-------------|---------|------------------------------------------------------------------------|
| Login       | Integer | The login of a dealer/gateway ID.                                      |
| RoutingName | String  | The name of the routing rule in which the dealer/gateway is specified. |
| Name        | String  | The name of the dealer/gateway.                                        |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_routing\_conds

mt5\_routing\_conds

Data about additional conditions specified in a routing rule is exported to this table.

| Name         | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Condition_ID | Integer | The unique identifier of the condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Name         | String  | The name of the routing rule to which the additional condition applies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Condition    | Integer | The type of the additional condition for the rule. Passed using the <a href="#">EnRouteCondition</a> enumeration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Rule         | Integer | A method for comparing a condition with the specified value. Passed using the <a href="#">EnConditionRule</a> enumeration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Type         | Integer | <p>The type of value of the additional condition for a routing rule. Based on this field, it is possible to determine the field in which the Condition value is contained: ValueInt, ValueUInt, ValueFloat or ValueString.</p> <p>Possible values:</p> <ul style="list-style-type: none"> <li>• 0 the current parameter does not have values (currently not used)</li> <li>• 1 the value is located in the ValueString field, and its type is string</li> <li>• 2 the value is located in the ValueInt field, and its type is int</li> <li>• 3 the value is located in the ValueUInt field, and its type is uint</li> <li>• 4 the value is located in the ValueFloat field, and its type is float</li> </ul> |
| ValueInt     | Integer | An int value for the condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ValueUInt    | Integer | An uint value for the condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ValueUInt    | Integer | An uint value for the condition. Used for volume with extended accuracy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ValueFloat   | Float   | A float value for the condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ValueString  | String  | A float value for the condition. For example, for the "Symbols" condition, the names of symbol (symbol groups) will be specified in this field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_feeders

mt5\_feeders

Data about data feed settings is exported to this table. The table contains the following fields:

| Name             | Type     | Description                                                                                                                                                                                                        |
|------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | String   | Get and set the data feed name.                                                                                                                                                                                    |
| Timestamp        | Integer  | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.                                          |
| Module           | String   | The name of the data feed module.                                                                                                                                                                                  |
| GatewayServer    | String   | The addresses at which the data feed will accept connections from the history server.                                                                                                                              |
| FeedServer       | String   | The addresses of the server to which the data feed is connected.                                                                                                                                                   |
| Enable           | Integer  | The data feed configuration status: 0 disabled, 1 enabled.                                                                                                                                                         |
| Mode             | Integer  | Data feed operation mode. Passed using the <u>EnFeedersMode</u> enumeration as a sum of flags. For example, 1 means that the data feed receives news, 9 receives news while working in the "remote datafeed" mode. |
| Timeout          | Integer  | Timeout of a data feed before reconnecting.                                                                                                                                                                        |
| TimeoutReconnect | Integer  | Timeout between attempts to reconnect to the source server.                                                                                                                                                        |
| TimeoutSleep     | Integer  | Timeout between the series of reconnections to the source server.                                                                                                                                                  |
| AttemptsSleep    | Integer  | The number of attempts in the series of reconnections to the source server.                                                                                                                                        |
| Symbols          | String   | The list of symbols for which the data feed provides quotes.                                                                                                                                                       |
| SysConnection    | Integer  | The status of the data feed connection to a source server. 0 no connection, 1 connected.                                                                                                                           |
| SysLastTime      | DateTime | The time of the last reconnection to the source server in the YYYY-MM-DD HH:MM:SS.MSC format.                                                                                                                      |
| Company          | String   |                                                                                                                                                                                                                    |

MetaTrader 5 trading platform

| <b>Name</b>    | <b>Type</b> | <b>Description</b>                                                                                                    |
|----------------|-------------|-----------------------------------------------------------------------------------------------------------------------|
|                |             | The name of the company who signed the executable file of the data feed.                                              |
| Issuer         | String      | The certification authority that issued the certificate of the above company.                                         |
| TickStatsCount | Integer     | The amount of price statistics received by the data feed from an external data source during the current session.     |
| TicksCount     | Integer     | The number of price changes received by the data feed from an external data source during the current session.        |
| BooksCount     | Integer     | The number of Market Depth changes received by the data feed from an external data source during the current session. |
| NewsCounts     | Integer     | The number of news items received by the data feed from an external data source during the current session.           |
| BytesReceived  | Integer     | The volume of traffic (in bytes) received by the data feed during the current session.                                |
| BytesSent      | Integer     | The volume of traffic (in bytes) sent by the data feed during the current session.                                    |
| StateFlags     | Integer     | Flags of states.                                                                                                      |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / [mt5\\_feeders](#) / Enumerations

## Enumerations

The following enumerations are used for passing information about data feed configurations:

- [EnFeederFlags](#)
- [EnFeedersFieldFlags](#)

### EnFeederFlags

Flags of predefined data feed settings are listed in EnFeederFlags.

| Identifier       | Value | Description                                    |
|------------------|-------|------------------------------------------------|
| FEED_FLAG_QUOTES | 1     | The data feed sends quotes.                    |
| FEED_FLAG_NEWS   | 2     | The data feed sends news.                      |
| FEED_FLAG_REMOTE | 8     | The data feed is running on a remote computer. |

### EnFeedersFieldFlags

Flags of editable fields are listed in EnFeedersFieldFlags.

| Identifier        | Value | Description             |
|-------------------|-------|-------------------------|
| FEED_FIELD_SERVER | 1     | The "Server" field.     |
| FEED_FIELD_LOGIN  | 2     | The "Login" field.      |
| FEED_FIELD_PASS   | 4     | The "Password" field.   |
| FEED_FIELD_PARAM  | 8     | The "Parameters" field. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_feeder\_translates

mt5\_feeder\_translates

Data about conversion settings of data feeds is exported to this table. The table contains the following fields:

| Name      | Type    | Description                                                                                   |
|-----------|---------|-----------------------------------------------------------------------------------------------|
| Symbol    | String  | The name of the symbol in the trading platform.                                               |
| Feeder    | String  | The name of the data feed, to which the conversion setting applies.                           |
| Source    | String  | The name of the symbol on the source server.                                                  |
| BidMarkup | Integer | Markup for the symbol's Bid price received from the data feed.                                |
| AskMarkup | Integer | Markup for the symbol's Ask price received from the data feed.                                |
| Digits    | Integer | The number of digits after the decimal point in the price of the symbol that receives quotes. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_feeder\_params

mt5\_feeder\_params

Data about additional settings of data feeds is exported to this table. The table contains the following fields:

| Name    | Type    | Description                                                                                                                                                                                                                                                  |
|---------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ParamID | String  | The unique identifier of the parameter.                                                                                                                                                                                                                      |
| Feeder  | String  | The name of the data feed, to which the setting applies.                                                                                                                                                                                                     |
| Type    | Integer | Parameter type: <ul style="list-style-type: none"> <li>• 0 string</li> <li>• 1 integer</li> <li>• 2 floating-point number</li> <li>• 3 time</li> <li>• 4 date</li> <li>• 5 date and time</li> <li>• 6 list of groups</li> <li>• 7 list of symbols</li> </ul> |
| Name    | String  | The name of the parameter.                                                                                                                                                                                                                                   |
| Value   | String  | The value of the parameter.                                                                                                                                                                                                                                  |



MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_feeder\_symbols

mt5\_feeder\_symbols

Data on symbol settings for data feeds is exported to this table. The table contains the following fields:

| Issued to  | Type    | Description                                                    |
|------------|---------|----------------------------------------------------------------|
| Symbol_ID  | Integer | Unique entry ID.                                               |
| FeederName | String  | The name of the data feed, to which the configuration applies. |
| Symbol     | String  | The name of the symbol in the trading platform.                |

## MetaTrader 5 trading platform

MetaTrader 5 Trading Platform  
/ Platform Components /  
Backup Server / SOL Export /  
mt5\_reports

mt5\_reports

Data about report settings is exported to this table. The table contains the following fields:

| <b>Name</b> | <b>Type</b> | <b>Description</b>                                                                                                                                                        |
|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name        | String      | The name of the report configuration.                                                                                                                                     |
| Server      | Integer     | The ID of the trade server, for which the report is configured.                                                                                                           |
| Timestamp   | Integer     | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed. |
| Module      | String      | The name of the report module.                                                                                                                                            |
| Mode        | Integer     | Report operation mode: 0 disabled, 1 enabled.                                                                                                                             |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_report\_params

mt5\_report\_params

Data about additional report settings is exported to this table. The table contains the following fields:

| Name    | Type    | Description                                                                                                                                                                                                                                                  |
|---------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ParamID | String  | The unique identifier of the parameter.                                                                                                                                                                                                                      |
| Report  | String  | The name of the report, to which the setting applies.                                                                                                                                                                                                        |
| Server  | Integer | The ID of the trade server, for which the report is configured.                                                                                                                                                                                              |
| Type    | Integer | Parameter type: <ul style="list-style-type: none"> <li>• 0 string</li> <li>• 1 integer</li> <li>• 2 floating-point number</li> <li>• 3 time</li> <li>• 4 date</li> <li>• 5 date and time</li> <li>• 6 list of groups</li> <li>• 7 list of symbols</li> </ul> |
| Name    | String  | The name of the parameter.                                                                                                                                                                                                                                   |
| Value   | String  | The value of the parameter.                                                                                                                                                                                                                                  |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_plugins

mt5\_plugins

Data about [plugin settings](#) is exported to this table. The table contains the following fields:

| Name      | Type    | Description                                                                                                                                                                                            |
|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name      | String  | The name of the plugin configuration.                                                                                                                                                                  |
| Server    | Integer | The ID of the trade server, for which the plugin is configured.                                                                                                                                        |
| Timestamp | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.                              |
| Module    | String  | The name of the plugin module.                                                                                                                                                                         |
| Enable    | Integer | Plugin operation mode: 0 disabled, 1 enabled.                                                                                                                                                          |
| Flags     | Integer | Plugin operation flags: <ul style="list-style-type: none"> <li>• 0 no flags</li> <li>• 1 permission to configure the plugin from a manager terminal</li> <li>• 2 the profiling mode enabled</li> </ul> |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_plugin\_params

mt5\_plugin\_params

Data about additional plugin settings is exported to this table. The table contains the following fields:

| Name    | Type    | Description                                                                                                                                                                                                                                                  |
|---------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ParamID | String  | The unique identifier of the parameter.                                                                                                                                                                                                                      |
| Plugin  | String  | The name of the plugin, to which the setting applies.                                                                                                                                                                                                        |
| Server  | Integer | The ID of the trade server, for which the plugin is configured.                                                                                                                                                                                              |
| Type    | Integer | Parameter type: <ul style="list-style-type: none"> <li>• 0 string</li> <li>• 1 integer</li> <li>• 2 floating-point number</li> <li>• 3 time</li> <li>• 4 date</li> <li>• 5 date and time</li> <li>• 6 list of groups</li> <li>• 7 list of symbols</li> </ul> |
| Name    | String  | The name of the parameter.                                                                                                                                                                                                                                   |
| Value   | String  | The value of the parameter.                                                                                                                                                                                                                                  |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_time

mt5\_time

Platform trading time settings are exported to this table. The table contains the following fields:

| Name          | Type    | Description                                                                                                                                                                      |
|---------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimeZone      | Integer | The time zone of a server in minutes from GMT. For example: 0 = GMT, -60 = GMT - 1, 60 = GMT + 1.                                                                                |
| Timestamp     | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has changed.             |
| TimeServer    | String  | The address of the current time synchronization server.                                                                                                                          |
| Daylight      | Integer | Daylight Saving Time mode: 0 off, 1 on.                                                                                                                                          |
| DaylightState | Integer | The presence of the daylight saving time in the platform time zone. 0 means no daylight saving time is applied in the platform time zone. Otherwise, any non-zero value is used. |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_time\_weekdays

mt5\_time\_weekdays

Platform's working time schedule by days is exported to this table. The table contains the following fields:

| Name     | Type    | Description                                                                                                                                                       |
|----------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimeZone | Integer | The time zone of a server in minutes from GMT. For example: 0 = GMT, -60 = GMT - 1, 60 = GMT + 1. Corresponds to the TimeZone value in the <u>mt5_time</u> table. |
| Day      | Integer | The ordinal number of the day of the week. For example, 0 is Sunday, 6 is Saturday.                                                                               |
| 00       | Integer | Flag of working time in the period from 00:00 to 00:59. 0 non-working time, 1 working time.                                                                       |
| 01       | Integer | Flag of working time in the period from 01:00 to 01:59. 0 non-working time, 1 working time.                                                                       |
| ...      | Integer | Further similar fields apply for each hour of the day.                                                                                                            |
| 23       | Integer | Flag of working time in the period from 23:00 to 23:59. 0 non-working time, 1 working time.                                                                       |

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Backup Server](#) / [SQL Export](#) / mt5\_gateways

mt5\_gateways

Data on the [gateway settings](#) is exported to this table. The table contains the following fields:

| Name          | Type    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name          | String  | Gateway configuration name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Timestamp     | Integer | A unique value within the table. Used by MetaTrader 5 servers for internal purposes. If the Timestamp of a record has changed, it means that the record has been changed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Module        | String  | Gateway module name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| GatewayServer |         | The address at which the gateway accepts connections from the history and trade servers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| TradingServer |         | Address of the server to which the gateway connects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Enable        | Integer | Gateway operation mode: 0 disabled, 1 enabled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Flags         | Integer | <p>Gateway operation flags:</p> <ul style="list-style-type: none"> <li>• 0x00000001 gateway works as a remote application.</li> <li>• 0x00000002 gateway is allowed to import symbol settings.</li> <li>• 0x00000004 do not broadcast quotes from the gateway in the system.</li> <li>• 0x00000008 gateway can manage clients' balances using IMTExecution::TE_BALANCE_CHANGE and IMTExecution::TE_BALANCE_CORRECT trade executions.</li> <li>• 0x00000010 if enabled, the gateway log receives additional operation data, including the results of measuring the trading operations handling speed. A more detailed information on extended logging is provided in the <a href="#">Journal of Gateways</a> section.</li> <li>• 0x00000020 gateway supports requesting the state of external trading system positions. The request is made from the <a href="#">Positions</a> tab of the gateway.</li> <li>• 0x00000040 collect advanced metrics related to request processing by the gateway.</li> </ul> |



# MetaTrader 5 trading platform

| Name               | Type    | Description                                                                                                                                                                                                                               |
|--------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |         | <ul style="list-style-type: none"> <li>• 0x00000100 gateway is running in demo mode. The mode is checked based on the license at the time the module is loaded.</li> <li>• 0x00000200 gateway is integrated into the platform.</li> </ul> |
| Gateway            | String  | Gateway module name.                                                                                                                                                                                                                      |
| TimeoutReconnect   | Integer | Timeout between attempts to reconnect to an external server in seconds.                                                                                                                                                                   |
| TimeoutSleep       | Integer | Timeout between the series of reconnections to an external server in seconds.                                                                                                                                                             |
| AttemptsSleep      | Integer | Number of attempts in a series of reconnections to an external server.                                                                                                                                                                    |
| ID                 | Integer | The gateway ID.                                                                                                                                                                                                                           |
| Symbols            | Array   | The list of symbols for which the gateway provides quotes and processes trading operations.                                                                                                                                               |
| SysConnection      | Integer | The state of gateway connection to an external trading system: 0 connected, 1 disconnected.                                                                                                                                               |
| SysLastTime        | Integer | The time of the last successful gateway connection to an external trading system in the YYYY-MM-DD HH:MM:SS.MS format.                                                                                                                    |
| Company            | String  | The company by which the gateway executable is signed.                                                                                                                                                                                    |
| Issuer             | String  | The certification authority that issued the certificate to the above company.                                                                                                                                                             |
| TickStatsCount     | Integer | The number of price statistics changes received by the gateway from the external system for the current session.                                                                                                                          |
| TicksCount         | Integer | The number of price changes received by the gateway from the external system for the current session.                                                                                                                                     |
| BooksCount         | Integer | The number of Market Depth changes received by the gateway from an external trading system for the current session.                                                                                                                       |
| TradeAverageTime   | Integer | Average time spent by the gateway to process one trading operation in milliseconds.                                                                                                                                                       |
| TradeRequestsCount | Integer | The number of trading operations processed by the gateway during the current session.                                                                                                                                                     |
| BytesReceived      | Integer |                                                                                                                                                                                                                                           |

# MetaTrader 5 trading platform

| Name       | Type    | Description                                                        |
|------------|---------|--------------------------------------------------------------------|
|            |         | Traffic volume received by the gateway during the current session. |
| BytesSent  | Integer | Traffic volume sent by the gateway during the current session.     |
| StateFlags | Integer | Flags of states. Currently not used.                               |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_gateways\_params

mt5\_gateways\_params

Data about additional report settings is exported to this table. The table contains the following fields:

| Name        | Type    | Description                                                                                                                                                                                                                                                  |
|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ParamID     | String  | The unique identifier of the parameter.                                                                                                                                                                                                                      |
| GatewayName | String  | The name of the gateway configuration the setting applies to.                                                                                                                                                                                                |
| Type        | Integer | Parameter type: <ul style="list-style-type: none"> <li>• 0 string</li> <li>• 1 integer</li> <li>• 2 floating-point number</li> <li>• 3 time</li> <li>• 4 date</li> <li>• 5 date and time</li> <li>• 6 list of groups</li> <li>• 7 list of symbols</li> </ul> |
| Name        | String  | Parameter name.                                                                                                                                                                                                                                              |
| Value       | String  | Parameter value.                                                                                                                                                                                                                                             |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_gateways\_translates

mt5\_gateways\_translates

Data about price translation settings in the gateway is exported to this table. The table contains the following fields:

| Name        | Type    | Description                                                                                             |
|-------------|---------|---------------------------------------------------------------------------------------------------------|
| Symbol      | String  | The name of the symbol in the trading platform.                                                         |
| GatewayName | String  | The name of the gateway configuration the setting applies to.                                           |
| Server      | Integer | The ID of the trade server, for which the plugin is configured.                                         |
| Source      | String  | The symbol name in the data feed, to which the gateway connects.                                        |
| BidMarkup   | Integer | Correction for the Bid price received for a symbol from the data source, to which the gateway connects. |
| AskMarkup   | Integer | Correction for the Ask price received for a symbol from the data source, to which the gateway connects. |
| Digits      | Integer | The number of digits after the decimal point in the price of the symbol that receives quotes.           |

MetaTrader 5 Trading Platform / Platform Components / Backup Server / SOL Export / mt5\_gateways\_symbols

mt5\_gateways\_symbols

Data on symbol settings for gateways is exported to this table. The table contains the following fields:

| Issued to   | Type    | Description                                                    |
|-------------|---------|----------------------------------------------------------------|
| Symbol_ID   | Integer | Unique entry ID.                                               |
| GatewayName | String  | The name of the data feed, to which the configuration applies. |
| Symbol      | String  | The name of the symbol in the trading platform.                |

MetaTrader 5 Trading Platform / Platform Components / Data Feeds

Data Feeds

Data feeds enable receipt of quotes and news in the online trading platform. They transmit information to the history server, from which they are translated to access points (data centers) and terminals. The trading platform includes several data feeds that enable receiving of quotes and news from the most popular feed providers:

- **Dow Jones Prime Tass News Feeder**
- **DJ News Feeder**
- **IOFeeder**
- **MetaTrader 4 Feeder**
- **MetaTrader 5 Feeder**
- **Trading Central News Feeder**
- **MetaTrader 5 UniFeeder**
- **Thomson Reuters Feeder**
- **RSS News Feeder**
- **IBTimes News Feeder**
- **ForexPros Feeder**
- **KnowledgeView News Feeder**
- **FXstreet Feeder**
- **Financial Source News Feeder**
- **Claws & Horns Feeder**
- **UniNewsFeeder**
- **Alliance News Feeder**
- **Newsquawk**
- **Remote Datafeed**
- **Universal DDE Connector**

Data feeds are executable files (with extension \*.exe) that are run as separate processes. These executable files must be located in the ***datafeed*** folder of the history server.

If the data feed file is copied to the above mentioned folder while the platform is operating, the history server needs to be restarted (select it in the "Network" section and execute the " Restart" command in the "Services" menu or in the context menu). After that refresh configurations in the administrator terminal by pressing " Refresh configuration".

In order to use a data feed, connection to the feed provider server needs to be established. Data delivery from a third party vendor is implemented based on a special agreement.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Dow Jones Prime Tass News Feeder

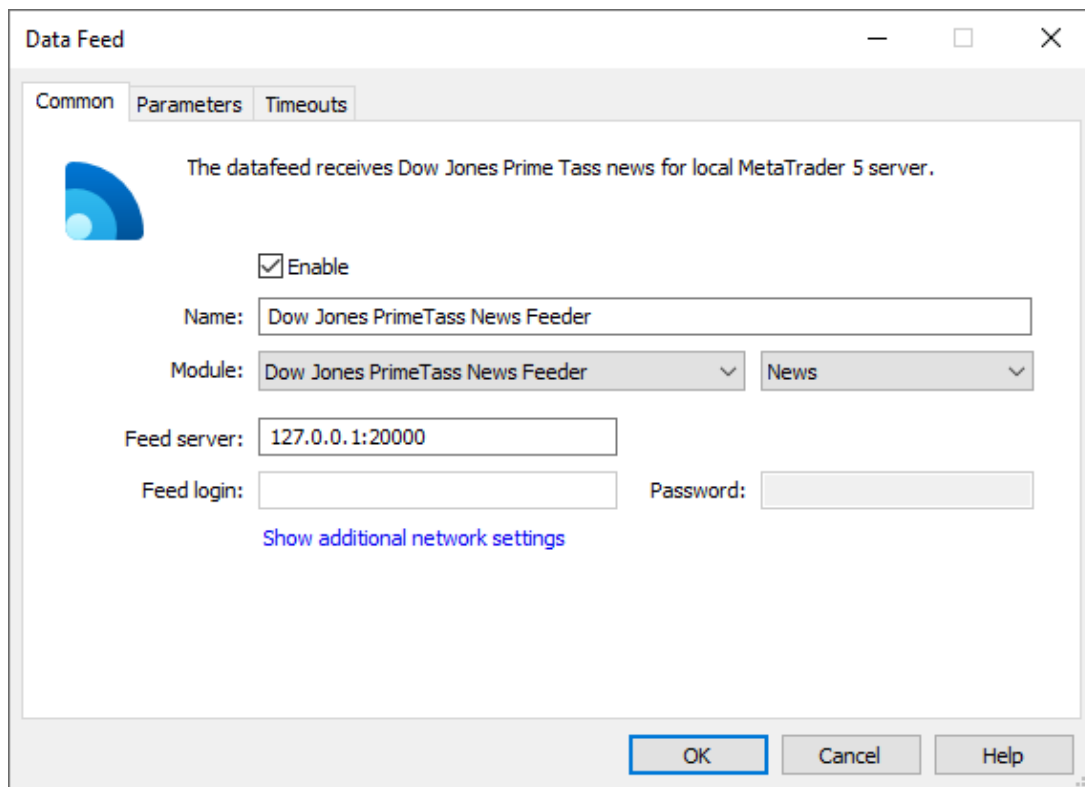
### Dow Jones Prime Tass News Feeder

Dow Jones Prime Tass News Feeder (DJPrimeTassNewsFeeder.exe) is the news feeder for receiving Dow Jones Newswires from the Prime agency.

#### How it works

To start receiving news, you should request a subscription at <https://www.1prime.ru/>. You should provide to Ptime Tass the IP address, at which the data feed will operate. It will be added to the list of addresses, to which the provider sends news. The data feed will enable external connections for this IP address and will start receiving incoming news. This connection scheme does not require additional data (such as login and password).

#### Setup



**Data Feed**

Common Parameters Timeouts

The datafeed receives Dow Jones Prime Tass news for local MetaTrader 5 server.

☒ Enable

Name: Dow Jones PrimeTass News Feeder

Module: Dow Jones PrimeTass News Feeder News

Feed server: 127.0.0.1:20000

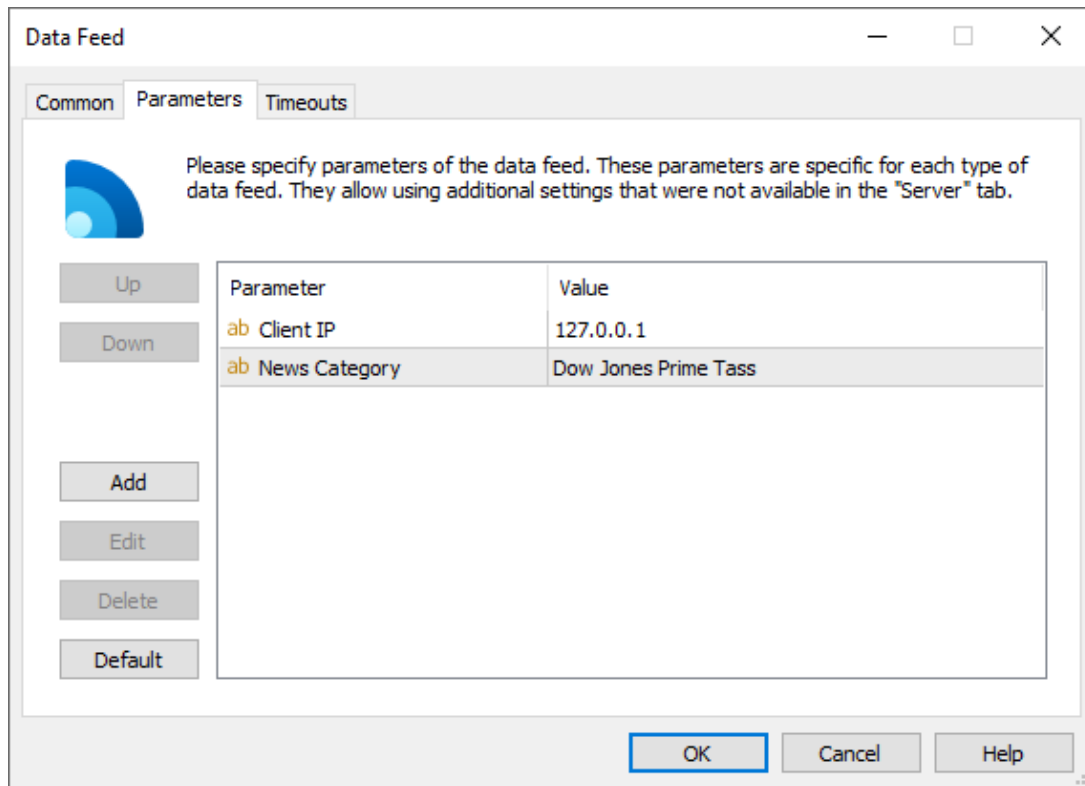
Feed login: Password:

[Show additional network settings](#)

OK Cancel Help

Specification of the following parameters is required on the "Common" tab of the data feed settings:

- **Module** DowJonesNewsFeeder;
- **Feed server** IP address (or domain name) and port that will be opened for receiving news from the news provider.





On the "Parameters" tab, you can set:

- **Client IP** IP address for the news provider, from which connection to the data feed will be allowed. If the parameter is absent, connections from all addresses are allowed.  
If you do not know the exact IP address of the news provider, you can find it out by requesting the journal of the history server or the data feed. The news provider sends information, and if their IP is not added to the list of allowed addresses, the data thread is blocked which is reflected in the journal. The address of the server the data is sent from is also specified in the entries.
- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.

It is recommended to specify an allowed IP address in the ClientIP parameter. It increases the security of operation.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / DJ News Feeder

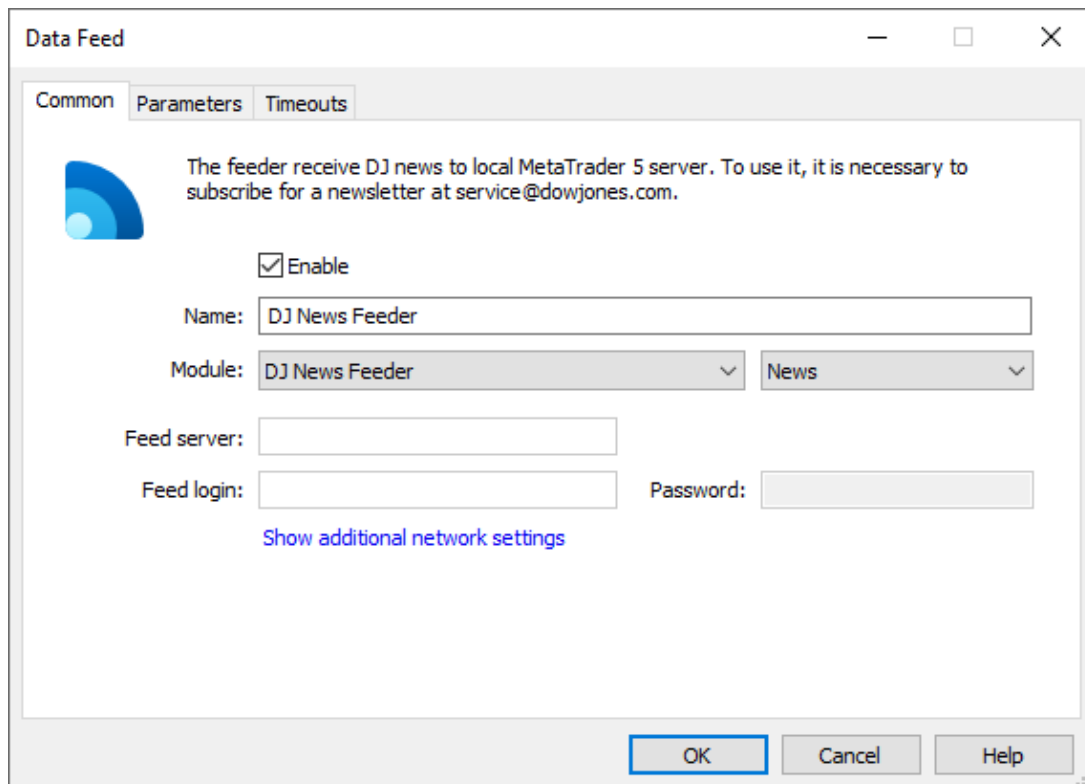
## DJ News Feeder

DJ News Feeder (DJNewsFeeder.exe) is the news feeder from the world-famous news agency Dow Jones. Dow Jones Newswires include comprehensive reviews and analytical materials, macroeconomic events and speeches of officials, rolling market commentary and expert analysis, company reports, breaking news and much more.

### How it works

To start receiving news, you should request a subscription at <https://www.dowjones.com>. In a contract with the news provider, you will receive the port number for the data feed to listen to. Specify this port number in data feed settings. You should also receive the list of IP addresses, from which the new content will be provided. These addresses must also be specified in the data feed settings for additional security. Otherwise, the data feed will accept connections from any addresses at the specified port.

### Setup



**Data Feed**

Common Parameters Timeouts

The feeder receive DJ news to local MetaTrader 5 server. To use it, it is necessary to subscribe for a newsletter at [service@dowjones.com](mailto:service@dowjones.com).

☒ Enable

Name: DJ News Feeder

Module: DJ News Feeder News

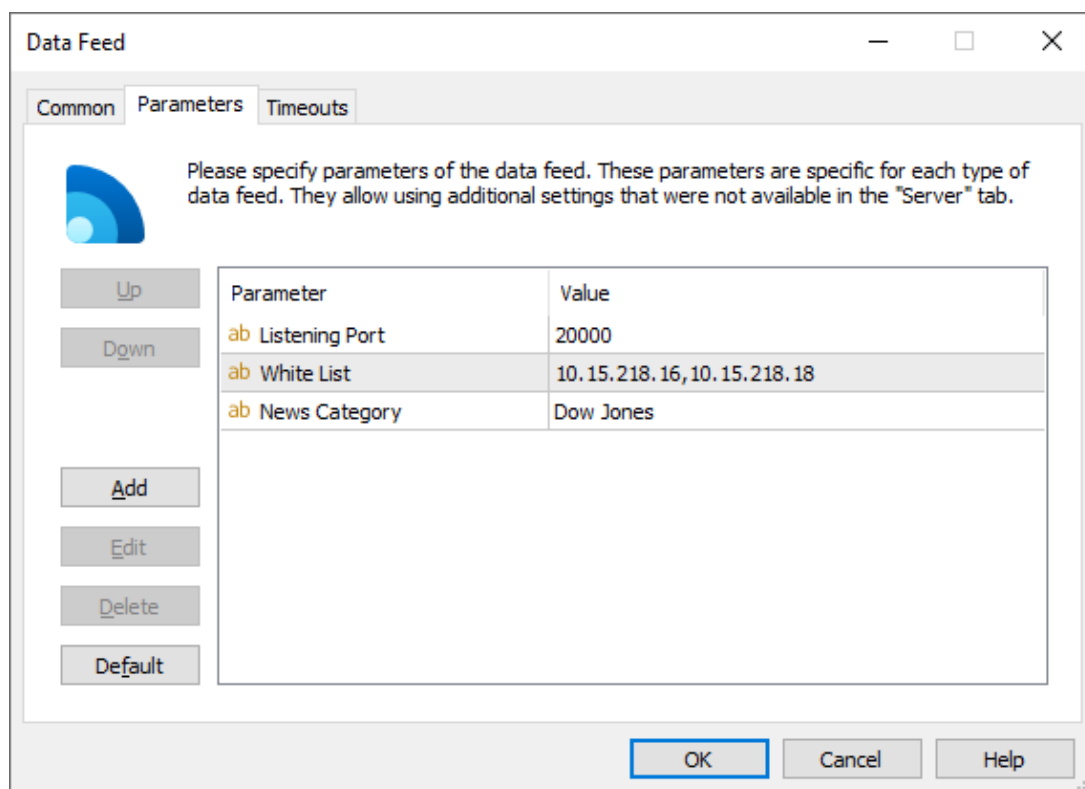
Feed server:

Feed login: Password:

[Show additional network settings](#)

OK Cancel Help

Open the Common tab of the data feed settings and set the DowJonesNewsFeeder module.



Additional settings should be configured in the Parameters tab:

- **Listening Port** the port number, at which the data feed will listen to connections from the data provider. Available in an agreement with the news provider.
- **White List** one or more IP addresses, from which connection to the data feed is allowed. If this parameter is not set, connection from any address is allowed. If the exact IP address is not known, you can identify it by requesting the logs of the history server or the data feed. The provider sends out news content, and if their address is not included in the list of allowed IPs, the data stream is blocked and an appropriate message is added to the journal. The logs also contain the address of the server, from which the data is sent. The log may look like this: address X.X.X.X is not white listed, disconnecting.
- **News Category** the category name for the newsletters received from the data feed. The category name can then be used for specifying news to be delivered to client groups.

It is recommended that you specify the provider's IP address in the WhiteList parameter. This will increase the safety of operation.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / IQFeeder

## IQFeeder

The IQFeeder is designed for receiving news and price data from IQFeed. The company offers brokers a wide range of services:

- Real-time quotes for financial instruments traded on the US and Canadian exchanges: NYSE, NASDAQ and Canadian Securities Exchange, among others
- Real-time quotes for Forex instruments
- Level II data (order book)
- More than 700 market stats/breadth indicators, most of which are updated every second
- Up to 180 calendar days of tick history
- More than 11 years of one-minute historical data
- Fundamental data on US stocks
- Real-time news from leading agencies

IQFeeder is free and is included in the standard platform delivery package. You only need to subscribe for the data by contacting IQFeed and then to configure the data feed in the platform.

## How It Works

IQFeed provides a special software (IQFeed Client) that is responsible for information delivery from the provider to the local server where this software is installed. These data are translated to the history server via the IQFeeder data feed. Interaction between the data feed and the IQFeed client is implemented through the local IP address 127.0.0.1.

IQFeed Client is already available in the data feed. During data feed configuration in the platform, the required components will be installed automatically and thus no additional installation will be required.

## Setup

IQFeed provides data for a variety of trading instruments. In order to be able to receive relevant data, you need to create corresponding symbols. If you want to receive the order book in addition to quotes, enable the appropriate option in symbol settings:

## MetaTrader 5 trading platform

Symbol: EURUSD

Common Quotes Trade Execution Margin Margin Rates Swaps Sessions

The setting up of main parameters of the symbol. Please specify its name, description, and other parameters.

Symbol: EURUSD Description: Euro vs US Dollar

ISIN: EUU0009652759 International: https://www.google.com/?finance?q=EURUSD

Basis: (empty) Page: (empty)

Source: (empty) Digits: 5 Base currency: EUR

Background: None Market depth: 32 Base currency digits: 2

Spread: off pt Chart mode: (empty) Profit currency: USD

Spread balance: 0 bid / 0 ask Profit currency digits: 2

Margin currency: EUR

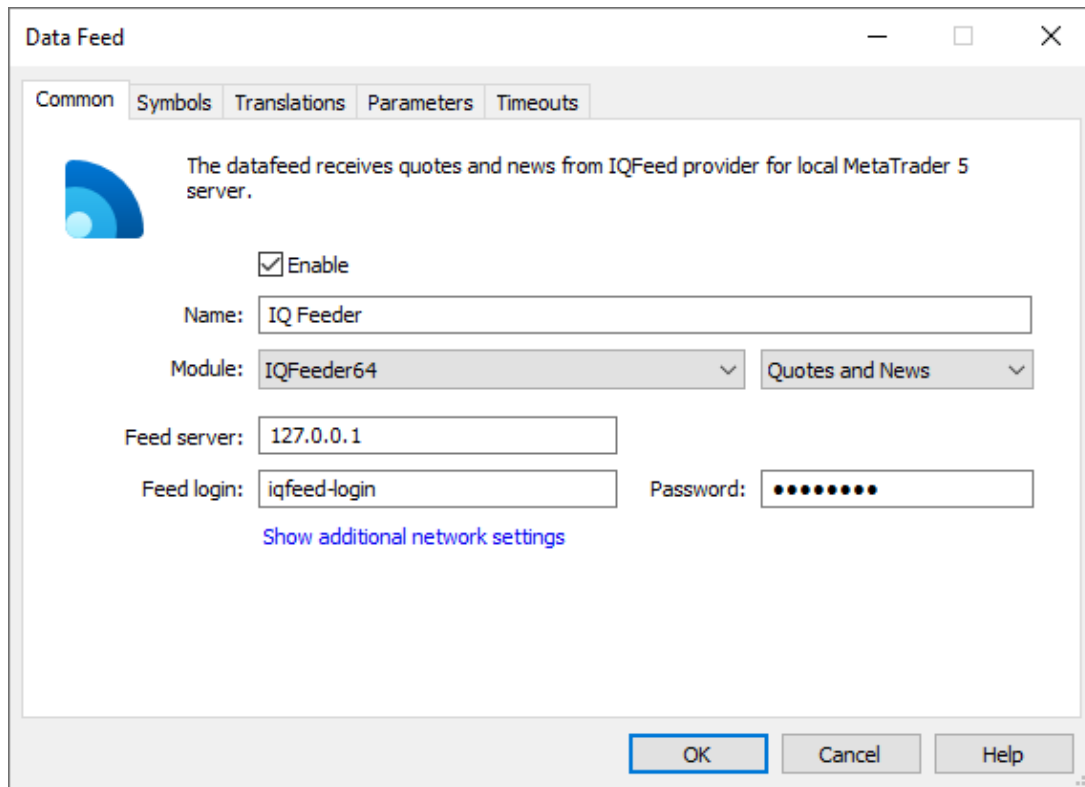
Margin currency digits: 2

OK Cancel Help

Set a value other than "off" for the Market Depth parameter. If the initial depth of the Market Depth is different, the data feed will automatically align it with the value specified in the symbol settings in the platform. The price accuracy in the Market Depth will be automatically adjusted to the value specified in symbol settings.

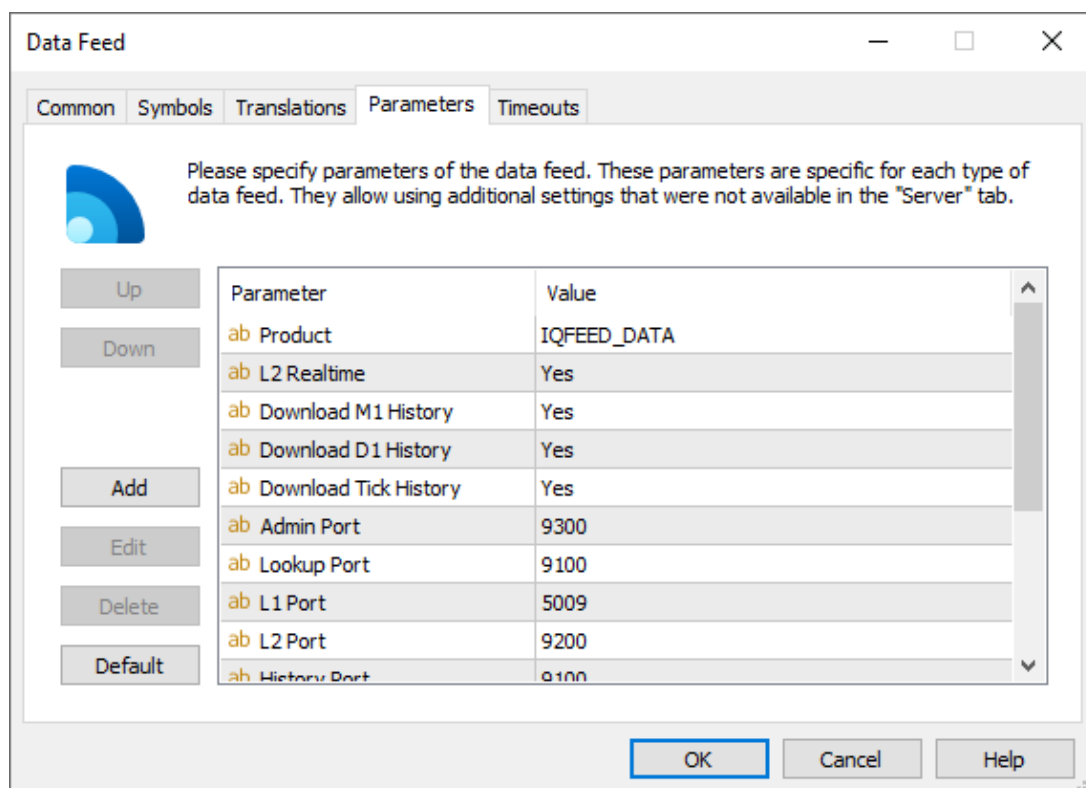
If symbol names on your platform side differ from those available in IQFeed, you can match corresponding symbols in the data feed settings, under the "Translation" settings.

Once you have prepared the symbols, create a new data feed configuration for IQFeeder.



The following parameters should be specified on the "Common" tab of the data feed:

- **Module** IQFeeder.
- **Feed server** 127.0.0.1, this address is used to implement connection between the data feed and the IQFeed client, it cannot be changed.
- **Feed login** login for connection, which is provided by IQFeed upon subscription
- **Password** password for connection, which is provided by IQFeed upon subscription.



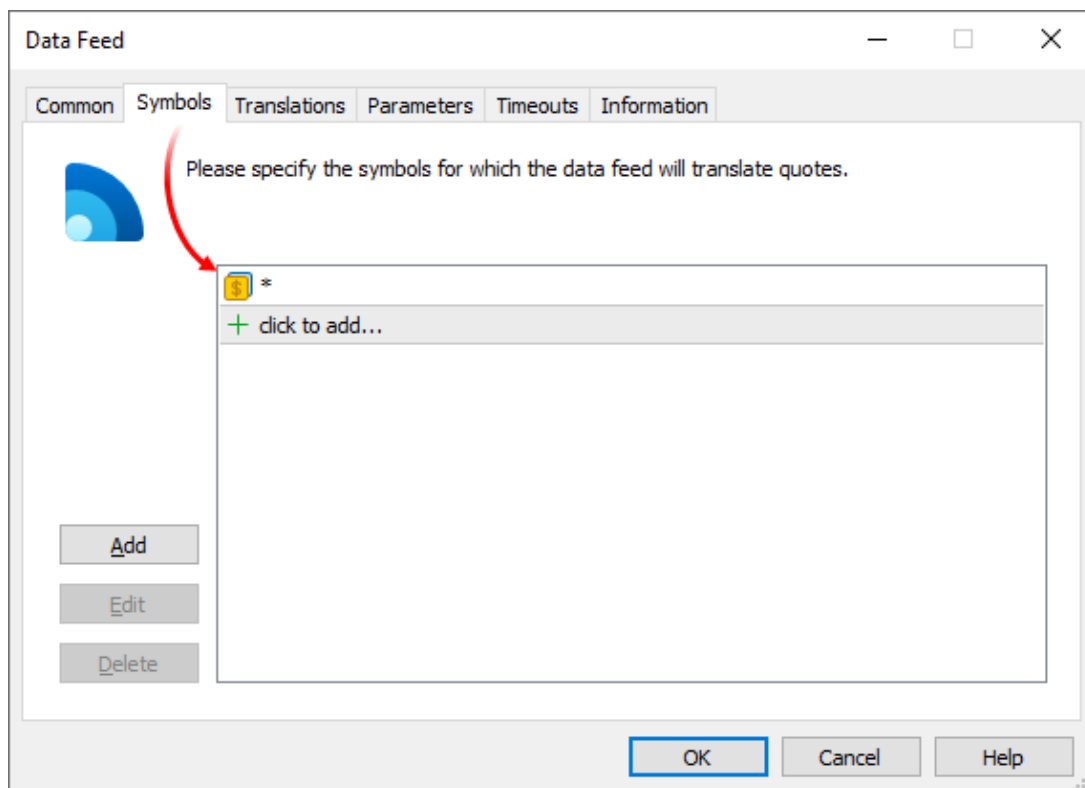
On the "Parameters" tab, specify "Product ID" that was received from the news provider and additional settings (if needed).

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Product** product identifier. It is provided by IQ Feed upon subscription.
- **L2 Realtime** set "Yes" to be able to receive the Market Depth data. Access to the relevant data must be granted by your IQFeed subscription and the Market Depth feature must be allowed in symbol settings on the platform side.
- **Download Tick History** if set to "Yes", the data feed will download the available tick history of trading instruments.
- **Download M1 History** if set to "Yes", the data feed will download the available one-minute history of trading instruments.
- **Download D1 History** if set to "Yes", the data feed will download the available daily history for trading instruments. If this parameter is enabled along with the "Download M1 History" option, the daily history will only be downloaded for the periods for which one-minute history is not available.
- **Admin Port, Lookup Port, L1 Port, L2 Port, History Port** the ports on which IQFeed Client waits for incoming connections to feed appropriate data (L1 and L2 prices, quoting history). The default values correspond to IQFeed Client ports.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay

is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.

- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

Before importing the price history, check the list of symbols configured in the data feed. If the data feed is allowed to download history, it will replace the history of all symbols within the platform, for which the data feed has permissions.





MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
MetaTrader 4 Feeder

## MetaTrader 4 Feeder

MetaTrader 4 Feeder is the data feed that enables receipt of quotes and news from MetaTrader 4 servers. This solution is built into the platform, which ensures minimal delays in quote delivery.

### How It Works

The data feed creates a common client connection with any MetaTrader 4 server, authorizing using a login and a password.

- Connection can be established using any account (demo or real).
- You can receive only quotes and news that are available to the account that is used for connection.

### Automatic Opening of Demo Accounts

The data feed is capable of automatic opening of demo accounts on the source server.

If the account details (login and password) are not specified in the data feed settings or the account has become invalid (expired), the data feed will open a new demo account and will use it for connecting.

### Setup

The data feed must be added via the corresponding section of the administrator terminal.

**Data Feed**

Common Symbols Translations Parameters Timeouts

The datafeed translates quotes and news from remote MetaTrader 4 server

☒ Enable

Name: MetaTrader 4 Feeder

Module: MetaTrader 4 Feeder Quotes

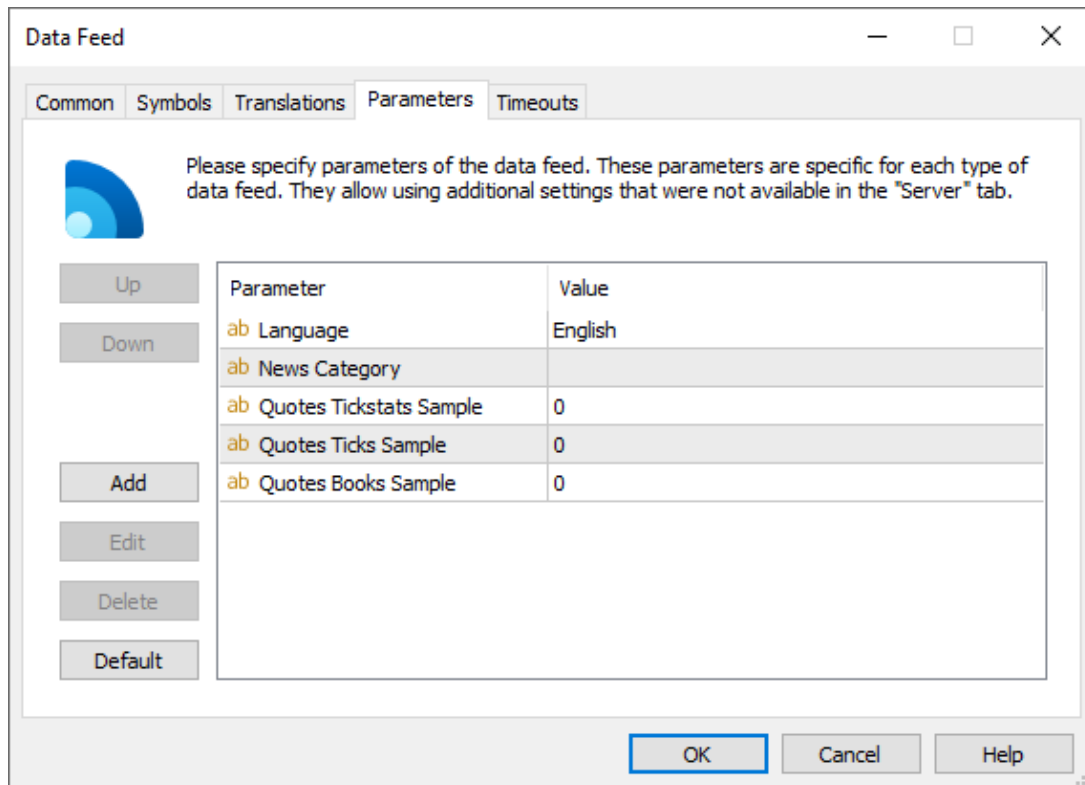
Feed server: demo.metaquotes.net:443

Feed login: 55555 Password: •••••

OK Cancel Help

The following parameters should be specified on the "Common" tab of the data feed:

- **Module** MetaTrader4Feeder;
- **Feed server** address of the MetaTrader 4 server and the port number to connect to it (separated by a colon);
- **Feed login** login (account number) for the authorization on the server;
- **Password** password of the account for the authorization on the server.



The data feed features the following additional parameters:

- **Language** the language to be used for the news with their language not specified on the source server. This is necessary for correct conversion and further sorting of such news. "English" is applied by default. The language should be specified in the format standard for the Windows operating systems, though without the regional dialectic specifications. E.g. English, Russian, etc.
- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
MetaTrader 5 Feeder

## MetaTrader 5 Feeder

MetaTrader 5 Feeder is a data feed which enables the delivery of news, quotes and Market Depth data from any MetaTrader 5 servers.

This solution is built into the platform, which ensures minimal delays in quote delivery. To speed up data delivery, the data feed operates works in multiple streams, and also it automatically selects the best access points for connection.

## Setup

Contact the desired broker and agree on data delivery terms and conditions. To connect to the source broker, you will need a regular trading account (demo or real), with access to the required news and trading instruments. The data feed will create a client connection and will deliver data through this connection.

The data feed must be added via the corresponding section of the administrator terminal.

**Data Feed**

Common Symbols Translations Parameters Timeouts

The datafeed translates quotes and news from remote MetaTrader 5 server

☒ Enable

Name: MetaTrader 5 Feeder

Module: MetaTrader 5 Feeder Quotes

Feed server: access.metatrader5.com:443

Feed login: 1000 Password: .....

OK Cancel Help

## MetaTrader 5 trading platform

The following parameters should be specified on the "Common" tab of the data feed:

- **Module** MetaTrader5Feeder;
- **Feed server** address of the MetaTrader 5 server and the port number to connect to it (separated by a colon);
- **Feed login** login (account number) for the authorization on the server;
- **Password** password for the authorization. If the extended authorization mode is enabled on the server you are going to receive data from, you should use an investor password instead of the main one. Otherwise connection will be impossible.

**Data Feed**

Common Symbols Translations **Parameters** Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

| Parameter                  | Value |
|----------------------------|-------|
| ab Quotes Time Original    | No    |
| ab Selfitled Translations  | No    |
| ab Symbols Path            |       |
| ab Symbols Update          | No    |
| ab News Category           |       |
| ab Quotes Tickstats Sample | 0     |
| ab Quotes Ticks Sample     | 0     |
| ab Quotes Books Sample     | 0     |

Up Down Add Edit Delete Default

OK Cancel Help

The data feed features the following additional parameters:

- **Quotes Time Original** if "Yes", the datafeed itself sets the time for ticks considering the time zone of the recipient trade server. If the parameter is absent or set to "No", the time for ticks is set by the history server according to the trade time used.
- **Selftitled Translations** this parameter is used to prevent quotes from looping when connecting to the server on which the datafeed is installed. The default value is No. For more details please see below.
- **Symbols Update** mode applied to the import of trading instruments to the platform from an external source. If set to No (default), the data feed will only add symbols from the source server that do not yet exist in the platform. If set to Yes, the data feed will also update the settings of already existing symbols, which may affect the broadcasting of quotes (general description, currency data, accuracy, order book depth, tick value and size, accrued interest on bond, settlement price, price limits, splicing data, and quoting sessions). Settings are updated in real time. Quoting sessions are updated taking into account server time zones.  
Symbol settings are only imported/updated, if the appropriate option is enabled in the data feed settings.
- **Symbols Path** path to import trading instruments to. If this parameter is absent (by default), the data feed will import trading instruments into the \Preliminary subgroup with the trading option disabled. If this parameter has a non-empty value, the data feed will import trading instruments into the specified location.
- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

### Automatic Opening of Demo Accounts

The data feed is capable of automatic of opening of demo accounts on the source server.

If the account details (login and password) are not specified in the data feed settings or the account has become invalid (expired), the data feed will open a new demo account and will use it for connecting.

### Transferring quotes

For symbols with the disabled market depth (on the external server side), MetaTrader 5 Feeder passes ticks with Bid and Ask prices.

## MetaTrader 5 trading platform

For symbols with the enabled market depth, the data feed transfers the market depth changes as well as ticks with Last prices and volumes. Ticks containing only Bid and Ask price changes are not transferred. If the best supply and demand prices change in the market depth, the history server generates the necessary tick with Bid and Ask prices and adds it to the flow.

The data feed features the built-in switching mechanism to prevent the quote flow from stopping in case the external server stops transmitting the market depth changes. If not a single market depth change for a symbol arrives from the external system within 30 seconds, the data feed stops sorting out ticks having only Bid and Ask prices. In other words, it starts transferring to the platform both Last/Volume and Bid/Ask ticks.

The following entries are shown in the data feed journal when the market depth is no longer transmitted:

|                         |        |                                                            |
|-------------------------|--------|------------------------------------------------------------|
| 2017.09.12 15:20:25.412 | Feeder | books stream <b>for</b> EURUSD stopped (no books during 31 |
| 2017.09.12 15:20:25.873 | Feeder | books stream <b>for</b> USDJPY stopped (no books during 31 |

If the flow is resumed:

|                         |        |                                        |
|-------------------------|--------|----------------------------------------|
| 2017.09.12 15:21:29.759 | Feeder | books stream <b>for</b> EURUSD resumed |
| 2017.09.12 15:21:30.060 | Feeder | books stream <b>for</b> USDJPY resumed |

Symbol masks and own price retranslation

In datafeed translation settings, the "\*" mask can be used as the source symbol and as the destination symbol in the platform. For example, the settings Symbol="\*", Source="\*" mean that the names of the symbols will be used as they are provided in the external system. If the symbol is entitled EURUSD in the external system, then its data will be feed to the symbol with the same name on the trading platform side. The only situation in which such settings cannot be used is the connection of the data feed to the platform on which it is installed. In this case, receiving and feeding of quotes by the datafeed into the same symbols will lead to looping.

To avoid such situations, the datafeed provides the parameter "Selftitled Translations". If it is set to "No" (default) and the datafeed has a translation setting "\*" <- "\*", the datafeed will not start. An appropriate entry will be added into the journal:

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| translation rule for symbols to themselves '* <- *' not allowed but exists, remove this rule or |
|-------------------------------------------------------------------------------------------------|

Before enabling this parameter, make sure that the data feed is not connected to the same cluster on which it is running. Otherwise, this can lead to quote looping in the platform.

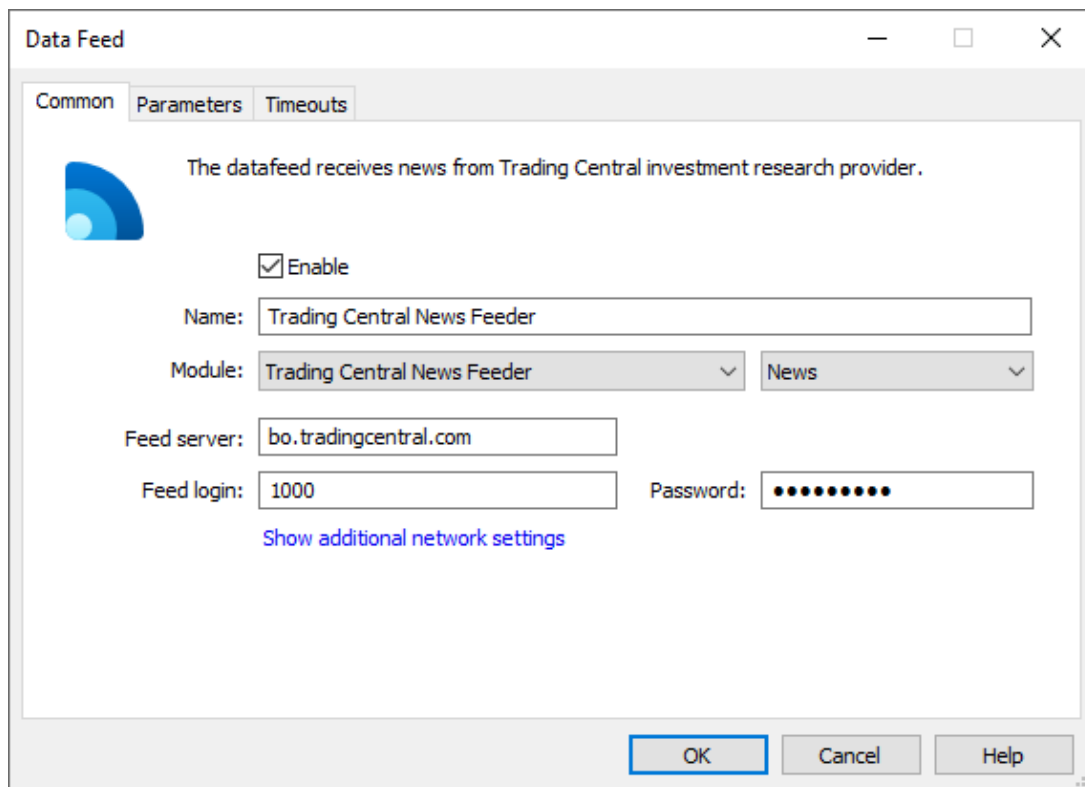
MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Trading Central News Feeder

## Trading Central News Feeder

Trading Central News Feeder is a data feed that enables receiving of analytical information on financial instruments from Trading Central ([www.tradingcentral.com](http://www.tradingcentral.com)). The basic activity of this company is the technical analysis trading recommendations sent by this company are based on.

Trading Central uses a FTP server to store data. In order to get access to these data, you should contact this company. To know the contact information, please visit the company's official website mentioned above. After you conclude an agreement, you are provided with the IP address of the FTP server, as well with the login and password for accessing it. Besides you will need to provide the IP address of your server (access server) in order for them to add it to the list of allowed addresses.

After that the data feed needs to be correctly set up via the administrator terminal:



**Data Feed**

Common Parameters Timeouts

The datafeed receives news from Trading Central investment research provider.

☒ Enable

Name: Trading Central News Feeder

Module: Trading Central News Feeder News

Feed server: bo.tradingcentral.com

Feed login: 1000 Password: .....

[Show additional network settings](#)

OK Cancel Help



The following parameters should be specified on the "Common" tab of the data feed:

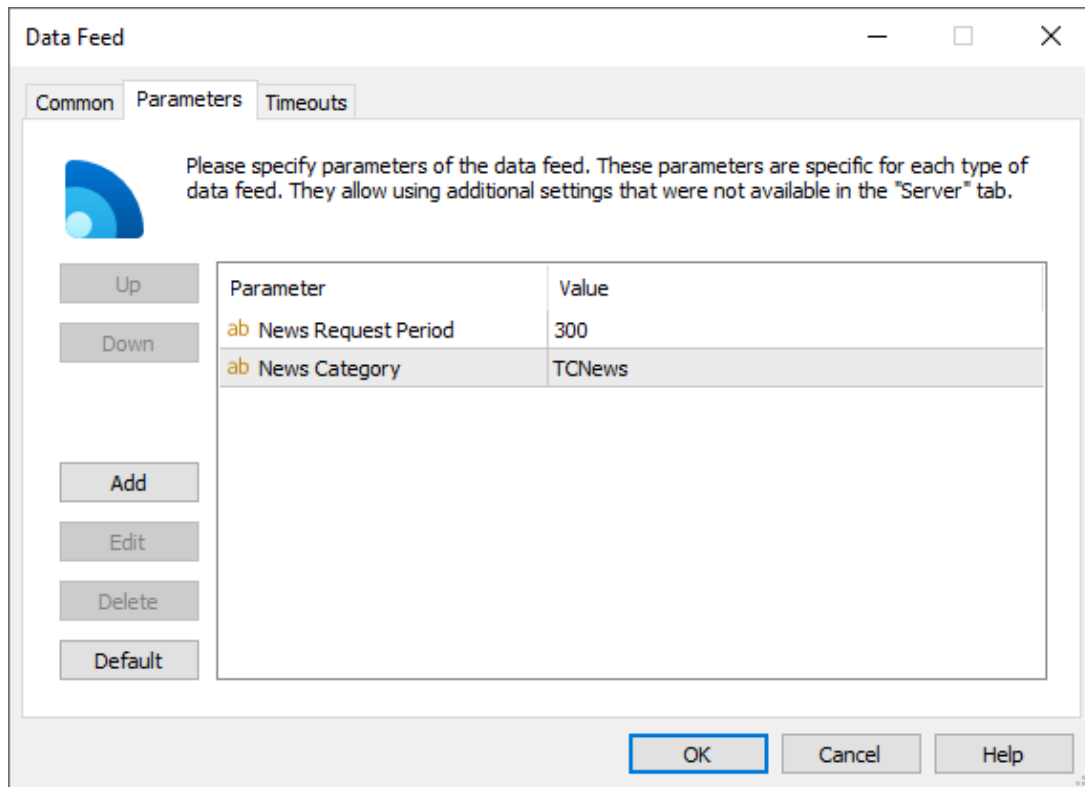
- **Module** TCNewsFeeder;
- **Feed server** address of the FTP server of Trading Central and a port to connect to it (separated by a colon). The data feed also supports connection via secure SFTP protocol. If you wish to use it, specify the protocol in the address, for example: <sftp://sftp.tradingcentral.com>. The connection address and the used protocol are determined by agreement with Trading Central.
- **Feed login** login to authorize on the server;
- **Password** password for the authorization.

For the data feed to work, one should allow ports 20 and 21 on the computer where it is installed. They are necessary to work with FTP server.

After you connect to the FTP server, the data feed starts downloading all available data. After the data is received, it removes them from the FTP server. Further connections to the server are established once in five minutes for checking and downloading new data.

Analytical information that comes from the data feed consists of several elements: the news body with a chart, signals on indicators and signals on Japanese candlesticks. The latter two elements are released not so often as news, so they are inserted only if necessary.

If the data feed operation is terminated, news can be received without the signals for some time, because the earlier signals have been deleted upon receipt already, while new haven't been received yet. While the data feed is operating, it keeps the last four signals in its memory, and inserts them to corresponding news on currency pairs.



The data feed features the following additional parameters:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **News Request Period** the period of checking and downloading new information in seconds.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
MetaTrader 5 UniFeeder

## MetaTrader 5 UniFeeder

MetaTrader 5 UniFeeder is the data feed that enables receipt of quotes from the special utility Universal DDE Connector. This solution is built into the platform, which ensures minimal delays in quote delivery.

### How It Works

Universal DDE Connector allows collecting quotes from different data feeds that support the DDE (Dynamic Data Exchange) protocol. The feeder translates quotes received from it to the MetaTrader 5 history server. More details about the Universal DDE Connector are given in a [separate section](#).

### Setup

The data feed must be added via the [corresponding section](#) of the administrator terminal.

**Data Feed**

Common Symbols Translations Parameters Timeouts

The datafeed connects to Universal DDE Connector for receiving quotes

☒ Enable

Name: MetaTrader 5 UniFeeder

Module: MetaTrader 5 UniFeeder Quotes

Feed server: localhost:2222

Feed login: localhost Password: .....

OK Cancel Help

## MetaTrader 5 trading platform

The following parameters should be specified on the "Common" tab of the data feed:

- **Module** UniFeeder;
- **Feed server** address of the server where the Universal DDE Connector is installed and port for connecting to it, separated by a colon;
- **Feed login** account, pre-created in the Universal DDE Connector;
- **Password** password of the account.

### Synthetic Quotes

MetaTrader 5 UniFeeder allows receiving quotes on symbols, whose data are not translated through Universal DDE Connector. This is achieved by way of mathematical conversion of other symbols' quotes translated through Universal DDE Connector. Such a calculation method can be applied to non-convertible currencies, whose exchange rate relative to convertible currencies is set by the central bank.

The calculation formulas for the quotes should be specified on the "Parameters" tab of the data feed:

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

| Parameter      | Value         |
|----------------|---------------|
| ab GOLDVND.ask | GOLD.ask*1950 |
| ab GOLDVND.bid | GOLD.bid*1950 |

Two fields are available here:

#### Parameter

In the "Parameter" field, you should specify the symbol name and its price (Bid or Ask), that will be calculated according to the formula specified in the next field. The symbol name and the price type should be separated by a point, for example: GOLVDVND.ask.

- Symbol whose quotes are calculated must be present in the "Symbols" section. Make sure that the two names match.
- Calculation of both Bid and Ask prices is required for all symbols.

#### Value

In this field the calculation formula should be specified. You may use:

- **Symbols** any symbol, whose quotes are received from Universal DDE Connector can be used for calculations. But only one symbol for one formula can be used. Do not forget to specify the price type to use (Bid or Ask). For example, EURUSD.bid.

A symbol that is used for calculation must exist in the trading platform. In addition, it must be added to the list of symbols, the quotes for which are translated by the data feed.

- **Simple mathematical calculations** multiplication (\*), division (/), addition (+) and subtraction (-).
- **Brackets** you can use brackets "(" to define the calculation order.

- You cannot use negative numbers in formulas. Thus "-3 + 5" is incorrect. The correct expression will be "5 - 3".
- The separator of the integer and fractional part in formulas is also a point, irrespective of the operating system settings.
- Only one symbol, whose quotes are received from Universal DDE Connector can be used in every formula.
- Synthetic quotes calculated from other symbols cannot be used in formulas.

#### Additional Settings

On the "Parameters" tab, you can specify additional settings:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History

Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.

- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Thomson Reuters Feeder

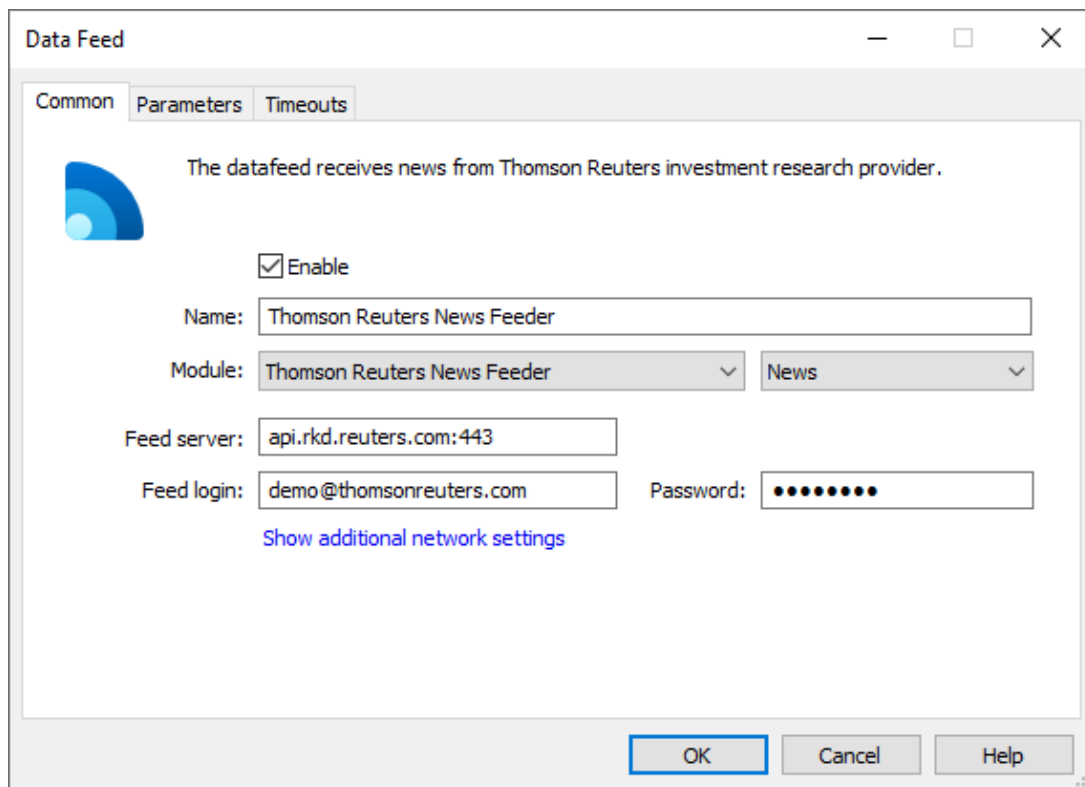
### Thomson Reuters Feeder

Thomson Reuters Feeder is a data feed that enables receiving of financial information from Thomson Reuters (<http://thomsonreuters.com>).

Information from this company is transferred through web services. In order to get an access to them, you will need to conclude a special agreement. Please contact the company representative in your region. The corresponding contact information can be found on the company's website mentioned above.

Once the agreement is concluded, a user is given an IP address, login and password to access the service, as well as a unique ID (AppID). All this information must be provided when setting up the data feed in the administrator terminal.

### Setup



The screenshot shows the 'Data Feed' dialog box with the 'Common' tab selected. The dialog contains the following fields and options:

- Enable:** A checked checkbox.
- Name:** A text field containing 'Thomson Reuters News Feeder'.
- Module:** A dropdown menu showing 'Thomson Reuters News Feeder'.
- News:** A dropdown menu showing 'News'.
- Feed server:** A text field containing 'api.rkd.reuters.com:443'.
- Feed login:** A text field containing 'demo@thomsonreuters.com'.
- Password:** A text field filled with dots.
- Show additional network settings:** A blue hyperlink.
- Buttons:** 'OK', 'Cancel', and 'Help' buttons at the bottom right.

On the "Common" tab of data feed, specify the following parameters:

- **Module** ThomsonReutersFeeder;
- **Feed server** address of the server of Thomson Reuters and port to connect to it separated with colon. A port that is used for connecting, must support connection through the secured https protocol (SSL), usually port 443 is used;
- **Feed login** login to authorize on the server;
- **Password** password for the authorization.

- IP address of the server, login and password are given when concluding the agreement.
- A port that is used for the connection must be allowed on the server where the data feed is installed.

**Data Feed**

Common Parameters Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

Up

Down

Add

Edit

Delete

Default

| Parameter              | Value           |
|------------------------|-----------------|
| ab Application ID      | demoid          |
| ab Allow Topics Codes  | Value           |
| ab Ignore Topics Codes | SPO,MSIC,BLG    |
| ab News Category       | Thomson Reuters |

OK Cancel Help



On the "Parameters" tab, specify one parameter "Application ID". This parameter is a unique identifier of an application that connects to the server of Thomson Reuters. Its value is also given when concluding the agreement.

News can be filtered by categories (topics) using additional optional parameters Ignore Topics Codes and Allow Topics Codes. To allow news from specified categories and disable all other news, specify the codes of the desired categories in the Allow Topics Codes parameter, separated by commas. To disable selected categories and enable all others, specify the categories to disable in Ignore Topics Codes, separated by commas. In the above example, the following codes are specified in Ignore Topics Codes:

- SPO: sport
- MSIC: music news
- BLG: blogs

In this case, the datafeed will receive all news letters, except those from the specified categories.

Please note that a news item can relate to several categories. If the news belongs to at least one of the categories specified in the parameter, it will be filtered out.

- The full list of topic codes can be obtained from Thomson Reuters.
  - News are filtered on Thomson Reuters side, the data feeds does not receive news by specified topics.

Do not use the Language parameter to filter news. It won't affect the data feed. The filtration of news by language is set up in the group configuration. Some news from Thomson Reuters may be transmitted without indication of the language. In this case the news will be passed to the terminals regardless of the languages specified in the group settings.

### Additional Settings

On the "Parameters" tab, you can specify additional settings:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.

## MetaTrader 5 trading platform

- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / RSS News Feeder

## RSS News Feeder

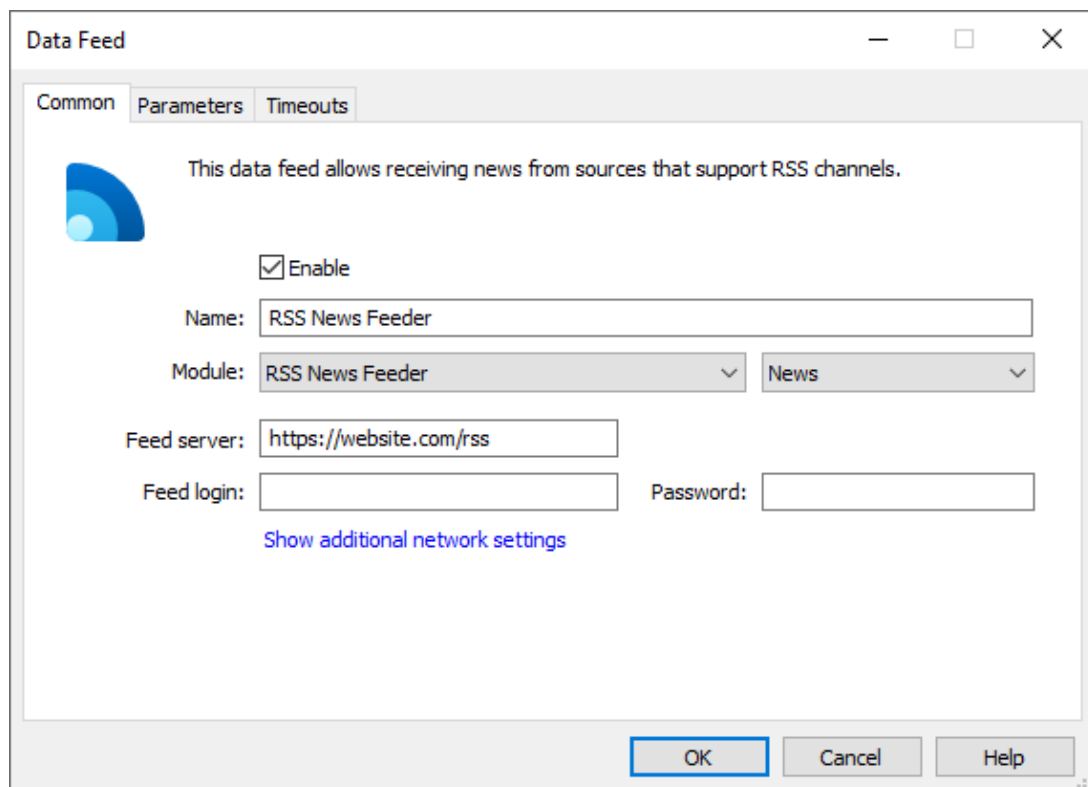
The RSS News Feeder data feed enables the receipt of news from any sources that support the RSS channels. Such sources are various websites, as a rule.

For each source of news (each URL), you need to set up a separate configuration of RSS News Feeder data feed. During the first connection to a source, all the available news are downloaded and transferred. Further, requests for information updates are performed once in five minutes. Obtained information is cached on the hard drive. Therefore, only newly coming news items are downloaded if the data feed is turned on after standing idle for a while.

RSS News Feeder supports categories of incoming news, if any are implemented in the data source. All news items will be divided into subcategories of a category specified on the "Server" tab of the data feed settings.

If you leave the "Category" field empty, the categories of incoming news will be written to the highest level. This may negatively affect the whole categorization of news received in the terminals.

## Setup



**Data Feed**

Common Parameters Timeouts

This data feed allows receiving news from sources that support RSS channels.

☒ Enable

Name: RSS News Feeder

Module: RSS News Feeder News

Feed server: https://website.com/rss

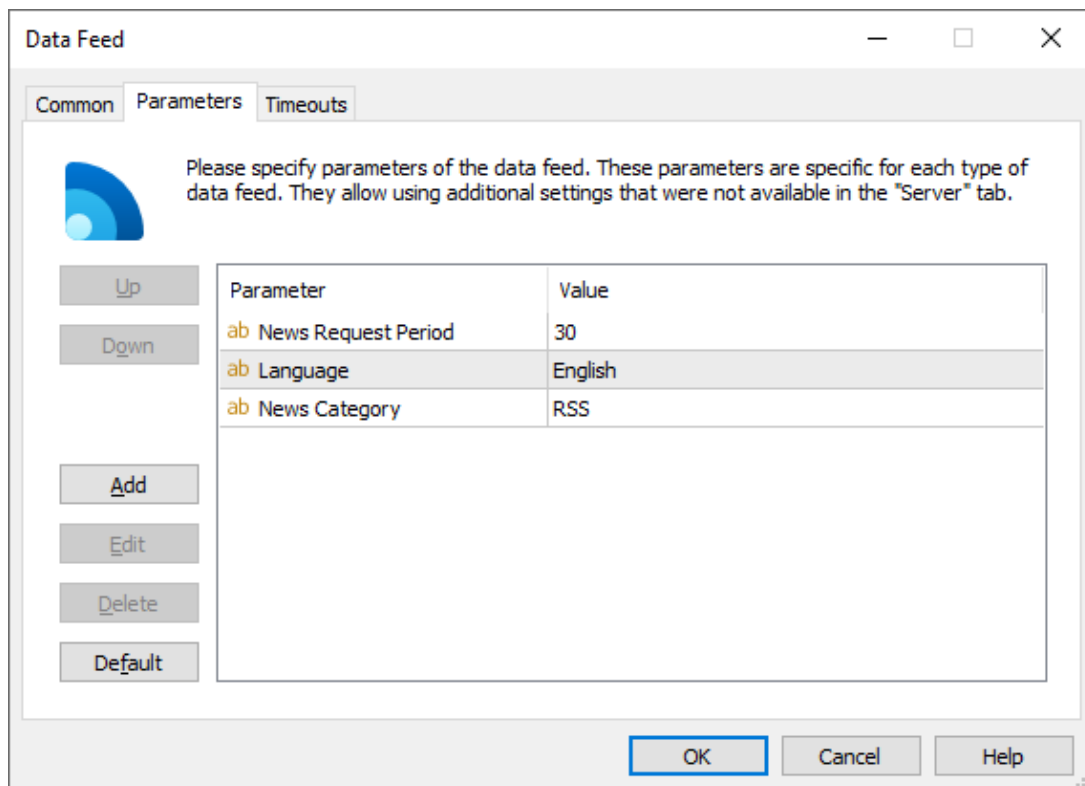
Feed login: Password:

[Show additional network settings](#)

OK Cancel Help

On the "Common" tab of the data feed, specify the following parameters:

- **Module** RSSNewsFeeder;
- **Feed server** address (url) of the data source. Additionally you can specify a port for connection separated with a colon from the address. If a port is not specified, port 80 is used on default.
- **Feed login** in case authentication is required for connecting to the data source, specify the login for connection in this field. Otherwise leave this field empty.
- **Password** in case authentication is required for connecting to the data source, specify the password for connection in this field. Otherwise leave this field empty.



**Data Feed**

Common Parameters Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

| Parameter              | Value   |
|------------------------|---------|
| ab News Request Period | 30      |
| ab Language            | English |
| ab News Category       | RSS     |

Up Down Add Edit Delete Default

OK Cancel Help

The data feed features the following additional parameters:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **News Request Period** the period of checking and downloading new information in seconds.
- **Language** the language of news received from the data source. This parameters is necessary for correct conversion and further sorting of news. "English" is applied by default. The language should be specified in the format standard for the Windows operating systems, though without the regional dialectic specifications. E.g. English, Russian, etc.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
IBTimes News Feeder

IBTimes News Feeder

IBTimes News Feeder delivers FxWire Pro Forex news from International Business Times.

With an existing network of 13 newsrooms worldwide, IBTimes has the adequate infrastructure necessary to create a fresh and new professional FX newswire service. FxWire Pro is a precise and timely newswire service for the modern day Forex trader. Each trading day FxWire Pro publishes over 600 news items in real time to help traders understand market-moving forces and make better and quicker trading decision.

Unique news feed for the modern-day FX trader

FxWire Pro was conceived to integrate perfectly into the most popular trading platforms

- **Succinct News: Short, Concise and To-The-Point**

FxWire Pro by IBTimes publishes news in an innovative format that significantly minimizes the information overload faced by modern day traders. The published items are sub categorized into News, Quotes and where applicable, Analysis. Rather than having to browse through lengthy paragraphs, the end-user will find clear and concise bullet points. The key objective in the development of FxWire Pro has been to transfer the maximum amount of information to the trader in the least possible time frame. IBTimes is proud to present a newswire that has been praised by several leading industry's analysts and traders for its unique and useful format of publishing.

- **Expert Forex Analysts and Editors Filter Information Noise**

The editorial process at publishing news on FxWire Pro involves Expert FX analysts and editors who act as a strong knowledge filter to remove any news item that is deemed irrelevant to the Forex trader.

FxWire Pro's 600+ a-day rolling news in an innovative format allow traders to be completely on top of the financial markets, economic and geo-political developments across the globe in real time.

News categories of FxWire Pro

The core areas covered by IBTimes' Professional Forex newswire service are:

- Macro-Economy News & Data
- Currencies Movement
- Geo Politics
- Real Time Economic Indicators
- Treasury
- Ratings
- Money Market

## MetaTrader 5 trading platform

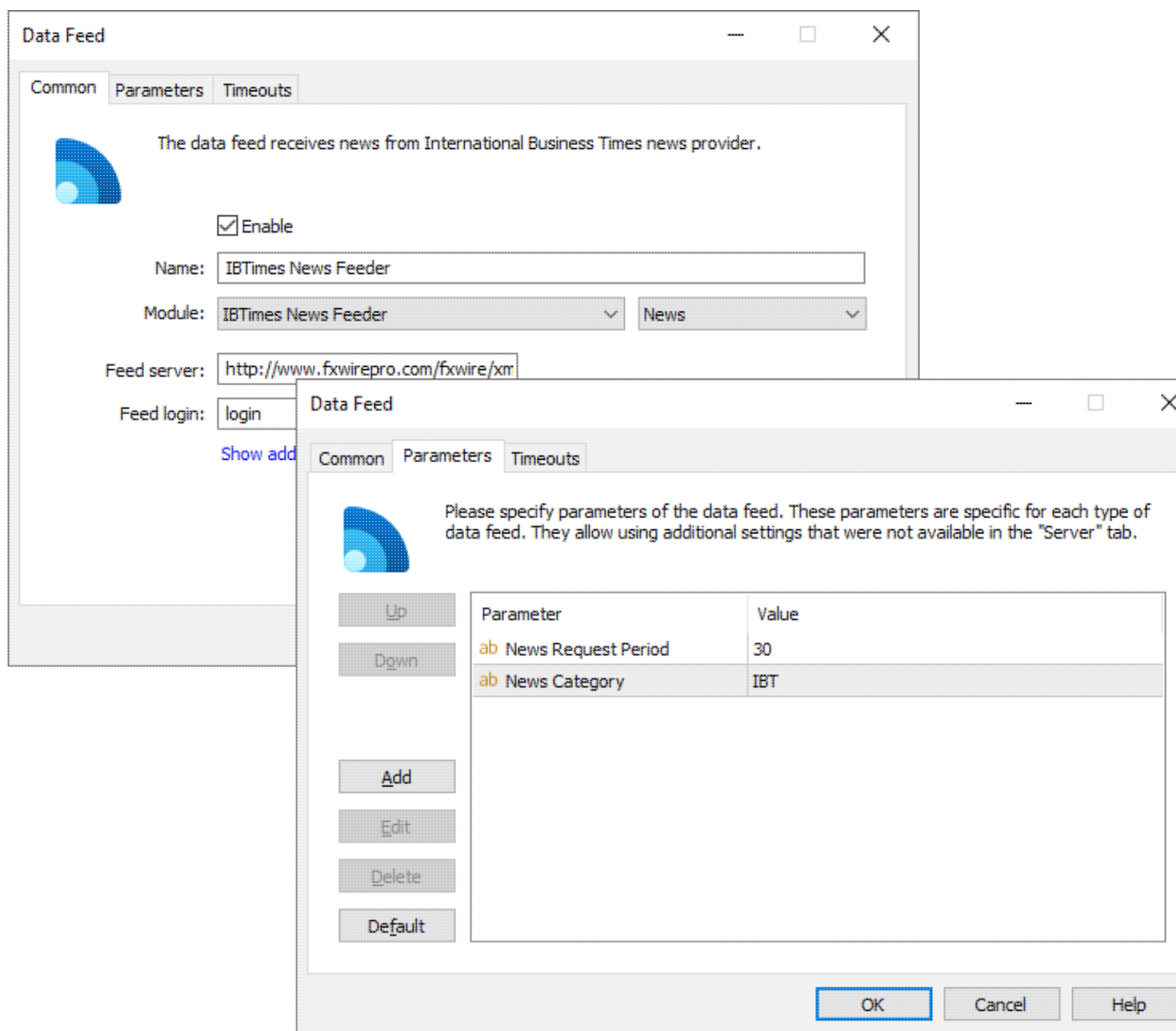
- Central Bank
- Market Moving Talks
- Media Round Ups & Picks
- Generic FX Relevant News
- Commodities
- Stocks & Indices
- Research Notes
- Currencies covered: Major, minor and exotic currencies are covered as part of this service along with economic indicators in real-time from the major economies.

### Pre-setup

Information is broadcast via web service. To gain access to the web service, you should sign an agreement with International Business Times by sending a request via [App Store](#). You will receive a special login and password which should be used to set up the data feed in the platform.

### Setup

## MetaTrader 5 trading platform





On the "Common" tab of the data feed specify the following parameters:

- **Module** IBTNewsFeeder;
- **Feed server** specify here one of the addresses of the data source, depending on the language of news:
  - **News in English** <http://www.fxwirepro.com/fxwire/xml/newswire.php>
  - **News in Chinese** <http://www.fxwirepro.com/fxwire/cn/xml/newswire.php>
  - **News in Japanese** <http://www.fxwirepro.com/fxwire/jp/xml/newswire.php>
- **Feed login** authorization login received after you've signed an agreement with the International Business Times company.
- **Password** authorization password received after you've signed an agreement with the International Business Times company.

On the "Parameters" tab, you can specify additional settings:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **News Request Period** the period of checking and downloading new information in seconds. During the first connection the data feed will receive all the available news items. Further connections are established at the intervals set in this parameter, default period is 30 seconds. Contact IB Times support service for the best news request interval as it may vary for different clients.

If you want to receive news in all the three languages, create three separate data feeds with different names in MetaTrader 5 Administrator. In the Server field specify the appropriate address, in the fields of login and password specify the details of your subscription allowing to receive news in the appropriate language.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
ForexPros Feeder

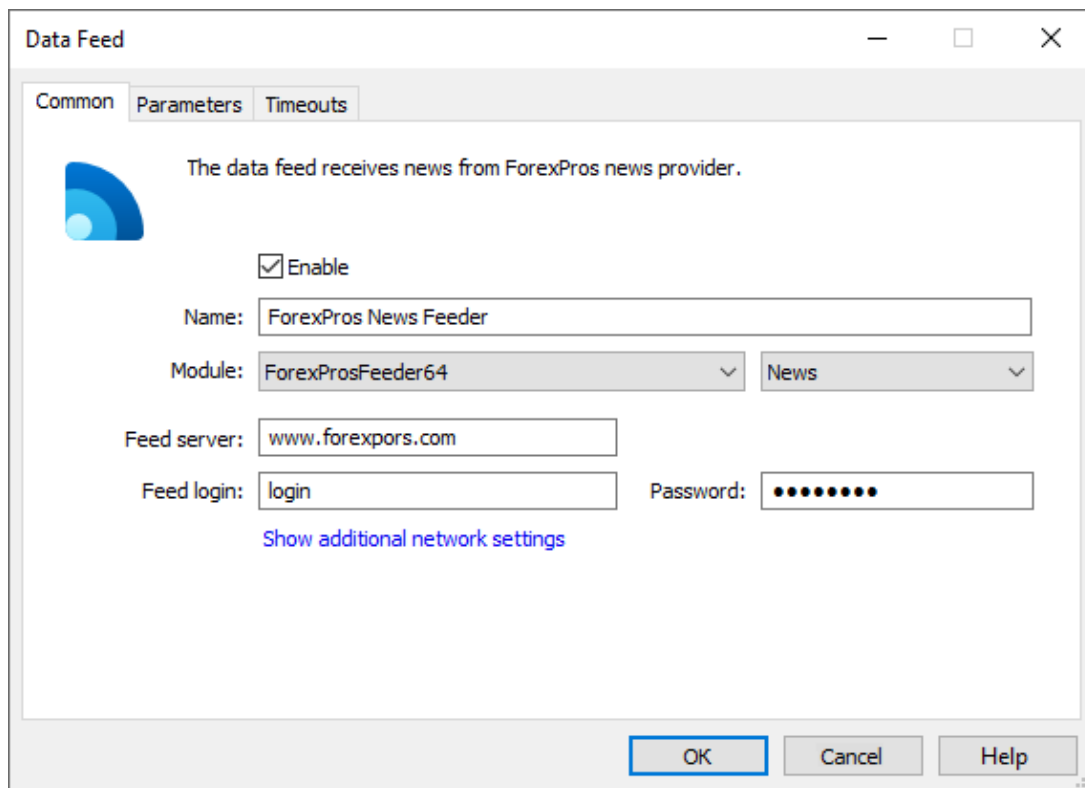
### ForexPros Feeder

ForexPros Feeder is a data feed that allows receiving financial news from ForexPros ([www.forexpros.com](http://www.forexpros.com)).

Information from this company is broadcast through web services. In order to get access to these services, it is necessary to conclude a special agreement with ForexPros. The corresponding contact information is published on the [official website](#) of this company.

After signing the contract, you will be given a login and password. Use these data to configure the data feed.

### Setup

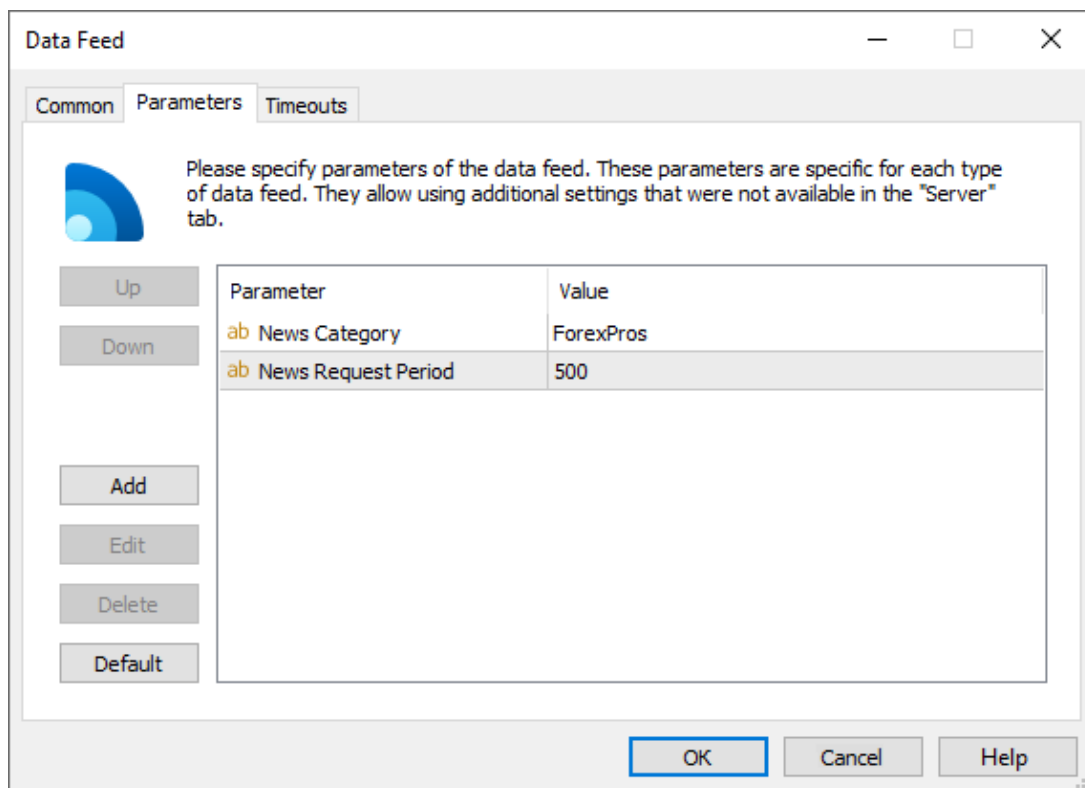


The screenshot shows the 'Data Feed' configuration window in MetaTrader 5. The window has three tabs: 'Common', 'Parameters', and 'Timeouts'. The 'Common' tab is selected. Inside the window, there is a blue circular icon on the left and a text box stating 'The data feed receives news from ForexPros news provider.' Below this, there is a checkbox labeled 'Enable' which is checked. There are three input fields: 'Name' with the value 'ForexPros News Feeder', 'Module' with a dropdown menu showing 'ForexProsFeeder64', and 'Feed server' with the value 'www.forexpros.com'. To the right of the 'Module' dropdown is another dropdown menu showing 'News'. Below the 'Feed server' field are two more fields: 'Feed login' with the value 'login' and 'Password' with a masked password represented by dots. A blue link 'Show additional network settings' is located below the 'Feed login' field. At the bottom of the window are three buttons: 'OK', 'Cancel', and 'Help'.

| Field       | Value                 |
|-------------|-----------------------|
| Name        | ForexPros News Feeder |
| Module      | ForexProsFeeder64     |
| Feed server | www.forexpros.com     |
| Feed login  | login                 |
| Password    | .....                 |

On the "Common" tab of the data feed specify the following parameters:

- **Module** ForexProsFeeder;
- **Feed server** specify here one of the addresses of the data source, depending on the language of news:
  - **News in English** www.forexpros.com;
  - **News in Spanish** www.forexpros.es;
  - **News in French** www.forexpros.fr;
  - **News in German** www.forexpros.de;
  - **News in Italian** www.forexpros.it;
  - **News in Russian** www.forexpros.ru;
  - **News in Arabic** www.forexpros.ae;
  - **News in Hebrew** www.forexpros.co.il;
  - **News in Dutch** nl.forexpros.com;
  - **News in Galician** www.forexpros.com.pt;
  - **News in Japanese** www.forexpros.jp;
  - **News in Chinese** cn.forexpros.com;
  - **News in Turkish** www.forexprostr.com;
- **Feed login** authorization login received after you've signed an agreement with the ForexPros company.
- **Password** authorization password received after you've signed an agreement with the ForexPros company.



**Data Feed**

Common Parameters Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

| Parameter              | Value     |
|------------------------|-----------|
| ab News Category       | ForexPros |
| ab News Request Period | 500       |

Up Down Add Edit Delete Default

OK Cancel Help

The data feed features the following additional parameters:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **News Request Period** the period of checking and downloading new information in seconds. During the first connection the data feed will receive all the available news items. Further checks for news are performed at the specified intervals.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / KnowledgeView News Feeder

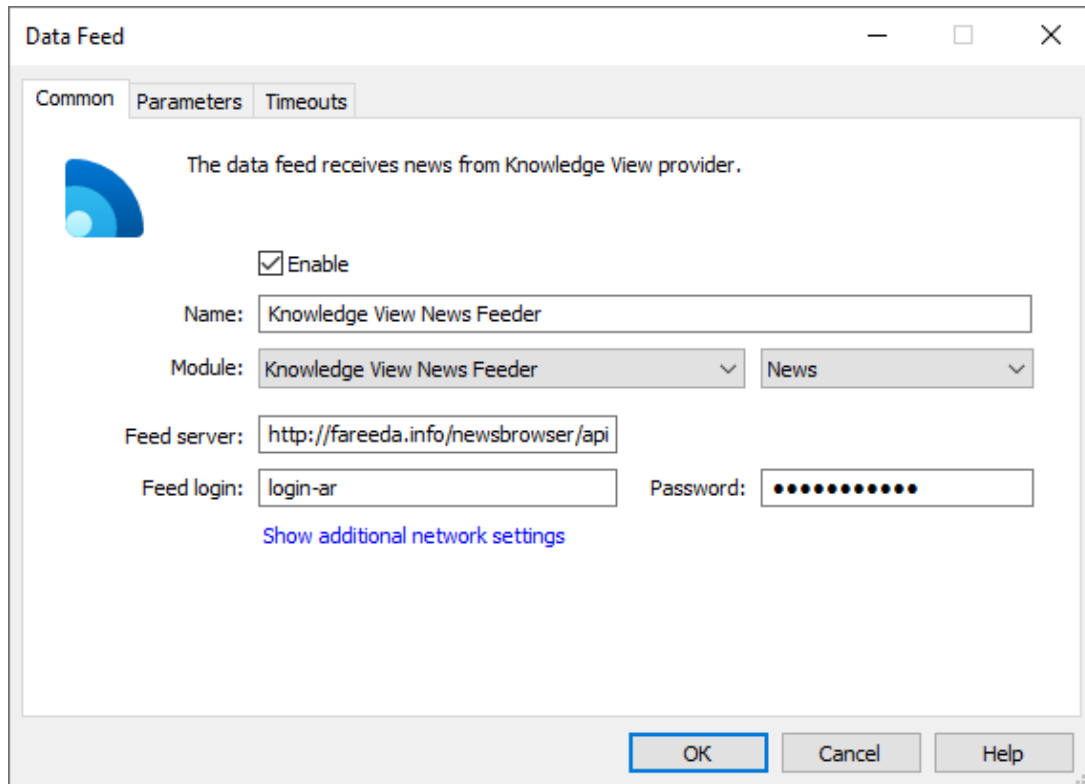
## KnowledgeView News Feeder

KnowledgeView News Feeder is designed to receive news from KnowledgeView, a partner of Dow Jones. It allows to receive news in real time in three languages: Arabic, Farsi and Turkish.

If you want to start getting the news, you need to contact KnowledgeView or Dow Jones to sign the agreement. The contact information can be found on the official websites of these companies: Offices section of the KnowledgeView official web site and Contact Us section of the Dow Jones official web site. After concluding the contract you will receive the parameters for connecting a data feed to KnowledgeView server: address, login, password and additional Filter parameter.

## Setup

Add the new KnowledgeView News Feeder data feed configuration via the corresponding section of the administrator terminal.



**Data Feed**

Common Parameters Timeouts

The data feed receives news from Knowledge View provider.

☒ Enable

Name: Knowledge View News Feeder

Module: Knowledge View News Feeder News

Feed server: http://fareeda.info/newsbrowser/api

Feed login: login-ar Password: .....

[Show additional network settings](#)

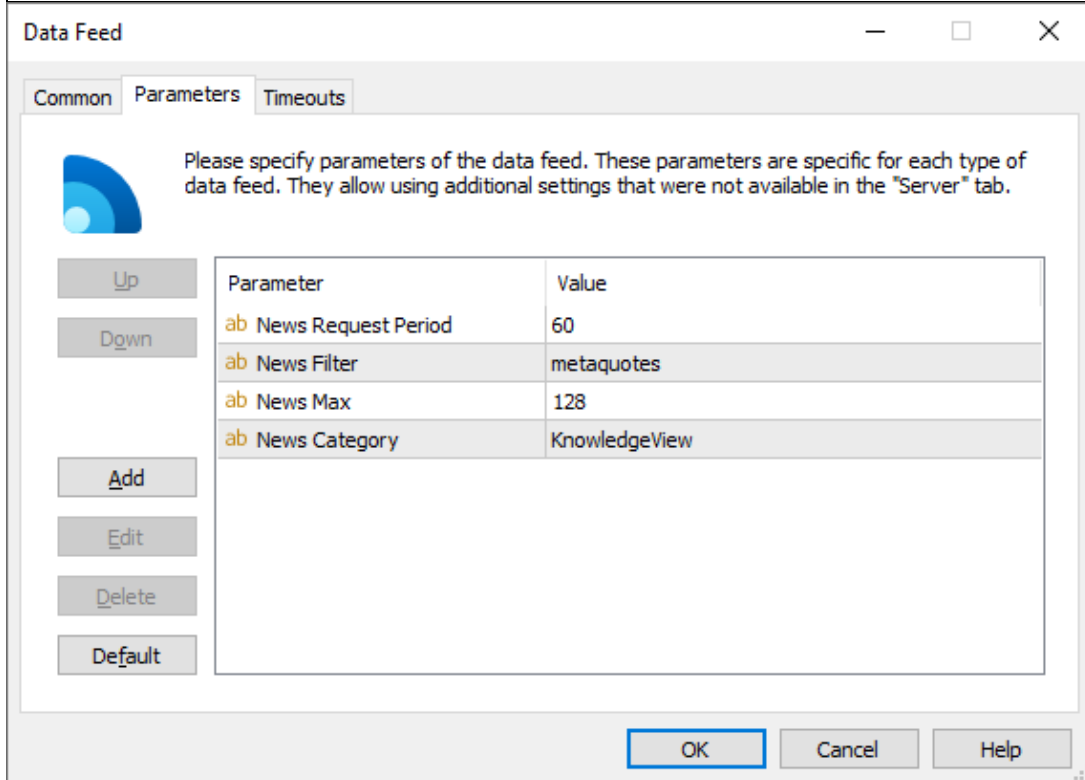
OK Cancel Help

## MetaTrader 5 trading platform

The following parameters must be specified on the "Common" tab of the data feed:

- **Module** KnowledgeViewNewsFeeder(64);
- **Feed server** KnowledgeView server web address. The default web address is <http://fareeda.info/newsbrowser/api>.
- **Feed login** login for connection to KnowledgeView server;
- **Password** password for connection to KnowledgeView server.

Address, login and password are submitted by KnowledgeView during the agreement conclusion. A separate login and password are submitted for each news language. Thus, you should create several configurations for that data feed to get news in several languages simultaneously.



**Data Feed**

Common Parameters Timeouts

Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

| Parameter              | Value         |
|------------------------|---------------|
| ab News Request Period | 60            |
| ab News Filter         | metaquotes    |
| ab News Max            | 128           |
| ab News Category       | KnowledgeView |

Up Down Add Edit Delete Default

OK Cancel Help

Specify the following parameters in the [Parameters](#) tab:

- **News Request Period** intervals to check and download new data in seconds. The data feed receives all available news during the first connection. After that, the latest news are checked at the specified intervals. The lower the value, the faster the news will appear in the platform. But this will increase the network load. The default value is 60 seconds.
- **News Filter** the value is provided by KnowledgeView upon the execution of an agreement. Separate Filter parameter value is specified for each news language in the same way as it is done with a login and a password.
- **News Max** the maximum number of news items which can be received by the data feed. If this parameter value is low while "News Request Period" is large, some of the news items provided by the source can be lost. The default value is 128.
- **News Category** the category name for the newsletters received from the data feed. The category name can then be used for specifying news to be delivered to client [groups](#).

If the Filter value is incorrect, the data feed will not be able to connect to the server. These parameters can be checked in the [data feed log](#):

|                     |         |                                                               |
|---------------------|---------|---------------------------------------------------------------|
| 2012.03.28 09:59:59 | Gateway | datafeed initialized, connecting to http://fareeda.info/newsb |
| 2012.03.28 09:59:59 | Gateway | available datafeed filter: 'Forex Turkish'                    |
| 2012.03.28 09:59:59 | Gateway | available datafeed filter: 'Forex Turkish - month'            |
| 2012.03.28 09:59:59 | Gateway | specified filter 'Turkish' is incorrect, check datafeed param |
| 2012.03.28 09:59:59 | Gateway | filter checking failed                                        |

In addition to the entries indicating the errors in the filter specification, the data feed log also displays available filter values (the entries of "available datafeed filter" type). The terminal will start receiving news right after the successful connection.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / FXstreet Feeder

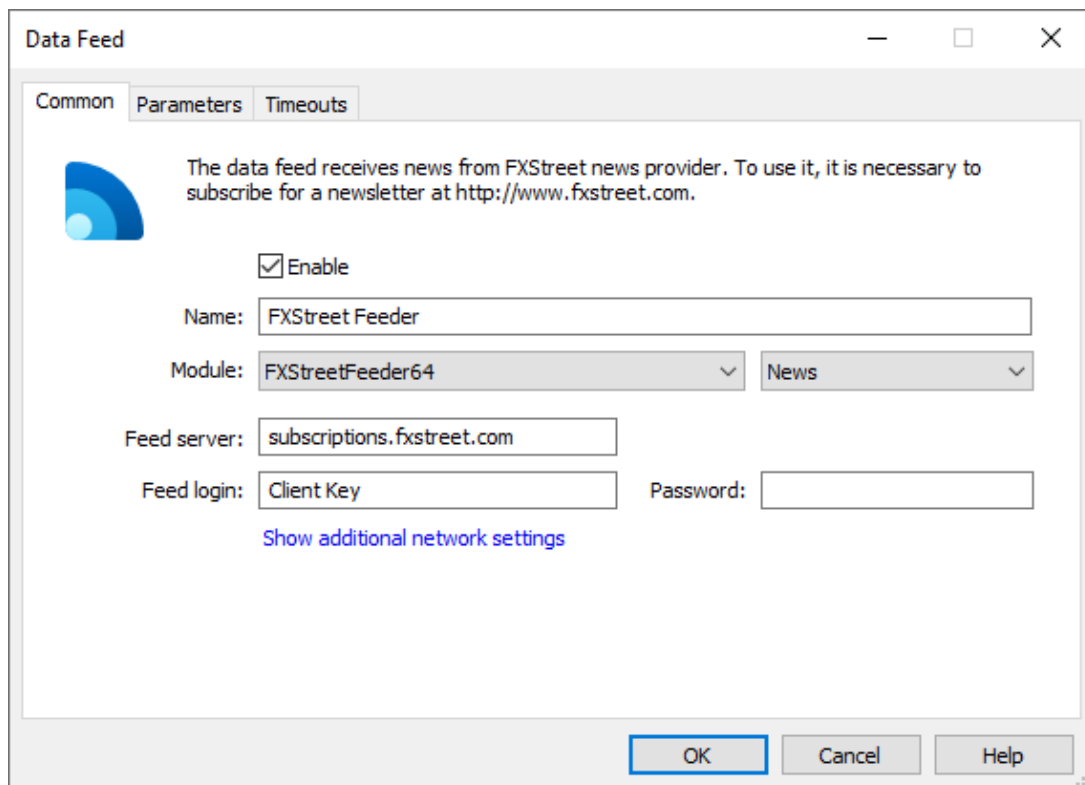
## FXstreet Feeder

FXstreet Feeder for MetaTrader 5 provides your clients with financial news, as well as the data on technical analysis from the [FXstreet.com](http://www.fxstreet.com) financial web portal. The news are provided in eight languages: English, Russian, Arabic, Chinese, German, Indonesian, Spanish and Turkish.

FXstreet Feeder is free and included in the platform standard delivery. Contact [FXstreet.com](http://www.fxstreet.com) to subscribe to the news. Contact details can be found on the web portal at "[Contact Us](#)" section.

## Setup

Add the new FXstreet Feeder data feed configuration via the corresponding section of MetaTrader 5 Administrator:



Data Feed

Common Parameters Timeouts

The data feed receives news from FXStreet news provider. To use it, it is necessary to subscribe for a newsletter at <http://www.fxstreet.com>.

☒ Enable

Name: FXStreet Feeder

Module: FXStreetFeeder64 News

Feed server: subscriptions.fxstreet.com

Feed login: Client Key Password:

[Show additional network settings](#)

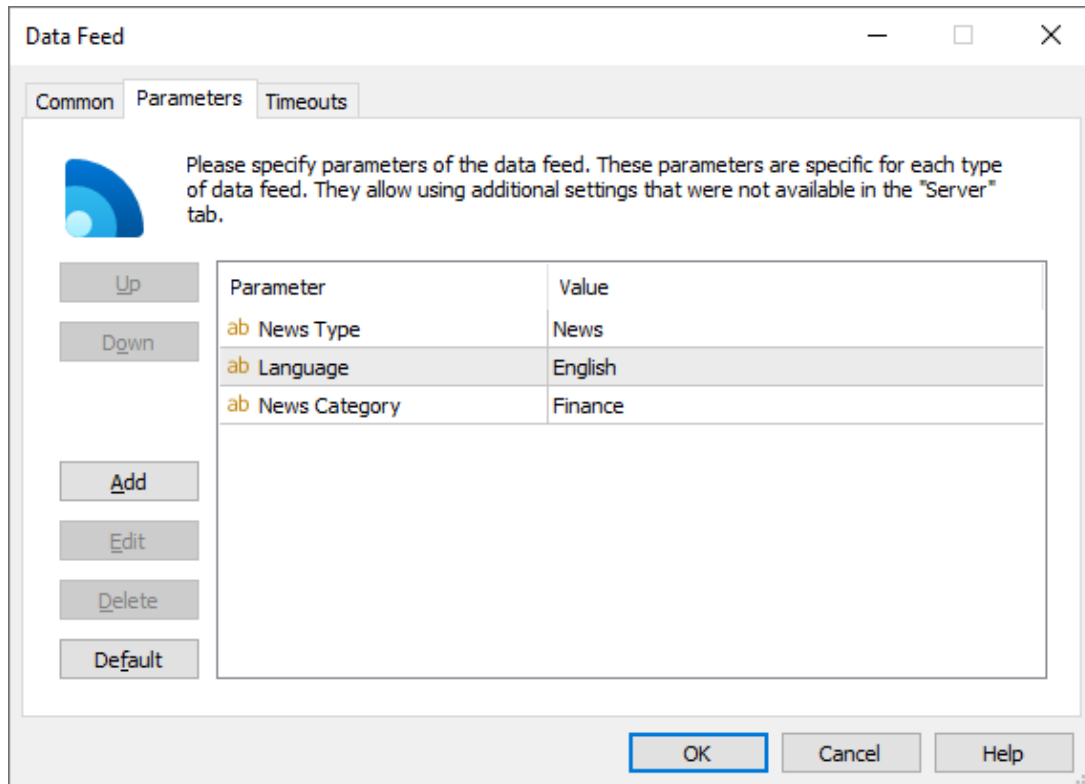
OK Cancel Help



## MetaTrader 5 trading platform

The following parameters must be specified on the "Common" tab of the data feed:

- **Module** FXstreetFeeder(64);
- **Feed server** FXstreet server address. subscriptions.fxstreet.com is set by default;
- **Feed login** login (or Client Key) for connection to FXstreet server. Submitted by FXstreet when concluding an agreement for providing news feed.



Specify additional settings in the Parameters tab:

- **Language** the language in which you wish to receive news. For each language, create a separate "Language" parameter with the appropriate value. For example, in order to receive news in English and Russian, you need to create two parameters: Language = english, Language = russian. If no language is specified, the datafeed will only deliver news in English. The following languages are supported:
  - ◆ arabic
  - ◆ chinese
  - ◆ english
  - ◆ french
  - ◆ german
  - ◆ indonesian
  - ◆ russian
  - ◆ spanish
  - ◆ thai
  - ◆ tradchinese
  - ◆ turkish
  - ◆ japanese
  - ◆ vietnamese
- **News Type** the type of news you wish to receive:
  - ◆ News standard subscription.
  - ◆ FXBeat news from famous Forex market participants featuring their personal opinion. To receive such newsletters, contact FXStreet.com for an appropriate subscription. FXStreet.com will provide a new Client Key or include the FXBeat subscription into the key your are using.
  - ◆ Crypto cryptocurrency market news.
- **News Category** the category name for the newsletters received from the data feed. The category name can then be used for specifying news to be delivered to client groups.

The client terminals will start receiving news right after enabling the data feed. The data feed logs can be requested to check its operation.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Financial Source News Feeder

## Financial Source News Feeder

Financial Source News Feeder enables the real-time delivery of news related to global currencies as well as trading analytics. In their <https://financialsource.co> website, Forex News state that they aim to be the first to deliver forex news to users and bring high probability trading opportunities. Their professional analysts scour global newswires so you catch every currency move.

Financial Source news items are divided into the following categories: Central Banks, Market Insights, Must Read, Order Flow Levels and Risk Events. The size of news items ranges from brief informative reports to copyright analytical articles, which describe the full picture of the global market.

The data feed module is free and is included in the platform standard delivery package. To start receiving news, you should request a subscription via the [official website](#). After that you will be provided with an Authorization Token for the data feed configuration.

## Setup

Add the new Financial Source News Feeder configuration via the [corresponding section](#) of MetaTrader 5 Administrator:

**Data Feed**

Common Parameters Timeouts Information

This data feed allows receiving Forex Source news. To receive this feed, please visit [www.forexsource.co/metatraderplugin](http://www.forexsource.co/metatraderplugin). After registering for the feed you will then receive your unique access key by email.

☒ Enable

Name: Financial Source News Feeder

Module: ForexSourceNewsFeeder64 News

Feed server: <https://server.financialsource.co>

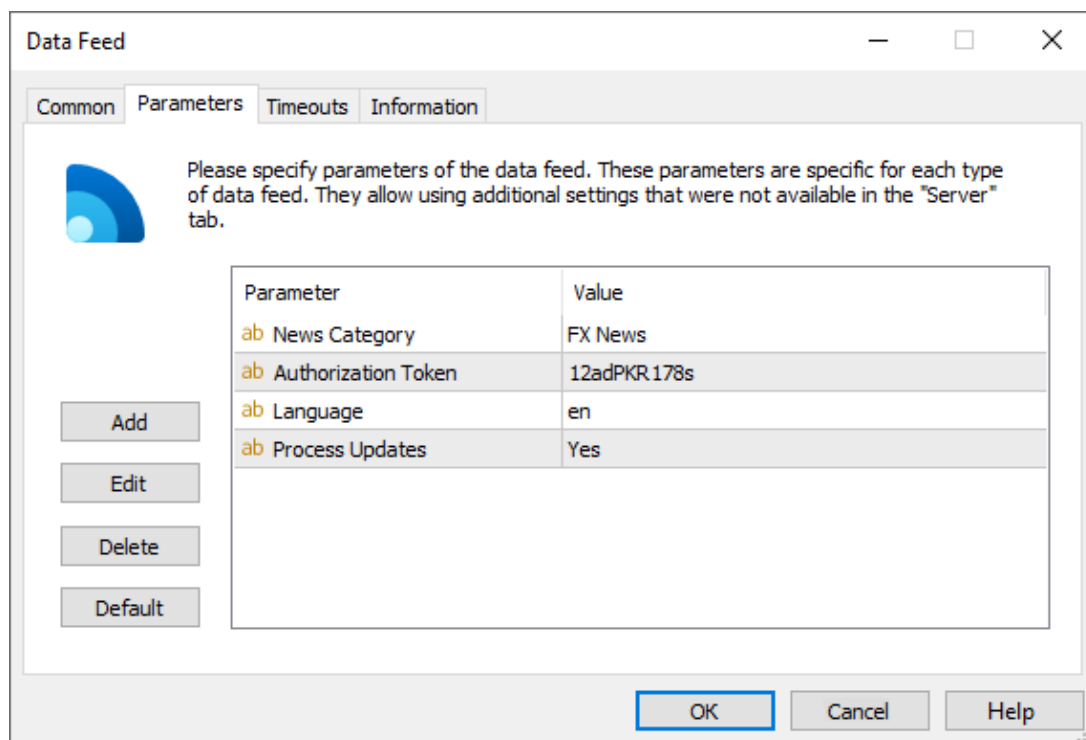
Feed login: Password:

[Show additional network settings](#)

OK Cancel Help

The following parameters should be specified on the Common tab of the data source:

- **Module** ForexSourceNewsFeeder64.
- **Feed server** Financial Source server address.  
https://server.financialsource.co is set by default.



Specify the following settings in the Parameters tab:

- **Authorization Token** API Key, a special key to connect to the server. provided after subscribing to the news.
- **Language** the news language. For example, en for English, es for Spanish, it for Italian, ar for Arabic, ru for Russian, de for German. Please contact Financial Source for the full list of supported language. English is used by default.
- **Process Updates** the news items broadcasted by Financial Source may change over time. For example, they may be adjusted or clarified. If the parameter is set to "Yes", the data feed will process these changes and send new versions of these news items to the platform (earlier received news letters cannot be changed). If the parameter value is "No", the data feed will not process changes and this only initial news versions will be available in the trading platform. The default value is "Yes".
- **News Category** the category name for the newsletters received from the data feed. The category name can then be used for specifying news to be delivered to client groups.

In order to receive news in several languages, create several data source configurations.

The client terminals will start receiving news right after enabling the data source. The data source journal can be requested to check its operation.



MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Claws & Horns Feeder

### Claws & Horns Feeder

This data feed allows you to provide your clients with financial news and analytics from Claws & Horns:

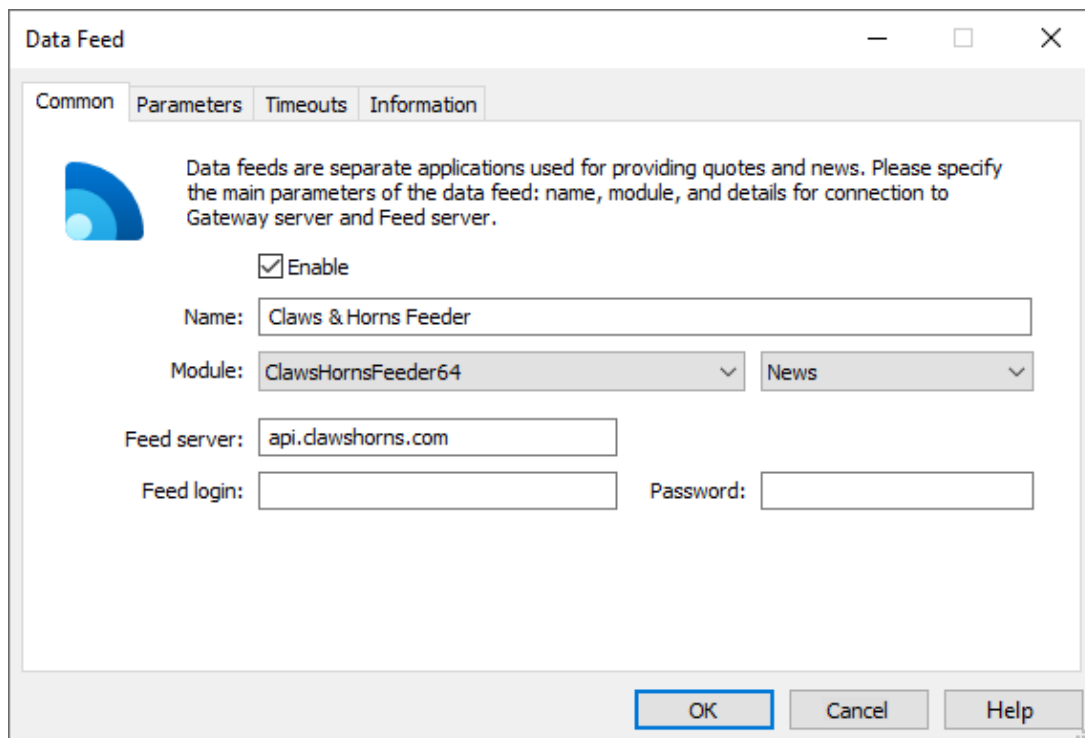
- economic calendar featuring all the important fundamental data
- technical analysis
- fundamental analysis
- signals (market entry recommendations with entry/exit points that are updated throughout the day)
- daily video podcasts with long-term outlook for the major currency pairs
- daily analysis of Russian shares traded on the Moscow Exchange

Materials are provided in 15 languages, including Russian, English, Chinese, Spanish, French, German, etc.

Claws & Horns Feeder is free and included in the platform standard delivery. You can purchase subscription on the Claws & Horns official website.

### Setup

Add the new Claws & Horns Feeder data source configuration via the corresponding section of MetaTrader 5 Administrator:



**Data Feed**

Common Parameters Timeouts Information

Data feeds are separate applications used for providing quotes and news. Please specify the main parameters of the data feed: name, module, and details for connection to Gateway server and Feed server.

☒ Enable

Name: Claws & Horns Feeder

Module: ClawsHornsFeeder64 News

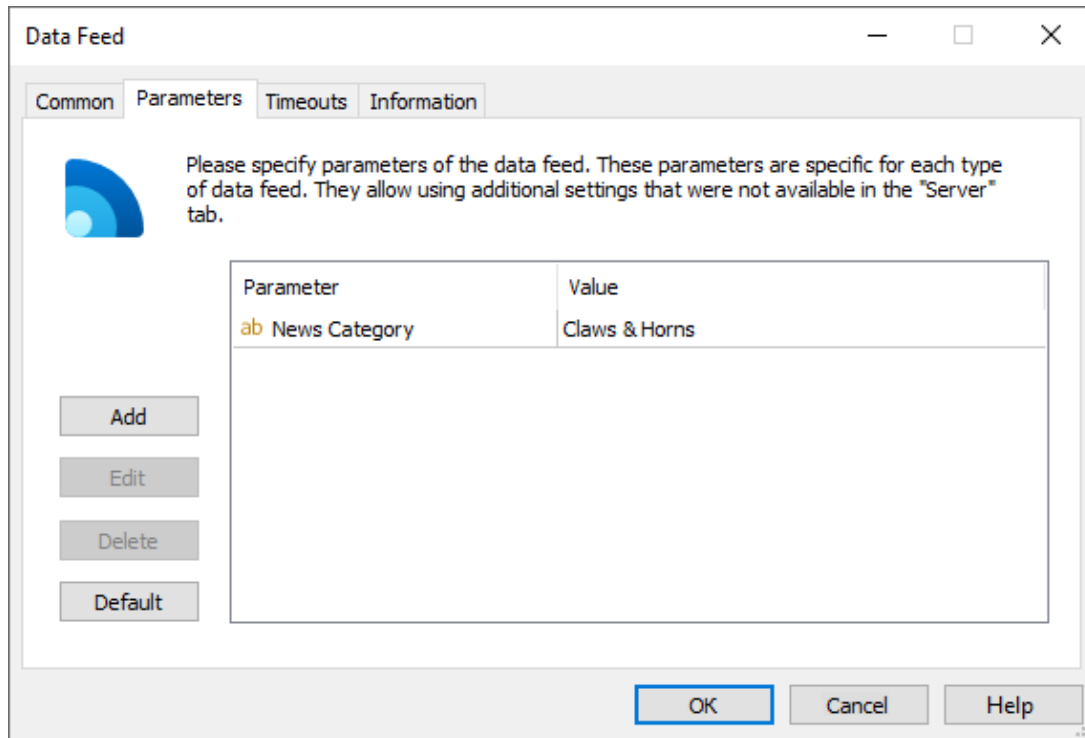
Feed server: api.clawshorns.com

Feed login: Password:

OK Cancel Help

The following parameters should be specified on the Common tab of the data source:

- **Module** ClawsHornsFeeder64.
- **Feed server** Claws & Horns server address. api.clawshorns.com is set by default. There is no need to change it.
- **Password** when purchasing the subscription, you receive a special ID (token) consisting of 32 symbols. Enter it here.



On the "Parameters" tab, you can also use an additional parameter "News Category" the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.

The client terminals will start receiving news right after enabling the data source. The data source journal can be requested to check its operation.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
UniNewsFeeder

## UniNewsFeeder

UniNewsFeeder is a universal news feed applying the new UniNews protocol.

MetaQuotes Software Corp. has developed its own binary news transfer protocol allowing for high speed of broadcasting news to the MetaTrader 4/5 platforms. Further on, it will allow using news in algorithmic trading by accessing them via MQL5 and the strategy tester.

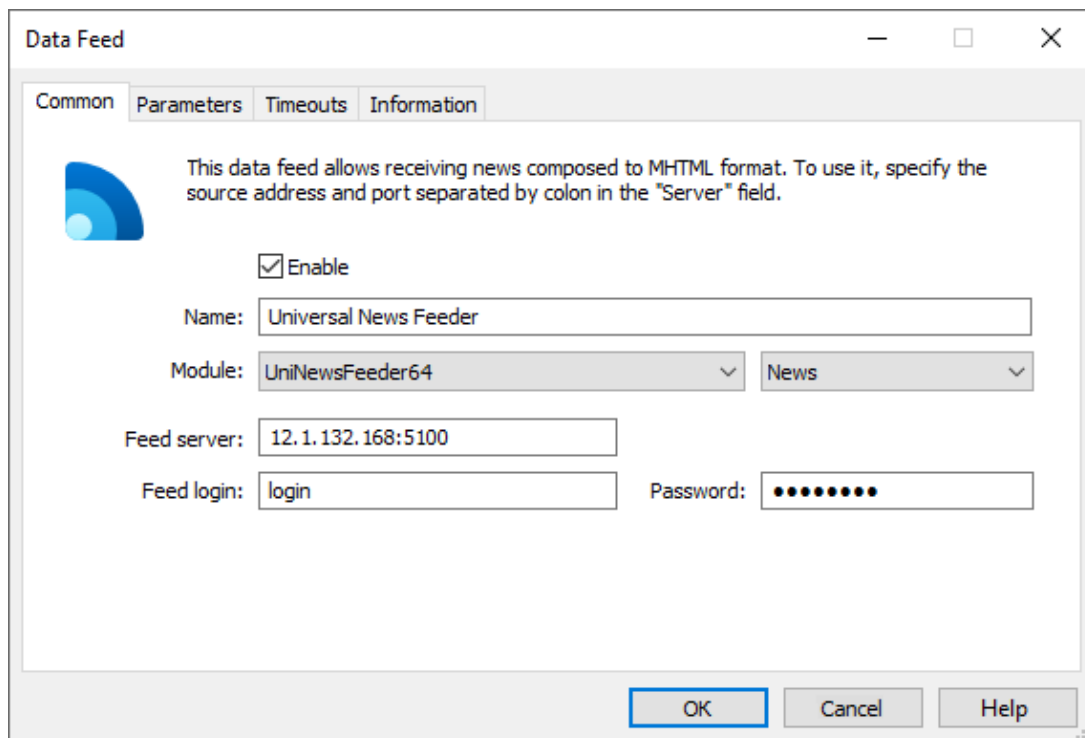
The new protocol will be of interest to news providers since the integration is completely ready on the side of all MetaTrader 4/5 brokers. The free UniNewsFeeder data feed is able to receive news from any source supporting the new protocol.

## Operation principles

UniNewsFeeder connects to the news provider server and requests news using keywords in multiple languages. First, the data feed receives news missed during its inactivity. After that, the news arrive in real time.

## Setup

Add the new UniNewsFeeder data feed configuration via the corresponding section of the MetaTrader 5 Administrator:



**Data Feed**

Common Parameters Timeouts Information

This data feed allows receiving news composed to MHTML format. To use it, specify the source address and port separated by colon in the "Server" field.

☒ Enable

Name: Universal News Feeder

Module: UniNewsFeeder64 News

Feed server: 12.1.132.168:5100

Feed login: login Password: .....

OK Cancel Help



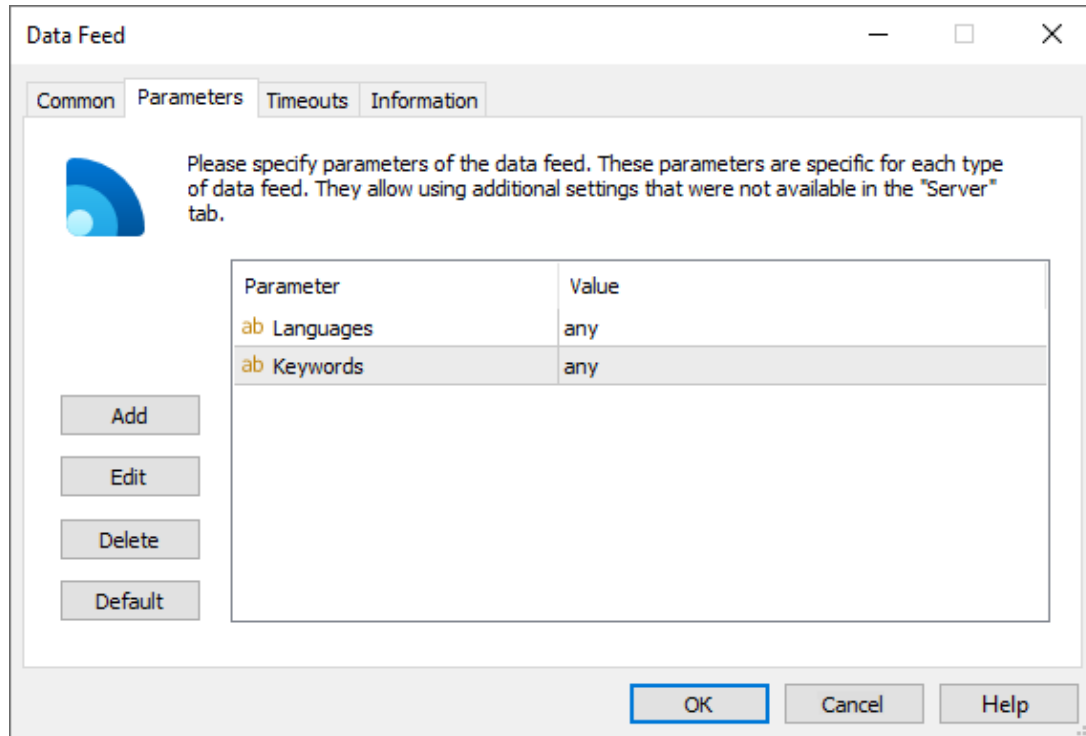
## MetaTrader 5 trading platform

The following parameters should be specified on the Common tab of the data feed:

- **Module** UniNewsFeeder64.
- **Feed server** news provider server address.
- **Login** login for connecting to the news provider's server.
- **Password** password for connecting to the news provider's server.

Connection data is provided by a news provider.

Set the news language and keywords for sorting the news on the Parameters tab.



## MetaTrader 5 trading platform

Set the incoming news language in the Languages parameter. The list of languages, in which you want to receive news, is comma-separated. For example, "russian,english,italian". The default value is "any". In this case, the data feed receives news in all languages in the MetaTrader 5 platform. Ask your news provider for the list of available languages.

In the Keywords parameter, set the keywords to be used by the data feed to request news. The default value is "any" (no filtration). Each news provider can sort out news using keywords. Ask your news provider for the list of keywords.

On the "Parameters" tab, you can also use an additional parameter "News Category" the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.

The client terminals will start receiving news right after enabling the data feed. The data feed journal can be requested to check its operation.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Alliance News Feeder

## Alliance News Feeder

The data feed allows receiving financial news and analytics from the British Alliance News agency. The service provides traders with extensive selection of news for making informed decisions when trading stocks, currencies, CFDs, options and futures. Alliance News covers over 2 400 companies, including FTSE giants, as well as smaller AIM participants and investment funds.

Blue-chip companies, British and international economic indicators, market reviews and broker ratings more than 500 daily news items provide a complete picture of the day. News articles are written in an easy-to-understand way, while maintaining the high level of information content.

## Why Alliance News?

- The core aim of the company is serving financial professionals and independent investors.
- Founders have well over 50 years of financial news and technology experience.
- Concise news service provides the most relevant information for trading without clutter.

## What The Alliance News Professional Service Offers Your Clients

- News articles written in an easy-to-understand way, with the informed but not expert, reader in mind.
- Blanket coverage of UK-listed companies and the global, political and macro-economic influences on their share prices.
- Coverage of global blue-chip companies, as well as the forex and commodities markets.
- Concise bullet point 'flash headline' reporting as the news unfolds, with full articles published soon thereafter.
- Helps your clients get ideas, uncover opportunities and check facts while there is still money to be made.

## Ordering Subscription

Order subscription right from the Buy page of the technical support website. After submitting your request, managers of Alliance News will contact you and provide all information.

Another way is to send an email to support@alliancenews.com or call +44 207 199 0347.

## Configuration of the Data Feed

After requesting a subscription and receiving connection details from Alliance News, add the new Alliance News Feeder configuration via the corresponding section of

MetaTrader 5 Administrator:

**Data Feed**

Common Parameters Timeouts Information

The data feed receives news from Alliance News news provider. To use it, it is necessary to subscribe for a newsletter at [integration@alliancenews.com](mailto:integration@alliancenews.com).

☒ Enable

Name: Alliance News Feeder

Module: AllianceNewsFeeder64 News

Feed server: <http://feeds2.alliancenews.info:244>

Feed login: login Password: .....

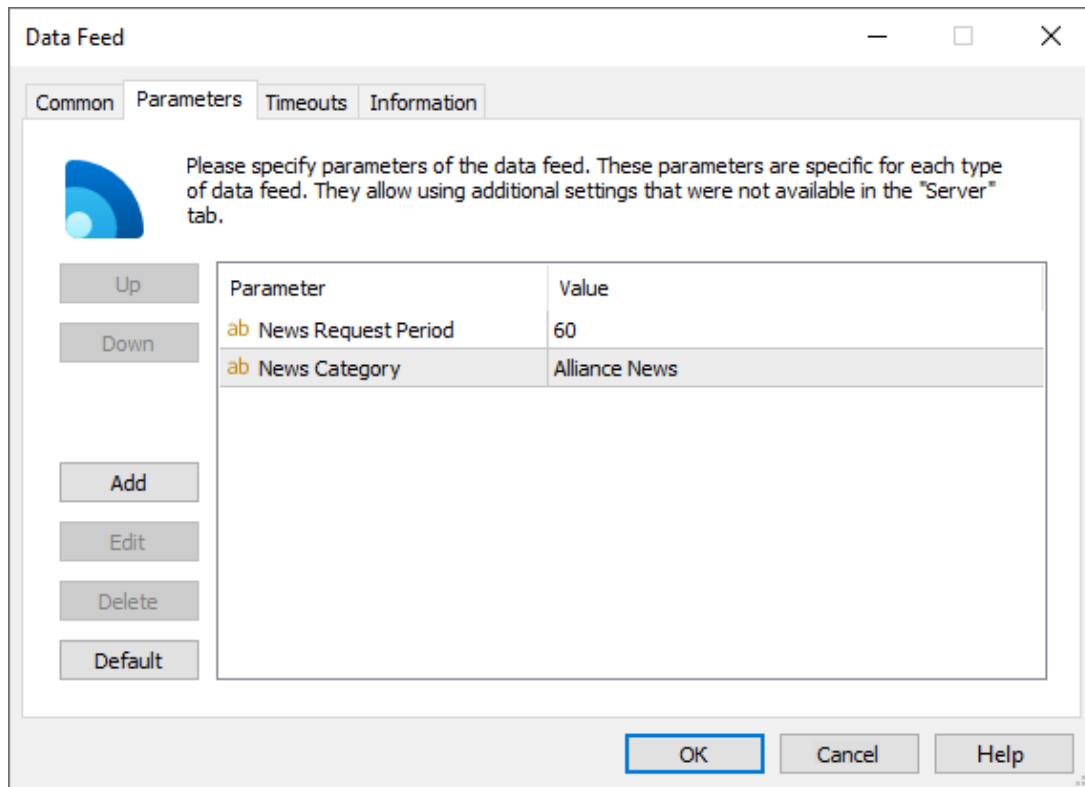
[Show additional network settings](#)

OK Cancel Help

The following parameters should be specified on the Common tab of the data feed:

- **Module** AllianceNewsFeeder.
- **Feed server** URL of news feed server provided by Alliance News representatives. By default it is <http://feeds2.alliancenews.info:244>
- **Feed login** authorization login acquired from Alliance News representatives.
- **Password** authorization password acquired from Alliance News representatives.

The port used for connection to Feed server must be allowed on the internal or external firewall of the server where the data feed is operating.



The data feed features the following additional parameters:

- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **News Request Period** the period of checking and downloading new information in seconds.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Newsquawk

### Newsquawk Feeder

Data feed for receiving financial newsletters from the Newsquawk company. Newsquawk provides real-time access to the most important financial news, filtering data streams from hundreds of sources, such as Reuters, Bloomberg and Daily Mail, among others. Additionally, you can receive daily market overviews across various categories of financial instruments, including forex, metals, stocks and commodities.

### Preparation

Fill out a form on the Newsquawk website. The company offers a free one-week trial subscription. After registration, you will receive a login/password or key to your email, which will be required to set up the data feed in the platform.

The built-in data feed module is already available in your platform. No fee is charged for the module use.

### Configuring the Data Feed

Add a new data feed configuration via the corresponding section of MetaTrader 5 Administrator:

## MetaTrader 5 trading platform

Data Feed

Common

Parameters

Timeouts



This data feed allows receiving Newsquawk news. To receive this feed, please visit <https://newsquawk.com/>. After registering for the feed you will then receive your unique access key by email.

☒ Enable

Name:

Module:

Feed server:

Feed login:  Password:

Show add

Data Feed

Common

Parameters

Timeouts



Please specify parameters of the data feed. These parameters are specific for each type of data feed. They allow using additional settings that were not available in the "Server" tab.

Up

Down

Add

Edit

Delete

Default

| Parameter              | Value   |
|------------------------|---------|
| ab Authorization Token |         |
| ab Language            | en      |
| ab Process Updates     | Yes     |
| ab News Category       | Finance |

OK

Cancel

Help

Specify the following parameters:

- **Feed server** news provider's server address. The default address is <https://newsquawk.com>. In most cases, there is no need to change it.
- **Feed login** login for connection provided by Newsquawk upon subscription.
- **Password** password for connection provided by Newsquawk upon subscription.
- **Authorization Token** key for connection provided by Newsquawk upon subscription. You can use either this parameter or Login/Password, depending on the credentials provided.
- **Language** a two-letter language code in ISO 639 format that will be included in incoming news. For example, en, es, pt, etc. It will be used to filter news by language on the client terminal side. Fill in this parameter in accordance with the language selected during subscription, that is, the language in which you actually receive the news. To receive news in different languages, create separate data source configurations.
- **Process Updates** Newsquawk may update and supplement previously published news. The parameter determines whether the data feed will transmit such updates to the platform. On the MetaTrader 5 side, each update will create additional news; previously transmitted news will not be changed. The default is "Yes" meaning updates are transmitted.
- **News Category** the category to which newsletters from the data source will be assigned. The category name can then be used for specifying news to be delivered to certain client groups.

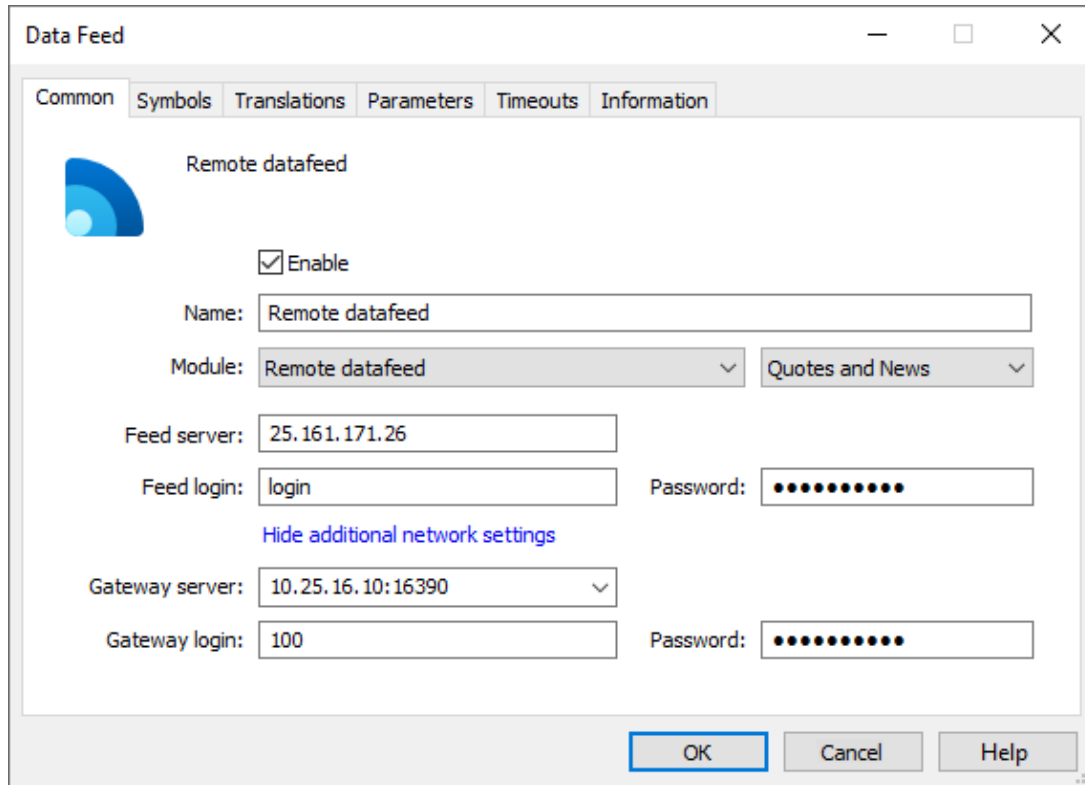
Once the data feed is enabled, client terminals will start receiving newsletters. To check the data feed operation, request its logs.



MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Remote Datafeed

## Remote Datafeed

This option in the data feed settings is intended for the connection of the server to a data feed located on a remote server. This data feed can be written using API.



The screenshot shows the 'Data Feed' dialog box with the 'Common' tab selected. The 'Remote datafeed' section is active, indicated by a blue icon. The 'Enable' checkbox is checked. The 'Name' field is set to 'Remote datafeed'. The 'Module' dropdown is set to 'Remote datafeed', and the 'Quotes and News' dropdown is set to 'Quotes and News'. The 'Feed server' field contains '25.161.171.26'. The 'Feed login' field contains 'login', and the 'Password' field is masked with dots. A blue link 'Hide additional network settings' is visible. The 'Gateway server' dropdown is set to '10.25.16.10:16390'. The 'Gateway login' field contains '100', and the 'Password' field is masked with dots. At the bottom, there are 'OK', 'Cancel', and 'Help' buttons.

In the parameters of connection to the data feed, the following needs to be specified:

- **Gateway server** the address of the server where the data feed is installed (module written using API, translating information from the provider), as well as the port number for connection, separated by a colon. For example, 192.168.0.180:443;
- **Gateway login** a login for connecting to the server;
- **Password** a password for connecting to the server;
- **Feed server** address of the data feed server;
- **Feed login** login to access the data feed server;
- **Password** password to access the data feed server.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds /  
Universal DDE Connector

Universal DDE Connector

Universal DDE Connector is a universal gateway that works through protocol DDE (Dynamic Data Exchange). It is used as an intermediary server that accepts quotes from different data sources and passes them to data feed MetaTrader 5 UniFeeder. Thus, using these two components, you can receive quotes from any data source that supports the DDE protocol.

UniDDE can process up to 1024 financial symbols and distribute quotes to the unlimited number of data feeds MetaTrader 5 UniFeeder tuned to it.

This section describes the following aspects of work with UniDDE:

- Installation and set up of data transmission to the MetaTrader 5 server;
- Set up of symbols for receiving quotes;
- Filtering of quotes.

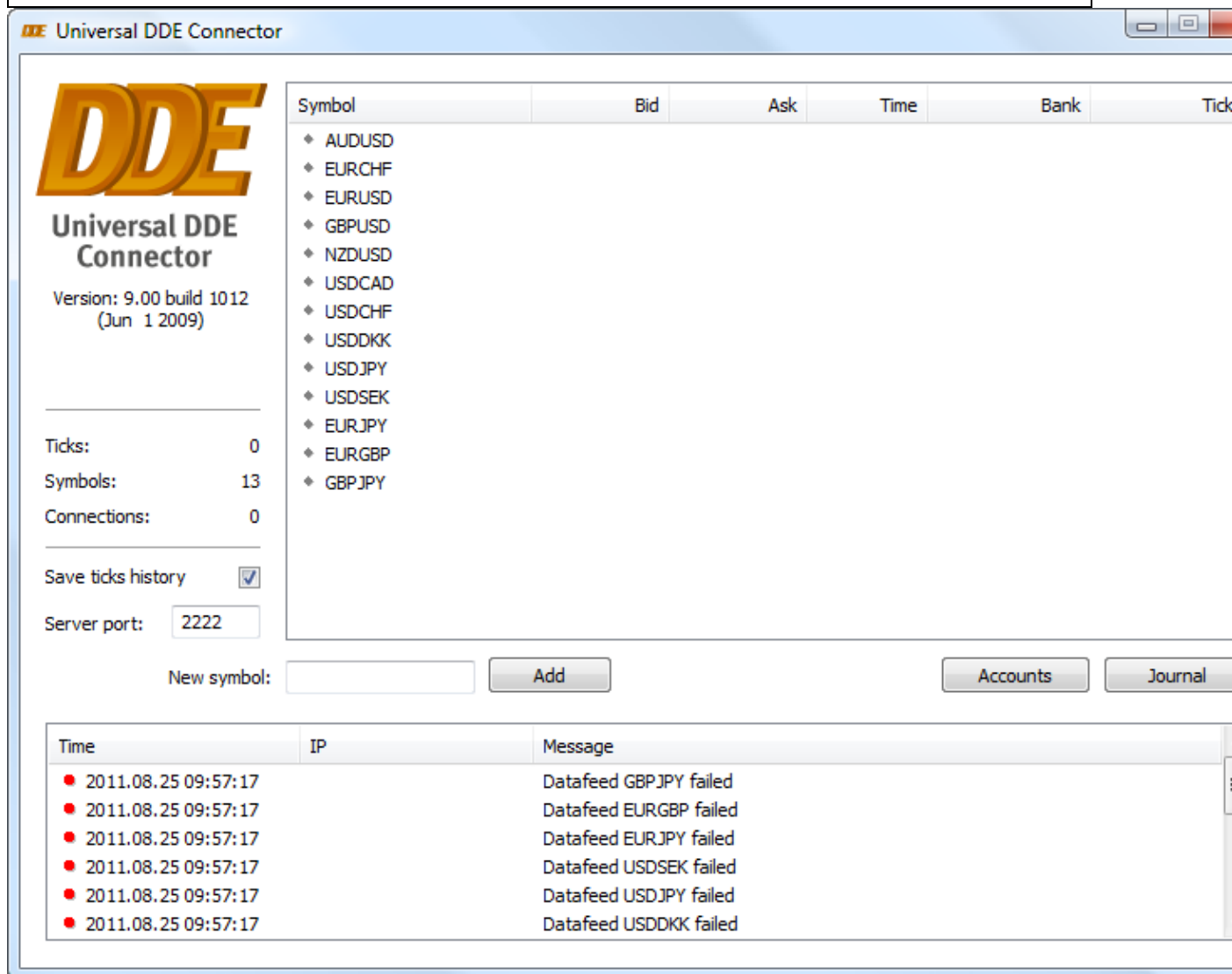
MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Universal DDE Connector / Installation and Setup

## Installing and Setting Up

In order to start installing the Universal DDE Connector, download it from the [technical support website](#) of MetaQuotes Software Corp. After that start the executable file and perform the simple installation procedure. As soon as the installation is over, start UniDDE.

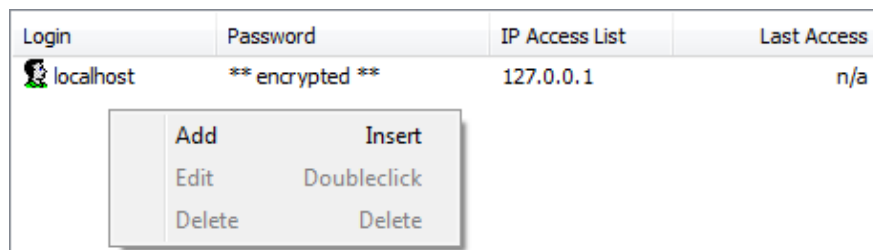
To prepare UniDDE for connecting MetaTrader5UniFeeder to it, a some configurations are required.

After you set up UniDDE, you'll need to [set up symbols](#) for translating quotes.



## Account Setup

During the program installation, one account for accessing from a local computer is created. To enable connection of MetaTrader5UniFeeder to UniDDE an account must be opened - execute the "Add" command in the context menu of the accounts block:



After you press it, a new line will appear, where you should enter the following data:

- **Login** login for access;
- **Password** password for access;
- **IP Access List** list of allowed IP addresses, separated by commas. Connection using the above specified login and password will be possible only from the specified IP addresses.

Date and time of the last access to UniDDE from a specified account is displayed in column "Last Access".

To provide higher security, besides accounts with the allowed IP addresses, UniDDE provides the permanent control of all connections with automatic blocking to prevent DoS attacks. In real-time mode, all network connections are analyzed, and if a non-standard activity is detected, the IP address is automatically closed for 5 minutes.

## Setup of Connection Port

Port specification is also required for connecting MetaTrader5UniFeeder. By default port 2222 is used, but you can change it in field "Server port".

The specified port must be open on the server where UniDDE is installed.

## Setup of Tick Storage

Universal DDE Connector allows storing tick data received from data sources. To do it, tick off field "Save Ticks History". Tick data are saved in file *universalddeconnector.ticks*, which is located in the UniDDE installation directory.

When selecting the option of saving of tick data, you should watch the file size they are stored in. If ticks are saved for a long time period and for a great number of symbols, the file size can reach tens of gigabytes.

## Journal

## MetaTrader 5 trading platform

Results of connecting to a quotes feeder or of the connection of MetaTrader5UniFeeder, as well as other important messages about the operation of UniDDE are reflected in the journal. To open it, press "Journal". All messages are displayed as a table with the following fields:

- **Time** date and time of message creation;
- **IP** IP address the message is connected with. For example, the IP address of MetaTrader5UniFeeder when it is connected to UniDDE;
- **Message** text of the message.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Universal DDE Connector / Setting Up Symbols

## Setting Up Symbols

Universal DDE Connector uses the standard DDE protocol (Dynamic Data Exchange) for accepting quotes from any data source that supports this protocol. During installation of UniDDE, a set of symbols is created, which are configured for receiving quotes from the MetaTrader 4 client terminal. In order to start translating quotes from a client terminal, start it and enable option "Allow DDE Server".

- **Adding a symbol**

In order to add a new symbol, enter its name in field "New Symbol", and then press "Add".

- **Editing a symbol**

In order to change symbol settings, click twice on it in the list or execute the "Edit" command of the context menu.

- **Deleting a symbol**

In order to delete a symbol, select it in the list and press "Delete" or execute the same command of the context menu.

After adding a symbol, restart UniDDE. Close it using the command of its context menu called by a right click on the icon in the area of notifications. Closing of the UniDDE window does not stop its running, and only minimizes it to the area of notifications.

## Symbol Setup

When adding or editing a symbol, a window of its setting appears:

## MetaTrader 5 trading platform

Symbol

Symbol: EURUSD

Forex: Euro vs US Dollar

Digits: 4

Spread: 0

☐ Force precision (12935 -> 1.2935)

DDE Link

Server: MT4

MetaTrader 4

Bid topic: Bid

Bid item: EURUSD

Ask topic: Ask

Ask item: EURUSD

☒ Use

Volume topic:

Volume item:

☐ Use

Bank topic:

Bank item:

☐ Use

List of banks:

☐ Auto

Filtration

Auto limit: 1.00

% of the current market price

☒ Automatic filter

☐ Accept bid/ask independently (use for futures only!)

☐ Use specific time session

Session from: 00 h 00 m

to: 24 h 00 m

Time zone:

OK

Cancel



Settings in this window are divided into several blocks:

### Common parameters

- **Symbol** name of the symbol in Latin letters up to 15 symbols long. A symbol of the same name must be created on the MetaTrader 5 server. The symbol name should not necessarily be the same as the symbol name on the data source;
- **Digits** the number of decimal places. The parameter value must be the same as the analogous symbol parameter on the MetaTrader 5 server;
- **Spread** automatically set spread. Setup of this option is possible only if one price Bid or Ask is received;
- **Force Precision** force transformation of prices like 12345 to 1.2345 if data are received without a separator.

A symbol with exactly the same name must be created on the MetaTrader 5 server.

### DDE Link

In this block parameters of quote receiving from a supplier are configured.

- **Server** name of the source server in the DDE protocol. The most popular data sources can be selected from the list here. Other symbol settings corresponding to it will be set up automatically.
- **Bid topic, Bid item** settings for receiving Bid prices;
- **Ask topic, Ask item, Use** settings for receiving Ask prices. In the "Use" field you can enable/disable direct collection of these data. In this case the Ask price will be calculated automatically based on the value specified in field "Spread";
- **Volume topic, Volume item, Use** parameters for receiving the Ask price. In the "Use" field you can enable/disable direct collection of these data;
- **Bank topic, Bank item, Use** parameters for receiving information about the bank that issued the prices. In the "Use" field you can enable/disable direct collection of these data;
- **List of banks** the list of banks, from which quotes are allowed (separated by commas). Tick off "Auto" to enable the automatic bank selection mode. The auto-selection of banks is described in the section about filtration.

- Data receiving parameters are set up automatically if you select a server of one of suppliers in the "Server" field.
- Contact the supplier for parameters of data receipt.
- Often, for futures instruments the Last price (price of the last executed deal) is translated. In this case symbols in UniDDE are configured manually so that the Last price is translated as Bid, and Ask is calculated based on this price and spread settings.

### Filtration

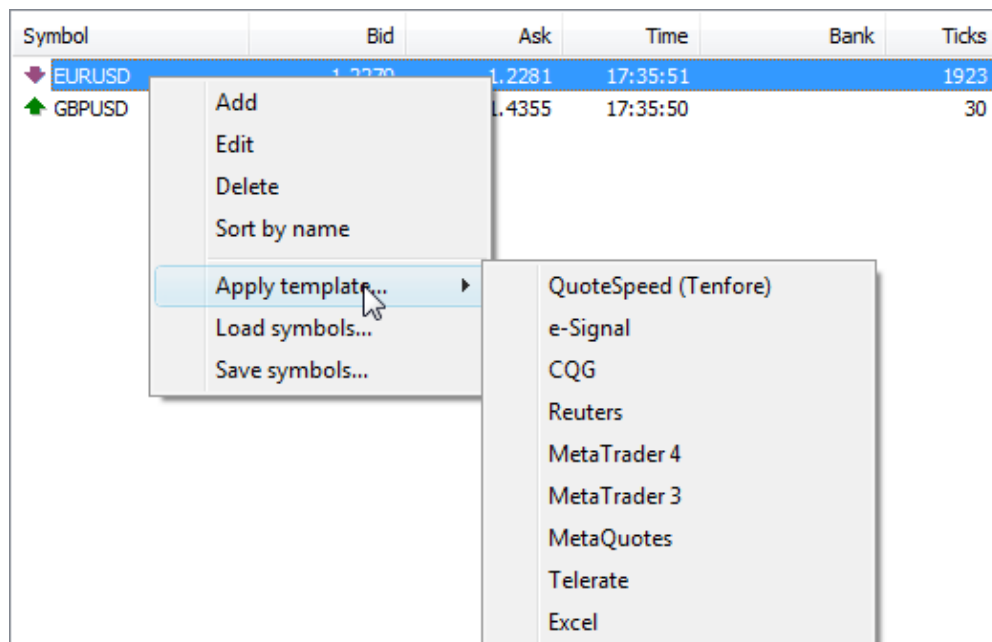
Filtration parameters for received quotes is set up here.

- **Auto limit** value of the maximally allowed deviation of a new price from the previous one in percents. If the difference between the new price and the previous one exceeds the specified limit, the new price is filtered away (is not passed to the server). This allows to prevent the accidental price substitution for different symbols (for example : USDJPY instead of USDCHF);
- **Automatic filter** enable/disable the automatically adjustable quote filtration. This mode is described in the section devoted to filtration.
- **Accept bid/ask independently** by default, quotes are sent to clients (MetaTrader5UniFeeder) only if Bid has changed. The change of Ask is preserved in such a case, but quotes are not sent to client connection until Bid is updated. This tick allows to disable this behavior and translate new prices when Ask is updated.
- **Use specific time session** use separate time periods. If this option is enabled, fields "Session from" and "To" become active. There you can set time limits for translating symbol quotes.

Option "Accept bid/ask independently" cannot be enabled for Forex symbols. Quotes where only Ask has changed are considered incorrect and are filtered out by the server. This can create additional load.

### Applying Profiles

A number of ready parameters for receiving quotes from different data feeders are available in UniDDE. To quickly switch between providers, you can use the special command "Apply template..." in the context menu of the symbols list.



Select here a necessary quote provider. To save the current set of symbols and their parameters, press "Save symbols...".

- When you change the provider, parameters of all symbols in the list are changed. So be careful using this procedure. In some systems, parameters of symbols may differ, and some of them can stop working when you change the provider.
- Each time you change provider, a backup copy of the previous state (file with SYM extension) is created in the "backup" folder located in the UniDDE installation directory. To restore the list of symbols, only use command "Load symbols..." and specify a required file.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Universal DDE Connector / Filtration of Quotes

## Filtration of Quotes

Universal DDE Connector has the built-in system of automatic filtration of received quotes before they are passed to the history server. The filtration system consists of several components:

### Selecting the Best Banks

One of the most effective ways of cleaning quotes is the automatic selection of the best banks based on statistic data on them. If you tick off "Auto" for the list of banks, UniDDE will start collecting statistics of symbol quotes from banks. Based on this statistics, once a day (except for holidays) several best banks are selected and automatically added to the "List of banks". Next day filtering of quotes will be performed according to this list. If the selected list of banks is satisfying, you can disable Auto.

The automatic selection of the list of banks according to the collected results is performed once a day. I.e. the list will be empty the first day.

### Cleaning from "White" Noise

For a milder cleaning of the quotes thread from the "white" noise, another new adaptive filtration mechanism is used - "Automatic filter". It analyzes the average deviation of received prices and sifts away questionable quotes that fit several template situations.

Very often the thread filtration requires analysis of the next quote for making a decision about the previous questionable price, which may slow down data delivery. But the automatic filter in UniDDE does not allow questionable prices stay longer than half a second. This condition can let some questionable quotes in, but there is one more filtration level described below.

### Rough Protection

For a rough protection from erroneous price substitution from other symbols the "Auto Limit" mode is used, which doesn't let in new prices if they differ from previous ones by more than the specified number of percents.

MetaTrader 5 Trading Platform / Platform Components / Data Feeds / Universal DDE Connector / UniFeeder Protocol

## UniFeeder Protocol

The general features of the UniFeeder protocol are provided below to help you understand its operation principles.

- The data feed establishes TCP connections with a data source.
- The data feed reads the data line by line. The "\r\n" sequence serves as a line separator.
- Authorization:
  - ◆ After connection is established, the data feed waits for a login request from the data source. The line containing "Login: " is expected, for example, "Login : \n\r". The "waiting login request" entry appears in the Journal while waiting for the request.
  - ◆ After the request is received, the "login request received" entry appears in the Journal, and the data feed sends the line with the login to the data source, for example, "user\_test\r\n".
  - ◆ The data feed waits for the password request from the data source. The line containing "Password: " is expected. The "login sent [%S], waiting password request" entry appears in the Journal while waiting for the request.
  - ◆ After the request is received, the "password request received" entry appears in the Journal, and the data feed sends the line with the password to the data source, for example, "password\r\n".
  - ◆ The data feed waits for the notification of successful authorization. The line containing "Access granted" is expected. The "password sent, waiting access confirmation" entry appears in the Journal while waiting for the request.
  - ◆ After receiving the notification, the "Login: '%S' successful" entry appears in the Journal.
  - ◆ The data feed sends the line containing the symbols, for which it is allowed to send quotes. The line looks as follows "> Symbols:USDRUB,EURUSD,EURRUB\r\n".
  - ◆ The data feed switches to the data receipt mode (quotes and news).
- Data receipt:
  - ◆ The data feed reads the data line by line. The "\r\n" serves as a line separator.
  - ◆ The lines beginning with ">" or "<" reserved characters are skipped, except for the lines beginning with "< News".
  - ◆ If the "< News\r\n" line is received, the data feed switches to news receipt mode:
    - ◇ After the "< News\r\n" line, the data feed reads the NewsTopic structure (read MetaTrader 4 API documentation for more details).
    - ◇ Next, the data feed reads the news body. The size of the news body is set in bytes in the NewsTopic::len field.
    - ◇ After reading the body, the news is sent to the platform, while the data feed continues the data receipt.
  - ◆ If the line does not begin with "<" or ">", it is deemed to be a tick line.

## MetaTrader 5 trading platform

- ◇ The tick line format: "<Symbol name> <bid> <ask>\r\n".  
Example: "USDJPY 1.0106 1.0099\r\n".
- ◇ If bid or ask is less than zero, or bid exceeds ask, the tick is skipped and the "failed to parse tick, invalid bid/ask" entry appears in the Journal.

Below is an example of the network exchange dump. The data sent by the data feed is shown in red:

```
Welcome
Login: test_user
Password: test_password
Access granted
USDJPY 0 4.0000 19.0000
GBPUSD 0 8.0000 19.0000
USDCHF 0 15.0000 16.0000
USDCHF 0 6.0000 25.0000
USDJPY 0 6.0000 17.0000
GBPUSD 0 6.0000 21.0000
USDCHF 0 7.0000 24.0000
USDJPY 0 9.0000 14.0000
GBPUSD 0 9.0000 18.0000
USDCHF 0 9.0000 22.0000
USDJPY 0 5.0000 18.0000
GBPUSD 0 4.0000 23.0000
USDCHF 0 7.0000 24.0000
USDJPY 0 7.0000 16.0000
GBPUSD 0 13.0000 14.0000
```

### MetaTrader 5 Trading Platform / Platform Components / Gateways

#### Gateways

Gateways are intended for integration of the MetaTrader 5 platform with external trade systems. Gateways allow bringing out trade operation to external systems as well as translating quotes and news from them. Gateways are executable files (\*.exe) that run as separate processes.

The trading platform includes many gateways.

#### Liquidity Gateways

- MetaTrader 5 Gateway to Currenex
- MetaTrader 5 Gateway to Integral
- MetaTrader 5 Gateway to Euronext FX
- MetaTrader 5 Gateway to Choe FX
- MetaTrader 5 Gateway to LMAX Global
- MetaTrader 5 Gateway to FXCM PRO
- MetaTrader 5 Gateway to MetaTrader 5
- MetaTrader 5 Gateway to MetaTrader 4

#### Exchange Gateways

- MetaTrader 5 Gateway to MOEXFX (Moscow Exchange, FX market)
- MetaTrader 5 Gateway to MOEX Securities (Moscow Exchange, Securities market)
- MetaTrader 5 Gateway to MOEX Derivatives (Moscow Exchange, Derivatives market)
- MetaTrader 5 Gateway to DGCX (Dubai Gold & Commodities Exchange, Derivatives market)
- MetaTrader 5 Gateway to Ukrainian Exchange (Stocks)
- MetaTrader 5 Gateway to ASX (Australian Securities Exchange, Futures and Stocks)
- MetaTrader 5 Gateway to Borsa Istanbul Futures (Derivatives market VIOP)
- MetaTrader 5 Gateway to Interactive Brokers

MetaTrader 5 Trading Platform / Platform Components / Gateways / MOEXFX

MetaTrader 5 Gateway to MOEXFX

MetaTrader 5 Gateway to MOEXFX enables one-click currency trading with the exchange "Depth Of Market" option and financial market analysis using built-in technical indicators. With the wide functionality of MetaTrader 5, traders can automate the trading process using robots, which they can develop themselves, order from professionals or buy in the trading application store MetaTrader Market. Financial news and important releases appear right in the terminal to enable traders keep the track of market events. The extremely popular market service of Copy Trading is also available to traders.

Moscow Exchange's FX Market is the oldest regulated domestic FX trading venue operating since 1992. The Exchange market is the center of liquidity for ruble related operations. The Bank of Russia sets the official ruble exchange rate on the basis of exchange trading results.

Moscow Exchange's FX Market is one of the most dynamically developing segments of Russia's financial market. The total trading volume on Moscow Exchange's FX Market increased 34% YoY and amounted to 156 trln rubles in 2013. Now Moscow Exchange is experiencing increased activity: in the third quarter of 2014 trading volumes increased in almost all markets, but the FX section showed the best result. The total trading volume of the FX Market has increased by 26.4% compared to the third quarter of 2013 and amounted to 55.86 trillion rubles.

USD, Euro, Chinese Yuan, Ukrainian hryvnia, Kazakh tenge, the Belarusian ruble, bi-currency basket, and FX swap transactions can be made via the market's electronic trading system.

Order MetaTrader 5 Gateway to MOEXFX

The foreign exchange market is about 30% of the total volume traded on Moscow Exchange and is one of the most dynamically developing segments of the Russian financial market. MetaTrader 5 Gateway to MOEXFX allows quick integration of the MetaTrader 5 platform with Moscow Exchange's FX Market. Order the gateway now to provide your clients with the advanced fast trading terminal offering wide opportunities including use of automated trading robots and mirror trading!

Common Trading Principles and their Implementation in MetaTrader 5

Trading in the FX Market is regulated by "FX and Precious Metals Market Trading Rules of the Moscow Exchange". Here are some of the key trading features and examples.

In the currency market, the result of a transaction is immediately shown on the client trading status. By opening a long position, a client receives assets, while obligations



arise from opening a short position.

At this point MetaTrader 5 provides two types of financial instruments of Moscow Exchange's FX Market:

- Settlement of obligations is due on the day of the transaction. Suffix TOD is used for such instruments. For example, USDRUB\_TOD. Each TOD symbol is traded during a certain session defined by the Exchange trading hours, obligations are settled at the end of the sessions. For example, USDRUB\_TOD trading hours are 10:00 to 17:15 (Moscow time).
- Settlement of obligations is due on the day following the day of the transaction. Suffix TOM is used for such instruments. For example, USDRUB\_TOM. TOM instruments are traded 10:00 to 23:50 (Moscow time). At the end of the trading session, all positions of TOM instruments turn into positions of TOD instruments. If there are TOD positions on the account in addition to the TOD positions, they are consolidated (position netting).

Consider opening a short and a long position of 1 lot for USDRUB\_TOD at the price of 52 RUR, the contract size is 1,000.

### Position Opening

- Short when a short position is opened, the client's obligation is to sell 1,000 of his US dollars for rubles at the price of the transaction.
- Long when a long position is opened, the client's obligation is to buy 1,000 US dollars for rubles at the price of the transaction.

### Margin Trading

Trade is marginal and contractual obligations are delayed in time. Until the time of settlement of obligations, the margin is locked on the client's account. The margin is defined by discounts relative to the current price of the instrument. Discounts are set by the broker, however they cannot be lower than the values set by National Clearing Centre (NCC). Due to the discounts, clients can make transactions with a leverage, i.e. traders can open positions investing a larger sum than is available on their account.

### Fulfillment of Obligations and Swap Operations

At the time of settlement, the broker takes the client's obligation in the fullest possible volume. If money on the client's account is not enough to settle a liability, then a swap transaction is conducted at the amount of the remaining position. Consider the examples of short and long positions.

- Long position the client has purchased 1,000 USD at 52 RUR. Thus, during settlement the client needs to take US dollars and pay 52,000 RUR. However, at the time of payment the client has only 30,000 RUR, so part of the position is uncovered. This part is swapped. The part of the position for which obligation was settled becomes a settled position - client's asset, its value at the current rate with liquidity ratio correction can be used as collateral for opening new positions.

## MetaTrader 5 trading platform

- Short position the client is obligated to sell his own 1,000 USD at 52 RUR. If the required currency amount is not available on the trader's account, the broker is committed to supply the required amount. This is a swap operation.

A swap consists of two transactions: the current position is closed and is re-opened for the appropriate TOM instrument. The difference between the position close price and the re-open price is called the "swap" price, it is determined by the broker's interest rate.

Thus, a trader can have two types of positions:

- Unsettled position the positions, for which the trader expects the settlement of obligations.
- Settled positions client's assets.

For customer convenience and ease of calculation, these positions are not differentiated in MetaTrader 5. The settled positions are displayed in the form of unsettled positions of corresponding TOD instruments.

### How the Gateway Works

MetaTrader 5 Gateway to MOEXFX is a separate "MOEXFXGateway64.exe" file that uses Gateway API for operation. Gateway module is originally included in the platform delivery set. After purchasing the gateway, the trading platform license will automatically update via LiveUpdate system. This allows you to start using MetaTrader 5 Gateway to MOEXFX without any limitations.

The gateway receives market data of trading on Moscow Exchange via Multicast. This is a UDP protocol that allows packets to be transmitted to multiple recipients. Find the details of the protocol in "[Market Data Multicast Ver3.3 User Guide](#)". The system administrator must configure the server where the gateway is running to receive data.

The UDP protocol does not have built-in methods to ensure reliability, ordering, or data integrity, which on the one hand makes it unreliable. But on the other hand, it saves time required to deliver data to clients. This provides data delivery to clients with minimum delay. All transmitted data are doubled to control the integrity of transmitted data. The gateway receives market data from the Exchange in four tables:

- Trades executed on the Exchange.
- Trade statistics.
- Depth of Market.
- List of instruments.

To deliver each data type, four connections are created: two connections for the incremental channel where only changes relative to the previous state are transmitted, and two for the snapshots (the state at a certain time).

For committing trade operations and receiving the trading status of clients, the gateway uses three FIX services, for each of which a separate connection is created:

- **MFIX Trade** entering and canceling orders and receiving order execution reports.
- **MFIX Drop Copy** receiving the stream of orders and deals from other terminals (other than MetaTrader 5).
- **MFIX Trade Capture** sending all trades of all accounts bound to the current FIX identifier of the broker.

Accordingly, three FIX IDs are required. Read more about the FIX services in "[Moscow Exchange public FIX 4.4 interface specification for Securities and FX markets](#)". If any of the FIX services is not required, it can be omitted. Simply do not specify its address in the gateway settings.

### Integration with Broker's Back Office

The relevant trading client state in MetaTrader 5 is maintained using the client limiting interface provided by the gateway. Limiting is implemented through import of limit files in the QUIK format. The detailed format description is available in the [QUIK WorkStation](#) reference. This format is used due to the fact that most of the brokers that provide access to Moscow Exchange have this platform, and therefore such an approach is the most simple and convenient one to move to MetaTrader 5.

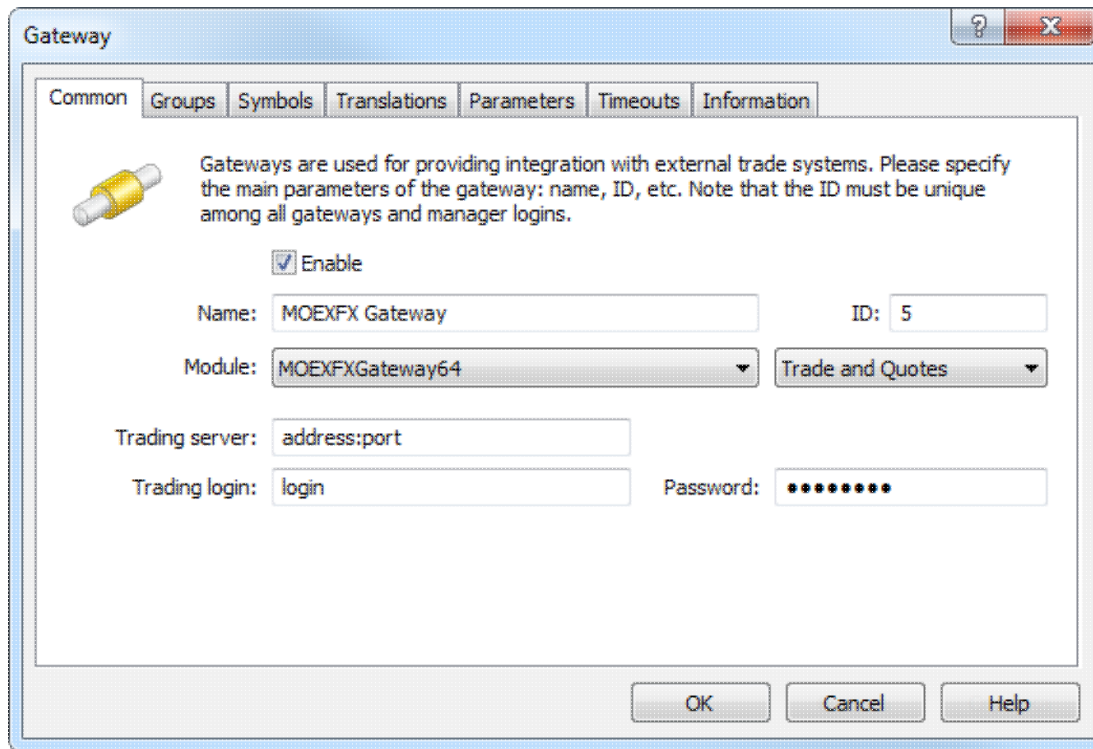
In [the gateway settings](#) specify the address (ftp or local address), from which the gateway shall automatically read the limit and correction files and synchronize client states in MetaTrader 5. The limit file can be used at any moment of gateway operation. Once the administrator adds a file to the specified directory, the gateway will process it and will set the client status in accordance with it. This behavior can also be used to recover client states in case of failure.

Limit correction files are used to reflect the limit changes in the broker's back-office system. The gateway processes them only during trading hours (10:00 to 23:50 Moscow time). Correction files are processed in accordance with the following algorithm:

- When the correction file is updated, the gateway starts analyzing it.
- It parses all the correction entries from the file one by one.
- The gateway finds the entry with LIMIT\_ID (the correction identifier is unique within a file for one trading day) equal to LIMIT\_ID\_LAST (the identifier of the last processed correction).
- All subsequent entries are processed and applied to client trading status. While each subsequent entry is processed, the value of LIMIT\_ID\_LAST is updated to avoid repeated correction.
- As soon as processing is over, the gateway uploads the file with the correction results. The file is of the QUIK system format. The file is uploaded at the address specified in the [gateway settings](#).
- At the beginning of each trading day, the ID of the last processed correction is reset.

### Configuring the Gateway

To start working, add [the new gateway configuration](#):

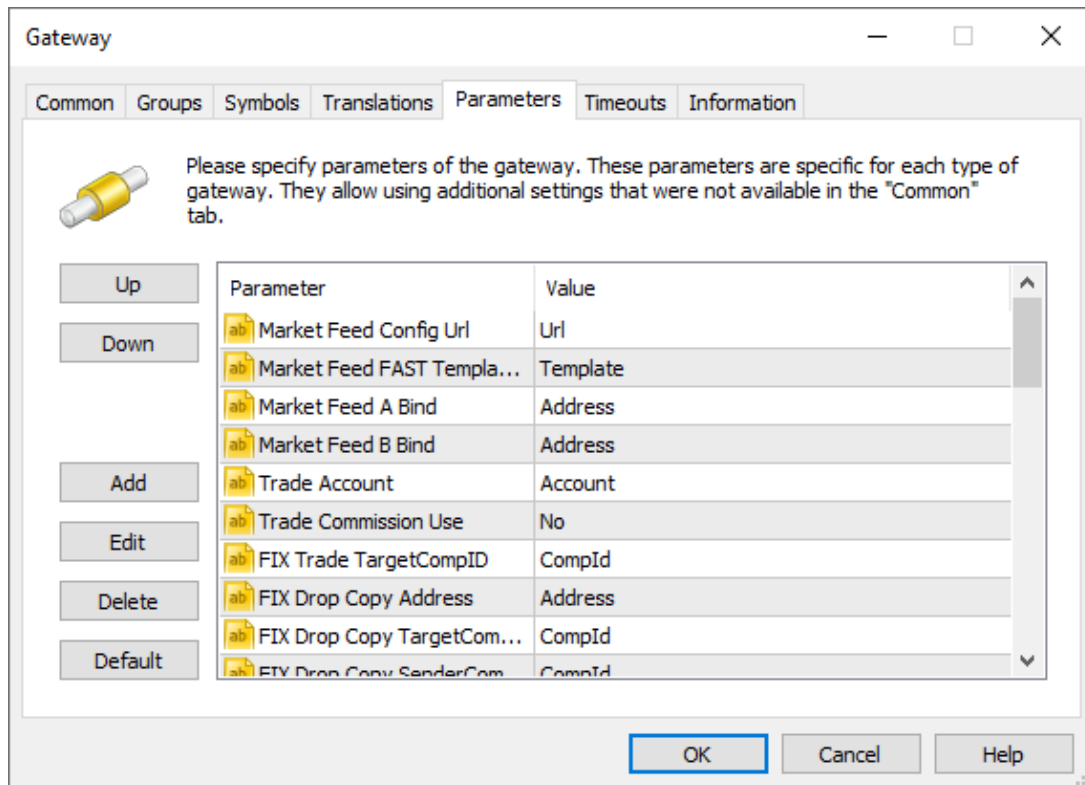


The following parameters on the "Common" tab must be configured:

- **Module** MOEXFXGateway64. After selection, use of the gateway default parameters must be allowed in the dialog request.
- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Trading server** address for connection to the MFIX Trade service.
- **Trading login** login (FIXTradeSenderCompID) for connection to MFIX Trade.
- **Password** password for connection to MFIX Trade.

- ID value must be unique in the field of manager logins and gateway identifiers.
- Connection data (address, login and password) are provided by the exchange.

Additional configuration parameters must be set on the Parameters tab:



The following additional parameters are available for the gateway:

- **Trading Calendar Holidays** each exchange works in accordance with its own trading calendar. By default, non-working days for the gateway are Saturday and Sunday. Using this option, you can override the working/non-working days for the gateway. To add a non-working day, specify a value of the form +DDMMM. The date is specified as two digits, the month is specified as the first three letters of the month name in English. For example, +01JAN. To add a working day on Saturday or Sunday, specify a value of type -DDMMM, for example, -07FEB. You can specify multiple working/non-working days separated by semicolons, for example, + 01JAN;-07FEB.
- **Market Feed ConfigUrl** parameters of channels for receiving market data from the exchange are described in the XML file. This file is available on the Exchange's FTP server. Specify the path to the file depending on what data you want to receive (real or test). Configuration file can be copied to a local server, the appropriate path should be specified in this field. The gateway reads data for connection to the following channels (set by the connection\_id parameter in the file):
  - ♦ **OBR and OBS** an incremental channel and a channel of snapshots for transmitting the depth of market.
  - ♦ **TLR and TLS** an incremental channel and a channel of snapshots for transmitting trading operations.
  - ♦ **MSR and MSS** an incremental channel and a channel of snapshots for transmitting trade statistics.
  - ♦ **IDF** a channel for transmitting the list of financial instruments.
- **Market Feed FAST Templates Url** all market data are transmitted through

the FAST protocol. Transmitted data are unpacked using the special FAST template. Specify here the path to the template file, it is available on the Exchange's FTP server. You can specify the FTP path or copy the file to a local server and specify the appropriate local path to it.

Market channel parameters and template files downloaded via FTP are saved in the directory [gateway folder]\downloads. If the gateway fails to download updated files, it will use earlier downloaded files stored in this folder. The following rules apply here:

- The gateway performs a checks that the file is not older than one week. If it was downloaded earlier, it is considered obsolete and will not be used.
- If path to a local file instead of FTP is specified in parameters Market Feed ConfigUrl and Market Feed FAST Templates Url, file opening will fail and the existing file from the 'downloads' directory will not be used.

- **Market Feed A Bind** the address of the local interface the connection will be bound to in order to receive market data through channel A (all data from the Exchange are duplicated on two channels).
- **Market Feed B Bind** the address of the local interface the connection will be bound to in order to receive market data through channel B (all data from the Exchange are duplicated on two channels).
- **Trade Account** the account used for registering the transaction performed through the gateway on the Exchange.
- **Trade Commission Use** this parameter sets the use of commissions from the Exchange. The commissions are specified in each transaction received from the Exchange. If the parameter is set to "Yes", this value is used in the platform. If set to "No" (default), the broker can configure commission calculation directly in MetaTrader 5 based on the commission calculation formula used on the Exchange.
- **FIX Trade TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Trade service. The remaining connection parameters are specified on the "Common" tab in the gateway settings.
- **FIX Drop Copy Address** address for connection to the MFIX Drop Copy service. The gateway can work without connection to the service. In that case, leave this parameter empty.
- **FIX Drop Copy TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Drop Copy service.
- **FIX Drop Copy SenderCompID** FIX messages heading standard parameter used for a message sender identification for the MFIX Drop Copy service.
- **FIX Drop Copy Password** password for connection to MFIX Drop Copy.
- **FIX Trade Capture Address** address for connection to MFIX Trade Capture.
- **FIX Trade Capture TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Trade Capture service.
- **FIX Trade Capture SenderCompID** FIX messages heading standard parameter used for a message sender identification for the MFIX Trade Capture service.

- **FIX Trade Capture Password** password for connection to the MFIX Trade Capture service.
- **Limits Url** local or FTP address of limit filed to be imported to MetaTrader 5.
- **Limits Corrections Url** local or FTP address of limit correction files to be imported to MetaTrader 5.
- **Limits Corrections Result Url** local or FTP address where the files of limit correction import results will be exported.

In order to download and import the limits and corrections, the gateway requires all three parameters filled: Limits Url, Limits Corrections Url and Limits Corrections Result Url. If any of them is not filled or absent, the gateway will not download and import the files.

- **Limits FTP Login** if limit files are received from the FTP server, specify the login and password for connection to it.
- **Limits FTP Password** if limit files are received from the FTP server, specify the login and password for connection to it.
- **Symbols Path** path for importing the exchange symbols. By default, if this parameter is absent, the gateway imports the exchange's trading symbols to \MOEX\FX\ subdirectory with trading ability disabled. A system administrator should manually allow trading for them. If this parameter is present, the gateway imports trading symbols following the specified path.
- **Limits Kind** this parameter is used for specifying a type of limits imported from the limits file (Limits Kind = LIMIT\_KIND from the file). For the foreign exchange market, the default value is "0" (corresponds to the T+0 limits).
- **Trades Mode** this parameter specifies the markets (for example, auction, repo deals, target deals) the gateway should work for. The mode codes are used to define the markets. The values can be comma-separated or entered via the "\*" and "!" masks.
- **Execution Reports Mode** this parameter specifies the markets, for which Execution Reports are processed. This parameter is reserved for future use. You do not need to specify it.
- **Limits Sync Exclude** allows disabling synchronization of positions and balances for the specified client groups. The platform allows transmitting trading operations from one account through different gateways. To do this, multiple accounts in external systems corresponding to different gateways are registered in the account. There are three gateways for the Moscow Exchange. Each of them synchronizes the trading state and limits. The possibility to disable synchronization allows to prevent overwriting of data about positions and limits during simultaneous operation through these gateways. Specify the list of groups for which the gateway should not synchronize positions and balances in this parameter. Groups can be specified using the "\*" mask.
- **FIX Heartbeat Interval** Heartbeat messages are used in the FIX protocol for monitoring the status of connection between the Exchange and the gateway. If the gateway does not receive any data from the Exchange or Heartbeat messages during the FIX Heartbeat Interval, the connection is considered to be lost and the gateway tries to restore it. The default value of FIX Heartbeat Interval is 30 seconds. The valid range of values is from 10 to 60 (the number of seconds).

- **External Account Required** if the external system account is not specified for the client on the MetaTrader 5 side, the client's requests are sent to the exchange on behalf of the broker, not on behalf of the client. To disable sending of requests from such clients to the exchange, set "Yes" for the External Account Required parameter. The gateway will immediately reject such requests. "No" is used by default, which means requests from clients without the external account are sent to the exchange.
- **Margin Calculation Mode** calculation type used for the financial instruments imported by the gateway into the platform. If "Common", symbols are imported with the "Exchange Stocks" type. If the value is "MOEX", symbols are imported with the "Exchange MOEX Stocks" type (margin calculation is performed according to the rules adopted at the exchange on July 1st, 2019).
- **Time To Start** by default, the gateway starts connecting to the exchange's server 70 minutes before the start of the main trading session (05:50 GMT+3). Some brokers may need to change this parameter. To do this, use Time To Start parameter to specify in how many minutes before the session the gateway should start connection. For example, if connection is to start at 6:00, the parameter's value should be 60. The maximum possible value for the parameter is 70.
- **Money Currency Codes** currencies of money limits for which import from the limits file is allowed. The default is "SUR,USD": money limits for all currencies except the Russian Ruble and the US Dollar are skipped. The value corresponds to the "CURR\_CODE" tag from the limits file.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

### How to Change Password for FIX Services

To change the password for connecting to the FIX services, enter the new password in the corresponding gateway settings. Make sure that the gateway is authorized - connected using the current valid password.

### Backup Servers for FIX Services



The gateway uses three FIX connections: MFIXTrade, MFIXDropCopy and MFIXTradeCapture. The related services operate on separate servers at the exchange. A parallel backup server is also provided for each of the services.

The gateway features the ability to automatically switch to a backup FIX server in case it is impossible to connect to the main one. Specify the addresses of the main and backup servers separated by commas in the gateway settings, for example:

- Trading server - 91.203.254.29:9212,91.203.253.29:9212
- FIXDropCopy - 91.203.255.21:9215,91.203.255.22:9215
- FIXTraderCapture - 91.203.255.21:9214,91.203.255.22:9214

The gateway will automatically switch to the specified backup servers when necessary.

### **Connecting to backup servers for FIX services with different TargetCompIDs**

The gateway allows you to connect to backup servers with TargetCompID different from the main one.

For MFIX Trade service, you can create up to 10 parameter pairs looking as follows:

- FIX Trade Address 1, FIX Trade Address 2, etc.
- FIX Trade TargetCompID 1, FIX Trade TargetCompID 2, etc.

For MFIX Drop Copy service, you can create up to 10 parameter pairs looking as follows:

- FIX Drop Copy Address 1, FIX Drop Copy Address 2, etc.
- FIX Drop Copy TargetCompID 1, FIX Drop Copy TargetCompID 2, etc.

For MFIX Trade Capture service, you can create up to 10 parameter pairs looking as follows:

- FIX Trade Capture Address 1, FIX Trade Capture Address 2, etc.
- FIX Trade Capture TargetCompID 1, FIXTradeCaptureCompID 2, etc.

Example:

Trade

Server="91.203.254.23:9120,91.203.255.24:9120,91.203.255.23:9120,91.203.254.24:9120"

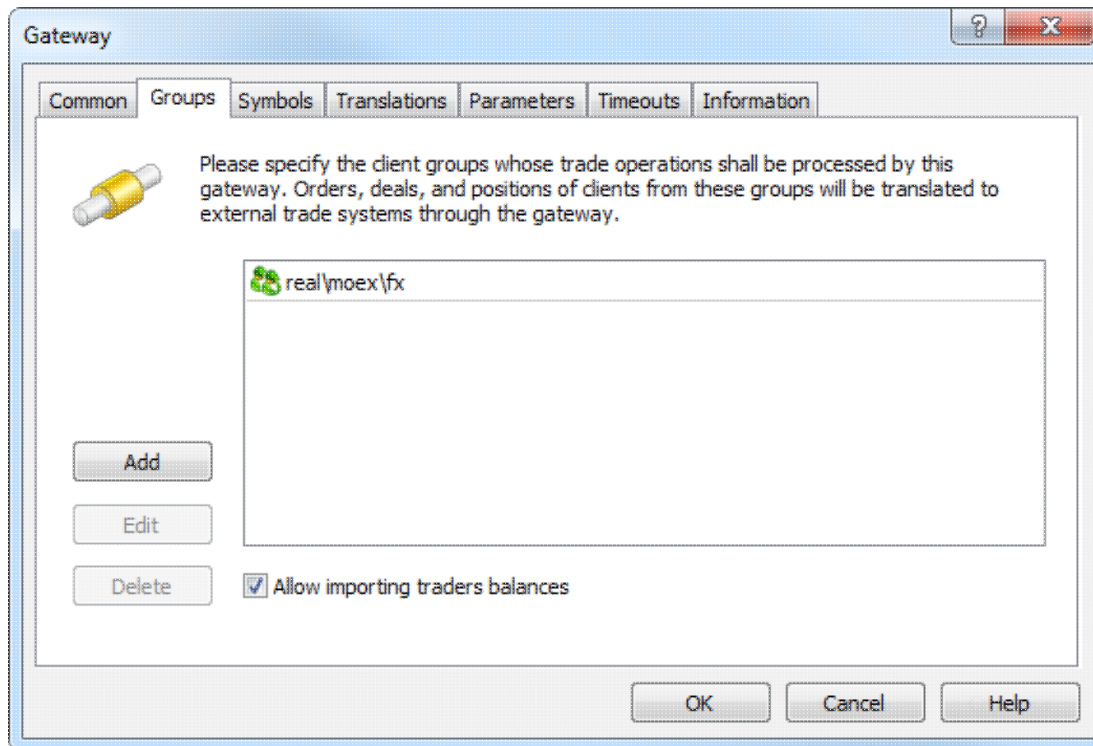
FIX Trade TargetCompID="J2FIXAEQ"

FIX Trade Address 1="91.203.255.30:9120,91.203.252.30:9120"

FIX Trade TargetCompID 1="G2FIXBEQ"

After a successful connection to the backup address, the gateway remembers it and connects to it later on. To return to the main address, temporarily delete the backup one from the gateway settings. After a successful connection to the main address, specify the backup one in the settings again.

**On the Groups tab**, specify the group of clients who will trade in the FX Market. The gateway will only receive trade requests from the clients included into the allowed groups list.

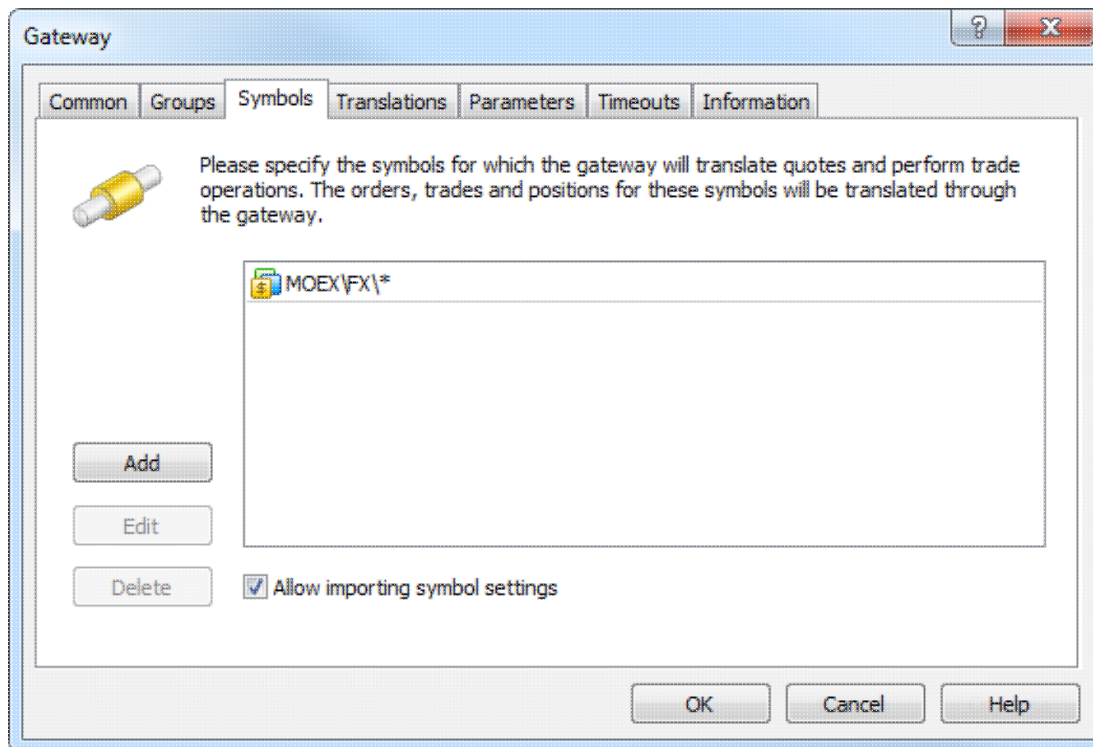


## MetaTrader 5 trading platform

To import the limits via the gateway, enable the "Allow importing traders balances" option. This will allow the gateway to synchronize clients' balances in MetaTrader 5 with exchange account limits.

Make sure to configure the group of clients trading in the FX Market correctly.

**On the Symbols tab**, specify the symbols available for the gateway. The gateway will receive quotes and perform trade operations for these symbols. Also, the settings will be automatically updated for them in case import of symbols is allowed:



## MetaTrader 5 trading platform

Specify here the group of symbols the gateway will operate with. Also, "Allow importing symbol settings" should also be enabled. MetaTrader 5 Gateway to MOEXFX adds necessary symbols and manages their settings on its own.

The symbols imported by the gateway are put to the "\MOEX\FX" symbols subgroup.

The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

### Configuration of Groups

Create a separate group for the clients working in the FX market. Specify RUR as the deposit currency in the group settings:

Group: real\moex\fx

Common Company News & Mail Permissions Margin Symbols Commissions Reports

 Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.

Name: real\moex\fx Currency: RUR

Trade server: Trade Main, 1

Authentication: Normal Minimum password length: 7

One-time password: Disabled ☐ Force one-time password usage

☒ Enable connections

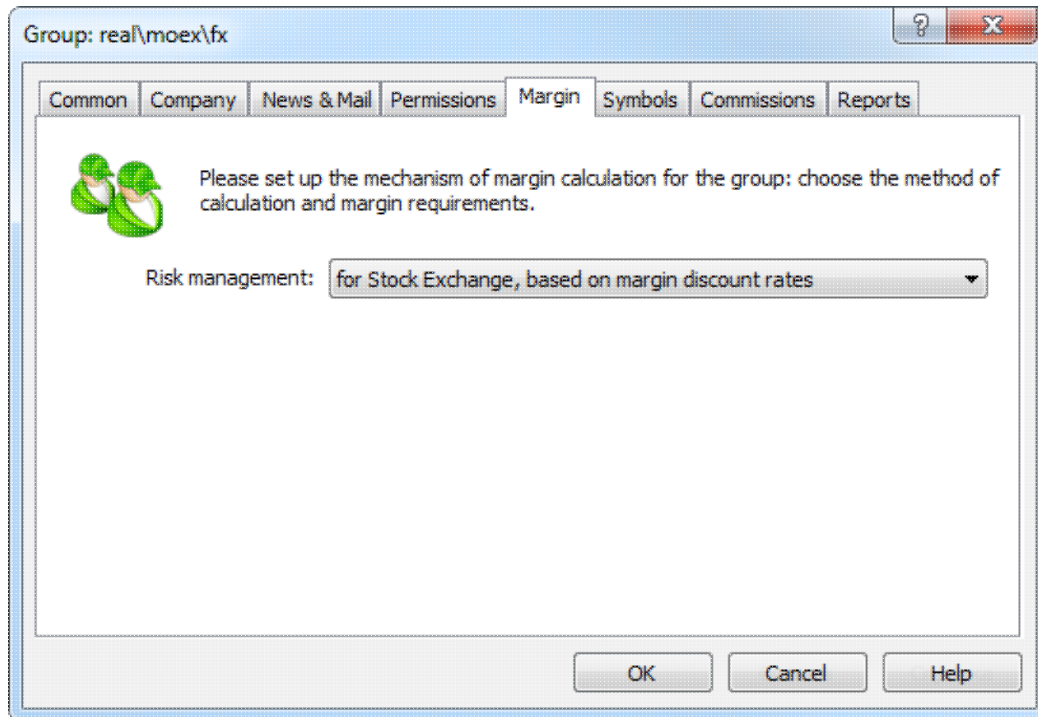
☐ Enable certificate confirmation

☐ Change password at first login

OK Cancel Help

## MetaTrader 5 trading platform

In the "Margin" tab choose the risk management mode "for Stock Exchange, based on margin discount rates". In this mode, pre-trade control is based on discounts specified in symbol settings. Discounts are set by the broker, however they cannot be lower than the values determined by National Clearing Centre (NCC).



To separate clients based on risk policies, create different groups for them. In these groups you can override symbol discount settings.

### Configuring Trade Requests Routing

To forward client requests concerning FX Market symbols to the gateway, the routing rules should be set. Examples can be found in the articles concerning the gateways to other trading systems, for example: GBOT, DGCX, etc.

### Configuring Trading Accounts

Each client on MetaTrader side should be provided with a corresponding separate trading account on the exchange. After an account is opened at the exchange, it should be connected with the account within the platform. Open the appropriate client account and move to Account tab:

## MetaTrader 5 trading platform

Account: 2013, John Smith, USD

Overview Personal **Account** Security

Group: real\moex\fx

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

Trade accounts:

| Gateway ID                                                                                          | Account  |
|-----------------------------------------------------------------------------------------------------|----------|
|  5, MOEXFX Gateway | 12345678 |

Synchronize

Add

Edit

Delete

Update Cancel Help

## MetaTrader 5 trading platform

A new entry should be added in "Trade accounts" section. Select MetaTrader 5 Gateway to MOEXFX in "Gateway ID" field. Specify the client code in "Account" field (client code on the exchange).

If the external system account is not specified for the client, the client's requests are sent to the exchange on behalf of the broker, not on behalf of the client. You may use the External Account Required parameter to disable sending of requests from such clients to the exchange.

The configuration of the gateway is complete. To control the gateway operation use the journal.

### Gateway Launch Specifics

To avoid an excessive load, the amount of attempts to connect the gateway to the exchange is limited. If the gateway is unable to connect the exchange immediately, it repeats connection attempts for approximately 5 minutes (about 15 attempts). If all the attempts fail, the gateway stops and writes the following entry in the journal:

reached the maximum number of connection attempts (16), gateway stopped

In order for the gateway to continue connection attempts, restart the gateway.

### Discrete Auction Mode

Discrete auction mode of trading the USDRUB pair can be launched on Moscow Exchange in case of ultra-high volatility. In this mode, USDRUB\_TOM trading stops and trading on the special USDRUB\_DIS symbol begins. To ensure the continuity of the trading process on the MetaTrader 5 side, a mechanism for automated mapping of quotes and trading operations between these symbols has been implemented in the gateway.

During the auction, all USDRUB\_TOM orders placed on the MetaTrader 5 side will be sent to the exchange as USDRUB\_DIS orders. All USDRUB\_DIS quotes provided by the exchange will be passed to the platform as USDRUB\_TOM quotes.

The start and stop of the discrete auction is displayed in the gateway log:

|                         |         |                                                        |
|-------------------------|---------|--------------------------------------------------------|
| 2018.04.18 12:37:00.355 | Gateway | discrete auction is started, name exchange: USDRUB_TOM |
| 2018.04.18 12:52:03.157 | Gateway | discrete auction is stopped, name exchange: USDRUB_TOM |

During the discrete auction time, all Stop Loss and Take Profit triggers for the USDRUB\_TOM instrument are rejected, in order to avoid closure of clients' positions by bids from the auction Market Depth. The following message is printed to the log in this case:

request rejected, because SL and TP are discarded during a discrete auction

Also, Stop orders on USDRUB\_TOM will not trigger during the auction.

### Running in Quote Receiving Mode

## MetaTrader 5 trading platform

This mode allows using the gateway only for receiving quotes from the exchange, without processing trading operations. Leave the value of the following parameters blank:

- Trade server
- FIX Drop Copy Address
- FIX Trade Capture Address

The gateway will only connect to FAST channels to receive a list of symbols and quotes.

In this mode, the gateway continues processing limit files, but does not process correction files.

### Gateway Service Operations

The gateway can perform various service operations on accounts during operation.

- Deals with the type "Correction" and with the comment "[synchronization]" are formed when synchronizing limits of the account with the QUIK limit file
- Deals with the comment "limit correction, 'XXXXX'" are formed when limits are corrected according to the corresponding QUIK file
- Deals with the comment "[swap close deal]" and "[swap reopen deal]" are formed when processing a swap deal arriving through the Trade Capture channel
- Deals with the comment "[auction]" are formed when processing a deal executed during an auction
- Deals with the comment "[technical deal]" are formed when processing a technical deal on the market



MetaTrader 5 Trading Platform / Platform Components / Gateways /  
MOEX Securities

MetaTrader 5 Gateway to MOEX Securities

MetaTrader 5 Gateway to MOEX Securities allows trading stocks and bonds on the Moscow Exchange Equity and Bond Market.

It includes two markets:

- **Equity Capital Market** Russian and foreign shares, depositary receipts, fund shares, ISUs, and ETFs. The main trading mode is T+2 order book (Main Trading Mode T+). Trading is performed with a central counterparty, partial collateral and deferred execution. Settlement and delivery are performed two days after a trade is conducted (T+2).  
The gateway supports only shares.
- **Debt Capital Market** federal loan bonds (OFZs), regional and municipal bonds, Russian corporate bonds (including exchange ones) denominated in rubles and foreign currency, corporate Eurobonds and sovereign Eurobonds of the Russian Federation.  
The gateway supports only OFZs. Trading is performed with a central counterparty, partial collateral and deferred execution. The main trading mode for OFZ is T+1 order book (Main Trading Mode T+). Settlement and delivery are performed the next day after a trade is conducted (T+1).

MetaTrader 5 Gateway to MOEX Securities

MetaTrader 5 Gateway to MOEX Securities is a separate MOEXSecuritiesGateway64.exe file that uses Gateway API for its operation. Gateway module is originally included in the platform delivery set. After purchasing the gateway, the trading platform license will automatically update via LiveUpdate system. This allows you to start using MetaTrader 5 Gateway to MOEX Securities without any limitations.

Order MetaTrader 5 Gateway to MOEX Securities

The gateway allows quick integration of the MetaTrader 5 platform with the Equity and Bond Market of the Moscow Exchange. Order the gateway now to provide your clients with the advanced fast trading terminal offering wide opportunities including use of automated trading robots and mirror trading!

Common trading principles and their implementation in MetaTrader 5

Trading in the market is regulated by the exchange rules: Trading Rules of the Closed Joint-Stock Company "MICEX Stock Exchange". Let's consider some of the key features and examples for more clarity.

In the stock market, the result of a transaction is immediately displayed on the client trading status: an open position appears, and an appropriate amount of funds is withdrawn from the balance. Depending on the instrument, the actual settlement may occur the next day or the second day after the position was opened (T+). However, unlike the FX section, positions are not moved between symbols with different calculation periods (for example, from USDRUB\_TOM to USDRUB\_TOD). Position on the same instrument is displayed on MetaTrader 5 side, while all operations related to moving and swapping are carried out on the exchange side. The resulting client status is then imported via the limit files.

### Groups of instruments

Securities market Instruments are divided by type and trading mode:

- TQBR T+, Shares and debt warrants (DW);
- TQDE T+, D Bonds;
- TQIF T+, Investment units;
- TQOB T+, Bonds;
- TQQI T+, mode for qualified investors;
- TQTF T+, foreign exchange traded funds (ETFs);
- AUCT Auction;
- SPEQ technical transactions.

The instrument groups the gateway is to work with are defined by the Trades Mode parameter. By default, all groups are enabled, except for AUCT. SPEQ transactions are transmitted by the gateway regardless of the settings.

The same instruments can be traded in different modes. Execution results are displayed on the same position. For example, it is possible to buy shares by auction and in the order book, and this affects the unified position for the instrument.

### Repo transactions

MetaTrader 5 Gateway to MOEX Securities does not allow repo deals, though it correctly imports the appropriate transactions to MetaTrader 5 from the external system. Thus, clients are able to see the current status and history of accounts in their terminals.

Repo deals are marked "repo leg 1" and "repo leg 2" in trading history for each transaction party. Target repo transactions are marked "target repo leg 1" and "target repo leg 2".

### Auction

Before the start and after the end of each trading session, an auction is performed in the securities market to determine the fair instrument price. Opening auction starts at 9:50 and ends at 10:00 (exchange time). Closing auction is held from 18:40 to 18:50.

Transactions performed during an auction on MetaTrader 5 side are not marked in any particular way.

## Bonds

The gateway supports trading OFZs only. The bonds have two key parameters:

- Face value    initial value set by an issuer.
- Accrued interest    accumulated coupon interest (part of the coupon interest calculated in proportion to the number of days since the coupon bond issuance or the last coupon interest payment).

Profit on MetaTrader 5 side is calculated as the difference between the prices of a position at the time of opening and closing. Position price is defined as follows:

$$\text{Price}/100 * \text{Face value} * \text{Volume in lots} * \text{Contract size} + \text{Accrued interest} * \text{Volume in lots} *$$

The price is divided by 100 since bond prices are passed as a face value percentage.

The current floating profit at bonds is calculated without an accrued interest:

$$\text{Price}/100 * \text{Face value} * \text{Volume in lots} * \text{Contract size}$$

At the end of a trading session, profit by positions can be corrected according to the exchange data.

## Margin trading

Trade is marginal and contractual obligations are delayed in time. Until the time of settlement of obligations, the margin is locked on the client's account. The margin is defined by discounts relative to the current price of the instrument. Discounts are set by the broker, however they cannot be lower than the values determined by National Clearing Centre (NCC). Due to the discounts, clients can make transactions with a leverage, i.e. open positions investing a larger sum than is available on their account.

How the gateway works

The gateway receives market data on trading on Moscow Exchange via Multicast. This is a UDP protocol that allows packets to be transmitted to multiple recipients. Find the details of the protocol in "Market Data Multicast Ver3.3 User Guide". The system administrator must configure the server where the gateway is running to receive data.

The UDP protocol does not have built-in methods to ensure reliability, ordering, or data integrity, which on the one hand makes it unreliable. But on the other hand, it saves time and money required to deliver data to clients. This provides data delivery to clients with minimum delay. All transmitted data are doubled to control their integrity. The gateway receives market data from the Exchange in four tables:

- Trades executed on the Exchange.
- Trade statistics.
- Depth of Market.
- List of instruments.

To deliver each data type, four connections are created: two connections for the

incremental channel where only changes relative to the previous state are transmitted, and two for the so-called snapshots (the state at a certain time).

For committing trade operations and receiving the trading status of clients, the gateway uses three FIX services, for each of which a separate connection is created:

- MFIX Trade - entering and canceling orders and receiving order execution reports.
- MFIX Drop Copy - receiving the stream of orders and deals from other terminals (other than MetaTrader 5).
- MFIX Trade Capture - sending all trades of all accounts bound to the current FIX identifier of the broker.

Accordingly, three FIX IDs are required. Read more about the FIX services in "[Moscow Exchange public FIX 4.4 interface specification for Securities and FX markets](#)". If any of the services FIX is not required, it can be omitted. Simply do not specify its address in the gateway settings.

### Integration with broker's back office

The relevant trading client state in MetaTrader 5 is maintained using the client limiting interface provided by the gateway. Limiting is implemented through import of limit files in the QUIK format. The detailed format description is available in the [QUIK WorkStation reference](#). This format has been selected due to the fact that the platform is currently used by most of the brokers that provide access to Moscow Exchange, and therefore such an approach is the most simple and convenient one to move to MetaTrader 5.

In the [gateway settings](#), specify the address (ftp or local address), from which the gateway shall automatically read the limit and correction files and synchronize client states in MetaTrader 5. The limit file can be used at any moment of gateway operation. Once the administrator adds a file to the specified directory, the gateway will process it and will set the client status in accordance with it. This behavior can also be used to recover client states in case of failures.

Limit correction files are used to reflect the limit changes in the broker's back-office system. The gateway processes them only in trading hours (10:00 to 23:50 Moscow time). Correction files are processed in accordance with the following algorithm:

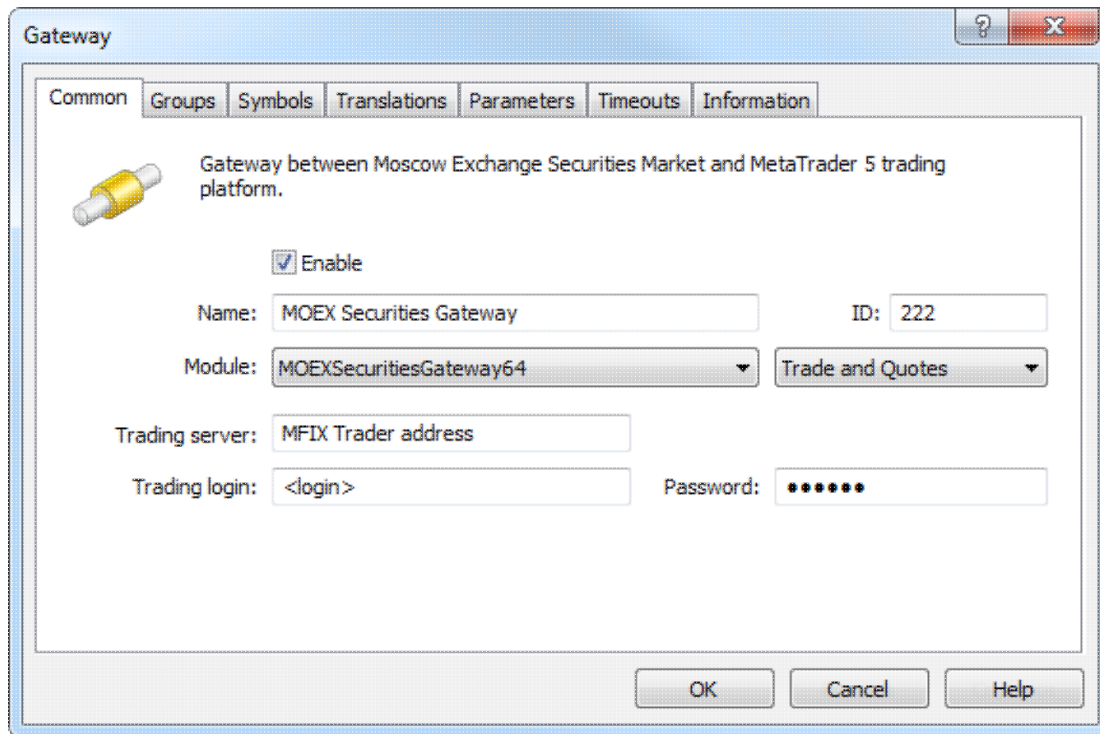
- When the correction file is updated, the gateway starts analyzing it.
- It parses all the correction entries from the file one by one.
- The gateway finds the entry with LIMIT\_ID (the correction identifier is unique within a file and within one trading day) equal to LIMIT\_ID\_LAST (the identifier of the last processed correction).
- All subsequent entries are processed and applied to the clients' trading states. While each subsequent entry is processed, the value of LIMIT\_ID\_LAST is updated to avoid repeated application of corrections.
- As soon as processing is over, the gateway uploads the file with the correction results. The file is of the QUIK system format. The file is uploaded at the address specified in the [gateway settings](#).

## MetaTrader 5 trading platform

- At the beginning of each trading day, the ID of the last processed correction is reset.

### Configuring the gateway

To start working, add the new gateway configuration:

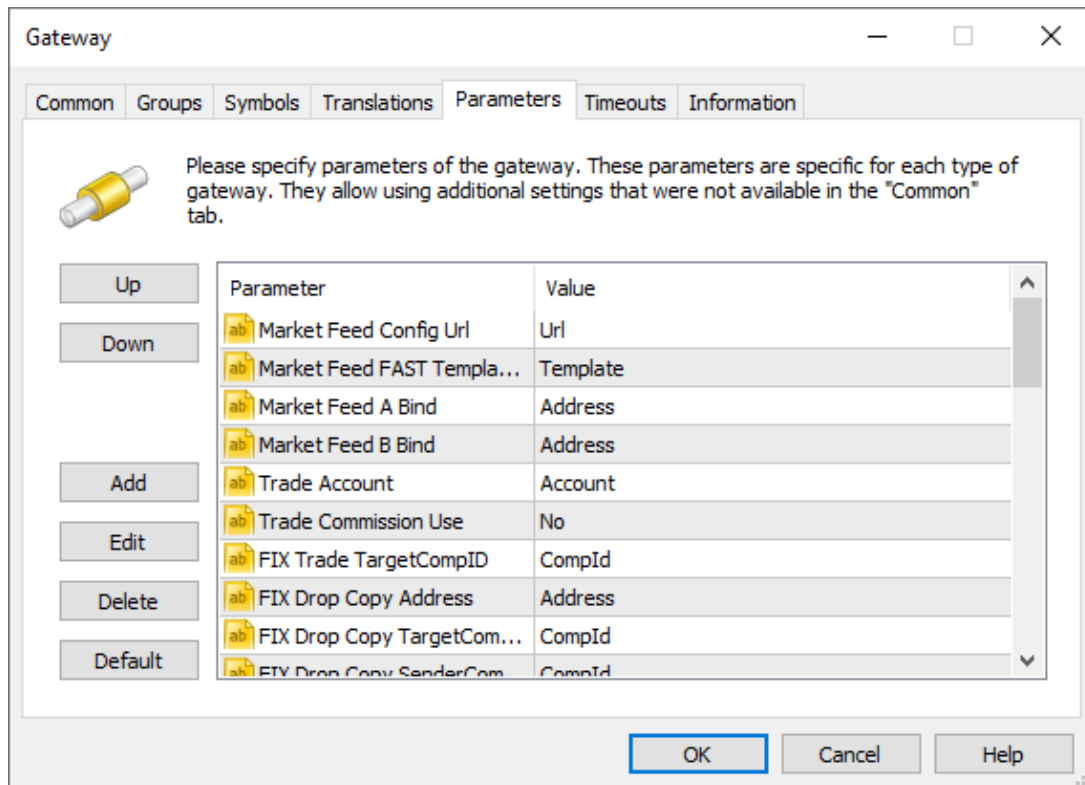


The following parameters on the "Common" tab must be set:

- **Module** MOEXSecuritiesGateway64. After that the gateway default parameters installation must be allowed in the dialog request.
- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Trading server** address for connection to the MFIx Trade service.
- **Trading login** login (FIXTradeSenderCompID) for connection to MFIx Trade.
- **Password** password for connection to MFIx Trade.

- ID value must be unique in the field of manager logins and gateway identifiers.
- Connection data (address, login and password) are provided by the exchange.

Additional configuration parameters must be set on the Parameters tab:



The following additional parameters are available for the gateway:

- **Trading Calendar Holidays** each exchange works in accordance with its own trading calendar. By default, non-working days for the gateway are Saturday and Sunday. Using this option, you can override the working/non-working days for the gateway. To add a non-working day, specify a value of the form +DDMMM. The date is specified as two digits, the month is specified as the first three letters of the month name in English. For example, +01JAN. To add a working day on Saturday or Sunday, specify a value of the type -DDMMM, for example, -07FEB. You can specify multiple working/non-working days, separated by semicolons, for example, +01JAN;-07FEB.
- **Market Feed ConfigUrl** parameters of channels for receiving market data from the exchange are described in the XML file. This file is available on the [Exchange's FTP server](#). Specify the path to the file, depending on what data you want to receive (real or test). Configuration file can be copied to a local server, the appropriate path should be specified in this field. The gateway reads data for connection to the following channels (set by the connection\_id parameter in the file):
  - ♦ **OBR and OBS** an incremental channel and a channel of snapshots for transmitting the depth of market.
  - ♦ **TLR and TLS** an incremental channel and a channel of snapshots for transmitting trading operations.
  - ♦ **MSR and MSS** an incremental channel and a channel of snapshots for transmitting trade statistics.
  - ♦ **IDF** a channel for transmitting the list of financial instruments.

- **Market Feed FAST Templates Url** all market data are transmitted through the FAST protocol. Transmitted data are unpacked using the special FAST template. Specify here the path to the template file, it is available on the Exchange's FTP server. You can specify the FTP path or copy the file to a local server and specify the appropriate local path to it.

Market channel parameters and template files downloaded via FTP are saved in the directory [gateway folder]\downloads. If the gateway fails to download updated files, it will use earlier downloaded files stored in this folder. The following rules apply here:

- The gateway performs a checks that the file is not older than one week. If it was downloaded earlier, it is considered obsolete and will not be used.
  - If path to a local file instead of FTP is specified in parameters Market Feed ConfigUrl and Market Feed FAST Templates Url, file opening will fail and the existing file from the 'downloads' directory will not be used.
- 
- **Market Feed A Bind** the address of the local interface the connection will be bound to to receive market data through channel A (all data from the Exchange are duplicated on two channels).
  - **Market Feed B Bind** the address of the local interface the connection will be bound to to receive market data through channel B (all data from the Exchange are duplicated on two channels).
  - **Trade Account** the account used for registering the transaction performed through the gateway on the exchange.
  - **FIX Trade TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Trade service. The remaining connection parameters are specified on the "Common" tab in the gateway settings.
  - **Trade Commission Use** this parameter sets the use of commissions from the exchange. The commissions are specified in each transaction received from the exchange. If the parameter is set to "Yes", this value is used in the platform. If set to "No" (default), the broker can configure commission calculation directly in MetaTrader 5 based on the commission calculation formula used on the exchange.
  - **FIX Drop Copy Address** address for connection to the MFIX Drop Copy service. The gateway can work without connection to the service. In that case, leave this parameter empty.
  - **FIX Drop Copy TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Drop Copy service.
  - **FIX Drop Copy SenderCompID** FIX messages heading standard parameter used for a message sender identification for the MFIX Drop Copy service.
  - **FIX Drop Copy Password** password for connection to MFIX Drop Copy.
  - **FIX Trade Capture Address** address for connection to MFIX Trade Capture.
  - **FIX Trade Capture TargetCompID** FIX messages heading standard parameter used for a message recipient identification for the MFIX Trade Capture service.
  - **FIX Trade Capture SenderCompID** FIX messages heading standard parameter used for a message sender identification for the MFIX Trade Capture service.

- **FIX Trade Capture Password** password for connection to the MFIX Trade Capture service.
- **Limits Url** local or FTP address of limit files to be imported to MetaTrader 5.
- **Limits Corrections Url** local or FTP address of limit correction files to be imported to MetaTrader 5.
- **Limits Corrections Result Url** local or FTP address where the files of limit correction import results will be exported.

In order to download and import the limits and corrections, the gateway requires all three parameters filled: Limits Url, Limits Corrections Url and Limits Corrections Result Url. If any of them is not filled or absent, the gateway will not download and import the files.

- **Limits FTP Login** if limit files are received from the FTP server, specify the login and password for connection to it.
- **Limits FTP Password** if limit files are received from the FTP server, specify the login and password for connection to it.
- **Symbols Path** - path for importing the exchange symbols. By default, if this parameter is absent, the gateway imports the exchange's trading symbols to \MOEX\Securities\ subdirectory with trading ability disabled. A system administrator should allow trading for them after the import. If this parameter is present, the gateway imports trading symbols following the specified path.
- **Limits Kind** specify a type of limits imported from the limits file (Limits Kind = LIMIT\_KIND from the file). For the securities market, the default value is "2" (corresponds to the T+2 limits).
- **Trades Mode** specify the markets (for example, TQBR, TQDE, TQIF) the gateway is to work for. The mode codes are used to define the markets. The values can be comma-separated or entered via the "\*" and "!" masks. Do not specify the "\*" mask (all markets) because the same instruments (the ones having similar names) with different quotes can be traded on different markets. In MetaTrader 5, symbol names are unique, and all quotes are merged into a single stream.
- **Execution Reports Mode** specify the markets, for which Execution Reports are processed. This parameter is reserved for future use. You do not need to specify it.
- **Short Limits Enabled** if "Yes", the gateway checks global limits by clients' short positions (by QUIK limit files). Brokers have certain assets for each trading instrument (for example, a certain number of shares). To avoid calculation issues, they should not allow their traders to sell more assets than they have. When the check is enabled, the gateway keeps internal statistics of short positions for each instrument opened by traders during the day. When reaching the limit, further requests for opening short positions are rejected. The gateway also considers traders' long deals if a long position is opened after a short one for the same instrument and with the same volume, the limit returns to its original state. If only a long position is opened, the total limit is not increased.

The gateway keeps an internal record of short positions during the session even if ShortLimitsEnabled is disabled. Thus, if the parameter is enabled during a trading session, the gateway does not need time for calculations. If the limit is already exceeded at the time of connection, further short operations are



immediately rejected. By default, the check is disabled ("No").

- **FIX Heartbeat Interval** Heartbeat messages are used in the FIX protocol for monitoring the status of connection between the Exchange and the gateway. If the gateway does not receive any data from the Exchange or Heartbeat messages during the FIX Heartbeat Interval, the connection is considered to be lost and the gateway tries to restore it. The default value of FIX Heartbeat Interval is 30 seconds. The valid range of values is from 10 to 60 (the number of seconds).
- **External Account Required** if the external system account is not specified for the client on the MetaTrader 5 side, the client's requests are sent to the exchange on behalf of the broker, not on behalf of the client. To disable sending of requests from such clients to the exchange, set "Yes" for the External Account Required parameter. The gateway will immediately reject such requests. "No" is used by default, which means requests from clients without the external account are sent to the exchange.
- **Margin Calculation Mode** calculation type used for the financial instruments imported by the gateway into the platform. If "Common", symbols are imported with the "Exchange Stocks" and "Exchange Bonds" types. If the value is "MOEX", symbols are imported with the "Exchange MOEX Stocks" and "Exchange MOEX Bonds" types (margin calculation is performed according to the rules adopted at the exchange on July 1st, 2019).
- **Time To Start** by default, the gateway starts connecting to the exchange's server 35 minutes before the start of the main trading session (9:25 GMT+3). Some brokers may need to change this parameter. To do this, use Time To Start parameter to specify in how many minutes before the session the gateway should start connection. For example, if connection is to start at 9:35, the parameter's value should be 25. The maximum possible value for the parameter is 250.
- **Money Currency Codes** currencies of money limits for which import from the limits file is allowed. The default is "SUR,USD": money limits for all currencies except the Russian Ruble and the US Dollar are skipped. The value corresponds to the "CURR\_CODE" tag from the limits file.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

To change the password for connecting to the FIX services, enter the new password in the corresponding gateway settings. Make sure that the gateway is authorized - connected using the current valid password.

### **Backup servers for FIX services**

The gateway uses three FIX connections: MFIXTrade, MFIXDropCopy, and MFIXTradeCapture. The related services operate on separate servers at the exchange. A parallel backup server is also provided for each of the services.

The gateway features the ability to automatically switch to a backup FIX server in case it is impossible to connect to the main one. Specify the addresses of the main and backup servers separated by commas in the gateway settings, for example:

- Trading server 91.203.254.29:9212,91.203.253.29:9212
- FIXDropCopy 91.203.255.21:9215,91.203.255.22:9215
- FIXTraderCapture 91.203.255.21:9214,91.203.255.22:9214

The gateway will automatically switch to the specified backup servers when necessary.

### **Connecting to backup servers for FIX services with different TargetCompIDs**

The gateway allows you to connect to backup servers with TargetCompID different from the main one.

For MFIX Trade service, you can create up to 10 parameter pairs looking as follows:

- FIX Trade Address 1, FIX Trade Address 2, etc.
- FIX Trade TargetCompID 1, FIX Trade TargetCompID 2, etc.

For MFIX Drop Copy service, you can create up to 10 parameter pairs looking as follows:

- FIX Drop Copy Address 1, FIX Drop Copy Address 2, etc.
- FIX Drop Copy TargetCompID 1, FIX Drop Copy TargetCompID 2, etc.

For MFIX Trade Capture service, you can create up to 10 parameter pairs looking as follows:

- FIX Trade Capture Address 1, FIX Trade Capture Address 2, etc.
- FIX Trade Capture TargetCompID 1, FIXTradeCaptureCompID 2, etc.

Example:

Trade

Server="91.203.254.23:9120,91.203.255.24:9120,91.203.255.23:9120,91.203.254.24:9120"

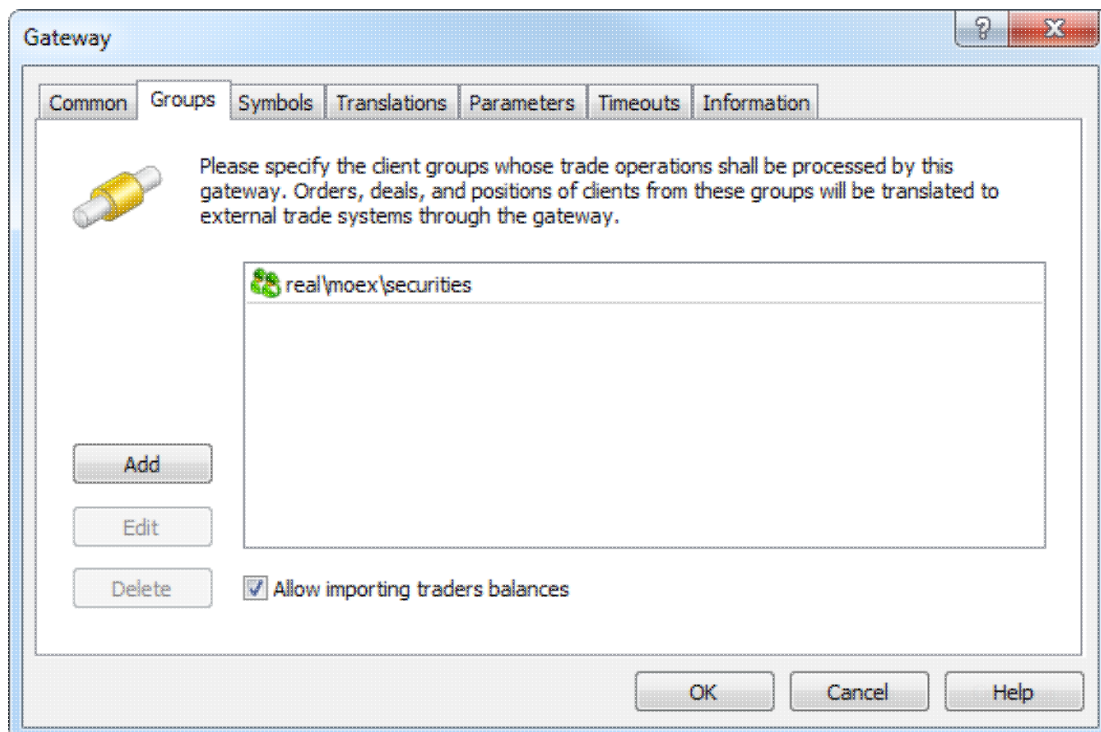
FIX Trade TargetCompID="J2FIXAEQ"

FIX Trade Address 1="91.203.255.30:9120,91.203.252.30:9120"

FIX Trade TargetCompID 1="G2FIXBEQ"

After a successful connection to the backup address, the gateway remembers it and connects to it later on. To return to the main address, temporarily delete the backup one from the gateway settings. After a successful connection to the main address, specify the backup one in the settings again.

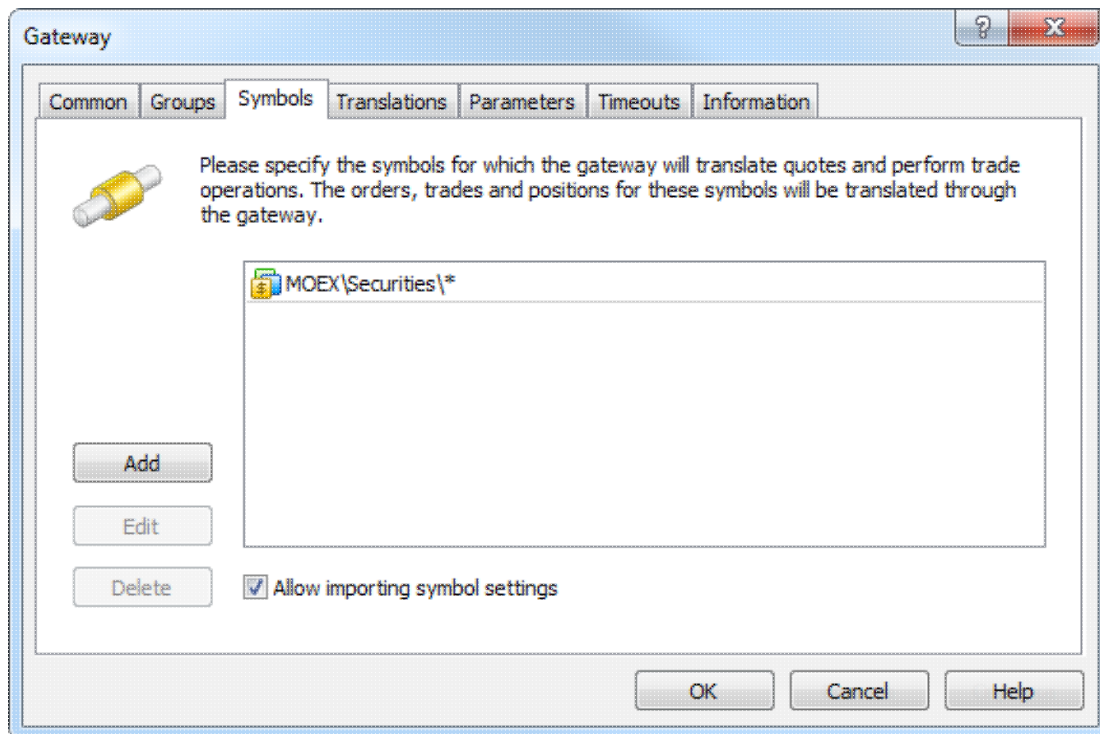
**On the Groups tab**, specify the group of clients who will trade in the FX Market. The gateway will only receive trade requests from the clients included in the allowed groups list.



To import the limits via the gateway, enable the "Allow importing traders balances" option. This will allow the gateway to synchronize clients' balances in MetaTrader 5 with exchange account limits.

A group of clients trading in the FX Market should be configured correctly.

**On the Symbols tab**, specify the symbols available for the gateway. The gateway will receive quotes and perform trade operations for these symbols. Also, the settings will be automatically updated for them in case import of symbols is allowed:



## MetaTrader 5 trading platform

Specify here the group of symbols the gateway will operate with. Also, "Allow importing symbol settings" should also be enabled. MetaTrader 5 Gateway to MOEX Securities adds necessary symbols and manages their settings on its own.

The symbols imported by the gateway are put to the "\\MOEX\\Securities" symbols subgroup.

The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

### Configuration of groups

A separate group should be created for the clients working in the securities market. Specify RUR as the deposit currency in the group settings:

Group: real\\moex\\securities

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authentication type.

Name: real\\moex\\securities Currency: RUR

Trade server: MetaTrader 5 Main Server, 1

Authentication: Normal Minimum password length: 7

One-time password: Disabled ☐ Force one-time password usage

☒ Enable connections

☐ Enable certificate confirmation

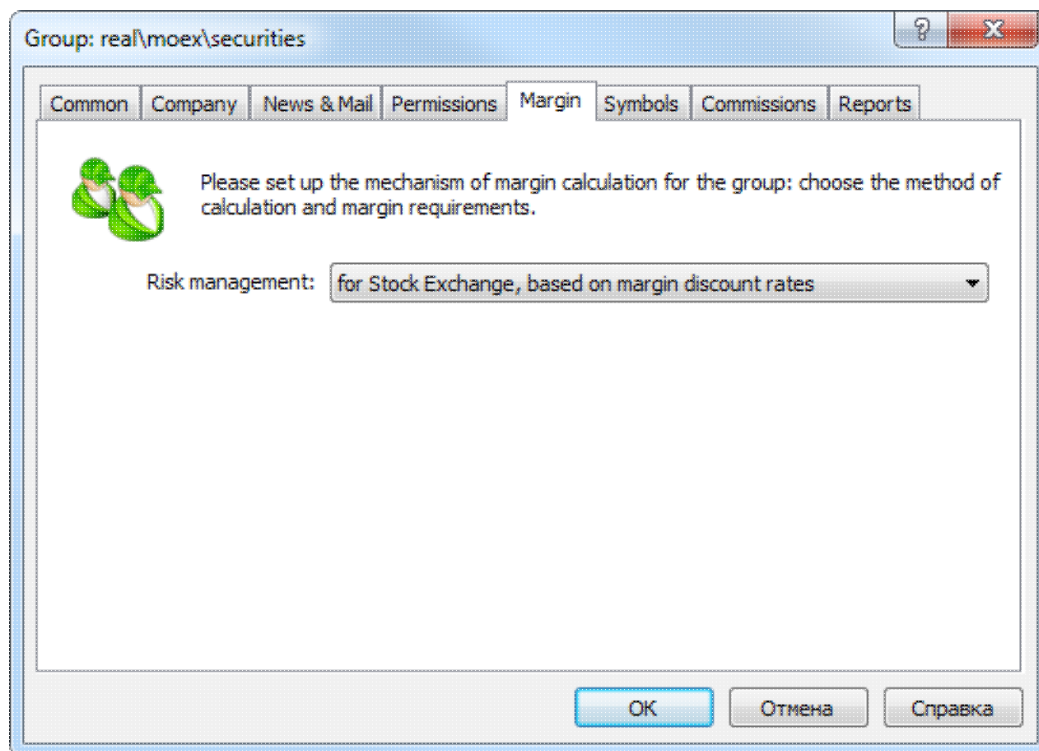
☐ Change password at first login

☐ Show the risk warning window after connection

OK Отмена Справка

## MetaTrader 5 trading platform

In the "Margin" tab, choose the risk management mode "for Stock Exchange, based on margin discount rates". In this mode, pre-trade control is based on discounts specified in symbol settings. Discounts are set by the broker, however they cannot be lower than the values determined by National Clearing Centre (NCC).



For separate clients based on risk policies, create different groups for them. In these groups, you can override symbol discount settings.

### Configuring trade requests routing

To direct client requests concerning the securities market symbols to the gateway, the routing rules should be set. The examples can be found in the articles about the gateways to other trading systems, like: GBOT, DGCX, etc.

### Configuring trading accounts

Each client on MetaTrader 5 side should be provided with a corresponding separate trading account on the exchange. After an account is opened at the exchange, it should be connected with the account within the platform. Open the appropriate client account and move to Account tab:

## MetaTrader 5 trading platform

Account: 2013, John Smith, USD

Overview Personal **Account** Security

Group: real\moex\securities

Color: ☐ None Leverage: 1 : 100

Bank account: Agent account:

☒ Enable this account ☒ Enable trading by Expert Advisors

☒ Enable trading ☒ Enable Web API connections

☒ Enable to change password ☒ Enable daily reports

☒ Enable trailing stops ☐ Change password at next login

Trade accounts:

| Gateway ID                 | Account  |
|----------------------------|----------|
| 5, MOEX Securities Gateway | 12345678 |

Synchronize

Add

Edit

Delete

Update Cancel Help

## MetaTrader 5 trading platform

A new entry should be added in "Trade accounts" section. Select MetaTrader 5 Gateway to MOEX Securities in "Gateway ID" field. Specify the client code in "Account" field (client code on the exchange).

If the external system account is not specified for the client, the client's requests are sent to the exchange on behalf of the broker, not on behalf of the client. You may use the External Account Required parameter to disable sending of requests from such clients to the exchange.

The configuration of the gateway is complete. To control the gateway operation, use the Journal.

### Gateway launch specifics

To avoid an excessive load, the amount of attempts to connect the gateway to the exchange is limited. If the gateway is unable to connect the exchange immediately, it repeats connection attempts for approximately 5 minutes (about 15 attempts). If all the attempts fail, the gateway stops and writes the following entry in the journal:

reached the maximum number of connection attempts (16), gateway stopped

In order for the gateway to continue connection attempts, the administrator should restart the gateway.

### Running in Quote Receiving Mode

This mode allows using the gateway only for receiving quotes from the exchange, without processing trading operations. Leave the value of the following parameters blank:

- Trade server
- FIX Drop Copy Address
- FIX Trade Capture Address

The gateway will only connect to FAST channels to receive a list of symbols and quotes.

In this mode, the gateway continues processing limit files, but does not process correction files.

### Gateway Service Operations

The gateway can perform various service operations on accounts during operation.

- Deals with the type "Correction" and with the comment "[synchronization]" are formed when synchronizing limits of the account with the QUIK limit file
- Deals with the comment "limit correction, 'XXXXX'" are formed when limits are corrected according to the corresponding QUIK file
- Deals with the comment "[swap close deal]" and "[swap reopen deal]" are formed when processing a swap deal arriving though the Trade Capture channel



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- Deals with the comment "[target repo leg 1]" and "[target repo leg 2]" are formed when processing a targeted repo trade arriving through the Trade Capture channel on the Securities Market
- Deals with the comment "[target deal]" are formed when processing a targeted deal arriving through the Trade Capture channel on the Securities Market
- Deals with the comment "[repo leg 1]" and "[repo leg 2]" are formed when processing a targeted repo trade arriving through the Trade Capture channel on the Securities Market
- Deals with the comment "[technical deal]" are formed when processing a technical deal on the Securities Market

MetaTrader 5 Trading Platform / Platform Components / Gateways /  
MOEX Derivatives

MetaTrader 5 Gateway to MOEX Derivatives

MetaTrader 5 Gateway to MOEX Derivatives allows trading futures and options on the derivatives market of the Moscow Exchange.

The futures and options market of OJSC "Moscow Exchange MICEX-RTS" is a leading trading venue for derivatives in Russia and Eastern Europe. The market combines the advanced infrastructure, reliability and guarantees of OJSC "Moscow Exchange MICEX-RTS" as well as state-of-the-art technologies for futures and options trading with more than ten years of the stable and successful market development.

The essential part of any transaction in derivatives is the settlement of a contract on a certain day in the future under fixed conditions. The market participants are reliable large cap investment companies and banks.

The derivatives market offers more than 150 different futures contracts. Their full list can be viewed on the Moscow Exchange's official website.

To start your activity on the market, you should connect to it, as described on the exchange's website. Brokerage and clearing companies can work on the market.

MetaTrader 5 Gateway to MOEX Derivatives

MetaTrader 5 Gateway to MOEX Derivatives is a separate MOEXDerivativesGateway64.exe file that uses Gateway API for its operation. Gateway module is originally included in the platform delivery set. After purchasing the gateway, the trading platform license will automatically update via LiveUpdate system. This allows you to start using MetaTrader 5 Gateway to MOEX Derivatives without any limitations.

Order MetaTrader 5 Gateway to MOEX Derivatives

MetaTrader 5 Gateway to MOEX Derivatives allows fast integration of MetaTrader 5 platform with the derivatives market of the Moscow Exchange, a leading trading venue for derivatives in Russia and Eastern Europe. By ordering the gateway now you provide your clients with new trade opportunities in the financial markets!

How the gateway works

MetaTrader 5 Gateway to MOEX Derivatives uses Plaza 2 protocol, which is the official software package for providing access to the exchange. Plaza 2 package has two main components:

- P2MQRouter module is a module providing connection to exchange servers, receiving and sending messages, data encryption and decryption, as well as

## MetaTrader 5 trading platform

participant's authentication in the exchange's network. In fact, this module is an intermediate component between the exchange and a third-party software connected to it.

- CGate library is an official program interface provided to third-party companies to create a software.

Both components are already included in MetaTrader 5 Gateway to MOEX Derivatives. Thus, no third-party software (P2MQRouter module) is required to use it. When the gateway is enabled in the terminal, P2MQRouter with necessary settings is launched on a local server. It provides a comprehensive interface for connecting MetaTrader 5 Gateway to MOEX Derivatives to the exchange.

By default, the gateway receives aggregated Market Depth from Plaza 2 streams: from FORTS\_FUTAGGR\*\_REPL with a frequency of 10 ms or from FORTS\_FUTAGGR\*\_FASTREPL with a frequency of 3 ms. Use the High Frequency Replication parameter to switch between them.

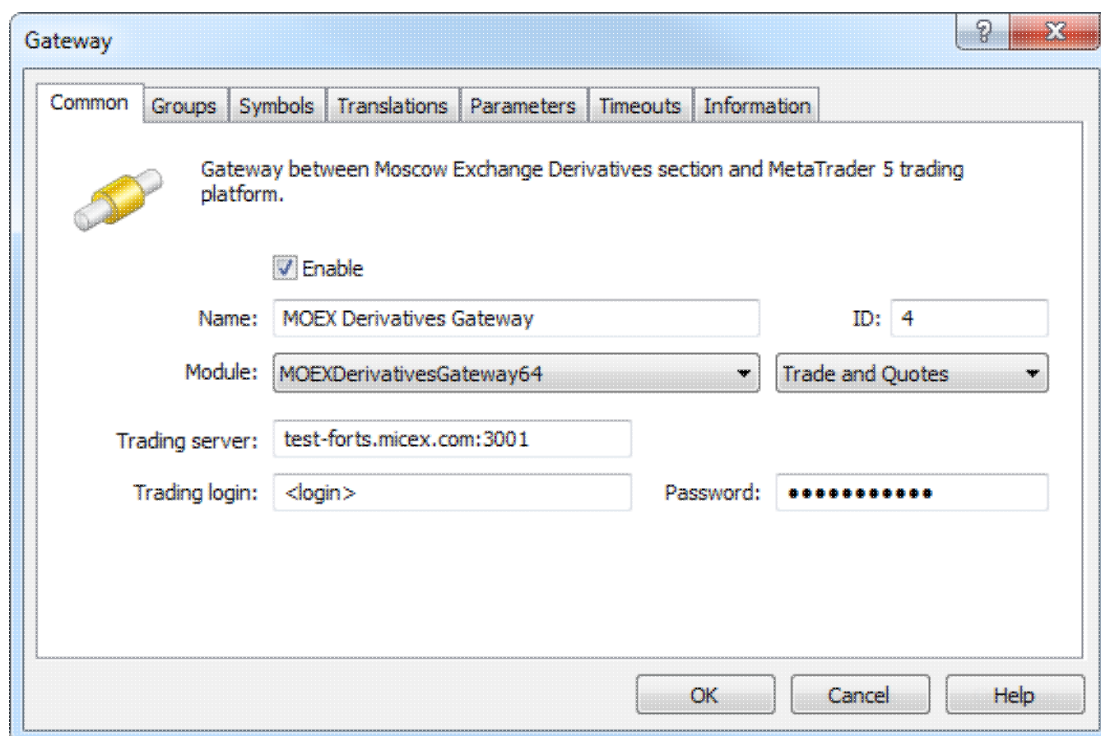
The gateway can optionally build the Market Depth using the journal of anonymous orders broadcast by Moscow Exchange through the FAST Multicast service in the ORDERS-LOG data channel. Unlike other data channels of the Exchange derivatives section, here the order book is broadcast in real time, which enables creation and display of the most up-to-date Market Depth. To enable this mode, fill in each of the four MarketFeed\* parameters.

- No third-party software is required to provide MetaTrader 5 Gateway to MOEX Derivatives operation.
- Please contact the exchange to connect FAST Multicast services of the derivatives market to your servers in the collocation zone.

### Configuring the gateway

To start working, add the new gateway configuration:

## MetaTrader 5 trading platform

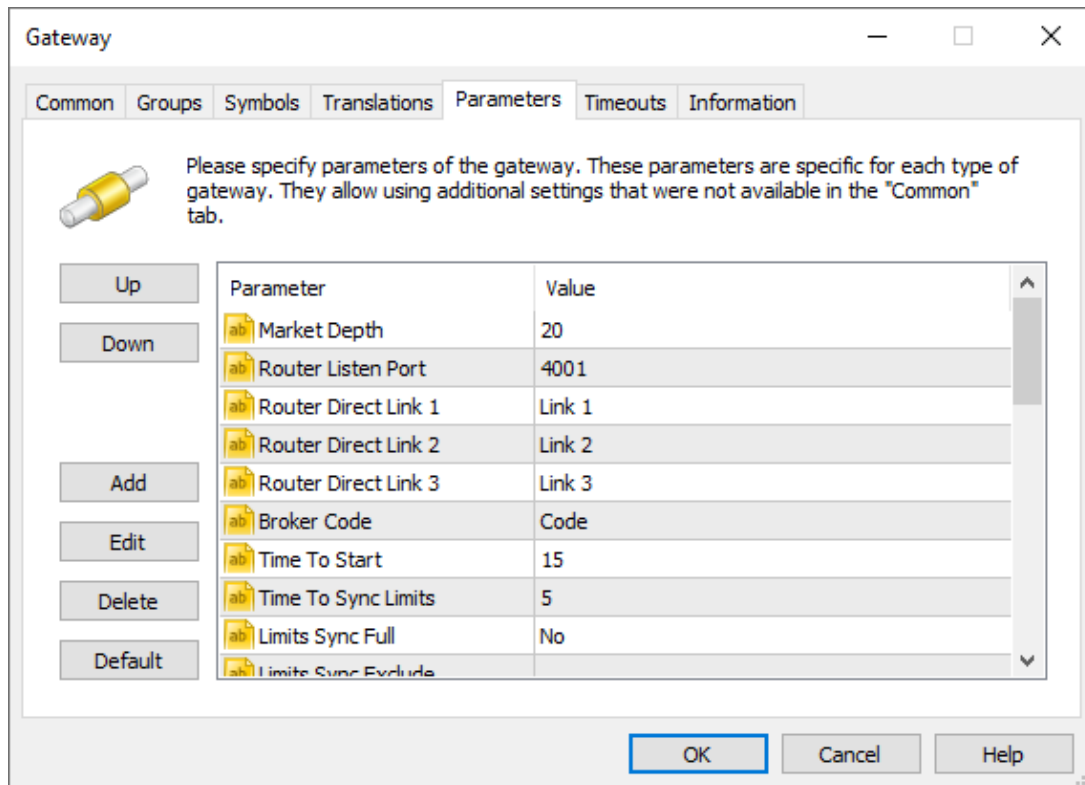


The following parameters on the "Common" tab should be set:

- **Module** MOEXDerivativesGateway. After that the gateway default parameters installation must be allowed in the dialog request.
- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Trading server** address for connection to the exchange (not to P2MQRouter module) should be specified here in "address:port" format. In the above example, it is "test-forts.micex.com:3001" address for developers.
- **Trading login** user ID for connection to the exchange.
- **Password** password for connection to the exchange.

- ID value must be unique in the field of manager logins and gateway identifiers.
- Connection data (address, login and password) are provided by the exchange during the agreement conclusion.

Set additional configuration parameters on the Parameters tab:



The following additional parameters are available for the gateway:

- **Trading Calendar Holidays** the exchange may have working and non-working days that are different from the MetaTrader 5 parameters for working days and holidays. This parameter redefines working/non-working days for the gateway. To add a non-working day, specify a value of the form +DDMMM. The date is specified as two digits, the month is specified as the first three letters of the month name in English. For example, +01JAN. To add a working day, specify a value of the form -DDMMM, for example, -07FEB. You can specify multiple working/non-working days, separated by semicolons, for example, + 01JAN;-07FEB.
- **Market Depth** depth of the transmitted depth of market. Depth of 5 and 20 requests are supported. The depth of 5 is used unless otherwise specified.
- **Router Listen Port** port on a local PC (127.0.0.1), where P2MQRouter module is launched by the gateway. This is the port where P2MQRouter module will accept connections from the gateway.
- **Router Direct Link1/2/3** additional IP addresses for connection to the exchange. They are used to receive data from different streams in parallel. All addresses are provided by the exchange. The order of specifying additional addresses does not matter. For test area, use addresses test-ports.micex.com:3000, test-ports.micex.com:3003 and test-ports.micex.com:3004.
- **Broker Code** unique alphanumeric identifier of your company as the market participant. The identifier consists of three parts XYZZZZ, where XX is a clearing company code, YY brokerage company code, ZZZ client code. When

representing a clearing company, broker's code is specified as 00, while client's code is removed (for example, Q100, where Q1 is a clearing company code). When representing a brokerage company, client's code is specified as 000 (for example, Q1DU000, where Q1 is a clearing company code, while DU is a brokerage company one). Broker Code is provided by the exchange and used to assign numbers to client accounts.

- **Time To Start** by default, the gateway starts connecting to the exchange's server 15 minutes before the start of the main trading session (9:45 GMT+3). Some brokers may need to change this parameter. To do this, use Time To Start parameter to specify in how many minutes before the session the gateway should start connection. For example, if connection is to start at 9:35, the parameter's value should be 25.
- **Time To Sync Limits** the gateway receives fund limits by clients from the exchange and synchronizes the client balances in MetaTrader 5 appropriately before the start of a trading session. By default, the limits start arriving 5 minutes before the start of the main trading session (9:55 GMT+3). Some brokers may need to change this parameter. To do this, use Time To Sync Limits parameter to specify in how many minutes before the session the gateway should start receiving the limits. For example, in order to receive the limits at 9:50, the parameter's value should be 10. If the exchange allocates an additional morning session on a certain day, the gateway will automatically determine such a session and will synchronize data 5 minutes (or the number of minutes specified in "Time To Sync Limits" if you use a custom value) before the session start.
- **Environment** the gateway uses Plaza 2 protocol, which is the official software system providing access to the exchange. The exchange has three areas - test, learning (for practice) and real (for trading) ones. Different versions of Plaza 2 protocols are applied at each of the areas. If you connect to the learning area, set Game value for this parameter to allow the gateway to use the appropriate version of the protocol. Select Real to connect to real trading area. Set Test to connect to test area.
- **FTP Statistic Process** after the end of a trading session, the gateway downloads the official report of the exchange concerning all performed deals from the special FTP server. This report is used for control and correction of historical data in MetaTrader 5. If you want to disable the monitoring of historical data by reports, set this parameter to No.
- **Spreads Allow** inter-month spreads (security deposit benefits) are applied to some futures contracts on the derivatives market. For example, when purchasing the closest quarterly futures and the futures for the next quarter, the funds limit is decreased by the maximum value of the security deposit for both contracts. The full list of futures can be found in the ["Futures inter-month spreads" table on the exchange's website](#). The gateway automatically imports the necessary spread settings if Yes is specified in Spreads Allow parameter.
- **Symbols Path** path for importing the exchange symbols. By default, if this parameter is absent, the gateway imports the exchange's trading symbols to \Preliminary subdirectory with trading ability disabled. A system administrator should manually relocate imported symbols to the proper group and allow trading for them. If this parameter is present, the gateway imports trading symbols following the specified path. The ability to trade the symbols is enabled immediately. Thus, the administrator has no need to additionally configure the symbols. For example, if MOEX is specified in the parameter, then futures

contracts are automatically added to MOEX\FORTS\\*, Standard section contracts are added to MOEX\Standard\\*, while expired contracts are automatically transferred to MOEX\FORTS\Expired\\*.

- **Symbols Path Expired** path for transferring the expired symbols. By default, if this parameter is not specified, all expired contracts are moved to \Expired subdirectory of the folder specified in Symbols Path parameter. In the above example, it is MOEX\FORTS\Expired\\*. This parameter allows re-configuring the path for transferring the expired contracts.
- **Options Allow** in addition to futures, the derivatives market allows trading options as well. If Yes, the gateway automatically imports available options to \FORTS\Options subdirectory of the directory set in the Symbols Path parameter. The gateway also handles trade requests for these symbols.
- **TP Coverage As Limit Order** the policy of passing the activation of a take profit level to the exchange. If set to Yes, an opposite limit order corresponding to the take profit activation level is passed to the exchange. If set to No, an opposite limit order at the minimum or maximum price (depending on the order direction) is passed to the exchange when a take profit is activated. For buy orders, current session's highest price is used; for Sell orders the lowest price is used. For example, when the Take Profit of a long position triggers, a limit sell order at the symbol's lowest price as of the current moment is sent.
- **Limits Sync Full** if Yes, clients' positions on the exchange are automatically imported to MetaTrader 5 in the morning, before the main trading session. Positions in MetaTrader 5, which differ from exchange positions, will be modified to conform to exchange positions (in case of a divergence, existing positions are completely closed and new ones are opened). This option has been added to automatically fix emergency possible out of sync state.
- **High Frequency Replication** Moscow Exchange provides the ability to receive market data with high frequency once per 3 ms instead of once per 10 ms. In order to use this ability, enable the appropriate additional service at the exchange and set the High Frequency Replication parameter to Yes. Also, replace the trade server address with the one, from which the high-frequency data is to be received. The address is provided by the exchange when enabling the service.
- **Limits Sync Exclude** allows disabling synchronization of positions and balances for the specified client groups. The platform allows transmitting trading operations from one account through different gateways. To do this, multiple accounts in external systems corresponding to different gateways are registered in the account. There are three gateways for the Moscow Exchange. Each of them synchronizes the trading state and limits. The possibility to disable synchronization allows to prevent overwriting of data about positions and limits during simultaneous operation through these gateways. Specify the list of groups for which the gateway should not synchronize positions and balances in this parameter. Groups can be specified using the "\*" mask.
- **Trade Limit** allows setting the overall limitation on the number of trades sent per second for the gateway. If 0 (default), no check is performed. The maximum possible value is 16384. In case of a greater value, no check is performed.
- **Trade Limit Default** allows setting the limitation on the number of trades sent by separate clients. For example, if 50 is specified, then each client can send no more than 50 trades per second via the gateway (unless the client limit is overridden by the Trade Limit <account> type parameter). If 0 (default), no check is performed.

- **Trade Limit <account>** parameters allow setting the limitation on the number of trades per second sent by certain clients. You can set several parameters of that kind. For example: Trade Limit 10008628 = 7, Trade Limit 10008589 = 5. The exchange account code without a broker's code (set in the Broker Code parameter) is used in <account> on the platform side, this code is set in the client record as an account number in the external trading system. If 0 (default), no check is performed for the specified client. The maximum possible value is 256 trades per second. Also, no check is performed if a larger value is specified.

Before issuing a trade request to the exchange, the gateway checks the limitation on the client account and throughout the entire gateway (brokerage account). If the check fails, the gateway does not send the client's trade and returns the MT\_RET\_REQUEST\_TOO\_MANY response code. The following entries are made in the gateway journal:

|                         |         |                                                        |
|-------------------------|---------|--------------------------------------------------------|
| 2017.11.27 18:26:21.224 | Gateway | account HO00001 trade transactions per second limit    |
| 2017.11.27 18:26:21.572 | Gateway | gateway trade transactions per second limit (7) exceed |

- **Password Change** this parameter determines whether the gateway should generate for the current login a new password for connecting to the exchange if the old password expires in less than 7 days. The parameter is set to No by default which means that the function is disabled. Please see Password Change for further details.
- **NewsCategory** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

This group of parameters is used for setting up the receipt of anonymous requests via the FAST Multicatst service:

- **Market Feed Config Url** channel parameters for receiving market data from the exchange are described in XML files. These files are available on the FTP server of the exchange. Specify here the path to the file, which describes connection to the ORDERS-LOG channel (separate configuration files are



available for real and testing data). Configuration file can be copied to a local server. In this case, the appropriate path should be specified in this field.

- **Market Feed FAST Templates Url** data transmitted via FAST protocol are unpacked using the special template. Specify here the path to the template file, it is available on the Exchange's FTP server. You can specify the FTP path or copy the file to a local server and specify the appropriate local path to it.

Market channel parameters and template files downloaded via FTP are saved in the directory [gateway folder]\downloads. If the gateway fails to download updated files, it will use earlier downloaded files stored in this folder. The following rules apply here:

- The gateway performs a checks that the file is not older than one week. If it was downloaded earlier, it is considered obsolete and will not be used.
- If path to a local file instead of FTP is specified in parameters Market Feed Config Url and Market Feed FAST Templates Url, file opening will fail and the existing file from the 'downloads' directory will not be used.

- **Market Feed A Bind** the address of the local interface the connection will be bound to, to receive market data through channel A (all data from the Exchange are duplicated on two channels).
- **Market Feed B Bind** the address of the local interface the connection will be bound to, to receive market data through channel B (all data from the Exchange are duplicated on two channels).

If at least one of the above parameters is not filled, the gateway will receive Market Depth data from Plaza 2 streams.

Also, the gateway automatically switches to receiving of Market Depth data from Plaza 2 in the following situations:

- If it is unable to subscribe to FAST Multicast newsletter
- In case of issues while reading the incoming data stream
- If the status of order journal of an instrument cannot be restored after an error, for more than 5 minutes

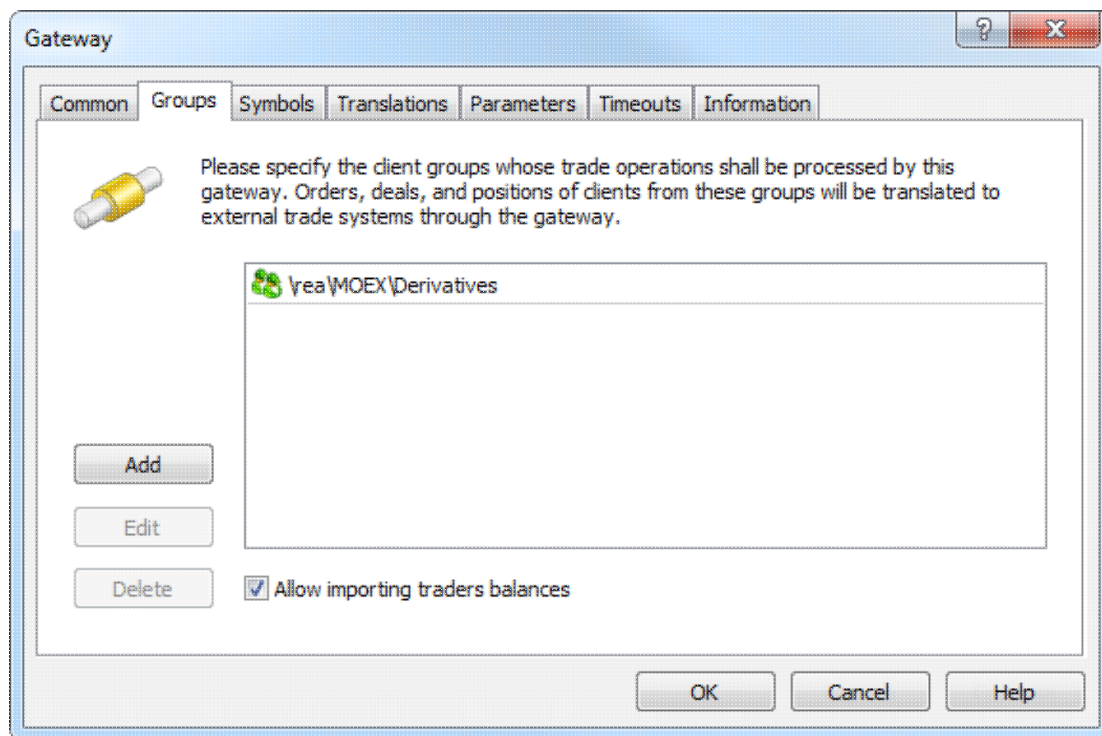
Appropriate messages are written to the gateway log in this case:

|                         |         |                                                          |
|-------------------------|---------|----------------------------------------------------------|
| 2019.02.07 09:58:12.350 | Gateway | failed to initialize FAST Market Data service, service v |
| 2019.02.07 09:58:48.856 | Gateway | FAST Market Data service stopped, market depth cha       |
| 2019.02.07 09:58:59.657 | Gateway | can not recover FAST market data state more than 30      |
| 2019.02.07 09:58:59.657 | Gateway | failed to apply FAST Market Data transaction, market     |

Please contact the exchange to connect FAST Multicast services of the derivatives market to your servers in the collocation zone.

**On the Groups tab**, specify the mask for the groups of clients who will trade in the derivatives market. The gateway will only receive trade requests from the clients

included in the allowed groups list.



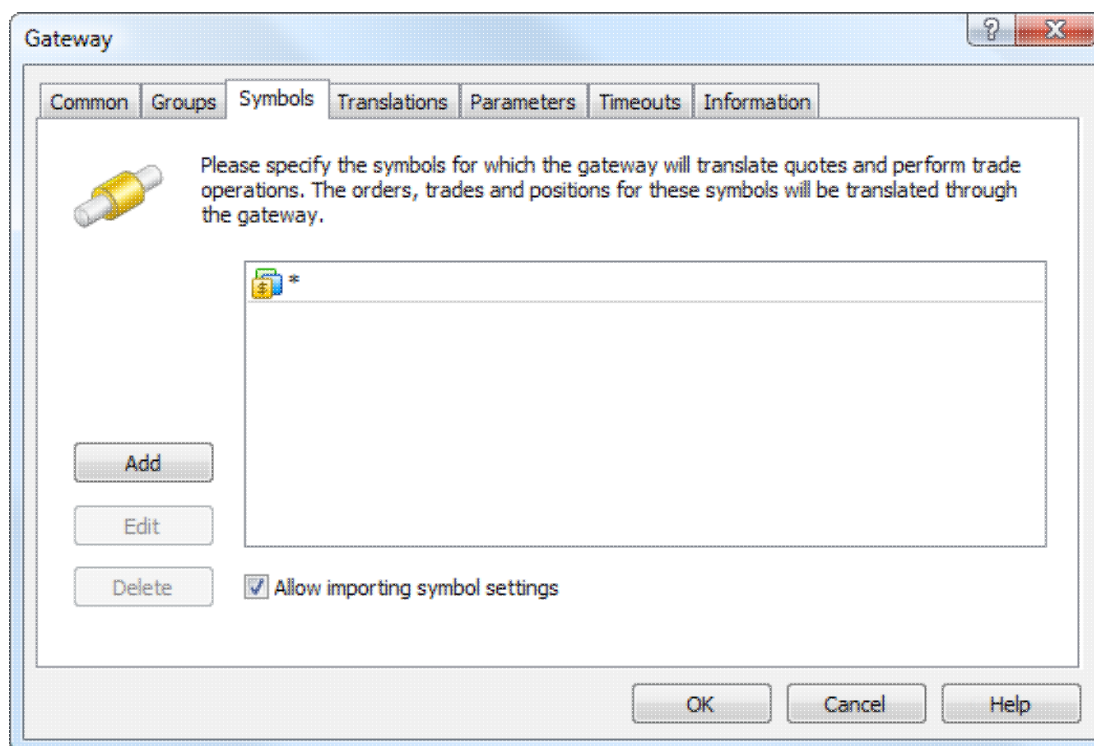
If you configure importing the limits via the gateway, make sure to enable "Allow importing traders balances" option. This will allow the gateway to synchronize clients' balances in MetaTrader 5 with exchange account limits.

Limits are synchronized in the following cases:

- launching/re-launching the gateway during a trading session;
- five minutes before the start of the main trading session;
- after the trading session when the clearing data is ready;
- in case funds are withdrawn/accrued on the client's stock account.

A group of clients trading in the derivatives market should be configured correctly.

**On the Symbols tab**, specify the symbols available for the gateway. The gateway will receive quotes and perform trade operations for these symbols. Also, the settings will be automatically updated for them in case import of symbols is allowed:



It is recommended to enable all symbols at this tab (\*). Also, make sure to enable "Allow importing symbol settings" option. MetaTrader 5 Gateway to MOEX Derivatives adds necessary symbols and manages their settings on its own.

- If the Symbols Path parameter is set, the gateway will import all new symbols to the subgroup specified in it. The trading ability is enabled immediately for all symbols.
- If the Symbols Path parameter is not set, the gateway will import symbols to the "\Preliminary" subgroup. In this case, the system administrator must move imported symbols to the proper subgroup and allow trading for them.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the configuration is updated. In this case the symbol is not transferred and its trading ability is not turned off.
- In case some changes are implemented to the Depth of Market parameter of the symbol settings, the gateway and a history server must be restarted to let the changes take effect. In fact, restart is required after any change in the symbol settings.

- On Moscow Exchange's Derivatives Market, an instrument cannot be traded during the evening session of the day on which it appeared. At an attempt to trade such an instrument, the gateway will return the "market closed" error.
- The gateway supports conversion of symbols and quotes. For details, please view the Symbol and Price Translation section.

#### Groups Configuration and Using Profit/Loss for the Current Trading Session

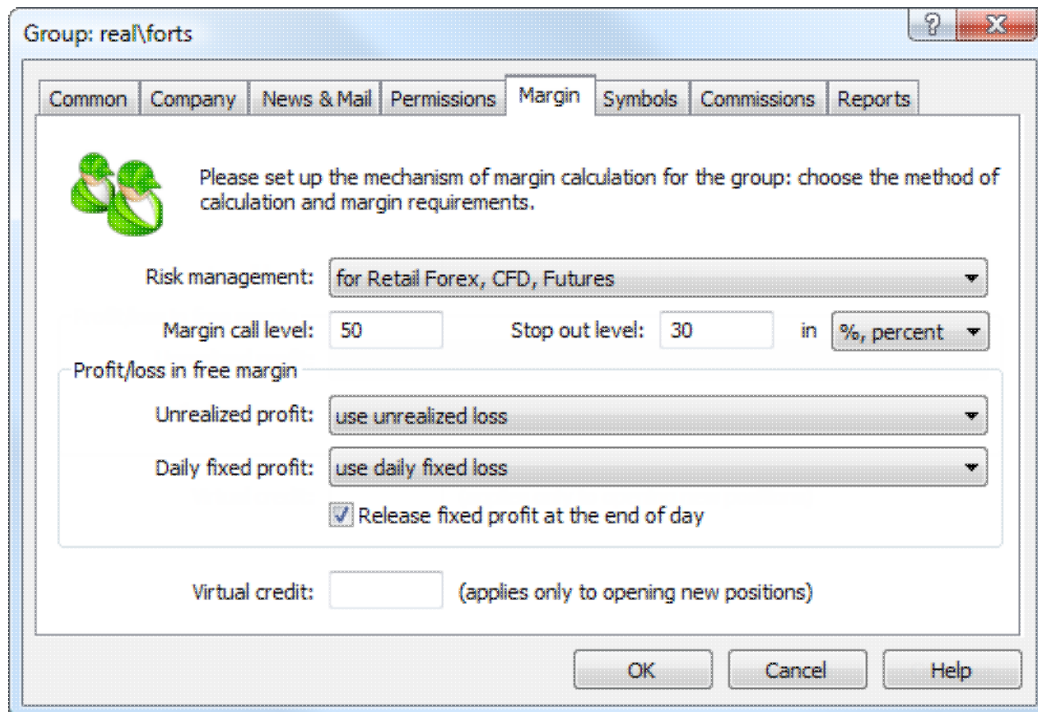
According to the exchange rules, profit obtained during a trading day cannot be used before clearing is performed. In fact, only clearing determines the financial results of

## MetaTrader 5 trading platform

all transactions during the day. The profit received by a client during the current trading session is accumulated during the trading session and released only after intermediate or main clearing becoming available for use by the client for further trade.

In order to avoid a situation in which a client who suffered losses will try to continue trading, fixed loss is withdrawn immediately.

In this regard, the following parameters should be specified in "Profit/loss in free margin" block of Margin tab in the settings of the group that will operate on the derivatives market:



The screenshot shows the 'Margin' tab of the 'Group: real\forts' settings window. The window has a title bar with a question mark and a close button. Below the title bar is a tabbed interface with tabs for 'Common', 'Company', 'News & Mail', 'Permissions', 'Margin' (selected), 'Symbols', 'Commissions', and 'Reports'. The main content area contains a green icon of two people and a text prompt: 'Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.' Below this, there are several settings: 'Risk management:' with a dropdown menu set to 'for Retail Forex, CFD, Futures'; 'Margin call level:' with a text input '50'; 'Stop out level:' with a text input '30'; and a unit selector 'in % , percent'. A section titled 'Profit/loss in free margin' contains 'Unrealized profit:' with a dropdown set to 'use unrealized loss', 'Daily fixed profit:' with a dropdown set to 'use daily fixed loss', and a checked checkbox 'Release fixed profit at the end of day'. At the bottom, there is a 'Virtual credit:' text input and a note '(applies only to opening new positions)'. The window ends with 'OK', 'Cancel', and 'Help' buttons.

Group: real\forts

Common Company News & Mail Permissions **Margin** Symbols Commissions Reports

Please set up the mechanism of margin calculation for the group: choose the method of calculation and margin requirements.

Risk management: for Retail Forex, CFD, Futures

Margin call level: 50 Stop out level: 30 in % , percent

Profit/loss in free margin

Unrealized profit: use unrealized loss

Daily fixed profit: use daily fixed loss

☒ Release fixed profit at the end of day

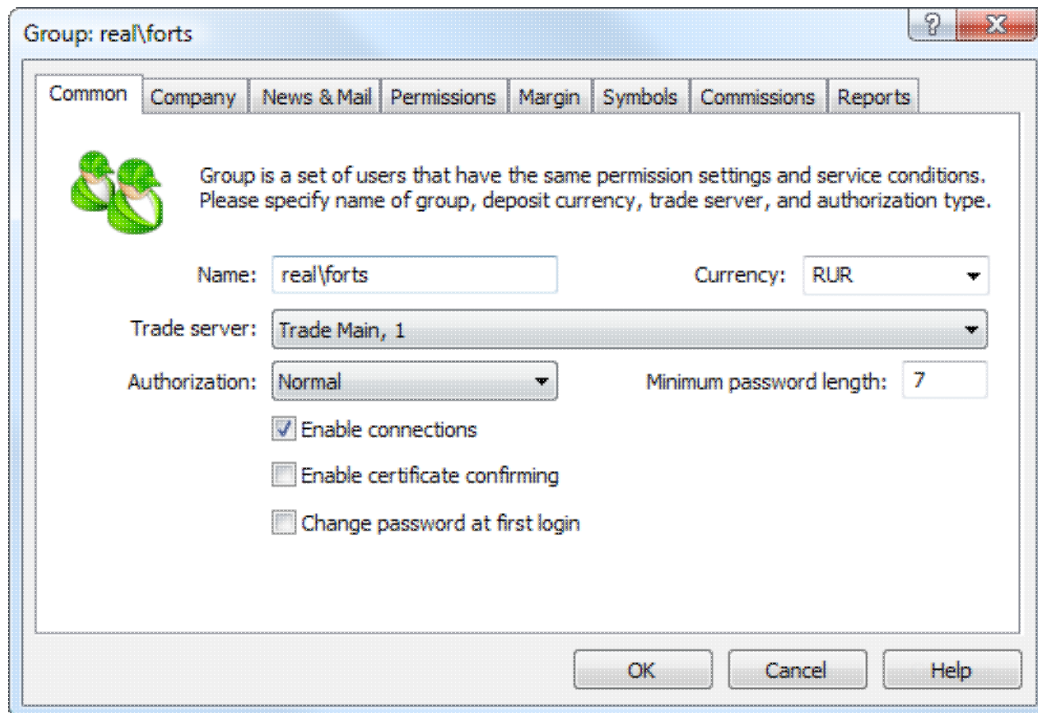
Virtual credit: (applies only to opening new positions)

OK Cancel Help

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Stop Out value specified in this tab does not matter. Orders and positions created via the gateway are not removed/not closed on MetaTrader 5 side when Stop Out is triggered. Find more information on margin settings in the [appropriate section](#).

All financial accounting on the derivative market is done in Russian rubles. Thus, Russian rubles (RUR) should be specified as deposit currency in the group settings:



### Configuring trade requests routing

To direct client requests concerning derivatives market symbols to the gateway, the [routing rules](#) should be set. Examples can be found in the articles concerning the gateways to other trading systems, for example: [GBOT](#), [DGCX](#), etc.

### Configuring trading accounts

Each client on MetaTrader's side should be provided with a corresponding separate trading account on the exchange. After an account is opened at the exchange, it should be connected with the account within the platform. Open the appropriate client account and move to Account tab:

## MetaTrader 5 trading platform

Account: 2013, John Smith, USD

Overview Personal **Account** Security

Group:

Color: ☐ None      Leverage:

Bank account:       Agent account:

☒ Enable this account      ☒ Enable trading by Expert Advisors

☒ Enable trading      ☒ Enable Web API connections

☒ Enable to change password      ☒ Enable daily reports

☒ Enable trailing stops      ☐ Change password at next login

Trade accounts:

| Gateway ID                                                                                                     | Account |
|----------------------------------------------------------------------------------------------------------------|---------|
|  40, MOEX Derivatives Gateway | 001     |

## MetaTrader 5 trading platform

A new entry should be added in "Trade accounts" section. Select MetaTrader 5 Gateway to MOEX Derivatives in "Gateway ID" field. Specify the client code in "Account" field. The prefix corresponding to Broker Code parameter in the gateway settings is removed from the client code. During the gateway operation, the final client's code is generated using Broker Code and the client code. For example, if Broker Code is Q1DU, while client code is 001, then the client account at the exchange is Q1DU001. Similarly, if Broker Code is Q1 and the client code is ST005, then the client account at the exchange is Q1ST005.

### Risk management system

Considering trading status of each client account allows the exchange to perform client-by-client risk management. Such control is an addition to the conventional risk management system of MetaTrader 5 platform. Thus, the ability of an end customer to perform each trading operation is controlled both on MetaTrader 5 and exchange's levels.

The gateway imports trading symbols with the appropriate settings, so that the collateral calculation algorithm for orders and positions fully corresponds to the exchange calculation algorithm.

MetaTrader 5 risk management system provides control of client positions and their collateral in real time both on client terminal's and trade server's sides. Besides, the platform allows accounting total positions of all clients of a brokerage company via MetaTrader 5 Manager terminal. The system controls occurrence of Margin Call and Stop Out status for each client sending appropriate notifications both to a client and brokerage company managers if necessary.

### Processing of Market orders

Since placing of market orders on the Moscow Exchange derivatives market is not supported, the gateway forwards market orders as limit orders with the minimum or maximum price depending on the order direction. The current session highest price is used for Buy orders, while the lowest price is used for Sell orders.

### Synchronizing trading status

If a client's exchange account already has a fund limit and/or open positions at the time of the gateway launch, the account status should be synchronized with MetaTrader 5. To do this, highlight the line of the gateway and the corresponding number of the account at the exchange in "Trade accounts" section. Then click Synchronize and Synchronize All.

After that, the balance and the client's open positions will be brought in line with the exchange. Synchronization of pending orders is not supported. Below is an example of a trading account in MetaTrader 5 client terminal after synchronization:

## MetaTrader 5 trading platform

| Toolbox                                                                                                                 |       |                  |      |        |        |       |       |        |        |         |         |
|-------------------------------------------------------------------------------------------------------------------------|-------|------------------|------|--------|--------|-------|-------|--------|--------|---------|---------|
| Symbol                                                                                                                  | Order | Time             | Type | Volume | Price  | S / L | T / P | Price  | Profit | Comment |         |
| si-6.13                                                                                                                 |       | 2013.03.20 13:11 | buy  | 3.00   | 31341  | 0     | 0     | 31358  | 51.00  | x       | HO53000 |
| rts-6.13                                                                                                                |       | 2013.03.20 13:11 | sell | 1.00   | 143540 | 0     | 0     | 143530 | 6.18   | x       | HO53000 |
| Balance: 99 996.50   Equity: 100 053.68   Margin: 15 069.64   Free Margin: 84 926.86   Margin Level: 663...   57.18     |       |                  |      |        |        |       |       |        |        |         |         |
| Trade   Exposure   History   News   Mailbox   Calendar   Market   Alerts   Signals   Code Base 1158   Experts   Journal |       |                  |      |        |        |       |       |        |        |         |         |

This example shows that two open positions have appeared at the account after the synchronization. Client's account at the exchange is shown in Comment column. The balance and account limit at the exchange have also been synchronized. The lower line of Trade tab displays the current status of the client's account:

- **Balance** client limit, not accounting for the results of currently open positions;
- **Equity** money on the client account regarding financial results of currently open positions;
- **Margin** amount of funds reserved on the client's account as a collateral;
- **Free Margin** amount of free funds that can be used for trading. This parameter is calculated as Equity - Margin;
- **Margin Level** percentage of the account equity to the margin volume (Equity / Margin \* 100).

As a result of the account synchronization with the exchange, deals marked by special comments appear in the client's history:

Toolbox

| Time /                                                                                               | Deal     | Order   | Symbol   | Type       | Direction | Volume | Price  | Profit  | Comment                      |         |         |
|------------------------------------------------------------------------------------------------------|----------|---------|----------|------------|-----------|--------|--------|---------|------------------------------|---------|---------|
|  2013.03.20 13:11 | 811192   |         |          | correction |           |        |        | -3.50   | [synchronization]            |         |         |
|  2013.03.20 13:11 | 811193   |         | si-6.13  | buy        | in        | 3K     | 31341  |         | [opened due synchronization] |         |         |
|  2013.03.20 13:11 | 811194   |         | rts-6.13 | sell       | in        | 1.00   | 143540 |         | [opened due synchronization] |         |         |
| Profit: 0.00 Credit: 0.00 Deposit: 100 000.00 Withdrawal: 0.00 Balance: 99 996.50                    |          |         |          |            |           |        |        | -3.50   |                              |         |         |
|                                                                                                      |          |         |          |            |           |        |        |         |                              |         |         |
| Trade                                                                                                | Exposure | History | News     | Mailbox    | Calendar  | Market | Alerts | Signals | Code Base 1158               | Experts | Journal |



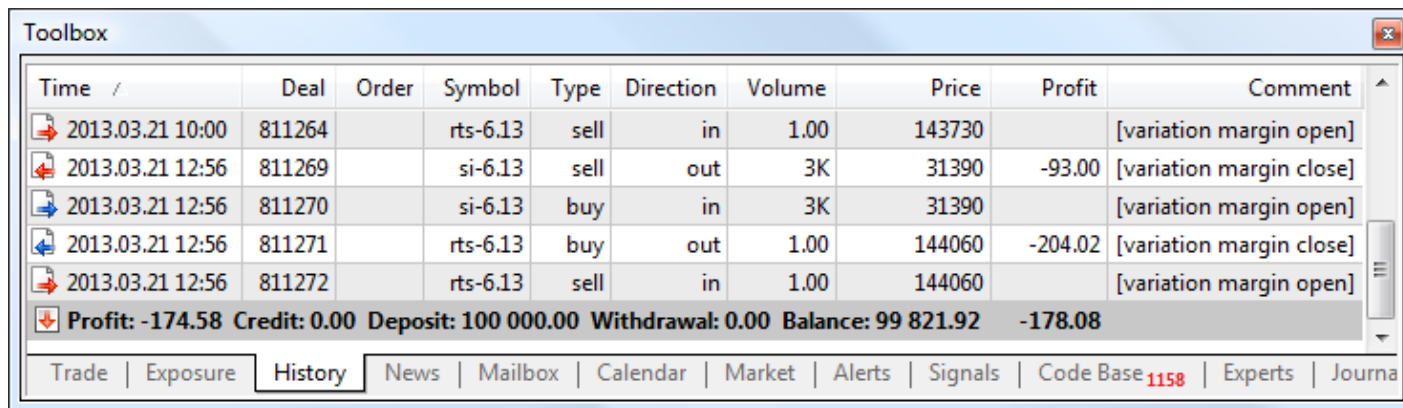
## MetaTrader 5 trading platform

In the example above, the client's exchange account had a limit equal to 100 000 RUR, as well as two positions. After synchronization, four entries have appeared in the history:

- The "balance" type deal used for synchronizing the client's balance and account limit at the exchange. The deal marked as "Synchronization".
- The "correction" type deal used for synchronizing the client's balance and account limit at the exchange. The deal marked as "Synchronization".
- The deal for buying si-6.13 symbol marked as "opened due synchronization".
- The deal for selling rts-6.13 symbol marked as "opened due synchronization".

### Charging variation margin

The variation margin is charged based on the results of intermediate and main clearing. The next example shows the deals in the client history that are created as a result of charging the variation margin.



| Time /                                                                                              | Deal   | Order | Symbol   | Type | Direction | Volume | Price  | Profit  | Comment                  |
|-----------------------------------------------------------------------------------------------------|--------|-------|----------|------|-----------|--------|--------|---------|--------------------------|
| 2013.03.21 10:00                                                                                    | 811264 |       | rts-6.13 | sell | in        | 1.00   | 143730 |         | [variation margin open]  |
| 2013.03.21 12:56                                                                                    | 811269 |       | si-6.13  | sell | out       | 3K     | 31390  | -93.00  | [variation margin close] |
| 2013.03.21 12:56                                                                                    | 811270 |       | si-6.13  | buy  | in        | 3K     | 31390  |         | [variation margin open]  |
| 2013.03.21 12:56                                                                                    | 811271 |       | rts-6.13 | buy  | out       | 1.00   | 144060 | -204.02 | [variation margin close] |
| 2013.03.21 12:56                                                                                    | 811272 |       | rts-6.13 | sell | in        | 1.00   | 144060 |         | [variation margin open]  |
| <b>Profit: -174.58 Credit: 0.00 Deposit: 100 000.00 Withdrawal: 0.00 Balance: 99 821.92 -178.08</b> |        |       |          |      |           |        |        |         |                          |

Trade | Exposure | **History** | News | Mailbox | Calendar | Market | Alerts | Signals | Code Base 1158 | Experts | Journal

Two pairs of deals are displayed in the example:

- closing positions at a settlement price for si-6.13 and rts-6.13 symbols with [variation margin close] comment;
- re-opening positions at a settlement price for si-6.13 and rts-6.13 symbols with [variation margin open] comment.

### Profit for the Current Session

The profit received by a client during the current trading session is unavailable for re-investing. It is accumulated during the trading session and released only after intermediate or main clearing becoming available for use by the client for further trade.

Below is an example of closing a profitable trading position during the trading session.

## MetaTrader 5 trading platform

| Toolbox                                                                                                                                                         |       |                  |      |        |        |       |       |        |        |         |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------|------|--------|--------|-------|-------|--------|--------|---------|--|
| Symbol                                                                                                                                                          | Order | Time             | Type | Volume | Price  | S / L | T / P | Price  | Profit | Comment |  |
| rt-6.13                                                                                                                                                         |       | 2013.03.21 13:40 | sell | 1.00   | 144040 | 0     | 0     | 143870 | 105.17 | x       |  |
| <b>Balance: 766 404.98 Equity: 766 510.15 Margin: 10 154.78 Free Margin: 756 250.20 Margin Level: ... 105.17</b>                                                |       |                  |      |        |        |       |       |        |        |         |  |
| <b>Trade   Exposure   History   News   Mailbox   Calendar   Market   Alerts   Signals   Code Base <span style="color: red;">1158</span>   Experts   Journal</b> |       |                  |      |        |        |       |       |        |        |         |  |

Profit received after closing the position will be shown in the balance but it will not be available for use till the intermediate or main clearing. The total accumulated profit is displayed in "Blocked" field of the trading status line:

| Toolbox                                                                                                                                                         |       |      |      |        |       |       |       |       |        |         |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|------|--------|-------|-------|-------|-------|--------|---------|--|
| Symbol                                                                                                                                                          | Order | Time | Type | Volume | Price | S / L | T / P | Price | Profit | Comment |  |
| <b>Balance: 766 576.20 Equity: 766 470.10 Blocked: 106.10 Free Margin: 766 470.10 0.00</b>                                                                      |       |      |      |        |       |       |       |       |        |         |  |
| <b>Trade   Exposure   History   News   Mailbox   Calendar   Market   Alerts   Signals   Code Base <span style="color: red;">1158</span>   Experts   Journal</b> |       |      |      |        |       |       |       |       |        |         |  |

"Blocked" field value is reset to zero after the clearing.

### Depositing/Withdrawing Funds from the Account

When depositing or withdrawing the funds from the account, the limit at the exchange is synchronized with the client's balance on MetaTrader 5 side. The appropriate balance operations with comments like "FORTS, account XXX", where XXX is the number of the client's account, appear in the client's deals history.

Updating the account limit in MetaTrader 5 does not lead to the account limit update at the exchange, as such a mechanism is currently not provided.

### Charging commissions

The exchange charges commission for each performed deal. Commission value is received by the gateway and added to the description of each deal performed at the exchange in Commission field.

Brokerage companies can take account of their own commissions in any way. In actual practice, commission is usually considered at the broker's back-office level and affects the client limits before the start of the trading session. However, it is impossible to identify this commission in the history of client deals, because there is no clear indication that this is exactly a broker's commission in balance operations performed during the synchronization of limits.

Besides, broker's commission can be additionally taken into account by means of MetaTrader 5. More details are available in the ["Commission settings"](#) section. In this case, commission will also be considered at the back-office level and affect client

## MetaTrader 5 trading platform

limits before the start of the trading session but the amount of the commission will be shown in real time in the client terminals.

Accounting of commissions in MetaTrader 5 should strictly comply with accounting at the broker's back-office level.

### Delivery of futures contracts

All futures contracts delivery deals are displayed in clients' history as special technical deals performed by the exchange.

- **Settlement contracts** positions are closed at a settlement price and the appropriate profit/loss is deposited/withdrawn.
- **Equity futures** upon delivery, technical deals of positions closing are performed. An appropriate position is opened for a client on the market.

### History Data Synchronization

During the conventional gateway operation, time and prices used to generate bars strictly correspond to the exchange data. All charts are generated using the prices and time of actual deals executed at the exchange.

After the end of a trading session, the exchange publishes the official report on the performed deals. This report is used by the gateway for data management. In case of any discrepancies, historical data is corrected in accordance with the report.

If the gateway is temporarily stopped and then relaunched again during the trading session, missing fragments of historical data are automatically restored (synchronized).

During the initial installation in MetaTrader 5, the gateway automatically imports historical data for all symbols for the last year since the launch of the gateway.

### Password change

In accordance with the Exchange policy, the account password for connection to MOEX must be changed every 90 days. Logging in with an expired password is impossible.

To avoid an unexpected interruption of your gateway due to an expired password, it can use a special mechanism for automatic password change. The mechanism is controlled via the "Password Change" parameter.

If the "Password Change" parameter is disabled (the value is "No"), then 30 days before the expiration of the current password the gateway starts displaying an appropriate message, prompting to change the password. This message will appear in the log every day, before the trading start time:

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| Gateway password for current exchange login 'TradingLogin' will expire in 15 days at 2020 |
|-------------------------------------------------------------------------------------------|

## MetaTrader 5 trading platform

If the parameter is enabled (the value is "Yes"), then 30 days before the password expires, the gateway starts displaying a warning every day before the start of trading, notifying that the password will be changed automatically 7 days before its expiration:

Gateway password for current exchange login 'TradingLogin' will expire in 14 days at 2020.1

7 days before password expiration, the gateway sends a password change request to the exchange. The new password is generated randomly. It will be displayed twice in the gateway log: before sending a request to the exchange and after receiving a password change confirmation:

Gateway password for current exchange login 'TradingLogin' will expire in 6 days at 2020.1

Gateway change password request sent, new password: NE@1fFtt

Gateway current password for login 'TradingLogin' on spectra-t1.moex.com:3001 changed,

After the password is changed, the gateway will start using the new password value for further connections, and this no additional action is required.

- The new password is saved to the settings.dat file in the gateway working directory. If this file is deleted, enter the new password manually in gateway settings. It can be found in the gateway log as described above.
- When the password is changed using third-party software or upon request to the exchange, specify a new password in gateway settings. This is enough for further gateway operation.

### Gateway Operation Management

To receive data on the gateway operation, request its logs via MetaTrader 5 Administrator. Select the gateway and move to Journal tab.

The following entries indicate that the gateway started its operation successfully:

Gateway synchronized with FORTS server

The following entry indicates successful synchronization of the clients' limits:

Gateway clients limits synchronized

Use "session" keyword to request data on trading session completion. Session ID (4293 in the example), as well as time of its completion (2013.04.09 12:45:00) are specified in such entries.

Gateway trade session 4293 on SEBM-4.13 finished at 2013.04.09 12:45:00

To receive data on the session's settlement price, request the journal using "variation margin" key phrase. In the example below, the settlement price for RTS-6.13 symbol is equal to 139750:

Gateway execution sending complete - variation margin **for** RTS-6.13, settlement price: 139

The following special entries are made in the gateway journal to analyze operations processing speed:

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| Gateway order #736398 - message process time: 31.322 ms (gateway time: 0.331 ms, exchange time: 31.0 ms) |
|----------------------------------------------------------------------------------------------------------|

The following parameters are shown here:

- message process time - total time between sending a message to the exchange and receiving an answer;
- gateway time - time spent by the gateway to send a message;
- exchange time - time spent by the exchange to process a message and send an answer to the gateway.

Apart from the gateway journal, P2MORouter module operation journal can also be analyzed in case of any issues. This module is launched by the gateway when starting its operation. Its operation journals are located at the history server in \Gateways\FORTS Gateway\[gateway configuration name]\Plaza 2\logs directory.

### Gateway Service Operations

The gateway can perform various service operations on accounts during operation.

Deals with the type "Correction" and with the comment "[synchronization]" are formed in the following cases:

- When you manually synchronize an account from the Administrator terminal
- When synchronizing limits Time To Sync Limits minutes before the start of the morning session, if the Limits Sync Full parameter is enabled

Deals with the type "Correction" and with the comment "FORTS, account XXXXX" are formed in the following cases:

- When synchronizing limits Time To Sync Limits minutes before the start of the morning session, if the Limits Sync Full parameter is enabled
- When synchronizing limits, if the gateway is launched during the trading session or later than "Time To Sync Limits" minutes before the start of the trading session
- When synchronizing limits after the completion of the main clearing

Deals with the type "Balance" and with the comment "FORTS, account XXXXX" are formed in case the balance on the exchange is changed during the trading session: In this case, you should contact the exchange to find out the reasons for balance changes. This is usually associated with replenishment/withdrawal of funds on the exchange account.

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Gateways](#) / DGCX

## MetaTrader 5 Gateway to DGCX

MetaTrader 5 Gateway to DGCX allows trading on [the Dubai Gold and Commodities Exchange](#).

### About Dubai Gold and Commodities Exchange

Dubai has historically been an international hub for the physical trade of not only gold, but also many other commodities. So the establishment of the [Dubai Gold & Commodities Exchange \(DGCX\)](#) was the next logical step for the region and the local economy. DGCX commenced trading in November 2005 as the region's first commodity derivatives exchange. Today it is the leading derivatives exchange in the Middle East.

### Benefits of Trading on DGCX

DGCX range of futures contracts offers participants of the physical commodities markets, such as producers, manufacturers and end users, a sophisticated means of hedging their price risk exposure. Such price risk management has previously been unavailable to producers in the Middle East. In addition, DGCX offers trading opportunities to financial communities and investment houses in both the Middle East and around the globe that wish to access the growing asset class of commodity and currency derivatives.

- Guaranteed settlement and reduced counterparty risk provided by Dubai Commodities Clearing Corporation (DCCC), a subsidiary 100% owned by DGCX
- The advantage of transacting and clearing business within the UAE and thus the local taxation and regulatory regimes
- A simple fee structure - one fee for all participants. All participants also pay the same margin, whether commercial or non-commercial entities
- Access to both regional and international liquidity pools
- Robust risk management and surveillance systems
- Uninterrupted trading hours from 7:00 to 23:30 (GMT +04:00)
- Regulated by the Emirates Securities & Commodities Authority (ESCA)

### Product Portfolio

The UAE enjoys an ideal location between the time zones of Europe and the Far East and DGCX offers a range of products from the precious metal, base metal, energy and currency sectors.

| Futures         |            |                  |             | Options |
|-----------------|------------|------------------|-------------|---------|
| Precious metals | Currencies | Energy resources | Base metals |         |
| Gold            |            |                  |             |         |

## MetaTrader 5 trading platform

| Futures |                             |                            |             | Options                 |
|---------|-----------------------------|----------------------------|-------------|-------------------------|
| Silver  | Australian Dollar/US Dollar | WTI Light Sweet Crude Oil  | Steel Rebar | Options on Gold Futures |
|         | British Sterling/US Dollar  | Brent Crude Oil            |             | Options on INR Futures  |
|         | Canadian Dollar/US Dollar   | Fuajairah 380 CSR Fuel Oil |             |                         |
|         | Euro/US Dollar              |                            |             |                         |
|         | Indian Rupee/US Dollar      |                            |             |                         |
|         | Japanese Yen/US Dollar      |                            |             |                         |
|         | Swiss Franc/US Dollar       |                            |             |                         |

How to become a broker at DGCX

All rules of the Dubai exchange can be found in [Regulatory section of the official web site](#). The information on how to become a member and start providing brokerage services on the exchange can also be found at [Membership](#) section of the official web site. Additionally, you can contact DGCX via e-mail [support@dgcx.ae](mailto:support@dgcx.ae). After obtaining the necessary information and registering as a market member, it is time to configure the gateway.

### Order MetaTrader 5 DGCX Gateway

MetaTrader 5 DGCX Gateway allows quick integration of MetaTrader 5 platform with Dubai Gold and Commodities Exchange. By ordering the gateway now you provide your clients with new trade opportunities in the financial markets!

### Preparing for the Gateway Launch, Configuring VPN

Before setting a brokerage company gateway, configure the virtual network channel to be able to connect to DGCX servers.

Configuration of a virtual network channel is performed together with DGCX technical specialists. To create a virtual channel to DGCX, a broker must request technical requirements and recommendations for connection to the DGCX virtual private network (VPN) from the exchange technical support team. After obtaining the necessary information a brokerage company should purchase a recommended router.

At the next stage the mercantile exchange technical support team will submit to the

## MetaTrader 5 trading platform

brokerage company description of the equipment settings on DGCX side: tunnel type, data encryption method, data encryption keys etc. The router must be configured according to the obtained data.

In addition, the brokerage company must specify its own equipment parameters and send this information to the mercantile exchange technical support team in the form received from DGCX. When all agreed equipment settings are configured at both a brokerage company and the exchange sides, the virtual channel for connection to DGCX will be ready for operation.

You can skip procedures related to establishing connection and to setting up secure access channels, by safely locating the gateway on a ready-made server provided by the exchange (collocation). It can be used in remote mode through a regular connection. For further collocation details please contact Service Desk.

### How the Gateway Works

Gateway operates mainly as a mediator between two systems connecting DGCX and MetaTrader 5 platform. The gateway establishes several connections to DGCX server for the data exchange via the created virtual channel:

- The first connection is used for performing trading operations and working according to FIX (Financial Information eXchange) protocol.
- The second one is used to obtain quotation data and utilizes a broadcast data transfer binary protocol (EMAPI). Data can be transferred both via TCP and UDP multicast protocol.
- In addition, the gateway can establish the third connection to DGCX FTP server to get special risk parameters files used for calculating margin requirements and files containing data on opened client positions.



The broker's servers are configured in such a way that the clients' trade requests are sent to MetaTrader 5 DGCX Gateway. The gateway checks correctness of the client requests. In case validation is successful, the relevant trade request is directed to DGCX server via a standard protocol for exchanging financial information - FIX. After receiving a response from the DGCX server, the gateway saves the trade request processing result in the platform, which in its turn reports this result to the trader.



## MetaTrader 5 trading platform

Orders are sent to MetaTrader 5 DGCX Gateway for processing in accordance with the set routing rules. Depending on the order type, each request is handled differently:

- **Market orders** are transferred to DGCX system directly. In case there is no liquidity for a requested financial instrument, the market order is rejected.
- **Limit orders** are processed at DGCX. Once a limit order has been placed, it is sent to DGCX server. There it is placed to the general requests queue awaiting for an opposite request with the same price to appear. Thus, clients can see their requests in the Depth of Market in the client terminal in real time.
- **Stop orders** are sent to DGCX immediately after being placed. When the specified stop price is reached, stop order is activated on a stock exchange and an appropriate market order is sent to MetaTrader 5 platform.
- **Stop-limit orders** are handled and stored on DGCX server until their stop price is reached. Once a stop-limit order has been triggered, the corresponding limit order with a specified price will be sent to MetaTrader 5 platform.
- **Take Profit and Stop Loss levels** for the current open positions are controlled by MetaTrader 5 platform. In case of Stop Loss activation, a market order to close a client's position is sent to the exchange. In case of Take Profit activation, a limit order at the Take Profit price is sent to the exchange.
- **Stop out** of positions and orders is controlled by the trading platform. In case a limit order must be removed after a Stop out level has been reached, an appropriate request for its removal is sent to DGCX server. In case a position must be closed because of Stop out, a market request to close a client's position is sent to the exchange.

### MetaTrader 5 DGCX Gateway Setup

MetaTrader 5 DGCX Gateway is a separate "DGCXGateway64.exe" file that uses Gateway API for its operation. The gateway module is included in the platform delivery set and available for use in test mode. In this mode the number of trade operations is limited - after the limit is reached they will not be processed. To switch to the full version, order the MetaTrader 5 Gateway to DGCX Gateway.

To start working, add the new gateway configuration:

## MetaTrader 5 trading platform

The screenshot shows the 'Gateway' dialog box with the 'Common' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar are tabs: 'Common', 'Groups', 'Symbols', 'Translations', 'Parameters', 'Timeouts', and 'Information'. The 'Common' tab contains a description of the gateway, an 'Enable' checkbox, and several input fields. The 'Name' field is 'DGCX Gateway', 'ID' is '20', 'Module' is 'DGCXGateway64', 'Trading server' is '10.30.30.126:7250', 'Trading login' is '<login>', and 'Password' is masked with dots. At the bottom are 'OK', 'Cancel', and 'Help' buttons.

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

This gateway allows connecting to remote DGCX server for performing trade operations, receiving quotes. The connection requires a DGCX provided client account on the target server.

☒ Enable

Name: DGCX Gateway ID: 20

Module: DGCXGateway64 Trade and Quotes

Trading server: 10.30.30.126:7250

Trading login: <login> Password: .....

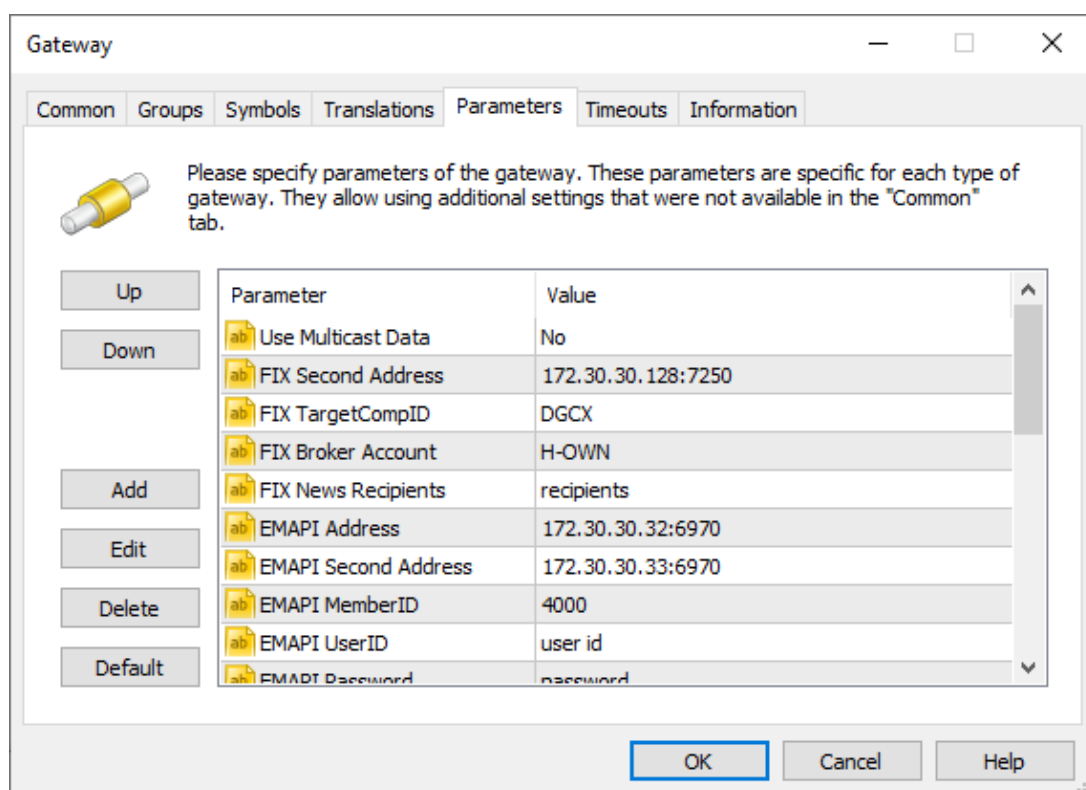
OK Cancel Help

The following parameters on the "Common" tab must be set:

- **Module** DGCXGateway64. After that the gateway default parameters installation must be allowed in the dialog request.
- **ID** unique dealer identifier, on whose behalf the trade requests will be processed. Requests are routed to the gateway according to this identifier.
- **Trading server** DGCX FIX server address must be entered in this field in the "address:port" format. The demo server address "10.30.30.126:7250" is shown in the provided example.
- **Trading login** user ID for FIX connection to DGCX (User id). The value 20041 is shown in the provided example.
- **Trading password** user password for FIX connection to DGCX.

- ID value must be unique in the field of manager logins and gateway identifiers.
- The data for connection to the DGCX server, where trade requests are processed, is submitted during the agreement conclusion.

Additional configuration parameters should be set on the Parameters tab:



Additional options should be provided by DGCX in the brokerage company connection specification.

- **Use Multicast Data** parameter defining the necessity of receiving quotes via UDP multicast connection. To do this, the value of the parameter should be set to "Yes" or "Y". Otherwise, TCP connection will be used to receive the quotes.
- **Bind Address** address of the network interface the gateway will interact with. The address is used in case there are several network interfaces on the computer, at which the gateway is operating.
- **FIX Second Address** address of the backup DGCX FIX server in "address:port" format. This server is automatically used in case it has been impossible to connect to the main FIX server after several attempts. The demo server address "10.30.30.127:7250" is shown in the provided example.
- **FIX TargetCompID** FIX messages heading standard parameter. This parameter is used to identify a recipient of trading messages. The parameter value is provided by DGCX.
- **FIX Broker Account** broker's trading account ID. It is used when all clients trade using a single general account.
- **FIX News Recipients** list of logins of MetaTrader clients separated by ";". These clients will receive e-mails containing news sent by DGCX.
- **EMAPI Address** main address for connecting to the DGCX "prelogon" channel in the "address:port" format. In the provided example, this is the address of the demo server "10.30.30.126:6970". The "prelogon" channel is used for a preliminary connection to the exchange. The gateway receives the exchange main working address from it (the "logon" channel). After that, the gateway connects to it for further operation.
- **EMAPI Second Address** backup address for connecting to the DGCX

"prelogon" channel in the "address:port" format. The system automatically switches to the server in case the main EMAPI server remains unavailable after several connection attempts. In the provided example, this is the address of the demo server "10.30.30.128:6970".

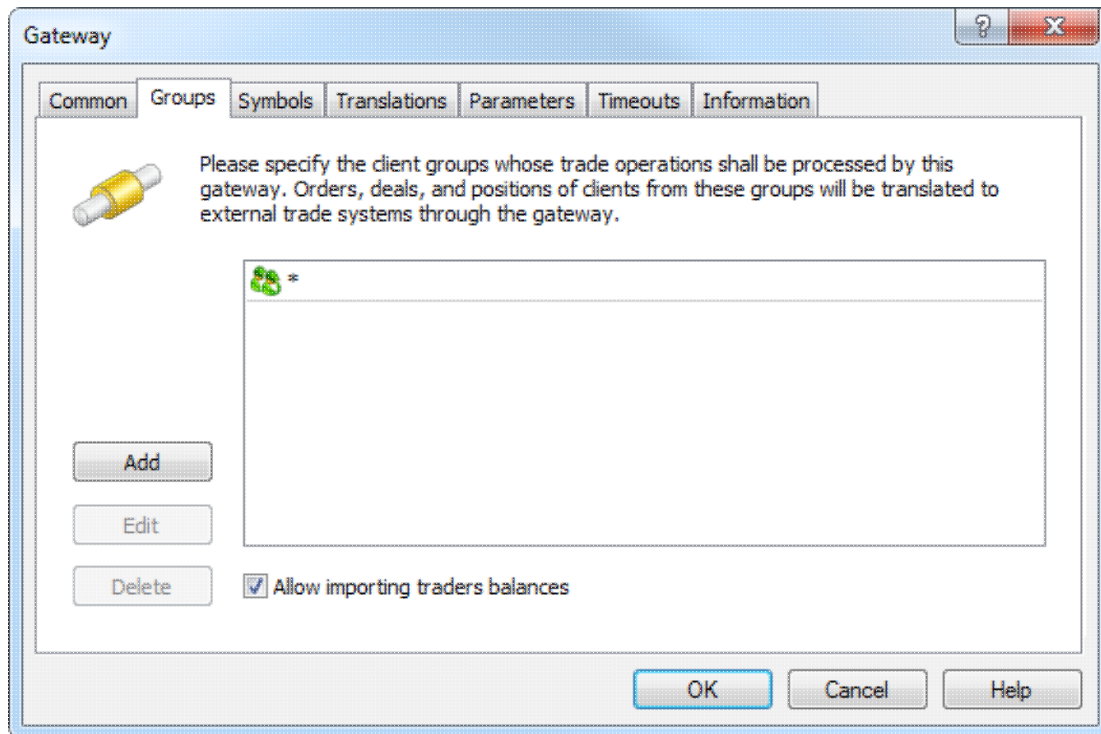
- **EMAPI Logon Address** "logon" address of the DGCX channel in the "address:port" format. This parameter allows redefining the address of the exchange main working server. If the parameter is set, connection is performed to the specified address rather than to the address obtained via the "prelogon" channel.
- **EMAPI MemberID** EMAPI member ID. The parameter value is provided by DGCX.
- **EMAPI UserID** EMAPI user ID. The parameter value is provided by DGCX.
- **EMAPI Password** EMAPI user password. The parameter value is provided by DGCX.
- **FTP Span Url** full path to the folder on the exchange's FTP server where the risk parameters files (SPAN risk parameter file) are stored. These files are used to calculate margin requirements. This connection is also used for receiving files containing data on opened client positions. The ftp or sftp protocols are supported. If the protocol is not specified explicitly in the address, ftp is used. The default value is sftp://eosftp.dgcx.ae:6022/Common/Parameter.
- **FTP Login** login for connection to the FTP server. The parameter is provided by DGCX.
- **FTP Password** password for connection to the FTP server. The parameter is provided by DGCX.
- **Import Span Spreads** if you set this parameter to "Yes", the gateway will import spreads. The default value is "No".
- **FTP Market Stats Url** an address of the folder on an FTP server, from which the gateway takes the market statistics files. The login and password used to receive span files are used for connection (the FTP Login and FTP Password parameters). The ftp or sftp protocols are supported. If the protocol is not specified explicitly in the address, ftp is used. The default value is sftp://eosftp.dgcx.ae:6022/Common/MarketStatistics.
- **TP Orders Coverage Timeout** Take Profit order expiration time in seconds. Take Profit orders execution is tracked on the MetaTrader 5 platform side. When triggered, an appropriate limit order for closing a position at the price of an activated Take Profit is sent to the market. The order lifetime is 20 seconds by default. If the exchange does not execute the order within this time, the order will be canceled. In this case, the Take Profit activation attribute is removed on the platform side. If the Take Profit is activated again on the next tick, another limit order is sent to the exchange. This behavior is implemented so that an order does not wait for its execution at an exchange for an indefinite period of time. The TP Orders Coverage Timeout parameter allows re-defining the lifetime of limit orders sent to an exchange when Take Profit is triggered. Set the necessary value in seconds. If there is no parameter, the value of 20 seconds is used.
- **Symbols Path** path to which trading symbols from the exchange will be imported. The parameter is absent by default and the gateway imports trading instruments to the \Preliminary\DGCX subgroup, with the disabled trading option. The system administrator should manually move the symbols to the proper group and enable trading for them. If the parameter is present, the

gateway will import trading instruments to the specified path. In this case, the trading option will be immediately enabled. Thus, there is no need for the administrator need to configure the symbols. For example, if "DGCX-Real" is specified in the parameter, the symbols will be automatically added to the DGCX-Real\\* group, while expired contracts will be moved to DGCX-Real\Expired\\*.

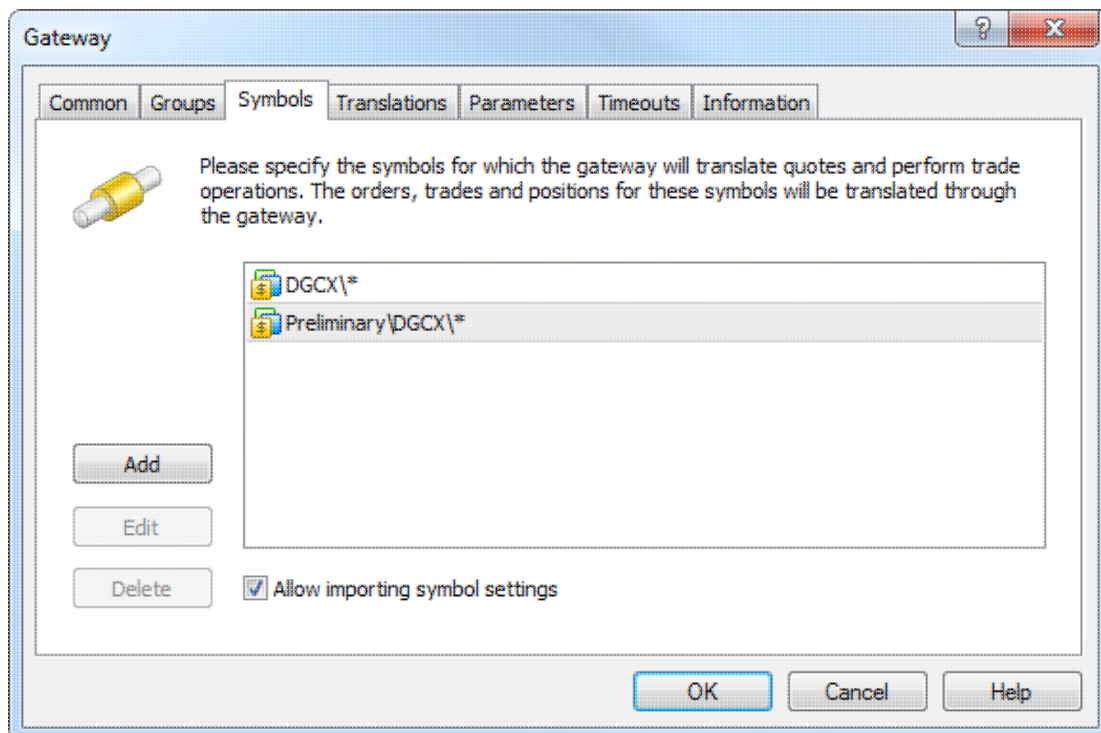
- **Symbols Path Expired** path to move expired symbols. The parameter is not specified by default, and the gateway moves all expired contracts to the \Expired subdirectory of the directory specified in "Symbols Path". In the above example, it is DGCX-Real\Expired\\*. By creating this parameter, you can override the path.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

The groups of the clients, whose requests will be processed via MetaTrader 5 DGCX Gateway, should be specified at the "Groups" tab.

## MetaTrader 5 trading platform



The "Symbols" tab allows configuring the list of symbols, according to which the gateway will process trade operations and transmit the quotes.



## MetaTrader 5 trading platform

Make sure to enable "Allow importing symbol settings" option. The gateway imports symbols from the DGCX to the MetaTrader 5 platform in accordance with the [Symbols Path](#) parameter. Then, in the future, the gateway will control the parameters of these symbols. In addition to various trading settings, trading and quotation sessions are also imported by symbols considering the time zones MetaTrader 5 platform and DGCX work in (GMT +04:00).

The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

### Configuring Trade Requests Routing

The next step is configuring the routing to let the clients requests to be transmitted to the DGCX gateway. To do this, add a [routing rule](#).

Select "Process to dealers" as an action in common settings. Assign this routing rule for all requests and orders. In additional conditions you may indicate the groups of clients, whose requests will be passed to the gateway, and the symbols, trading operations with which will be also passed to the gateway.

In the figure below all client orders by symbols from the DGCX\\* group will be sent to the gateway for processing.

Routing: DGCX Gateway

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: DGCX Gateway

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

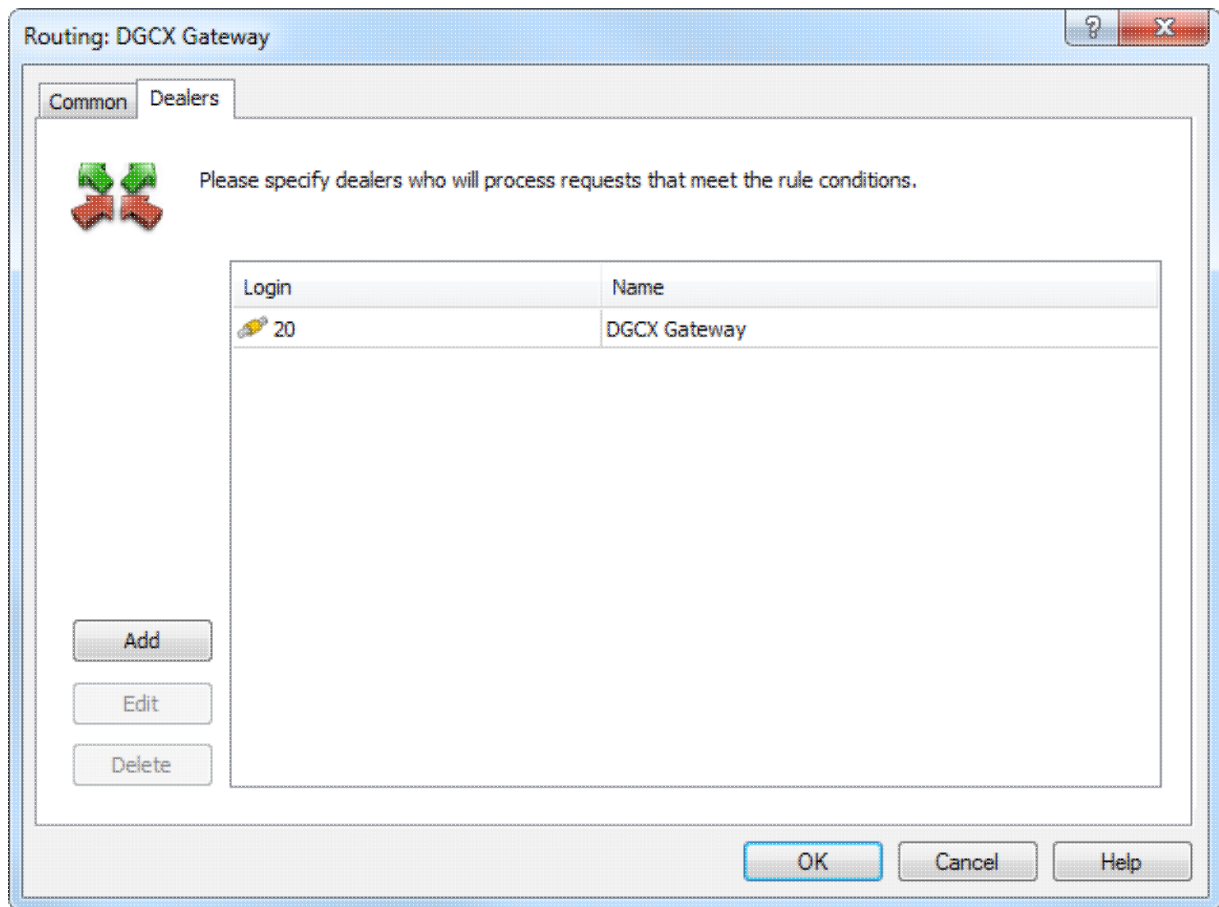
| Type       | Condition | Value  |
|------------|-----------|--------|
| ab Symbols | Equal (=) | DGCX\* |

Add Edit Delete

OK Cancel Help

## MetaTrader 5 trading platform

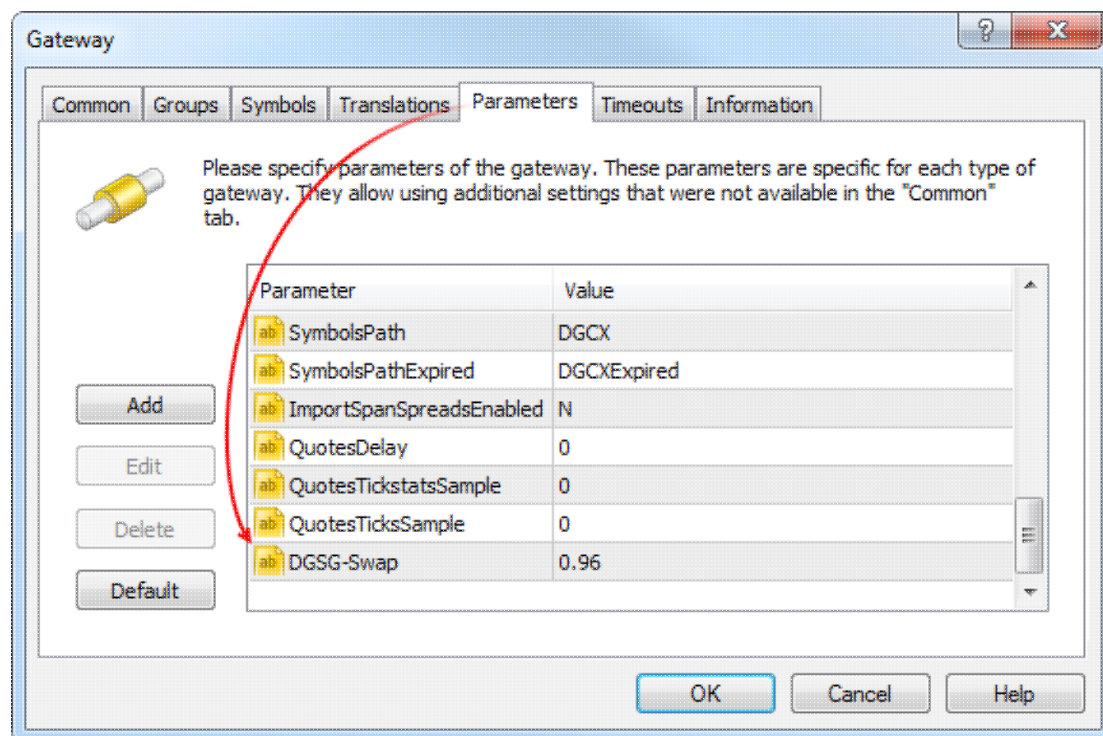
Only DGCX gateway should be added at the "Dealers" tab:





## Swap Setup

Swap size can be adjusted for specific symbols via the gateway parameters in the format <Symbol\_name>-Swap = <Swap value>. The screenshot shows configuration of the DGSG symbol as an example:



## Gateway Launch

After the application of all settings described above, your gateway is almost ready to go. It will connect to DGCX server.

|                     |         |                                                         |
|---------------------|---------|---------------------------------------------------------|
| 2013.04.01 11:06:22 | Gateway | 'DGCX Gateway' starting                                 |
| 2013.04.01 11:06:23 | Gateway | license check succeeded                                 |
| 2013.04.01 11:06:23 | Gateway | emapi: establishing prelogon connection to DGCX EMAPI s |
| 2013.04.01 11:06:24 | Gateway | emapi: prelogon completed                               |
| 2013.04.01 11:06:24 | Gateway | emapi: establishing logon connection to DGCX EMAPI serv |
| 2013.04.01 11:06:24 | Gateway | emapi: logon completed                                  |
| 2013.04.01 11:06:26 | Gateway | emapi: reference data synchronized                      |
| 2013.04.01 11:06:28 | Gateway | emapi: subscription groups data synchronized            |
| 2013.04.01 11:06:28 | Gateway | connecting to DGCX FIX server (10.30.30.126:7250)       |
| 2013.04.01 11:06:29 | Gateway | fix: logged on                                          |
| 2013.04.01 11:06:29 | Gateway | synchronized with DGCX exchange                         |

The last entry means that the gateway has successfully connected to DGCX and is now ready for operation. In case of connection failure, the entries concerning the failure reasons appear in the gateway journal.

New symbols are imported after the first connection. As noted above, the newly added symbols must be relocated from "Preliminary" group and trading for them must be

## MetaTrader 5 trading platform

allowed. After that the MetaTrader 5 platform main trade server must be restarted. These actions should be performed only during the gateway first launch.

After the main trading server is restarted, the gateway installation and configuration is complete. Traders can perform transactions on DGCX using their MetaTrader 5 client terminal.

### Gateway Operation

This section describes the standard procedures performed by the gateway during its operation.

#### Updating the Symbols Settings

The gateway automatically updates the symbols settings during its operation, if necessary. Appropriate entries appear in the journal in that case:

|                     |             |                                     |
|---------------------|-------------|-------------------------------------|
| 2013.04.01 11:06:30 | SymbolsBase | symbol 'DS-20130906' config updated |
|---------------------|-------------|-------------------------------------|

#### Multi-leg Instruments

DGCX provides possibility to trade multi-leg instruments. These are the instruments consisting of several subinstruments allowing to implement definite hedging strategies in trading. Such symbols are not supported by MetaTrader 5 platform and it is impossible to import them.

#### Risk Parameters

During a trading day and at the end of it DGCX publishes special span files on its FTP servers. These span files contain risk management parameters. The gateway downloads the files to \Gateway\DGCXGateway\DGCX Gateway\ftp\ directory on the history server and uses the obtained data for recalculation of margin requirements and accrual of the variation margin at the end of the trading session.

- The files having dgcxrpf-20130320-0600-01-i.spn format contain intraday risk management parameters.
- The files having dgcxrpf-20130321-2345-01-e.spn contain risk management parameters for the end of a trading session.

The following entries are shown in the gateway journal when downloading and processing risk parameters files:

|          |         |                                                               |
|----------|---------|---------------------------------------------------------------|
| 10:51:22 | Gateway | span: downloaded file dgcxrpf-20130326-1500-02-i.spn (267 Kb) |
| 10:51:22 | Gateway | span: data processing completed, 95 span data records found   |

MetaTrader 5 Trading Platform / Platform Components / Gateways / MetaTrader 5

MetaTrader 5 Gateway to MetaTrader 5

The MetaTrader 5 Gateway is a gateway that allows bringing up trade operations to external MetaTrader 5 servers. Trade operations performed on the server will be processed and stored in an external system. Interaction between the platforms is performed through a client connection.

Order MetaTrader 5 Gateway to MetaTrader 5

With this gateway every brokerage firm can integrate with the larger brokers and liquidity providers working on the MetaTrader 5 trading platform. Using MetaTrader 5 Gateway, the brokers minimize their own risk and receive the potential profit from each trade operation of a client. All you need to do is select a liquidity provider and set up the gateway.

The executable module of the gateway is included into the platform package and can run in a demo mode, allowing you to perform 100 operations.

How the Gateway works

The main task of the MetaTrader 5 gateway is to convert trade requests of a broker's clients into requests to the external MetaTrader 5 trading platform.



All the requests of the broker's clients are translated to the liquidity provider. The response from the liquidity provider is transmitted to the client. The broker's server is configured in such a way that the clients' trade requests are sent to the MetaTrader 5 gateway. The gateway checks the clients' requests and after the successful check creates new trade requests to a remote platform. After receiving a response from the remote MetaTrader 5 platform, the gateway returns a response to the client's request to the broker's server.

Trading Operations

## MetaTrader 5 trading platform

Orders are sent to MetaTrader 5 Gateway to MetaTrader 5 for processing in accordance with the set routing rules. Depending on the order type, each request is handled differently:

- Market orders are transferred to the gateway directly.
- Limit orders are not transferred to the gateway by default. They are handled within the trading platform instead. An appropriate market request is sent to the gateway immediately after activation. Limit orders output mode can be changed using Limit Orders Coverage Mode parameter in the gateway settings.
- Stop orders are not transferred to the gateway immediately. Such orders are handled and stored on the MetaTrader 5 platform side. They stay in the MetaTrader 5 platform until a specified price level is reached. An appropriate market order is sent to the external system immediately after reaching this level.
- Stop-limit orders are handled and stored on the MetaTrader 5 platform side. Such orders stay in the MetaTrader 5 platform until their stop price is reached. Once it has been reached, the corresponding limit order with a specified price will be created in MetaTrader 5 or sent to an external server depending on the gateway settings.
- Take Profit, Stop Loss and Stop Out orders are handled and stored on the MetaTrader 5 platform side. An appropriate trading request is sent to an external platform immediately after the order triggering.

The gateway handles operations on symbols with Market Execution mode similar to exchange execution symbols. Handling does not depend on how operations are actually executed on the external server in the market or exchange execution mode.

The gateway translates only to the external system only the requests that are lossless for the broker

In the instant execution mode, some client trade requests with maximum price deviation specified may be unprofitable for the broker. If the operation is unprofitable for the broker, the gateway automatically generates a reply with a requote, without transmitting the trade operation to the liquidity provider. For example, if the client makes a Buy request at a price less than the price of the liquidity provider, such a request will be requoted. If a client makes a Buy request at a price greater than the price of the liquidity provider, the client's request will be executed at the price of the liquidity provider, but the client's position will be opened at the price specified by the client. The broker earns a profit from the price difference. If the client's price and the price of the liquidity provider are equal, the request is transmitted to the liquidity provider and the broker has the zero profit. The broker can increase the earned profit by correcting prices.

- Gateway delivered together with standard installation of MetaTrader 5 platform works in trial mode, full-functional edition is purchased additionally.
- In trial mode the gateway performs only 100 trading operations for a work session (until restart).
- The gateway translates quotes from the external MetaTrader 5 platform.

- The gateway ensures that a response with one of the possible return codes will be generated for every selected request.

## Requirements

For proper operation of the gateway and correct accounting of funds, it is necessary to ensure that the symbol on the broker's server have the same trading settings as the remote MetaTrader 5 platform. The symbol's trading parameters can be set in the symbols setup dialog.

In particular, it is necessary to match the following setting of symbols:

- Execution mode for the given symbol. If the Request or Instant Execution mode is used on a remote platform, exactly the same execution mode should be set on your server. If the Market or Exchange Execution mode is used on a remote platform, any of these two modes can be used on your side (both modes are valid).
- For the Instant Execution mode, you should have the same maximum order volume, above which the order is switched to the Request Execution mode.
- For the Request execution mode, the mode of order confirmation should be enabled
- Contract size
- Number of digits after the decimal point. If the symbol price accuracy on the source server is higher than that on the receiving server, the gateway will not provide the Depth of Market for that symbol, while it will only stream tick data.
- The price of one point of the price change, except for instruments with the Forex calculation mode
- Size of one point, except for instruments with the Forex calculation mode

If both platform use hedging, then the external platform must provide the 'Fill or Kill' fill policy for the symbols. It is used when forwarding pending order activations to ensure that they are executed as one order on the external server. This way, the total number of positions on the two servers will match. If the Fill or Kill policy is not available, the gateway will reject order activation requests.

Do not use the expiration mode "Good till today including/excluding SL/TP" when forwarding orders to an external system via the gateway. This can lead to issues in the synchronization of operations, in which case "unknown execution report" errors will be printed to logs. Expired intraday orders are canceled in accordance with the end-of-day time of the local platform, without any interaction with gateways. Even if the relevant expiration mode is enabled on a remote server, the end-of-day time may differ on that server, which will also lead to unsynchronized operation.

The gateway does not check the matching of symbol settings, if it is used as the liquidity provider for ECN. In this case, the compatibility of trade orders with symbol settings on the remote server is provided by the ECN.

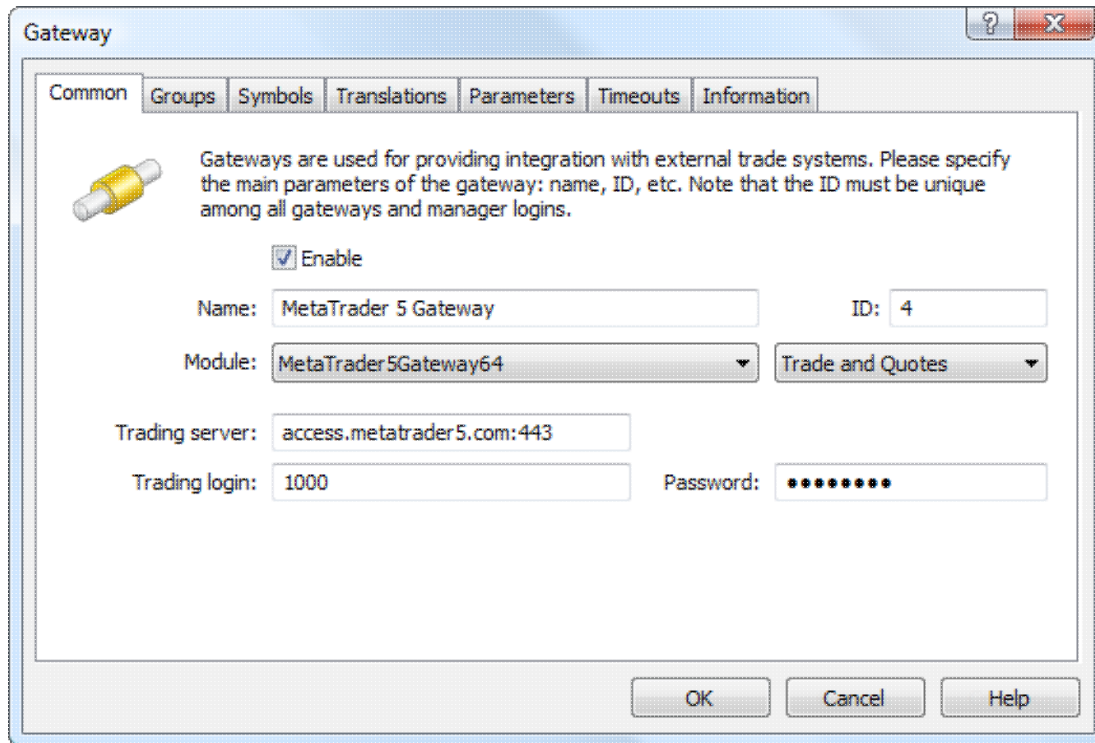
## Installation

MetaTrader 5 Gateway is a single executable file MetaTrader5Gateway64.exe, and it uses Gateway API. The gateway is included in the MetaTrader 5 platform, and after

## MetaTrader 5 trading platform

the installation it is located in the "gateway" subdirectory of the History Server directory.

To start working, add the new gateway configuration:

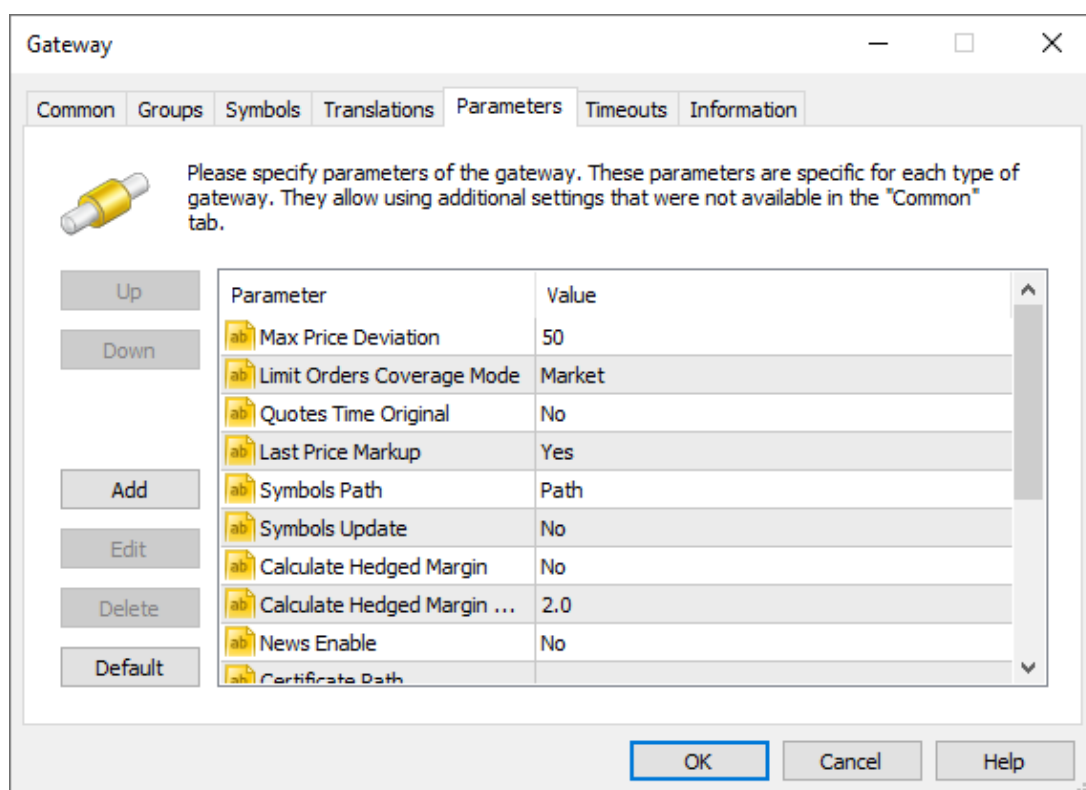


Set the following parameters on the "Common" tab:

- **ID** in the ID field, set the identifier of the dealer, from whose name the requests routed to the gateway will be confirmed.
- **Module** in this field, select MetaTrader5Gateway64 in the list of available modules and load its default settings.
- **Trading server** external MetaTrader 5 server IP-address and port, where trade requests are sent.
- **Trading login** login for connection to the external MetaTrader 5 server.
- **Password** password for connection to the external MetaTrader 5 server.

- ID value must be unique in the field of manager logins and gateway identifiers. Usually, this field is filled out automatically by an acceptable default value.
- the gateway will perform all trading operations with the liquidity provider using this trading account.

Now, go to the "Parameters" tab.



The following parameters values must be specified here:

- **Max Price Deviation** the maximum allowed deviation of the current price from the price requested in an order. The order will not be executed in case that deviation has been exceeded. This parameter is used when a pending order is activated on the MetaTrader 5 side and transferred to an external system as a market order. This parameter is applied only to the symbols having "Instant" execution type.  
If a common market order is transferred to an external system, a client should specify the maximum allowable deviation. A trader cannot specify a deviation when setting pending orders. Thus, Max Price Deviation protects a trader from executing an order at considerably deviated price.
- **Limit Orders Coverage Mode** the mode of handling limit orders by the gateway. This parameter simplifies configuration of trade requests routing to the gateway, since there is no need to create a separate rule for routing the appropriate order types. Three processing modes are available:
  - ♦ **Market** limit orders are processed on the MetaTrader 5 platform side. If an order is activated, an appropriate market order is sent to the external system.
  - ♦ **Limit** limit orders are processed on the MetaTrader 5 side. Once a limit order is triggered, an equivalent limit order is sent to the external system. If the "Specified time" expiration type is allowed for a symbol in the external system, the expiration of 1-2 minutes is set for this limit order. Otherwise the order validity is set to the current day; in the worst case the order will be valid until canceled.  
Since the price specified in the order is already present in the market, the

order will be executed with that price – a market deal will be performed. If the necessary volume of the financial instrument is not available in the market at the specified price, the order will be executed partially. In this case, a client will have a market position as well as a limit order with a residual volume, which will be further processed in a similar way on the side of MetaTrader 5.

Limit mode allows to protect against slippage, as a limit order is sent to the the external system with a specified price rather than a market order for execution by the current price. In addition, this mode allows you not to reserve margin on a client's account before sending an order to the external system.

- ◆ **Gateway** limit orders are processed on the the external system side. Once a limit order has been placed by a client, an appropriate order is sent to the external system.

- **Symbols Path** path for importing the symbols. By default, if this parameter is absent, the gateway imports the trading symbols to \Preliminary subdirectory with trading ability disabled. A system administrator should manually allow trading for them. If this parameter is present, the gateway imports trading symbols following the specified path.

- **Symbols Update** if Yes, the gateway updates the symbol settings if they have been changed on the source server. The default is No. Settings are updated in real time. Trading sessions are updated considering the servers' time zones. Regardless of the value of "Symbols Update", the gateway does not update the following settings of symbols that exist in the platform:

- group and symbol background color
- quote processing and filtration parameters
- trading, margin calculation, execution and GTC mode
- spread and spread difference
- stop and freezing levels, maximum quote delay
- swap settings
- symbol time limits
- instant and request execution settings

- **Quotes Time Original** if "Yes", the gateway itself sets the time for ticks considering the time zone of the recipient trade server. If the parameter is absent or set to "No", the time for ticks is set by the history server according to the trade time used.

- **News Enable** if "Yes", allows transmitting news from the liquidity provider server. The default value is "No".

- **Last Price Markup** if Yes value is set, price markup settings specified in an external platform (source) will be applied to streamed Last prices. Bid price spread difference values (from the Spread balance parameter) will be applied to Last prices. If you need to disable the Last price markup, set the parameter to No. The default value is Yes.

The parameter affects Last prices only. For bid/ask prices, the gateway always applies price markup settings from an external platform.

- **Calculate Hedged Margin** when set to No (default) the gateway copies the hedged margin of the source symbol. If Yes, the gateway will calculate the hedged margin. If the initial margin is specified, the hedged margin is calculated as the initial margin multiplied by the Calculate Hedged Margin Rate value. If the initial margin is not specified, the hedged margin is calculated as



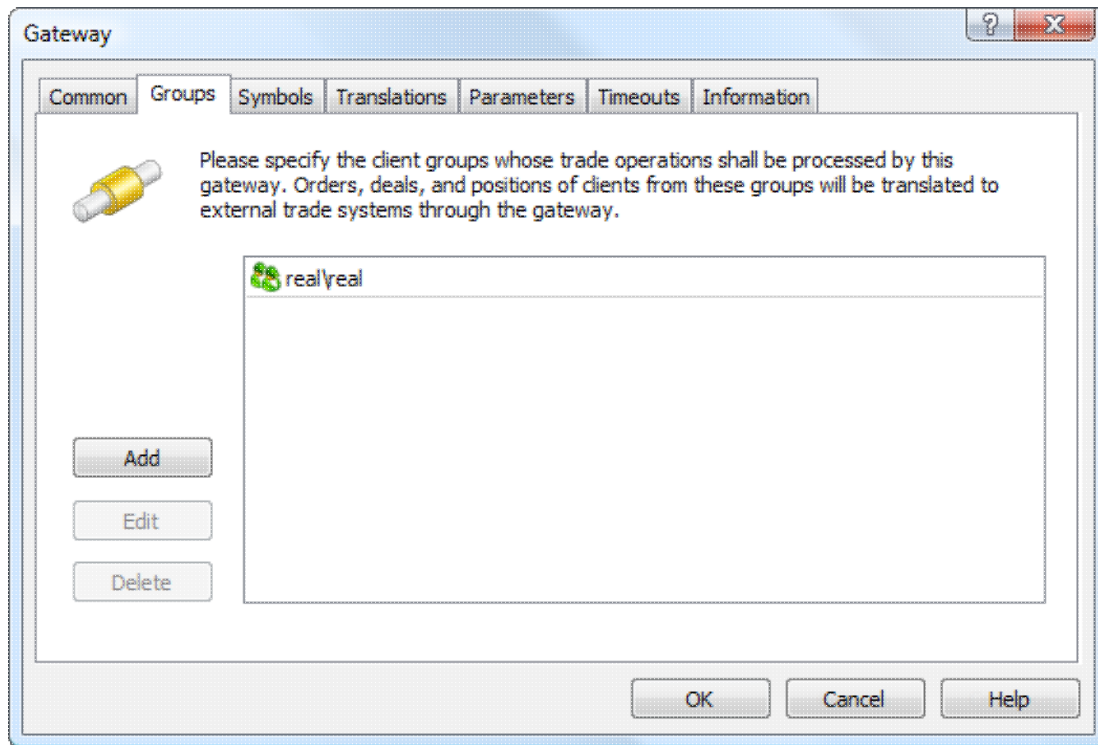
the contract size multiplied by Calculate Hedged Margin Rate. The calculated value of the hedged margin is rounded to the number of decimal places specified in the symbol settings.

- **Calculate Hedged Margin Rate** hedged margin calculation multiplier used if Calculate Hedged Margin = Yes. Default is 2.0.
- **Certificate Path** path to the PFX file of the certificate, which the gateway will use for advanced authentication on an external MetaTrader 5 server.
- **Certificate Password** password to open the PFX file of the certificate used for advanced authentication.
- **Selftitled Translations** this parameter is used to prevent quotes from looping when connecting to the server on which the gateway is installed. The default value is No. For more details please see below.
- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

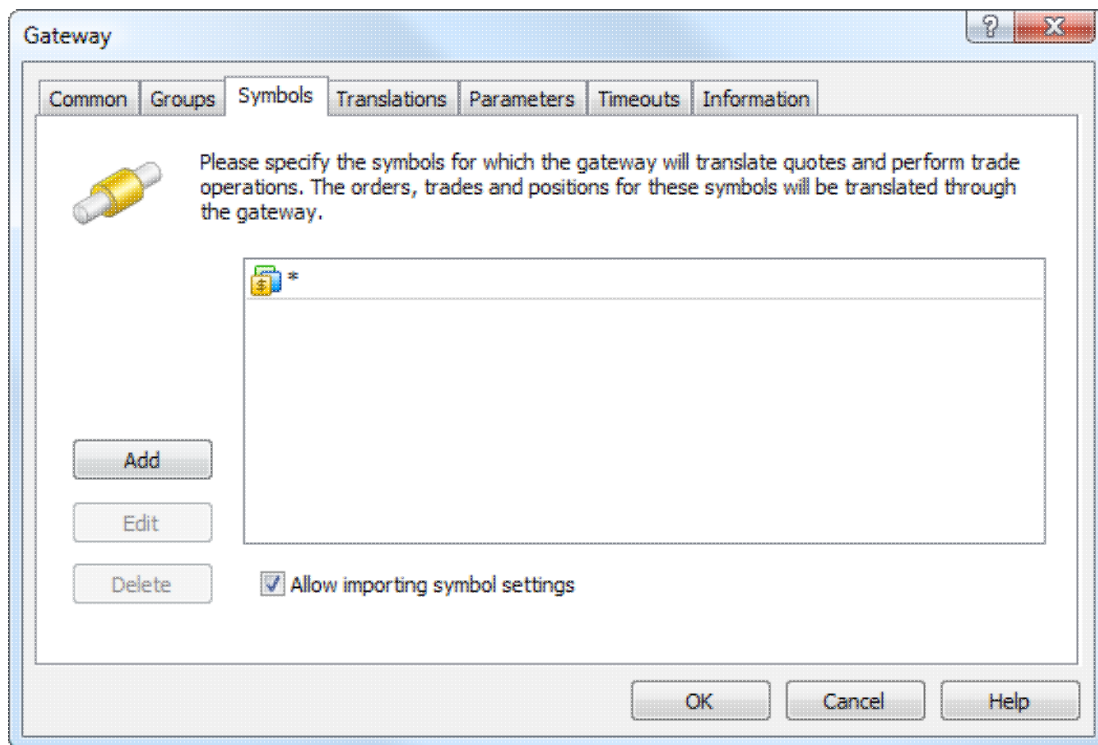
Margin reservation of the clients should be configured for the appropriate order type in case Limit orders are directly transferred to the external system.

On the "Groups" tab select the group of clients, whose orders and positions will be available to the gateway.

## MetaTrader 5 trading platform



The "Symbols" tab allows to configure the list of symbols, according to which the gateway will process trade operations and transmit the quotes.



## MetaTrader 5 trading platform

MetaTrader 5 Gateway to MetaTrader 5 supports the import of symbols and their settings from an external trading server. If the option "Allow importing symbols settings" is enabled, the import of symbols from an external server will be performed. All symbols available for an account used for connection (specified in "Trading server" field of the "Common" tab) are imported. The symbols are imported to Symbols/Preliminary/ subgroup according to their hierarchy at the external server.

- The symbols imported by the gateway are put to the "\Preliminary" symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After the symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the existing symbol settings are not updated. Configuration process of such symbol is skipped.

The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

### Margin Setup

Margin reservation of the clients should be configured for the appropriate order type in case Limit orders are directly transferred to the external system.

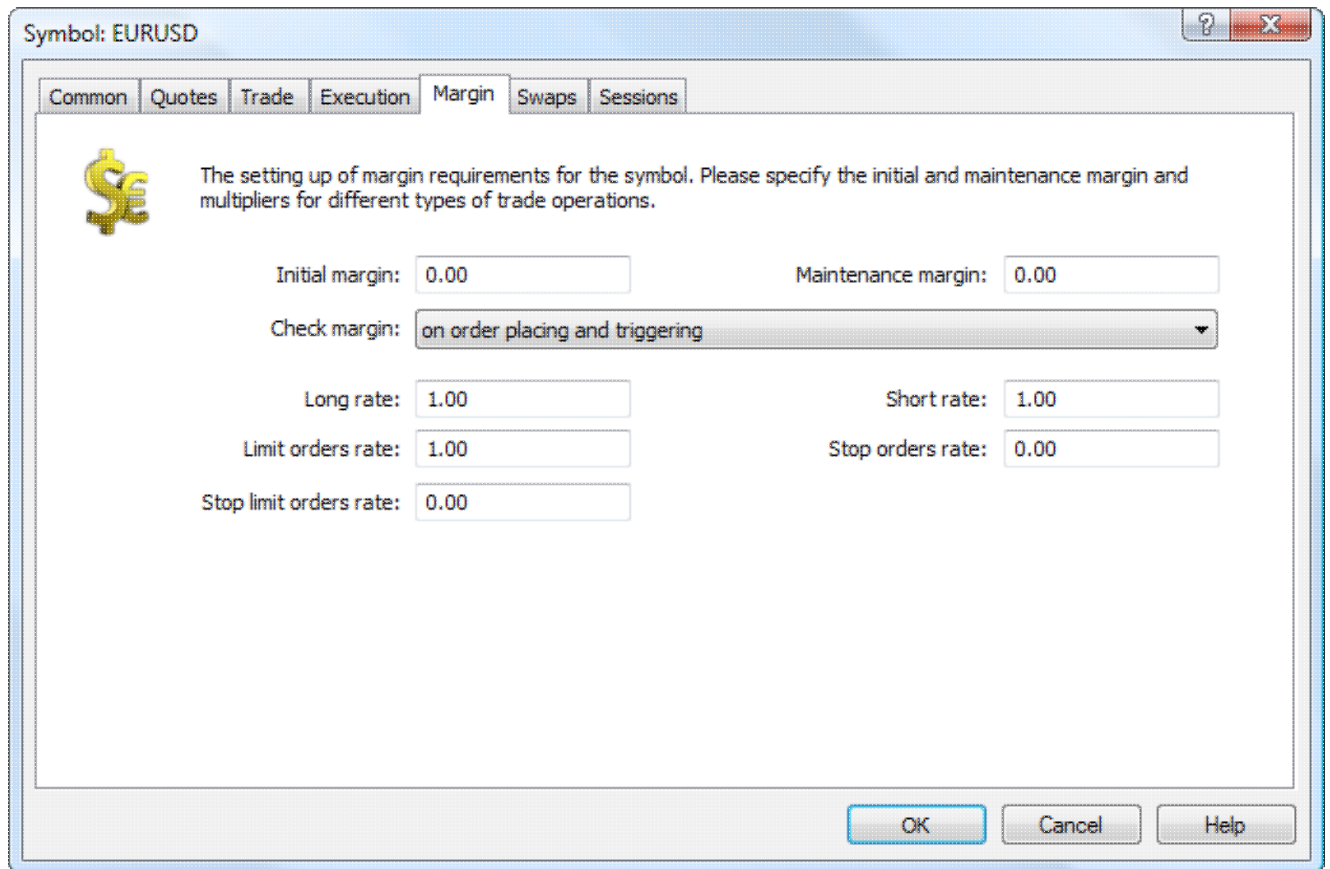
The external system checks sufficiency of the funds that are necessary to provide any type of order placed via the gateway. However, the check is performed on the broker's general account, on behalf of which the work is carried out.

By default, the margin is charged on the side of MetaTrader 5 only when market orders are placed. When placing pending orders in an external system, the client's available funds should be controlled on the side of MetaTrader 5 before the orders are transferred to avoid using all broker's funds by the client.

After an order has been transferred to the external system, MetaTrader 5 platform is not able to check the client's margin sufficiency any more. After the order has been executed in the external system, the gateway cannot ignore that fact. Therefore, the appropriate trading operation is performed in the platform.

Set non-zero coefficients for the orders directly transferred to the external trading system in symbol settings for the appropriate symbols to configure margin collection:

## MetaTrader 5 trading platform



The screenshot shows the 'Margin' tab of the MetaTrader 5 settings window for the EURUSD symbol. The window has a title bar with 'Symbol: EURUSD' and standard window controls. Below the title bar are tabs for 'Common', 'Quotes', 'Trade', 'Execution', 'Margin' (selected), 'Swaps', and 'Sessions'. The main area contains a yellow dollar and euro symbol icon and a text box explaining the purpose of the settings. Below this are several input fields and a dropdown menu for configuring margin requirements. At the bottom right are 'OK', 'Cancel', and 'Help' buttons.

Symbol: EURUSD

Common Quotes Trade Execution **Margin** Swaps Sessions

 The setting up of margin requirements for the symbol. Please specify the initial and maintenance margin and multipliers for different types of trade operations.

Initial margin: 0.00 Maintenance margin: 0.00

Check margin: on order placing and triggering

Long rate: 1.00 Short rate: 1.00

Limit orders rate: 1.00 Stop orders rate: 0.00

Stop limit orders rate: 0.00

OK Cancel Help

### Configuring Trade Requests Routing

After adding a new gateway, you must configure routing of requests so that the clients' requests are routed to this gateway. To do this add a routing rule in the "Routing - Add..." section.

In general settings, select "Process to dealers" as the action. Specify that all requests and orders must be routed according to this rule. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

## MetaTrader 5 trading platform

Routing: MetaTrader 5 Gateway Execution

Common Dealers

 Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☐ Enable this rule

Name: MetaTrader 5 Gateway Execution

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

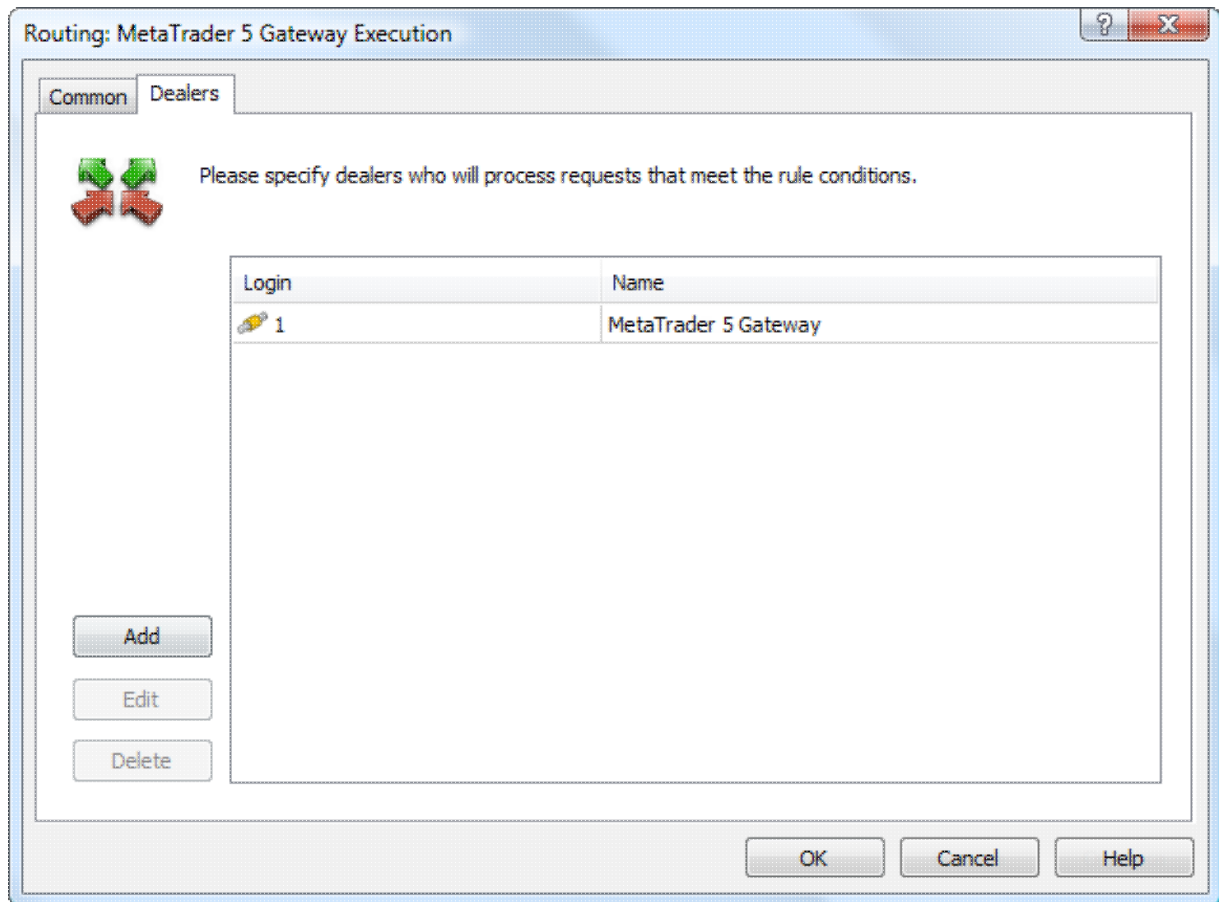
Where conditions are:

| Type                                                                                           | Condition | Value     |
|------------------------------------------------------------------------------------------------|-----------|-----------|
|  Client group | Equal (=) | real\real |

Add Edit Delete

OK Cancel Help

In the settings of the "Dealers" tab, add the previously created gateway.



After the correct execution of the steps described above, the MetaTrader 5 Gateway is started and ready to work. The result of the gateway operation is reflected in its journal. Go to the "Network" section and choose the Main Trade Server. In the appeared information dialog, in the tab "Gateways" request the logs for the corresponding period.

### Transferring quotes

MetaTrader 5 Gateway transfers the price flow from an external MetaTrader 5 platform.

For symbols with the disabled market depth (on the external server side), the gateway passes ticks with Bid and Ask prices.

For symbols with the enabled market depth, the gateway transfers the market depth changes as well as ticks with Last prices and volumes. Ticks containing only Bid and Ask price changes are not transferred. If the best supply and demand prices change in the market depth, the history server generates the necessary tick with Bid and Ask prices and adds it to the flow.

The gateway features the built-in switching mechanism to prevent the quote flow from stopping in case the external server stops transmitting the market depth changes. If not a single market depth change for a symbol arrives from the external system within 35 seconds, the gateway stops sorting out ticks having only Bid and Ask prices. In

## MetaTrader 5 trading platform

other words, it starts transferring to the platform both Last/Volume and Bid/Ask ticks.

The following entries are shown in the gateway journal when the market depth is no longer transmitted:

|                         |         |                                                           |
|-------------------------|---------|-----------------------------------------------------------|
| 2017.09.12 15:20:25.412 | Gateway | books stream <b>for</b> EURUSD stopped (no books more th  |
| 2017.09.12 15:20:25.873 | Gateway | books stream <b>for</b> USDJPY stopped (no books more tha |

If the flow is resumed:

|                         |         |                                        |
|-------------------------|---------|----------------------------------------|
| 2017.09.12 15:21:29.759 | Gateway | books stream <b>for</b> EURUSD resumed |
| 2017.09.12 15:21:30.060 | Gateway | books stream <b>for</b> USDJPY resumed |

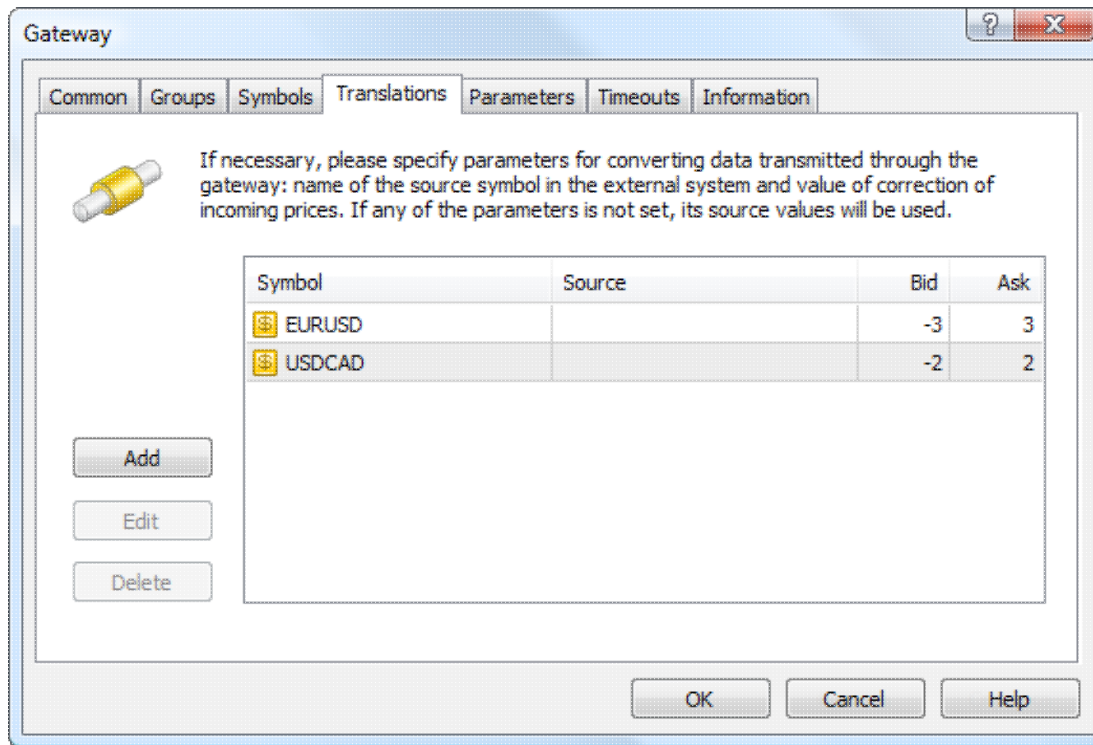
|                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The gateway transmits the original stream of quotes from the external MetaTrader 5 platform, without spread markup settings specified for instruments/groups in the external platform. Only the relevant <u>translation settings specified on the gateway</u> are applied to the prices. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Price Markups

The gateway receives prices from the MetaTrader 5 external server and transmits them to clients taking into account transformation settings (markups). The clients perform trade operations using transformed prices. However, while processing trading operations on the gateway and their transmission to the external system, initial, not transformed prices are automatically used.

Thus, a brokerage company receives its profit share from each deal performed at the external system.

Markup values are set separately for Bid and Ask prices by each symbol:



In this example, the following corrections are specified for EURUSD: for each tick the Bid price will be reduced by 3 points, and the Ask price will be increased by 3 points.

If no correction is set for a symbol, the clients will work with the original prices of the liquidity provider.

### Symbol masks and own price retranslation

In translation settings, "\*" mask can be used as the source symbol and as the destination symbol in the platform. For example, the settings Symbol="\*", Source="\*" mean that the names of the symbols will be used as they are provided in the external system. If the symbol is entitled EURUSD in the external system, then its data will be feed to the symbol with the same name on the trading platform side. The only situation in which such settings cannot be used is the connection of the gateway to the platform on which it is installed. In this case, receiving and feeding of quotes by the gateway into the same symbols will lead to looping.

To avoid such situations, the gateway provides the parameter "Selftitled Translations". If it is set to "No" (default) and the gateway has a translation setting "\*" <- "\*", the gateway will not start. An appropriate entry will be added into the journal:

translation rule for symbols to themselves '\* <- \*' not allowed but exists, remove this rule or

Before enabling this parameter, make sure that the gateway is not connected to the same cluster on which it is running. Otherwise, this can lead to the looping of transactions, configurations and quotes in the platform.

### Example of MetaTrader 5 Gateway



## MetaTrader 5 trading platform

All trade operations performed by the gateway are included in the gateway journal. For example, using the client terminal we buy EURUSD 1.0. After processing the request, the following entries will appear in the journal of the MetaTrader 5 Gateway:

```
'1002': request #1179132 received (#2073 instant buy 1.00 EURUSD at 1.31237)
'1002': request #1179132 answered - Done at 1.312370 (#1002 instant buy 1.00 EURUSD at 1.312370)
```

Consider the contents of the logs in more detail.

A request with the ID # 1179132 is received from the broker's client with the account 1002. Corresponding to this request, on the broker's server there is the client's order with the ID # 2073 to Buy one lot of EURUSD at the price of 1.31237.

To the request ID #1179132, the broker's client with the account 1002 receives a reply that the request has been executed (Done), with the request execution price - 1.312370. In addition the original order of the client is specified. The last part of the message contains the parameters of the trading operation on the remote platform. In particular, upon the client's request, generated order # 2448959, deal # 2448959, with the actual execution prices: Sell - 1.31198, Buy - 1.31212. In this case, this means that the client's Buy has been executed at 1.31212 on the external trading platform.

On this basis we can calculate the profit from the price difference:

The broker's profit =  $1.31237 - 1.31212 = 0.00025$ .

### Operation in the advanced authentication mode

The advanced authentication mode can be enabled for the account used by the gateway for connection to an external MetaTrader 5 server. In this case, the PFX certificate file is required for connection, in addition to the login and password. The file path and a password for file opening must be specified in appropriate gateway parameters: Certificate Path and Certificate Password.

To receive the certificate, please contact the broker or generate a certificate using the client terminal. In the latter case, connect to the account and go through the standard generation procedure. The resulting PFX file should be transferred to the computer, on which the gateway is running. Save it to a disk or install it in Windows storage. When installed to the storage, the 'Certificate Path' parameter can be left empty; thus the gateway will request the certificate from the storage.

When installing the certificate to the Windows storage, make sure to select the Local Computer storage, and not that of the Current User. Otherwise the gateway will not be able to access the certificate.

If the advanced authentication is required, but the gateway cannot find the certificate at the path specified in 'Certificate Path', the following error will be added in its log:

```
'2098': loading of standard SSL certificate from 'C:\{path}\2098_ServerName.pfx1' failed
```

If the certificate cannot be found in the Windows storage, the following error will be added:

'2098': standard SSL certificate for advanced authorization is not found

If an invalid password is specified in 'Certificate Password', the following error will be written in the log:

'2098': invalid standard SSL certificate password

Coverage using MetaTrader 5 Gateway to MetaTrader 5

The MetaTrader 5 trade platform allows covering the positions of clients on other trade servers protecting own company from financial risk. Coverage of positions can be performed using the MetaTrader 5 Gateway to MetaTrader 5.

Coverage of client positions is performed using special coverage accounts. Such accounts are those created in groups whose names start with symbols "coverage" (case sensitive). For example, coverage\forex. Performing trade operations on a coverage account, a manager covers the position of clients on another trade server.

In order to make covering possible, an administrator of the trade platform should make the corresponding settings:

Creating a Group

Create a group, whose name starts with "coverage":

Group: coverage

Common Company News & Mail Permissions Margin Symbols Commissions Reports

Group is a set of users that have the same permission settings and service conditions. Please specify name of group, deposit currency, trade server, and authorization type.

Name: coverage Currency: USD

Trade server: MetaTrader 5 Main Server, 1

Authorization: Normal Minimum password length: 5

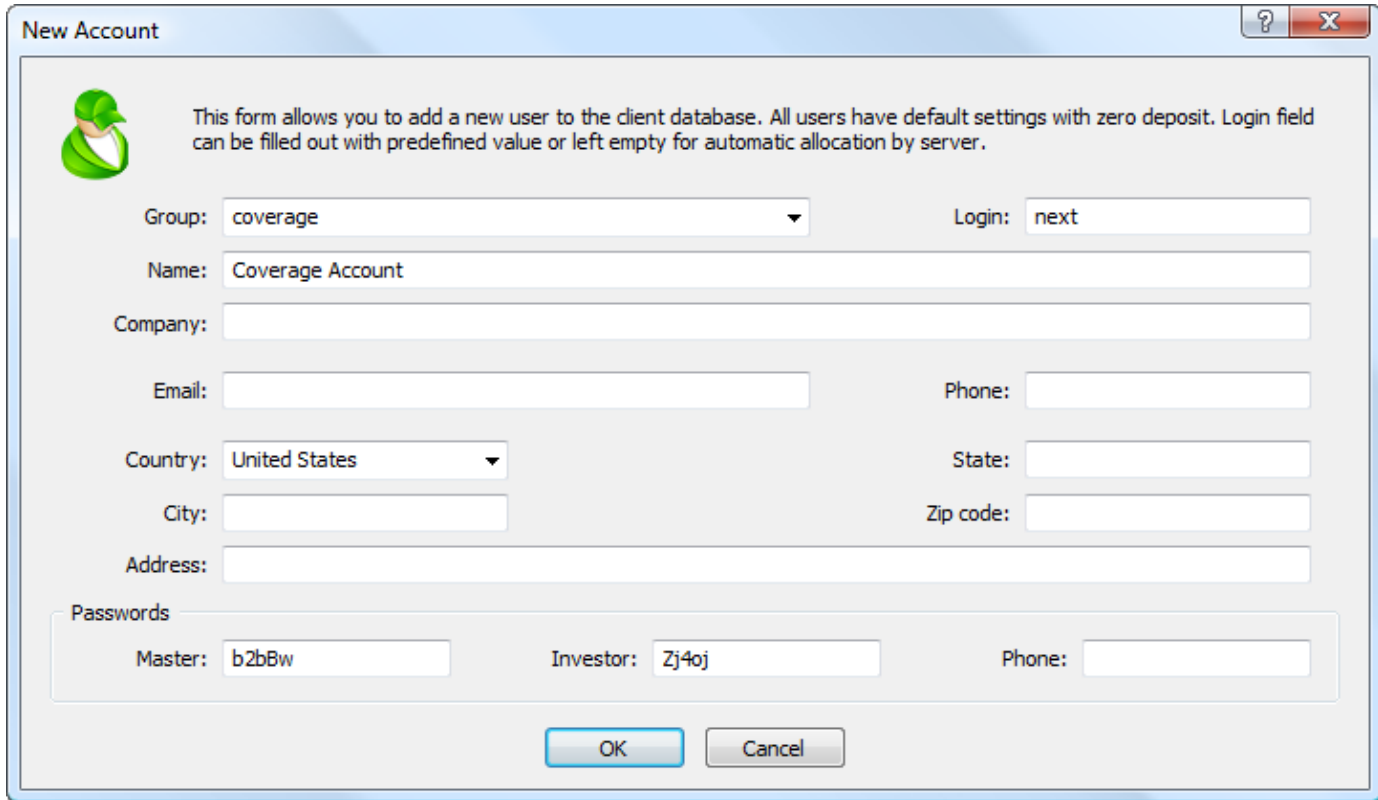
☒ Enable connections

☐ Enable certificate confirming

OK Cancel Help

## Creating a Coverage Account

In the coverage group, create an account. This account will be used by a manager for performing trade operations that cover client positions.



**New Account**

This form allows you to add a new user to the client database. All users have default settings with zero deposit. Login field can be filled out with predefined value or left empty for automatic allocation by server.

Group: coverage Login: next

Name: Coverage Account

Company:

Email: Phone:

Country: United States State:

City: Zip code:

Address:

**Passwords**

Master: b2bBw Investor: Zj4oj Phone:

OK Cancel

## Opening a Trade Account

You need to open a trade account on an external trade server, where coverage positions will be opened. The account should have enough deposit for covering client positions.

## Setting up the Gateway

The next step is setting up the MetaTrader 5 Gateway using the authorization details of the account opened at the external trade server. Then you need to set up the routing of trade operations from the coverage account to the external trade system through the gateway:

Routing: Hedging

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: Hedging

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: Instant Execution

Where order is: Buy, Sell

Where conditions are:

| Type         | Condition | Value    |
|--------------|-----------|----------|
| Client group | Equal (=) | coverage |

Add Edit Delete

OK Cancel Help

After that, specify the MetaTrader 5 Gateway to MetaTrader 5 in the Dealers tab.

Once the coverage account is set up, a manager can perform trade operations using it in the manager terminal. The summary rates for all coverage accounts on the trade server are displayed in the Summary positions and Exposure tabs of the Toolbox window in the manager terminal.

[MetaTrader 5 Trading Platform](#) / [Platform Components](#) / [Gateways](#) /  
MetaTrader 4

## MetaTrader 5 Gateway to MetaTrader 4

The gateway allows any brokerage firm to integrate with larger brokers and liquidity providers using the MetaTrader 4 trading platform. You will be able to minimize your risk and receive potential profit from each trade operation of a client due to markups. All you need to do is select a liquidity provider and perform a simple gateway configuration.

### [Order MetaTrader 5 Gateway to MetaTrader 4](#)

#### How the Gateway Works

The gateway acts as an intermediary between the two platforms by converting trade requests of brokerage clients into requests to the external MetaTrader 4 trading platform, receiving answers and sending them back to clients.

The [routing rules](#) allow configuring the broker's server in such a way that the clients' trade requests are sent to the gateway. Depending on the order type, each request is handled differently:

- **Market orders** are passed to the gateway directly.
- **All pending orders** are not passed to the gateway by default. They are handled within the trading platform instead. An appropriate market request is sent to the gateway immediately after activation.
- **Take Profit, Stop Loss and Stop Out orders** are handled and stored on the MetaTrader 5 platform side. The appropriate market operation is sent to the external platform right after activation.

- The gateway is already included in the MetaTrader 5 platform and can be used in demo mode, which allows performing not more than 100 trading operations for a work session (until restart). The full version should be purchased separately.
- The gateway transmits quotes from the external MetaTrader 4 platform.

#### Requirements

For proper operation of the gateway and correct accounting of funds, it is necessary to ensure that the symbol on the broker's server have the same trading settings as the remote MetaTrader 4 platform. The symbol's trading parameters can be set in the symbols setup dialog. The following symbol settings should be the same:

- The mode of [order execution](#) by a symbol
- [Number of digits](#) after the decimal point
- [Type of calculation](#) of margin requirements and profit on a symbol
- [The price of one point](#) of the price change, except for instruments with the Forex calculation mode

## MetaTrader 5 trading platform

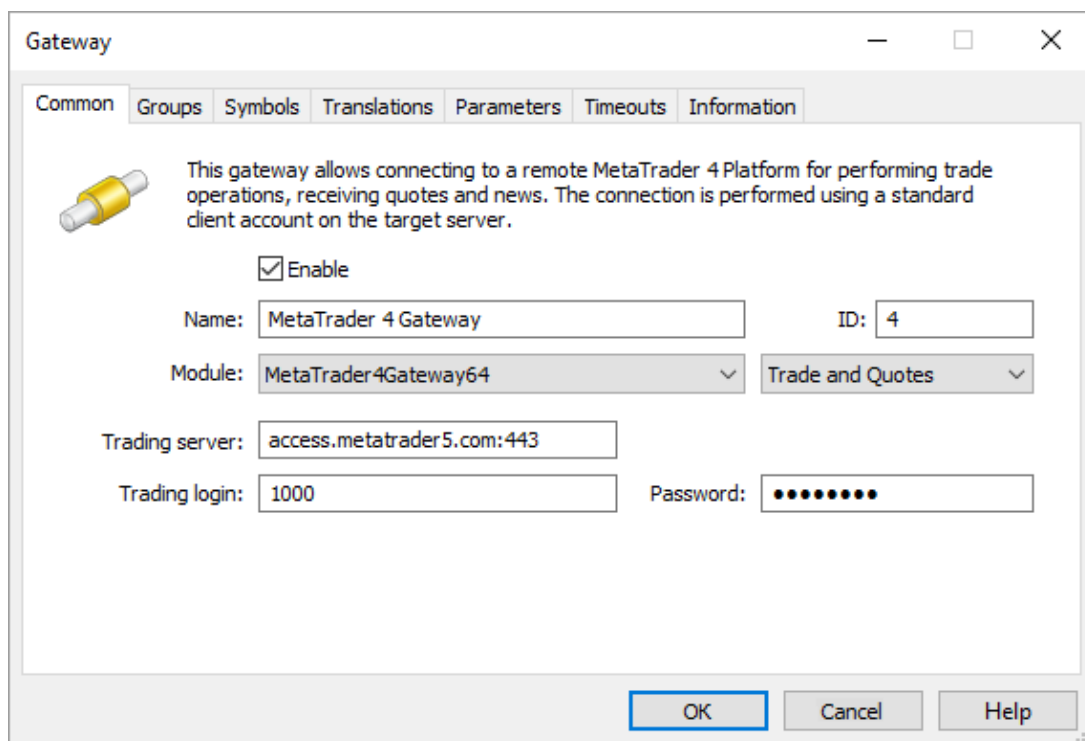
- Size of one point, except for instruments with the Forex calculation mode
- For the Request execution mode, the mode of order confirmation should be enabled

A trade account should be created on MetaTrader 4 external server, on behalf of which all client trade operations passed through the gateway are performed.

### Gateway Configuration

MetaTrader 5 Gateway to MetaTrader 4 is a separate MetaTrader4Gateway64.exe file that uses Gateway API for its operation. The gateway is distributed as part of the MetaTrader 5 platform and located in [history server installation directory]\gateway\.

To start working, add the new gateway configuration:



The screenshot shows the 'Gateway' configuration window with the 'Common' tab selected. The window title is 'Gateway'. The 'Common' tab contains a description of the gateway, an 'Enable' checkbox, and several input fields. The 'Name' field is 'MetaTrader 4 Gateway', the 'ID' field is '4', the 'Module' dropdown is 'MetaTrader4Gateway64', the 'Trade and Quotes' dropdown is 'Trade and Quotes', the 'Trading server' field is 'access.metatrader5.com:443', the 'Trading login' field is '1000', and the 'Password' field is masked with dots. The 'OK', 'Cancel', and 'Help' buttons are at the bottom right.

Gateway

Common Groups Symbols Translations Parameters Timeouts Information

This gateway allows connecting to a remote MetaTrader 4 Platform for performing trade operations, receiving quotes and news. The connection is performed using a standard client account on the target server.

☒ Enable

Name: MetaTrader 4 Gateway ID: 4

Module: MetaTrader4Gateway64 Trade and Quotes

Trading server: access.metatrader5.com:443

Trading login: 1000 Password: .....

OK Cancel Help

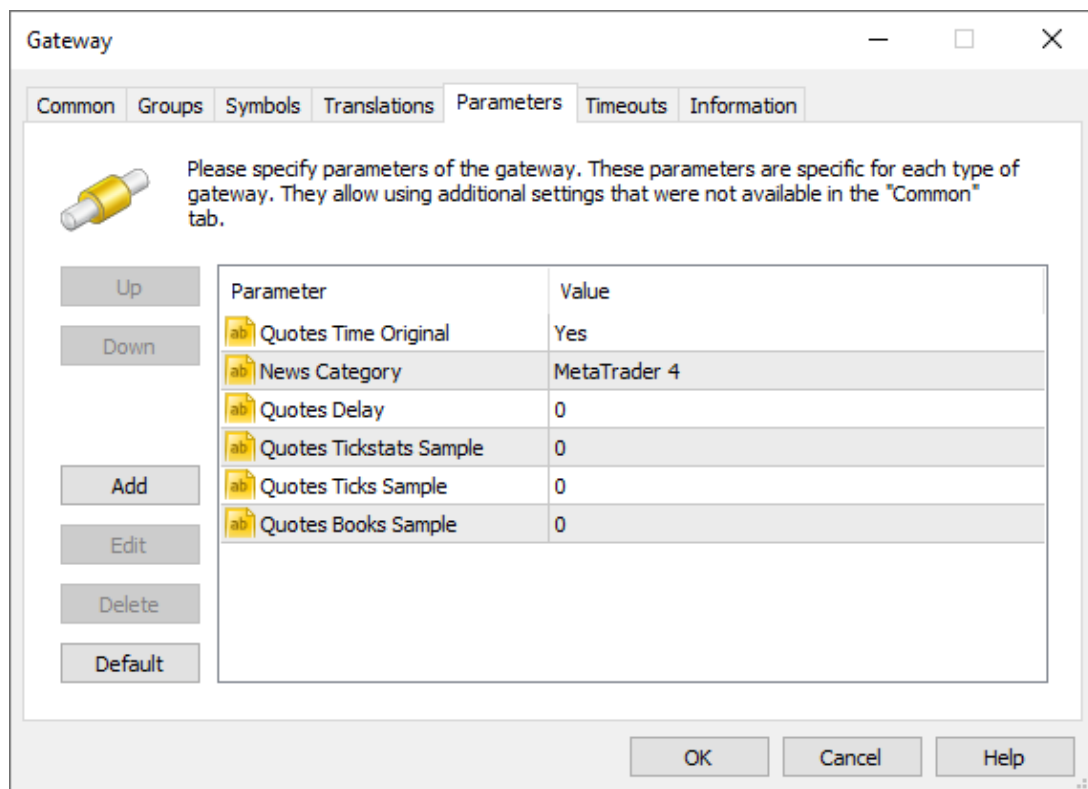
## MetaTrader 5 trading platform

Set the following parameters on the "Common" tab:

- **ID** set the identifier of the dealer, from whose name the requests routed to the gateway are confirmed.
- **Module** select MetaTrader4Gateway64 in the list of available modules and load its default settings.
- **Trading server** MetaTrader 4 external server IP address and port trade requests are sent to.
- **Trading login** index of a trade account opened at MetaTrader 4 external server. This is the account, at which client trading operations passed through the gateway are performed.
- **Password** password for connecting to an account at MetaTrader 4 external server.

ID value must be unique in the field of manager logins and gateway identifiers. Usually, this field is filled out automatically by an acceptable default value.

Now, go to the "Parameters" tab.



Gateway

Common Groups Symbols Translations **Parameters** Timeouts Information

Please specify parameters of the gateway. These parameters are specific for each type of gateway. They allow using additional settings that were not available in the "Common" tab.

Up Down Add Edit Delete Default

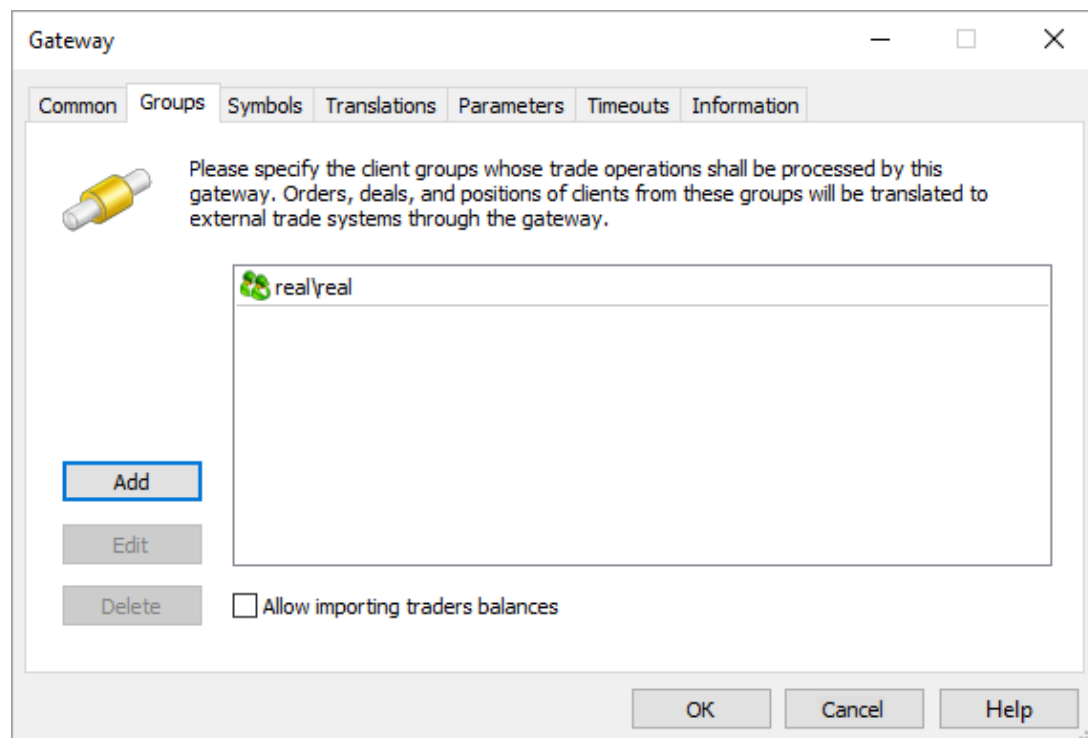
| Parameter               | Value        |
|-------------------------|--------------|
| Quotes Time Original    | Yes          |
| News Category           | MetaTrader 4 |
| Quotes Delay            | 0            |
| Quotes Tickstats Sample | 0            |
| Quotes Ticks Sample     | 0            |
| Quotes Books Sample     | 0            |

OK Cancel Help

The following additional parameters are available for the gateway:

- **Quotes Time Original** if the value is Yes, the gateway sets the time of ticks on its own considering a time zone of a recipient trading server. If No, or the parameter is absent, the time of ticks is set by the history server according to its own trading time.
- **News Category** the name of the category of news received from this data feed. Further this category name can be used to specify news to be received by separate groups.
- **Quotes Delay** delay of transmitted quotes in seconds. The maximum duration of quotes delay is 20 minutes (1200 seconds). The flow of delayed quotes is neither thinned out, nor changed. A quote is passed to MetaTrader 5 History Server only after the expiration of delay period since the quote has arrived to Gateway API. If the temporary delay parameter is not defined, the quote delay is not used. Changes in depth of market and price statistics are delayed together with the quotes flow.
- **Quotes Tickstats Sample** the minimum frequency of sending price statistics in milliseconds. This parameter allows thinning out updates of price statistics reducing the traffic.
- **Quotes Ticks Sample** the minimum frequency of sending quotes in milliseconds. This parameter allows thinning out updates of quotes reducing the traffic. It is recommended for use on demo servers only.
- **Quotes Books Sample** the minimum frequency of depth of market updates in milliseconds. This parameter allows thinning out updates of the depth of market reducing the traffic. It is recommended for use on demo servers only.

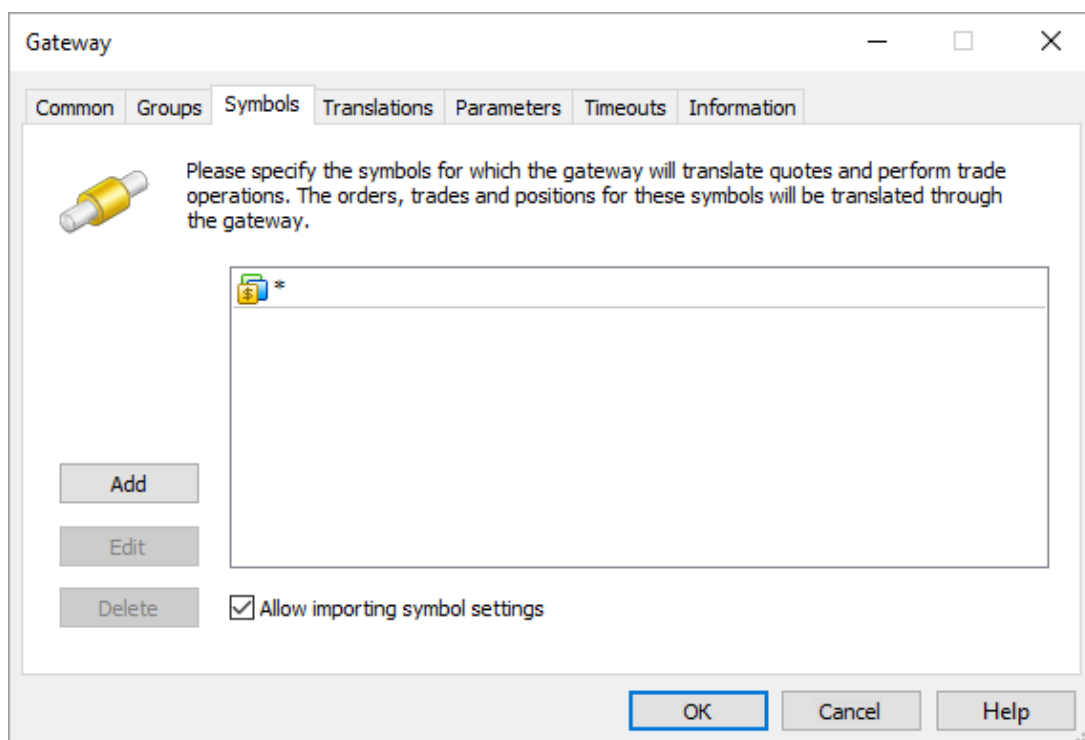
Use the Groups tab to select the group of clients, whose orders and positions will be available to the gateway.





## MetaTrader 5 trading platform

The Symbols tab allows you to configure the list of symbols, according to which the gateway will process trade operations and transmit the quotes.



MetaTrader 5 Gateway to MetaTrader 4 supports the import of symbols and their settings from an external trading server. If the option "Allow importing symbol settings" is enabled, the import of symbols from an external server will be performed. All symbols available for an account used for connection (specified in "Trading server" field of the Common tab) are imported. The symbols are imported to Symbols/Preliminary/ subgroup according to their hierarchy at the external server.

- The symbols imported by the gateway are put to the \Preliminary symbols subgroup. All symbols have trading ability disabled. System administrator must relocate imported symbols to the proper subgroup and allow trading for them.
- After symbols are relocated and trading abilities are enabled, the main trading server must be restarted.
- In case the gateway transmits configuration for the symbol that is already present in the platform, the existing symbol settings are not updated. Configuration of such a symbol is skipped.

The gateway supports conversion of symbols and quotes. For details, please view the [Symbol and Price Translation](#) section.

### Configuring trade requests routing

After adding a new gateway, configure [routing](#) so that the clients' requests are routed to this gateway. Select "Process to dealers" as an action in the common rule settings. Assign this routing rule for all requests and orders. In additional conditions, indicate groups of clients, whose requests will be passed to the gateway.

## MetaTrader 5 trading platform

Routing

Common Dealers

Please specify dealers who will process requests that meet the rule conditions.

| Login | Name                 |
|-------|----------------------|
| 1     | MetaTrader 4 Gateway |

Add Edit Delete

Routing

Common Dealers

Using routing rules, one can adjust processing of trade requests by different conditions. Please specify conditions, for which the rule will be applied.

☒ Enable this rule

Name: MetaTrader 4 Gateway

Perform action: Process to dealers ☐ skip this rule if no dealers online

Where request is: All

Where order is: All

Where conditions are:

| Type         | Condition | Value     |
|--------------|-----------|-----------|
| Client group | Equal (=) | real/real |

Add Edit Delete

OK Cancel Help

## MetaTrader 5 trading platform

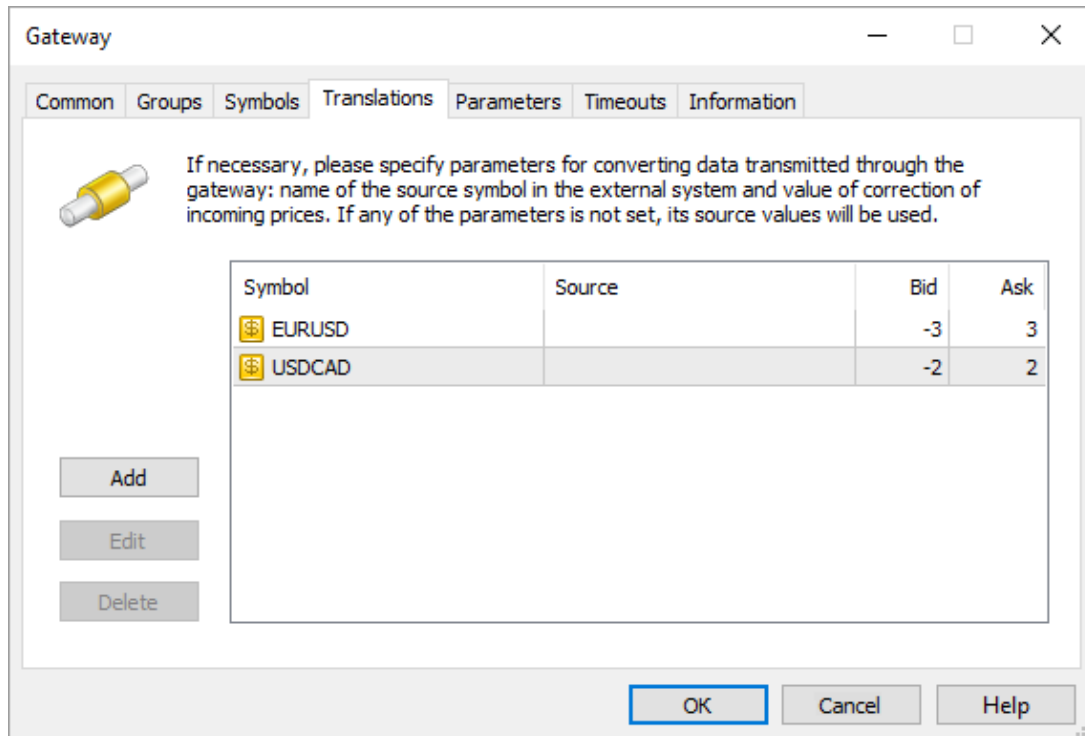
Add the previously created gateway at the Dealers tab.

After the correct execution of the steps described above, the gateway is launched and ready to work. The result of the gateway operation is reflected in its journal.

### Formation and Correction of Prices

MetaTrader 5 Gateway to MetaTrader 4 translates the price flow from an external MetaTrader 5 platform, just like the data feeds do. The priority of quotes from the gateway is higher than the priority of quotes from a data feed. If quotes for a symbol are available both from the gateway and data feed, a client will receive the quotes from the gateway.

The gateway performs all trading operations considering the price correction. To set correction for each individual symbol, go to the Translations tab in the gateway settings and enter the appropriate values:



Gateway

Common Groups Symbols **Translations** Parameters Timeouts Information

If necessary, please specify parameters for converting data transmitted through the gateway: name of the source symbol in the external system and value of correction of incoming prices. If any of the parameters is not set, its source values will be used.

| Symbol | Source | Bid | Ask |
|--------|--------|-----|-----|
| EURUSD |        | -3  | 3   |
| USDCAD |        | -2  | 2   |

Add Edit Delete

OK Cancel Help

## MetaTrader 5 trading platform

In this example, the following corrections are specified for EURUSD: for each tick the Bid price will be reduced by 3 points, and the Ask price will be increased by 3 points.

A price is given to the broker's client only after the correction, so the clients work only with the corrected prices. If no correction is set for a symbol, the client will work with the original prices of the liquidity provider.

### Example

All trade operations performed by the gateway are included in the gateway journal. For example, using the client terminal we buy EURUSD 1.0. After processing the request, the following entries will appear in the journal of the MetaTrader 5 Gateway:

```
'1002': request #1179132 received (#2073 instant buy 1.00 EURUSD at 1.31237)
'1002': request #1179132 answered - Done at 1.312370 (#1002 instant buy 1.00 EURUSD at
```

Let's consider the contents of the logs in more detail.

A request with the ID # 1179132 is received from the broker's client with the account 1002. Corresponding to this request, on the broker's server there is the client's order with the ID # 2073 to Buy one lot of EURUSD at the price of 1.31237.

To the request ID #1179132, the broker's client with the account 1002 receives a reply that the request has been executed (Done), with the request execution price 1.312370. In addition, the original order of the client is specified. The last part of the message contains the parameters of the trading operation on the remote platform. In particular, upon the client's request, generated order # 2448959, deal # 2448959, with the actual execution prices: Sell - 1.31198, Buy - 1.31212. In this case, this means that the client's Buy has been executed at 1.31212 on the external trading platform.

On this basis, we can calculate the profit from the price difference:

The broker's profit =  $1.31237 - 1.31212 = 0.00025$ .

## Simulating the Netting Accounting System on MetaTrader 4

MetaTrader 4 trading platform supports only the hedging accounting system. This system allows you to have unlimited number of trading positions on a single financial instrument. In addition to hedging, MetaTrader 5 supports the netting system. This system is used on exchanges allowing a trader to have only one cumulative position at a symbol. Making a trade on the same instrument may change the existing position's volume, close it completely or reverse it.

If netting is used on MetaTrader 5 side, the gateway will try to simulate it on MetaTrader 4 external server.

The gateway automatically assigns an identification number to an account in the external server in the account settings of each client whose trades are passed to the external server (in case the number has not been already specified manually).